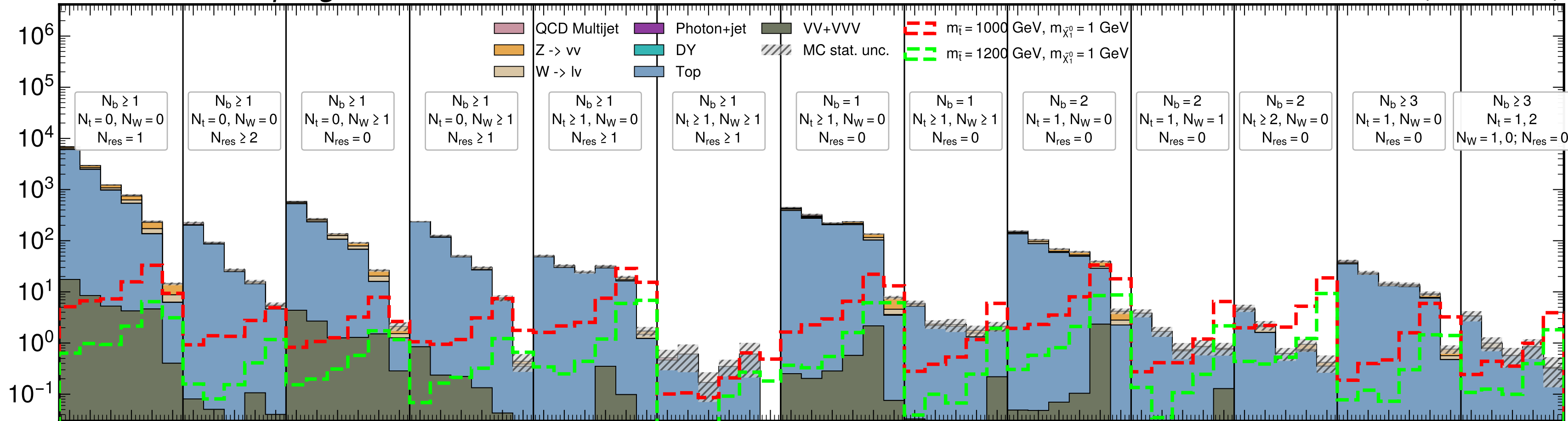
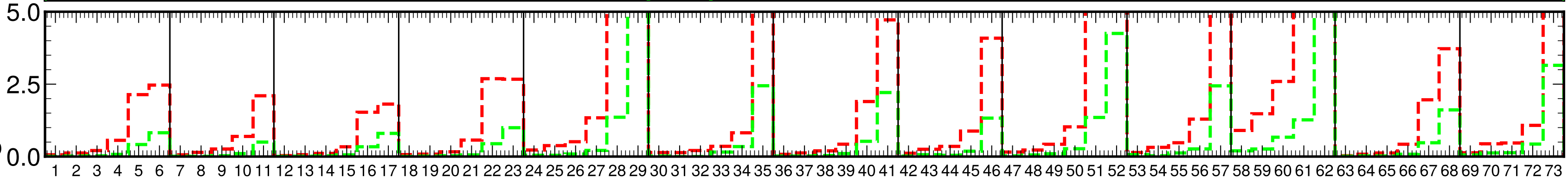


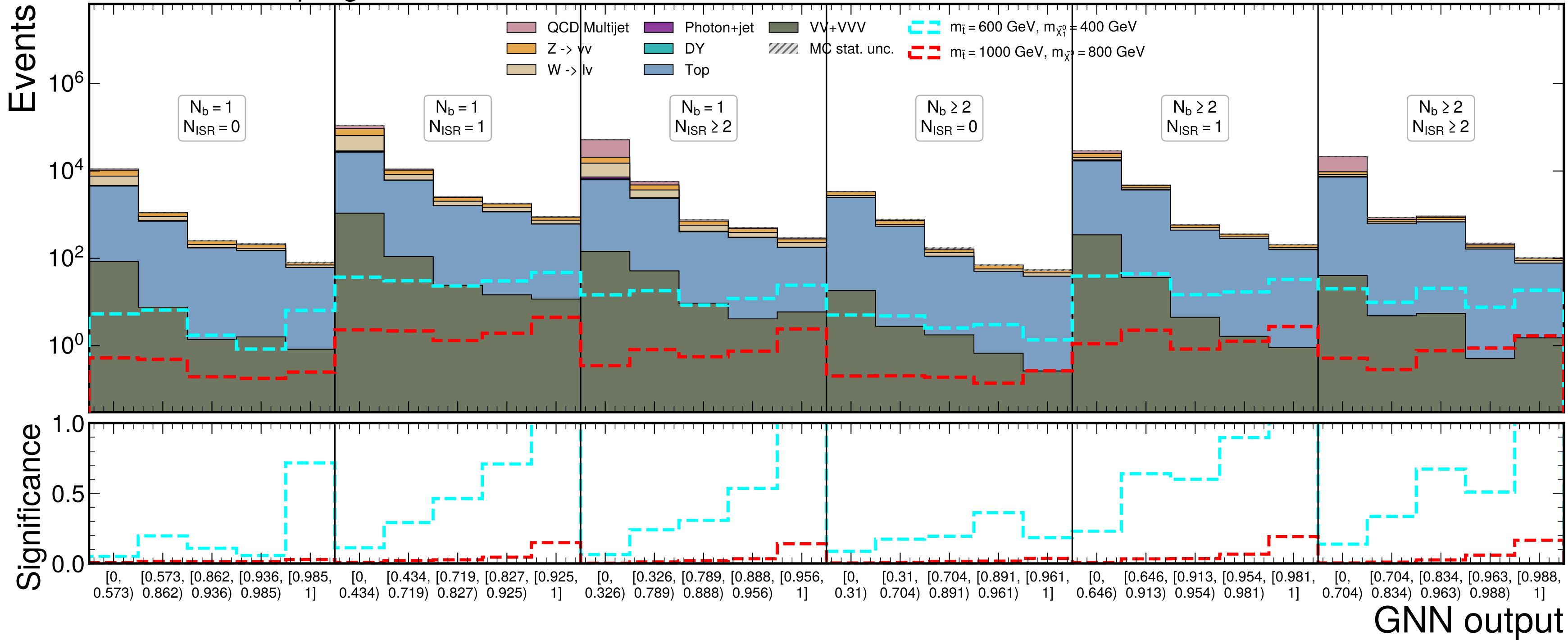
Events



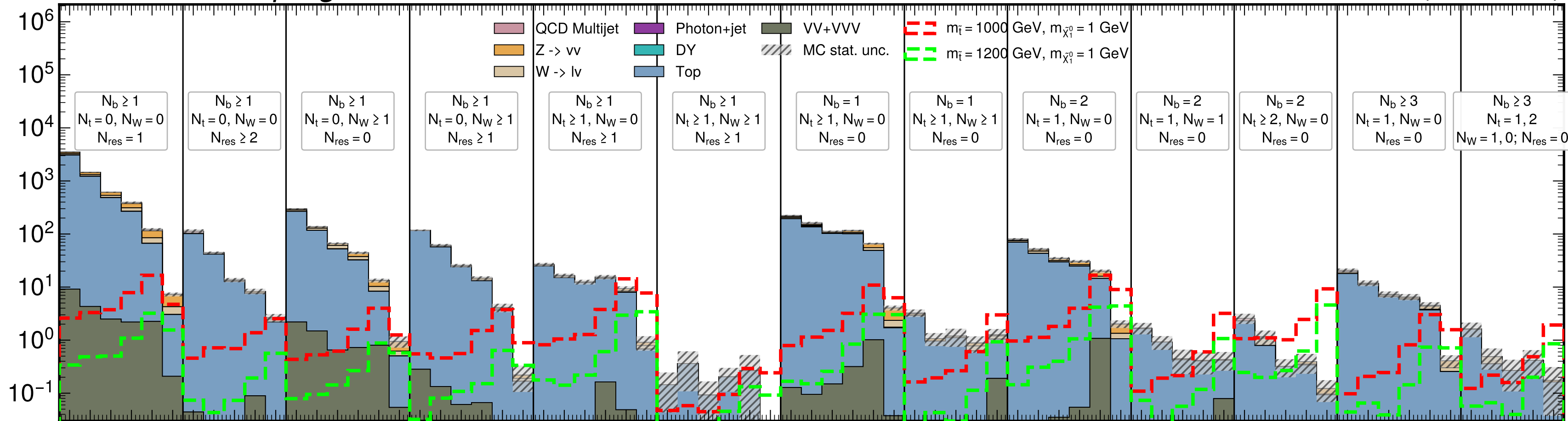
Significance



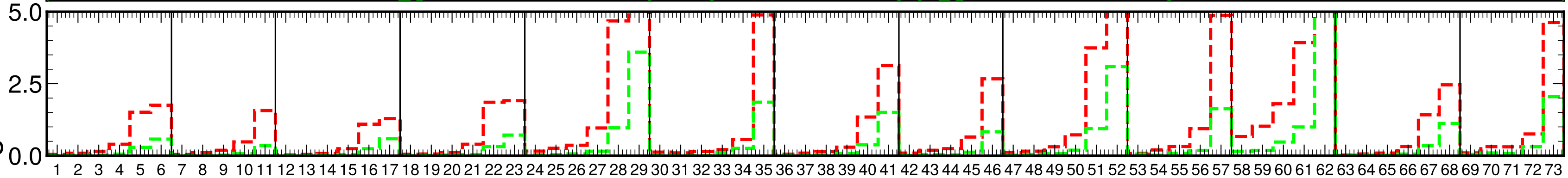
Search bin



Events

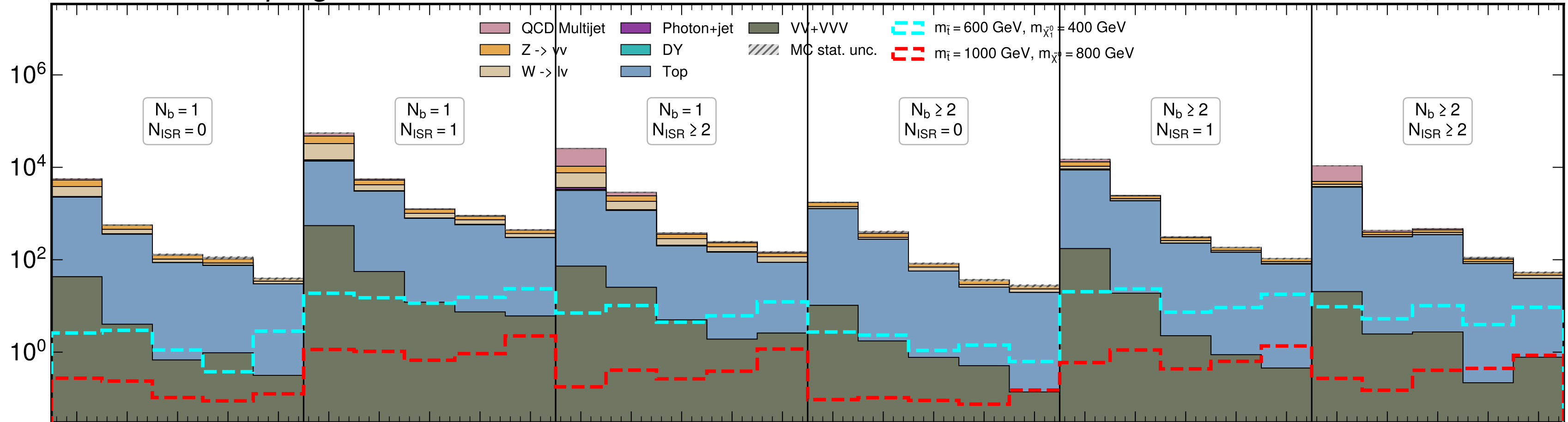


Significance

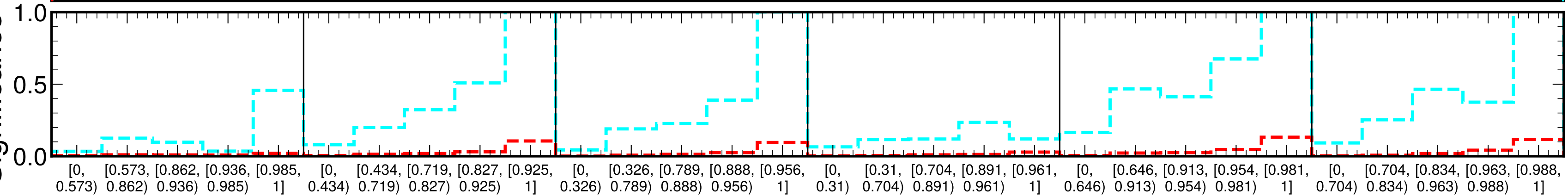


Search bin

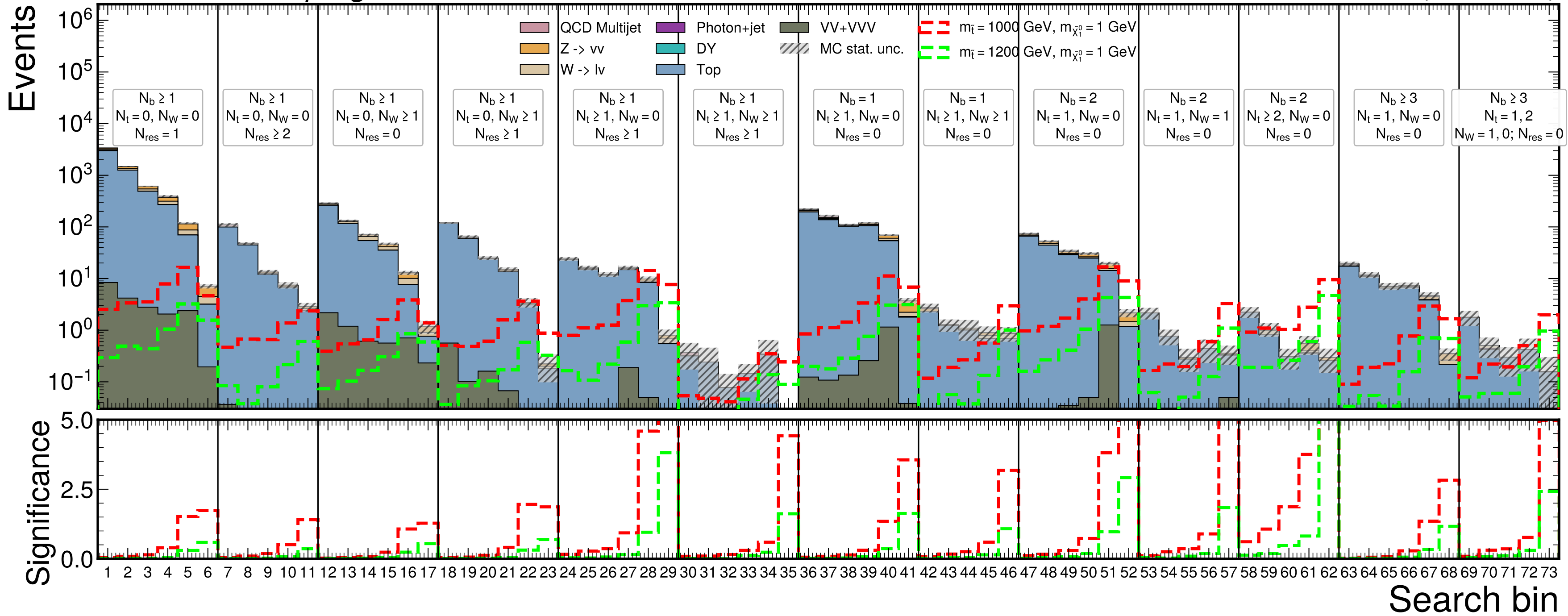
Events



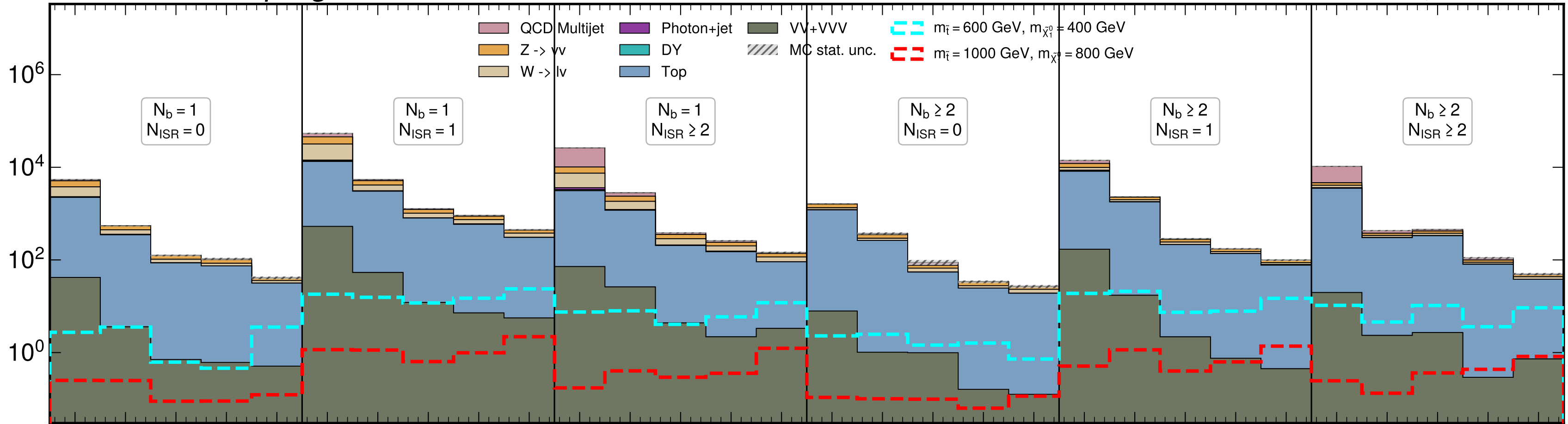
Significance



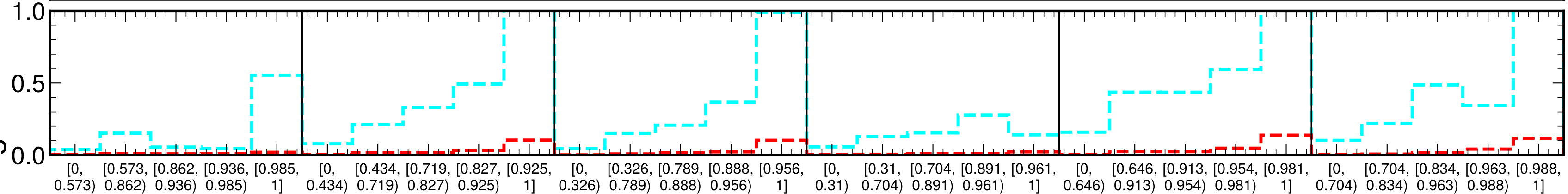
GNN output



Events

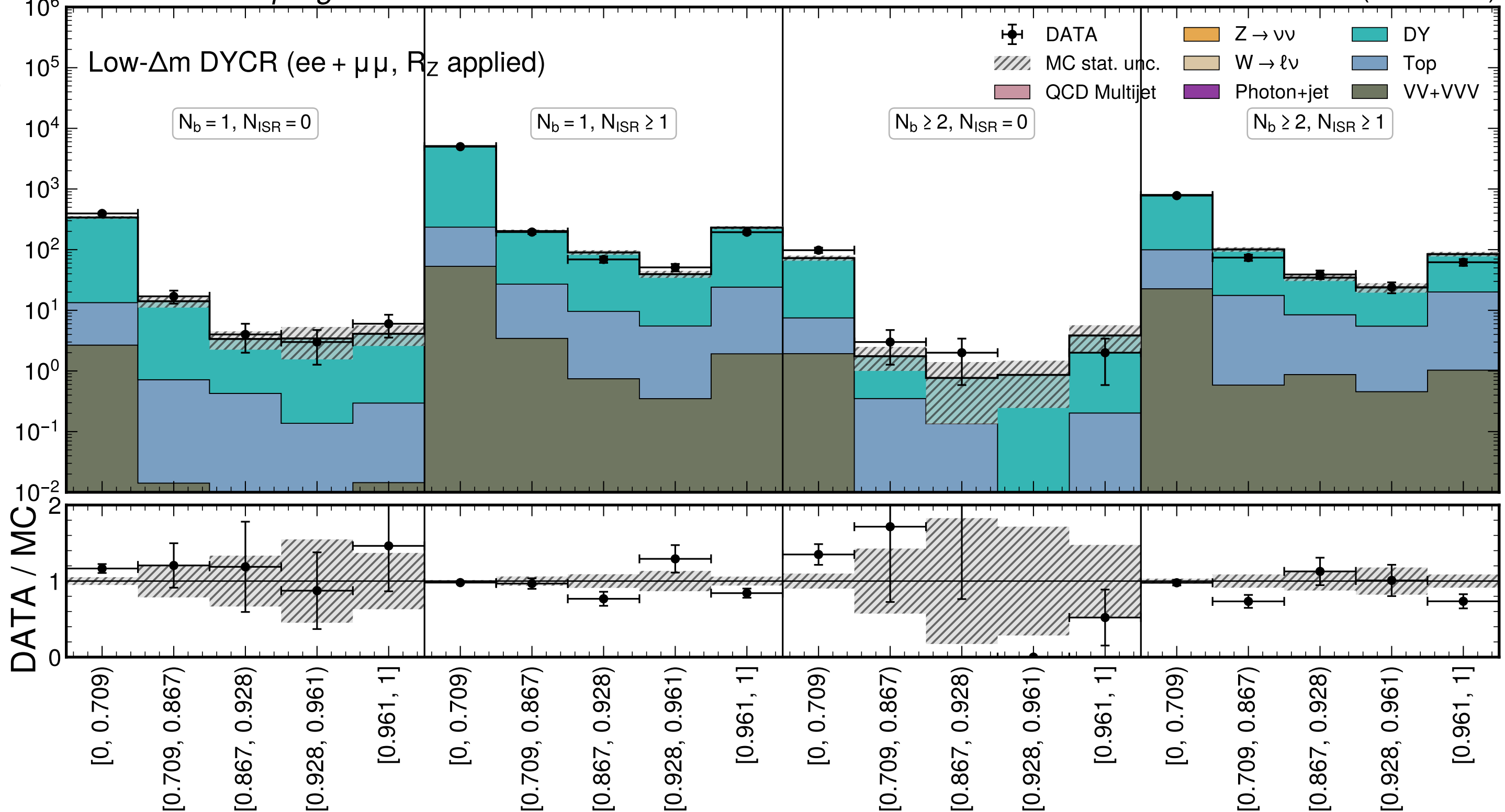


Significance

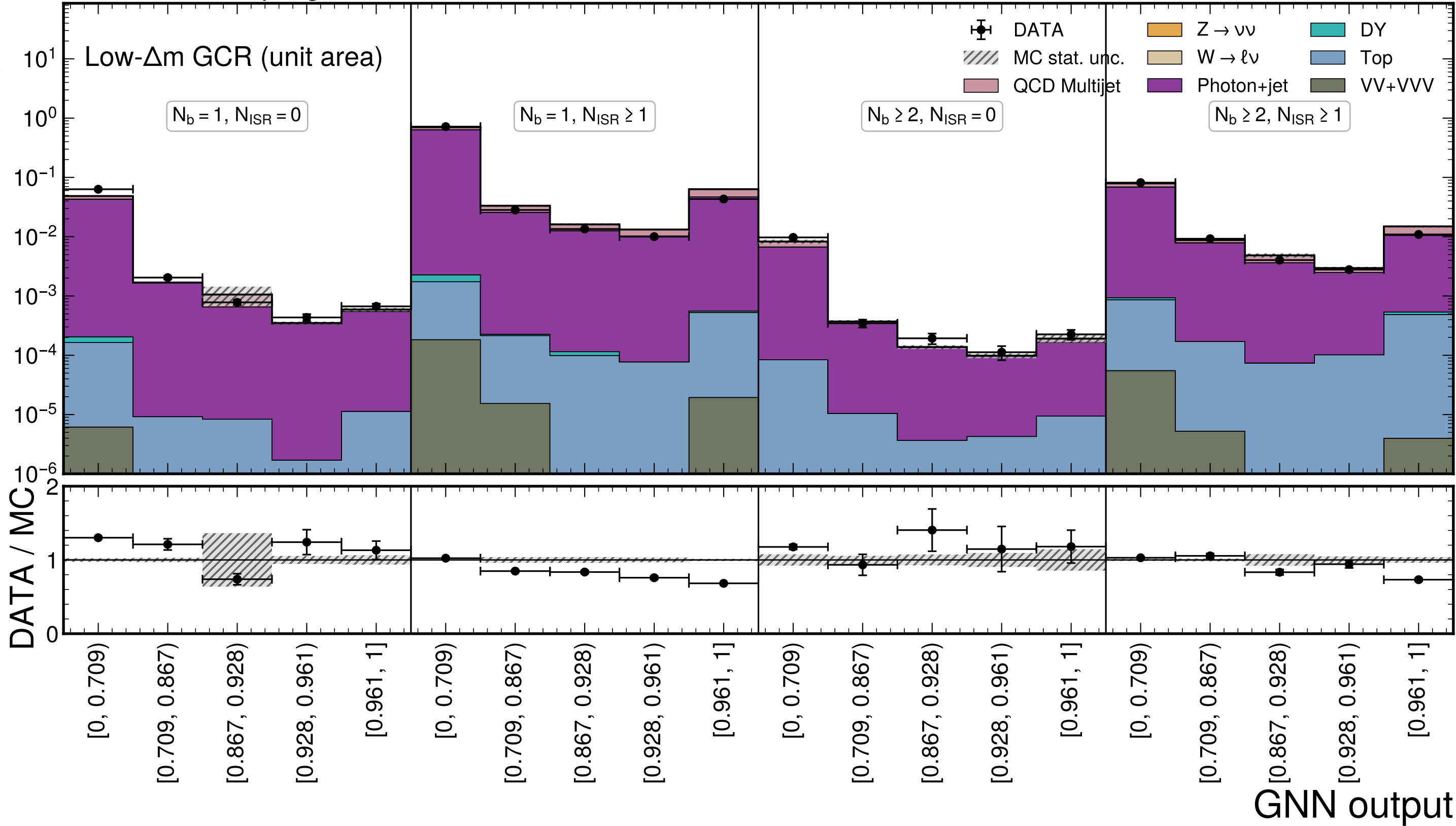


GNN output

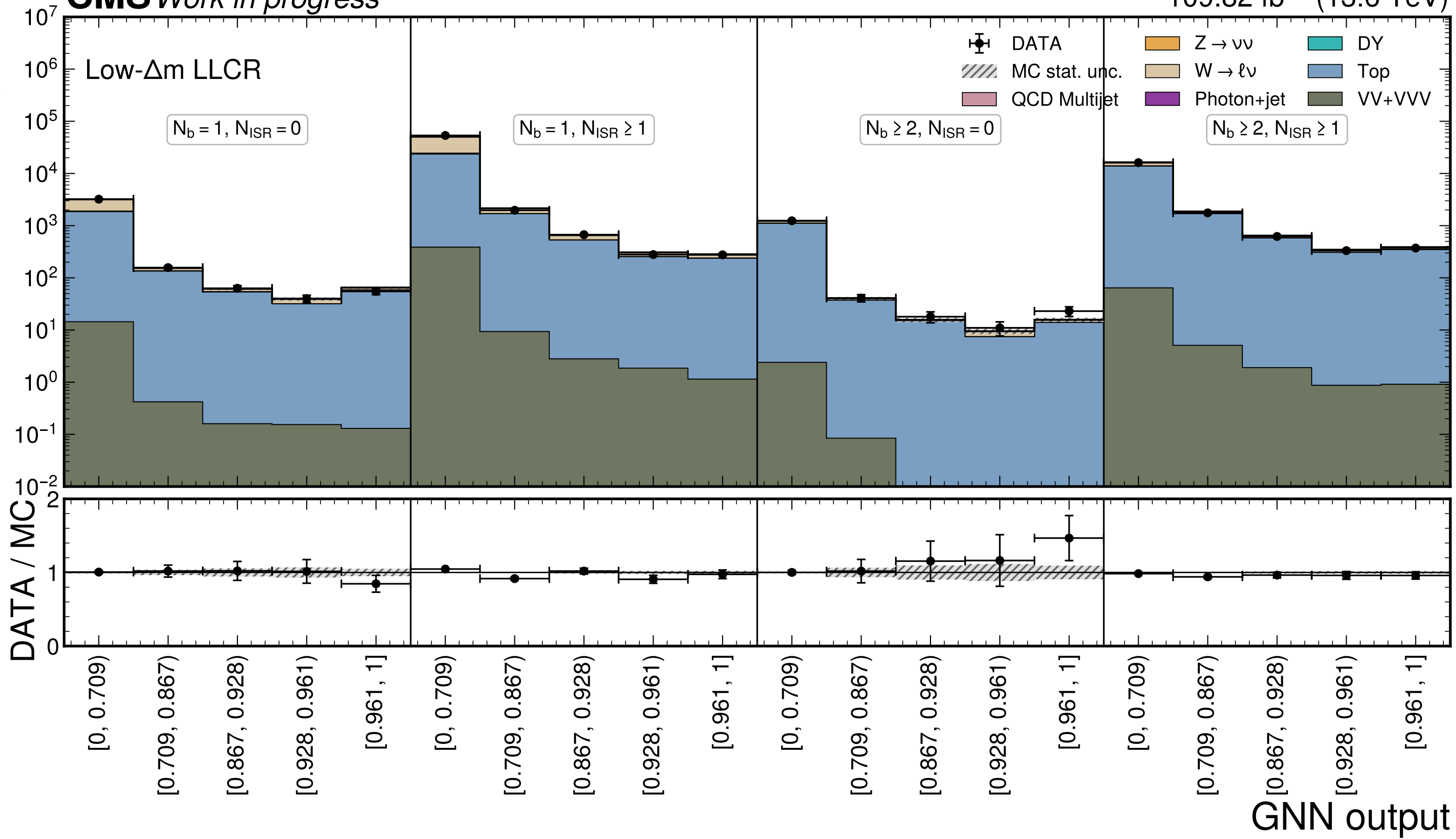
Events / bin



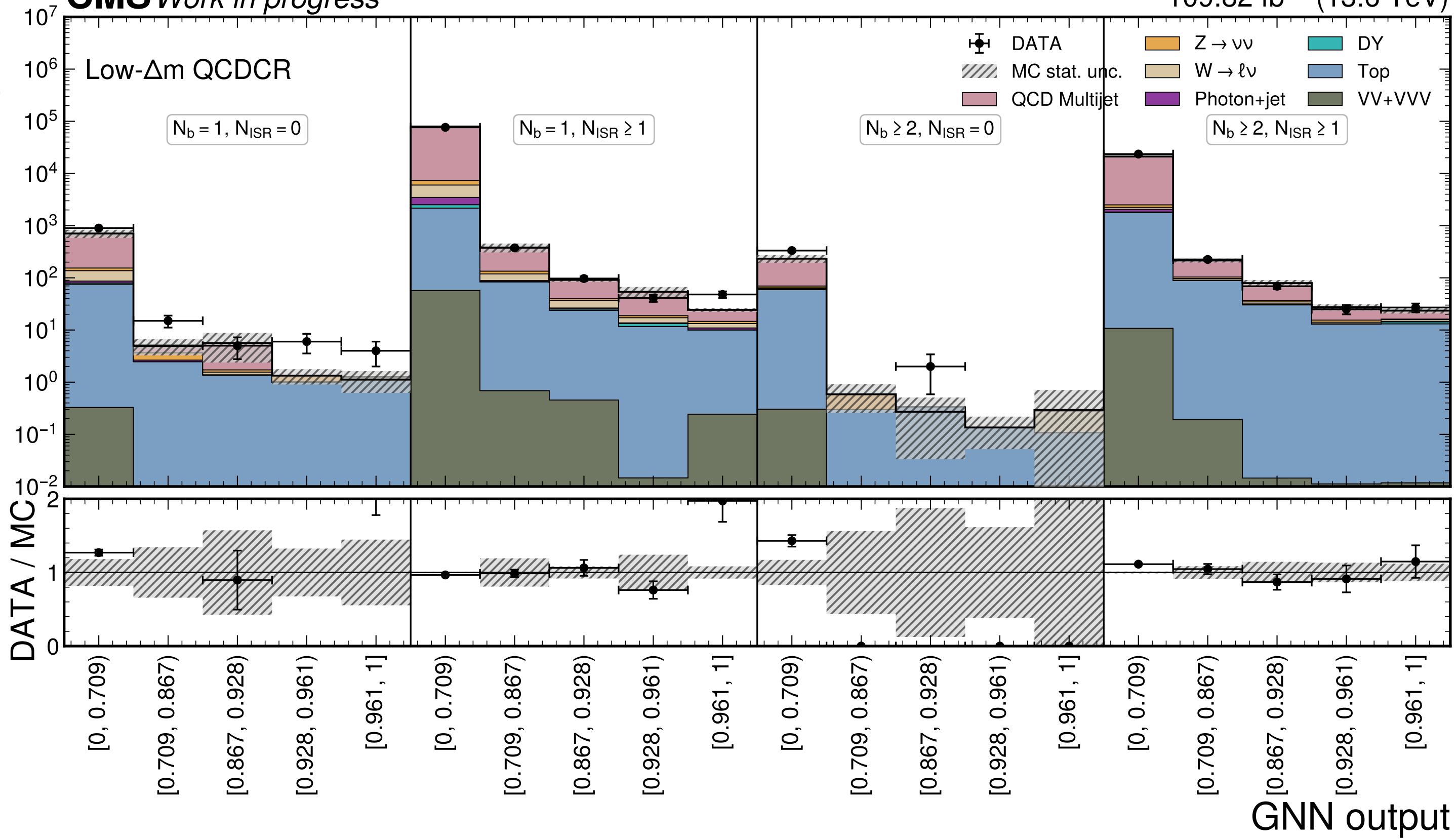
Normalized events / bin



Events / bin



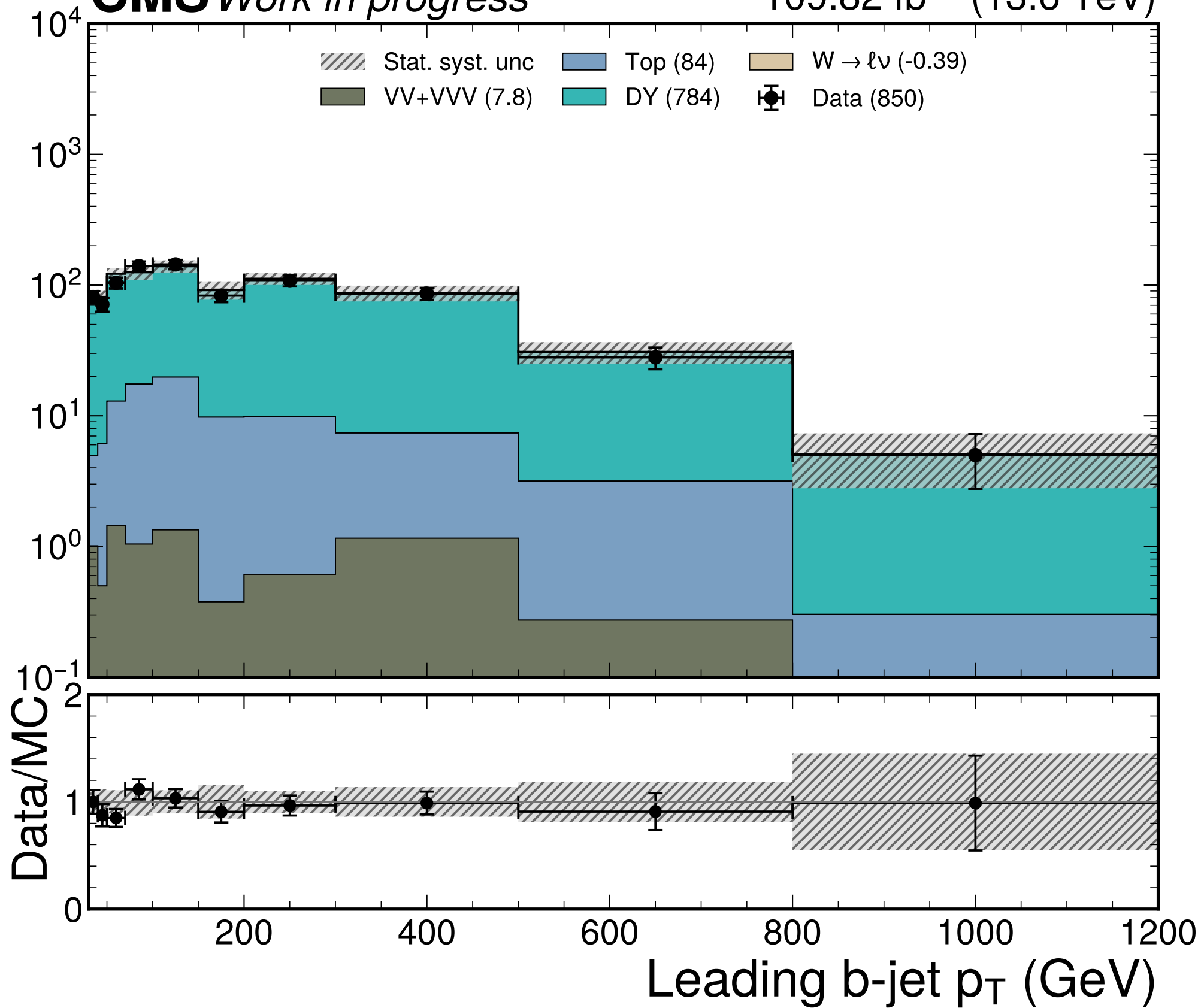
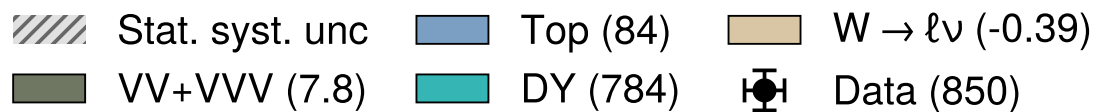
Events / bin



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

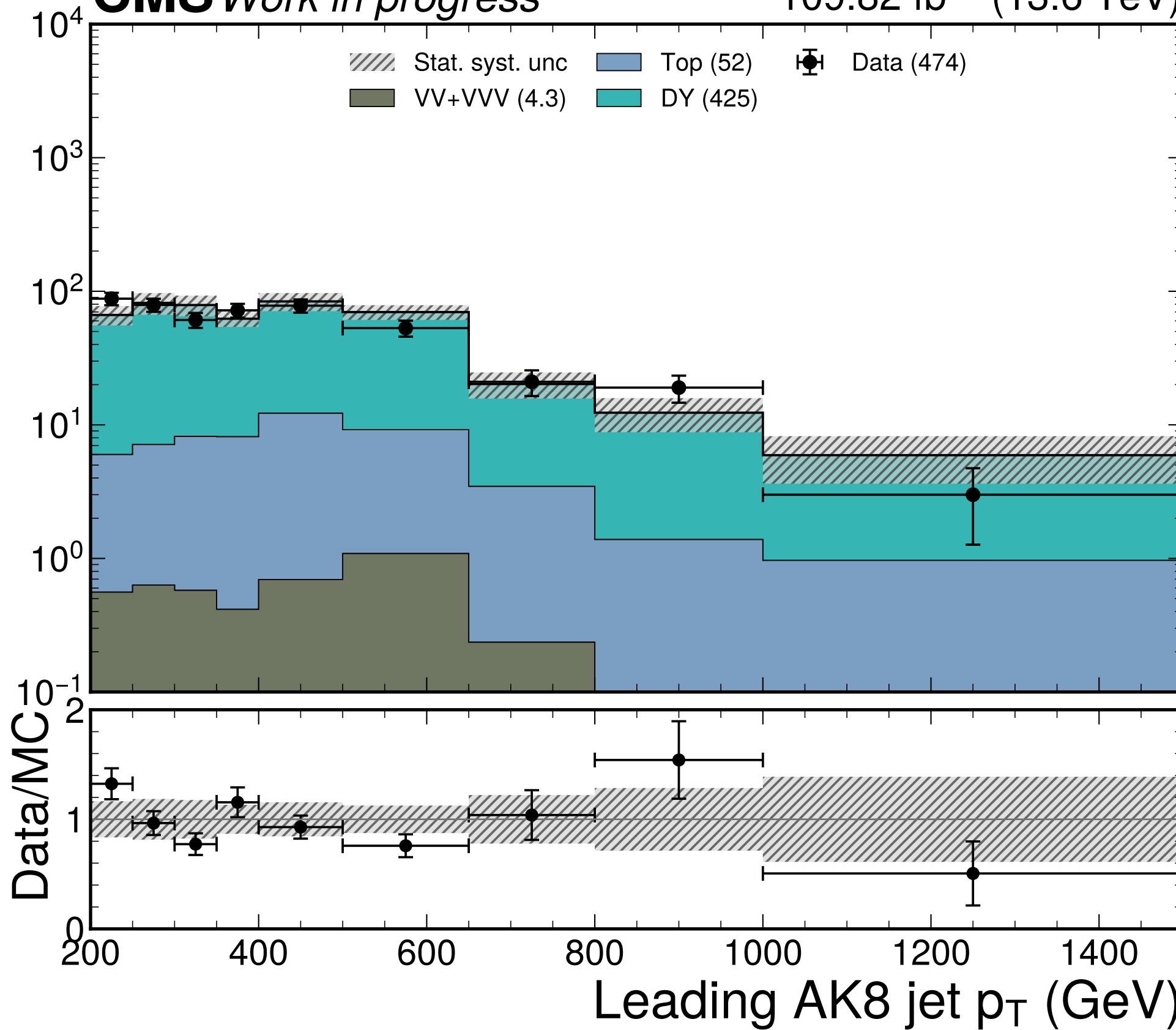
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

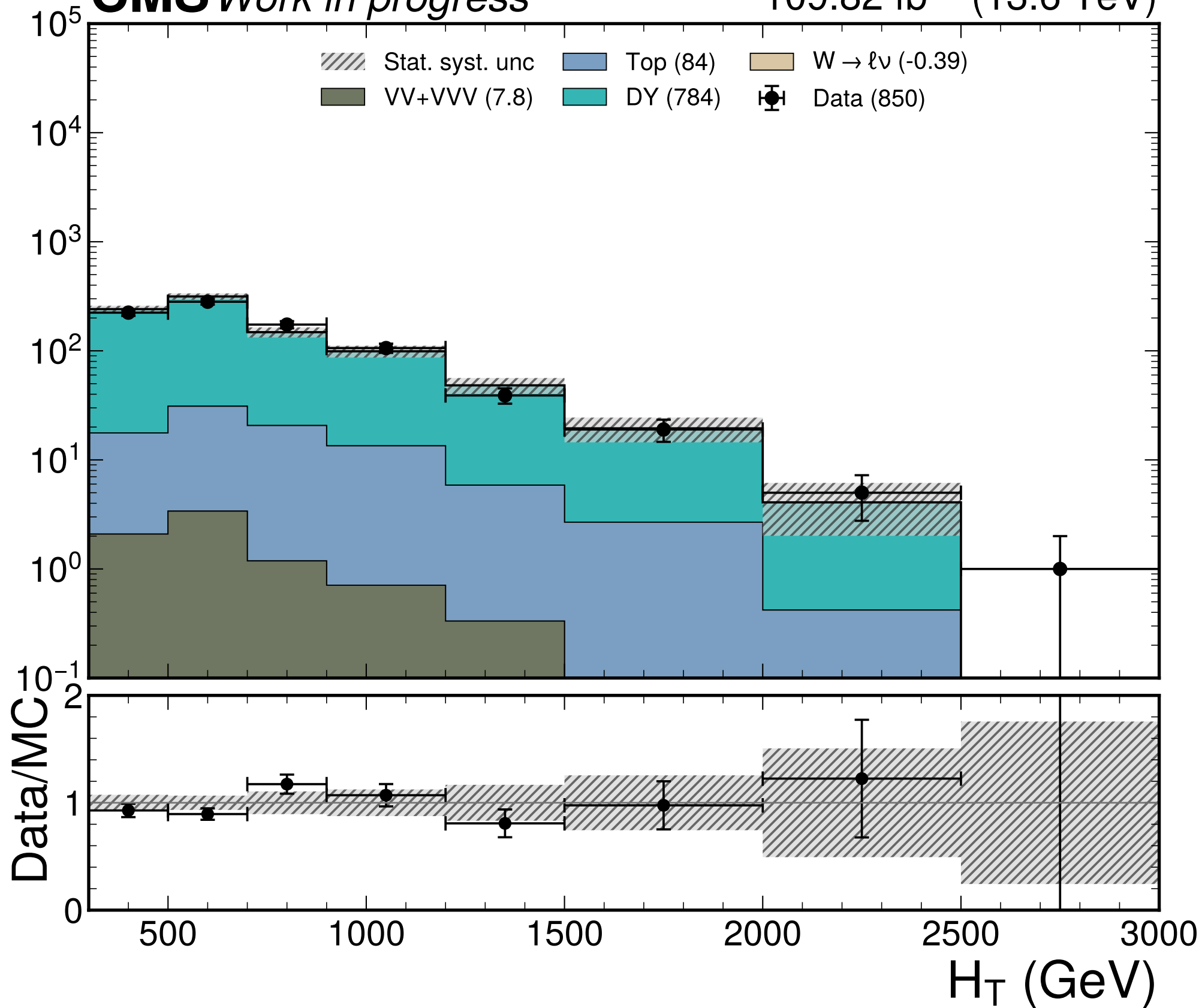
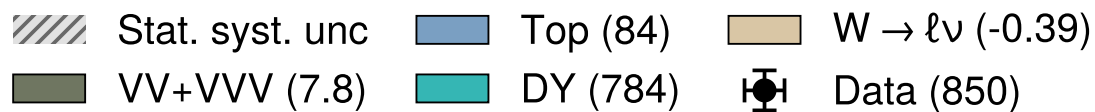
Events



CMS *Work in progress*

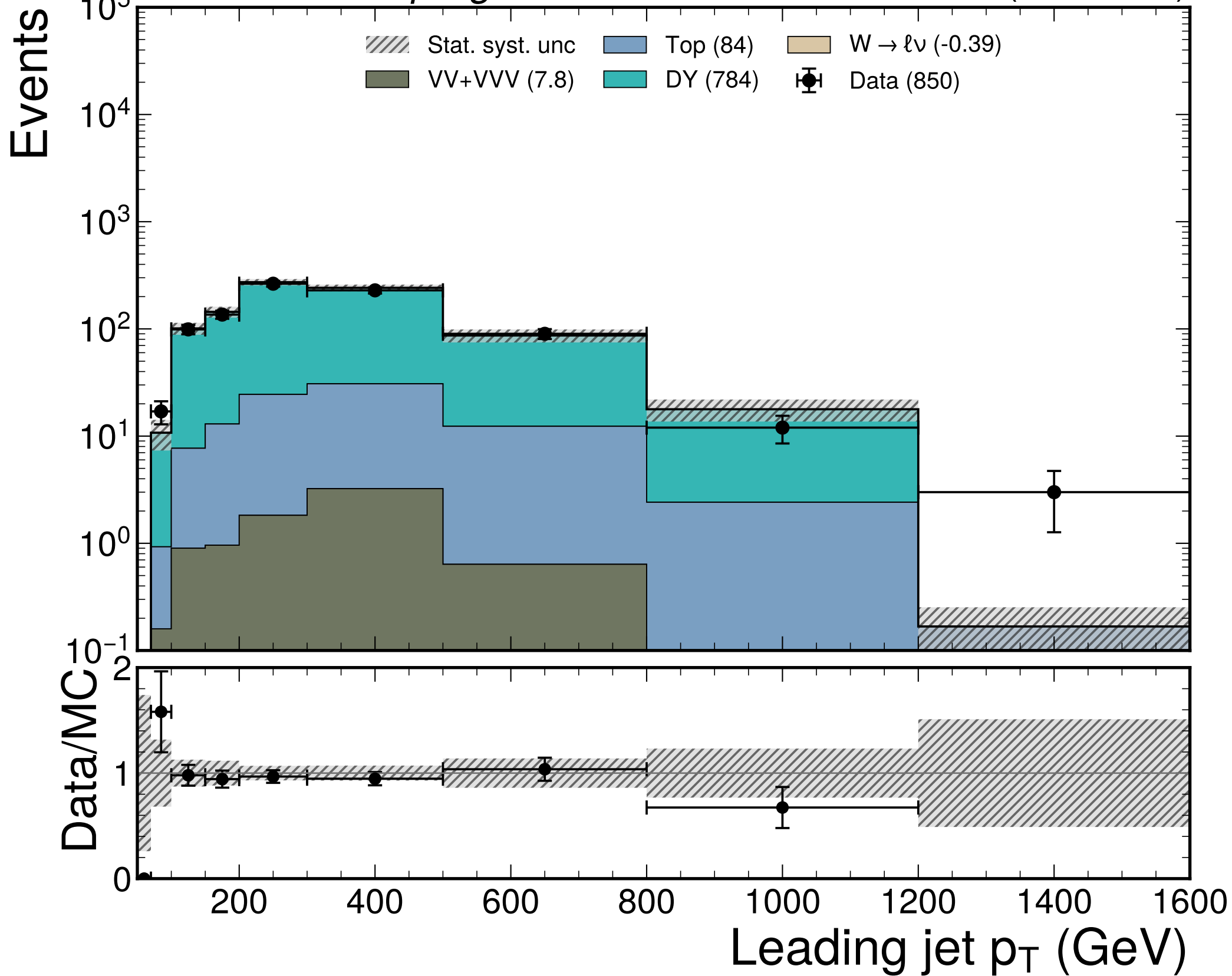
109.82 fb⁻¹ (13.6 TeV)

Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

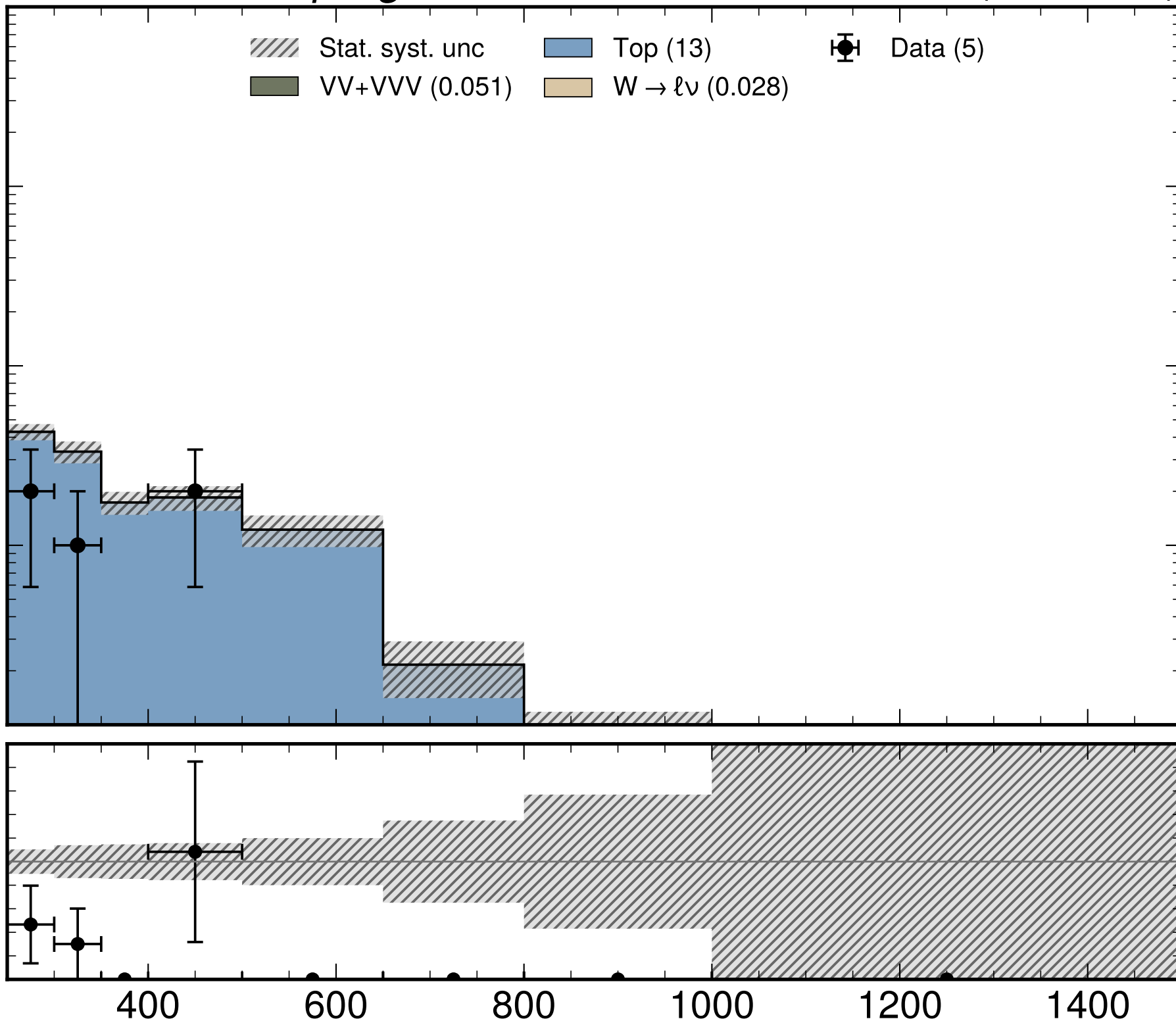
109.82 fb⁻¹ (13.6 TeV)

Events

Stat. syst. unc Top (13) Data (5)
VV+VVV (0.051) W → ℓν (0.028)

Data/MC

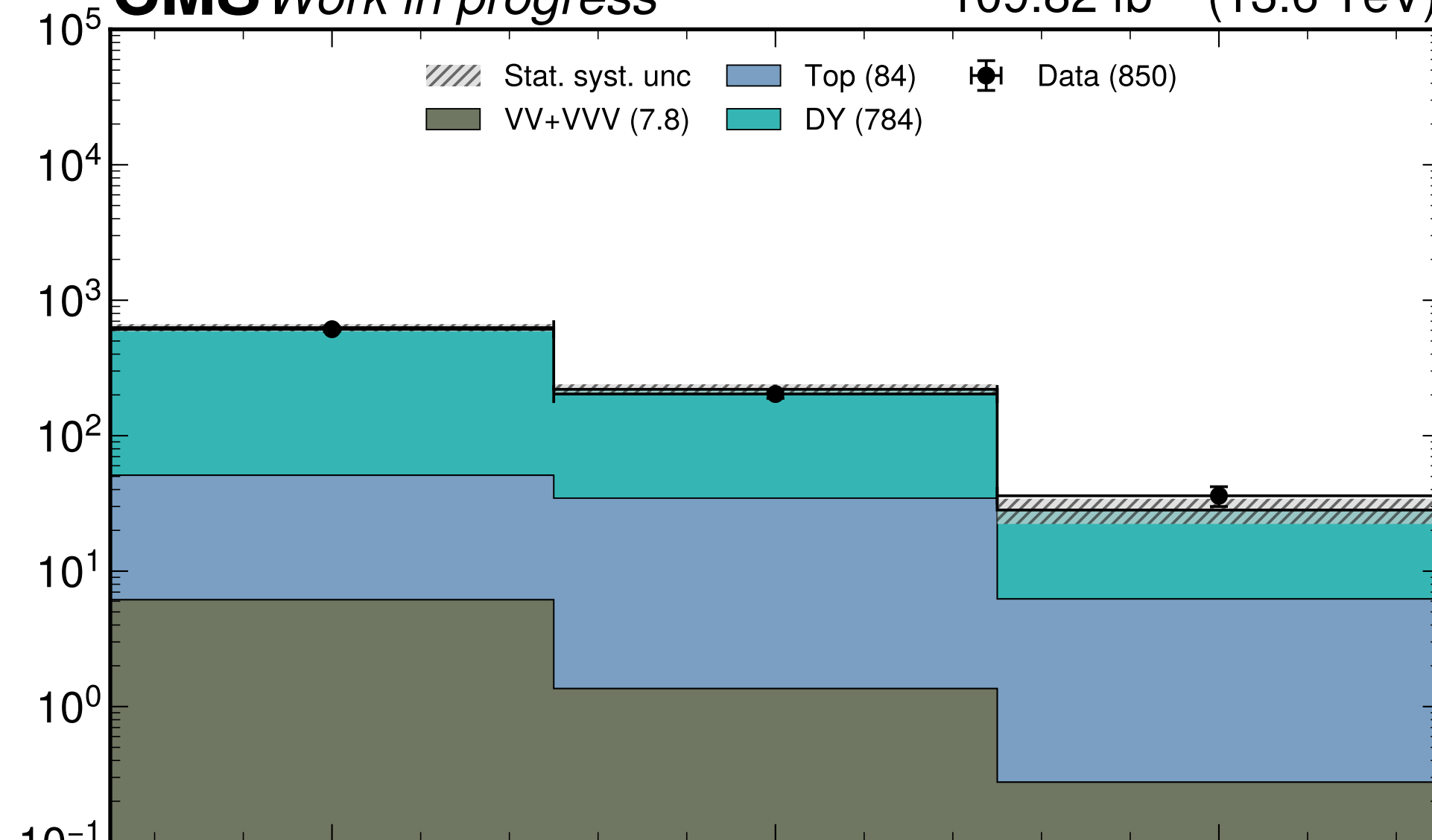
p_T^{miss} (GeV)



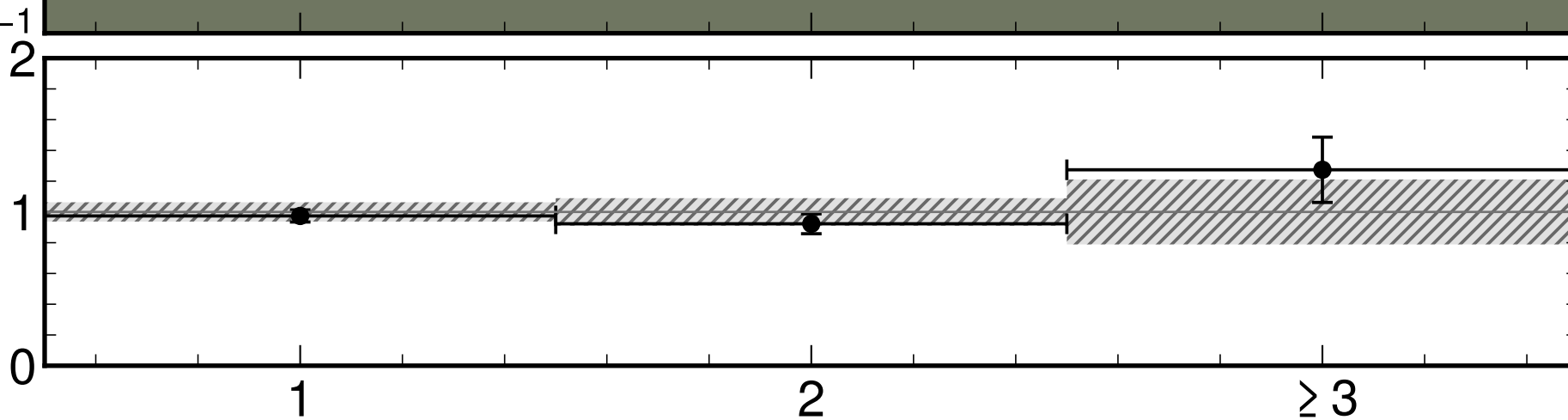
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

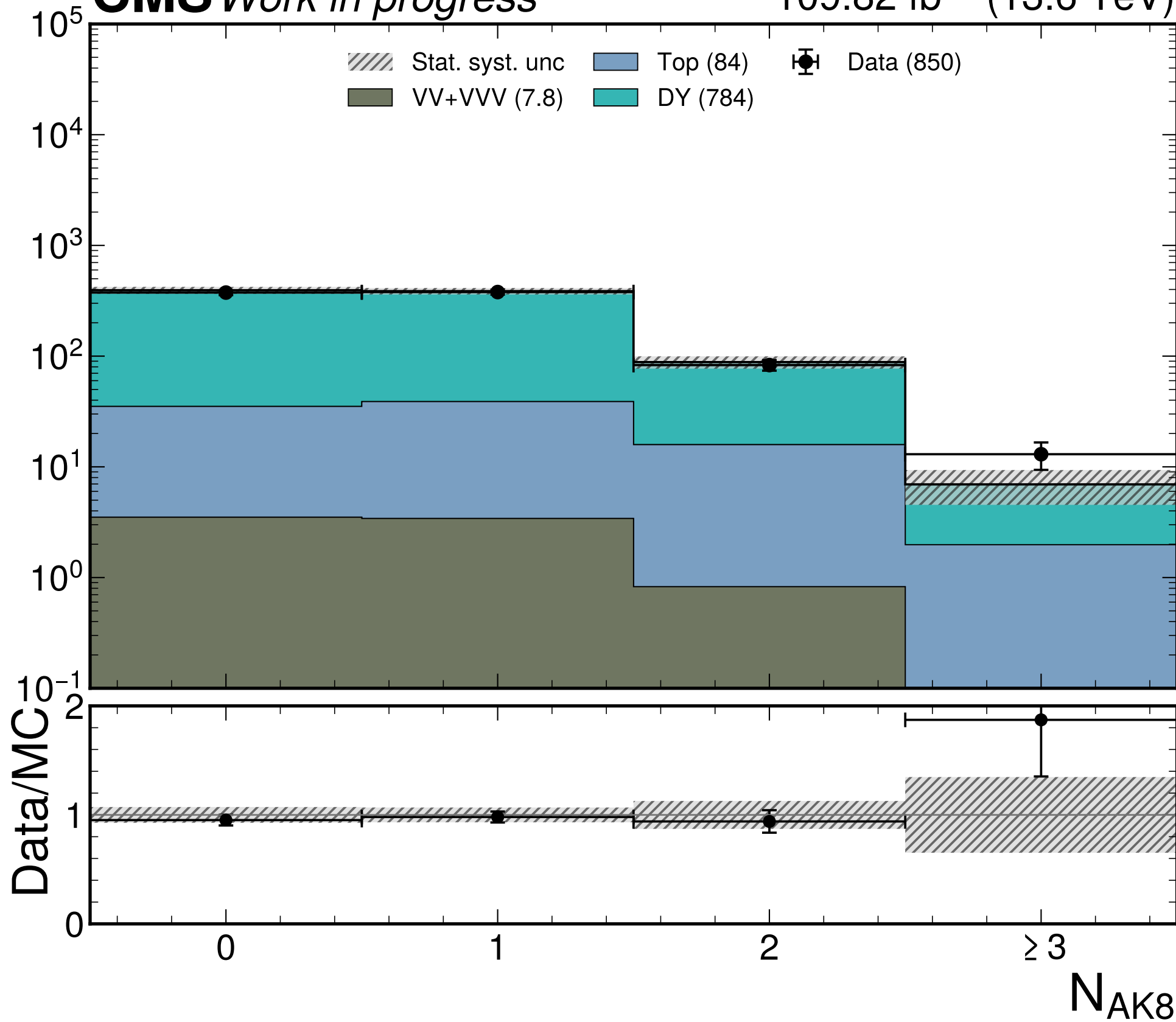


N_b

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

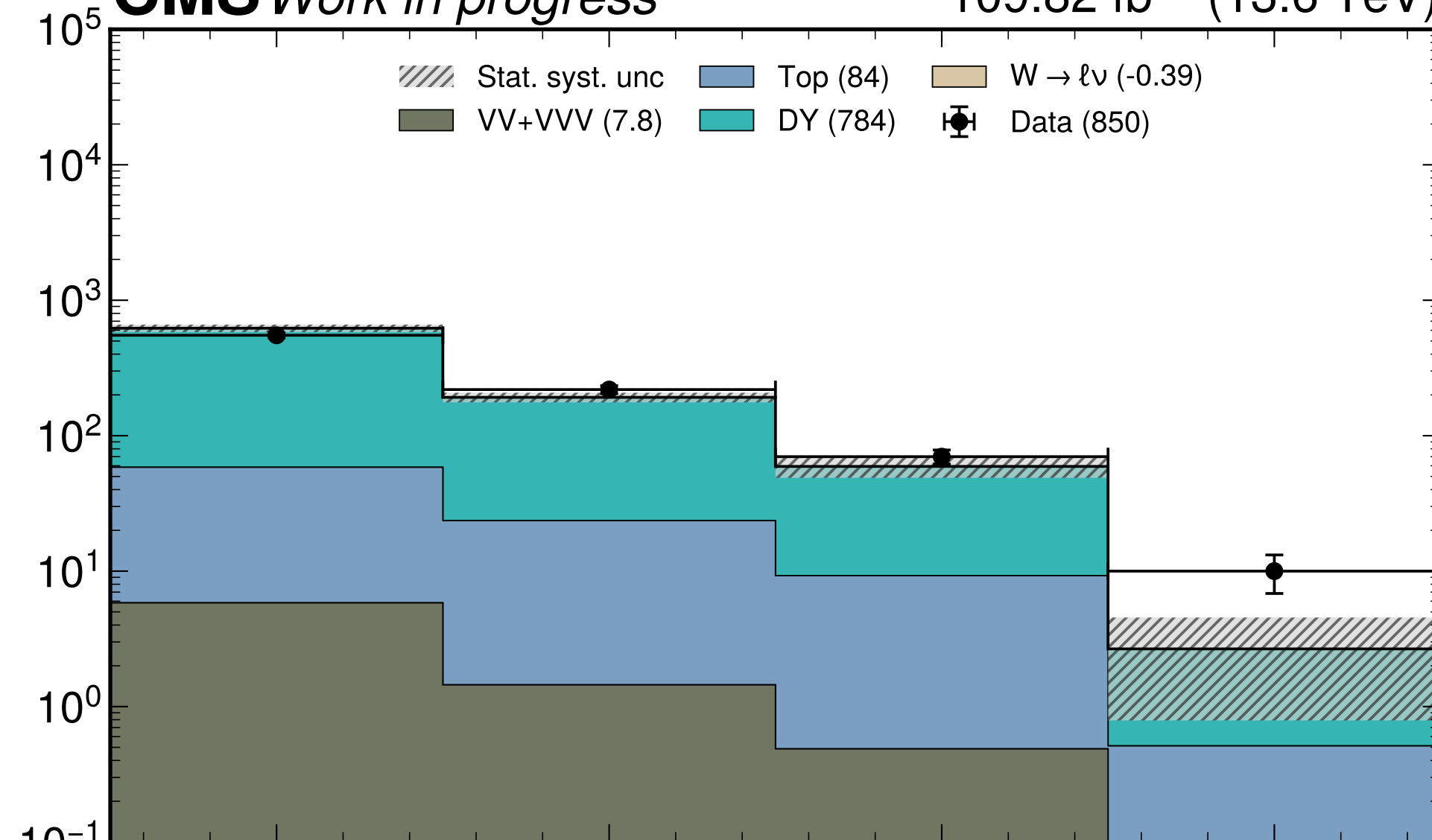
Events



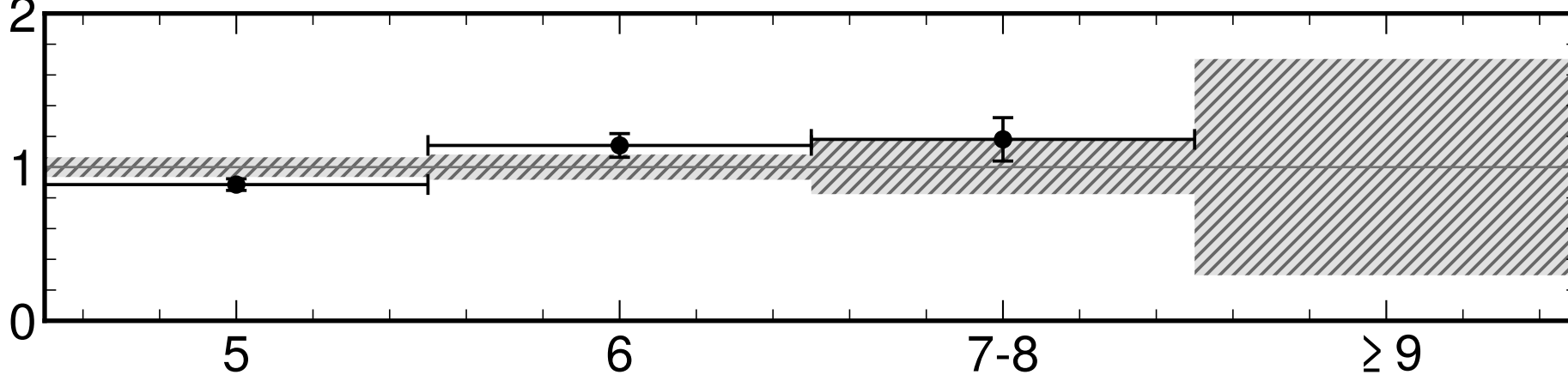
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

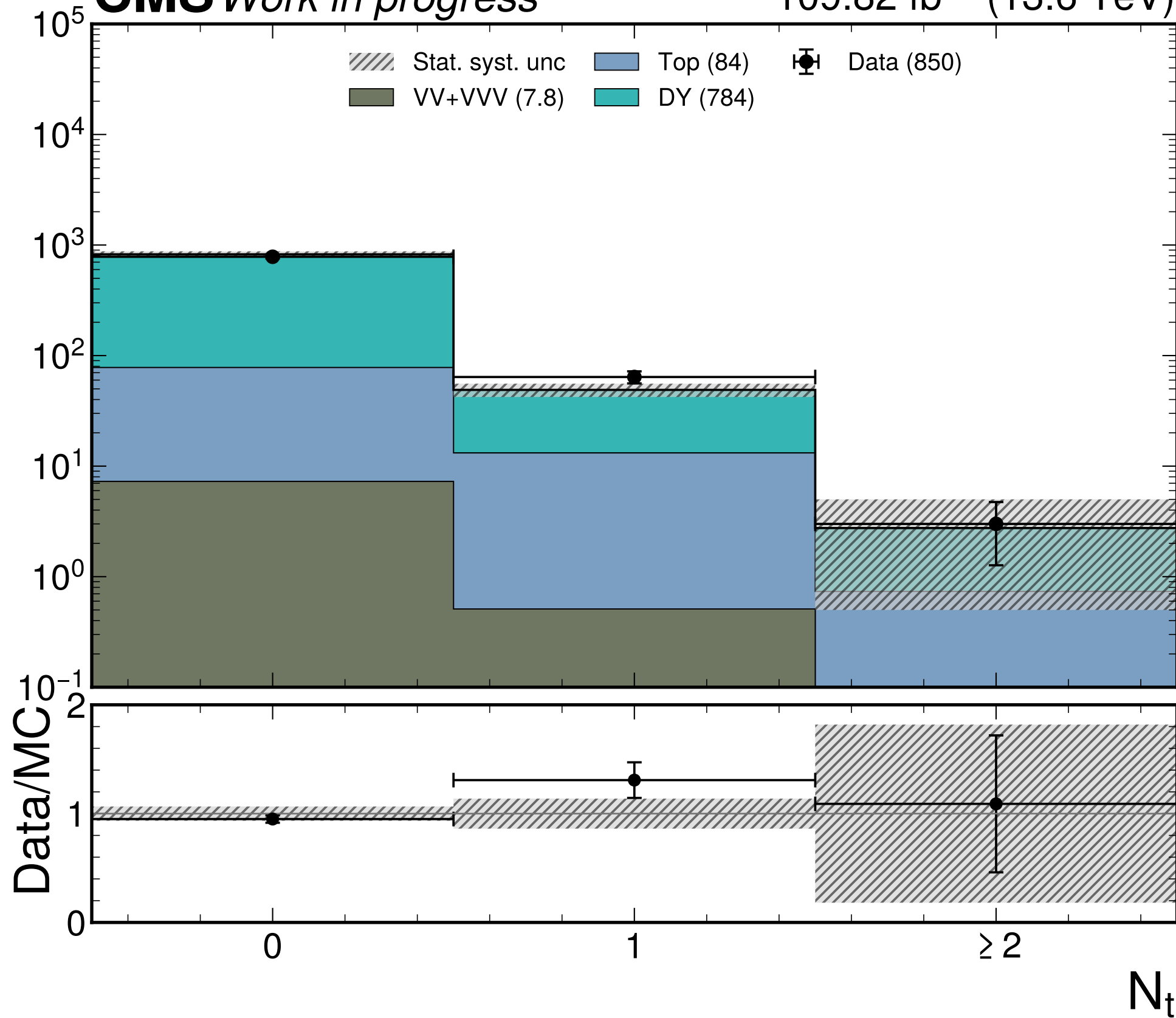


N_j

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

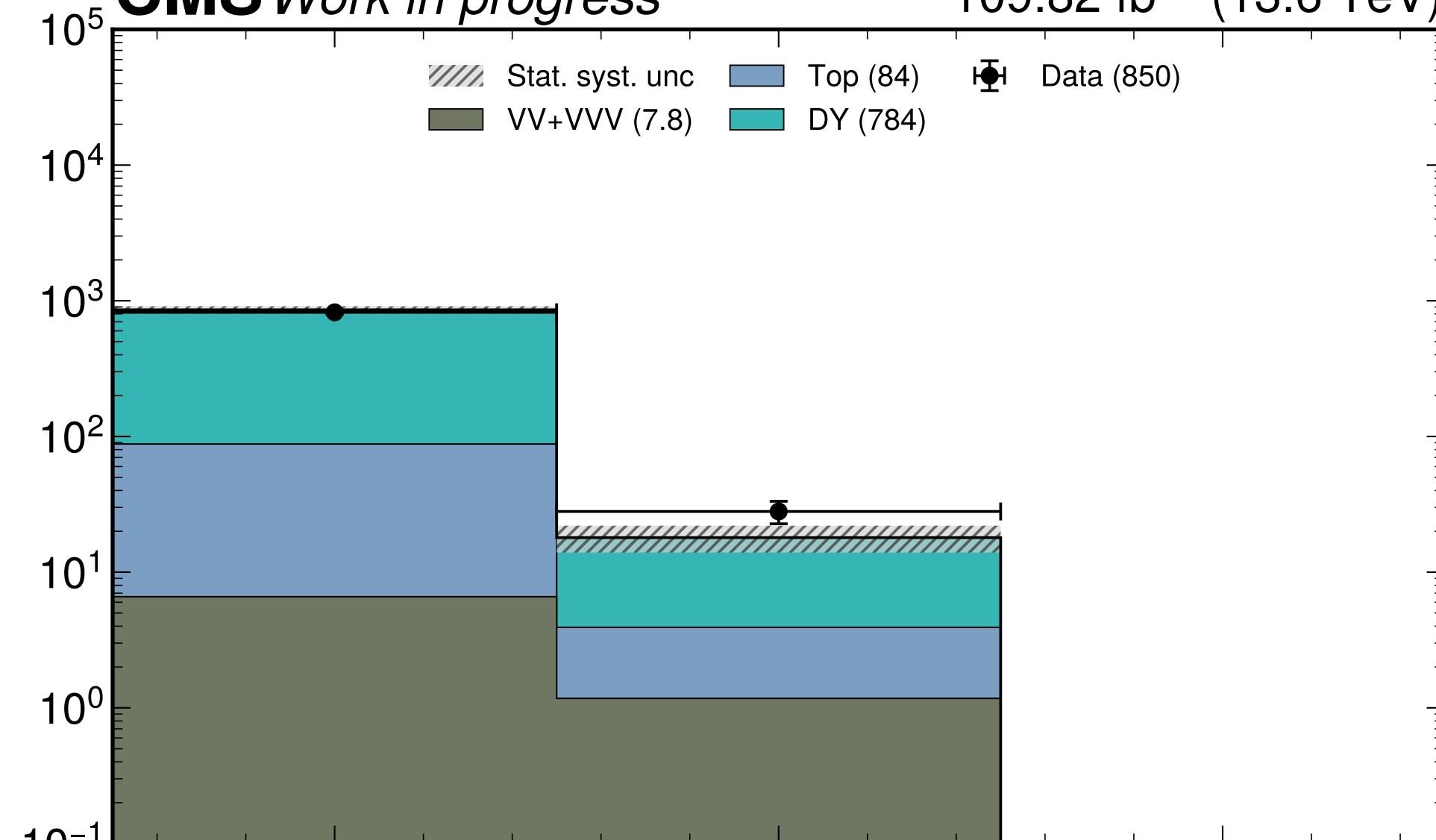
Events



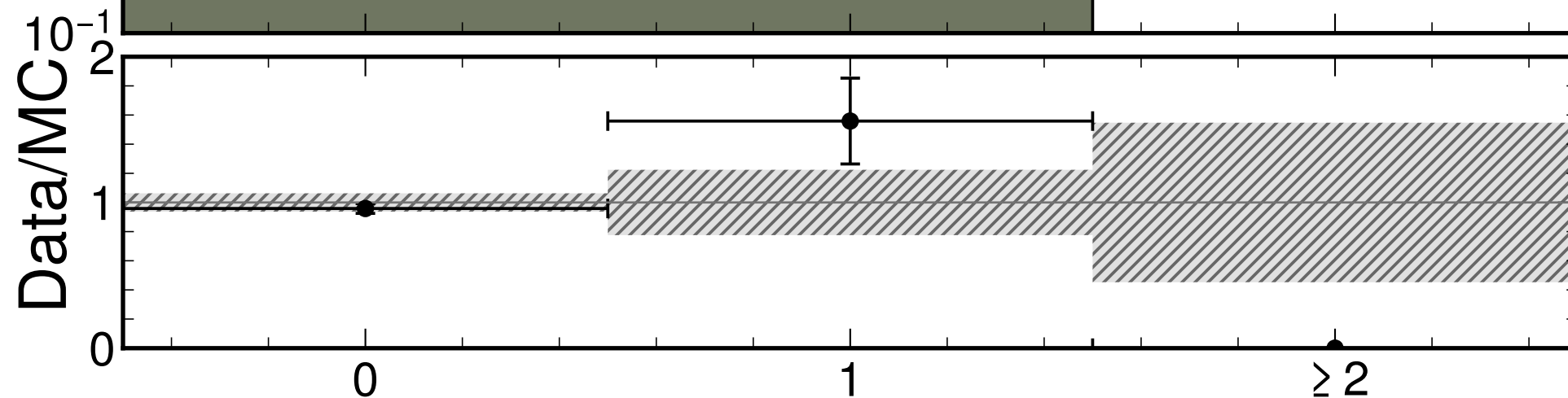
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC



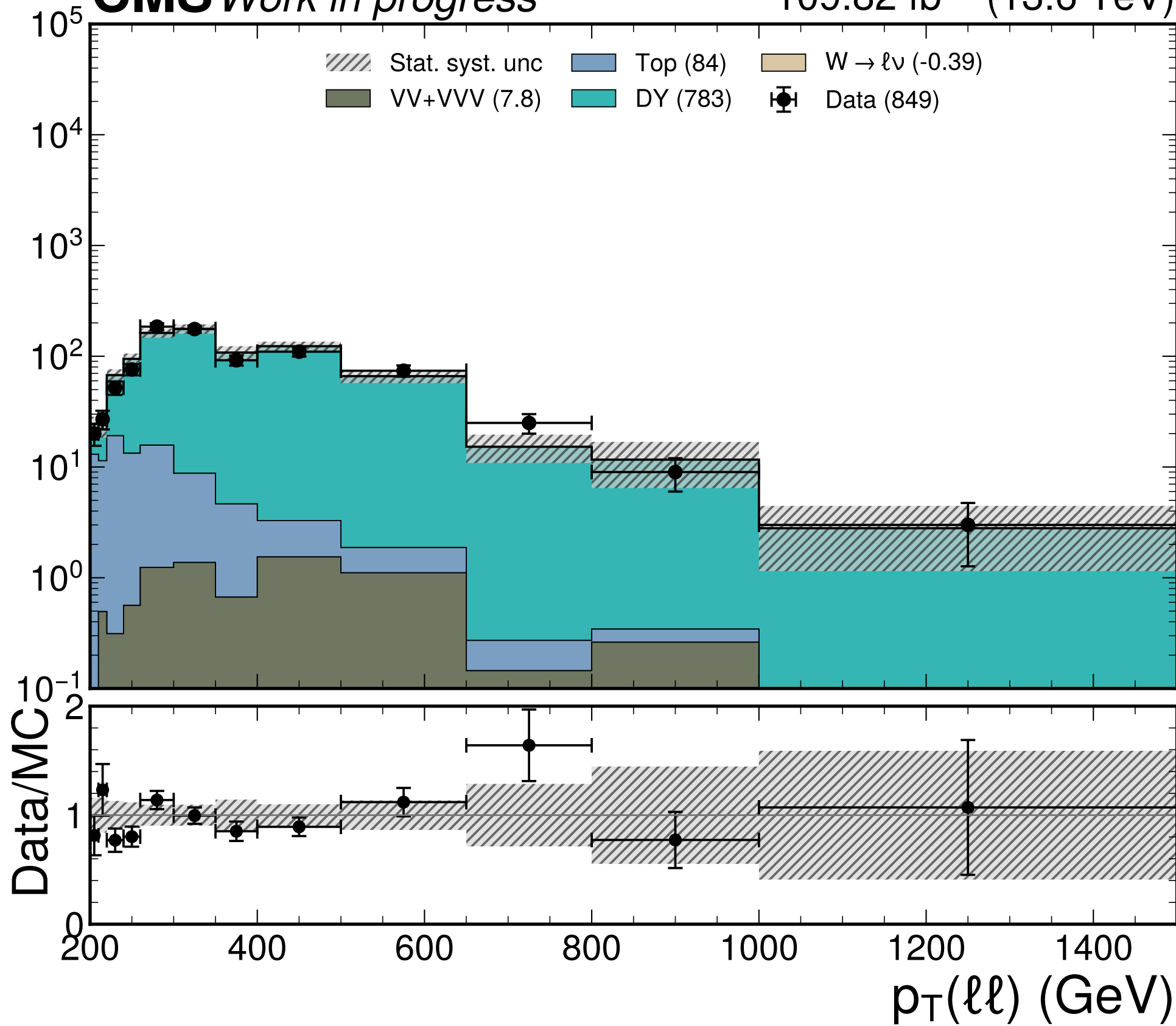
N_W

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events

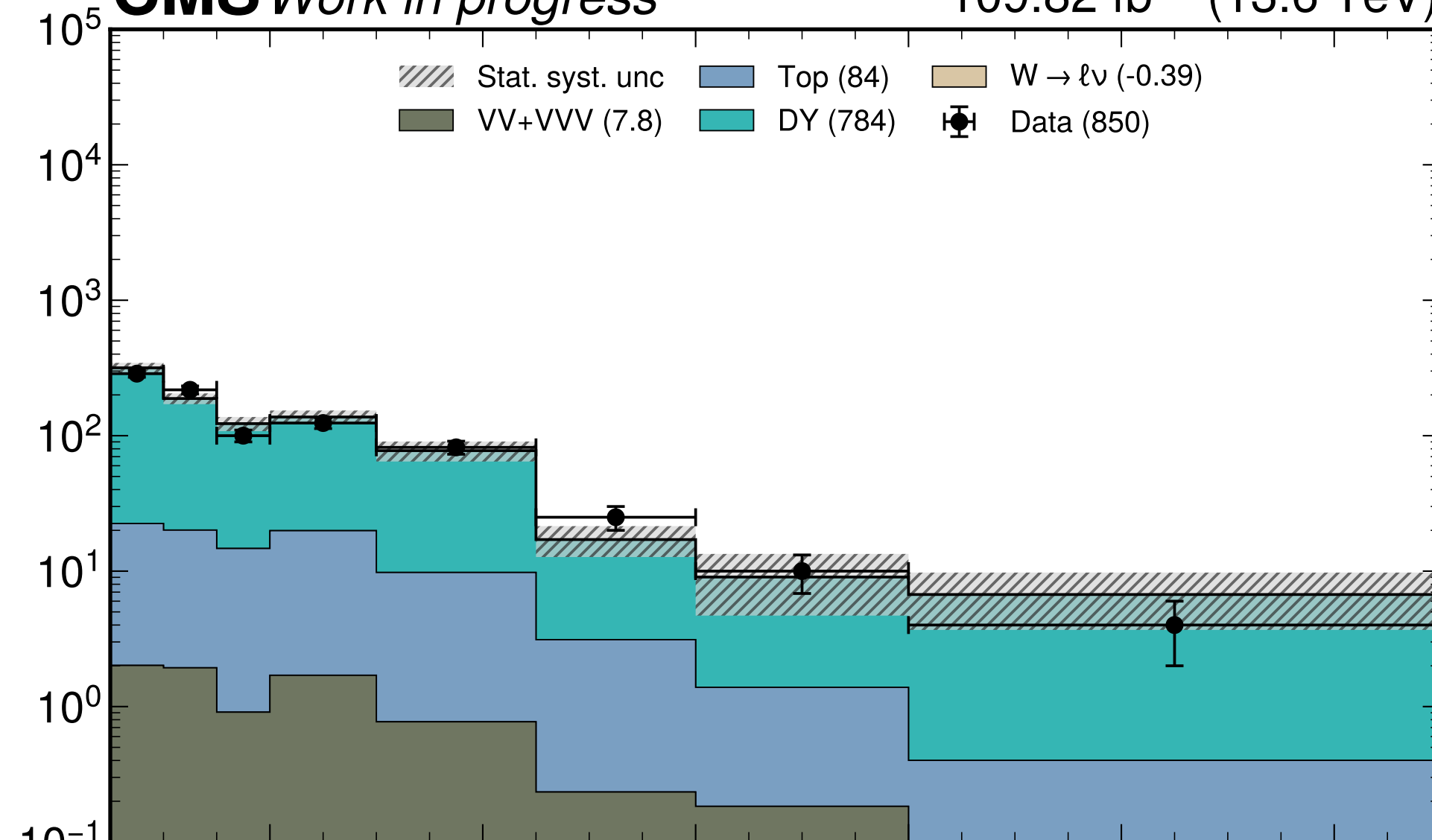
Stat. syst. unc Top (84) W → ℓν (-0.39)
VV+VVV (7.8) DY (783) Data (849)



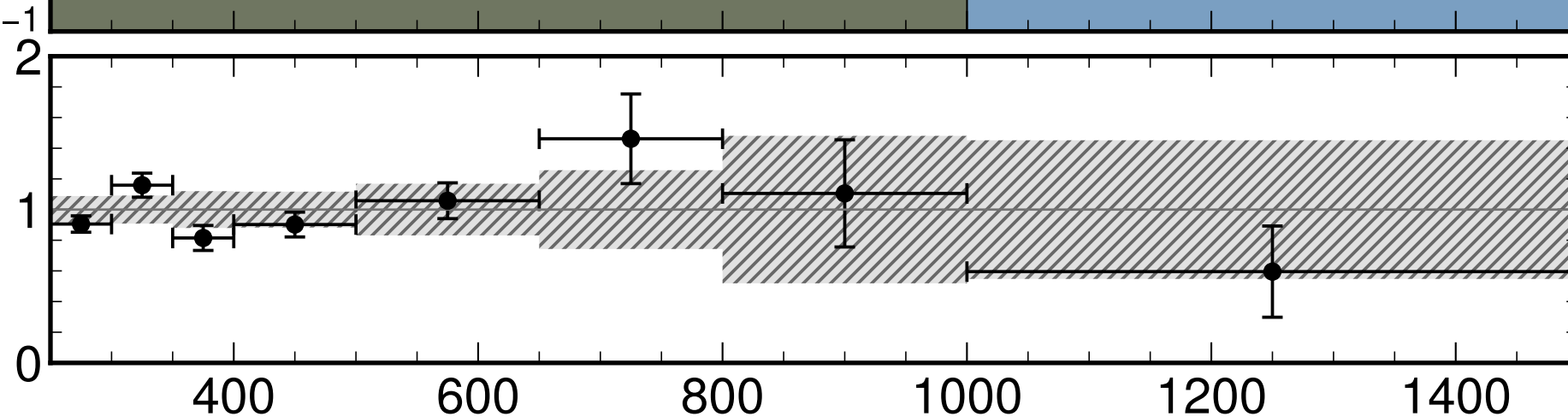
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



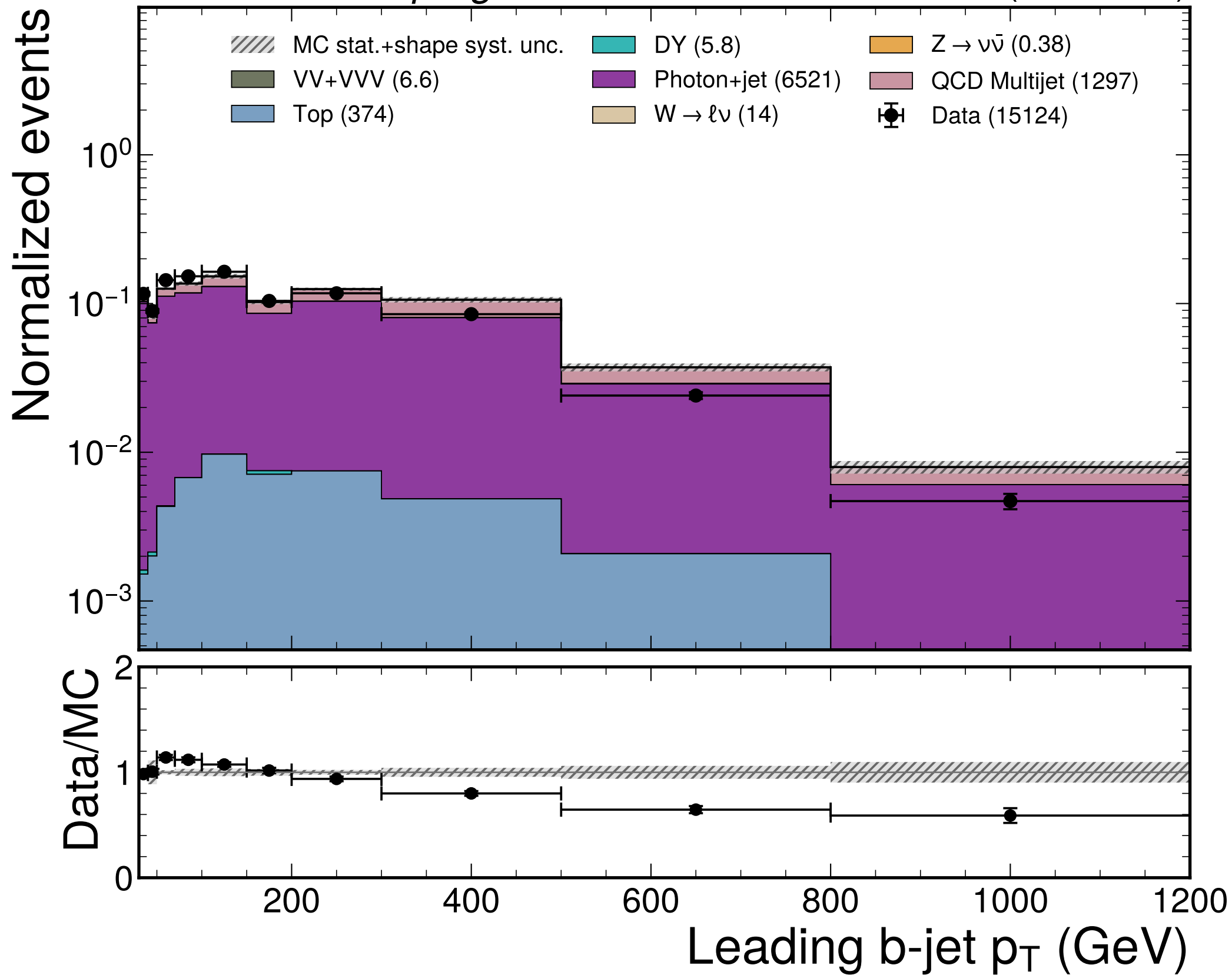
Data/MC



U_T (GeV)

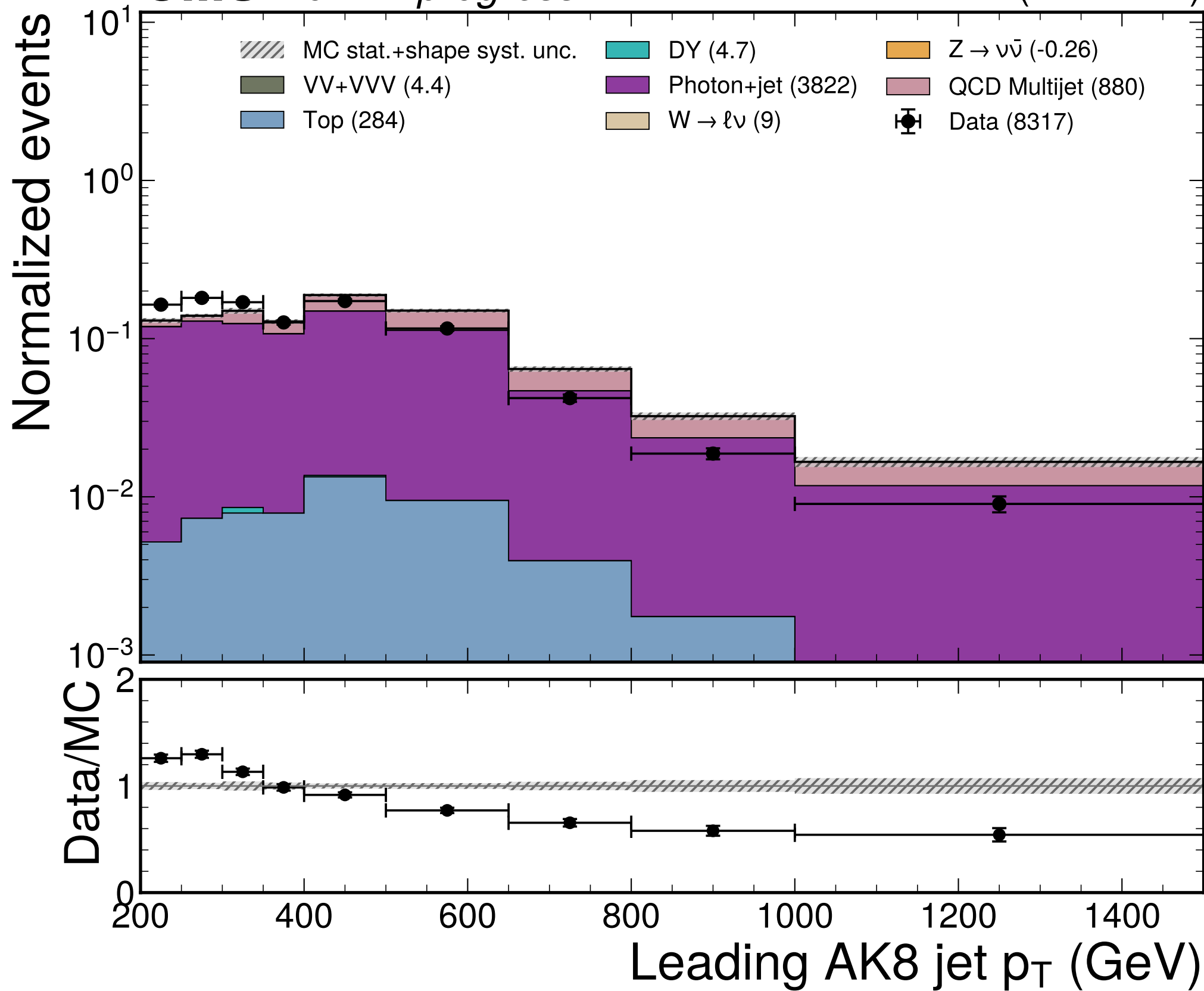
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



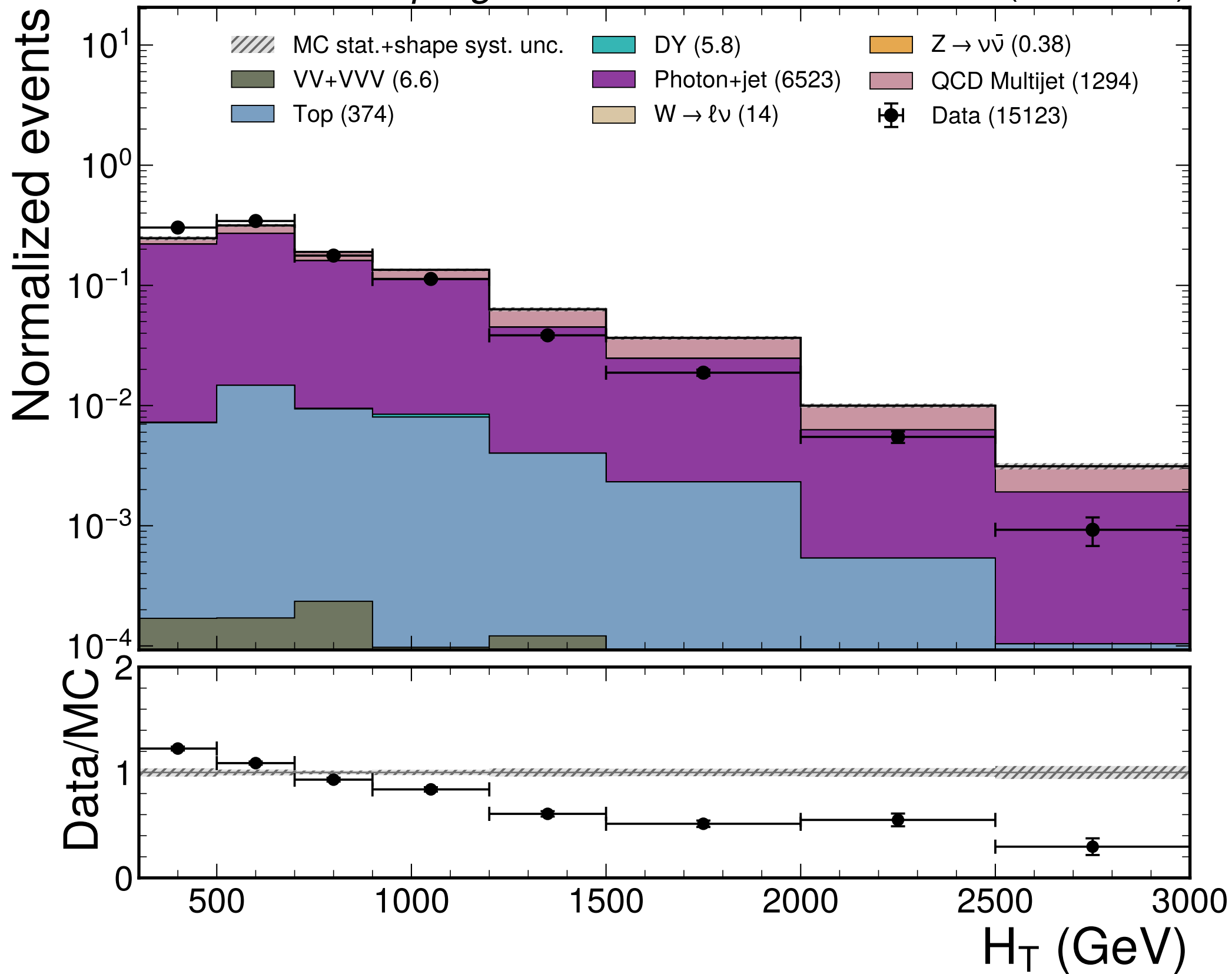
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



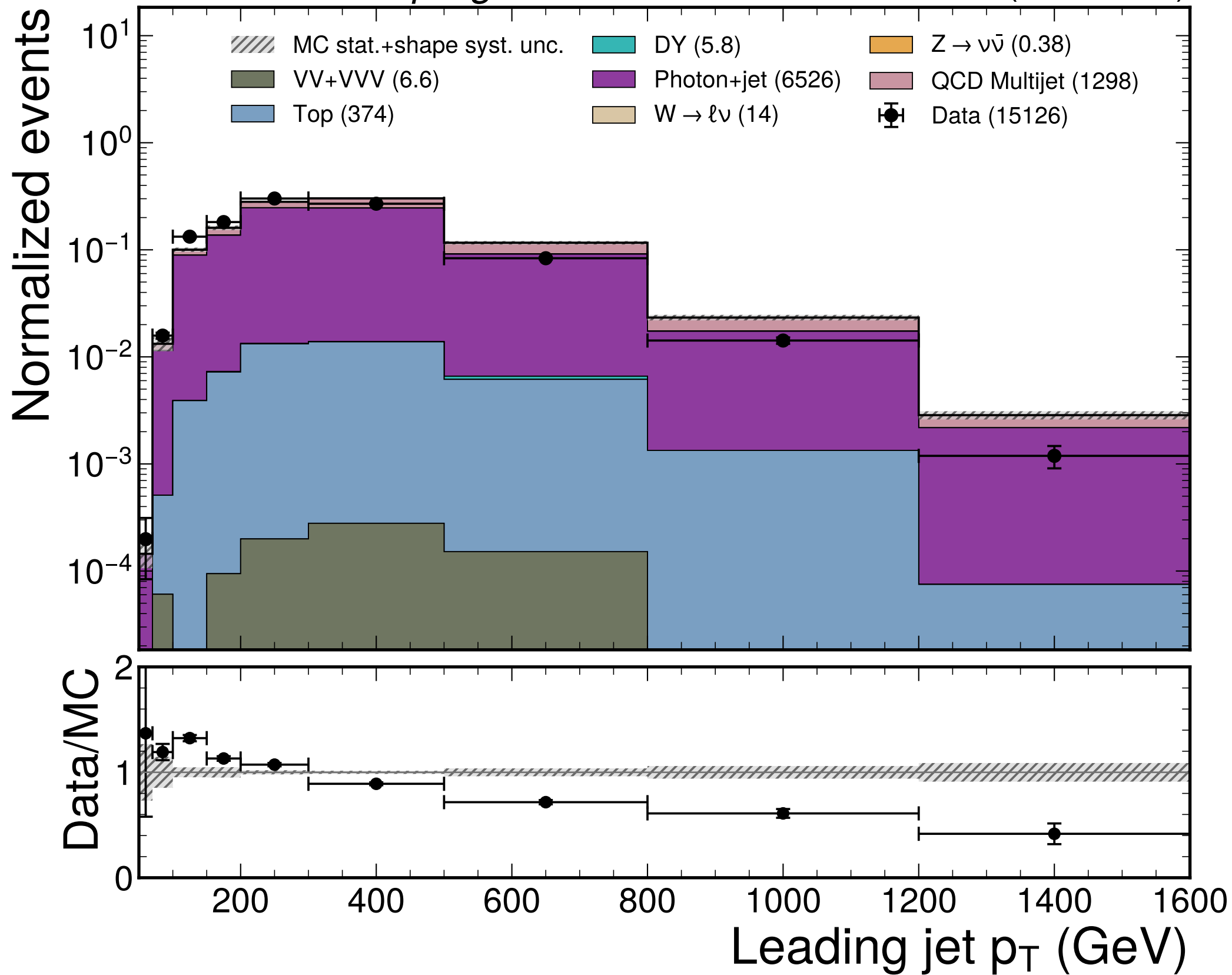
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



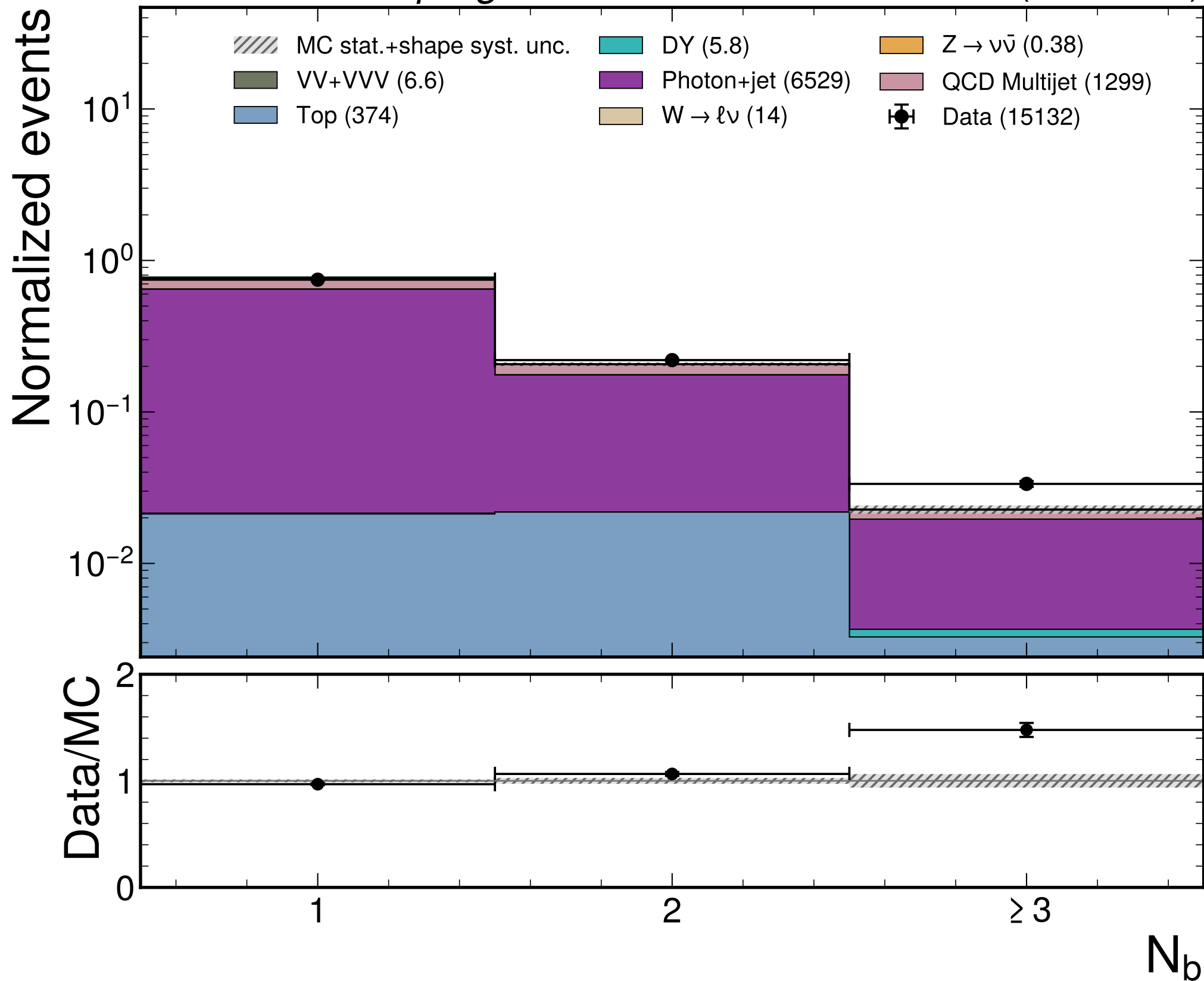
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



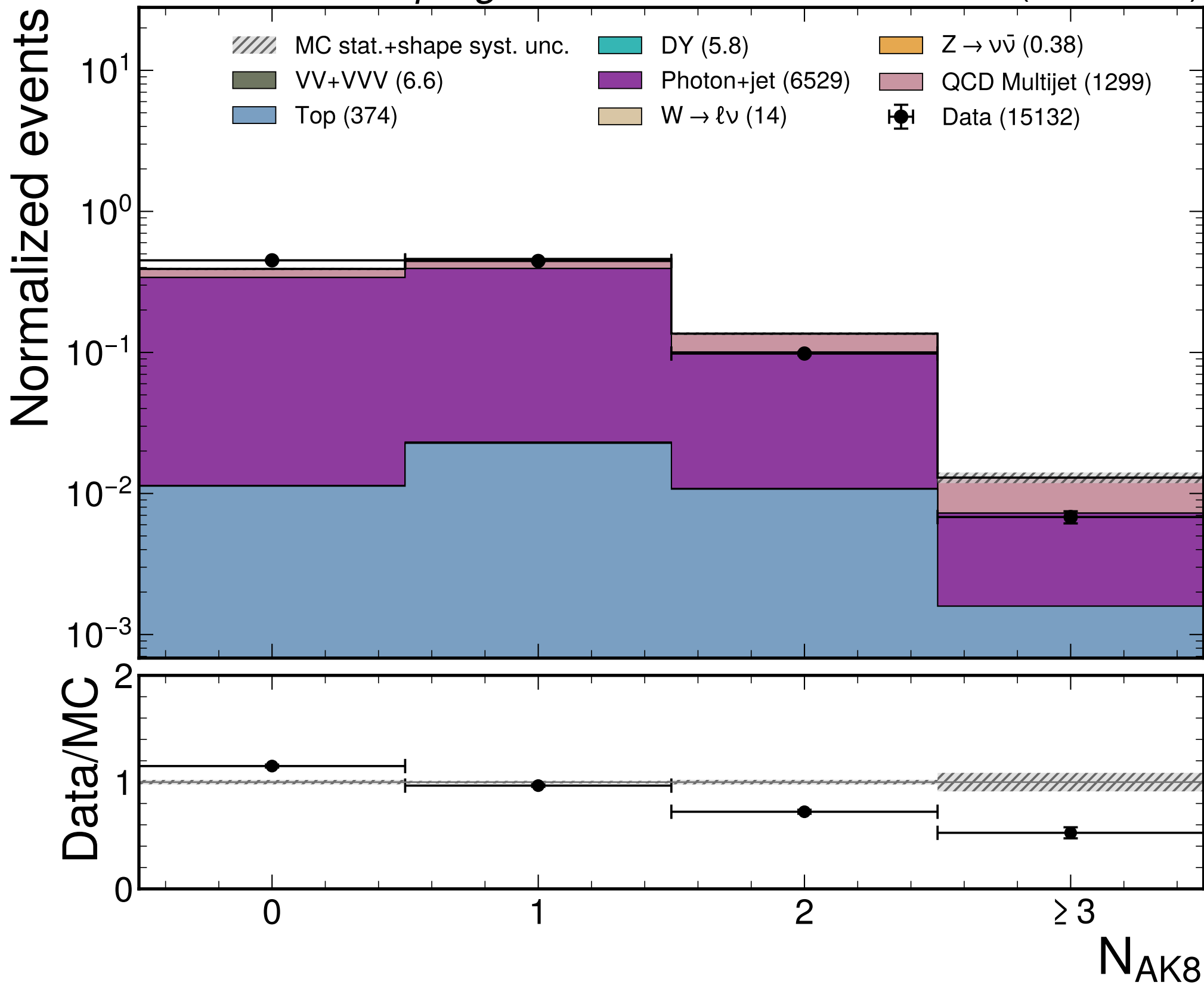
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



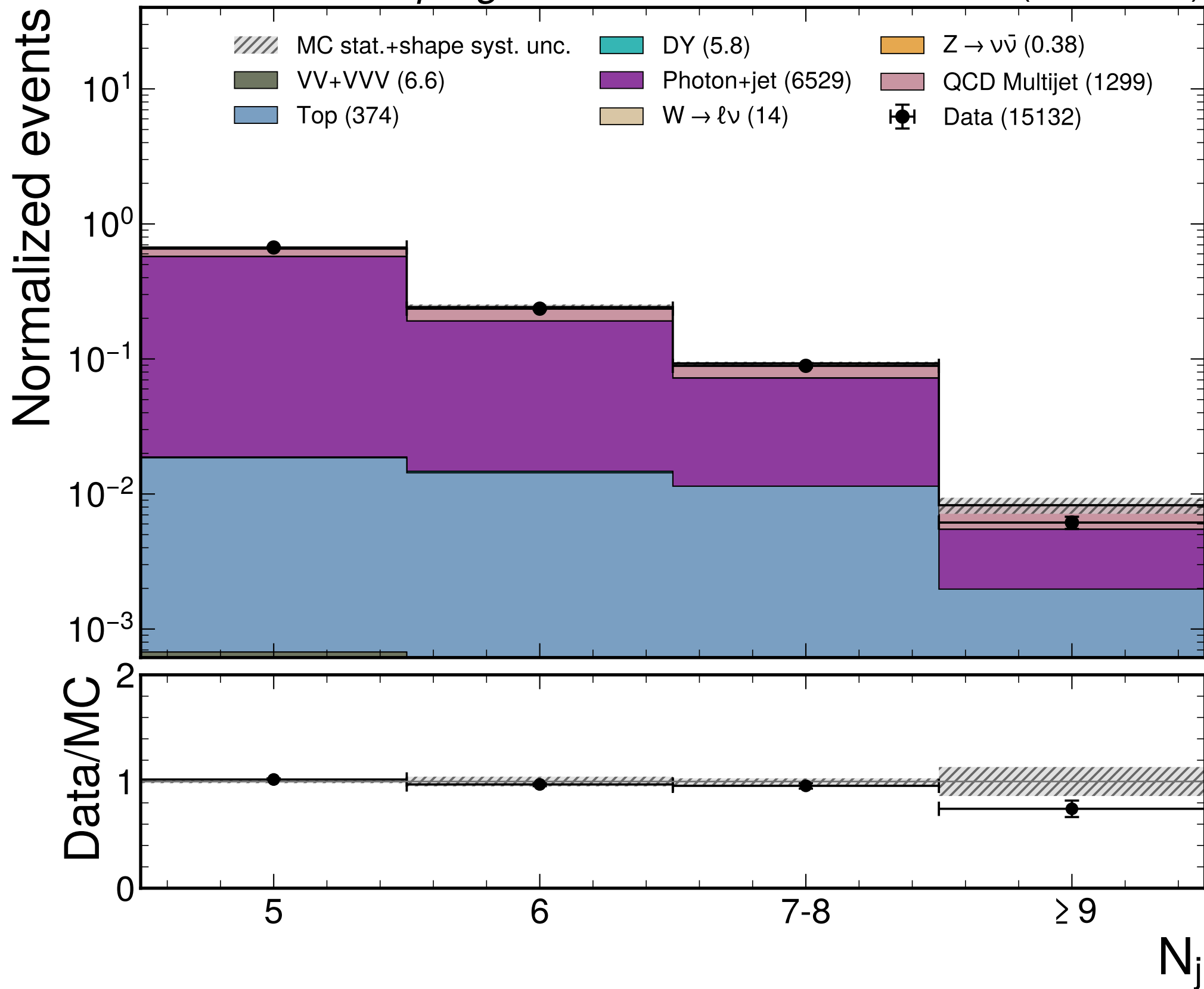
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



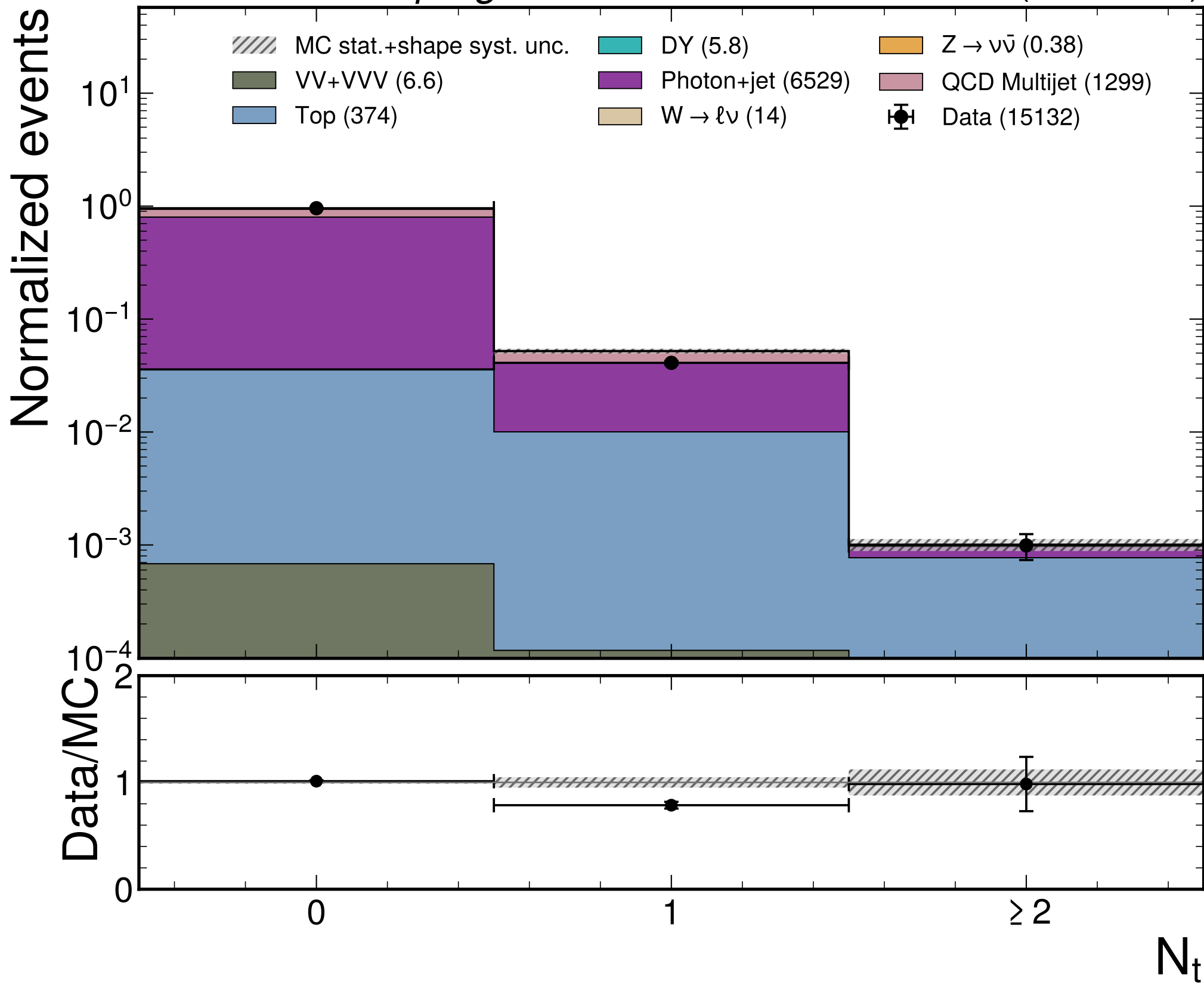
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



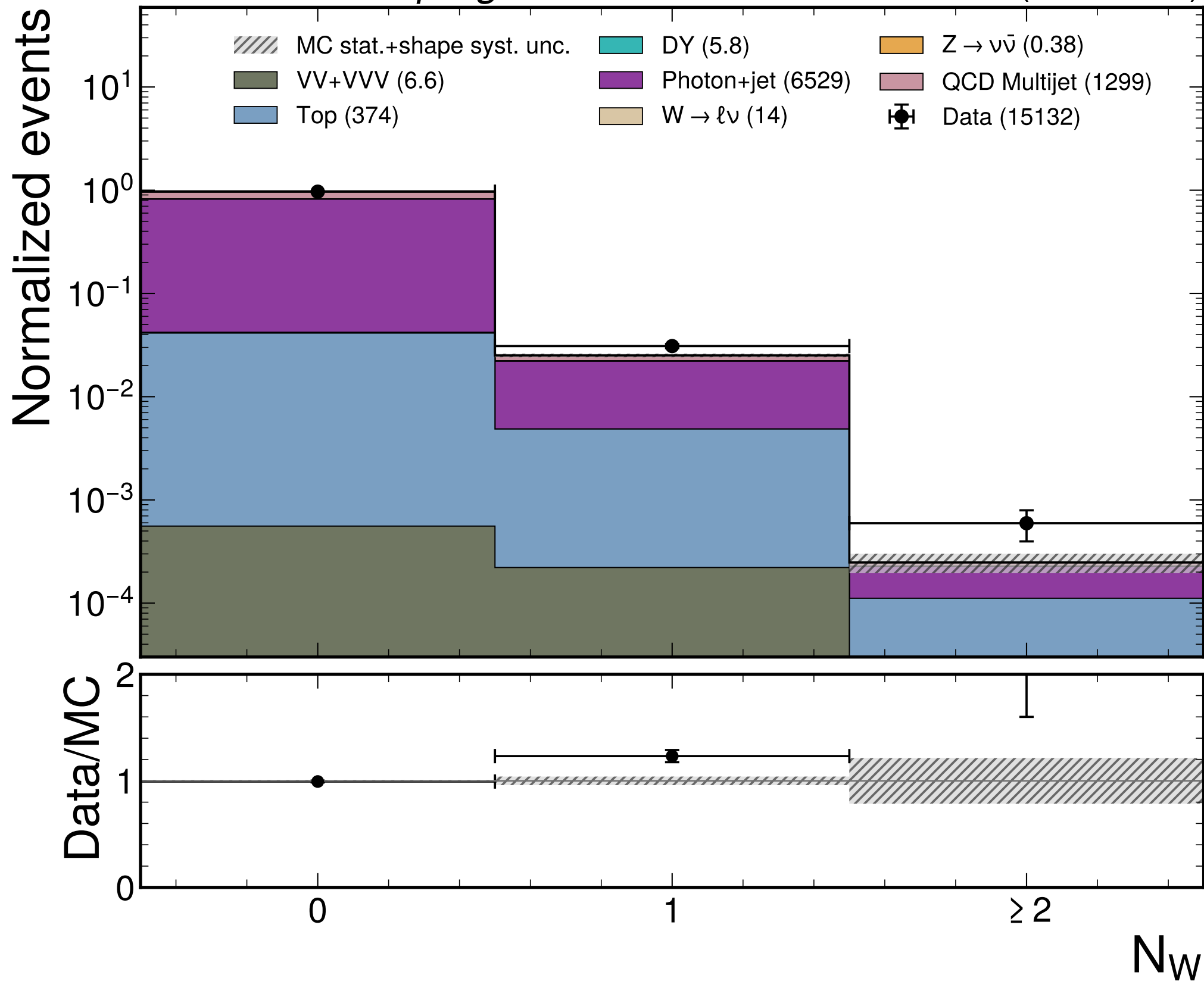
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



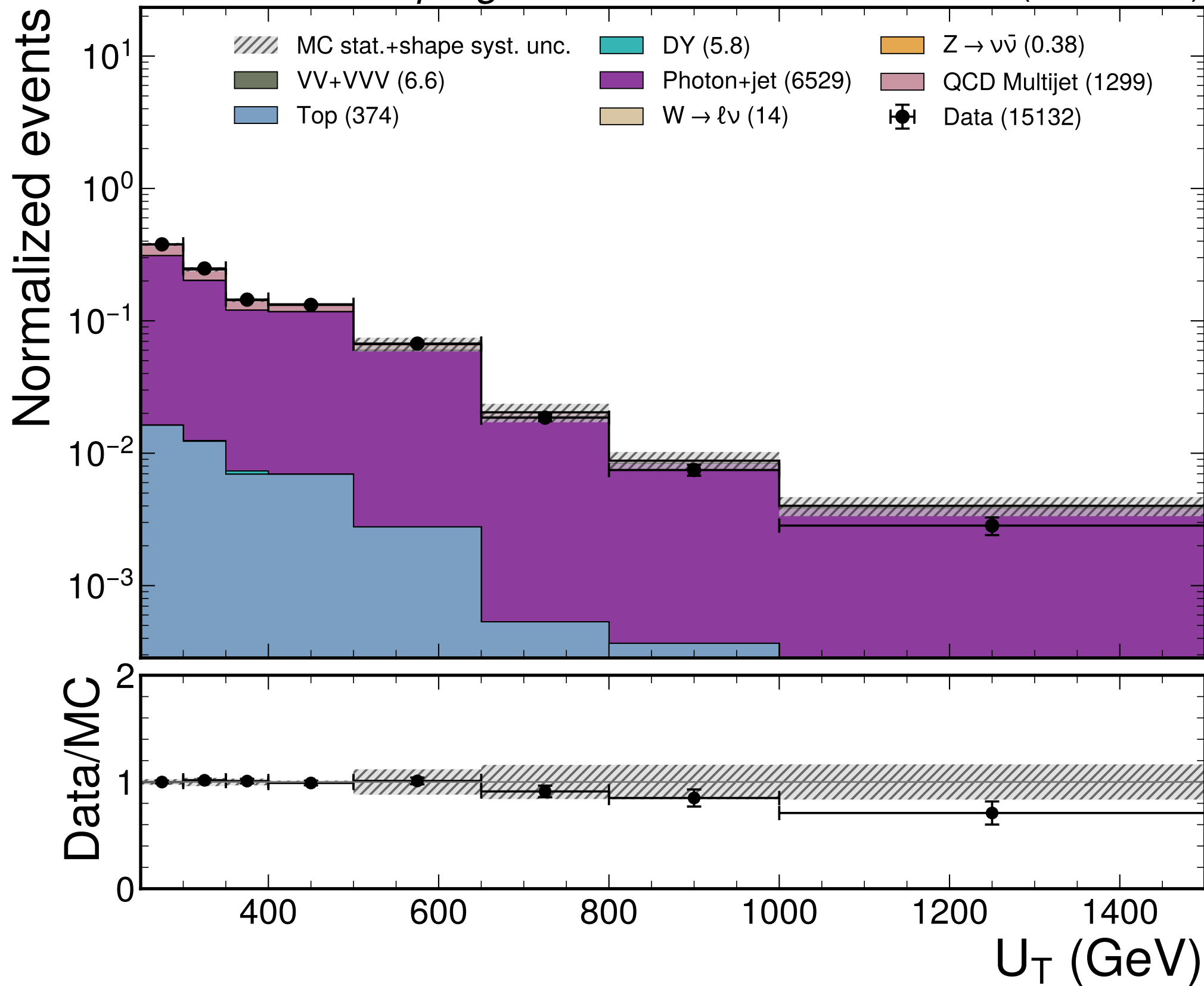
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



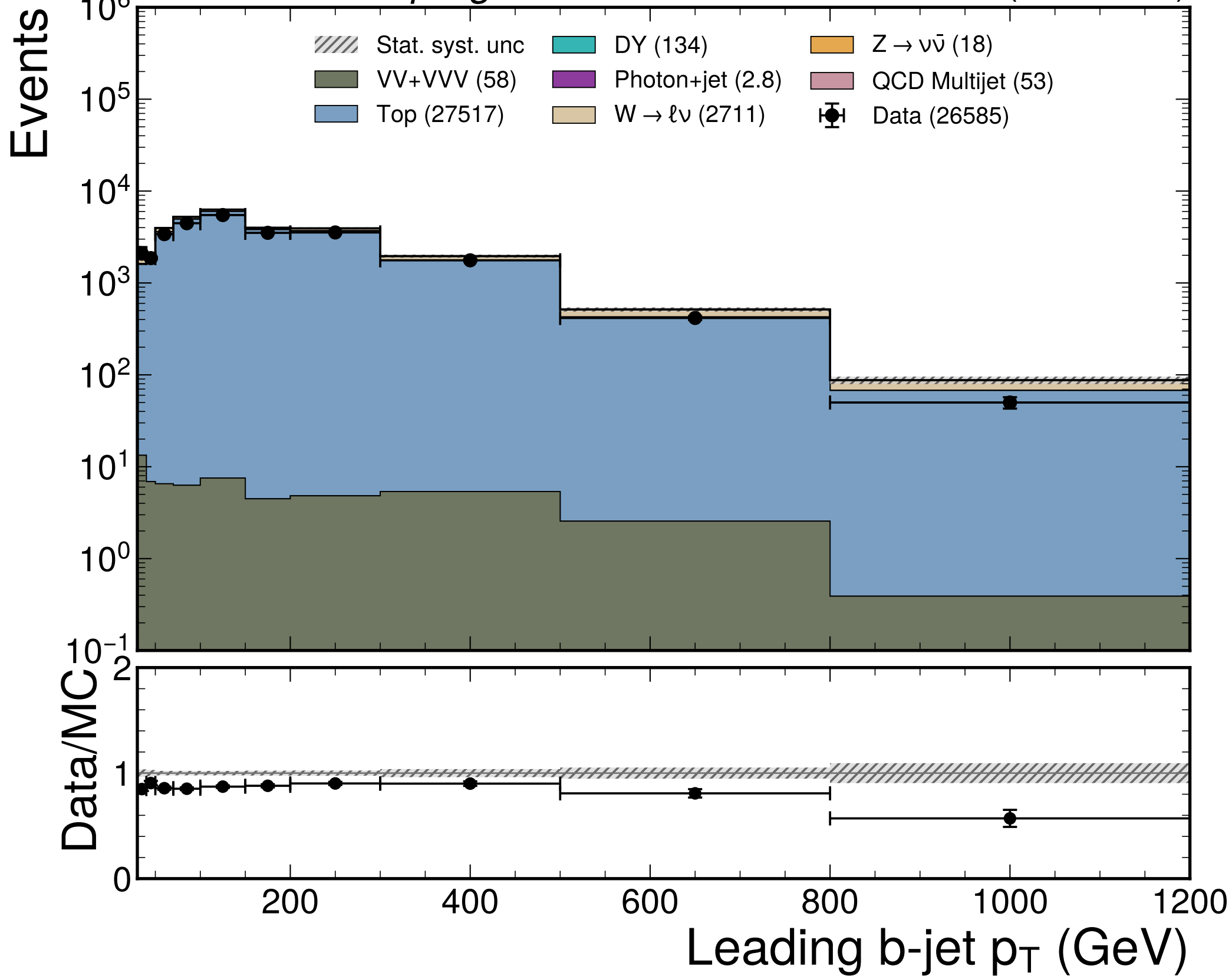
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

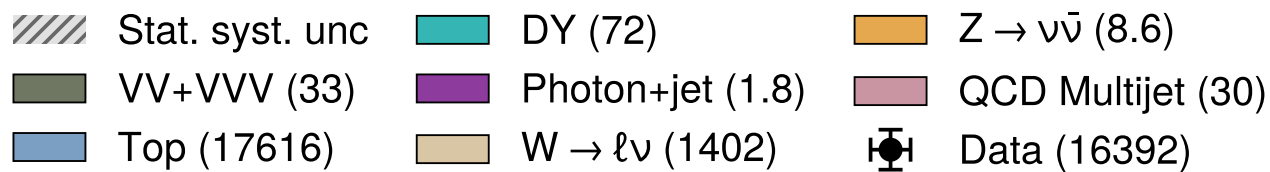
109.82 fb⁻¹ (13.6 TeV)



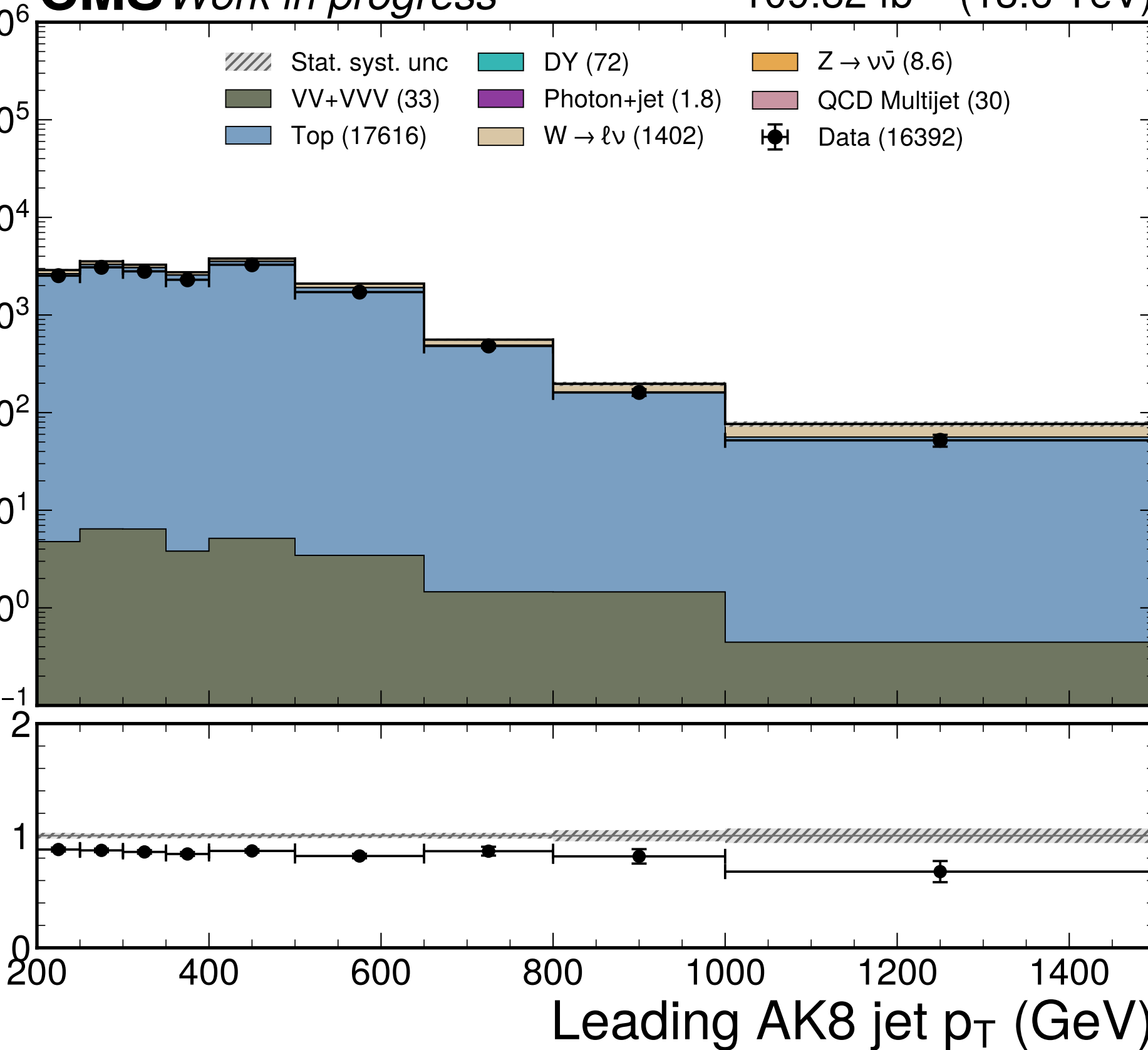
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



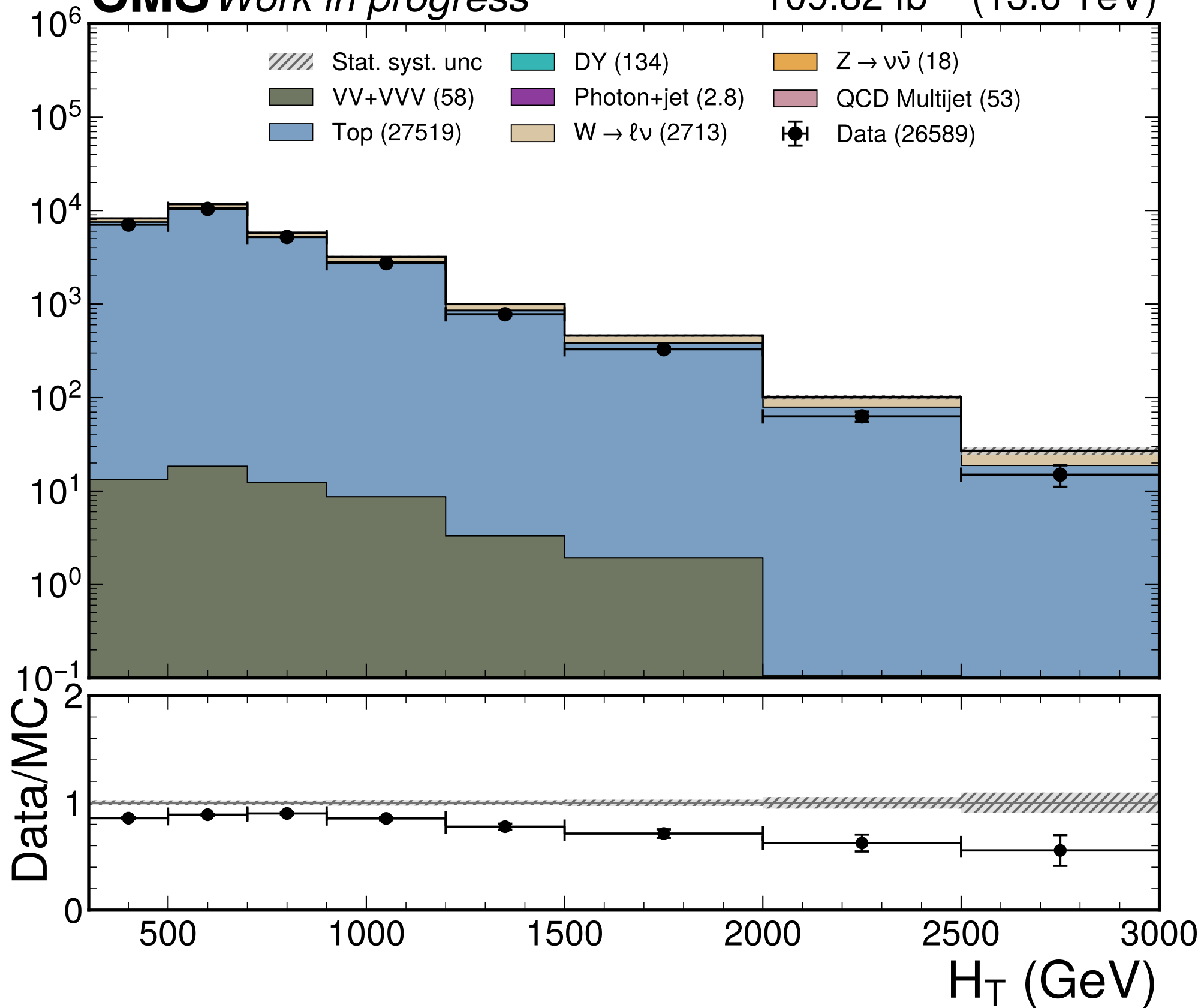
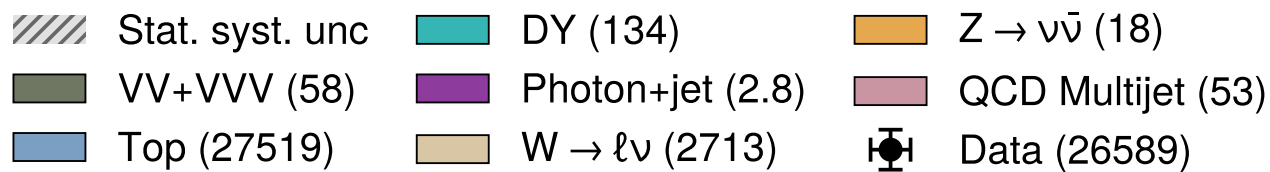
Data/MC



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

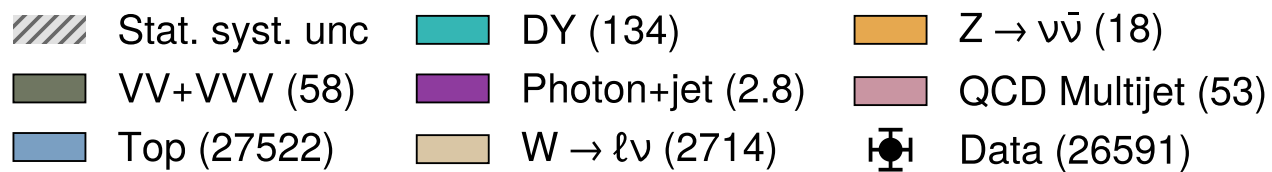
Events



CMS *Work in progress*

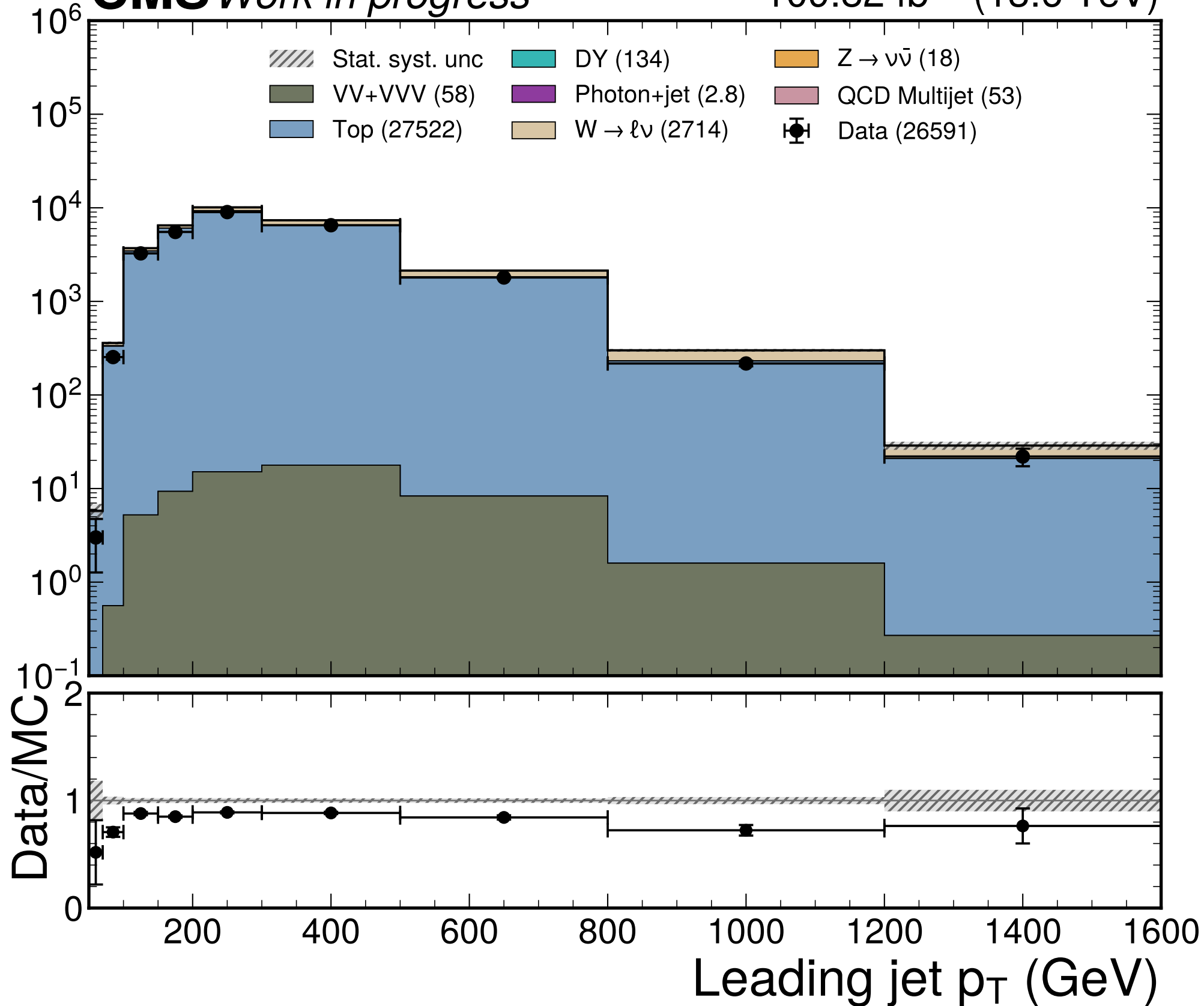
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

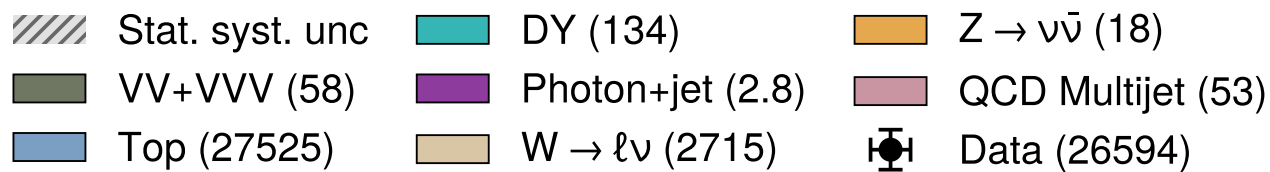
Leading jet p_T (GeV)



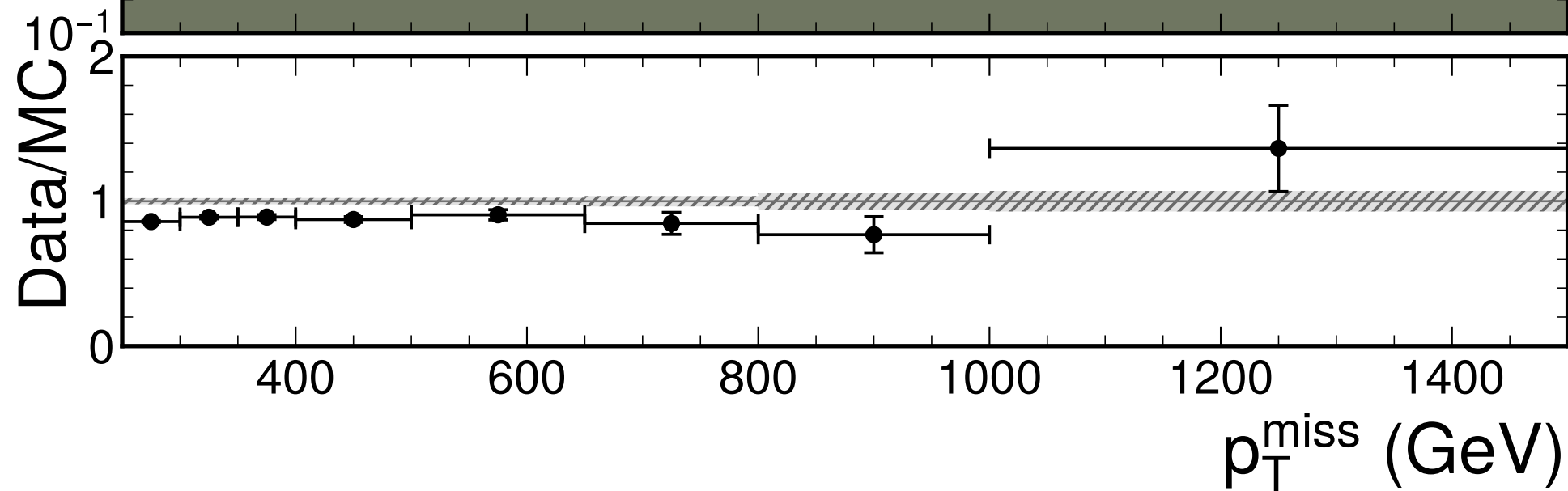
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



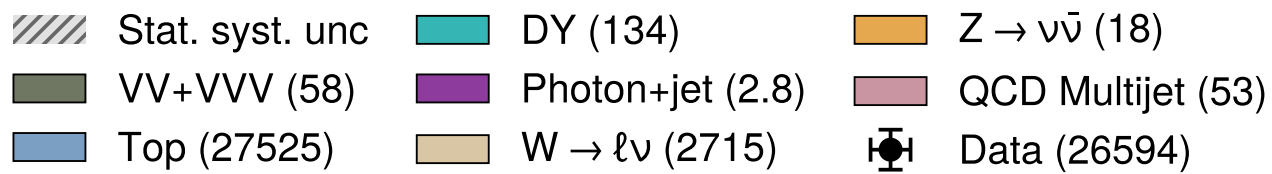
Data/MC



CMS *Work in progress*

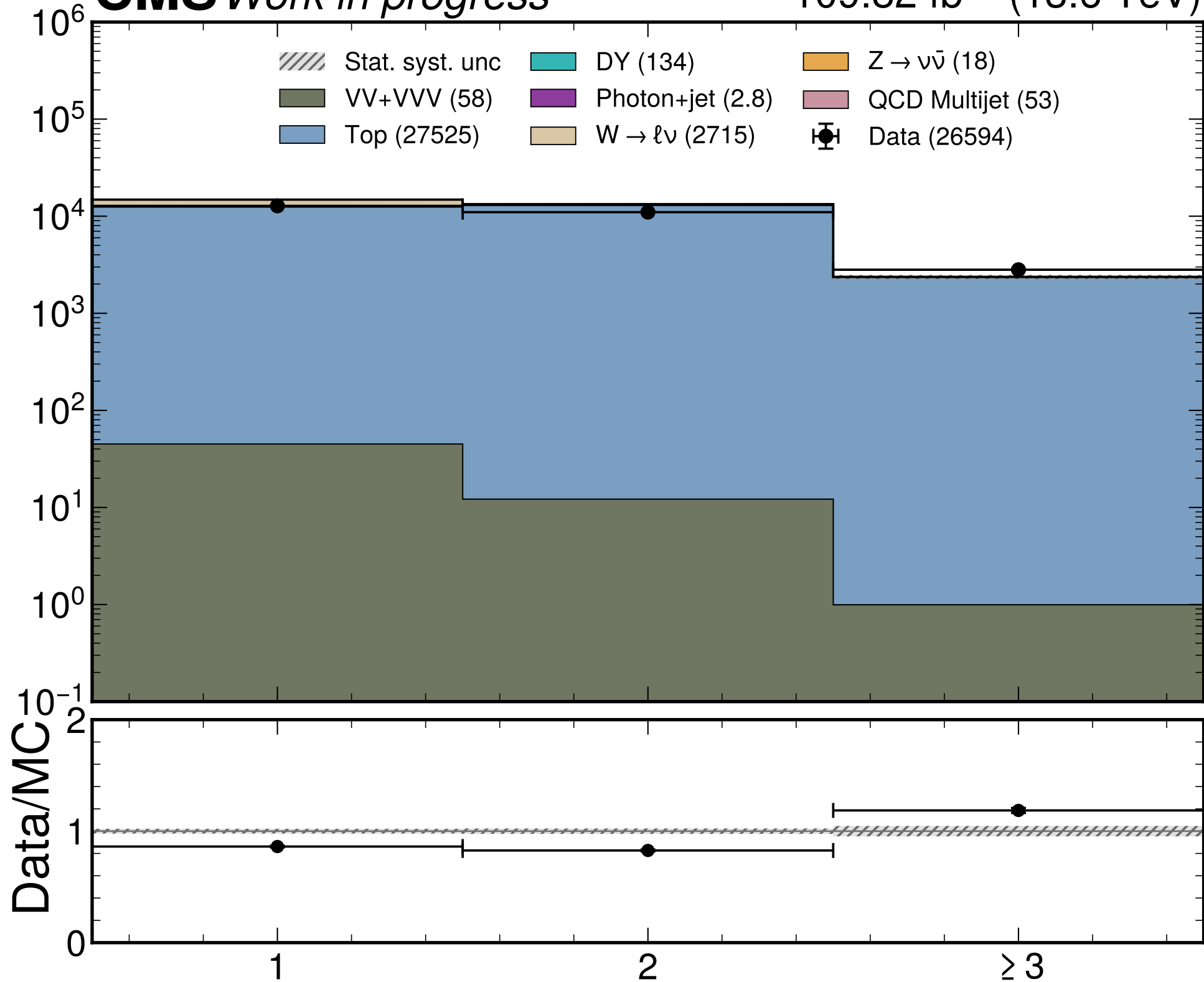
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

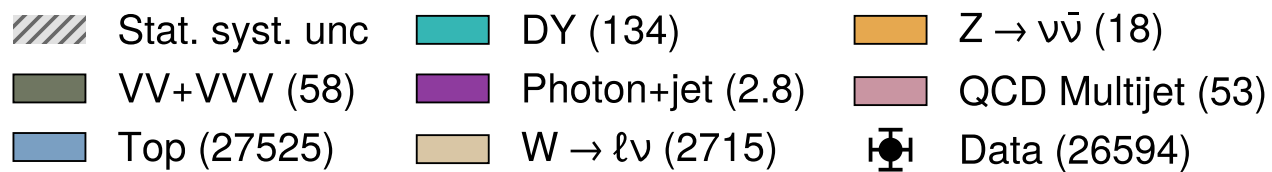
N_b



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

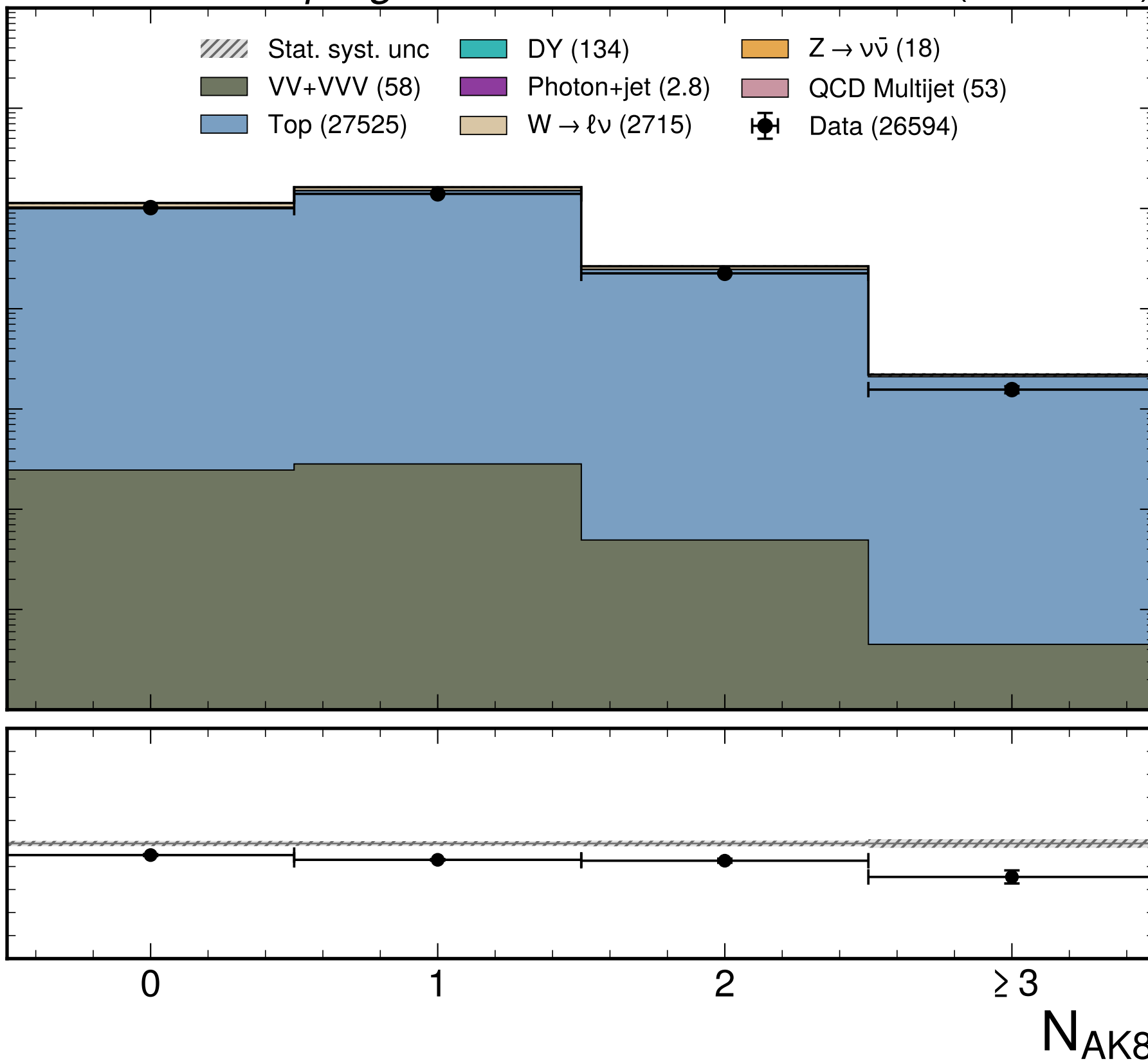
0

1

2

≥ 3

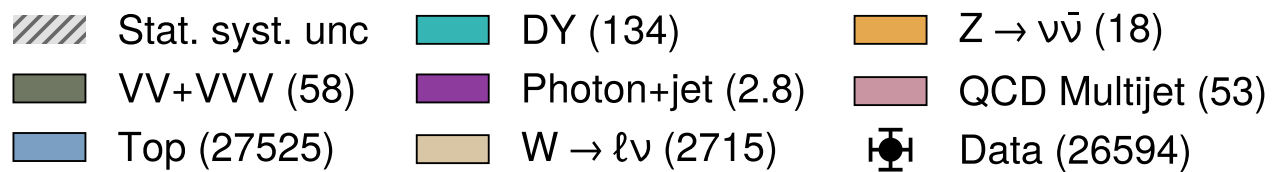
N_{AK8}



CMS *Work in progress*

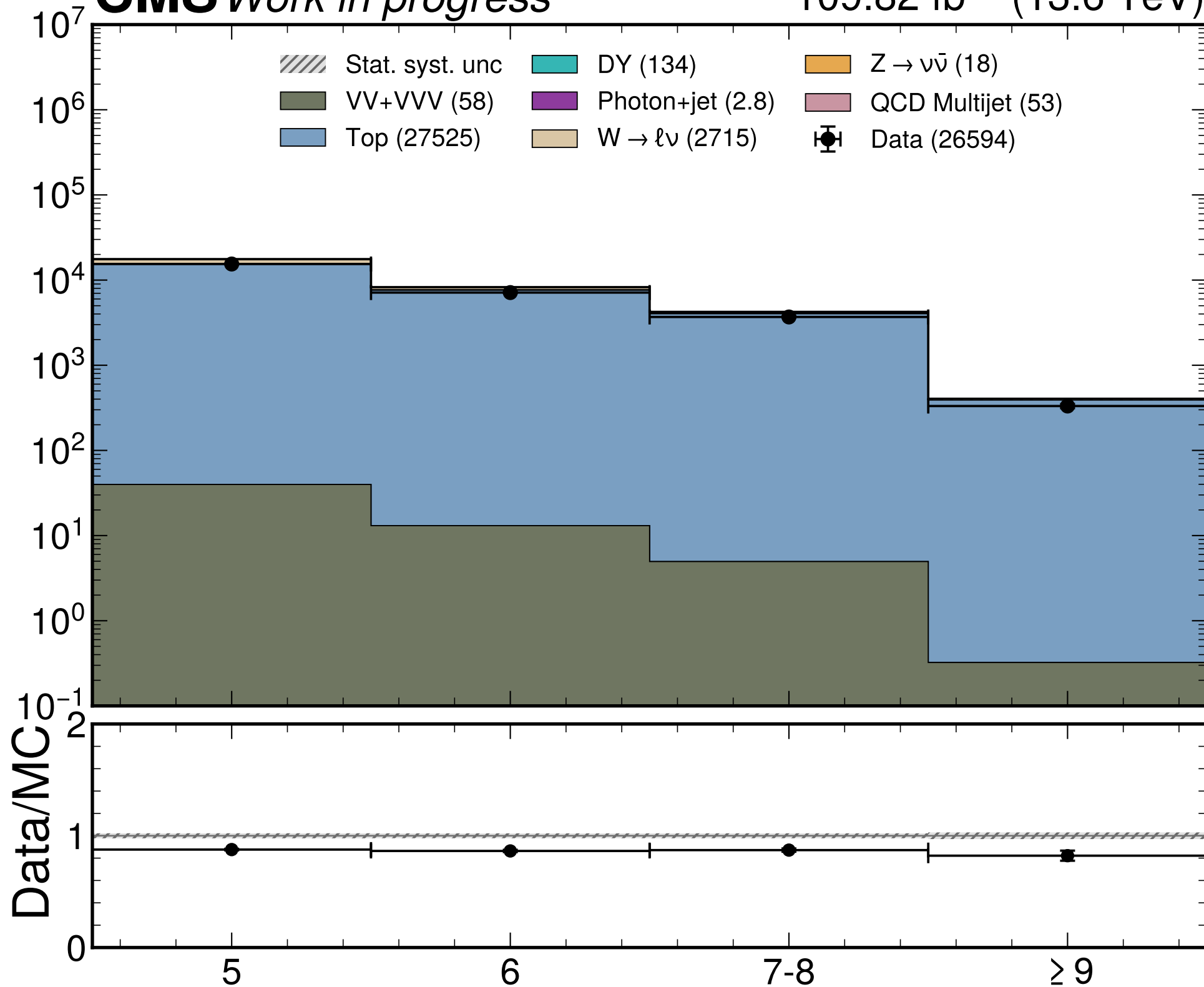
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

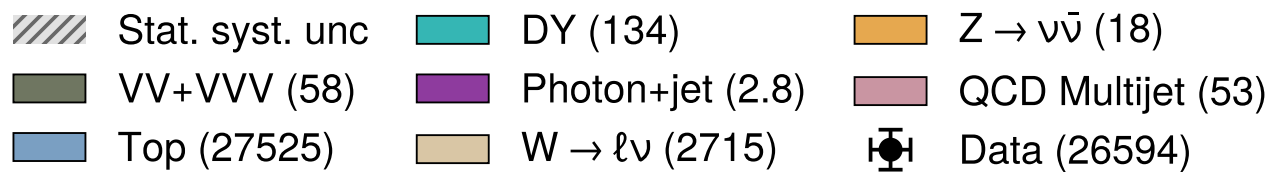
N_j



CMS *Work in progress*

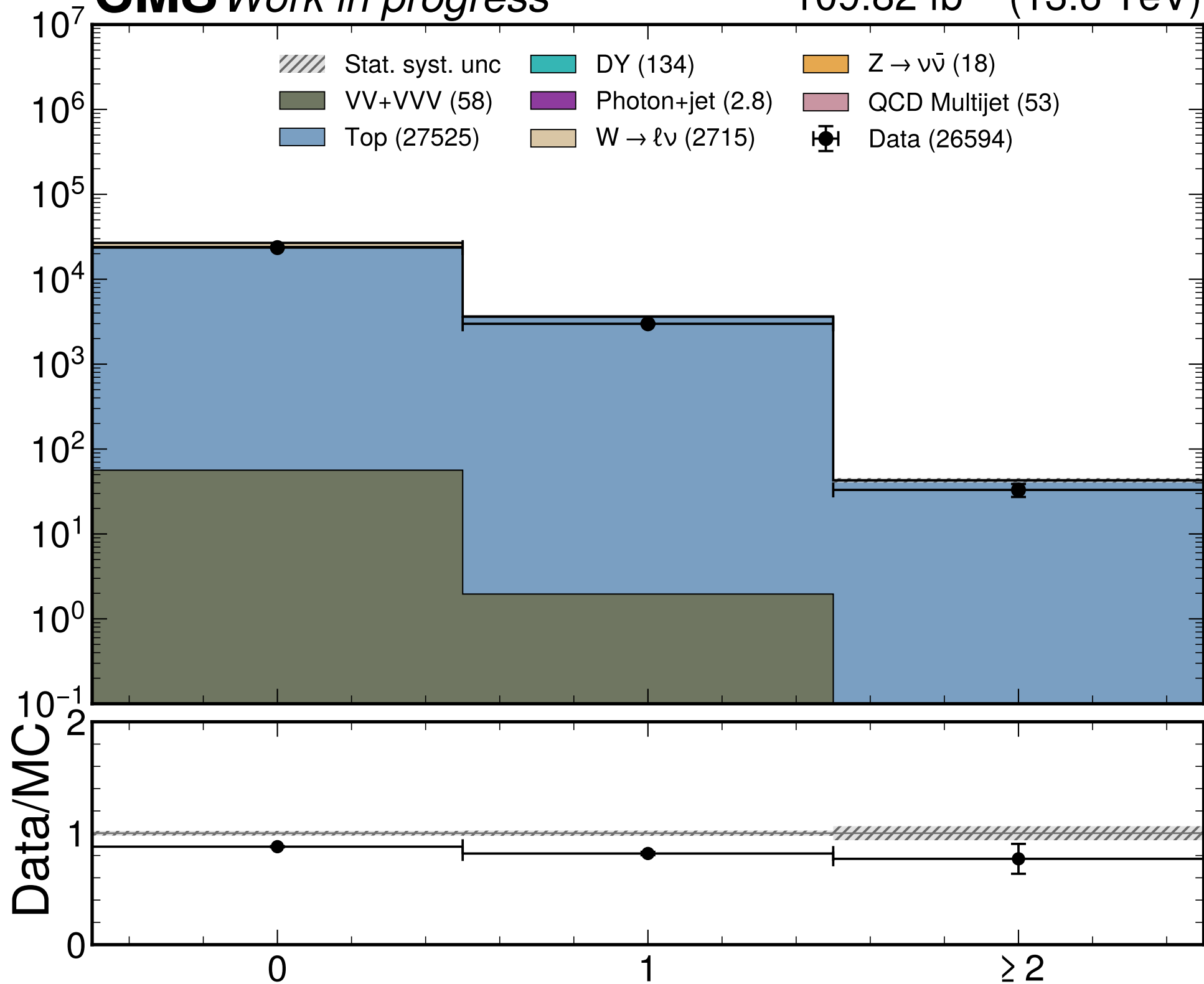
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

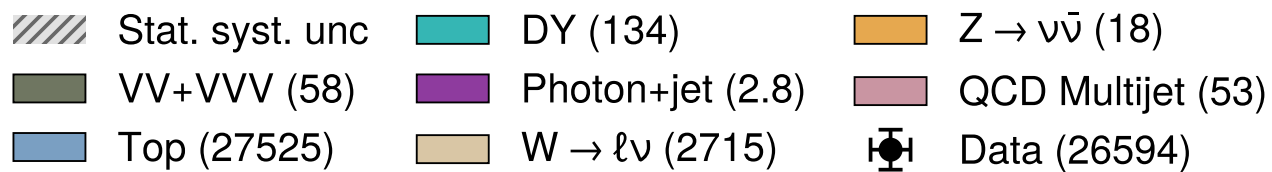
N_t



CMS *Work in progress*

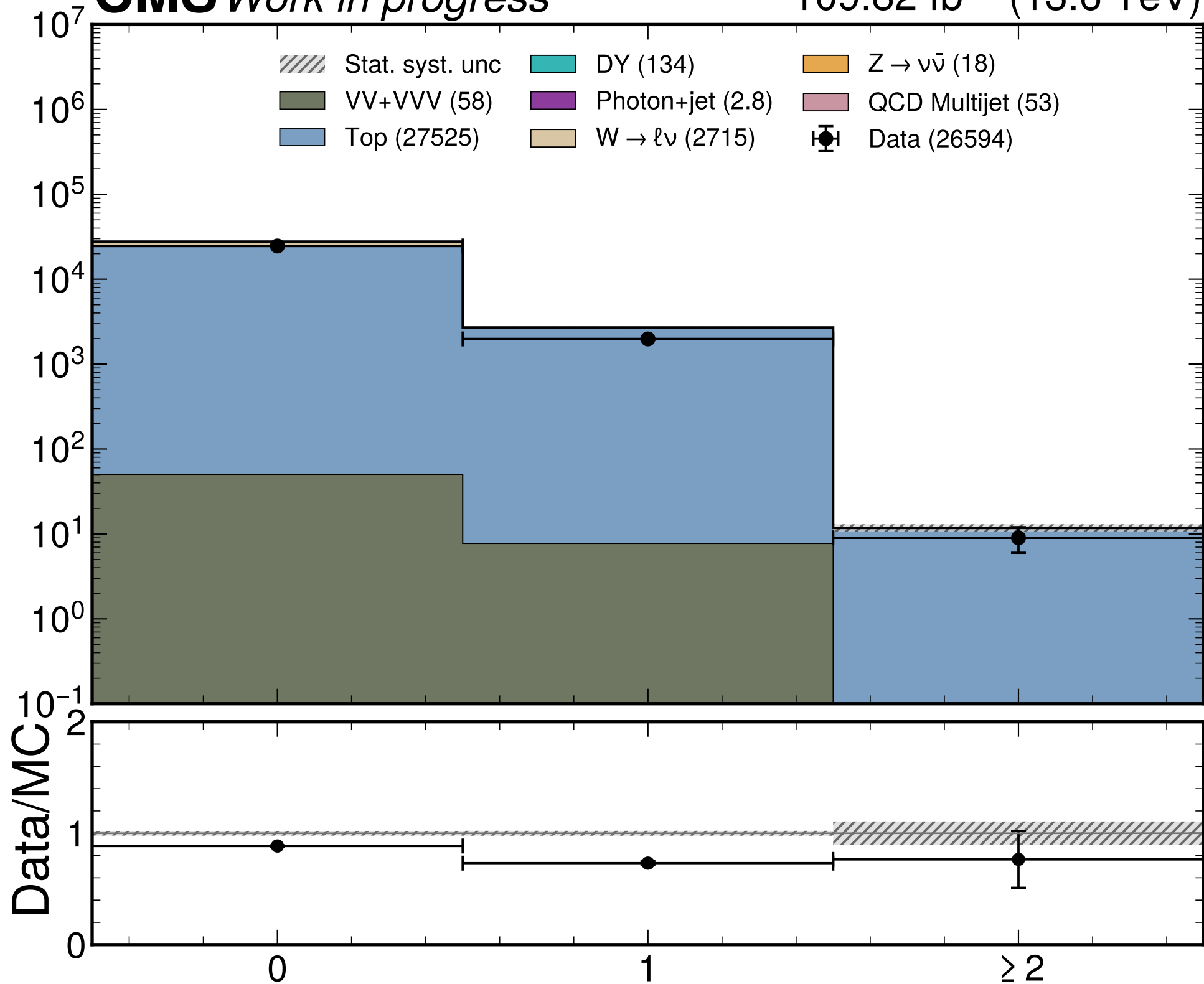
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

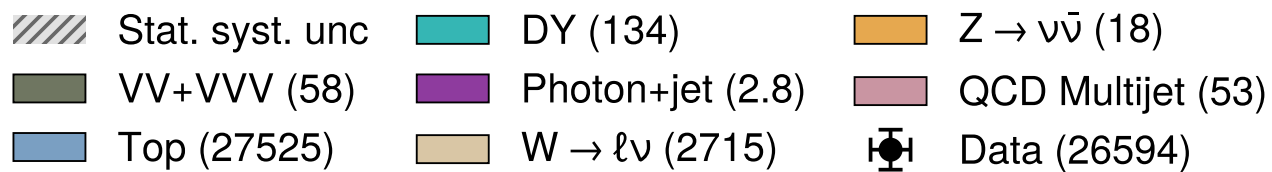
N_W



CMS *Work in progress*

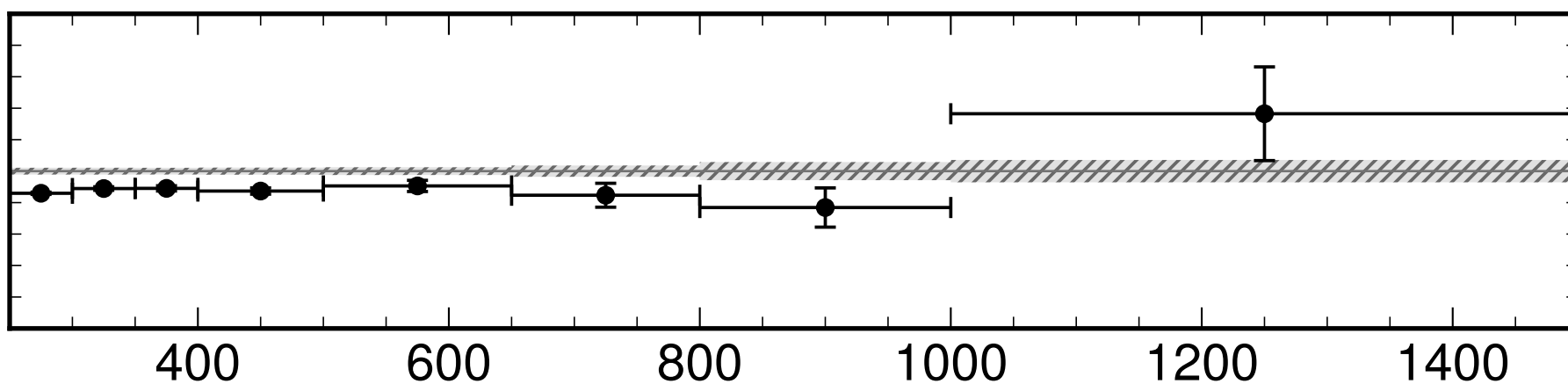
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

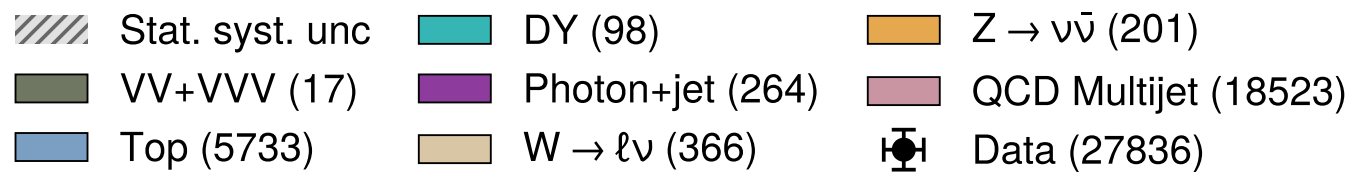


U_T (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



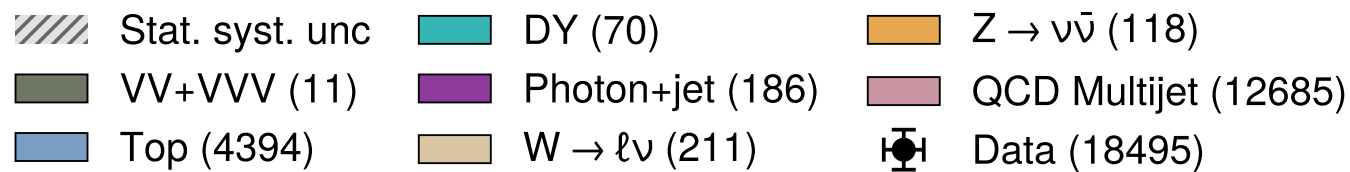
Data/MC

Leading b-jet p_T (GeV)

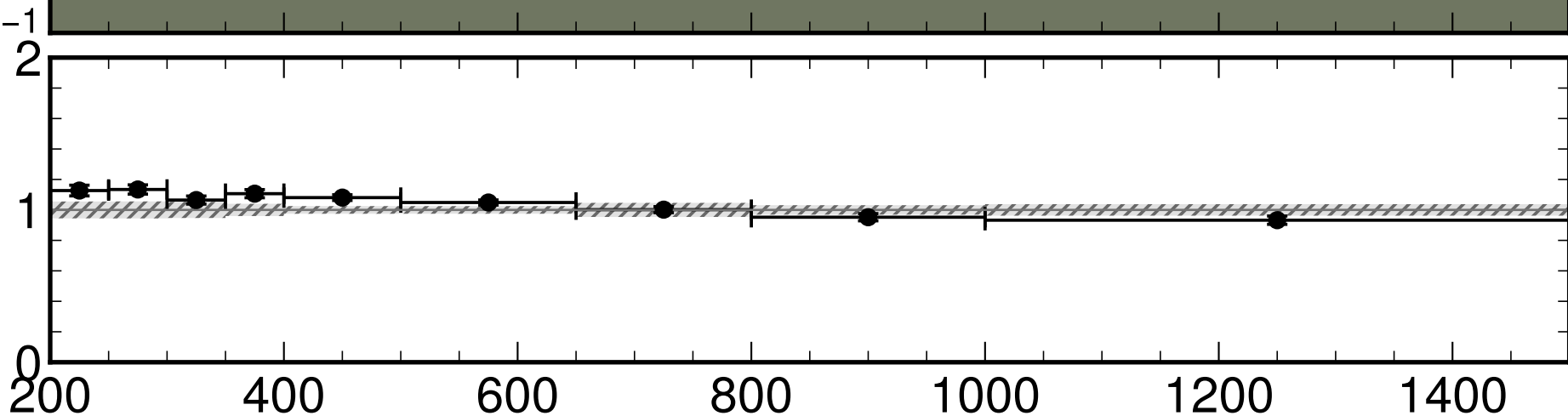
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

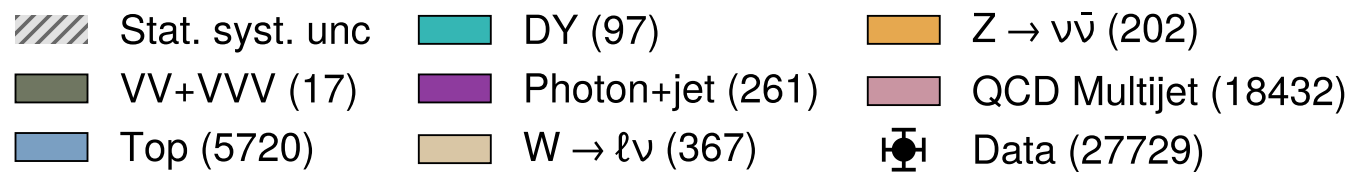


Leading AK8 jet p_T (GeV)

CMS *Work in progress*

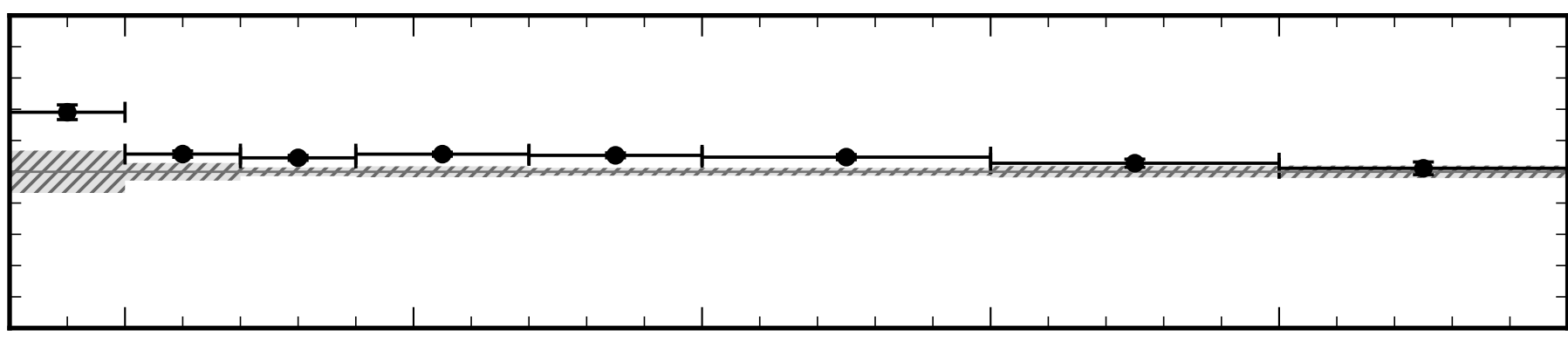
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0



500

1000

1500

2000

2500

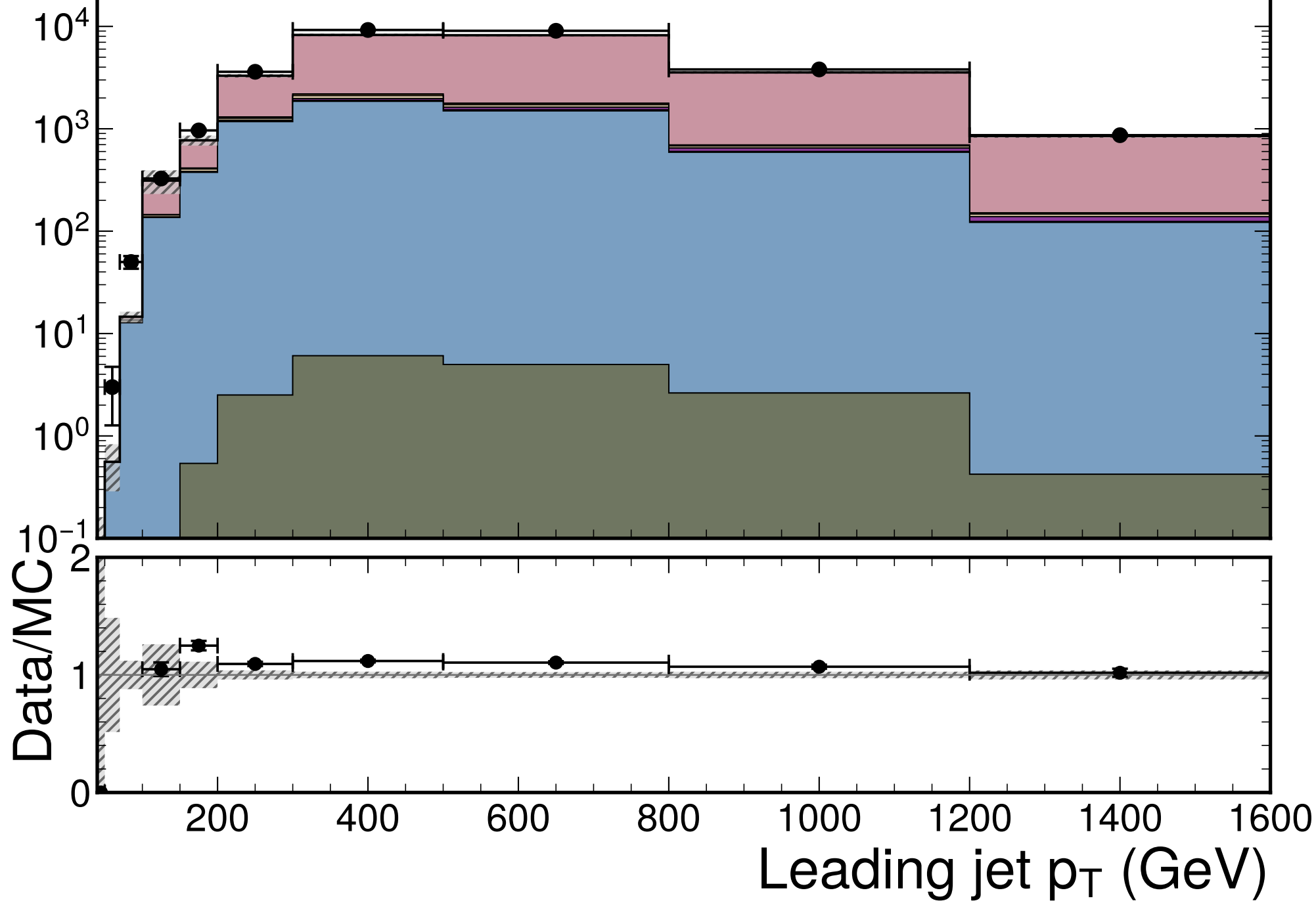
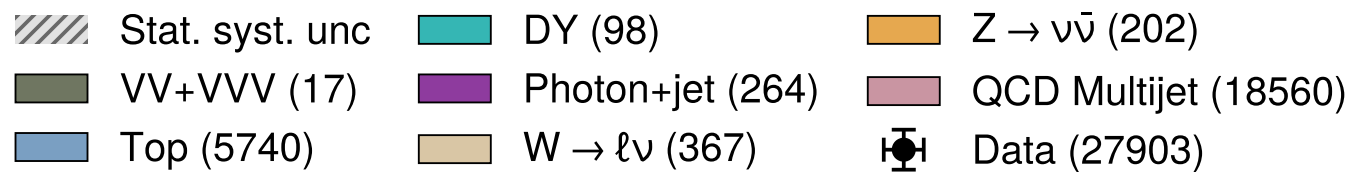
3000

H_T (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

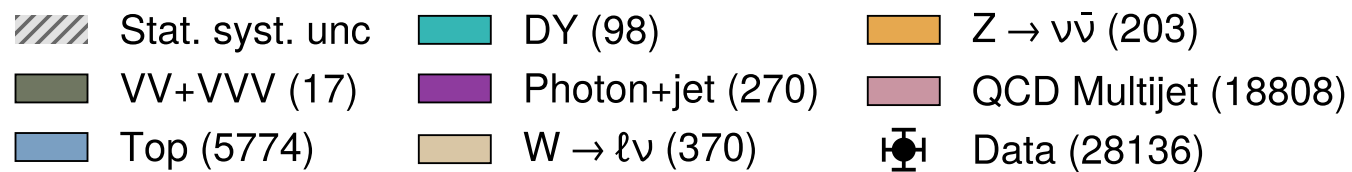
Events



CMS *Work in progress*

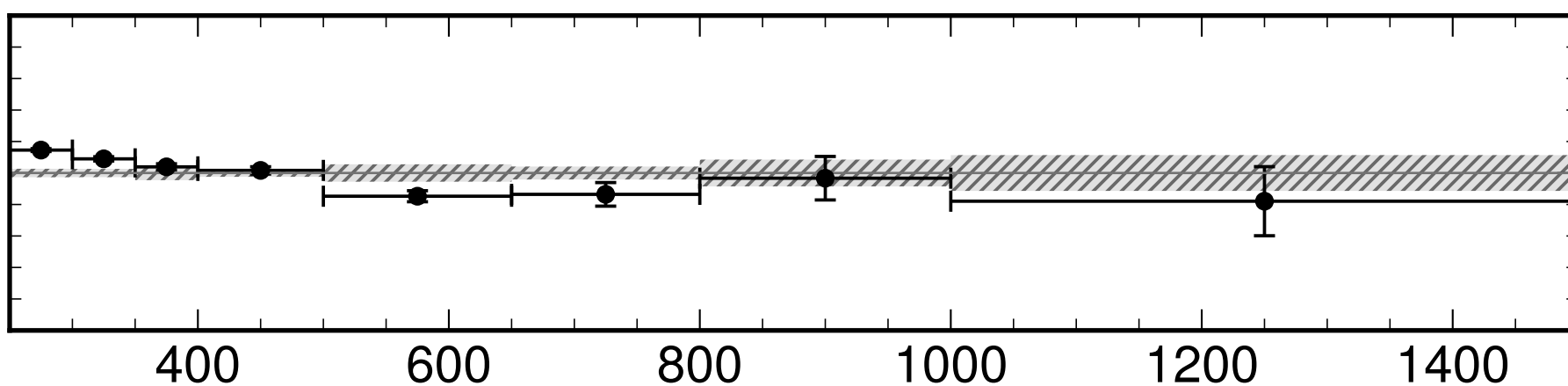
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2

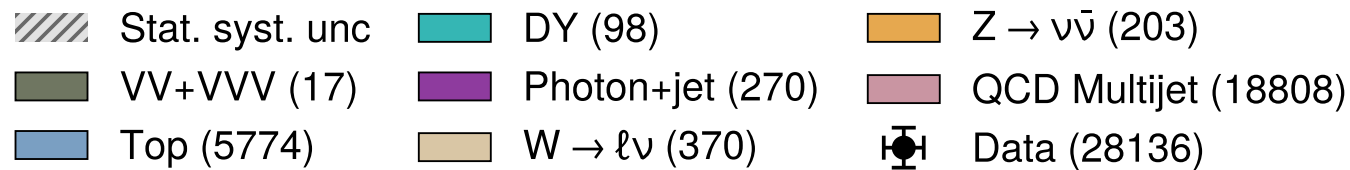


p_T^{miss} (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

1

2

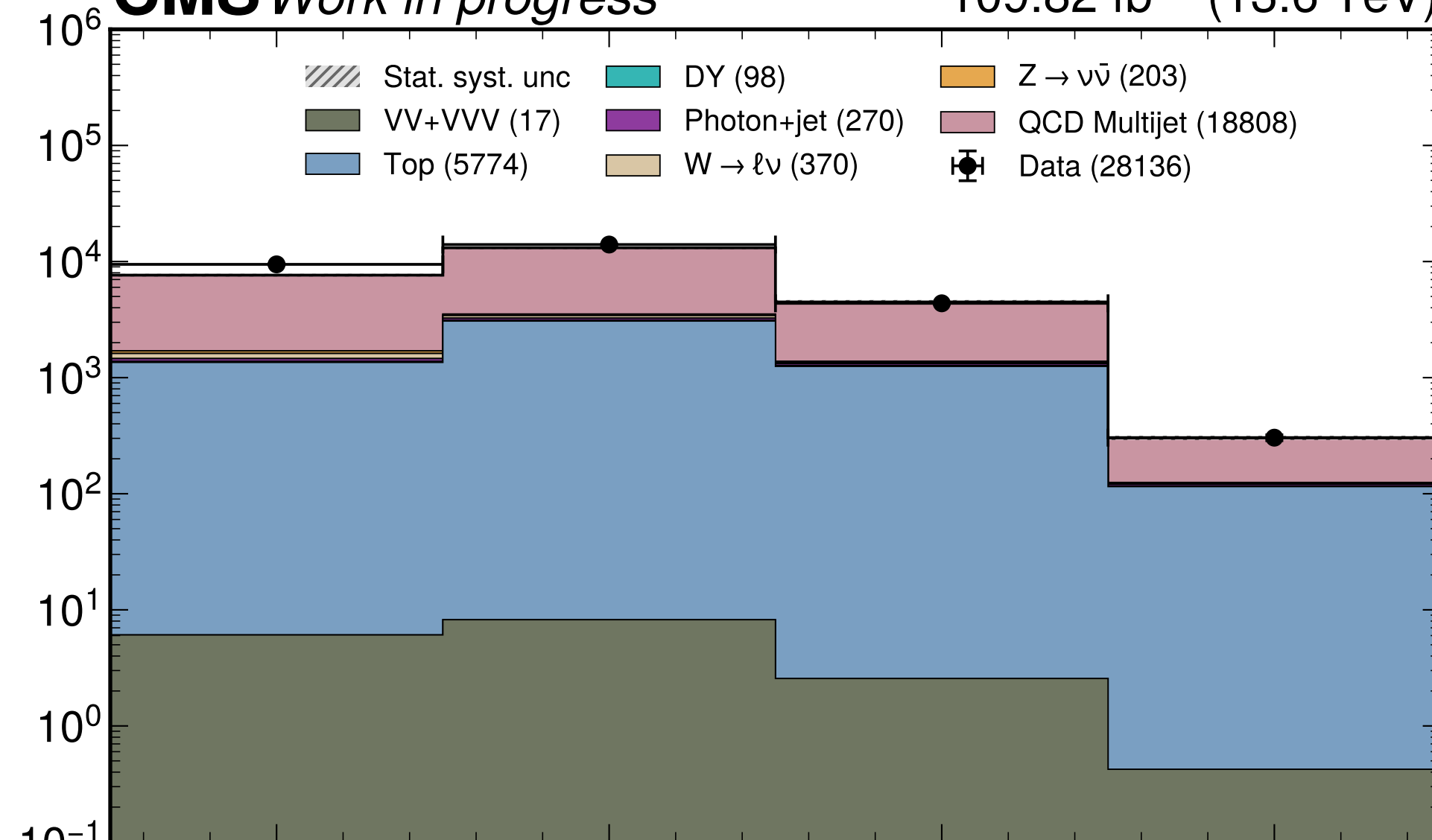
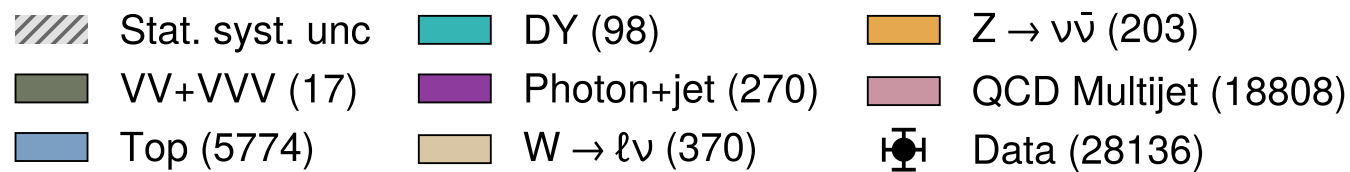
≥ 3

N_b

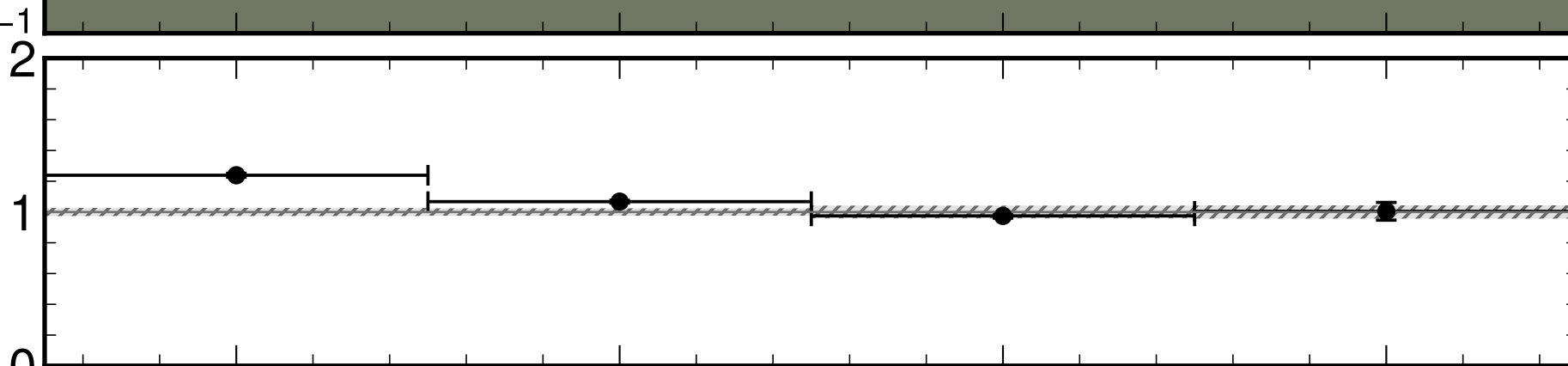
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC



0

1

2

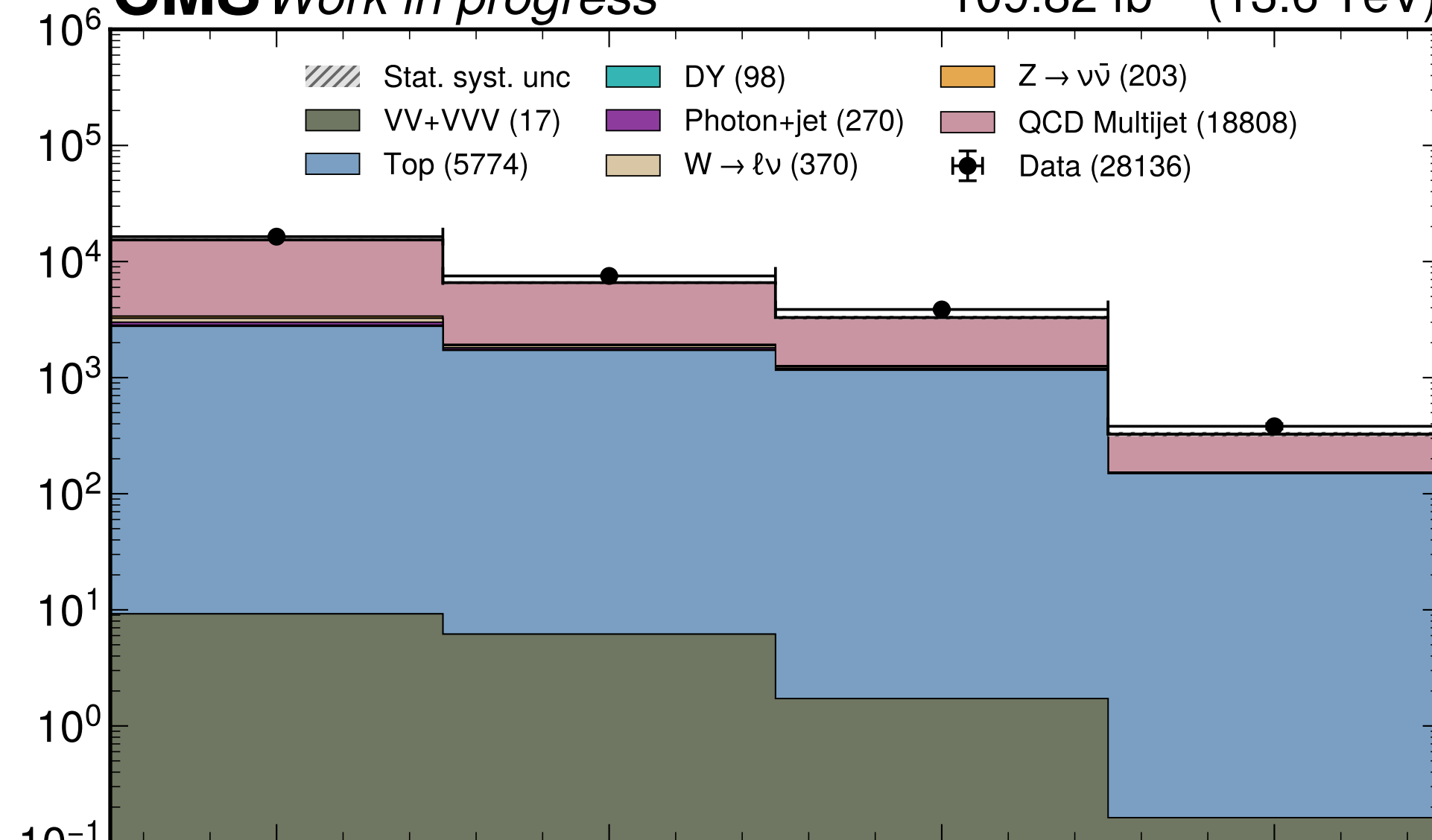
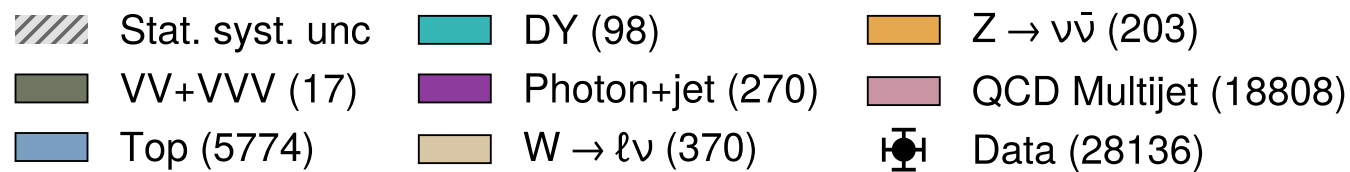
≥ 3

N_{AK8}

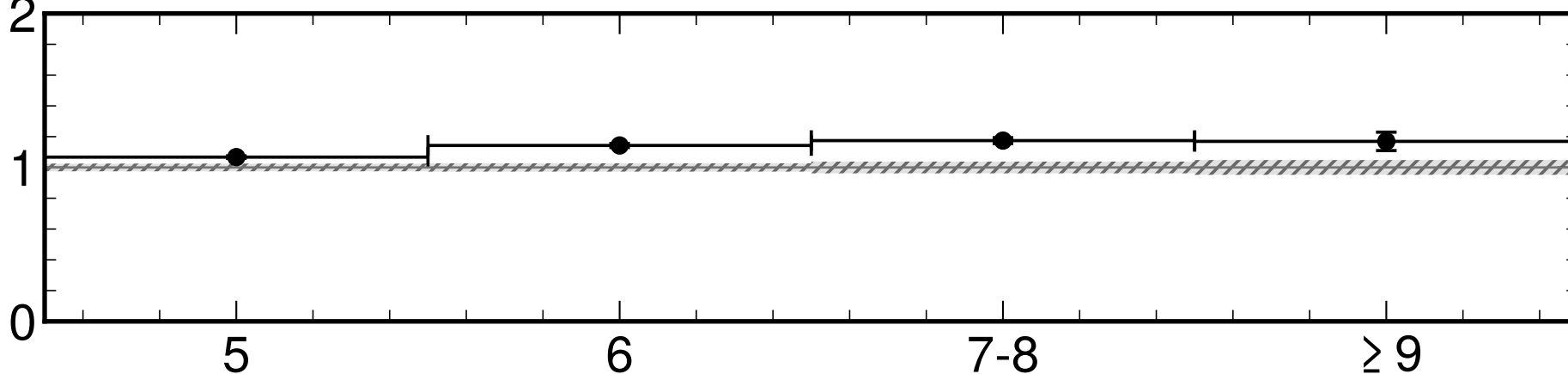
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

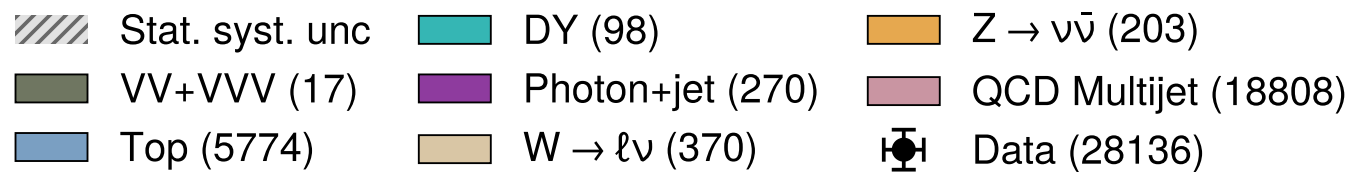


N_j

CMS *Work in progress*

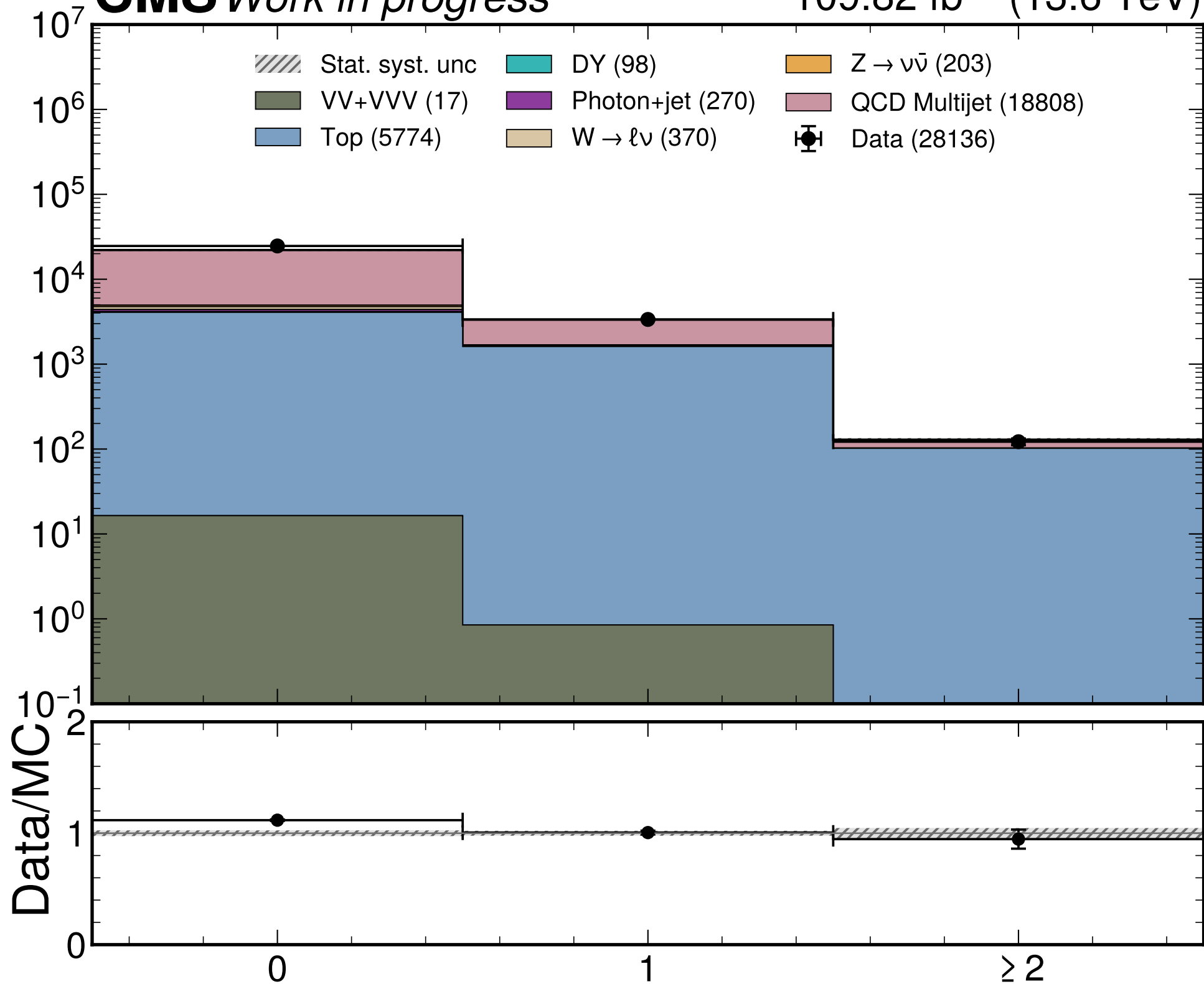
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

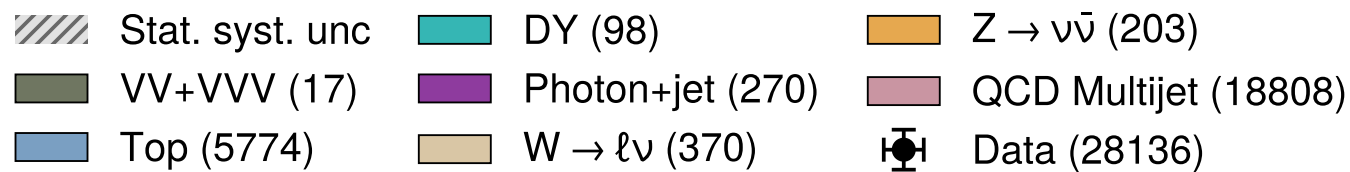
N_t



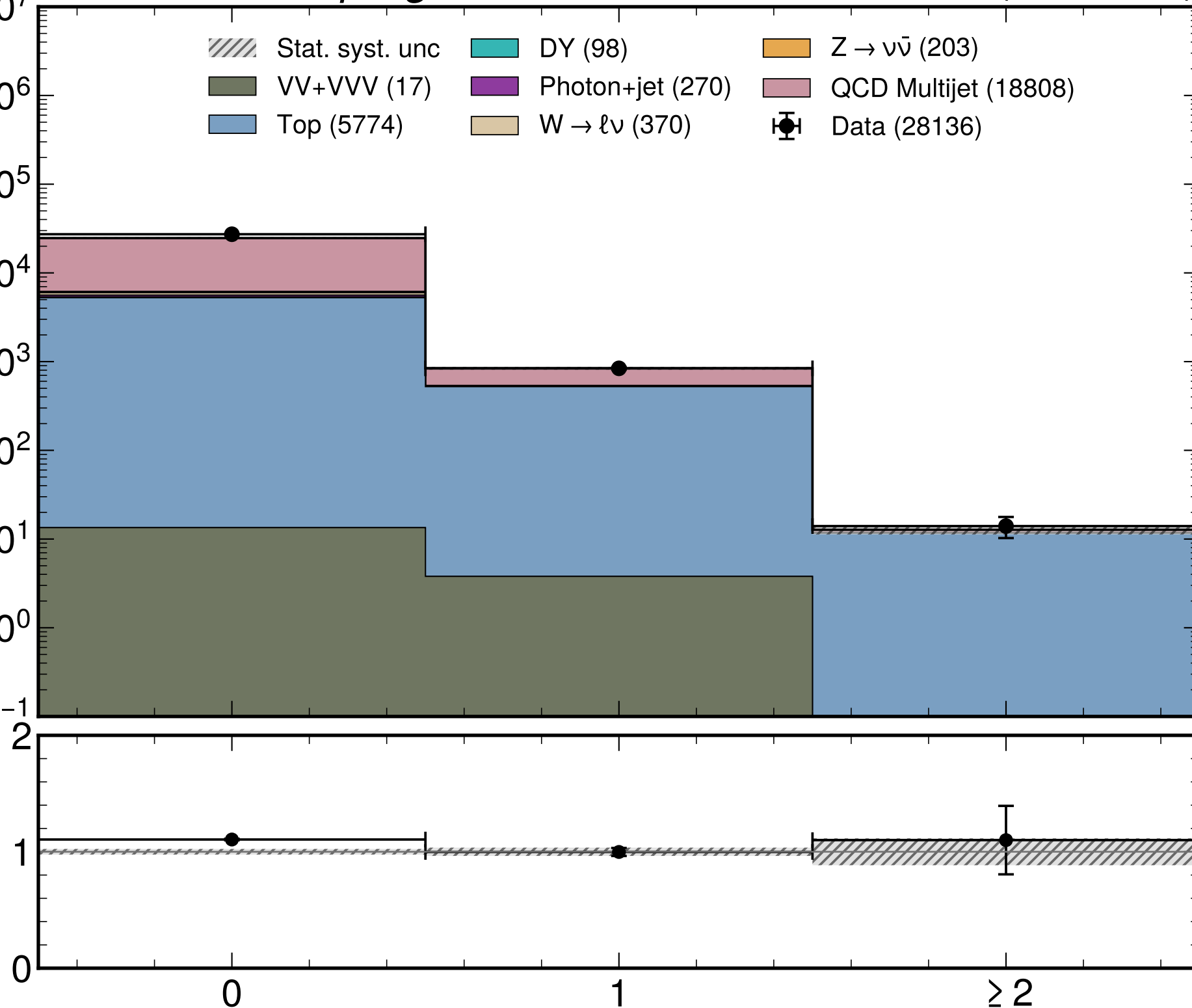
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

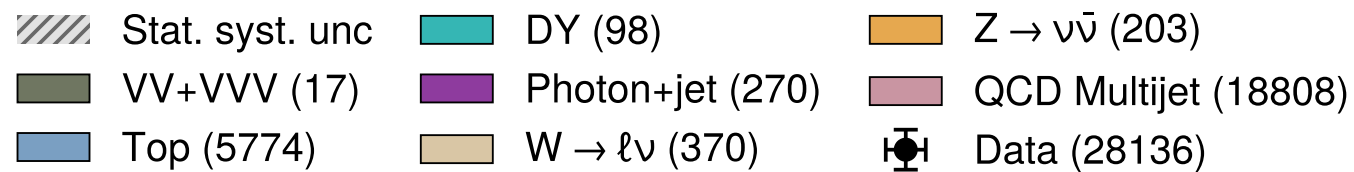


N_W

CMS *Work in progress*

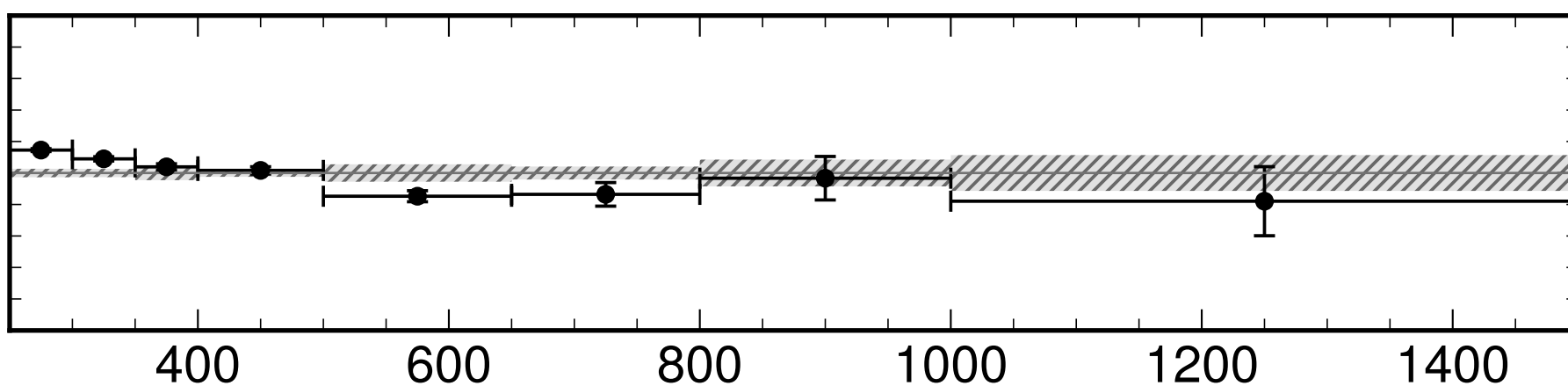
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2

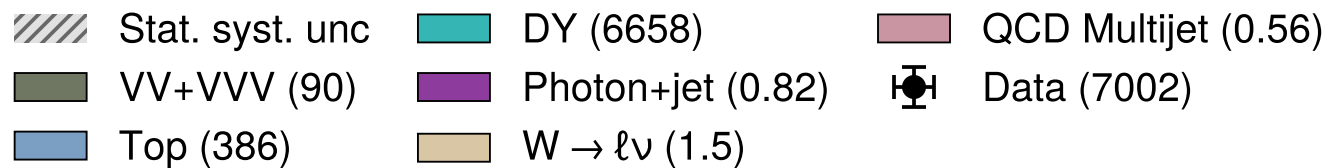


U_T (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

500

1000

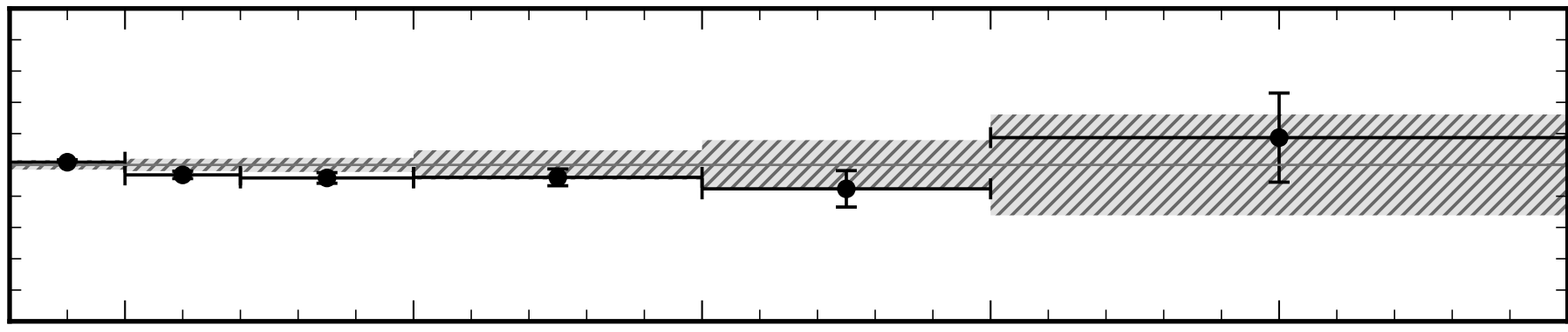
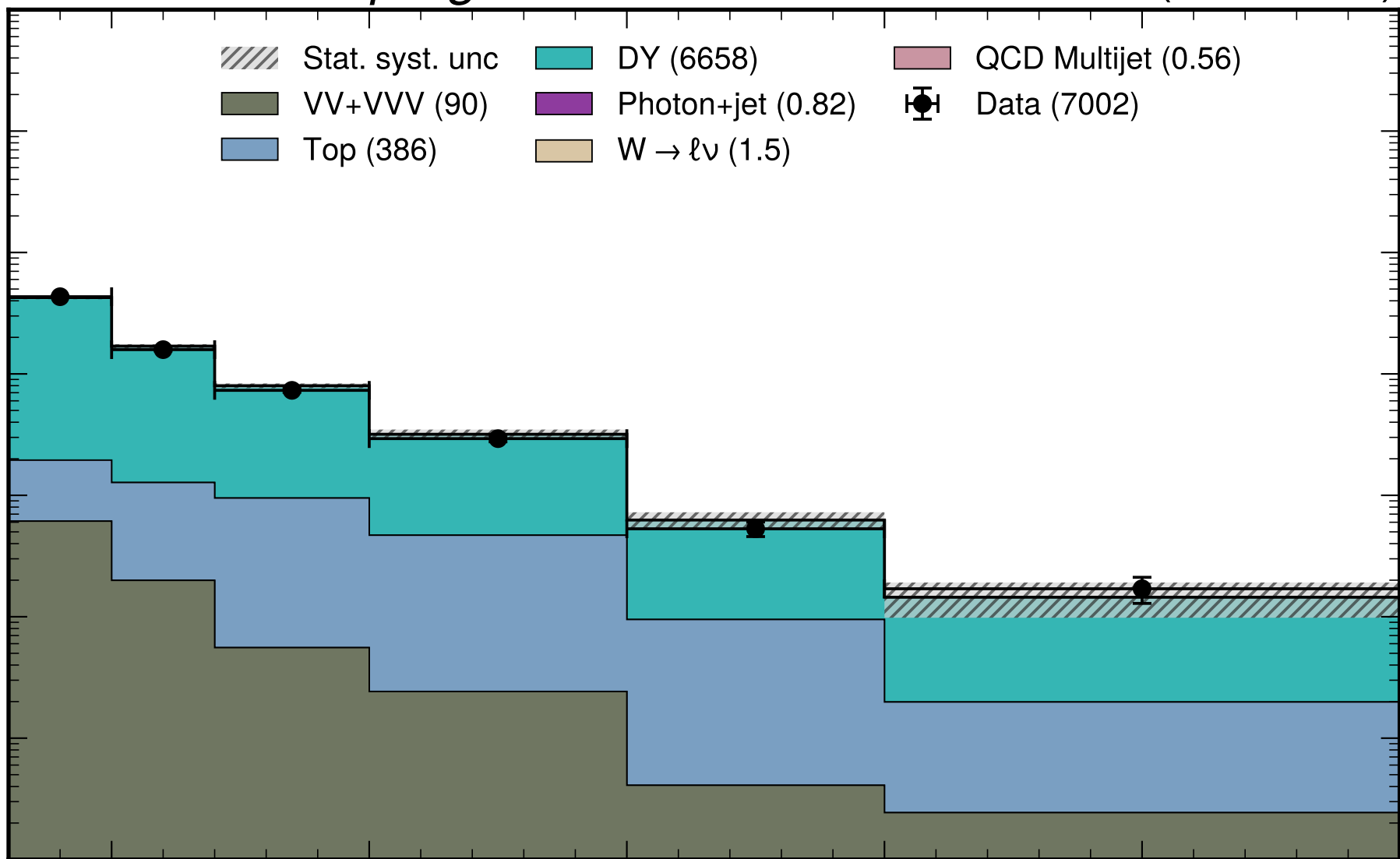
1500

2000

2500

3000

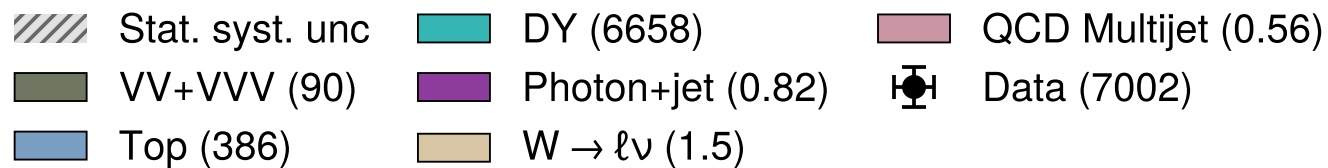
H_T (GeV)



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

500

1000

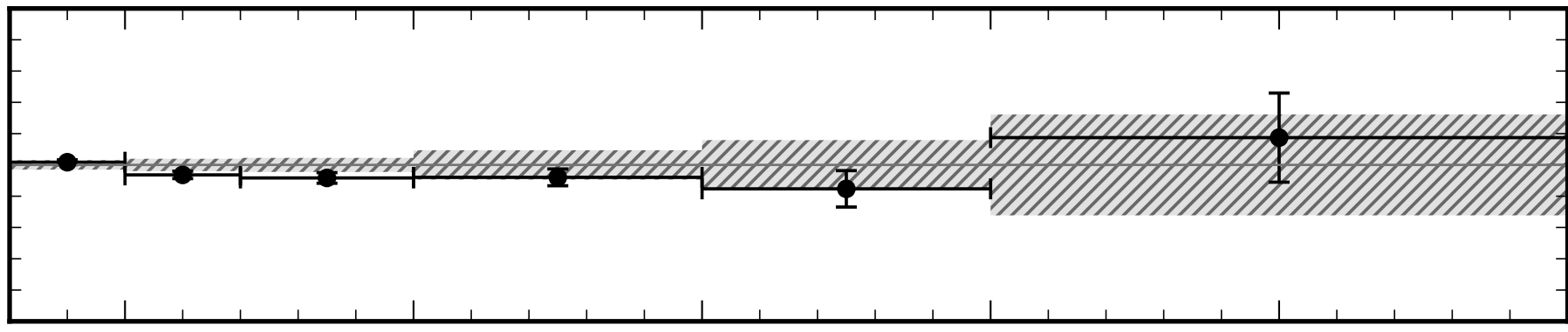
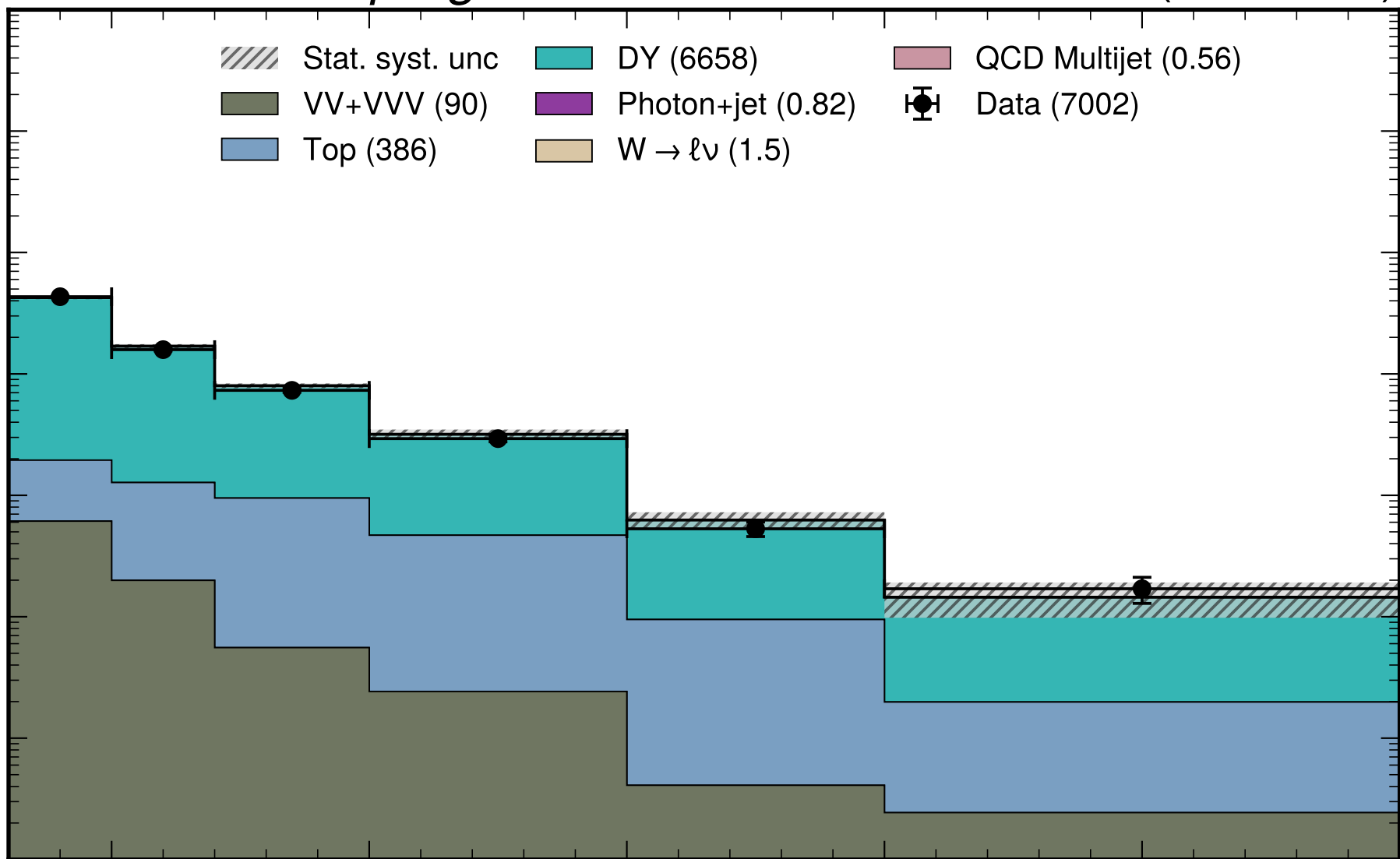
1500

2000

2500

3000

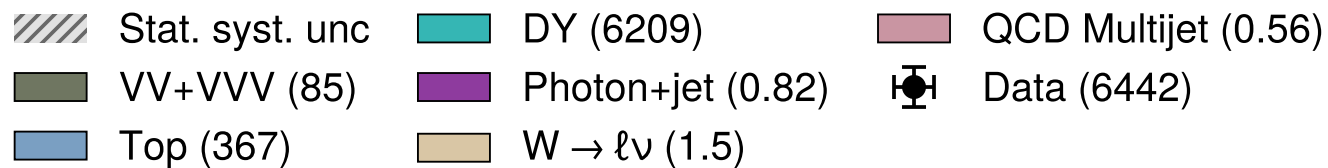
H_T (GeV)



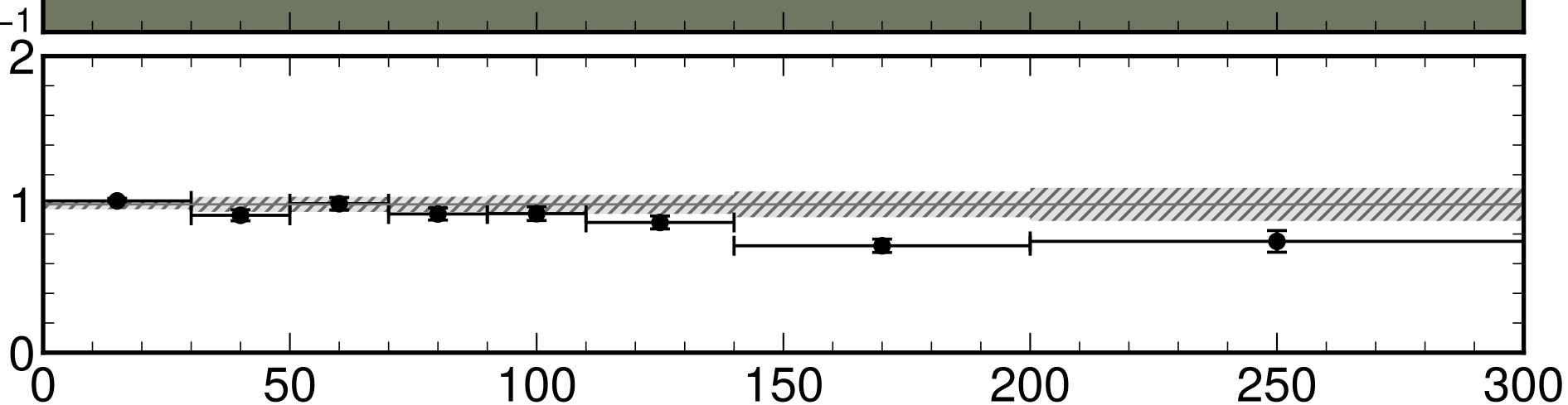
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

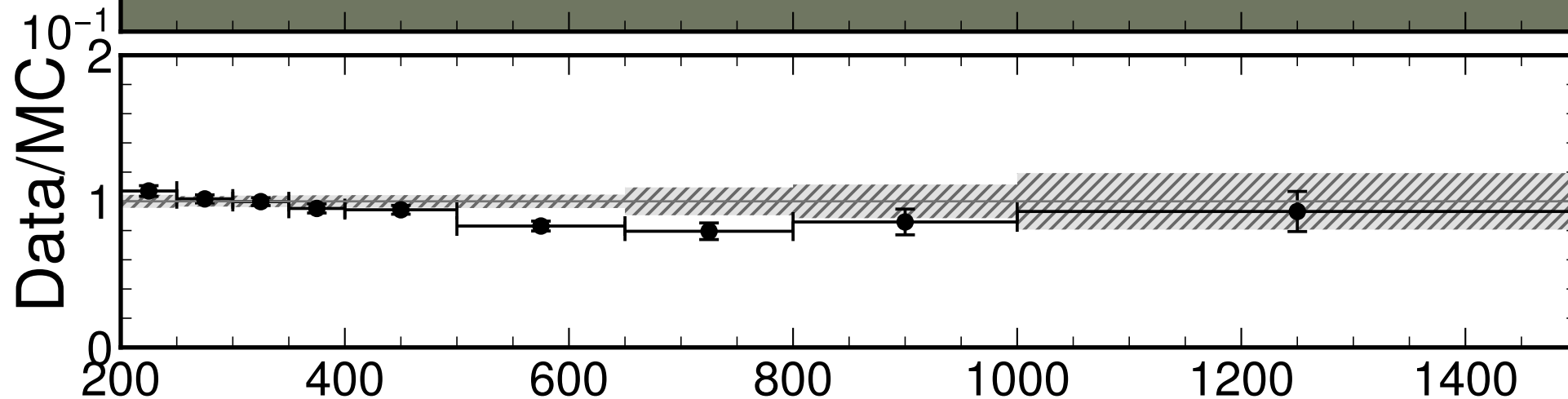
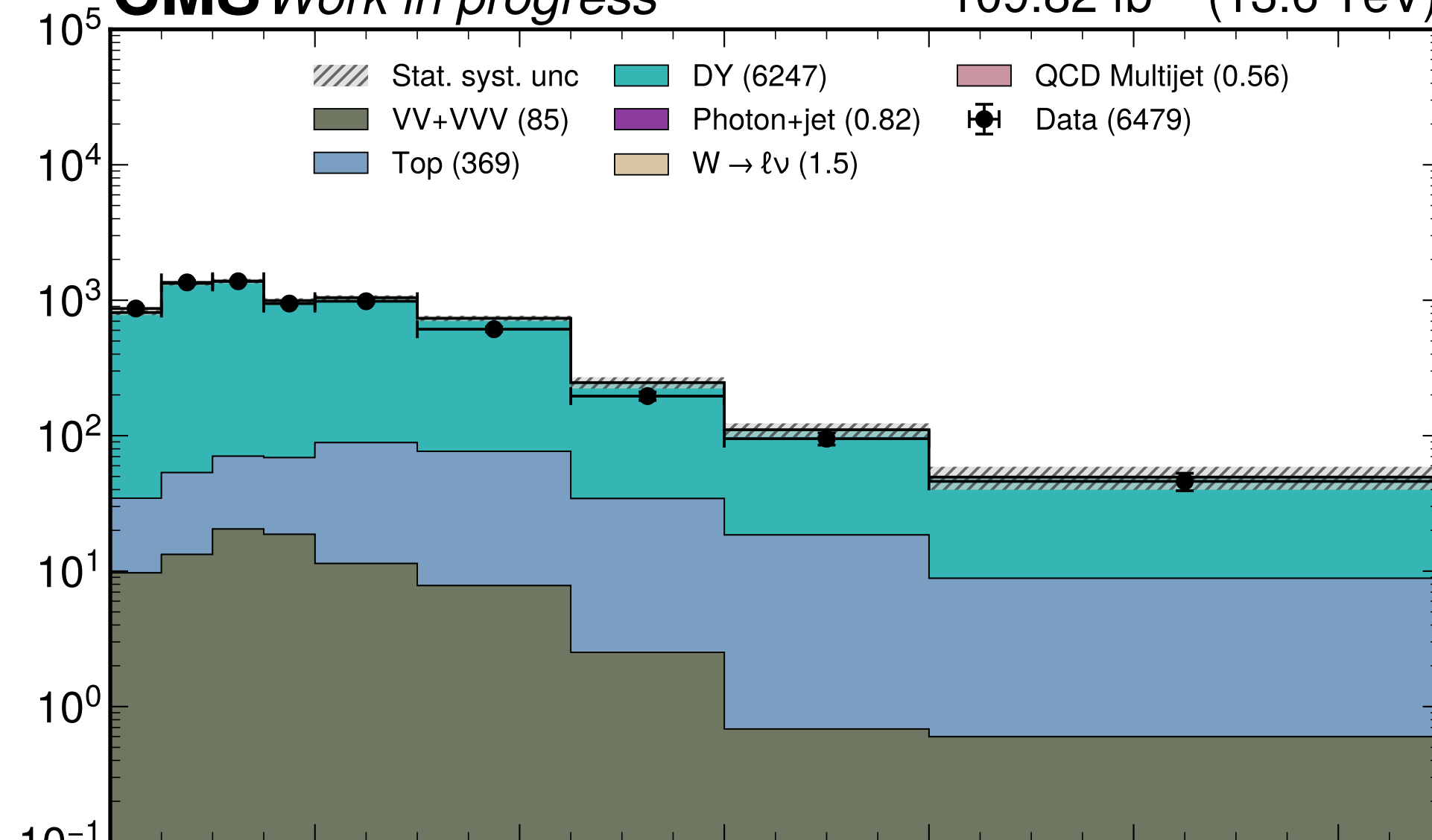
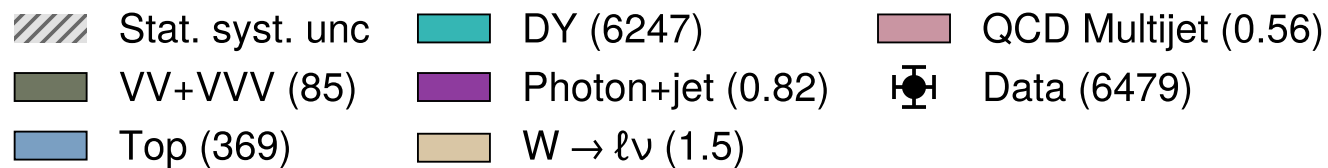


Leading AK8 jet m_{SD} (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events

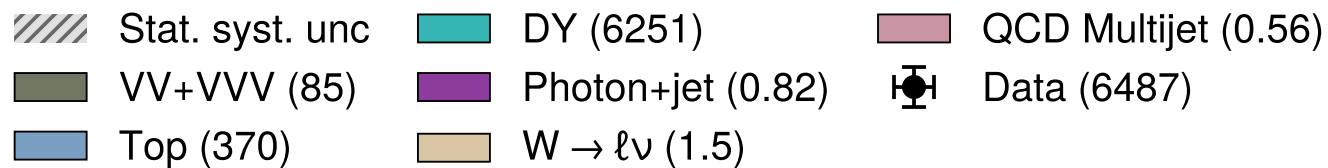


Leading AK8 jet p_T (GeV)

CMS *Work in progress*

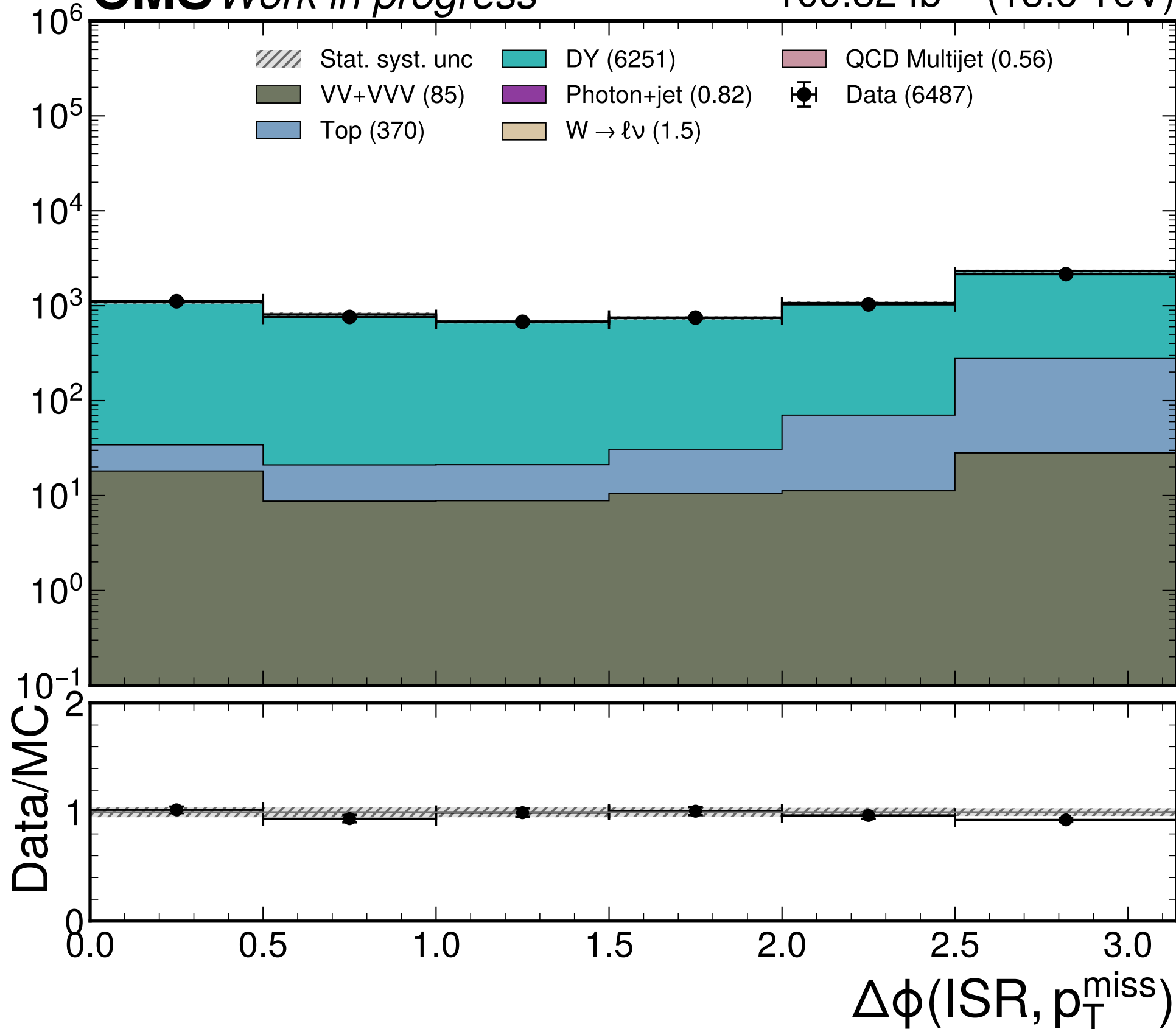
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

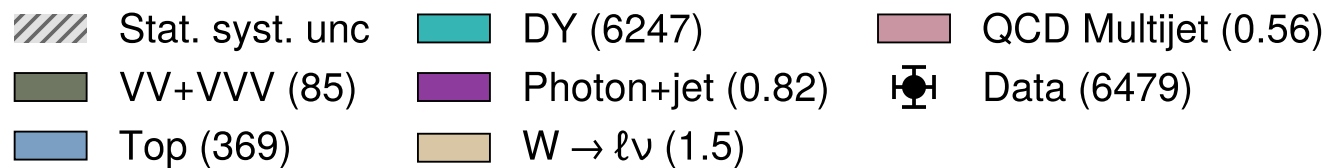
$\Delta\phi(\text{ISR}, p_{\text{T}}^{\text{miss}})$



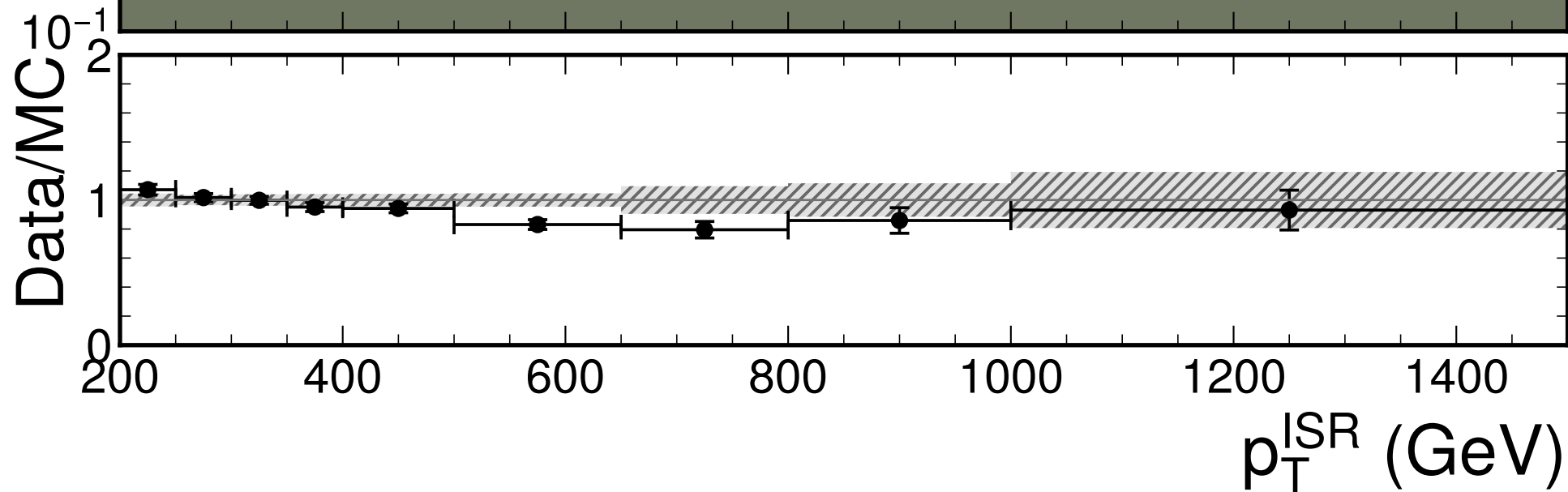
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



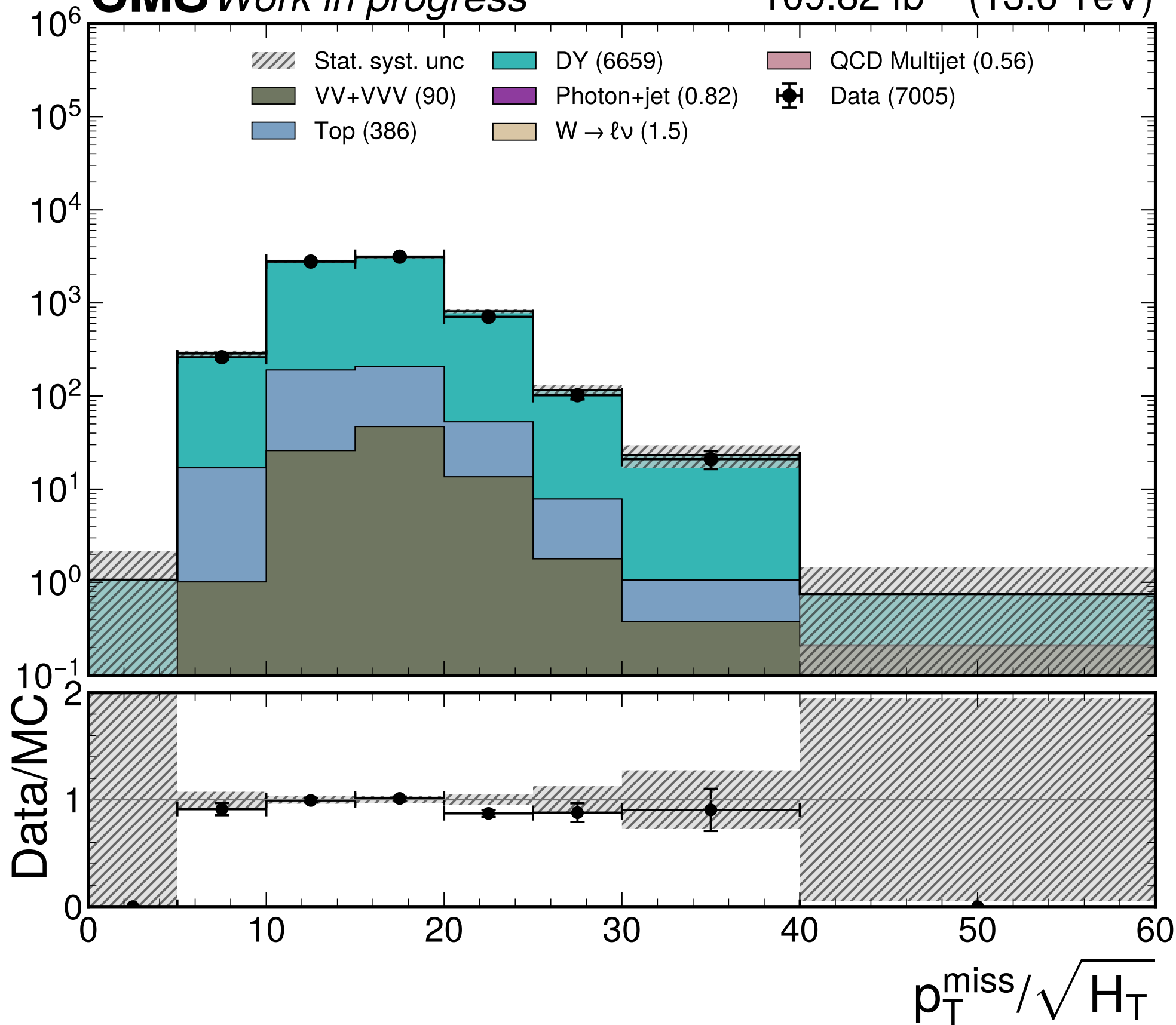
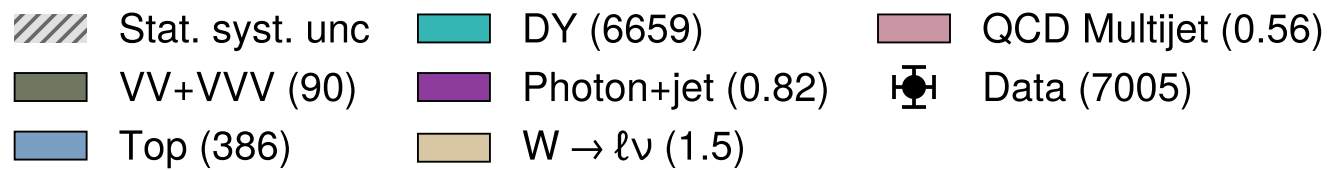
Data/MC



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

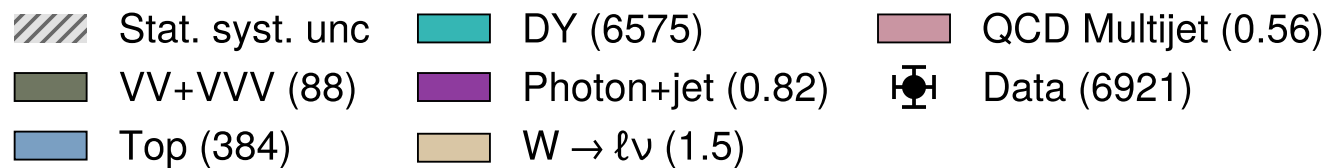
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2

0

1

2

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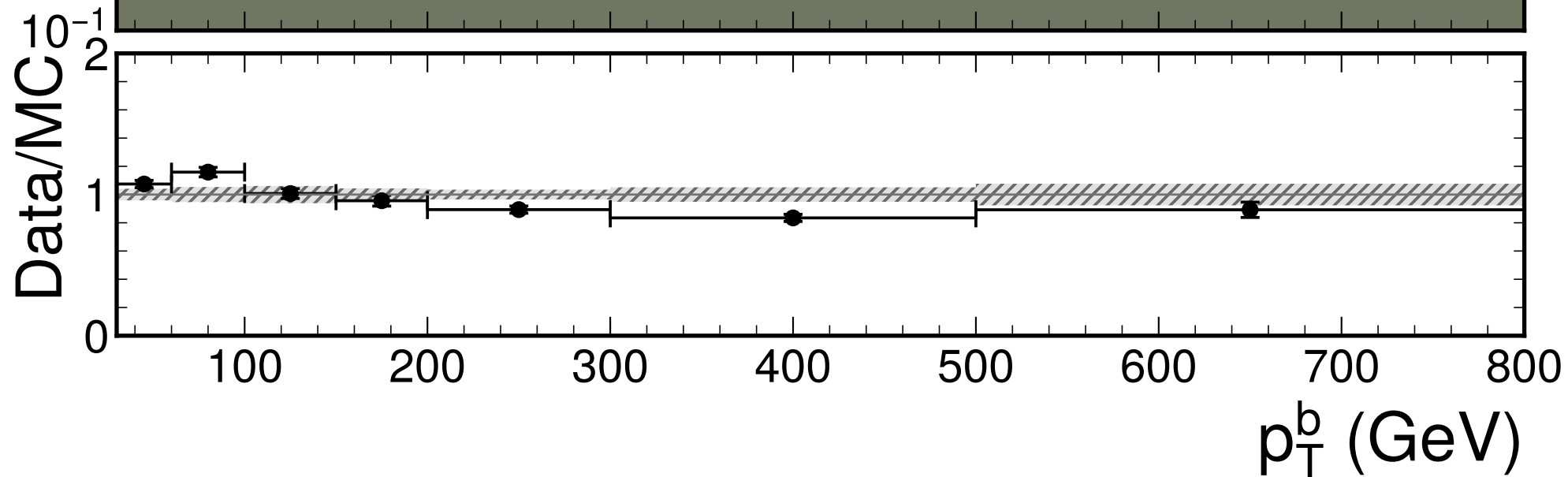
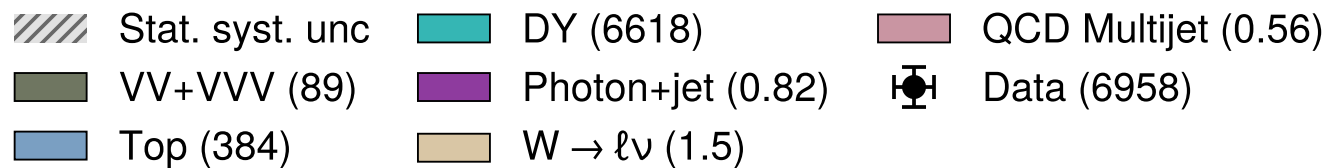
330

331</

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

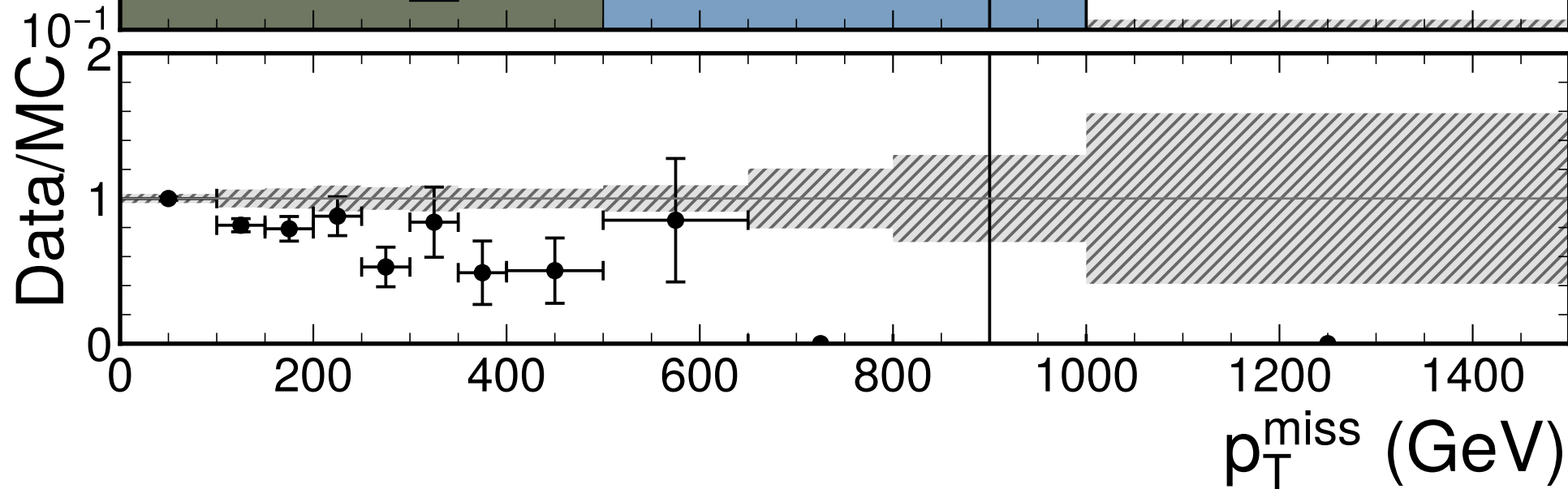
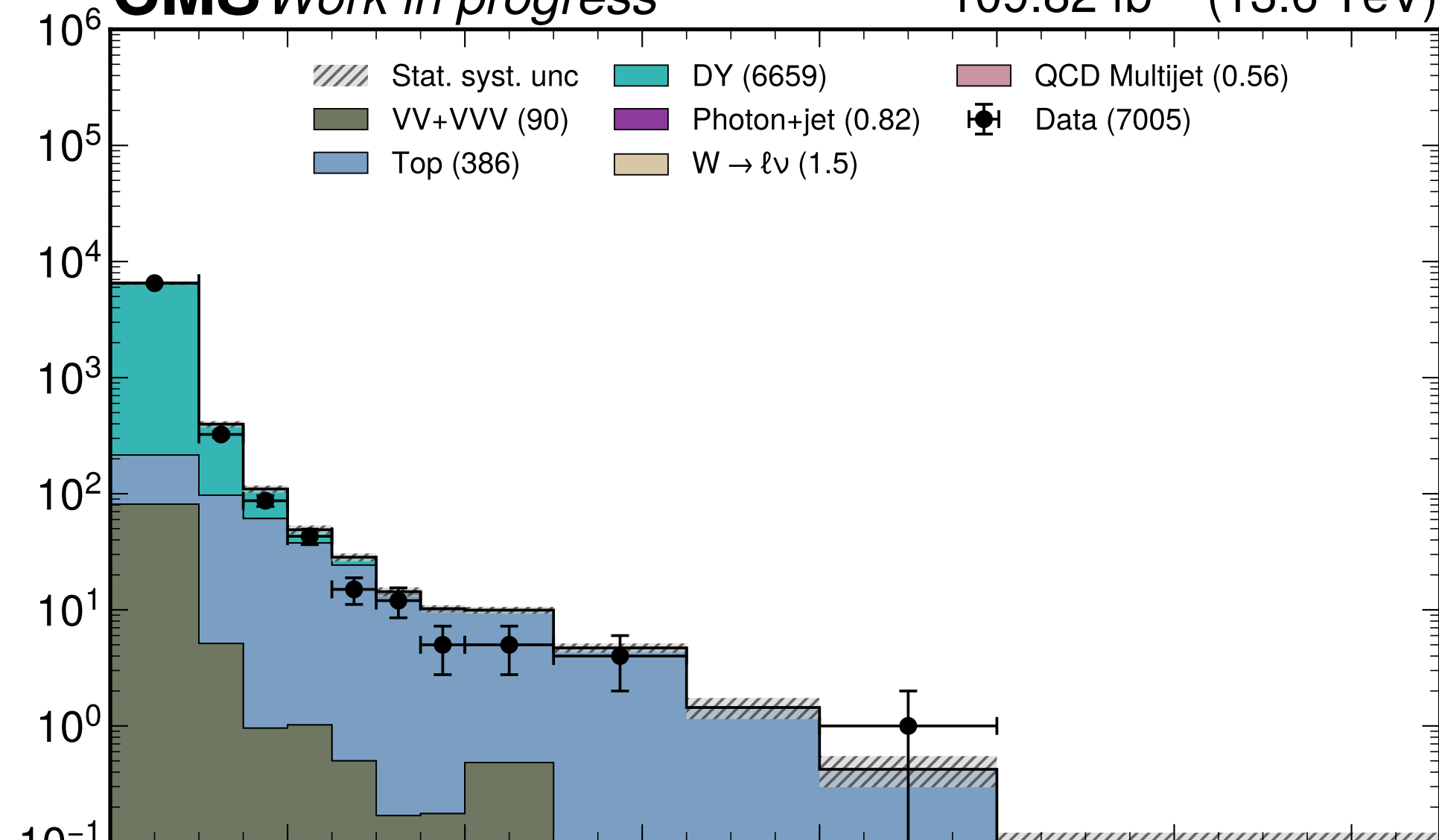
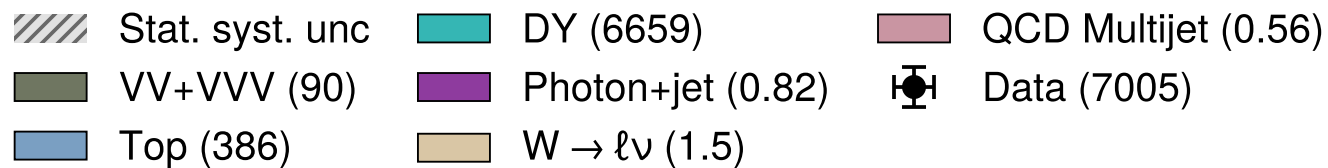
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events

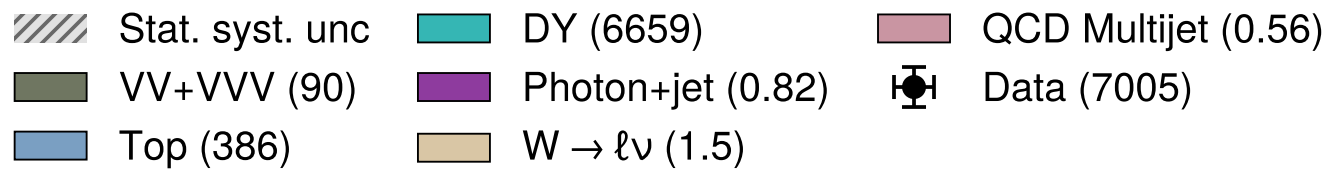


p_T^{miss} (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

70

80

90

100

110

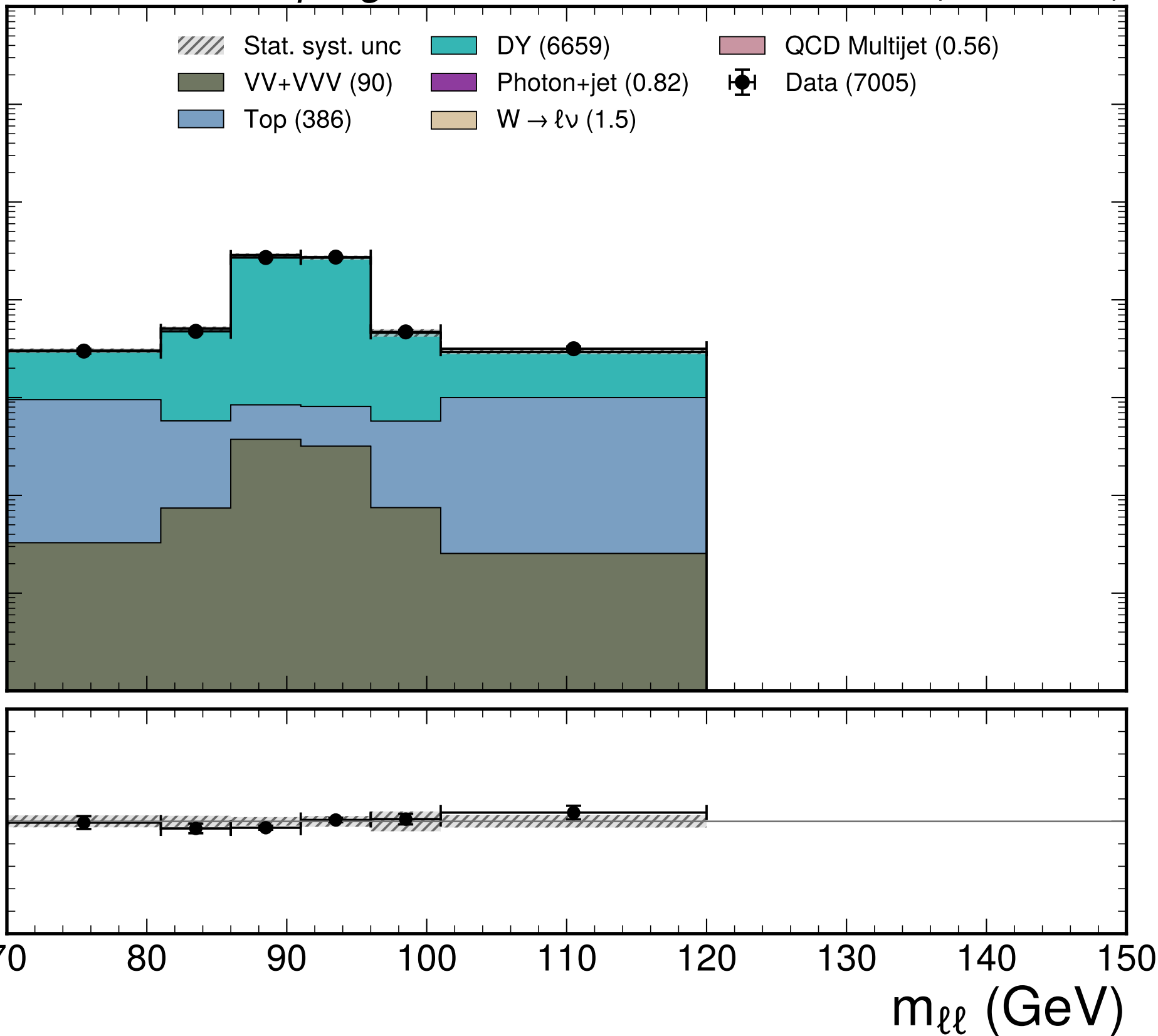
120

130

140

150

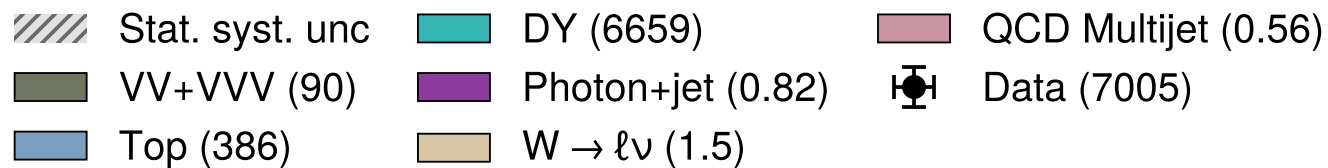
$m_{\ell\ell}$ (GeV)



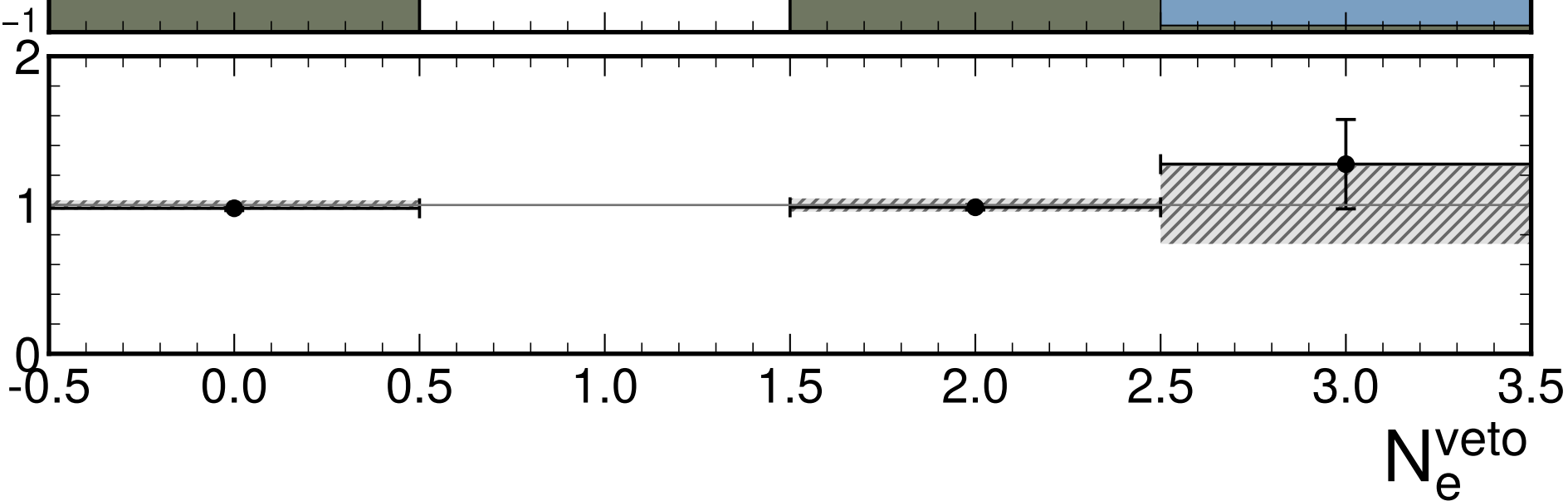
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



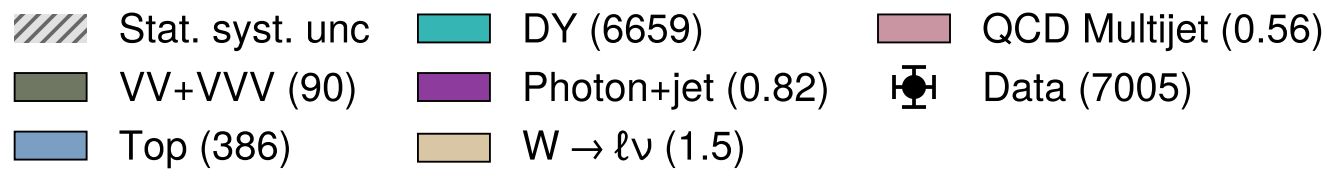
Data/MC



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

0

0

1

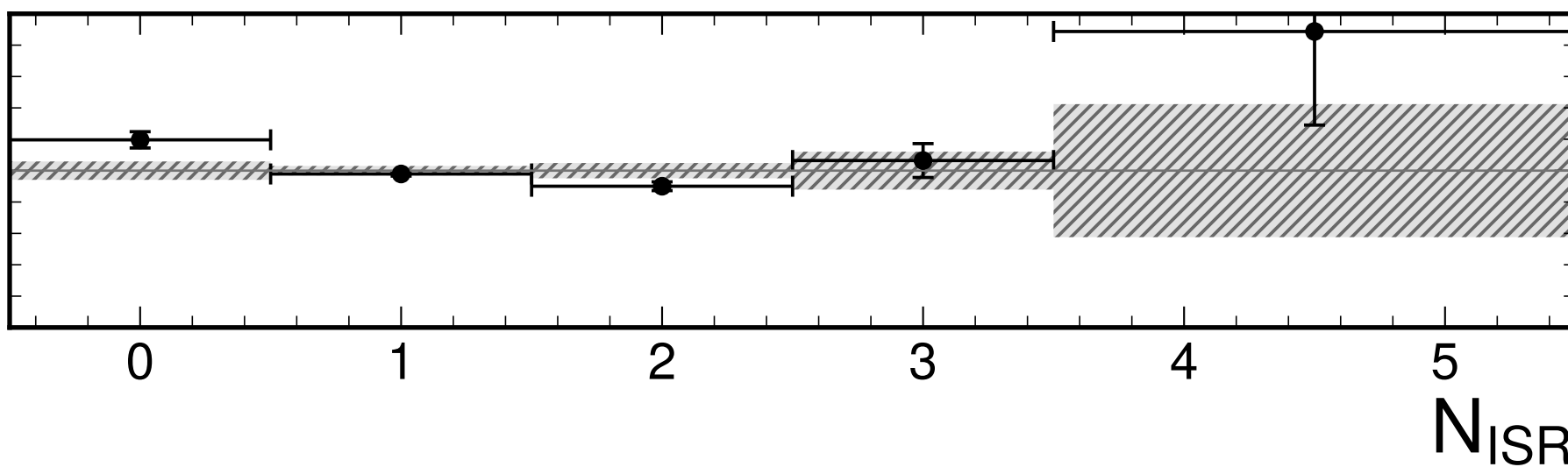
2

3

4

5

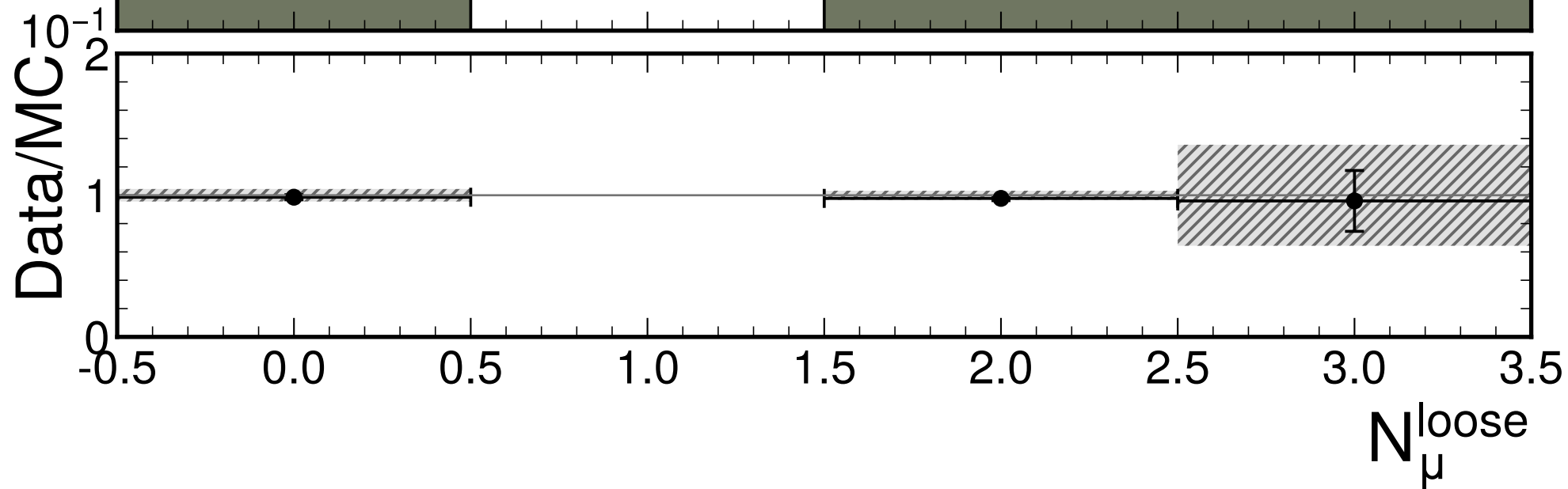
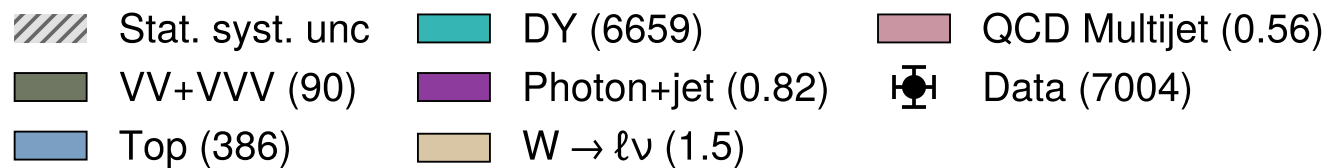
N_{ISR}



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

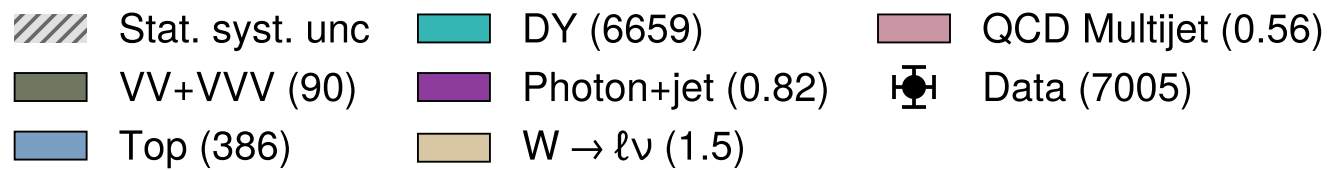
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

1

2

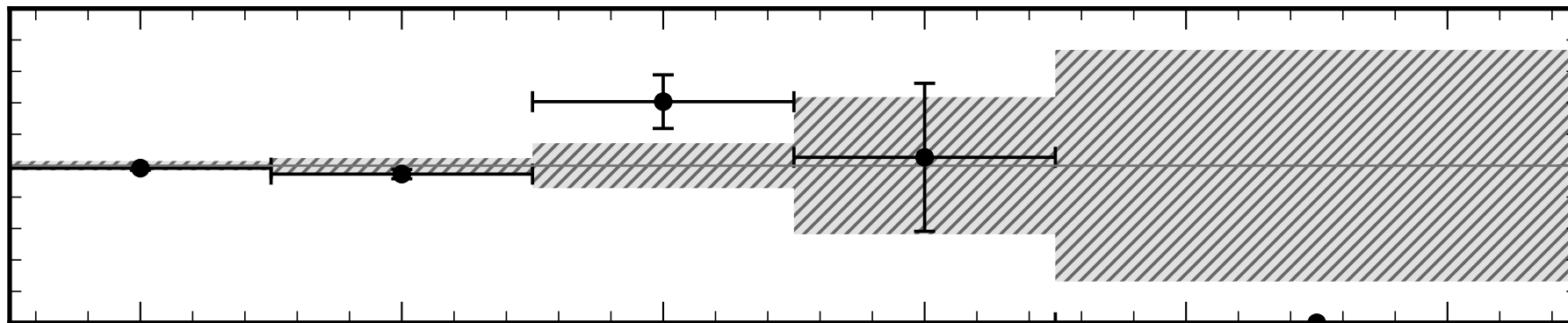
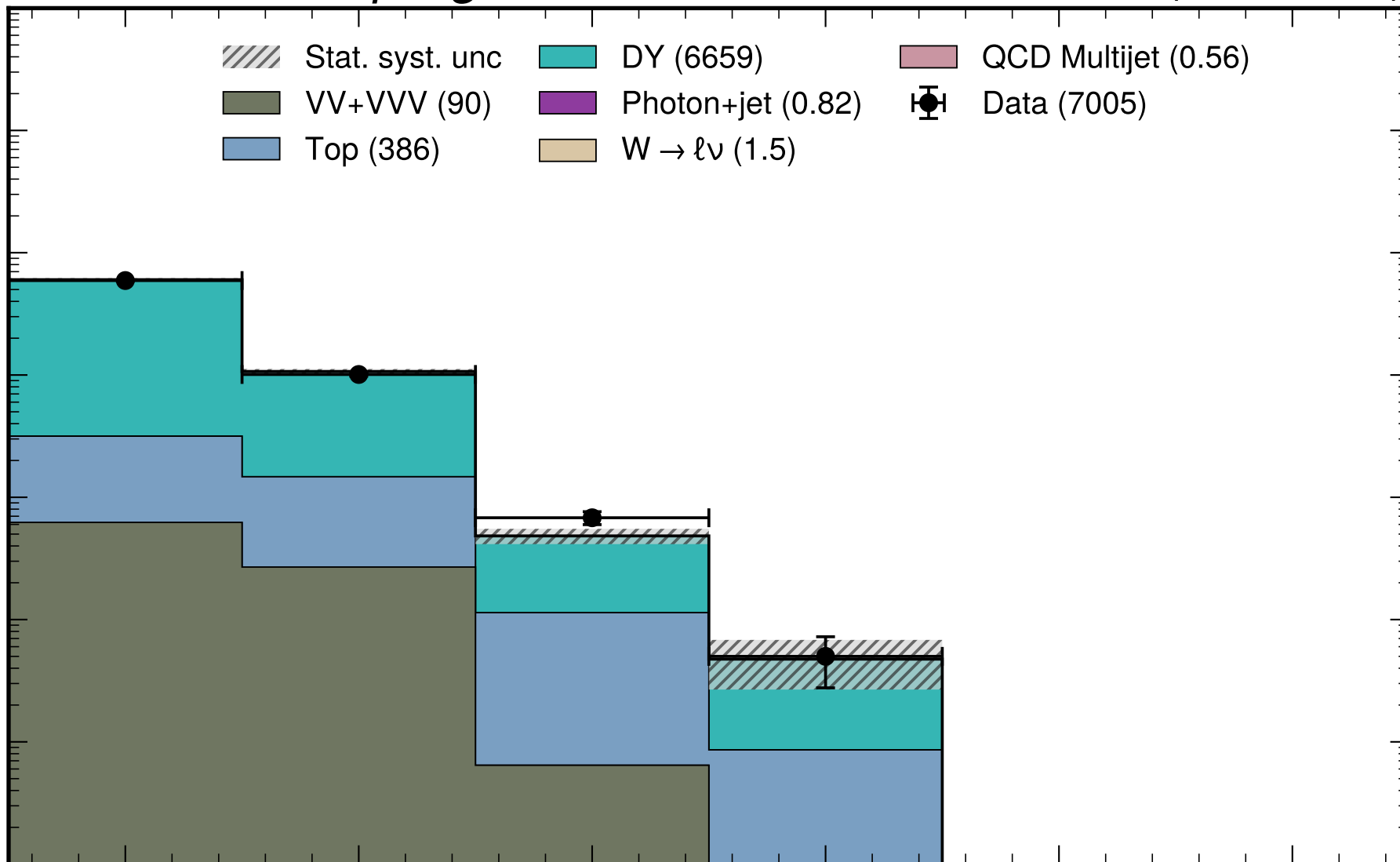
3

4

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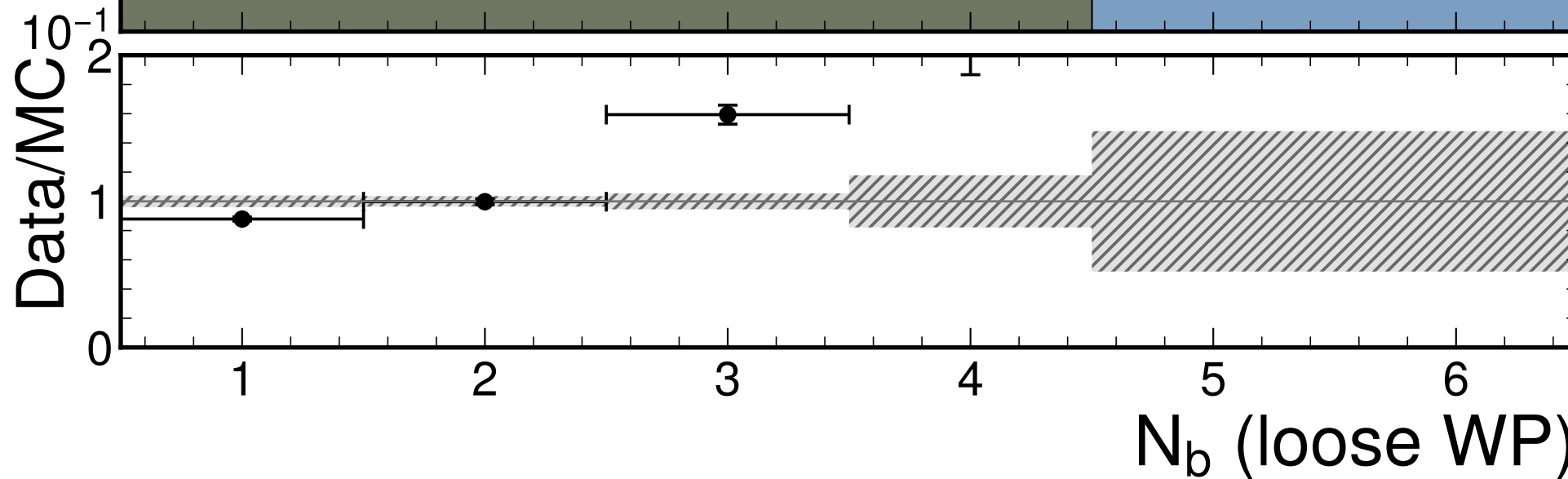
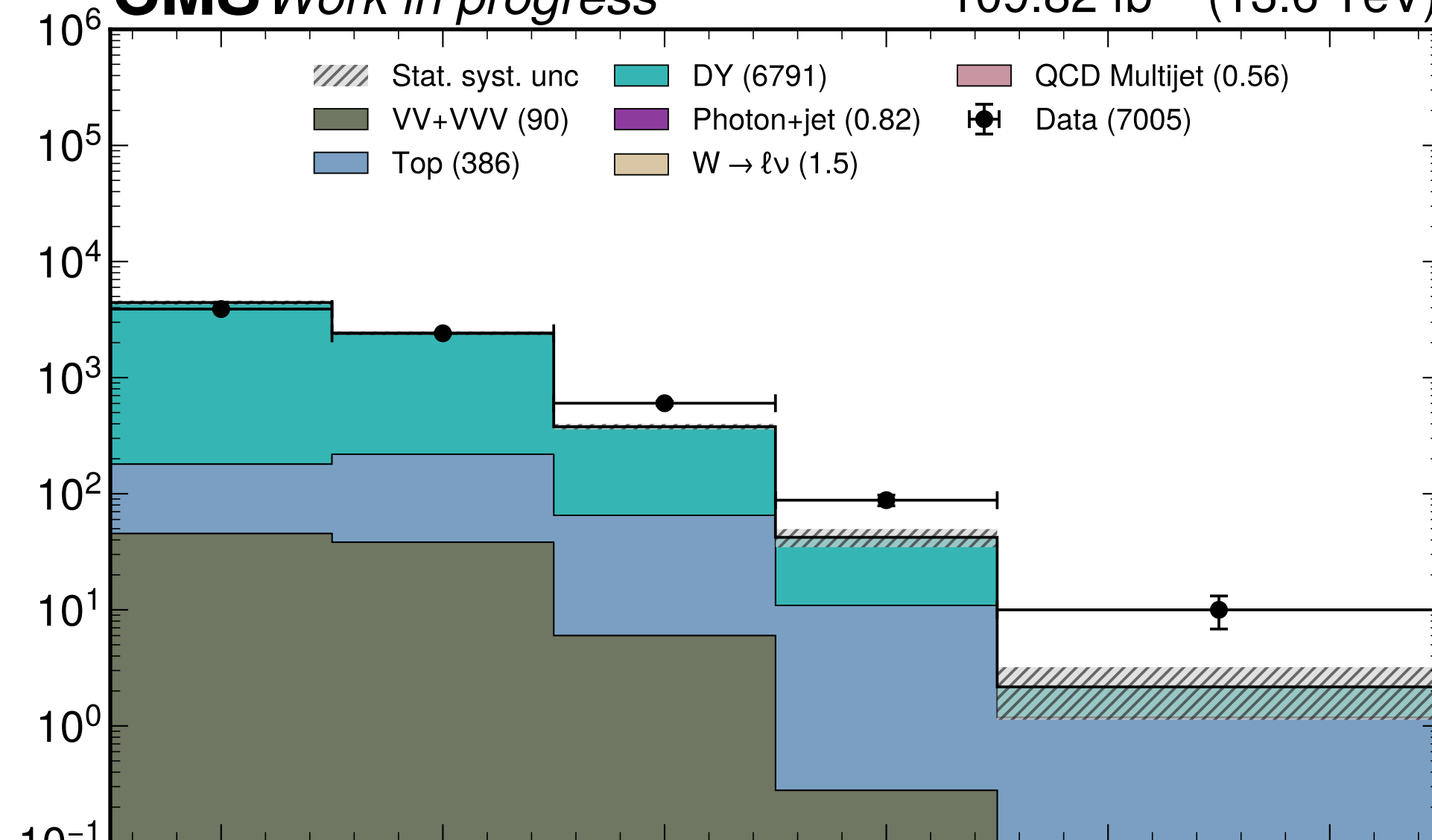
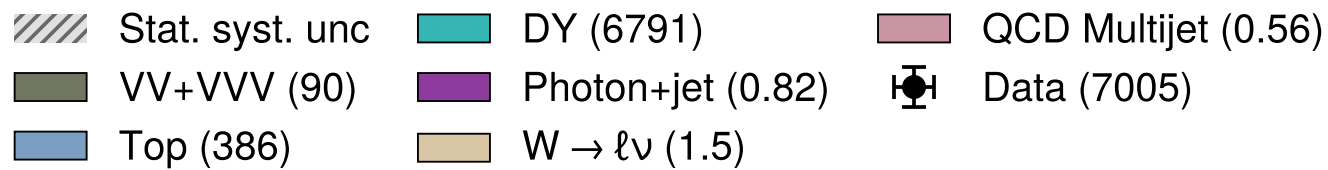
N_b



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

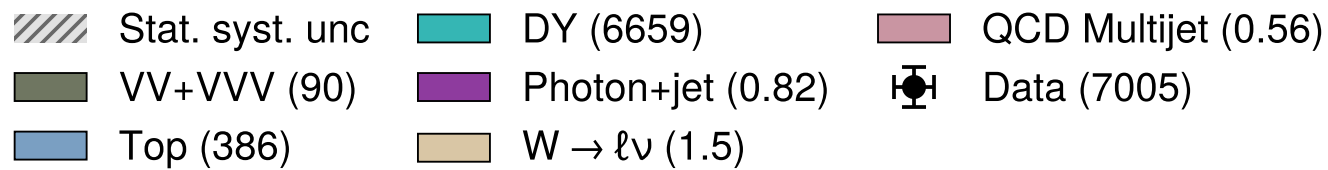
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

1

2

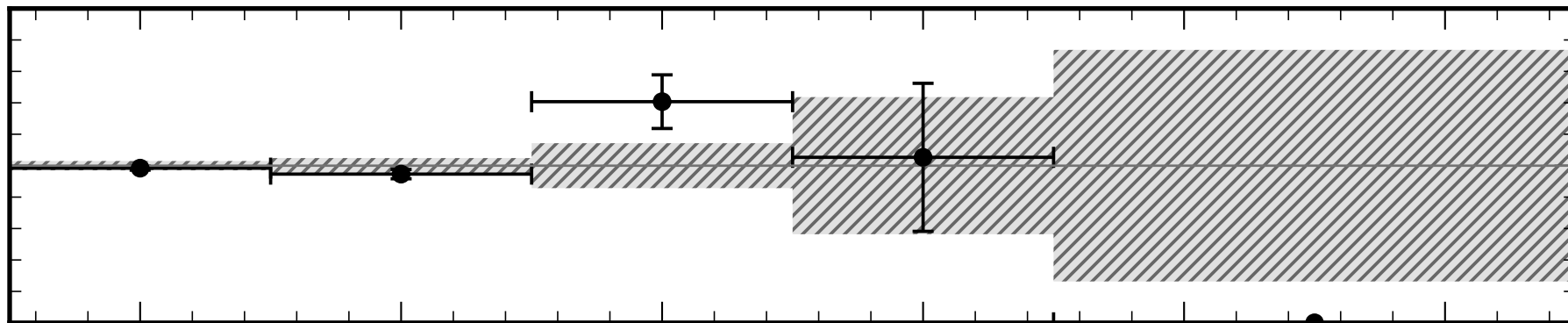
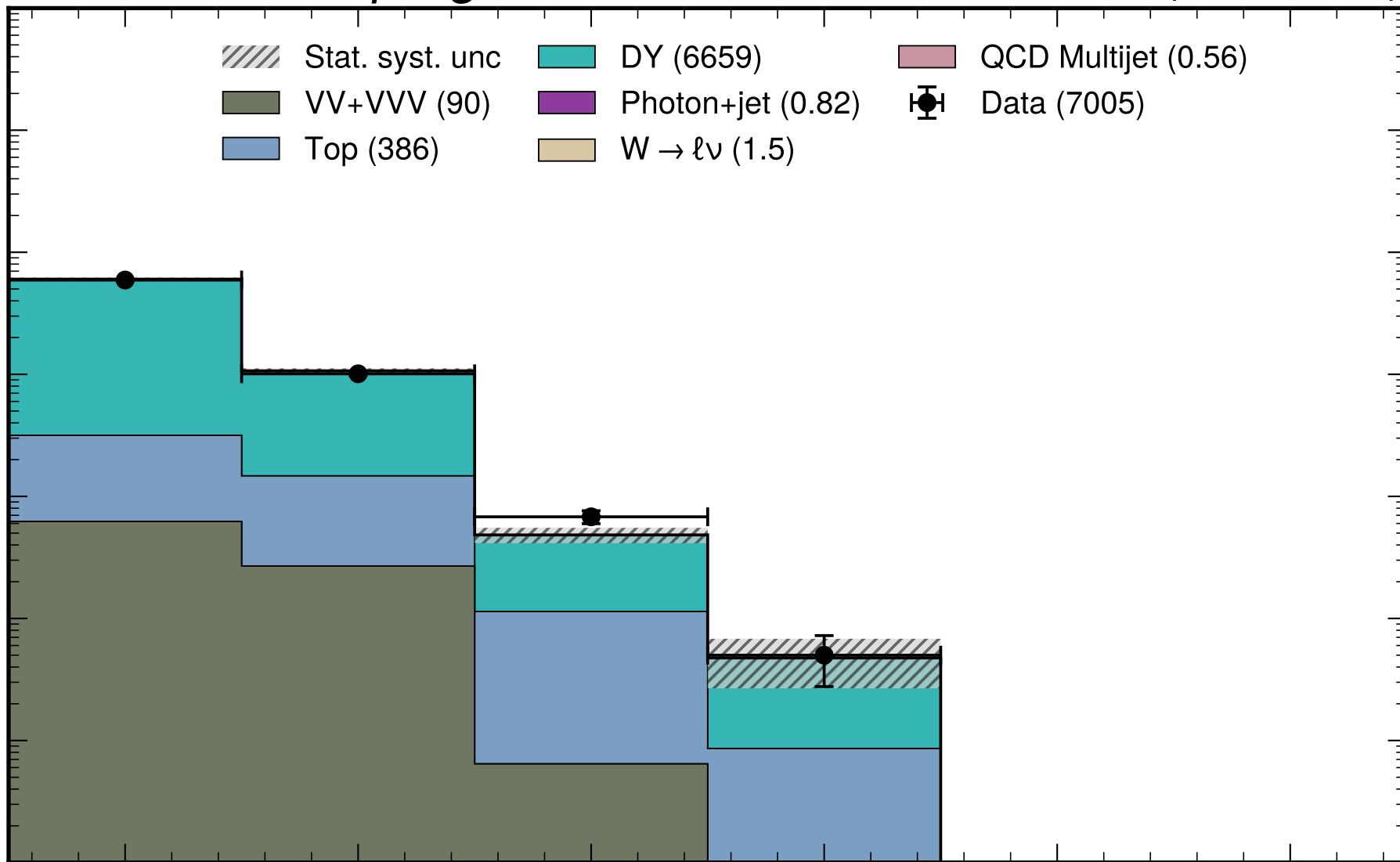
3

4

5

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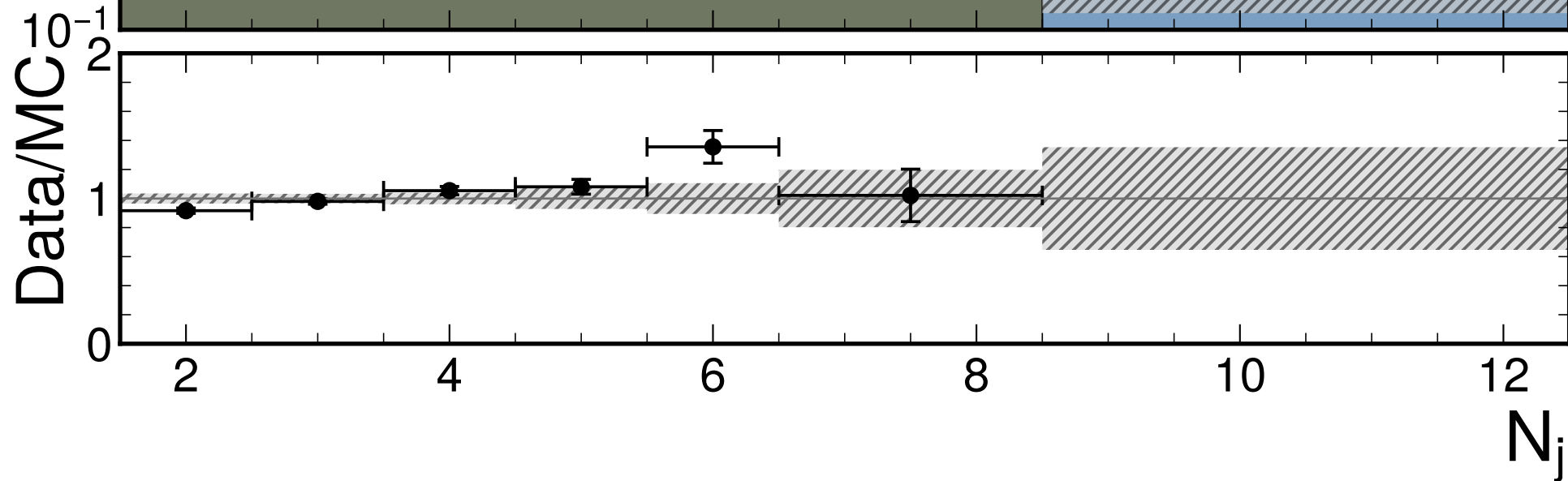
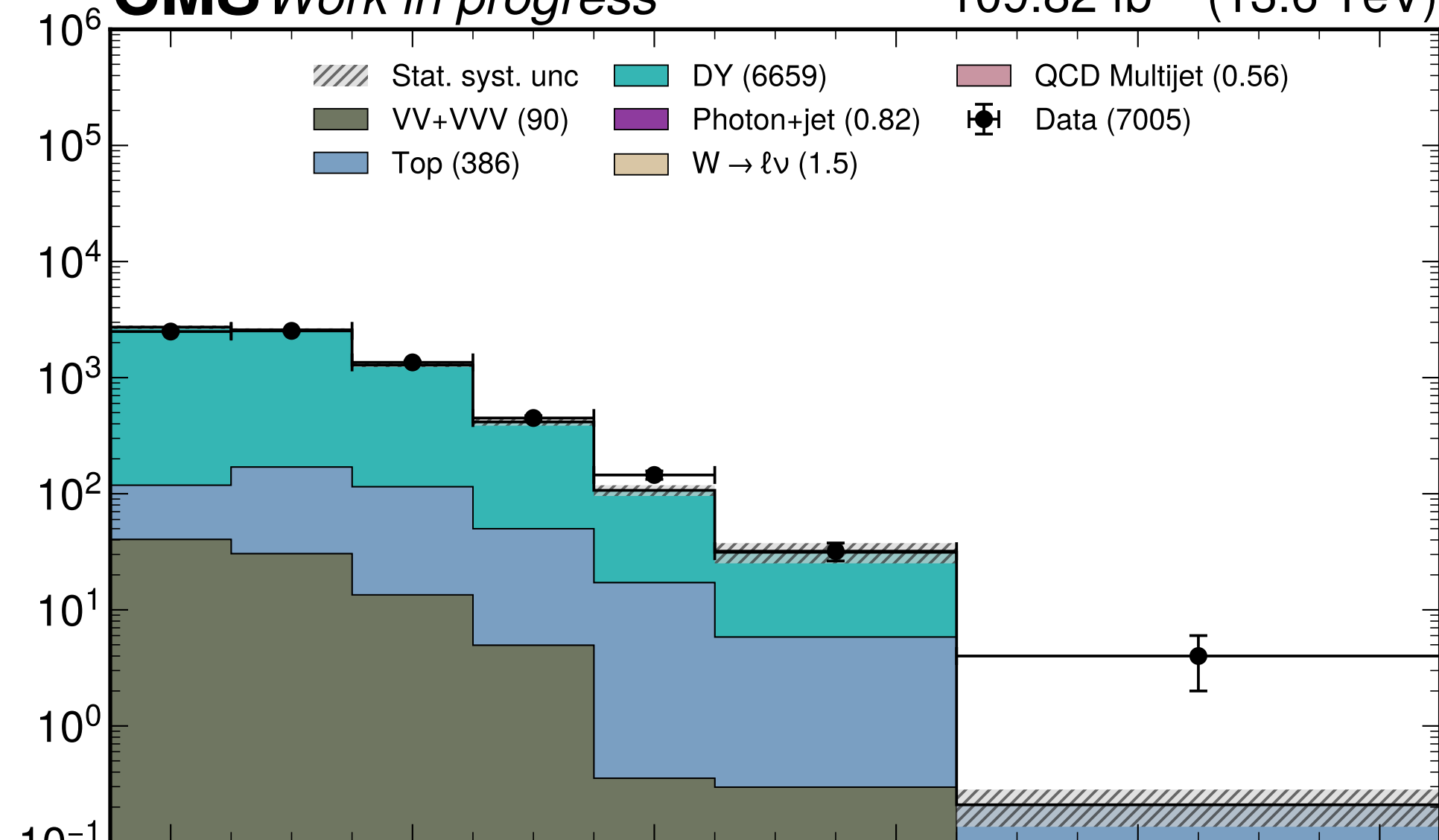
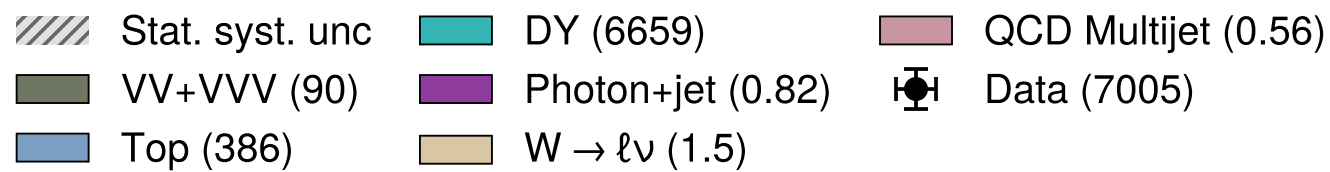
N_b



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

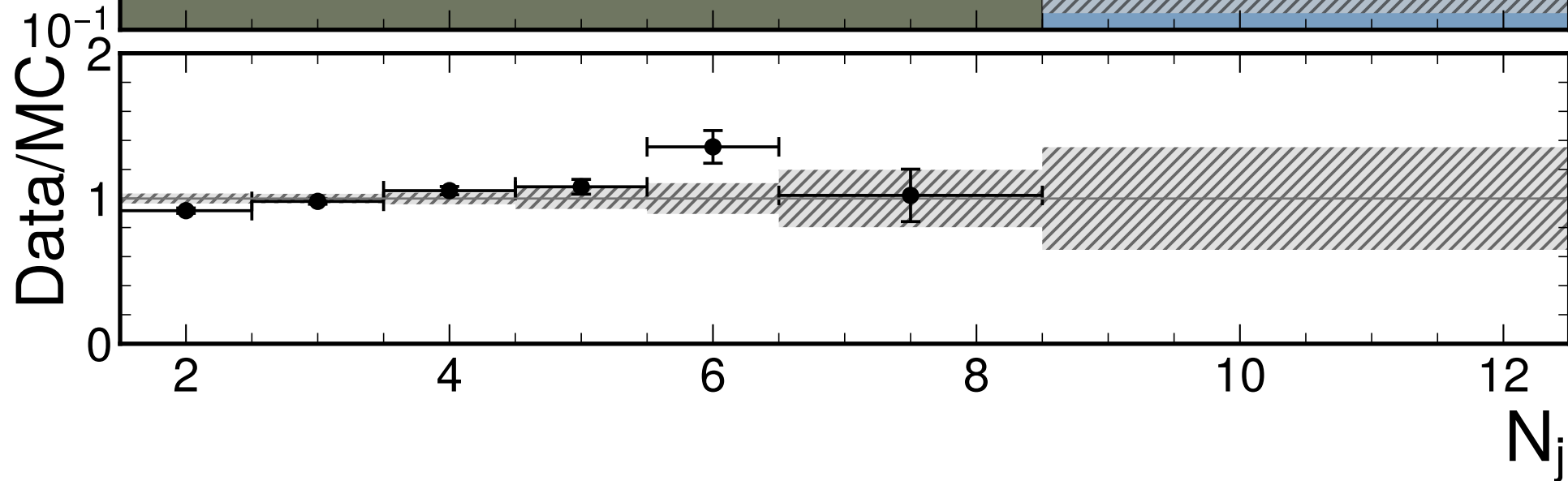
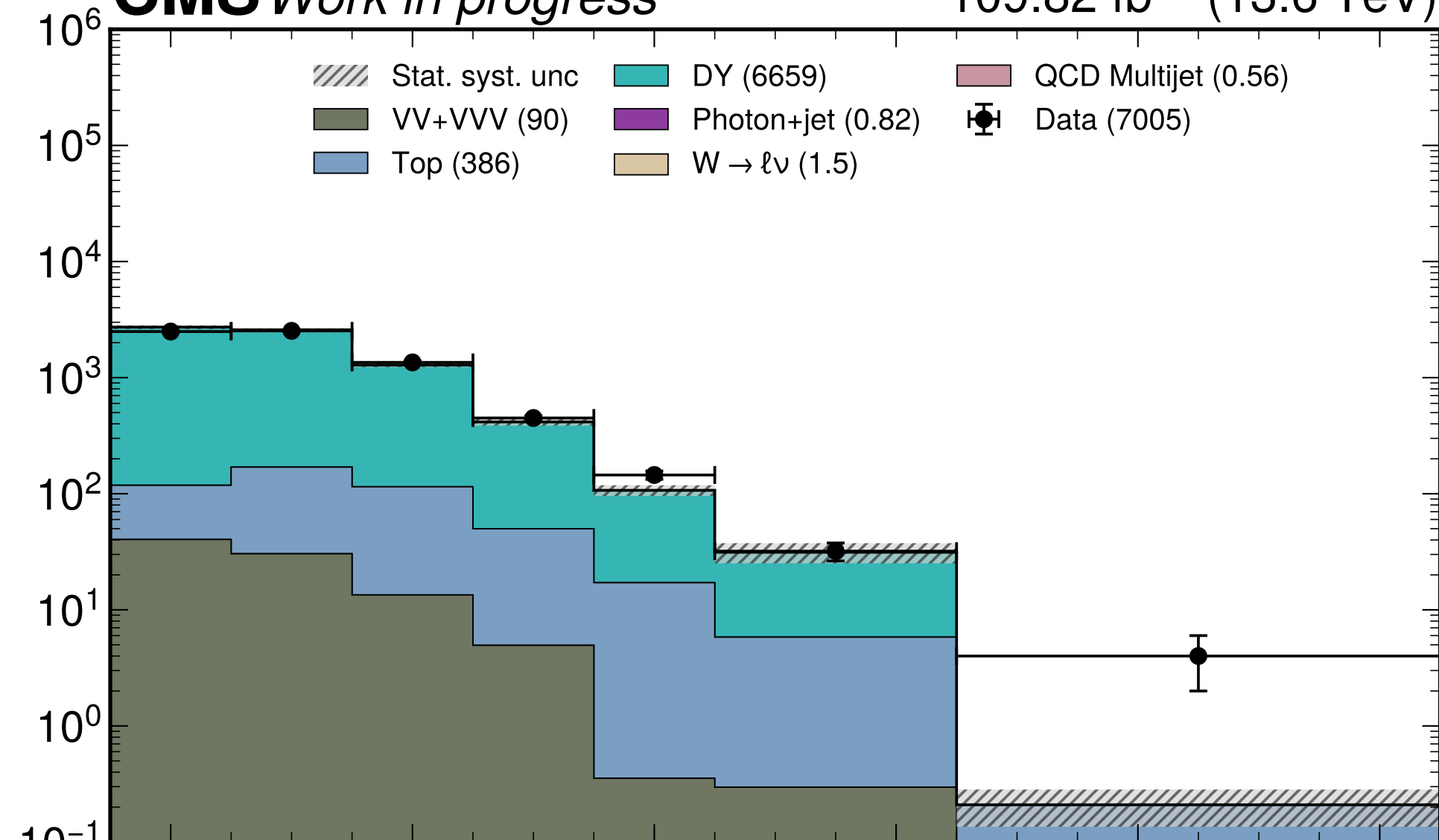
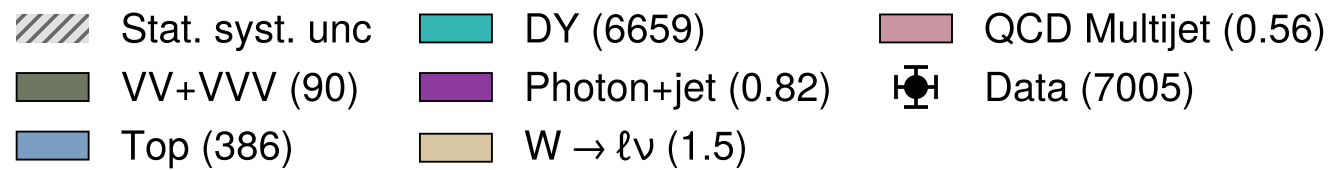
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

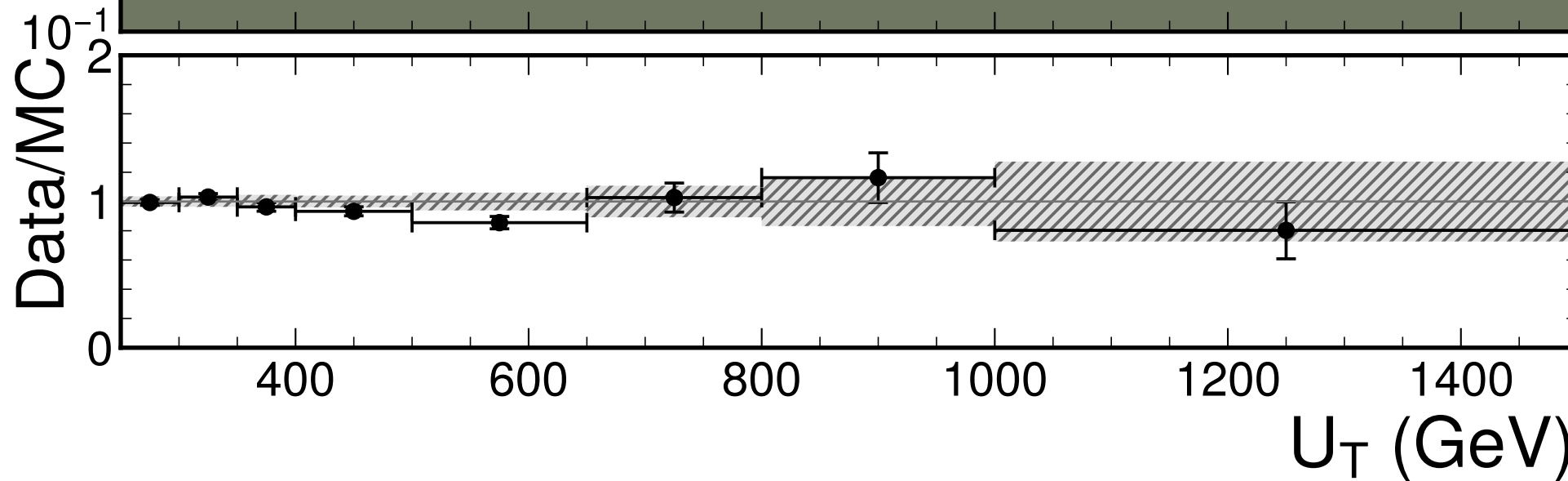
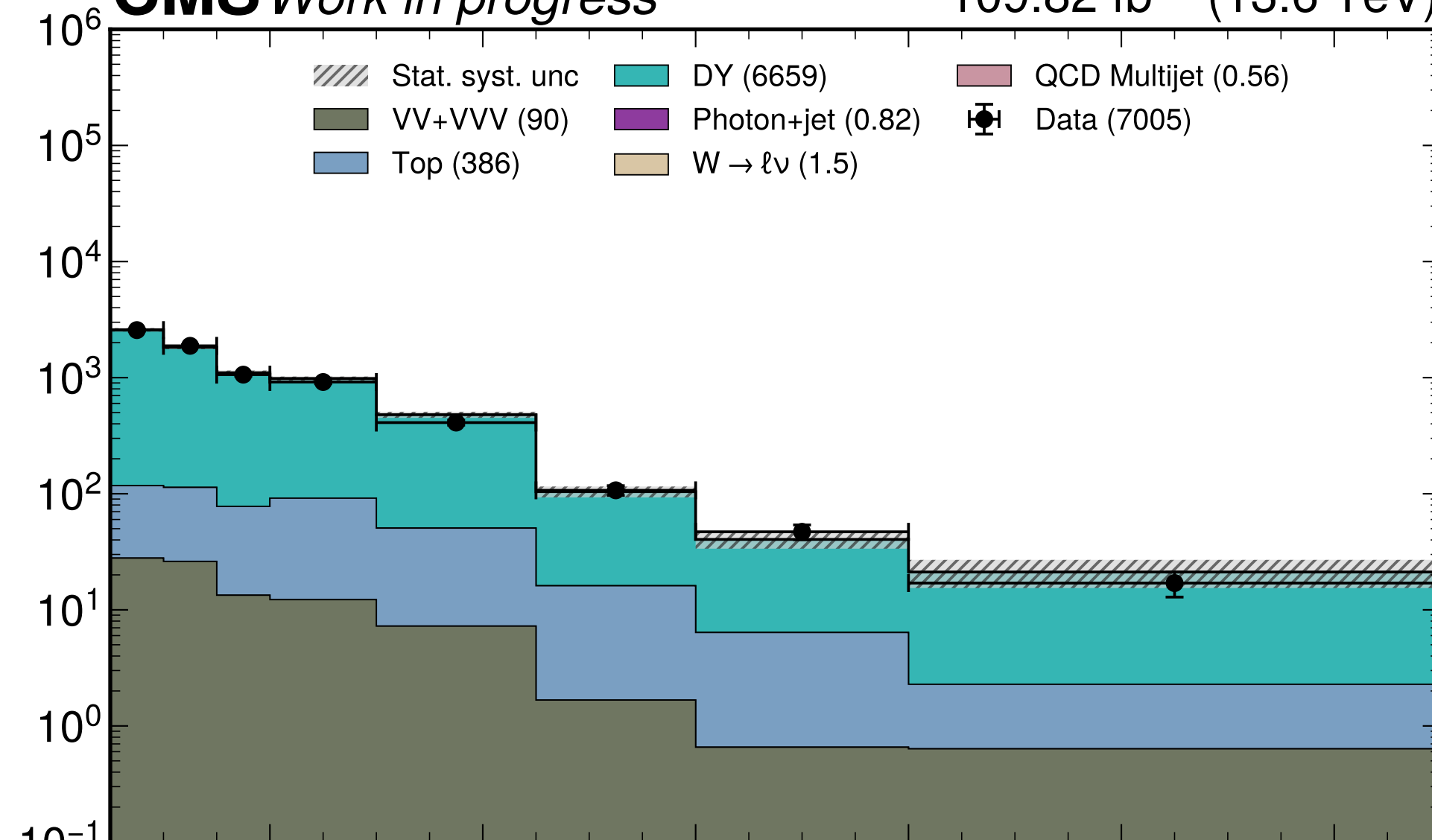
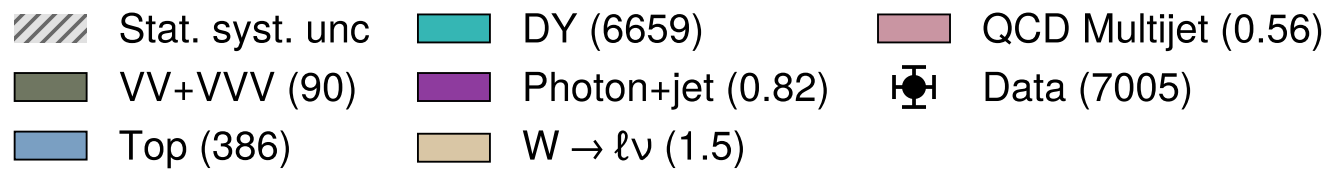
Events



CMS *Work in progress*

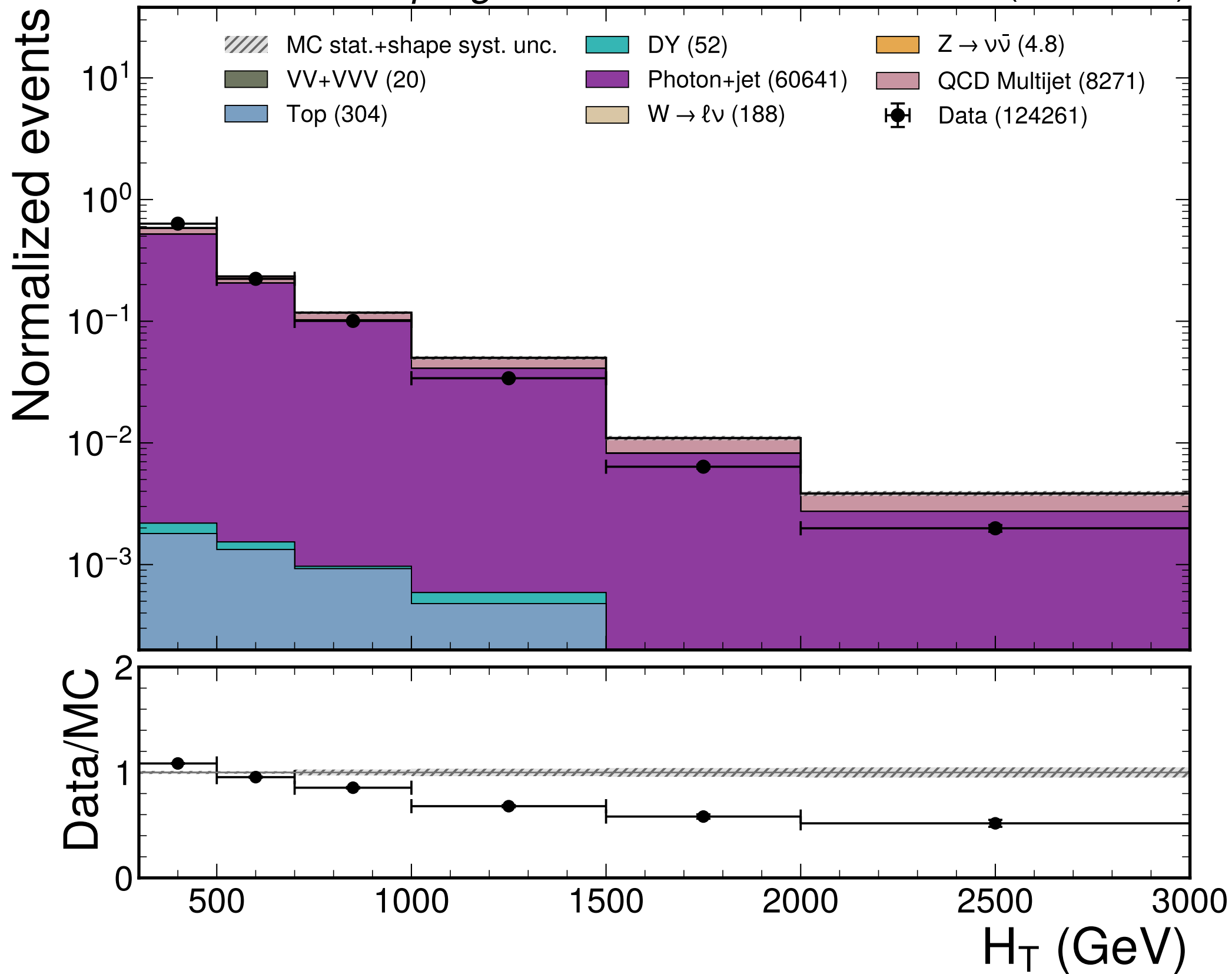
109.82 fb⁻¹ (13.6 TeV)

Events



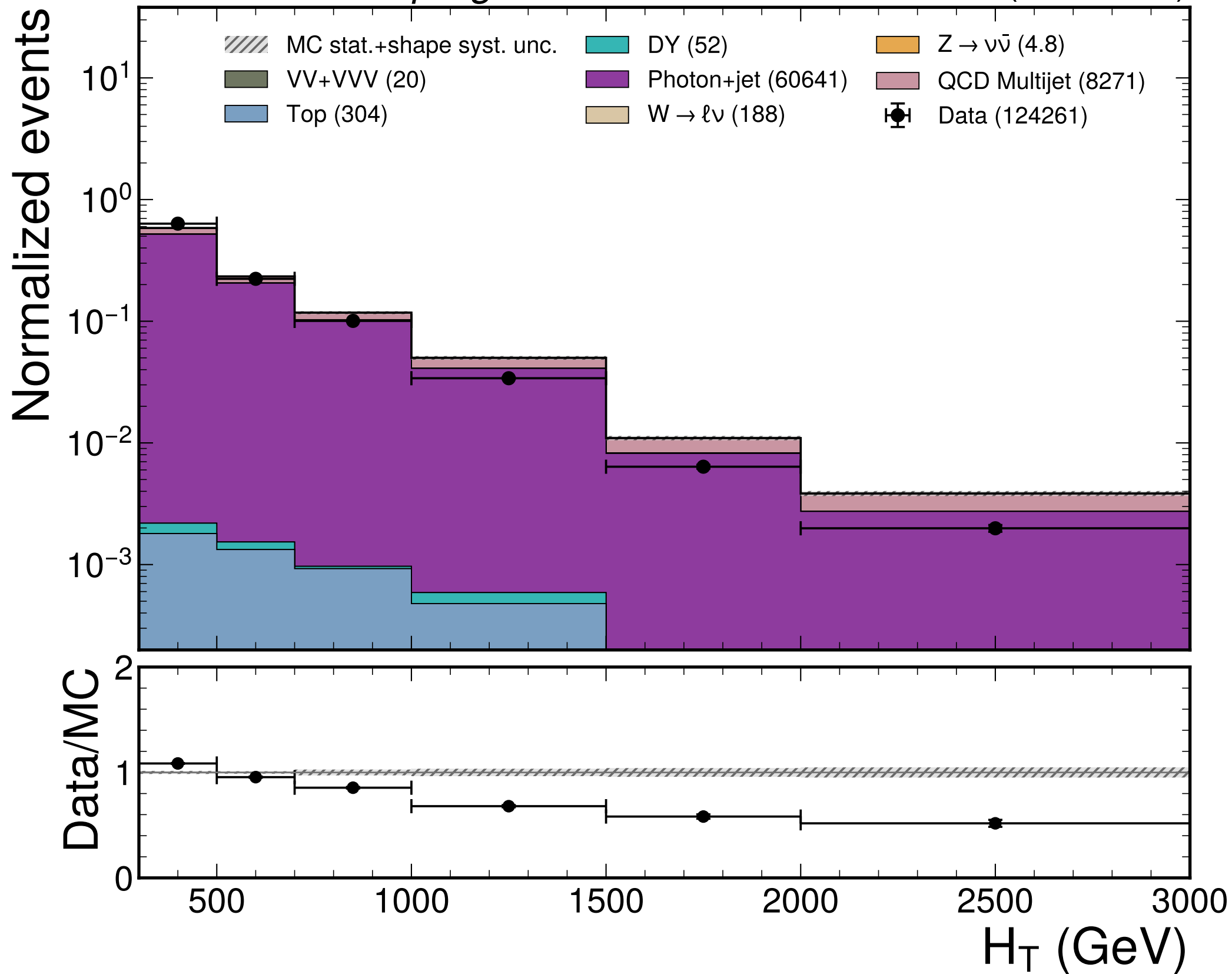
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



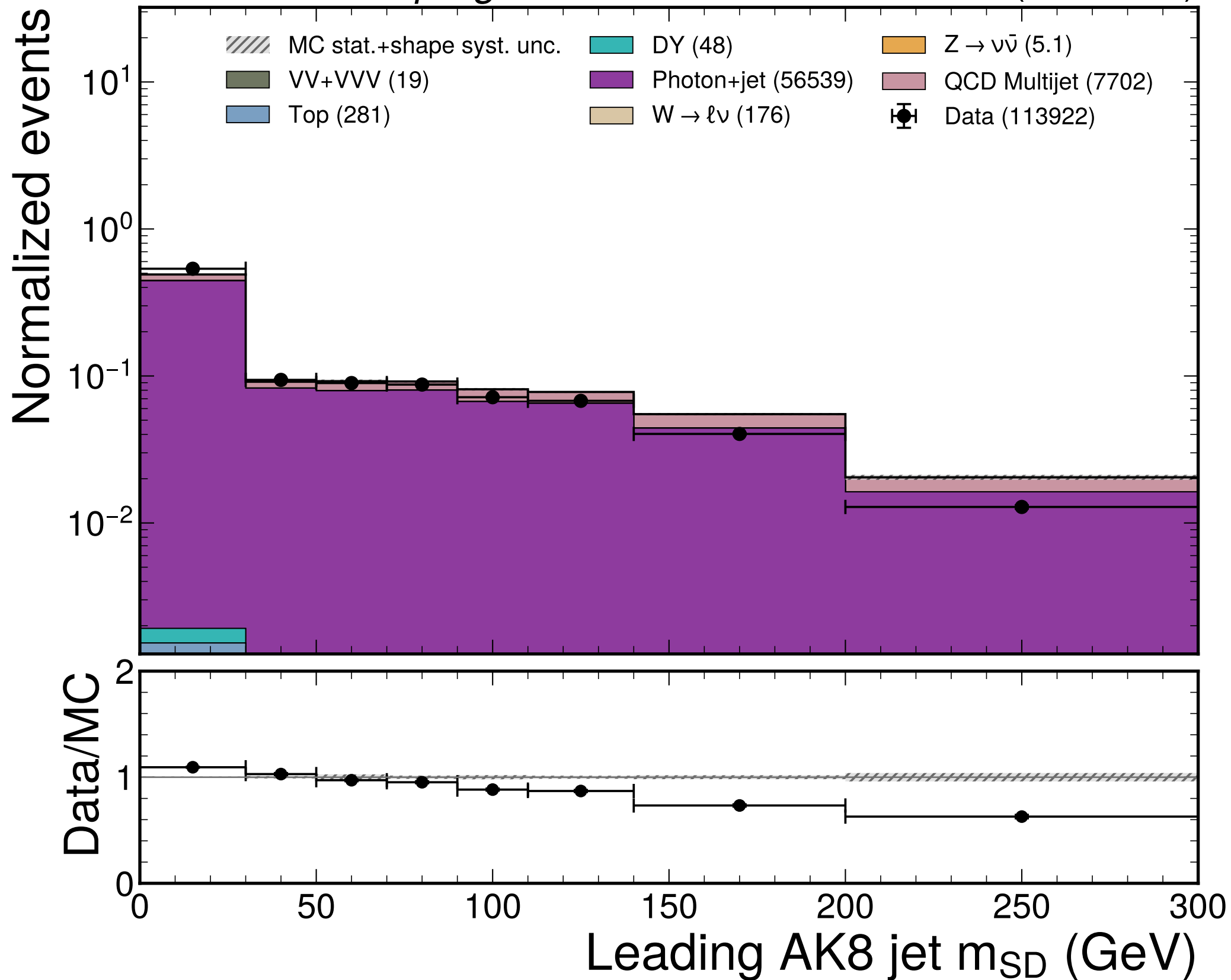
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



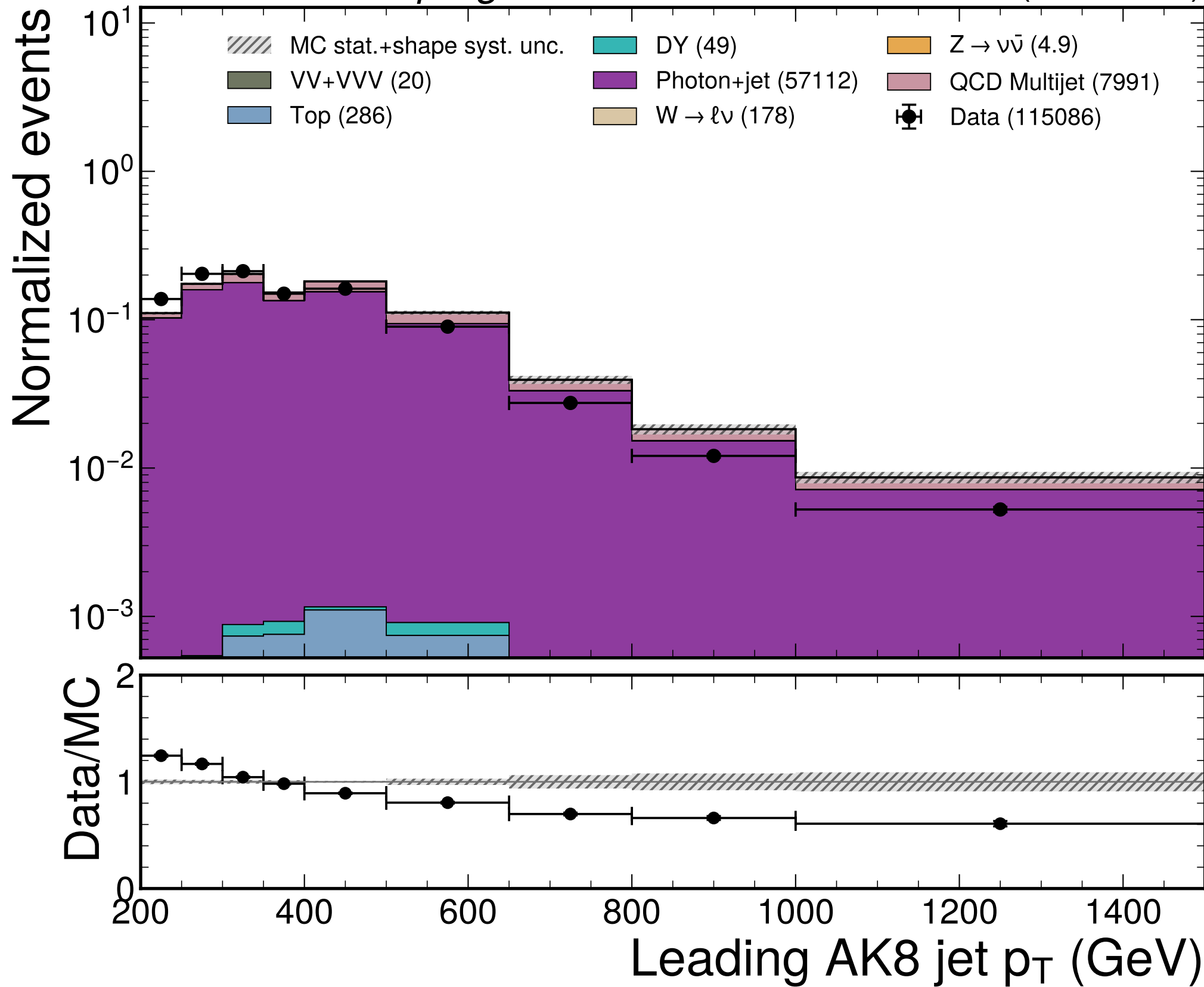
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



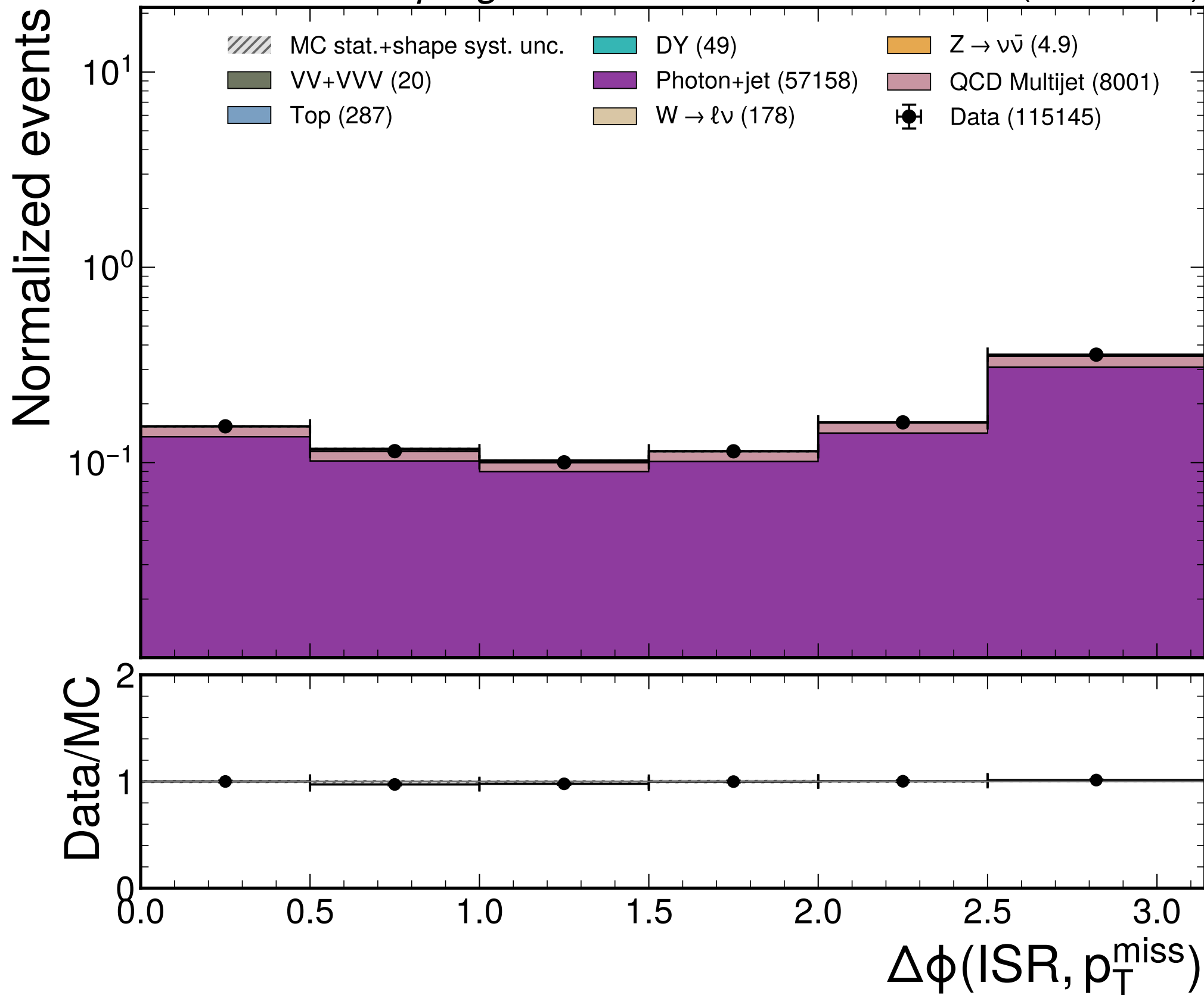
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



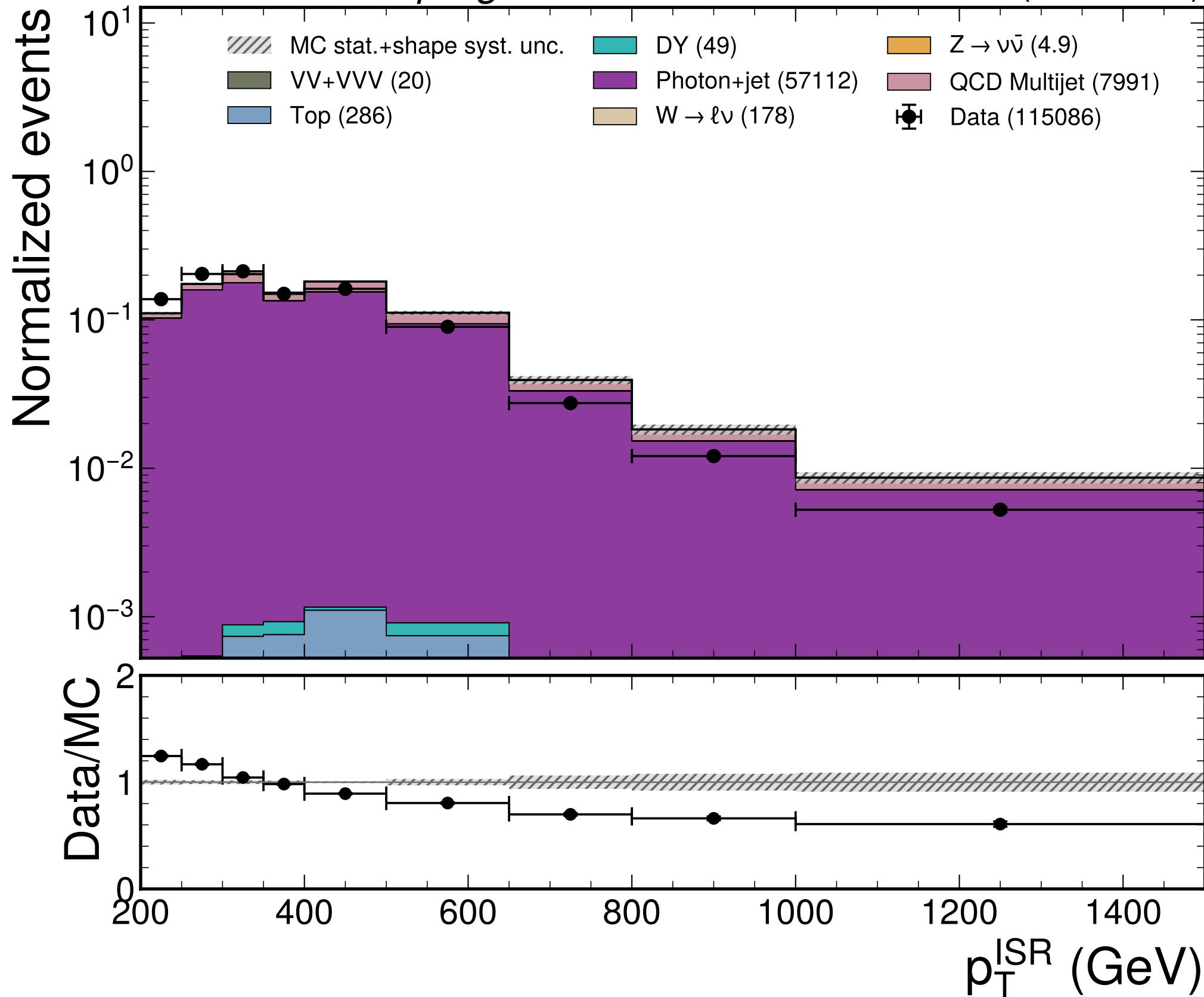
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



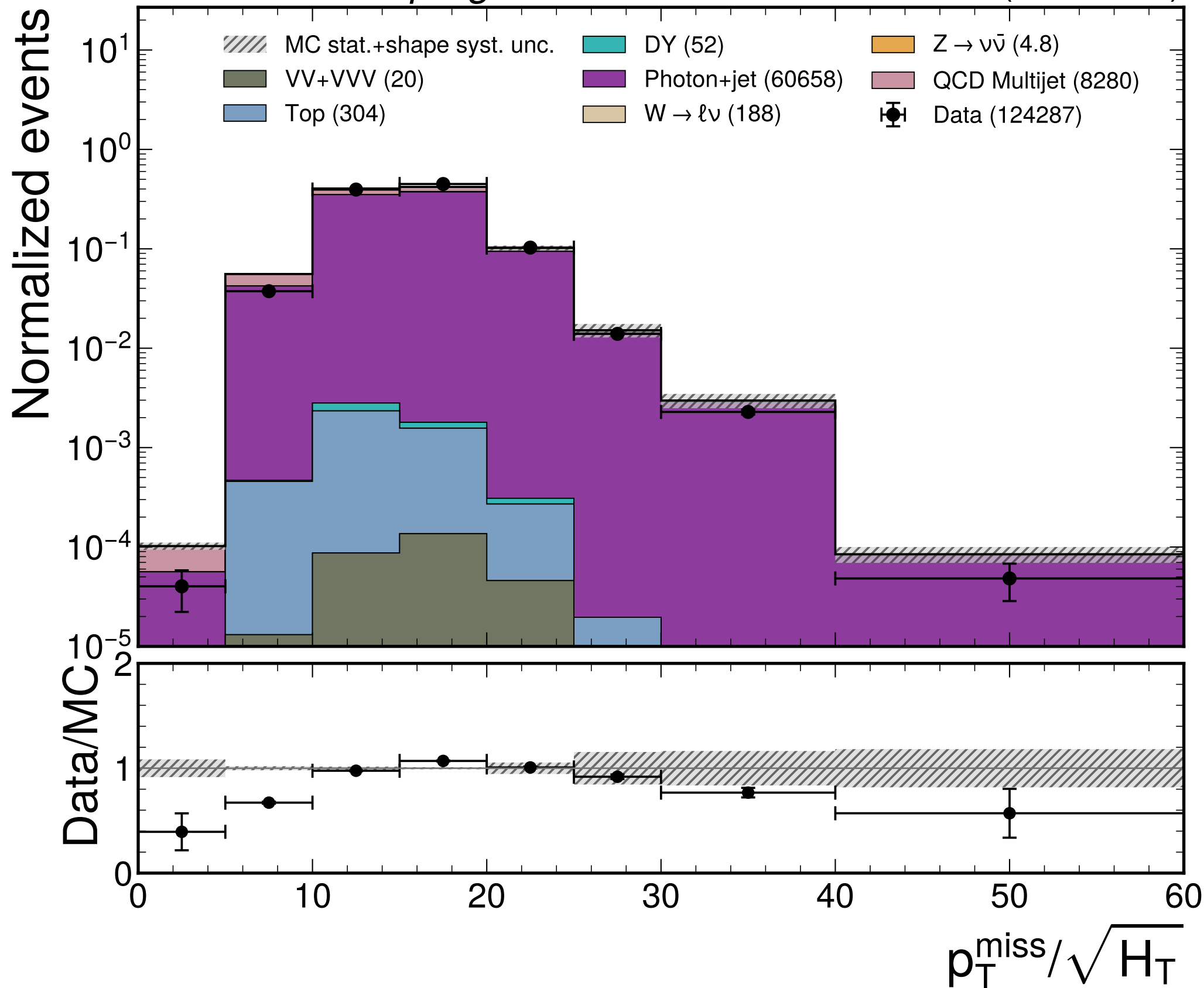
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



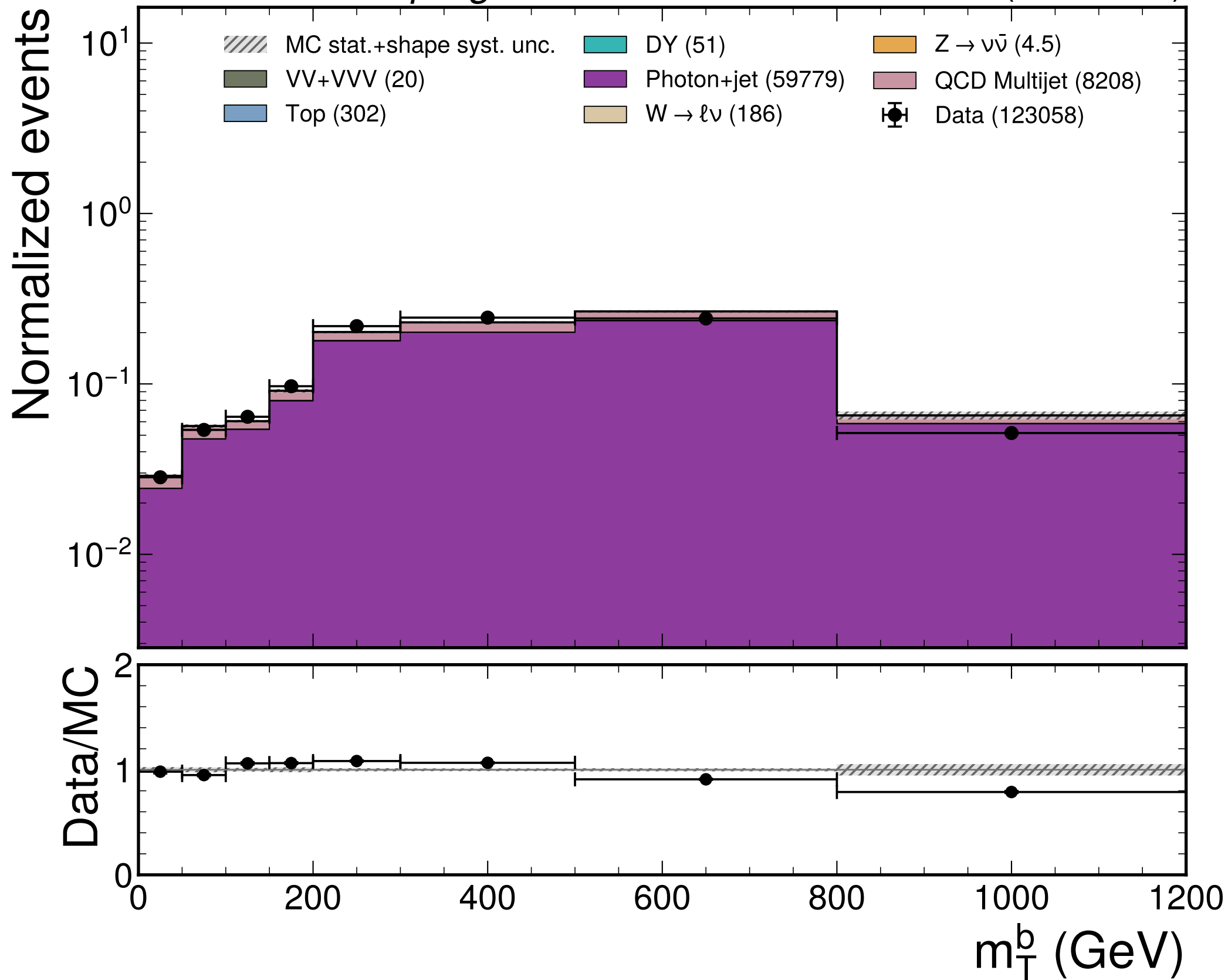
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



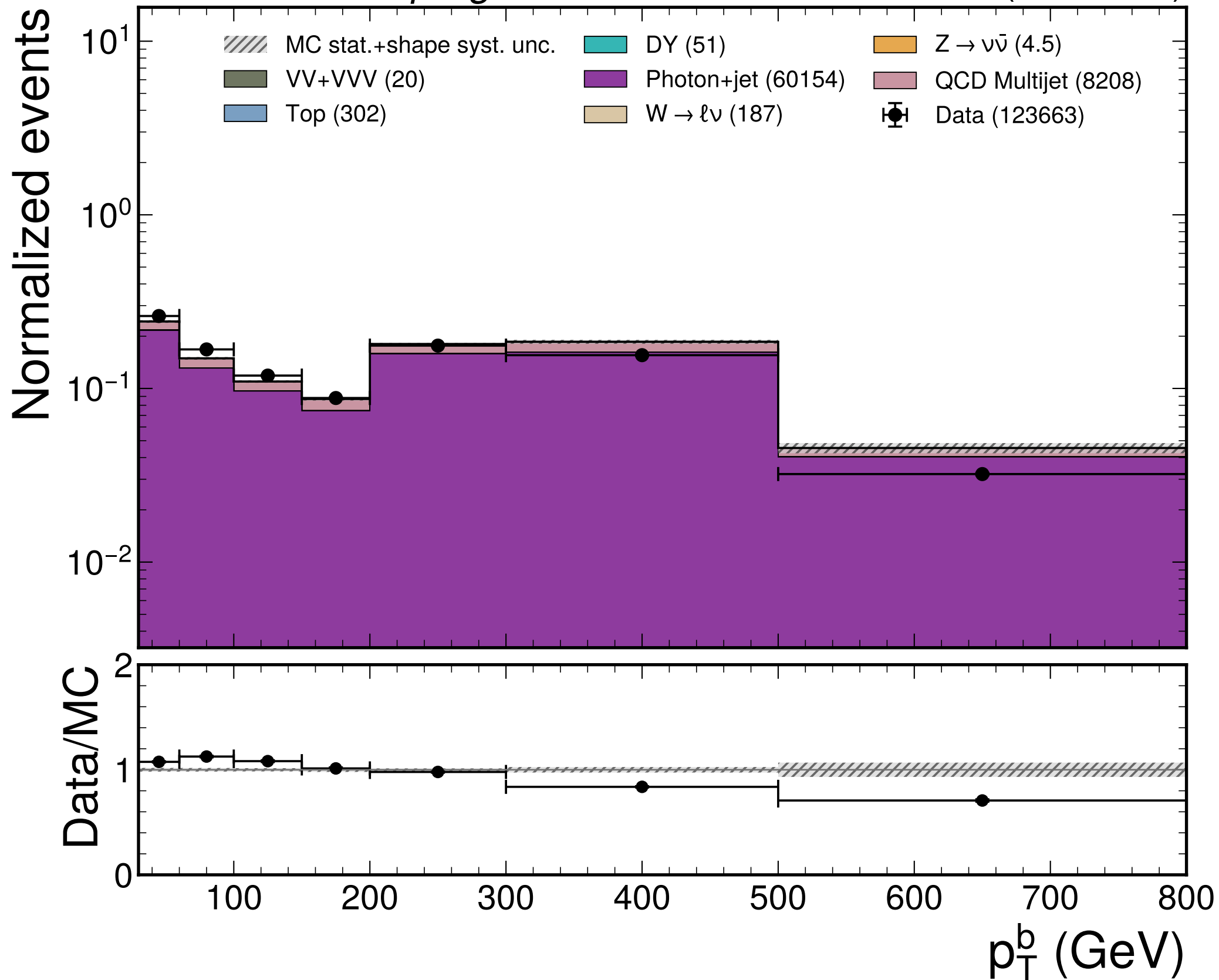
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



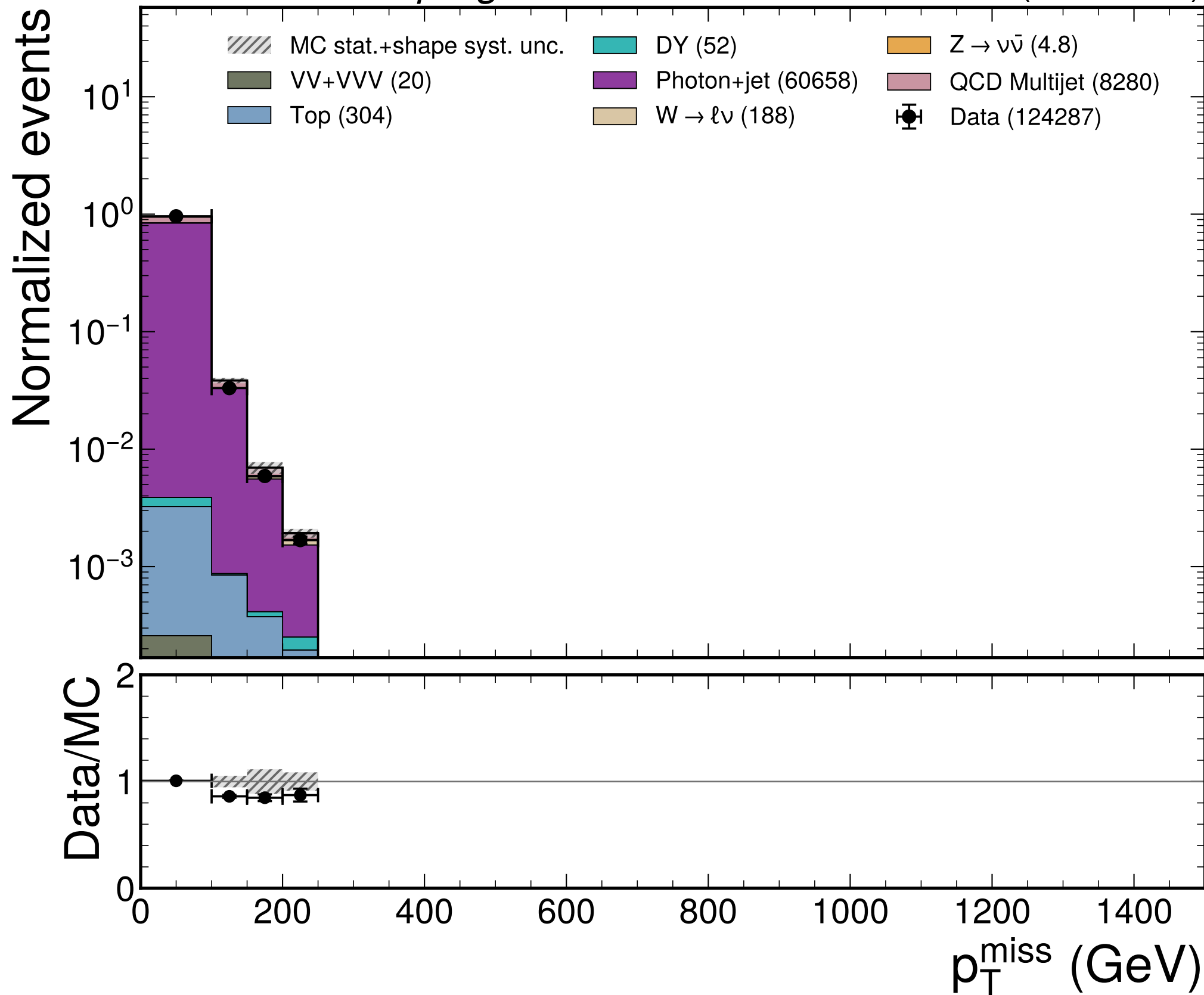
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



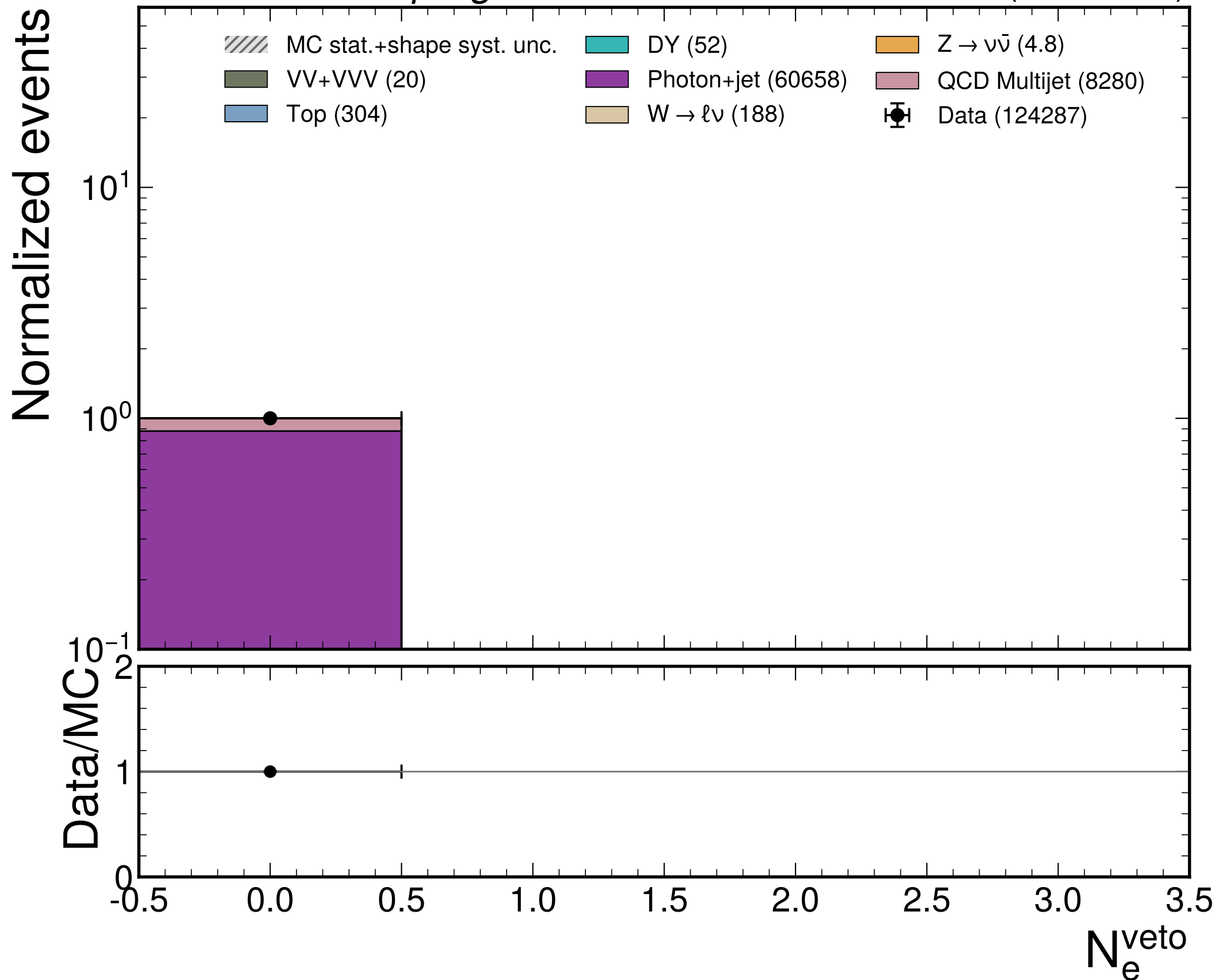
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



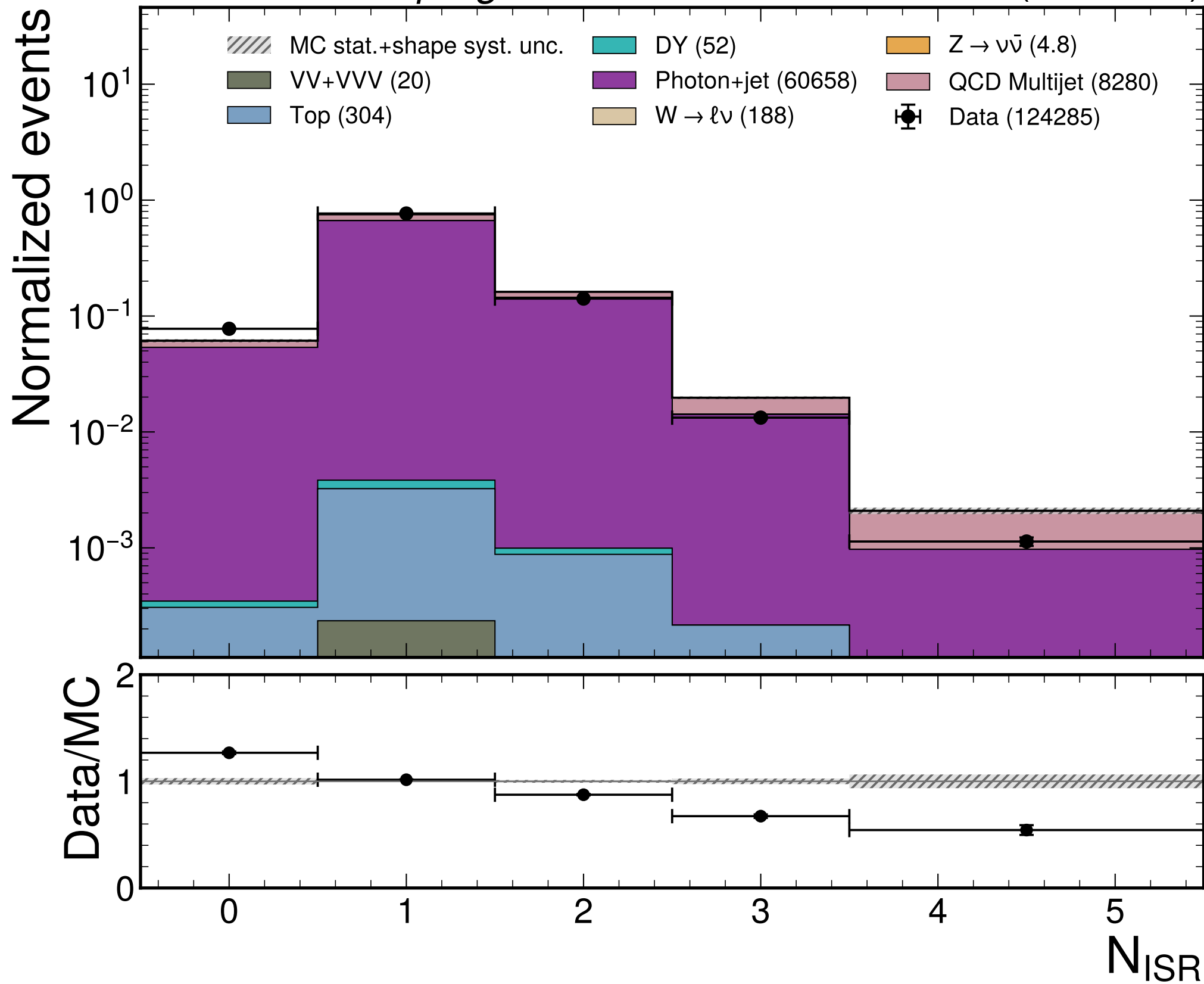
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



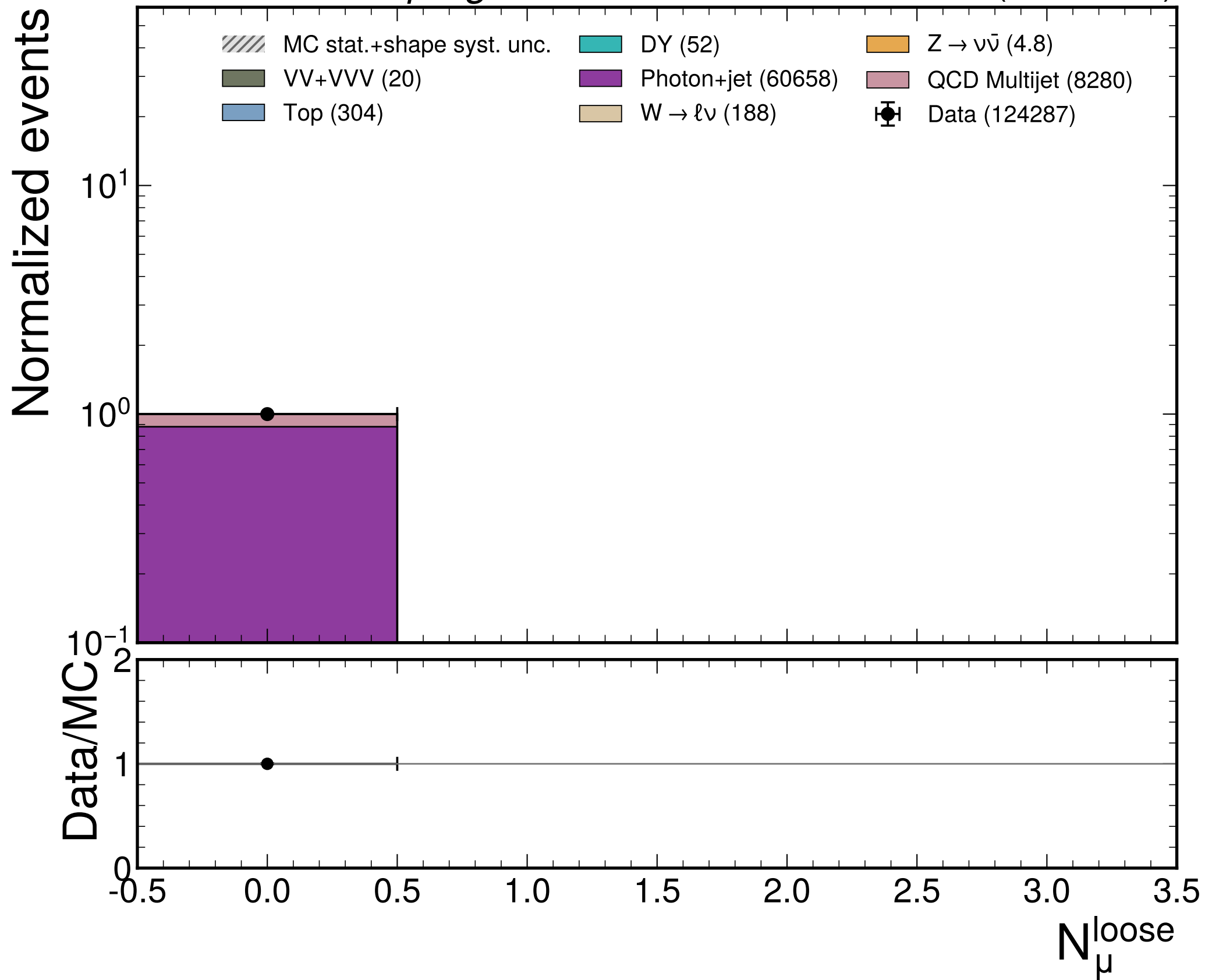
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



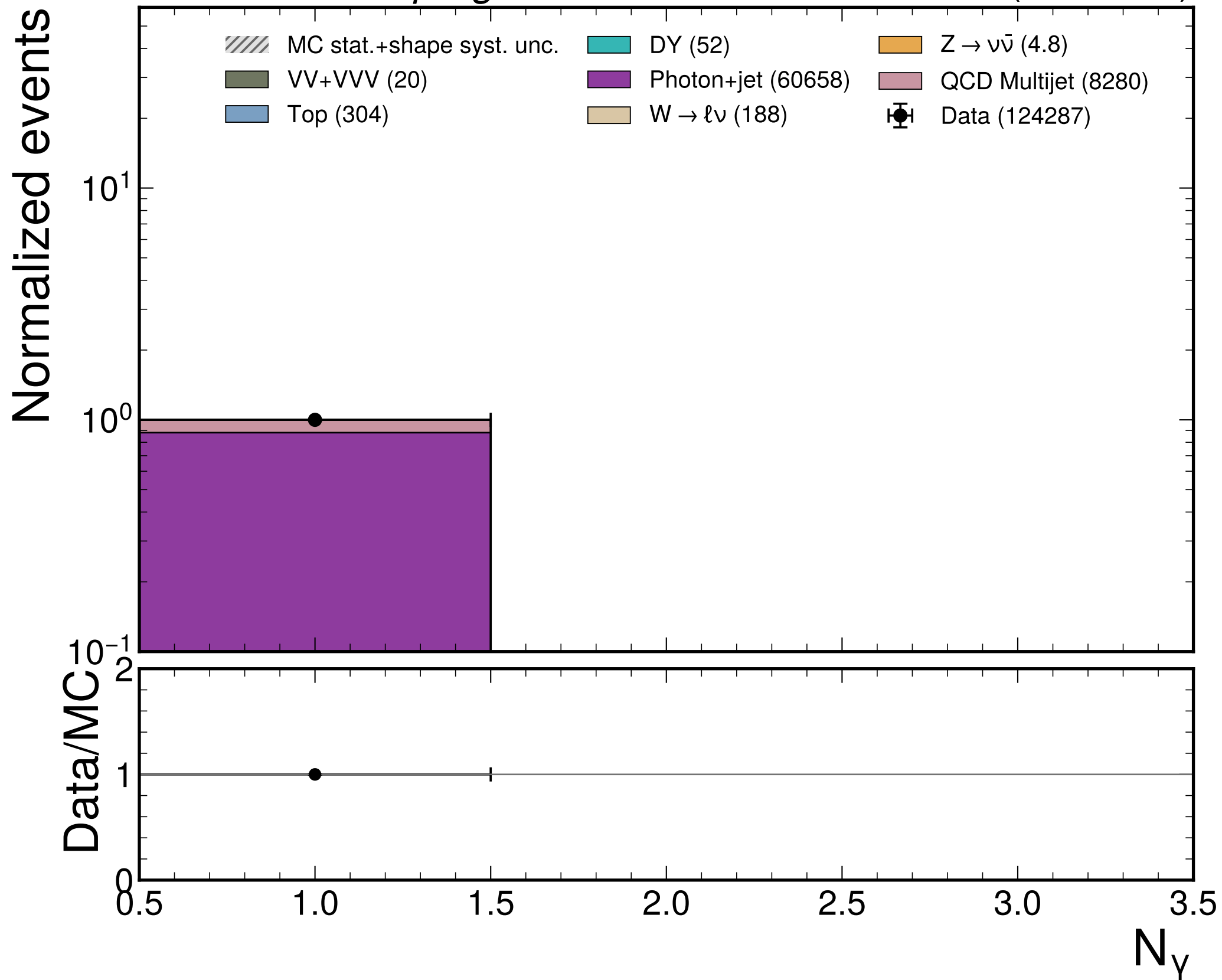
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



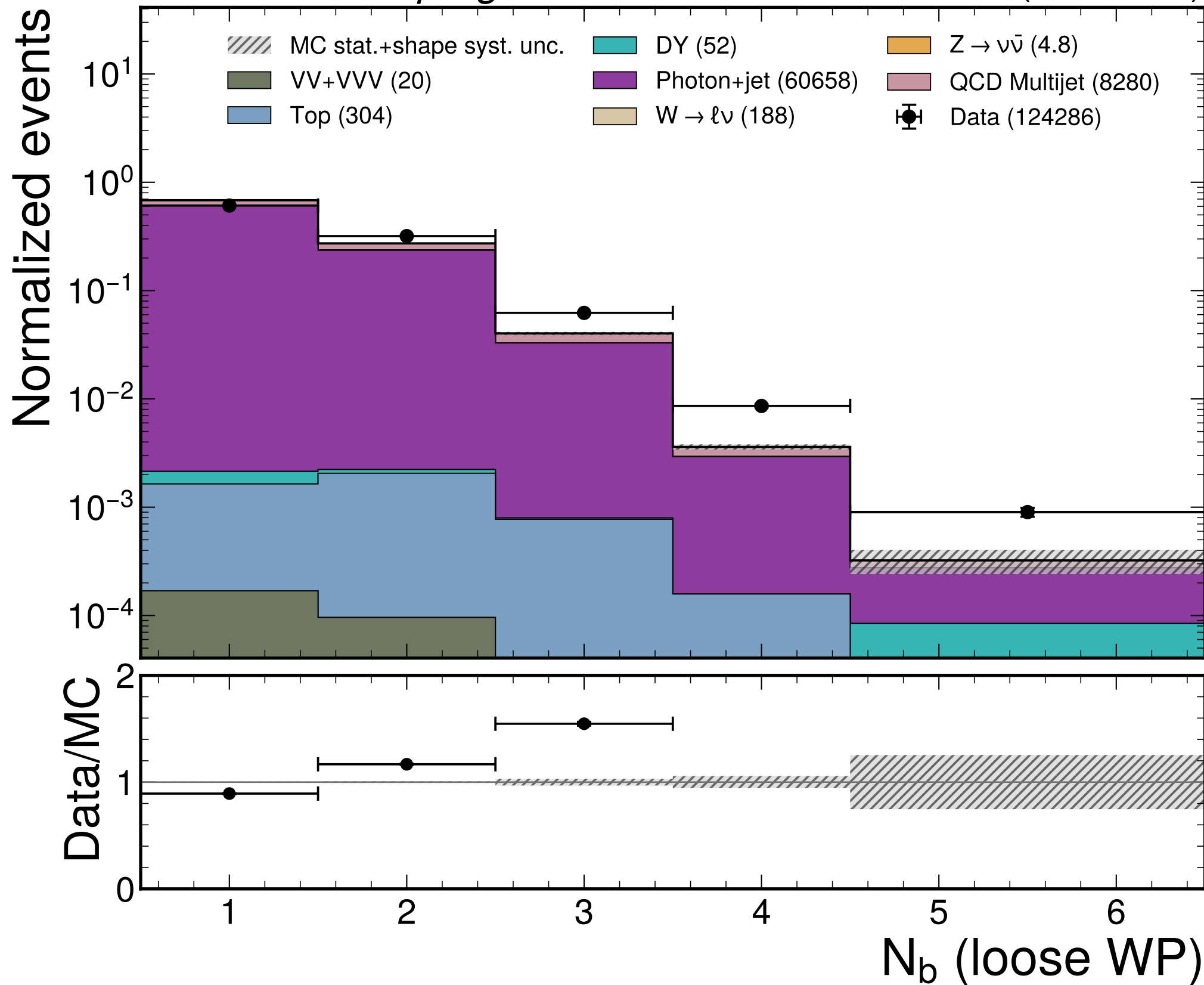
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



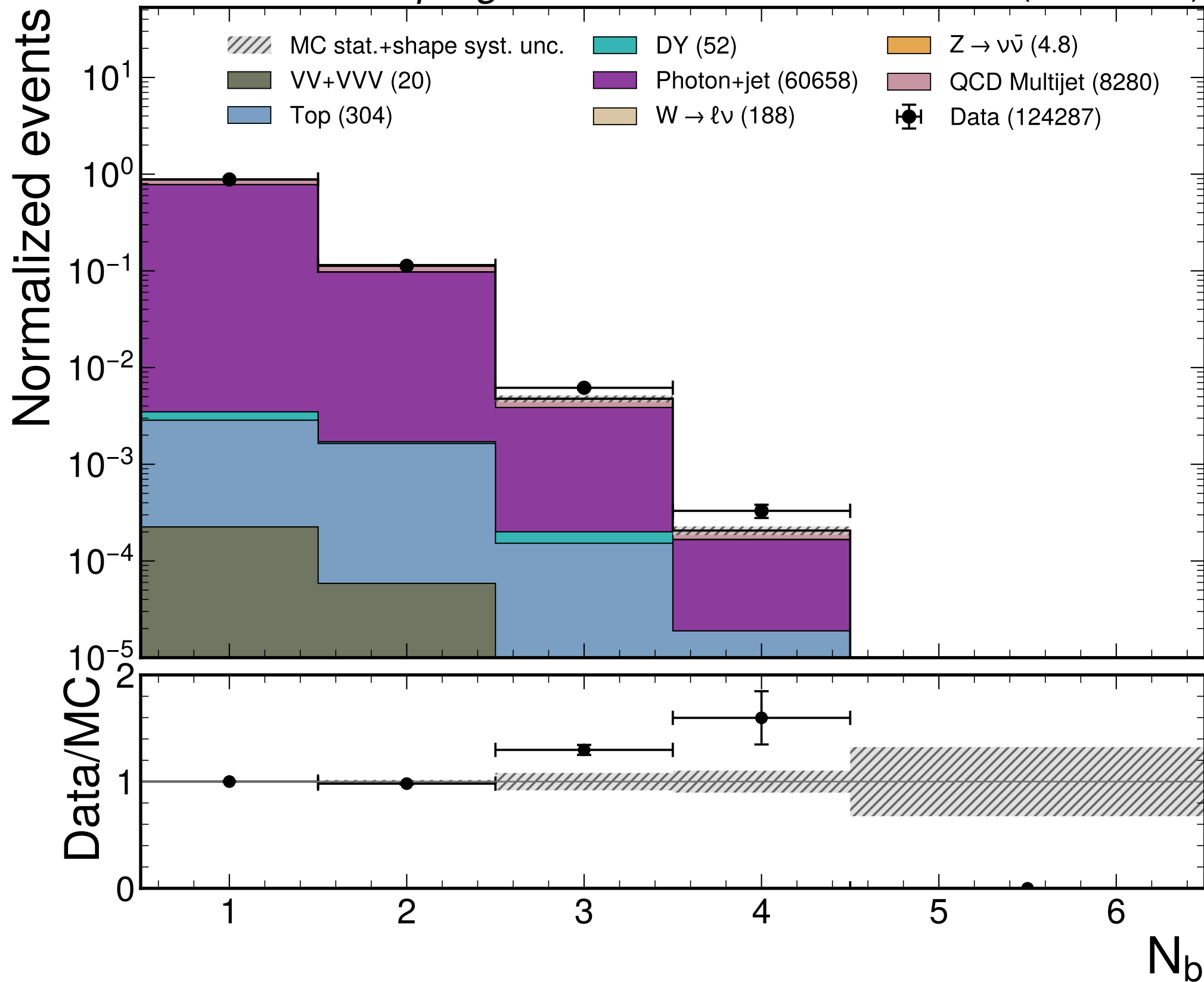
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



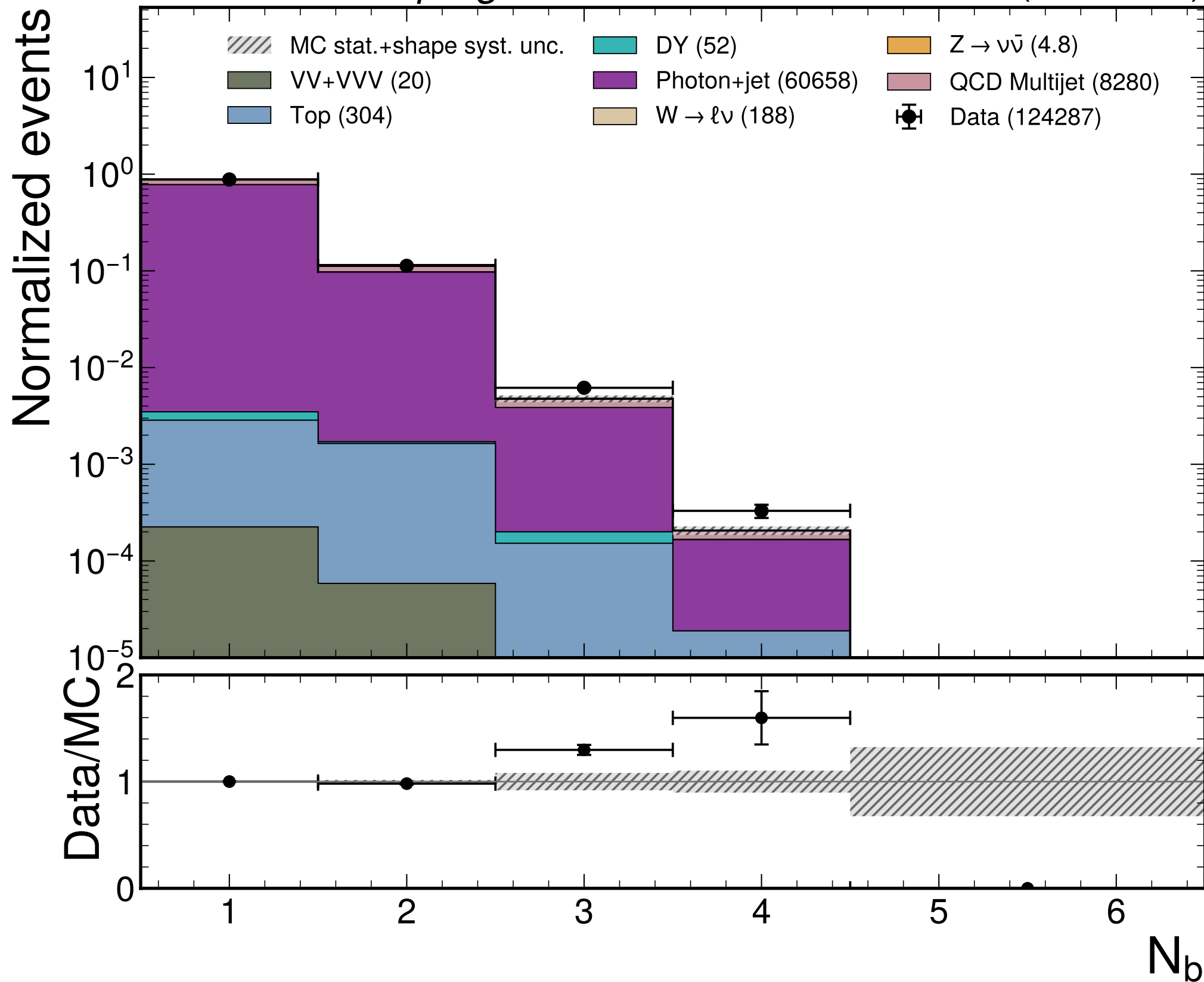
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



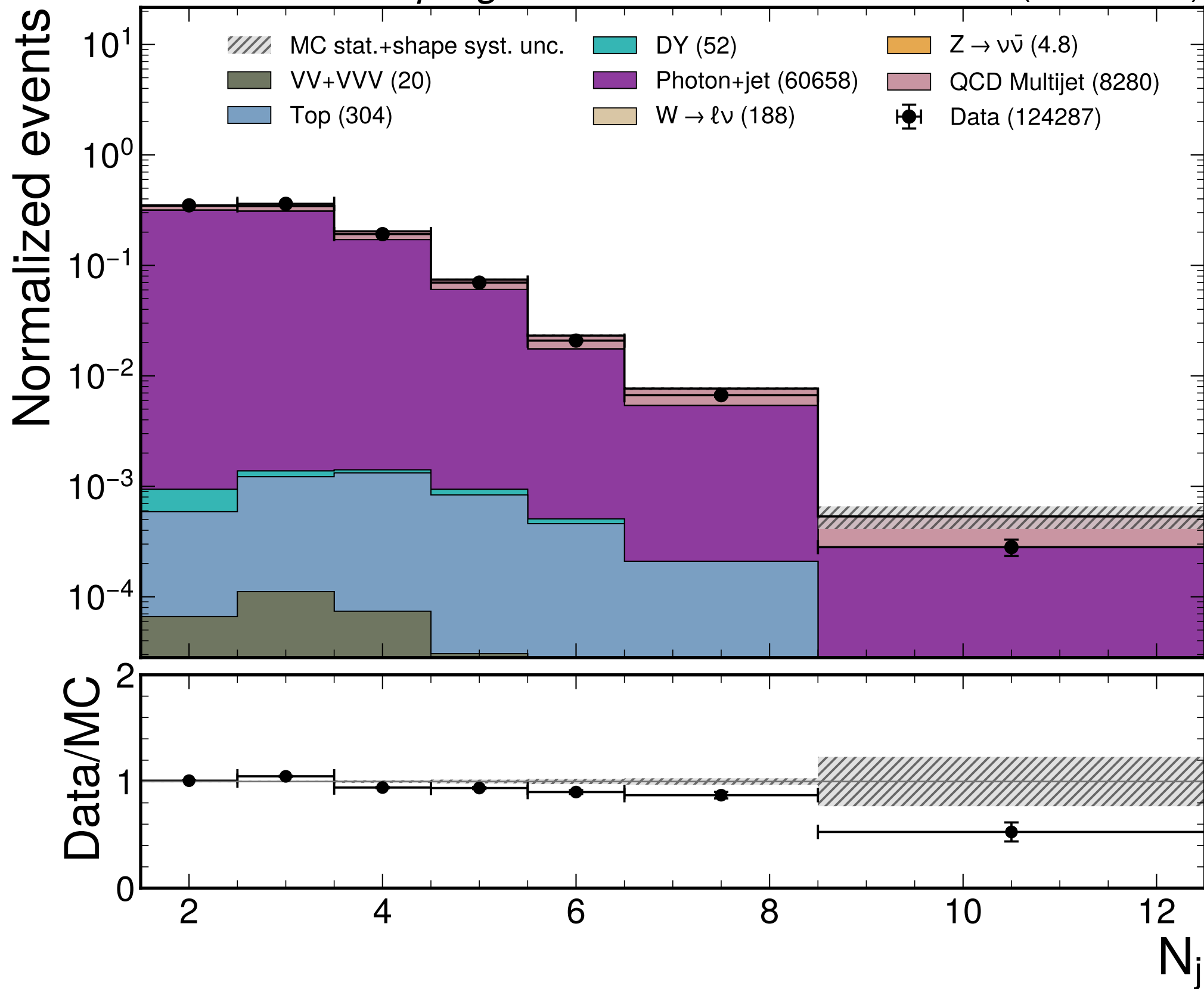
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



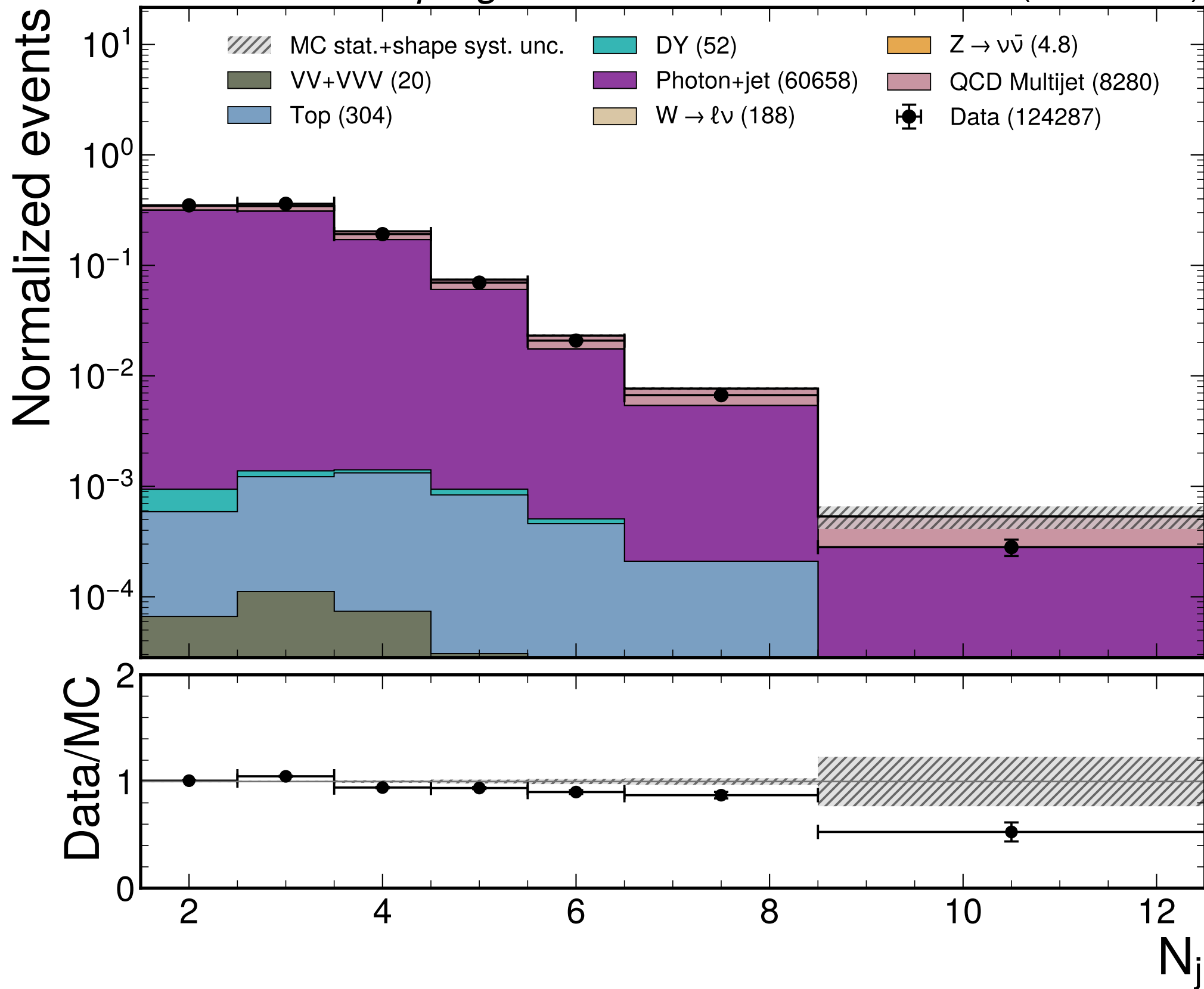
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



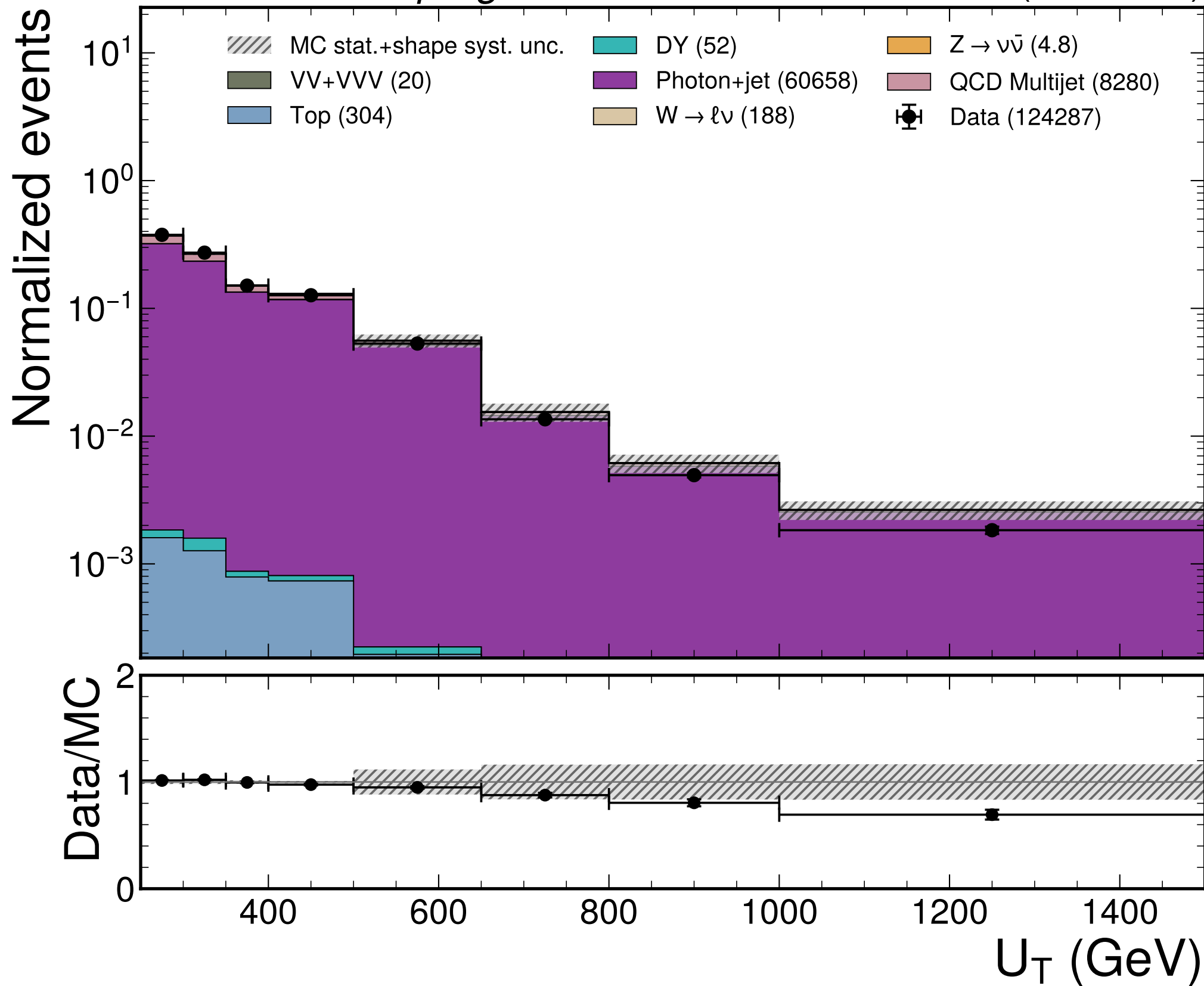
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



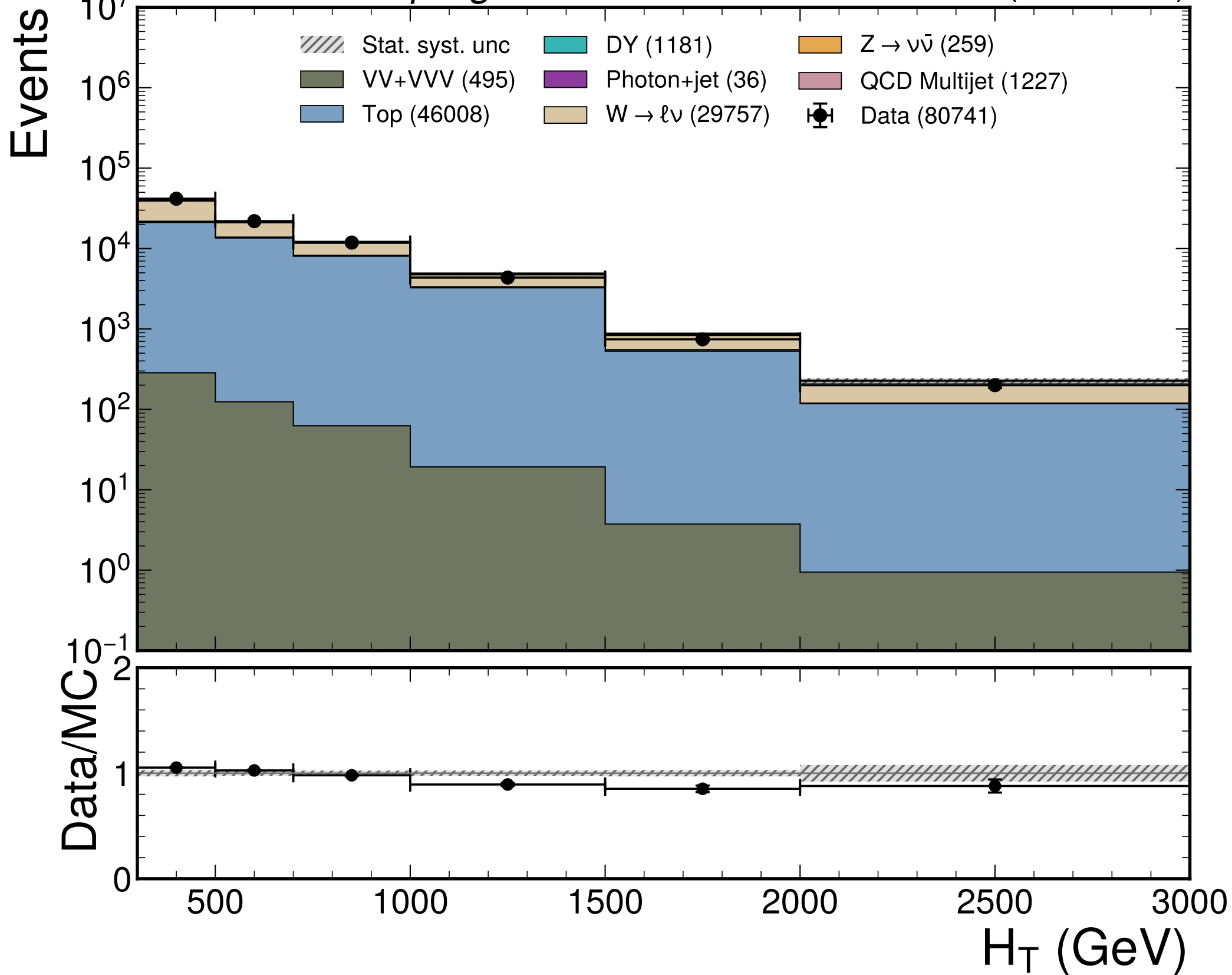
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

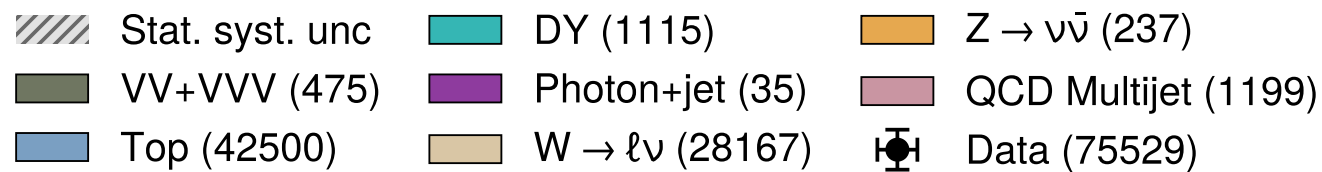
109.82 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

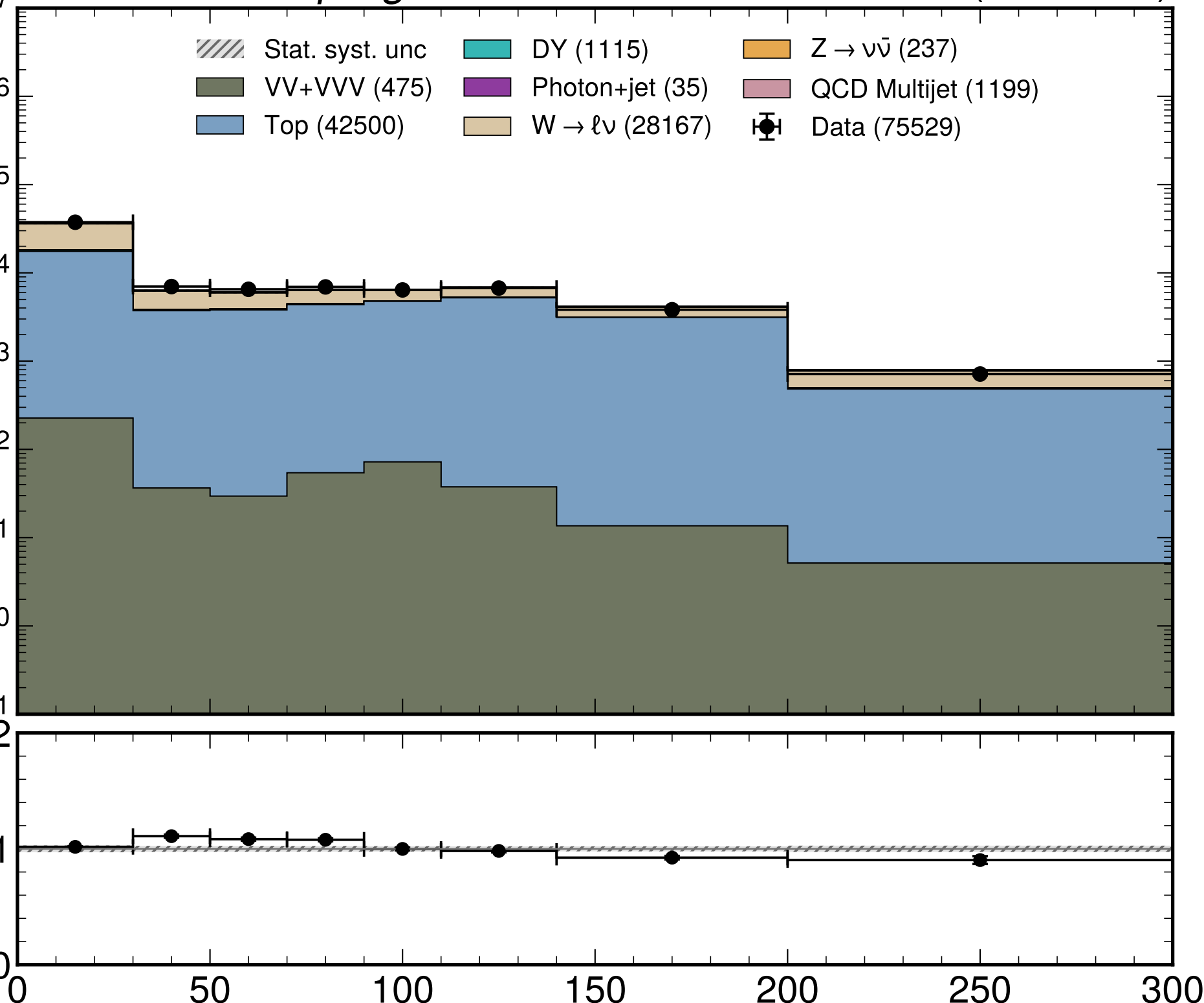
Events



Data/MC

10⁻¹
2
1
0

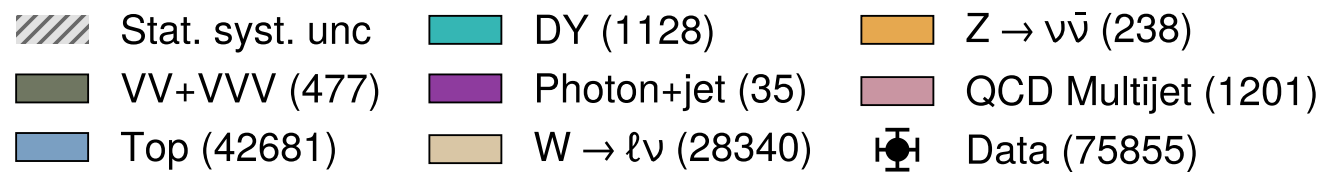
Leading AK8 jet m_{SD} (GeV)



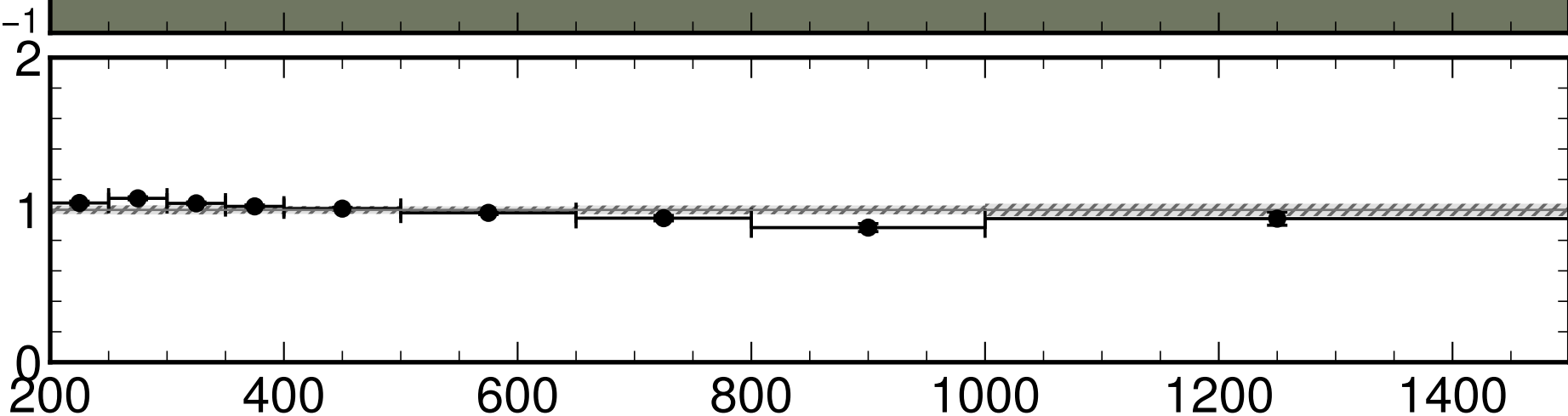
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

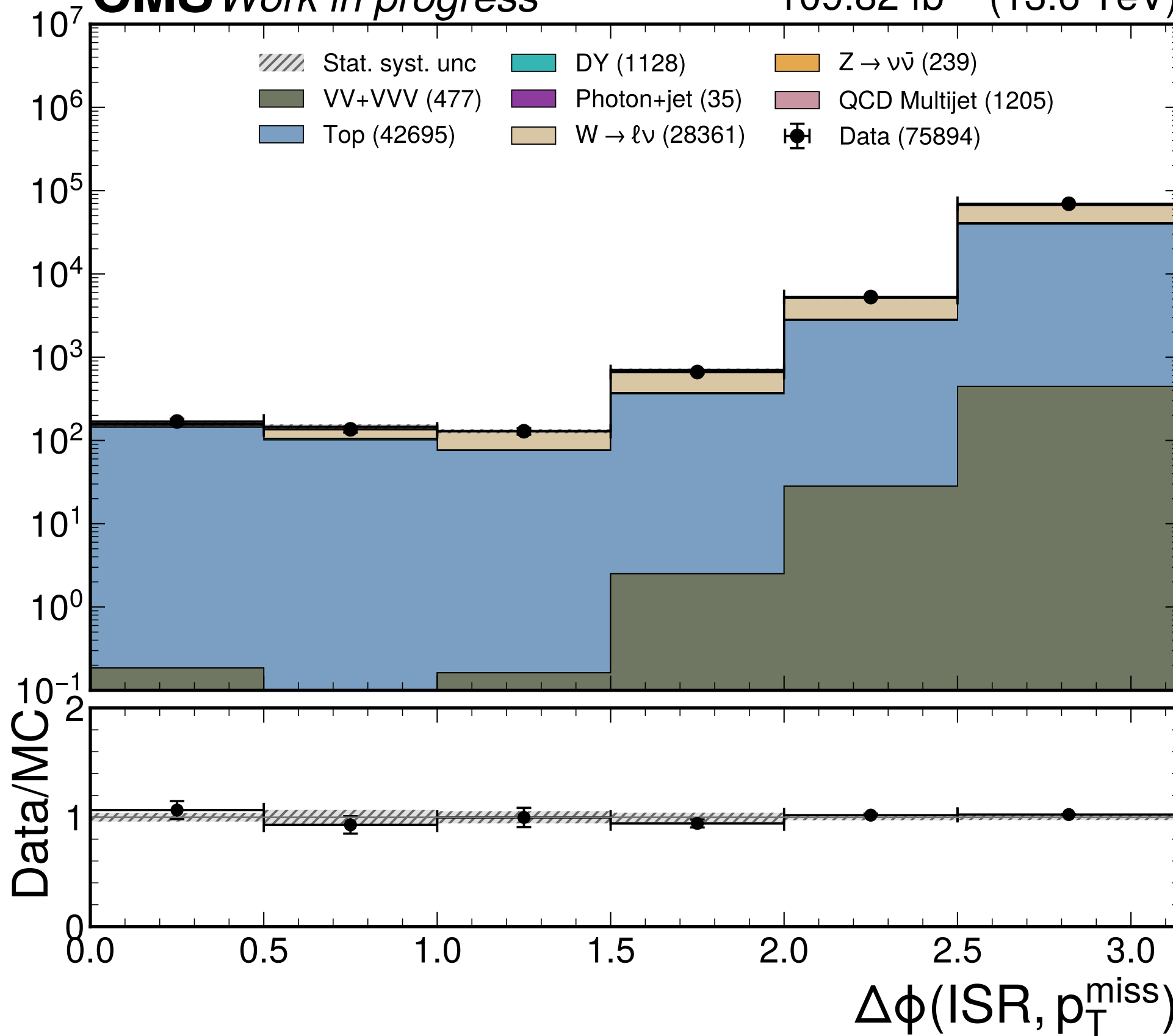
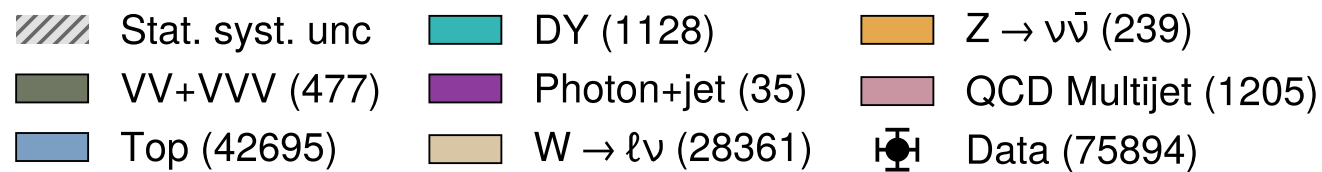


Leading AK8 jet p_T (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

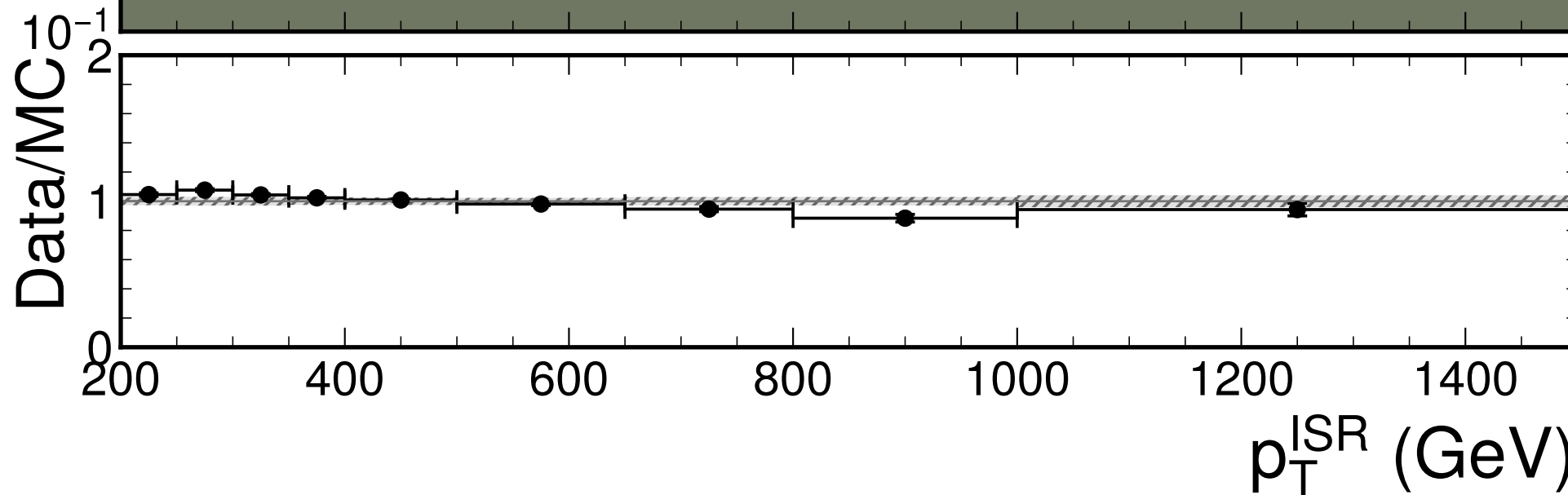
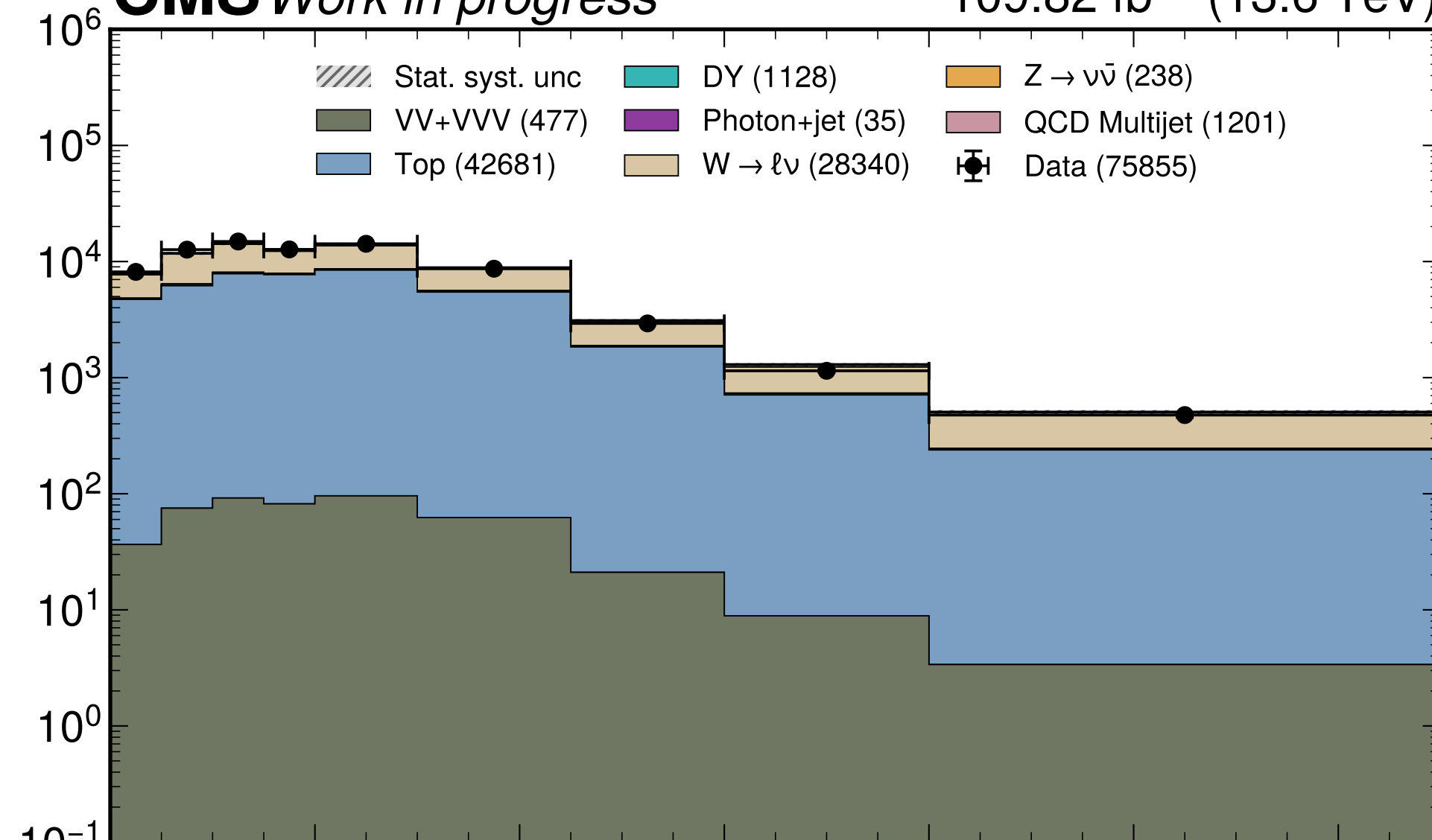
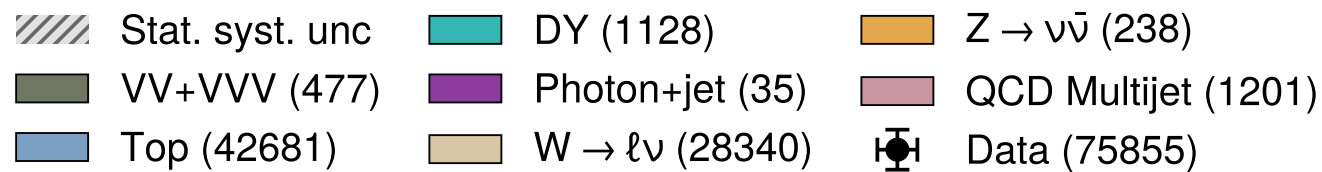
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

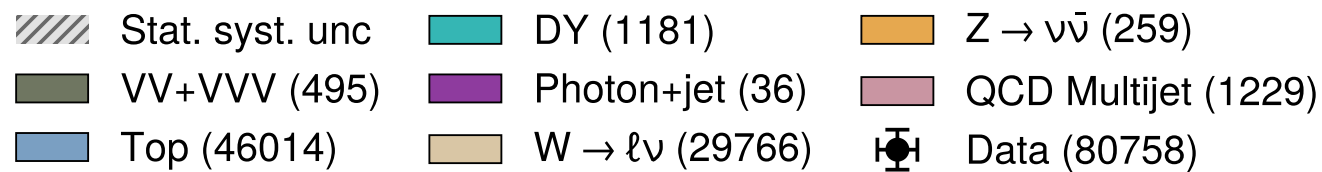
Events



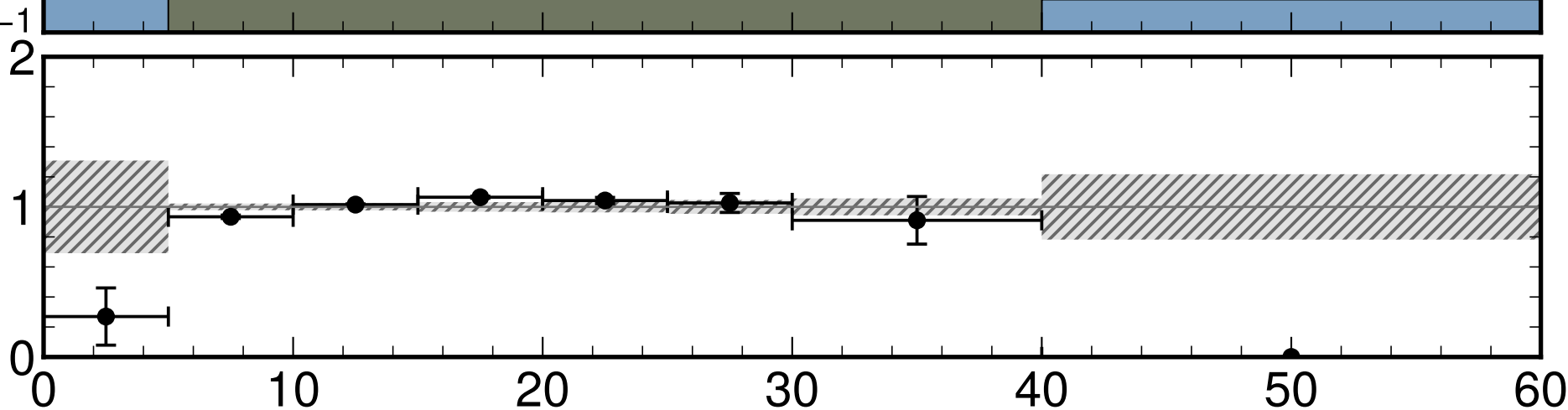
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

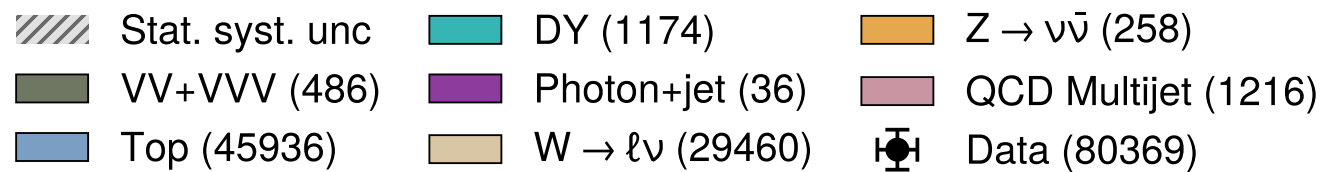


$p_T^{\text{miss}}/\sqrt{H_T}$

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events

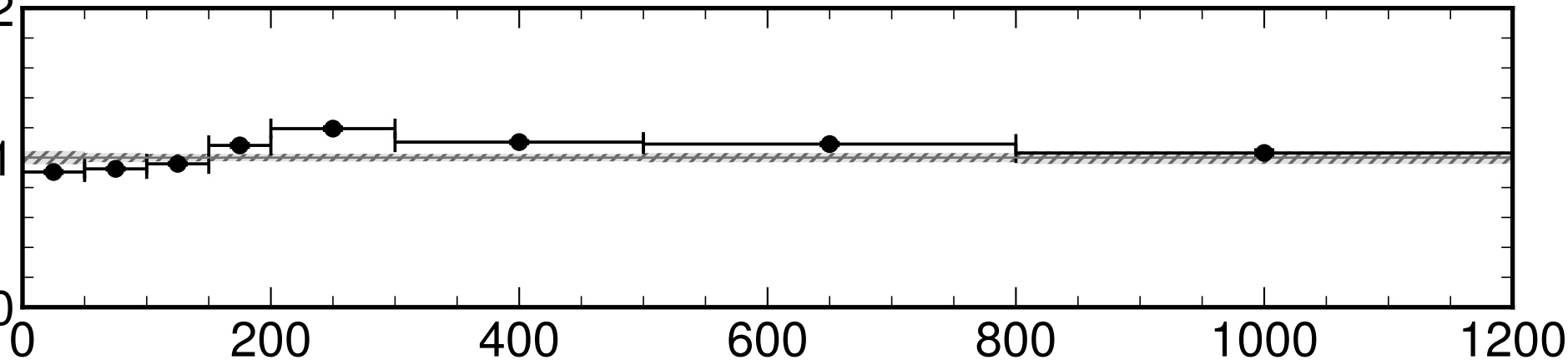


Data/MC

10⁻¹

2

0

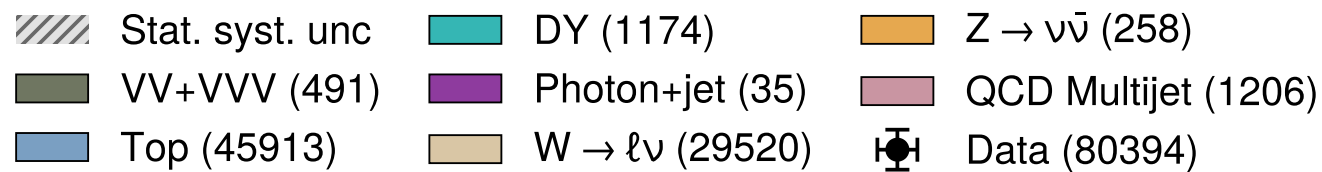


m_T^b (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



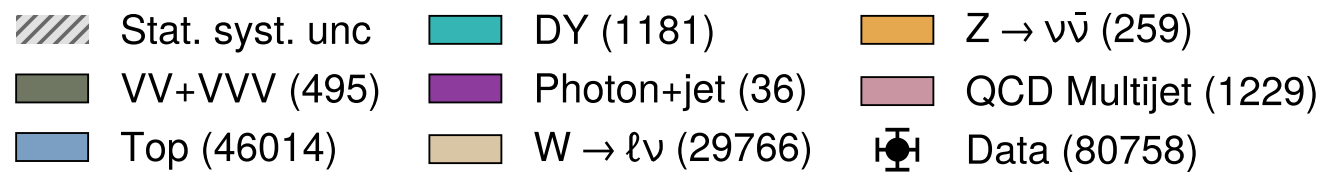
Data/MC

p_T^b (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

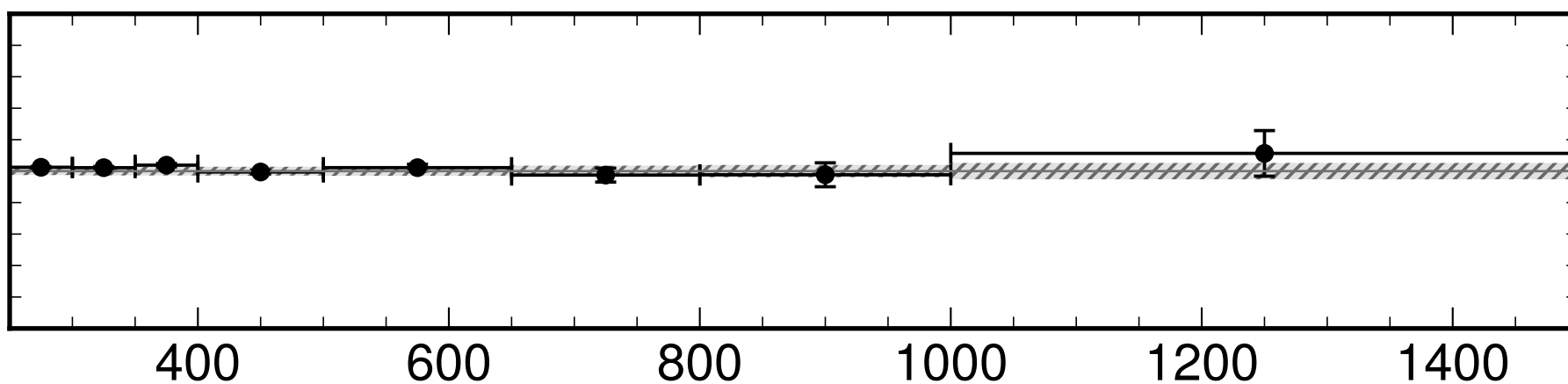
Events



Data/MC

10⁻²

0

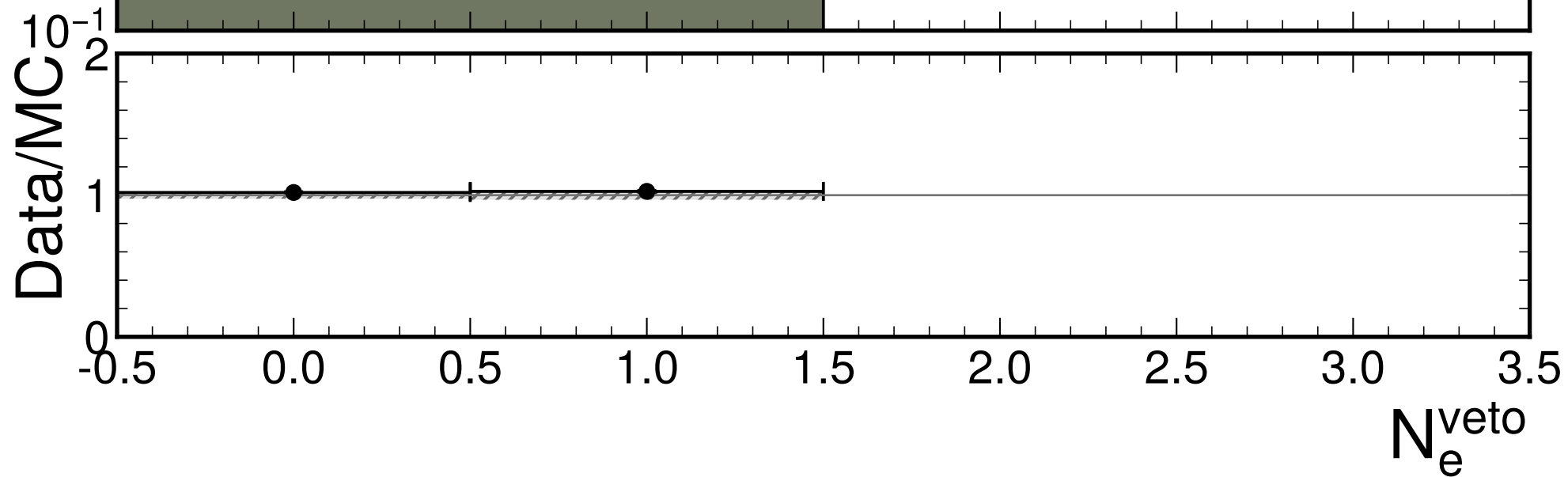
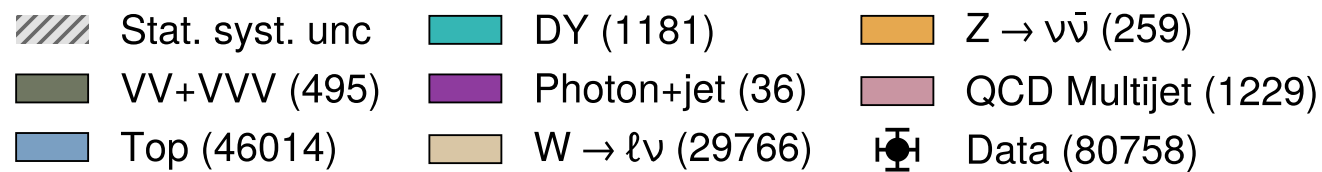


p_T^{miss} (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

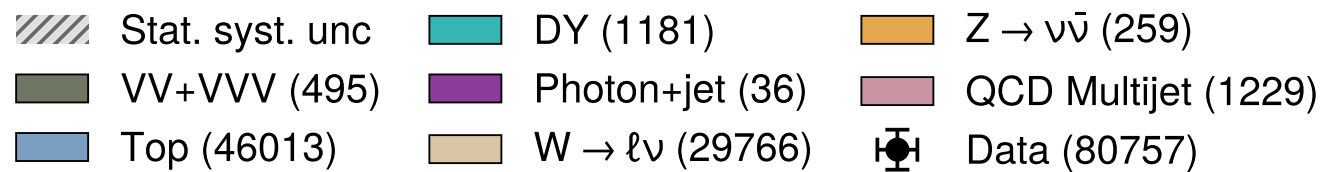
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

0

1

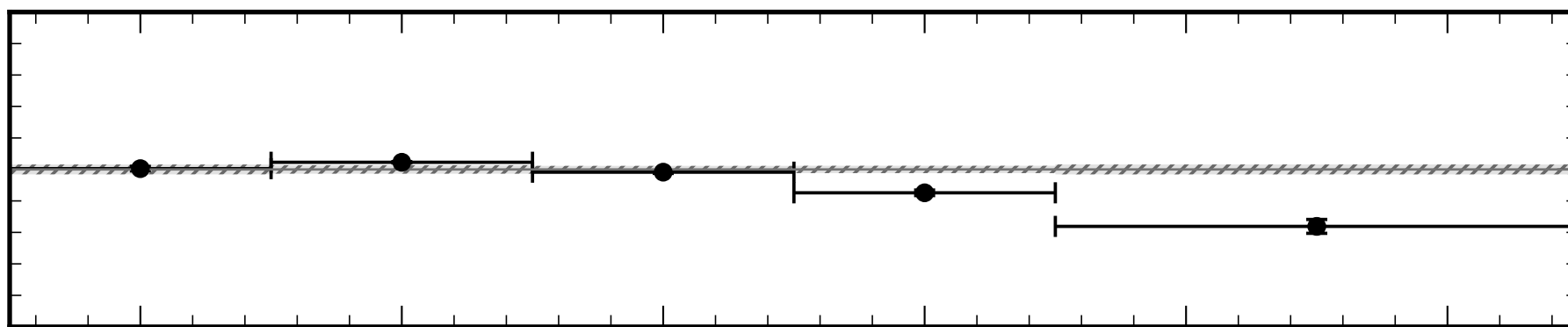
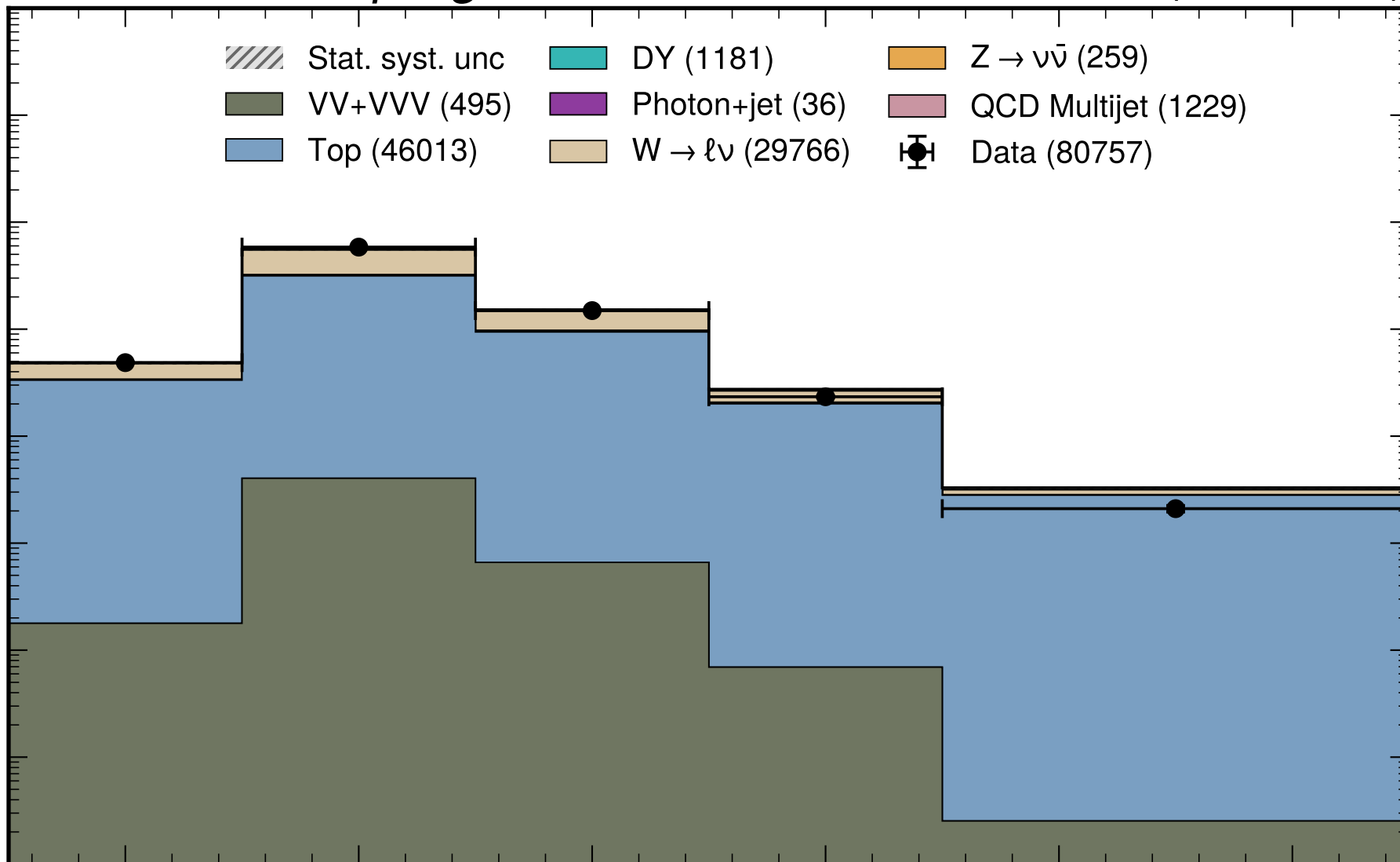
2

3

4

5

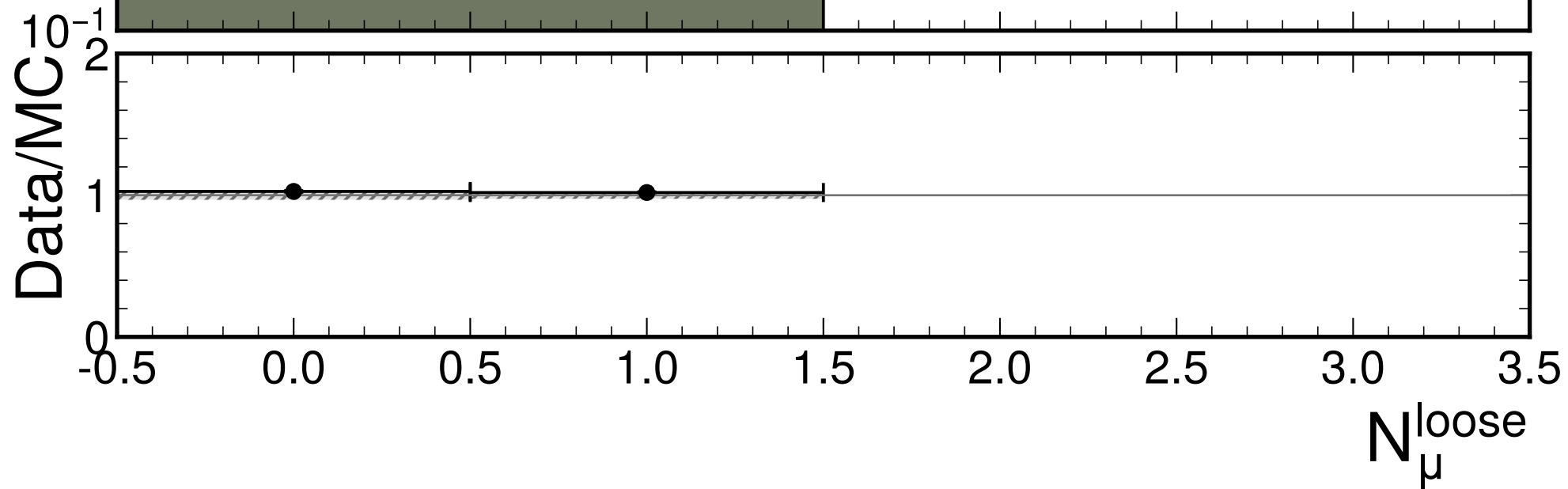
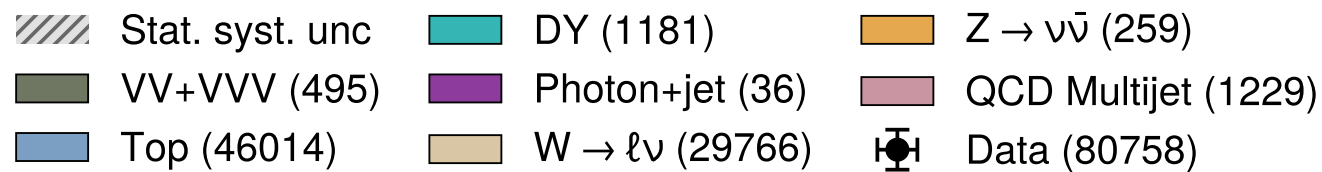
N_{ISR}



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

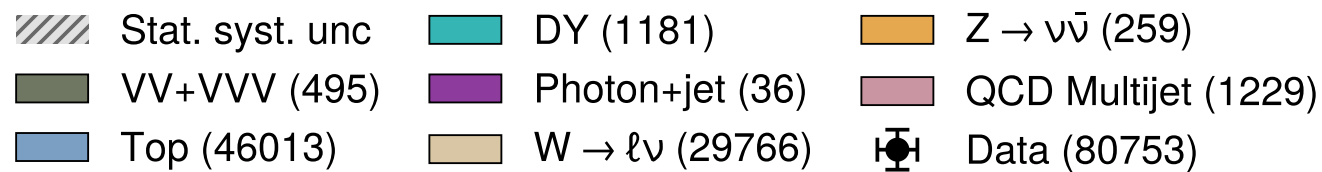
Events



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

1

2

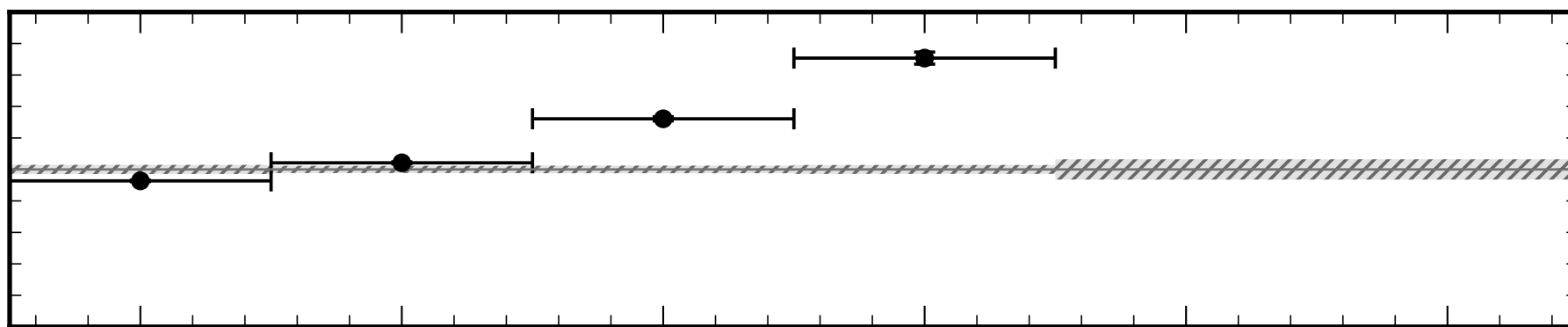
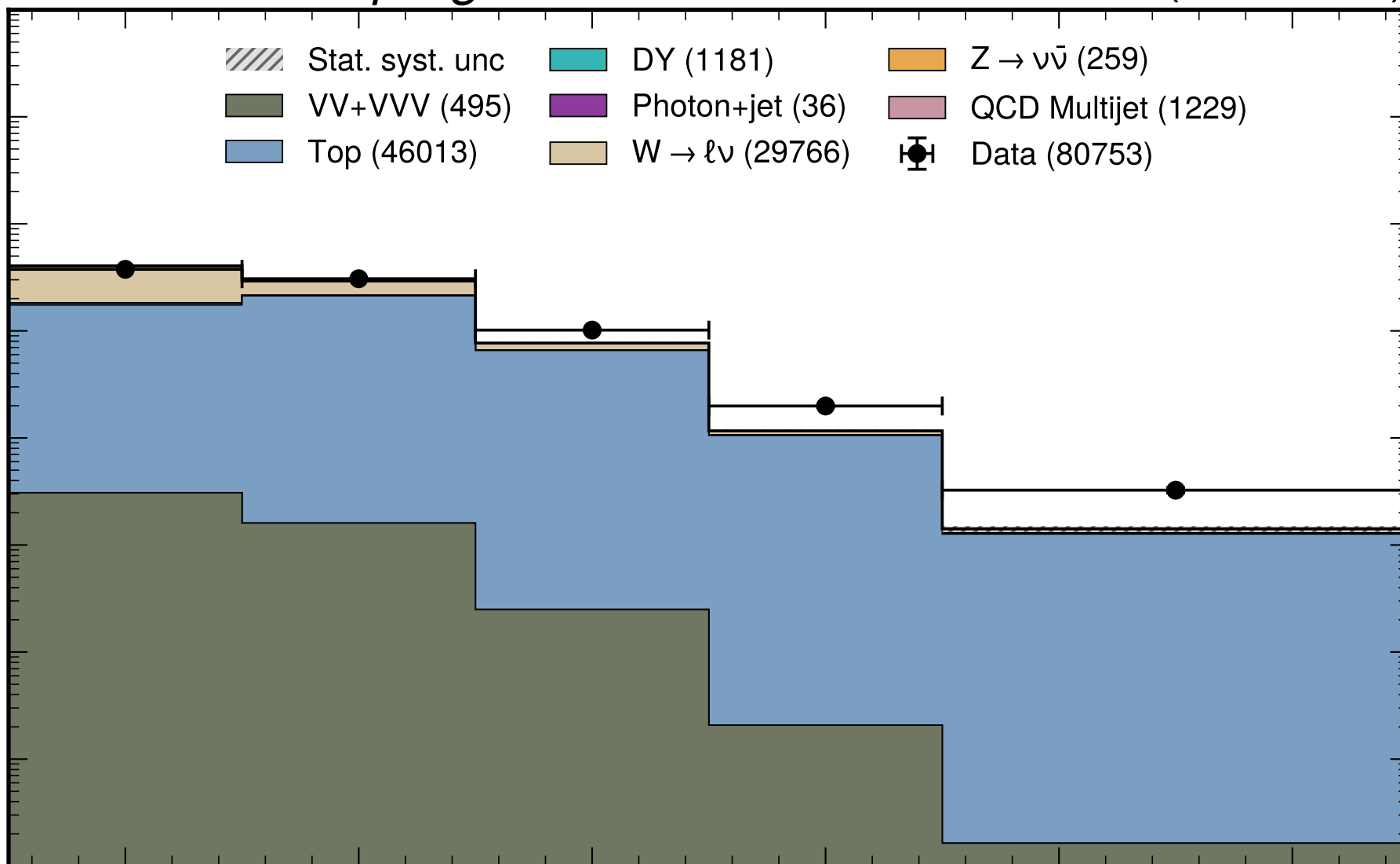
3

4

5

6

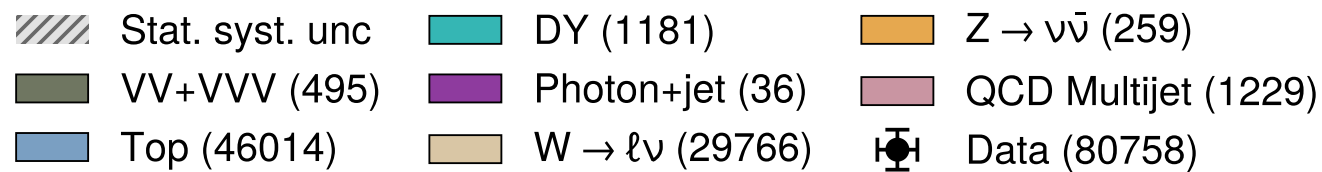
N_b (loose WP)



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

1

2

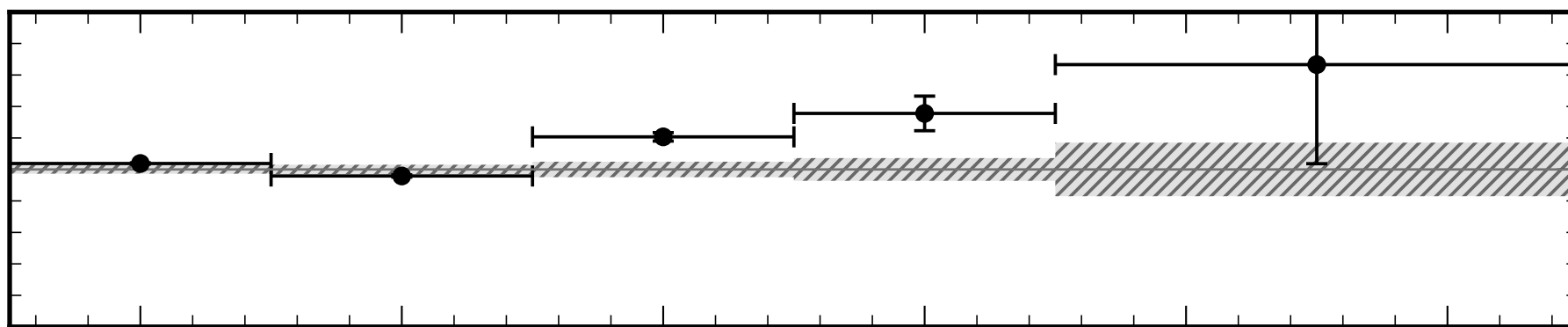
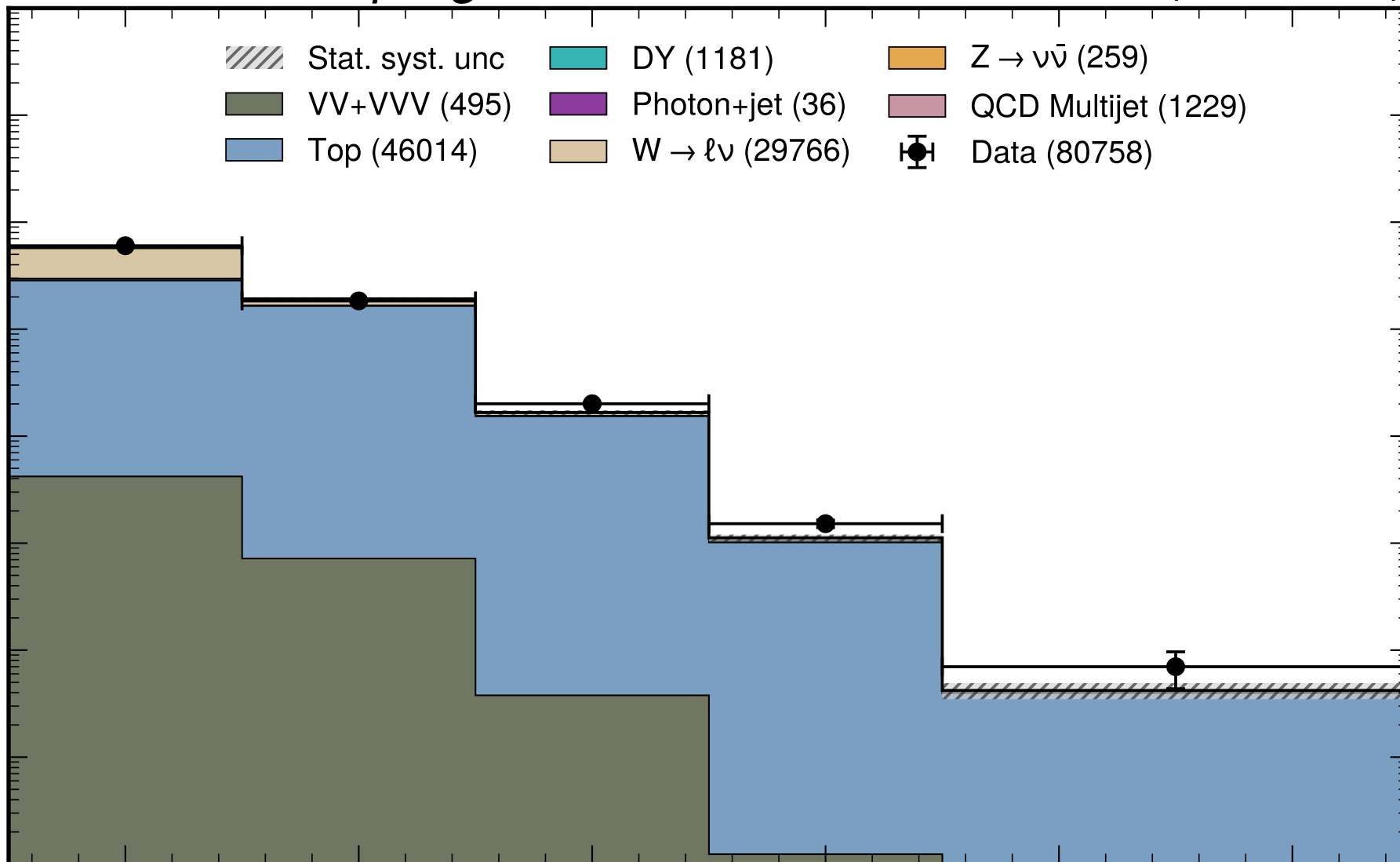
3

4

5

6

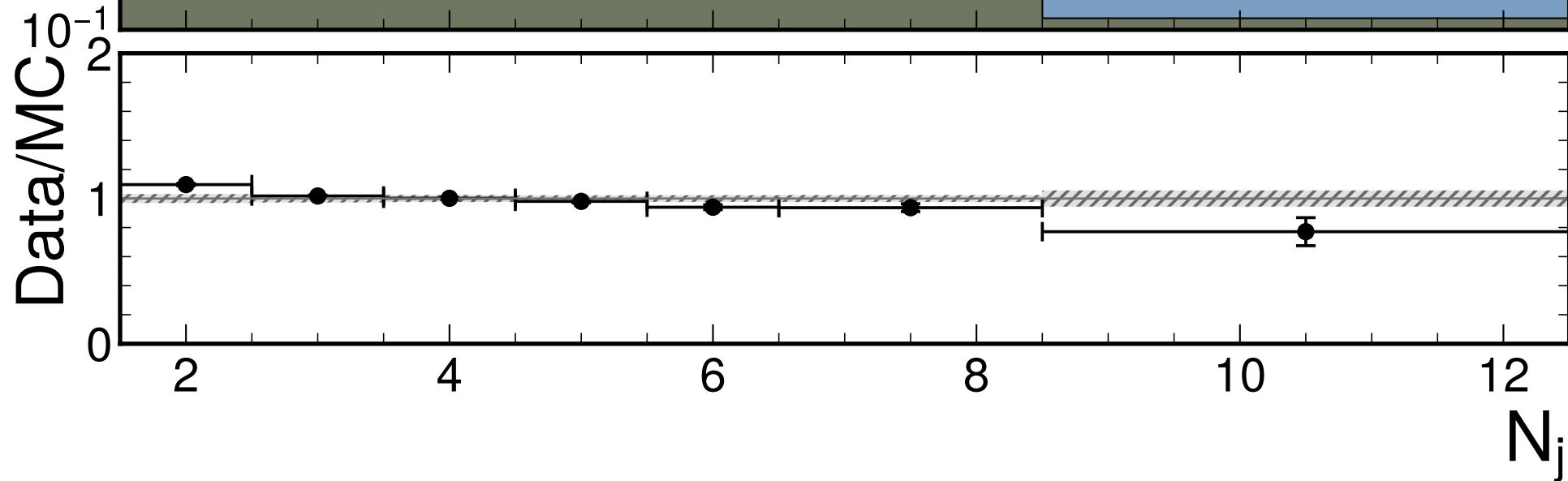
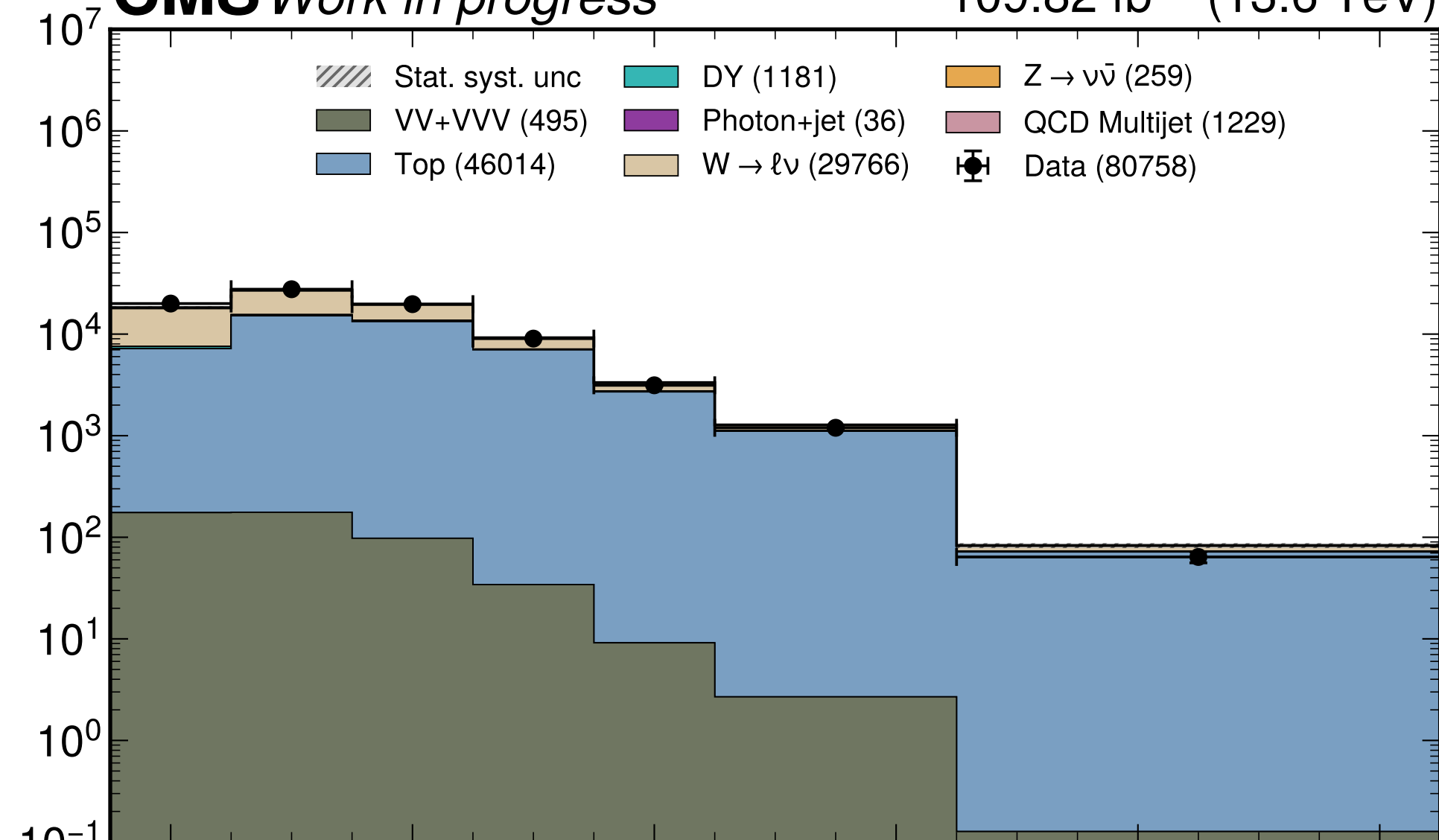
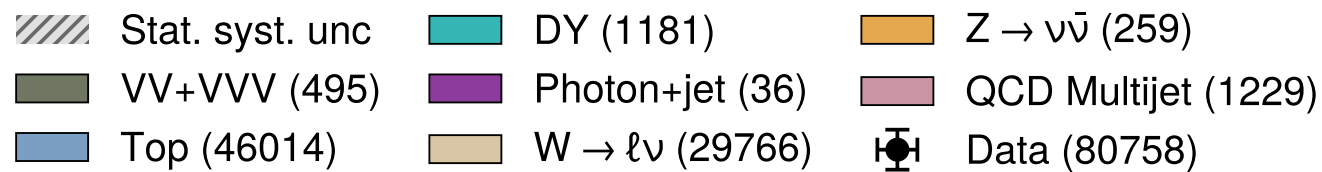
N_b



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

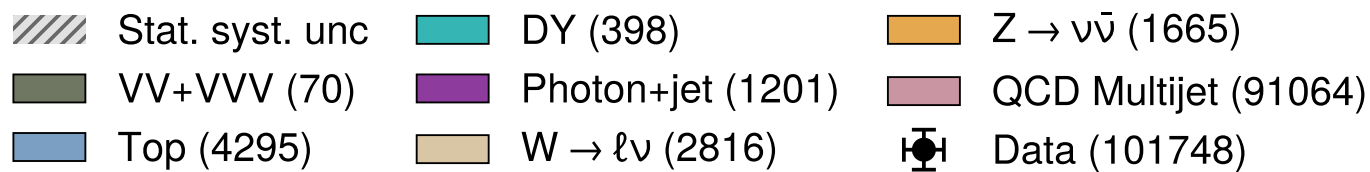
Events



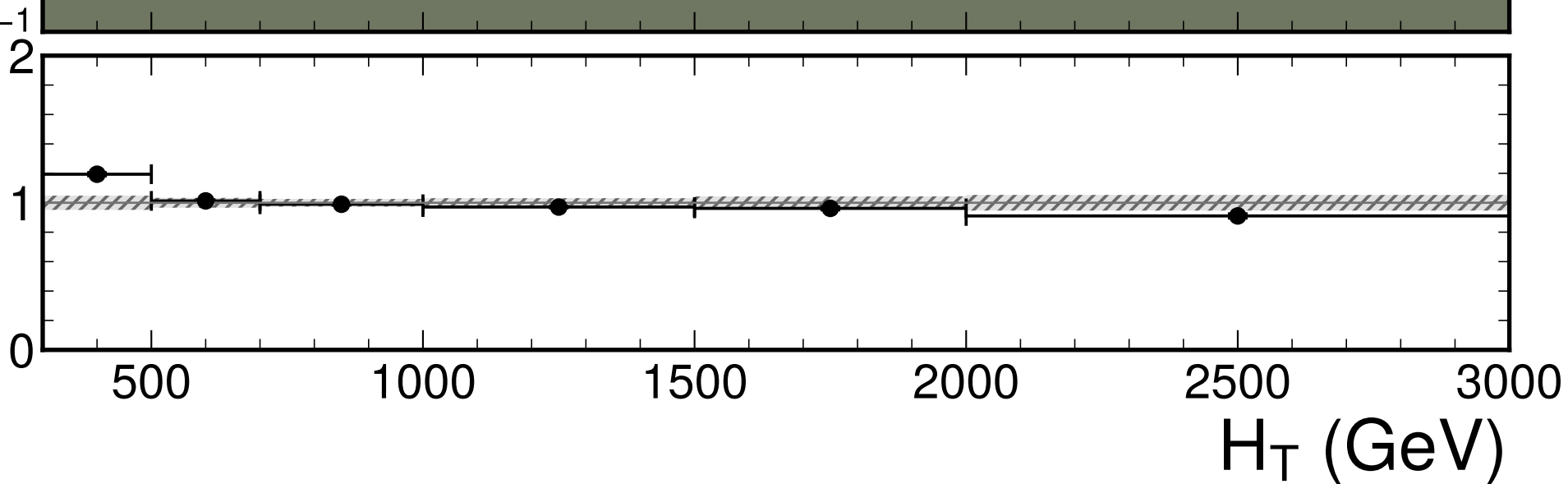
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



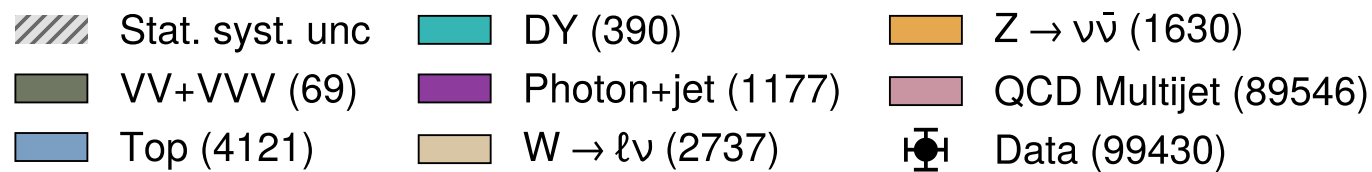
Data/MC



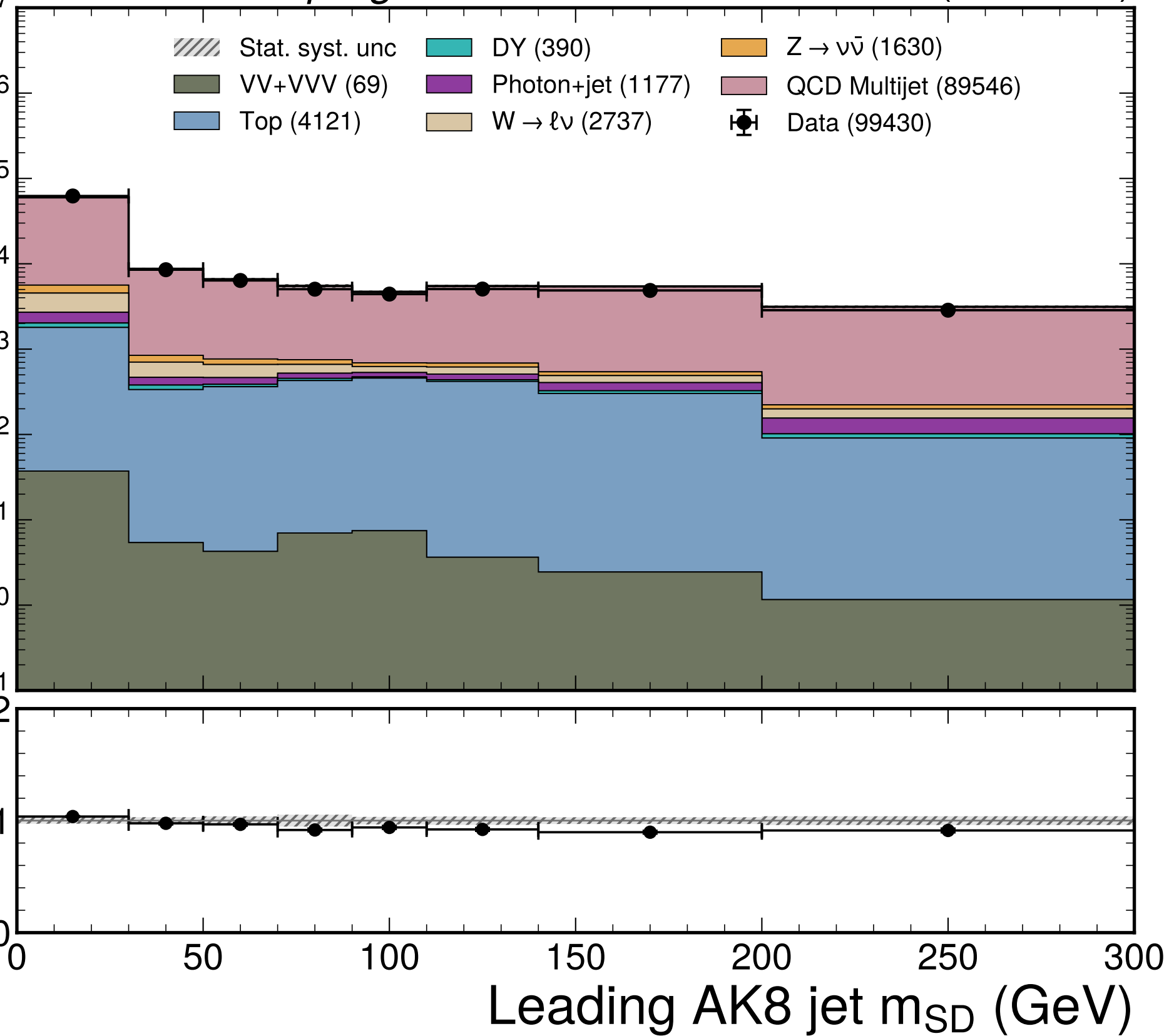
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



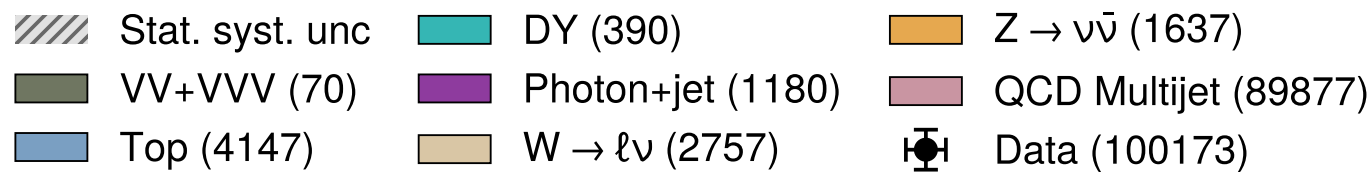
Data/MC



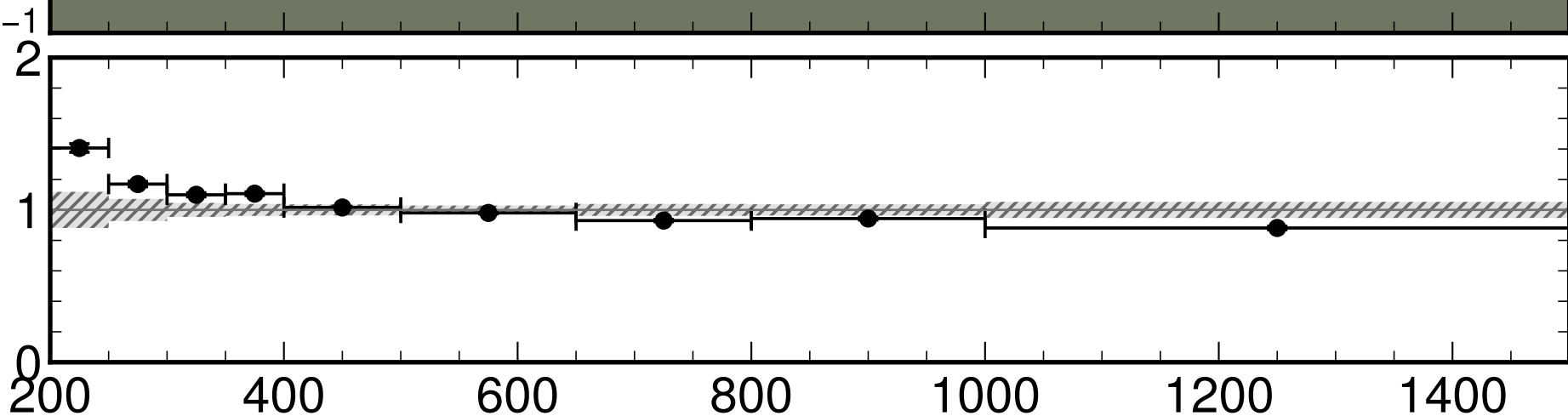
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

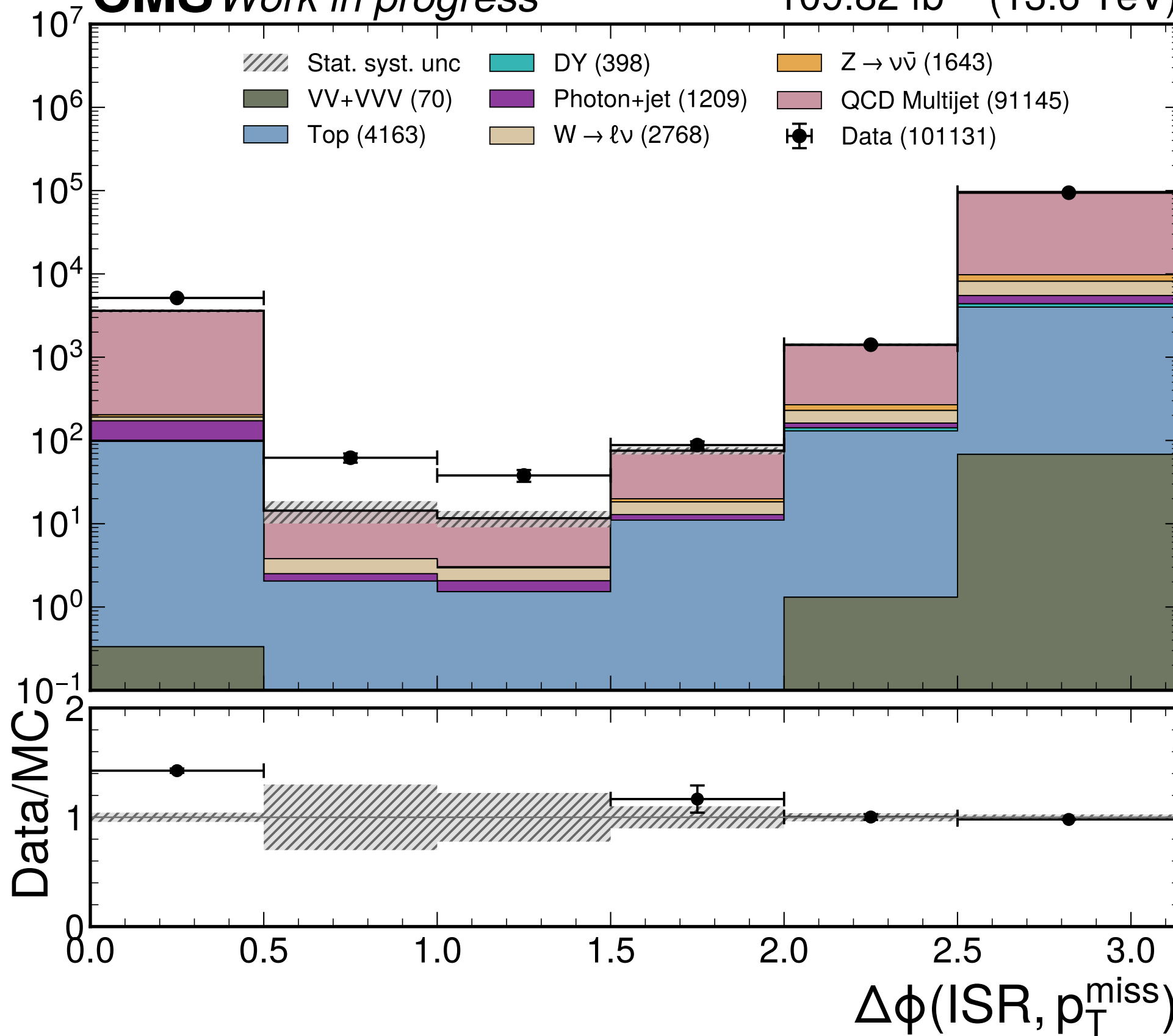
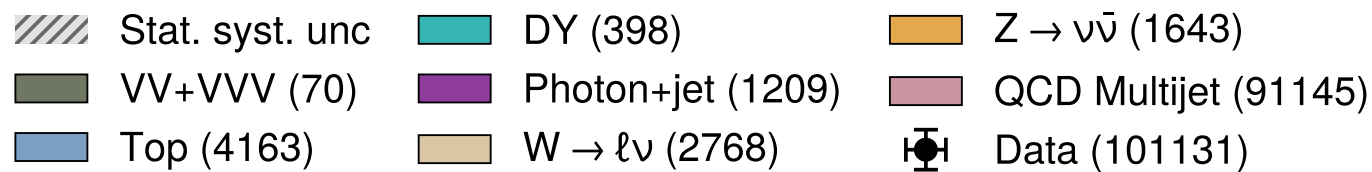


Leading AK8 jet p_T (GeV)

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

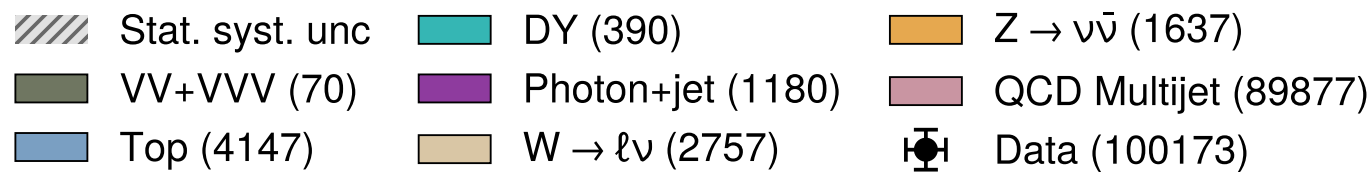
Events



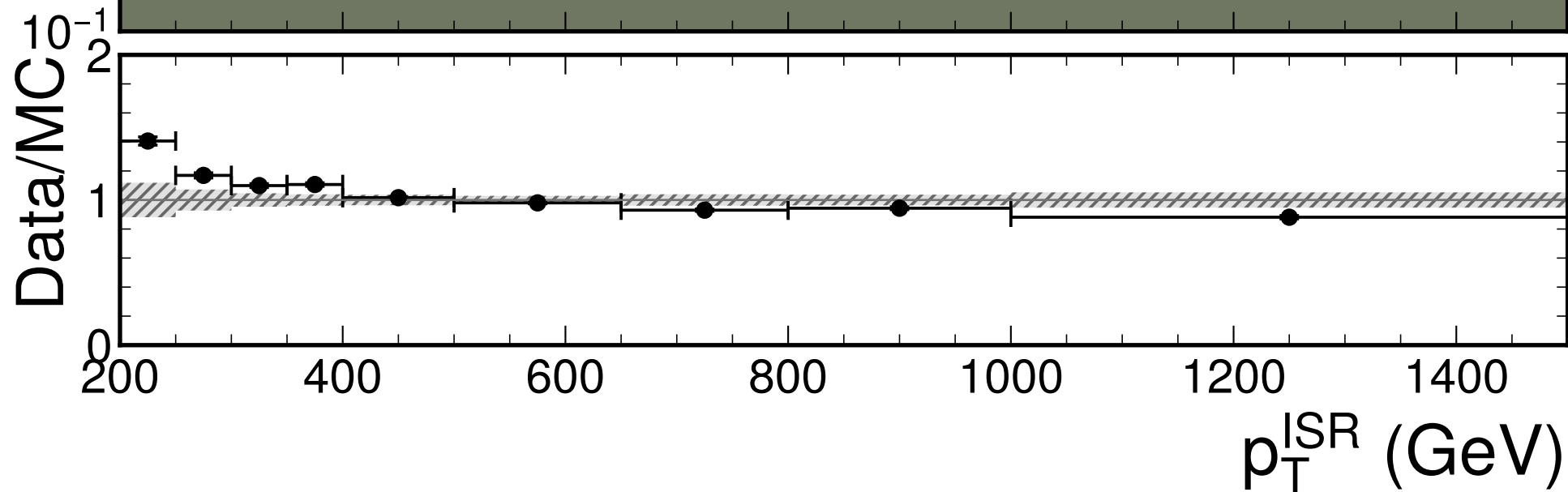
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

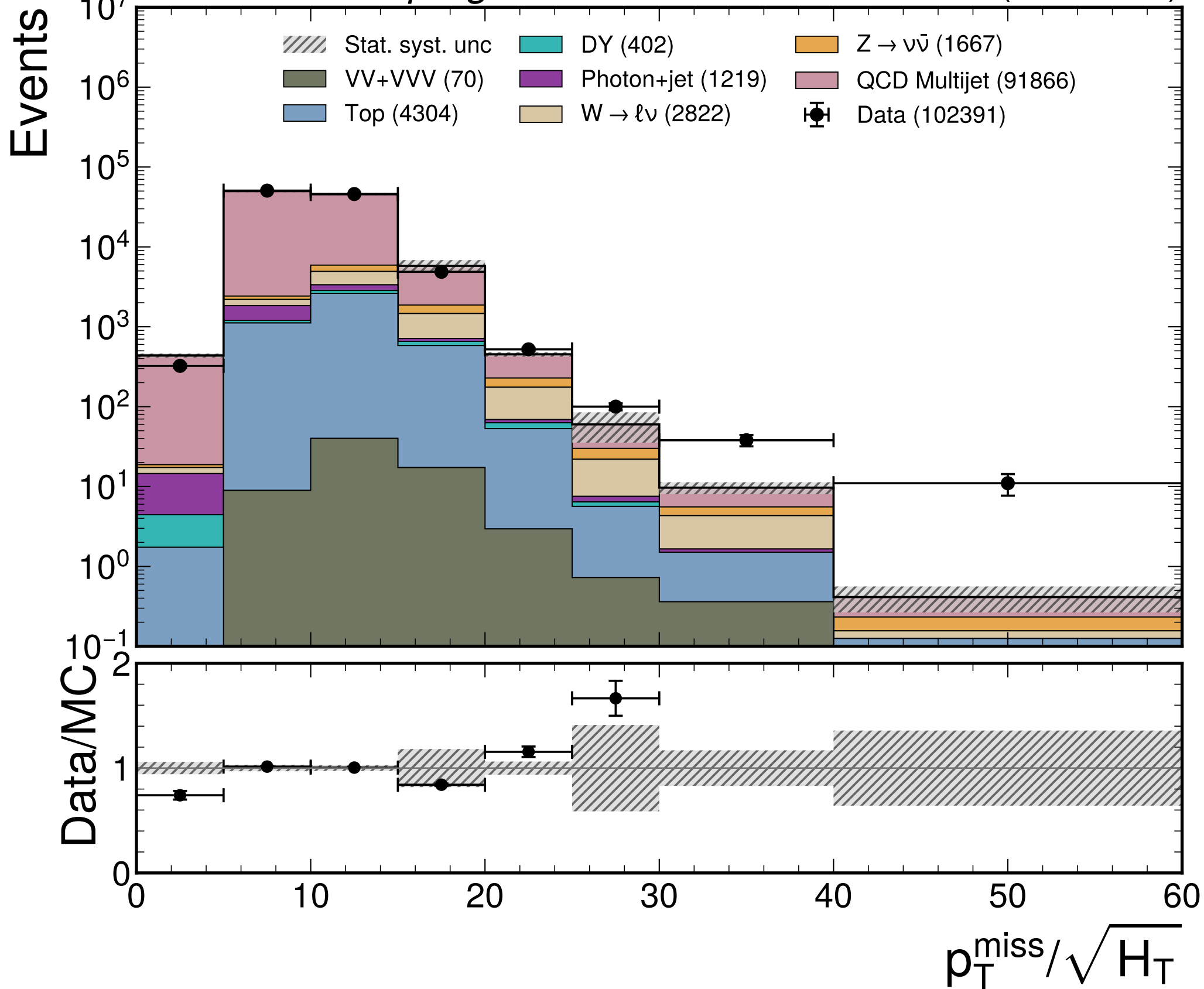
Events



Data/MC



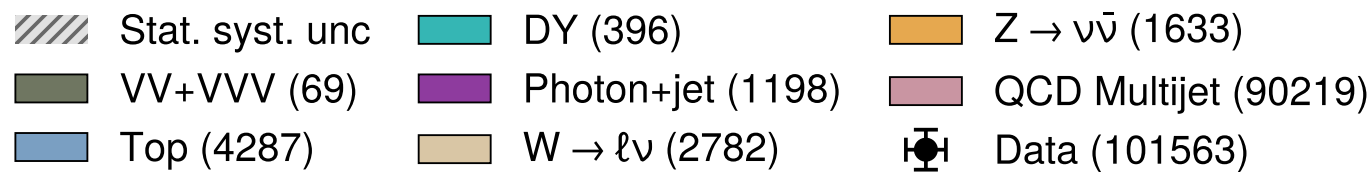
109.82 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

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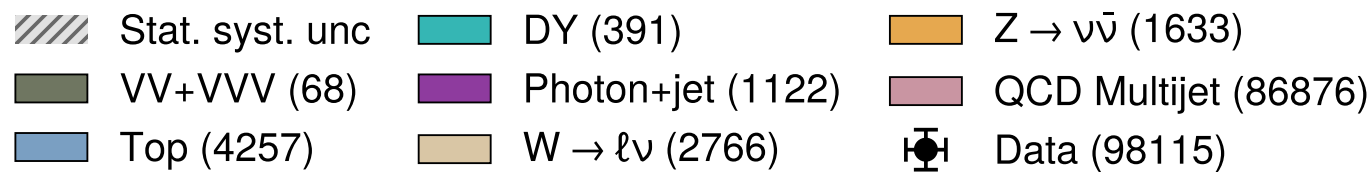
2

10⁻¹

CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



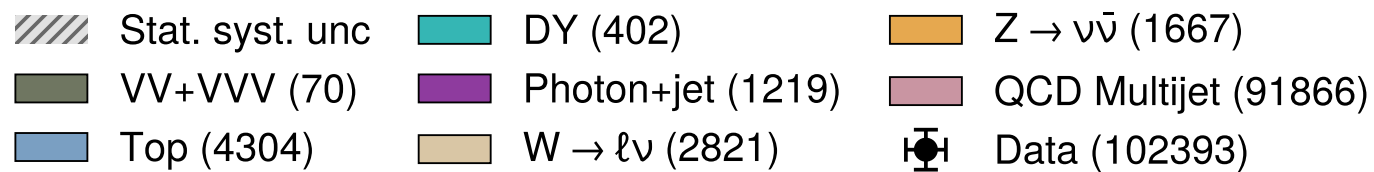
Data/MC

p_T^b (GeV)

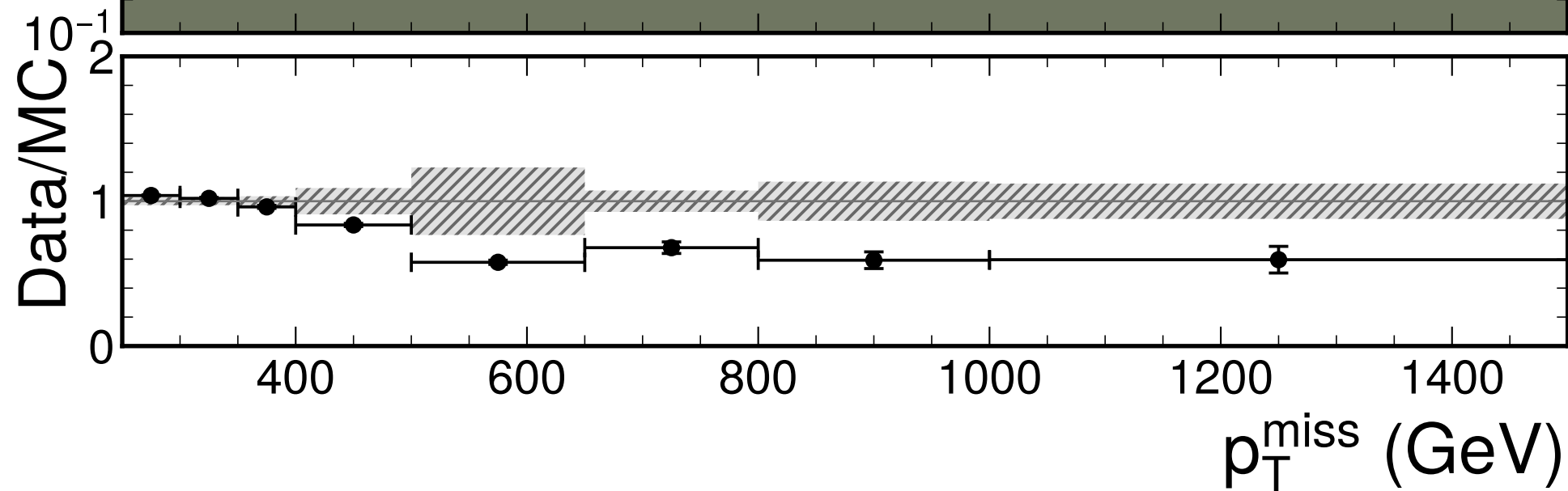
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



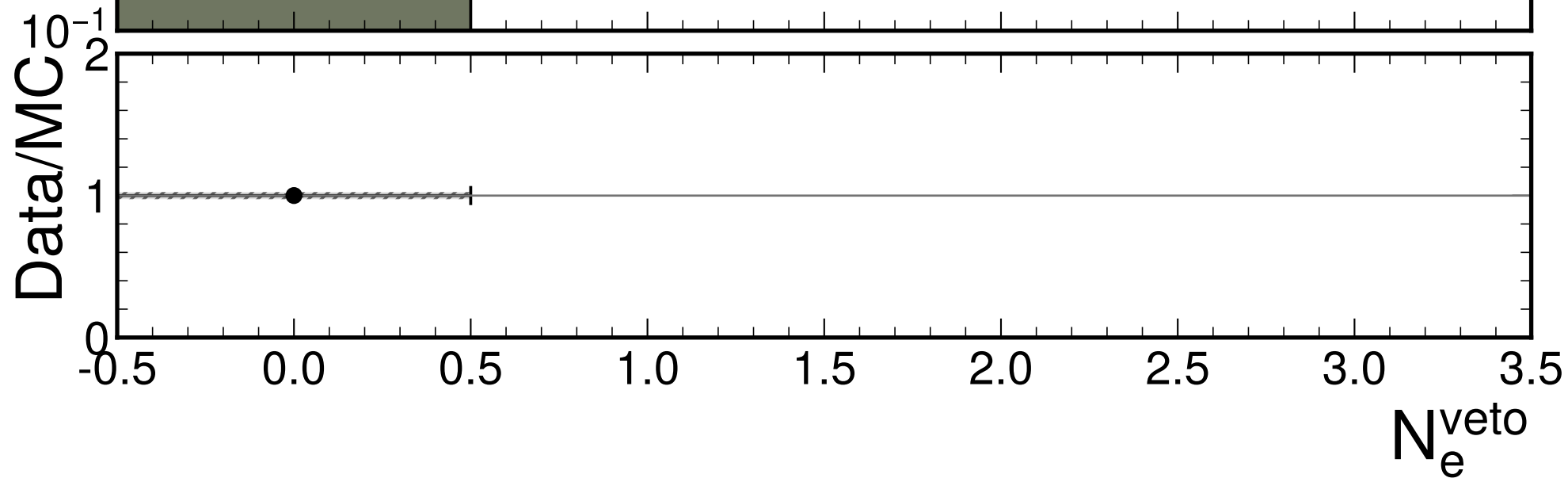
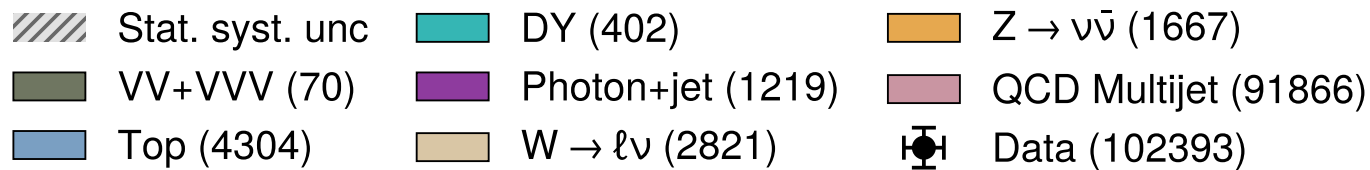
Data/MC



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

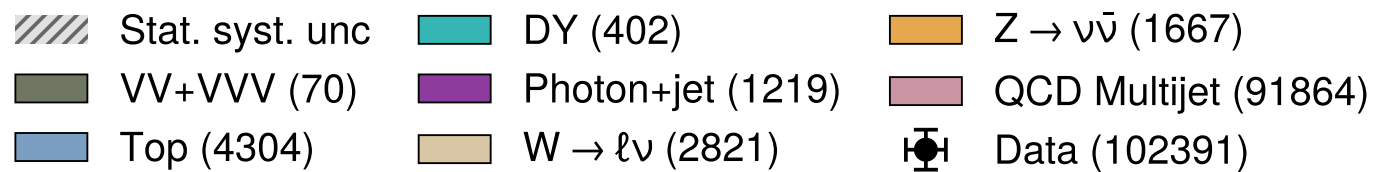
Events



CMS *Work in progress*

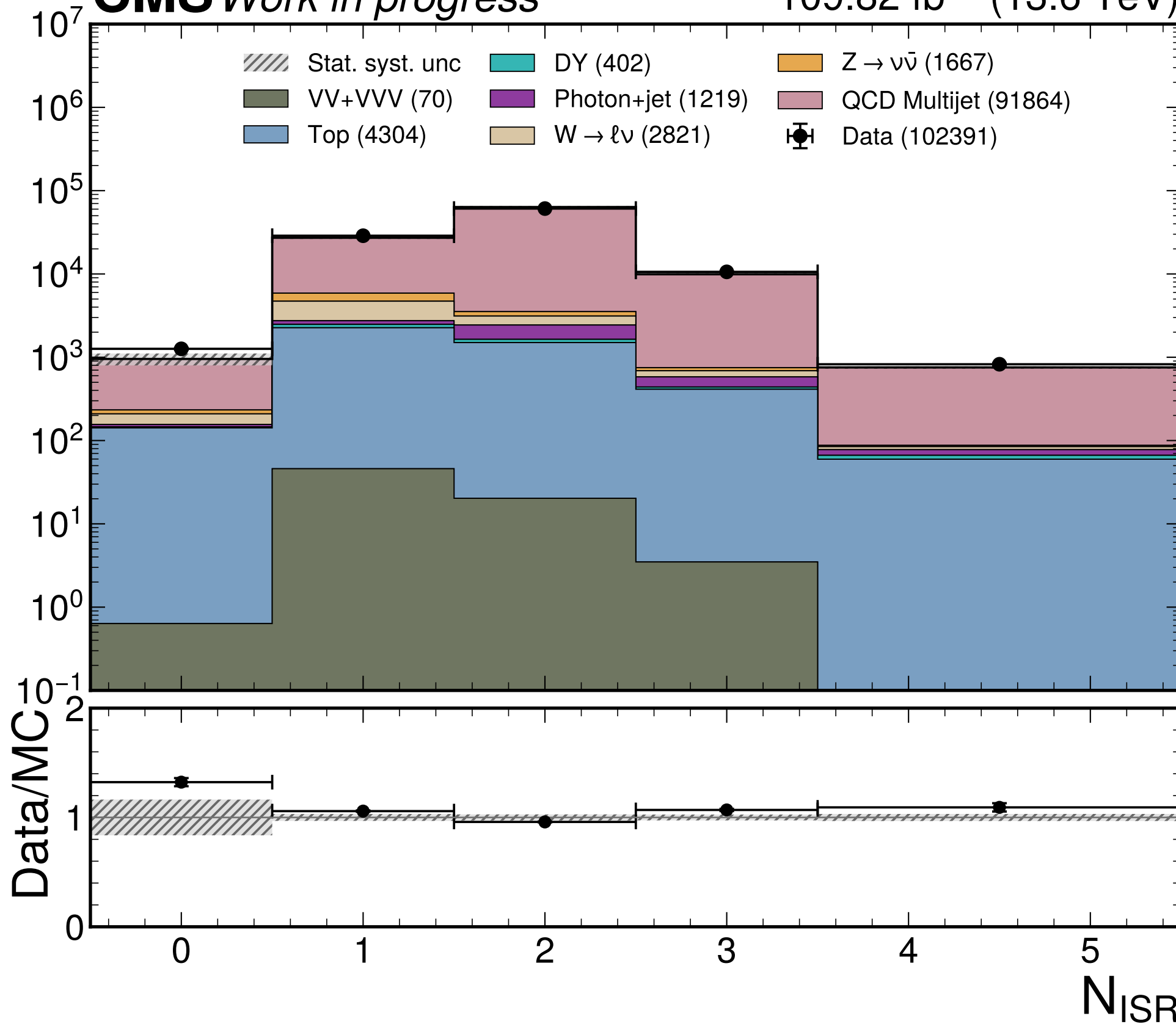
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

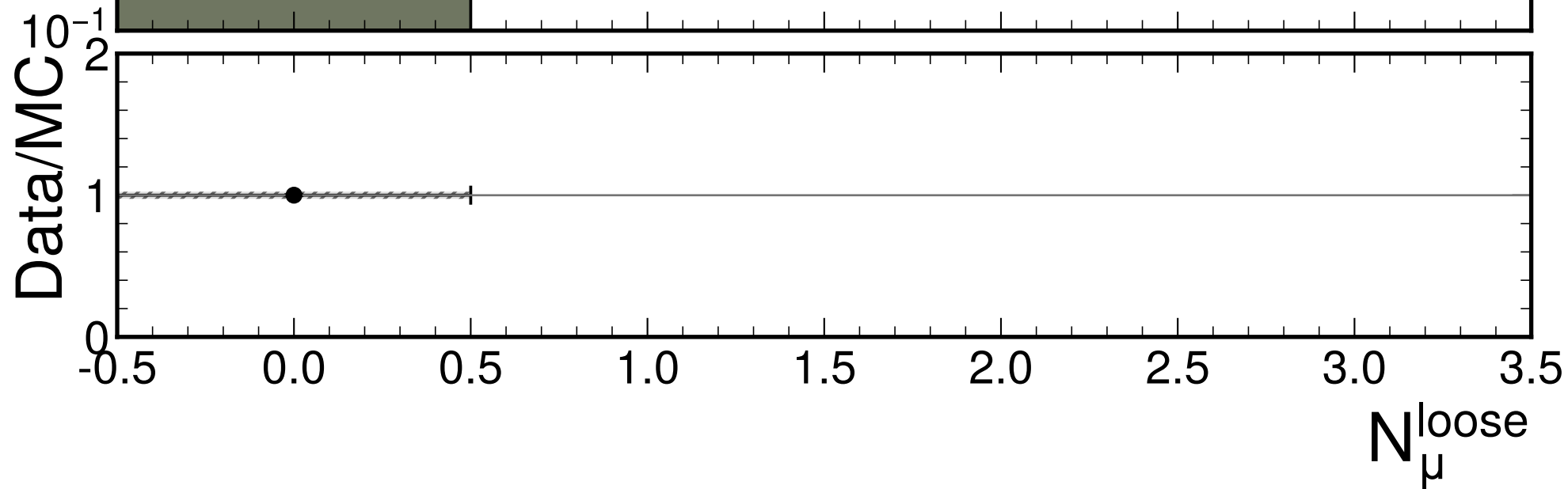
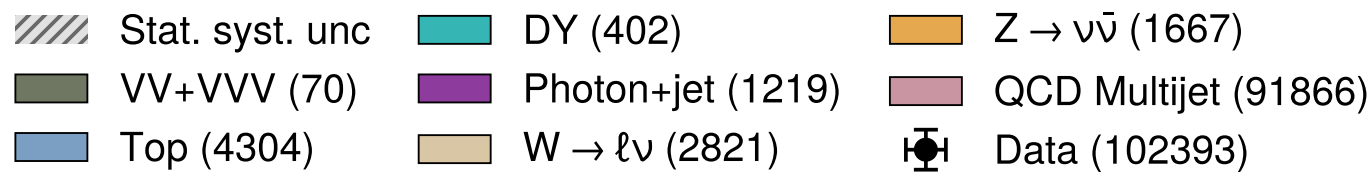
N_{ISR}



CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

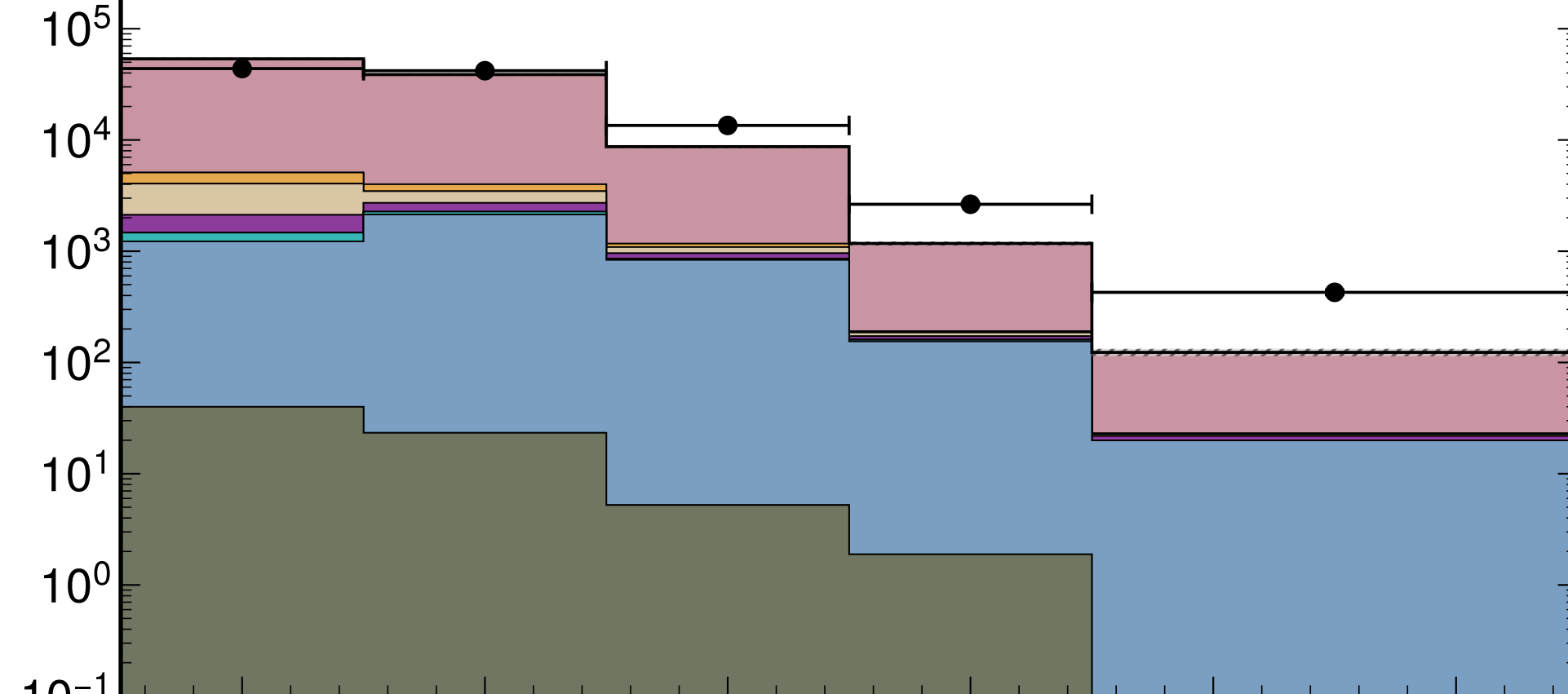
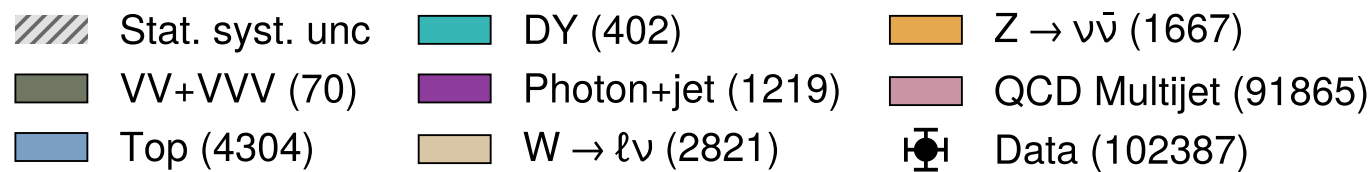
Events



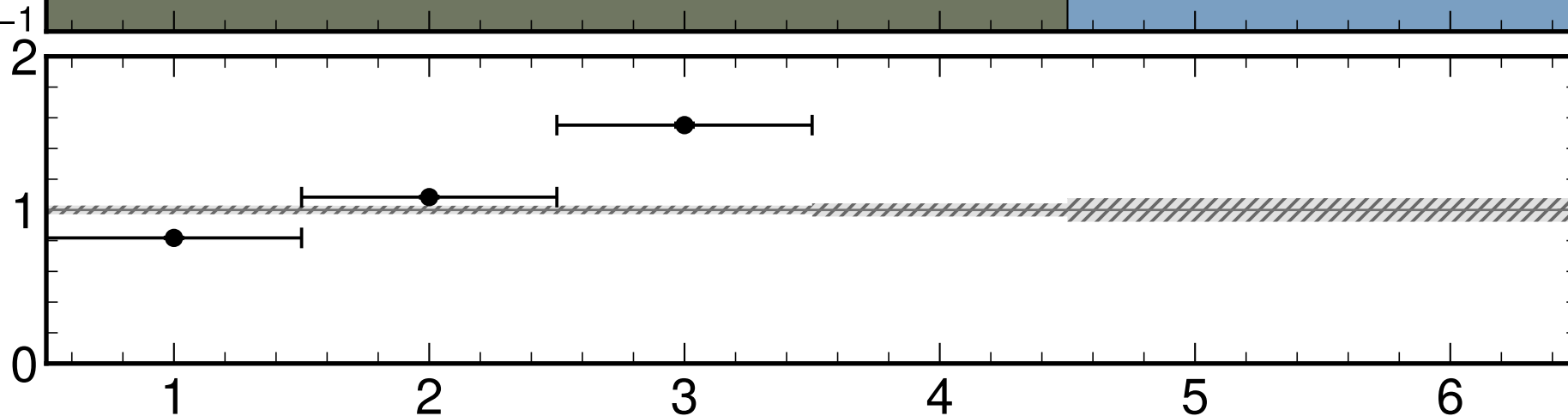
CMS *Work in progress*

109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

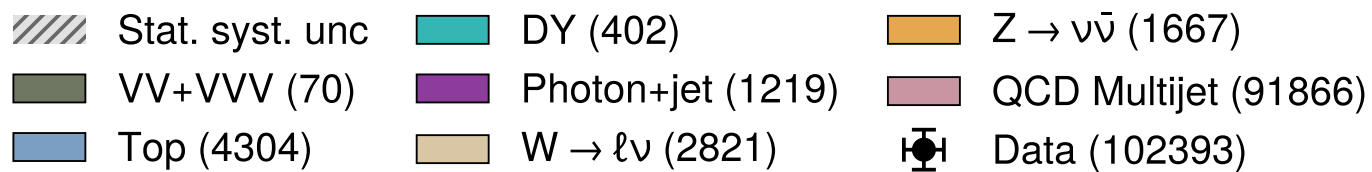


N_b (loose WP)

CMS *Work in progress*

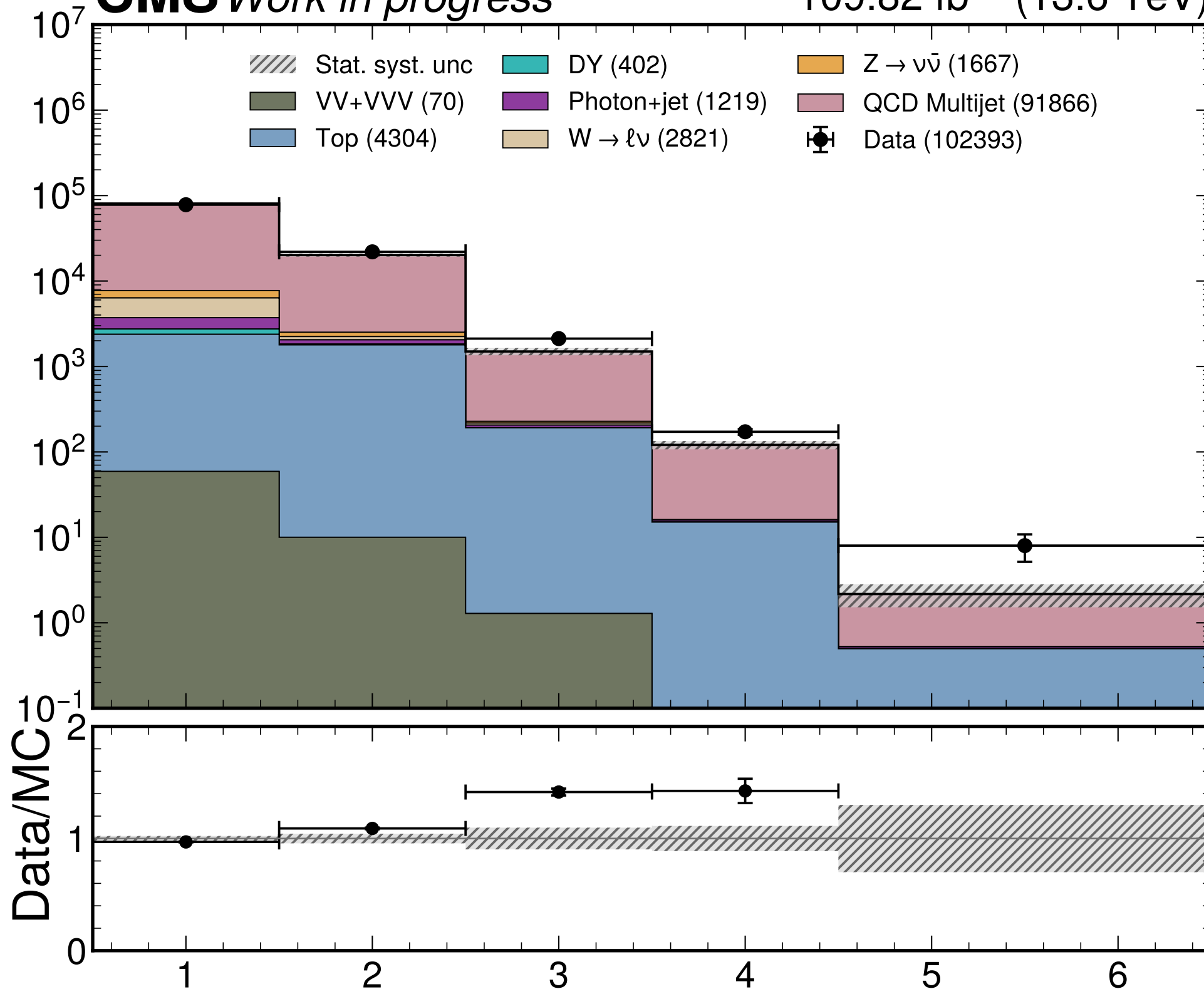
109.82 fb⁻¹ (13.6 TeV)

Events



Data/MC

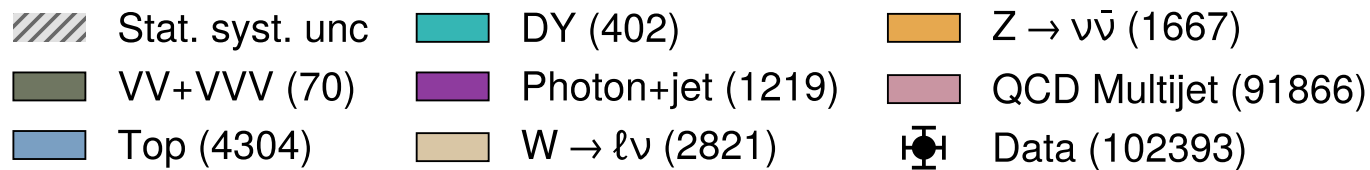
N_b



CMS *Work in progress*

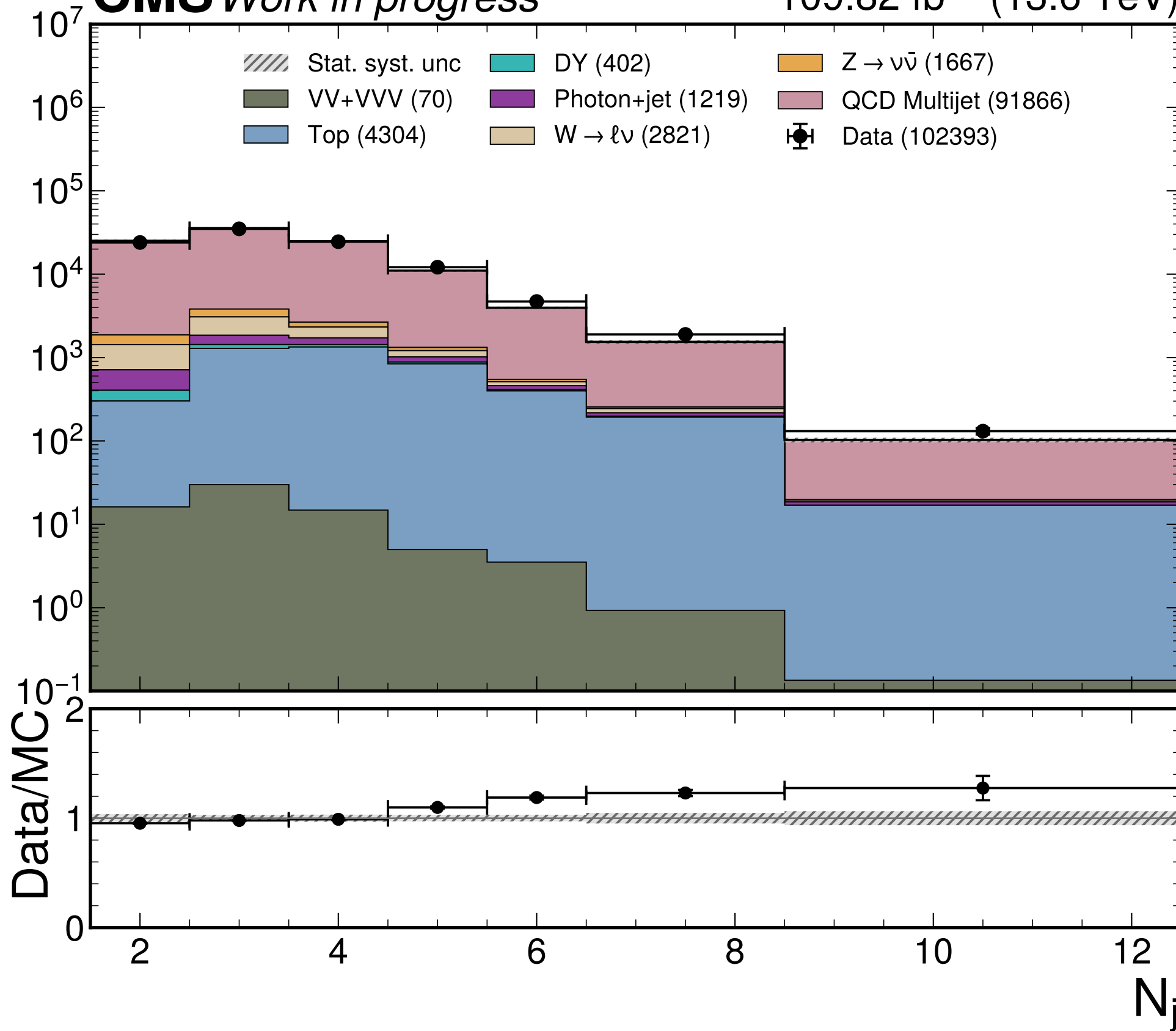
109.82 fb⁻¹ (13.6 TeV)

Events

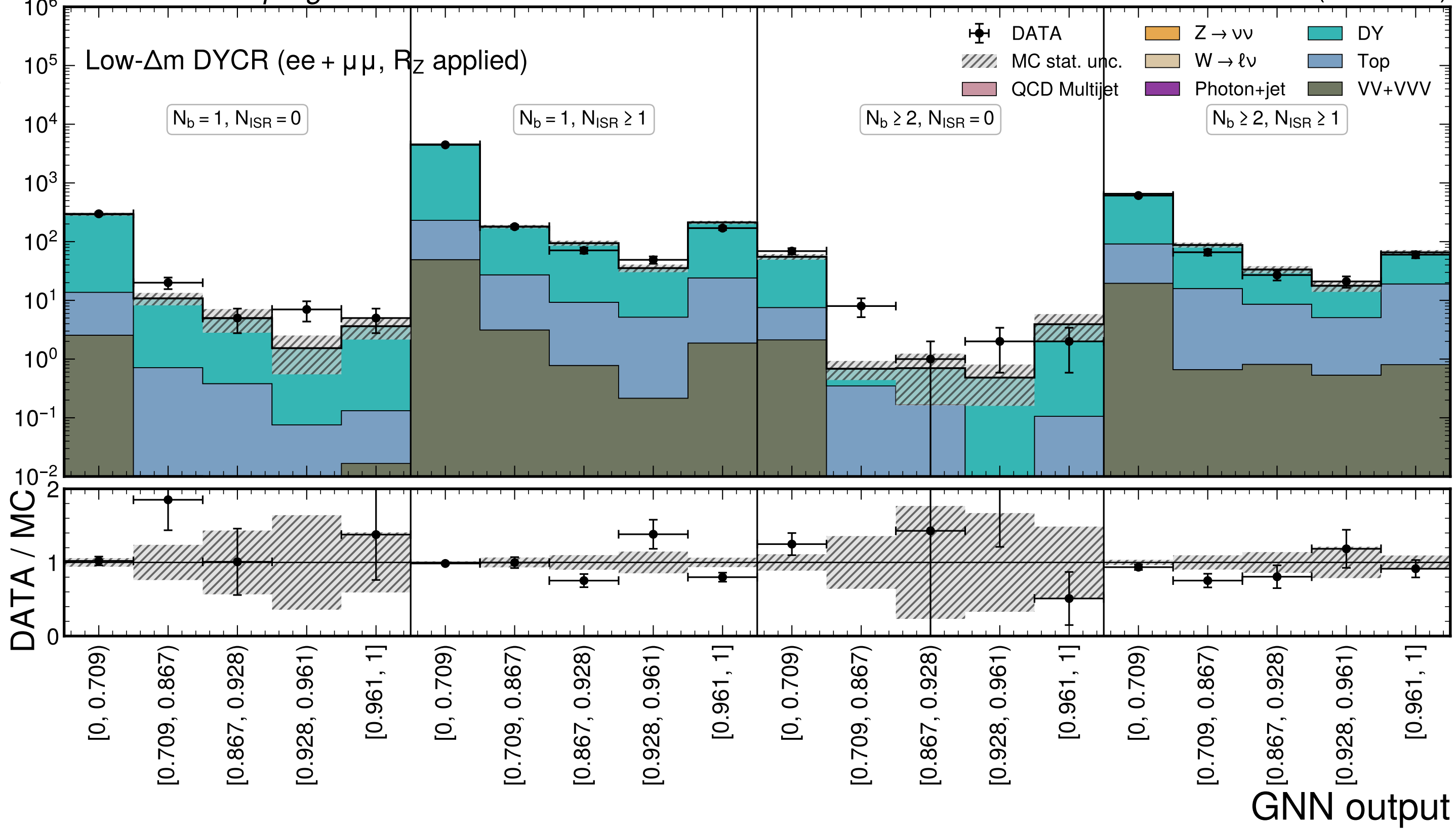


Data/MC

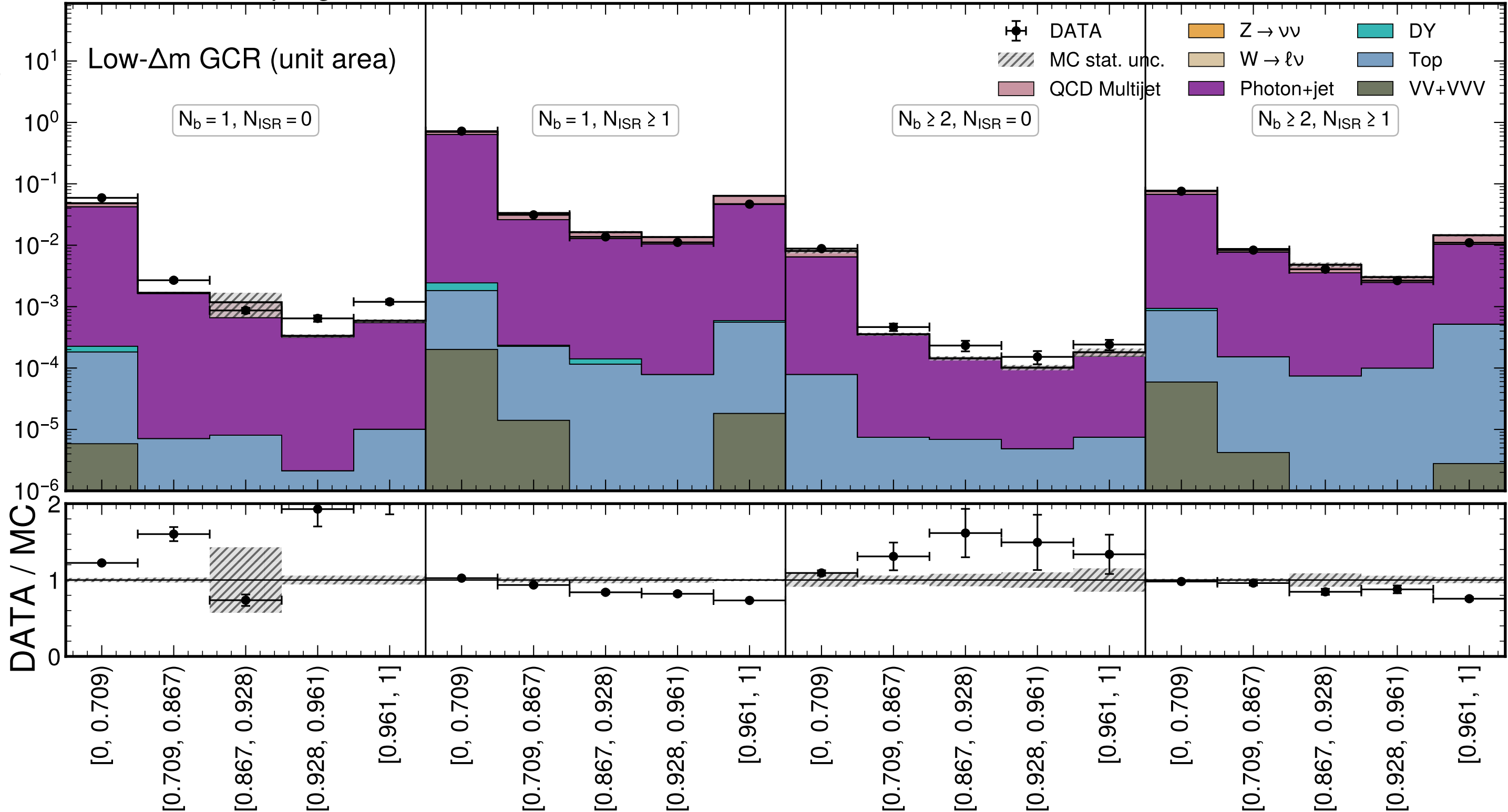
N_j



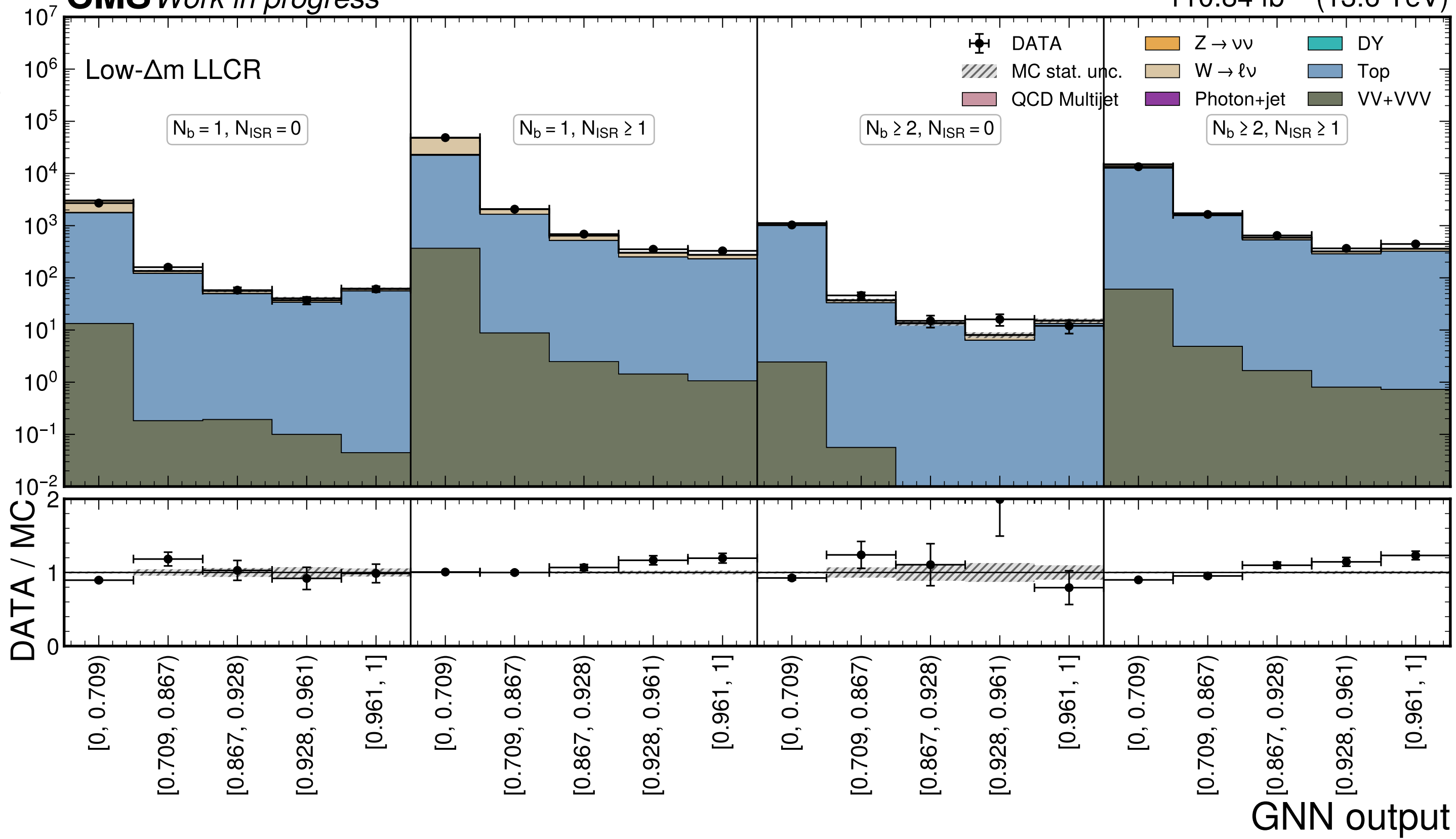
Events / bin



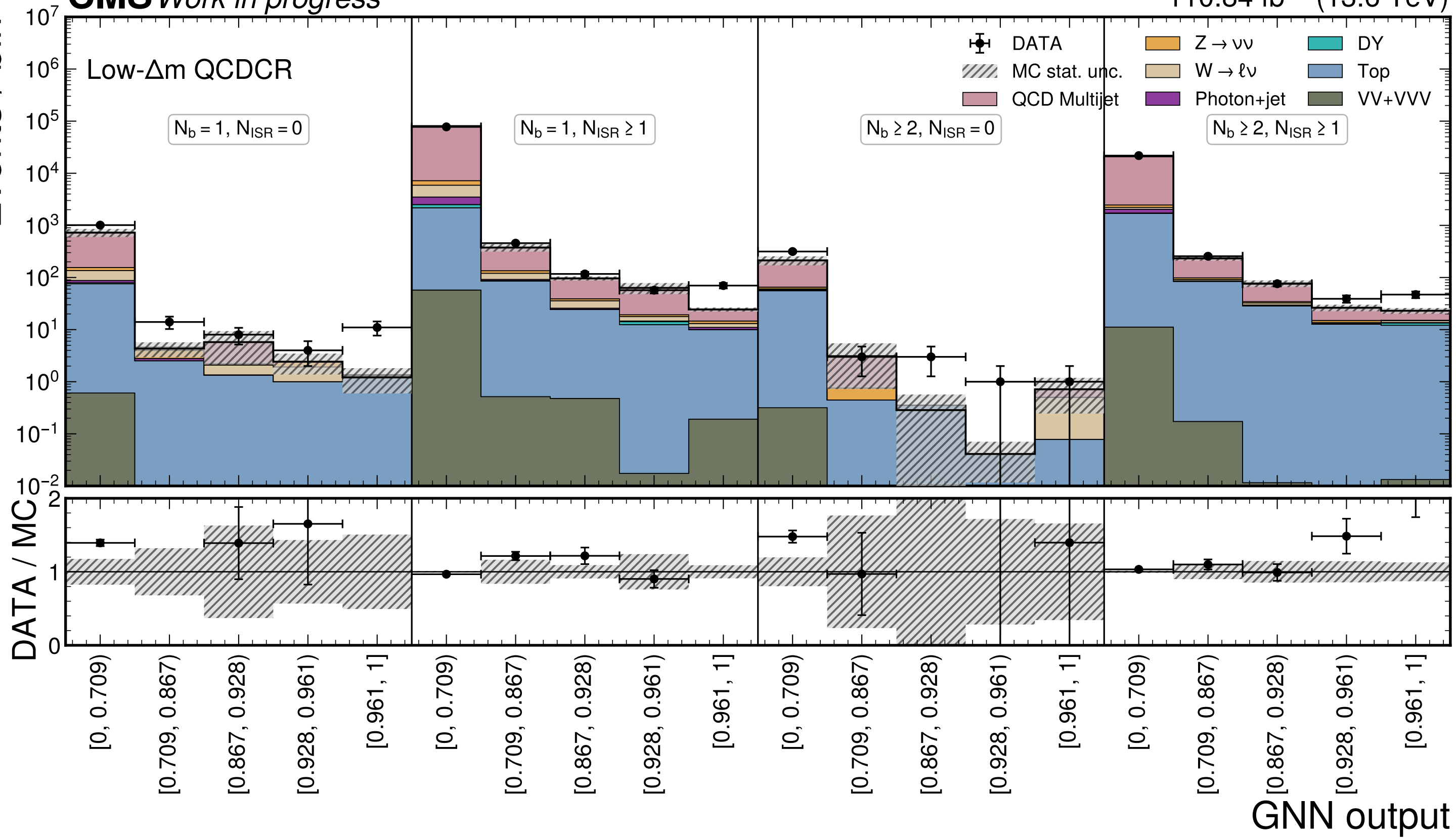
Normalized events / bin



Events / bin



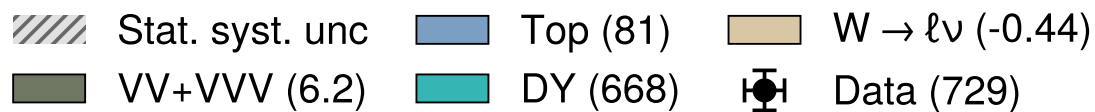
Events / bin



CMS *Work in progress*

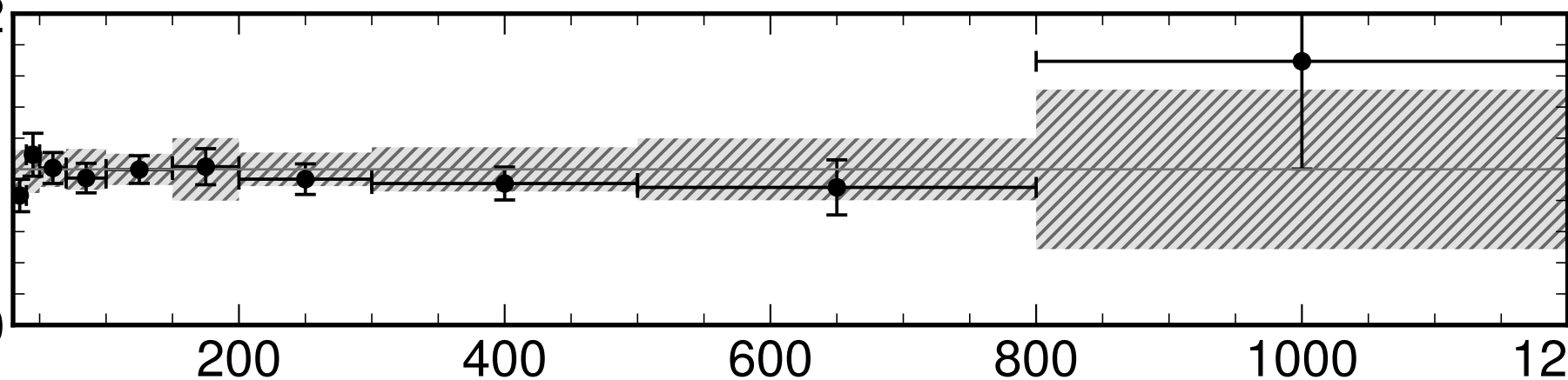
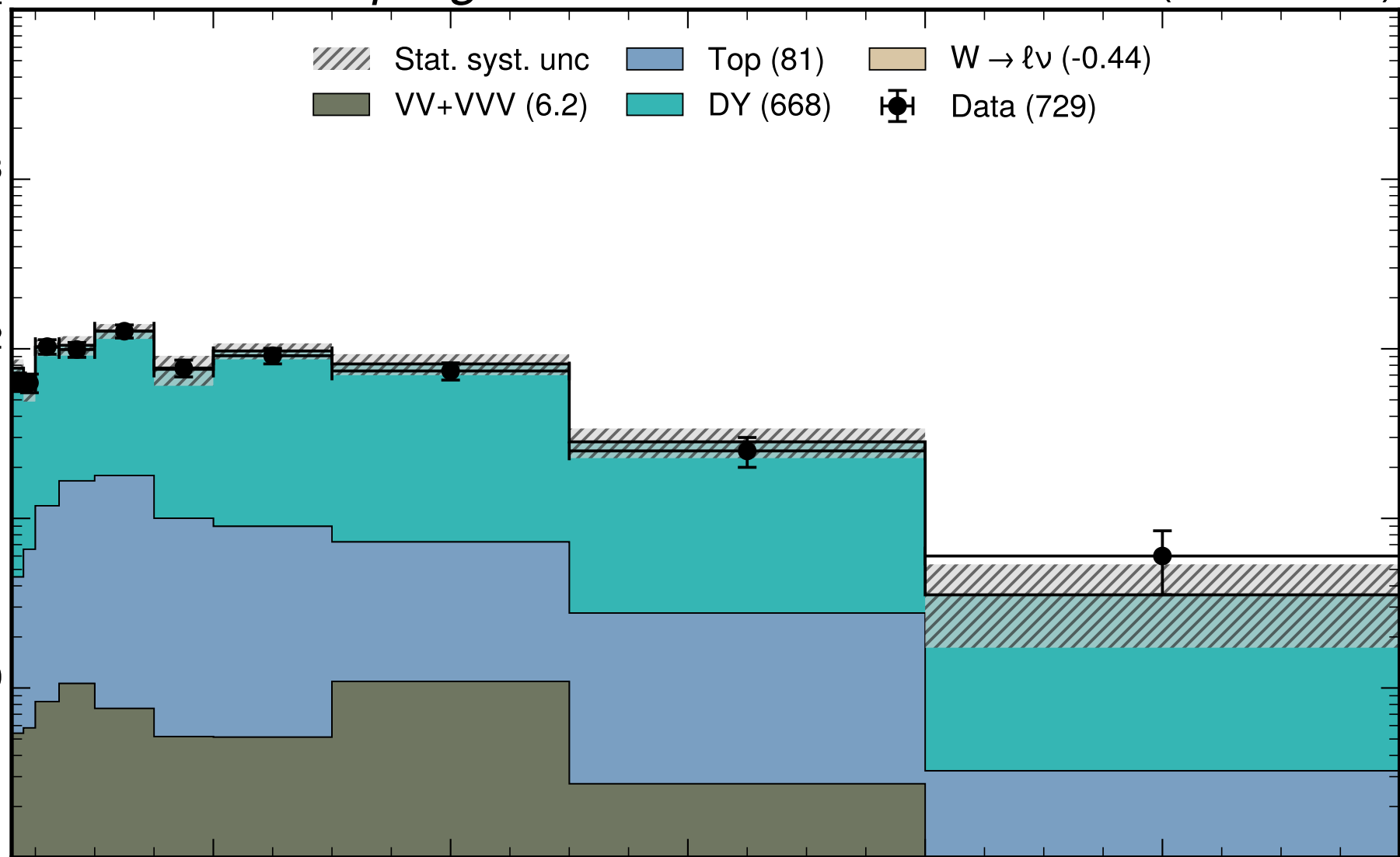
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

Leading b-jet p_T (GeV)

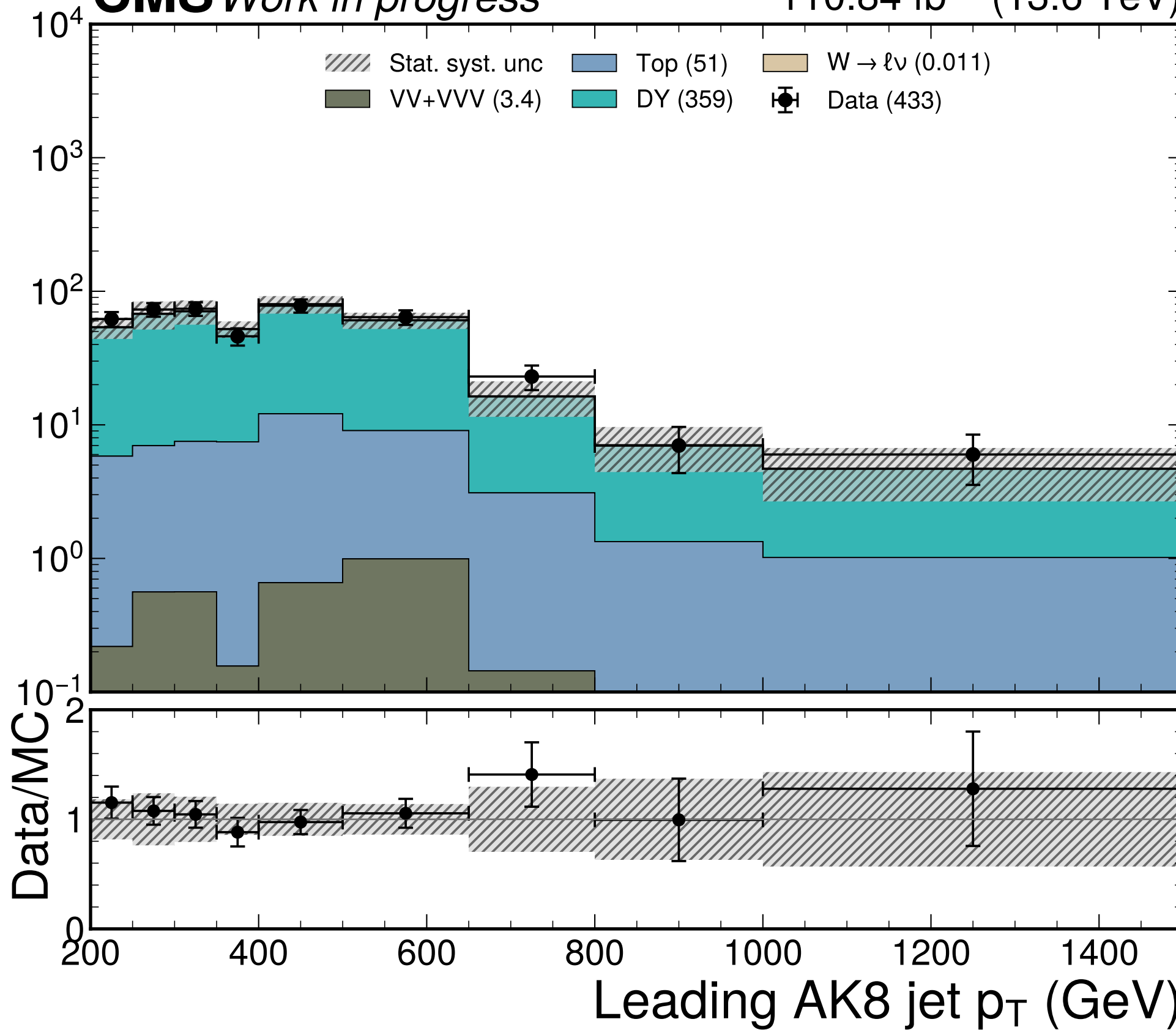


CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

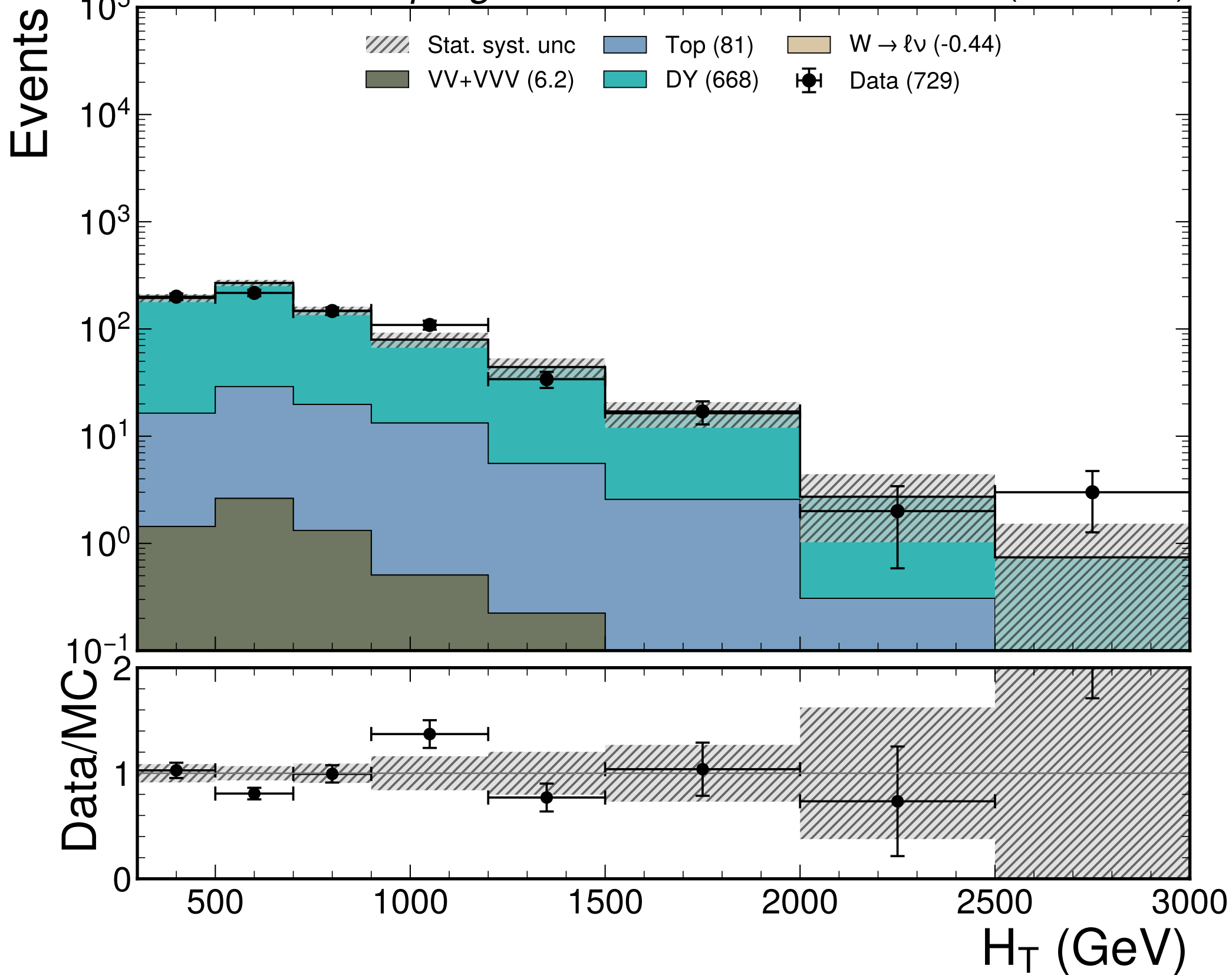
Events

Stat. syst. unc Top (51) $W \rightarrow \ell\nu$ (0.011)
VV+VVV (3.4) DY (359) Data (433)



CMS *Work in progress*

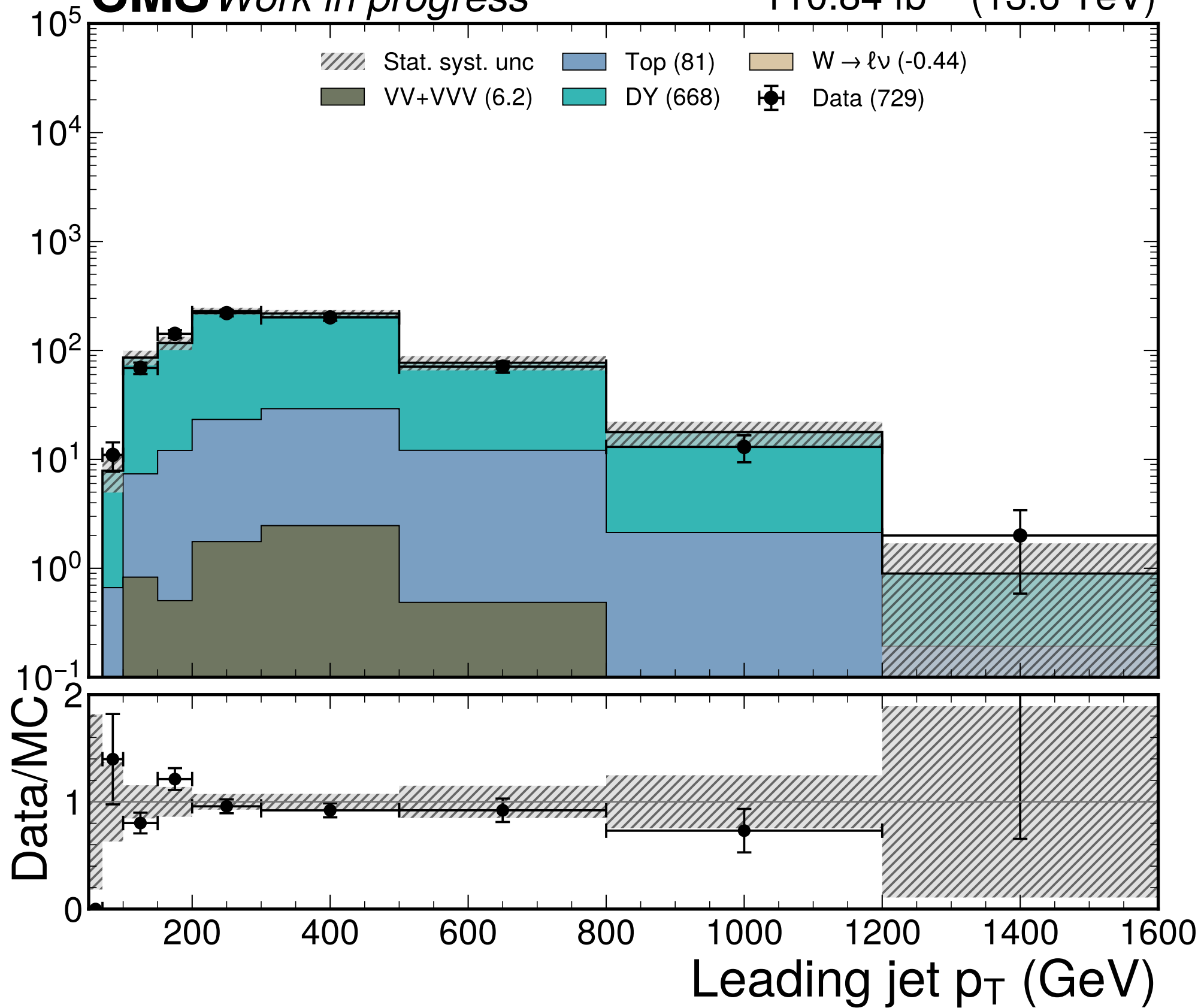
110.84 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

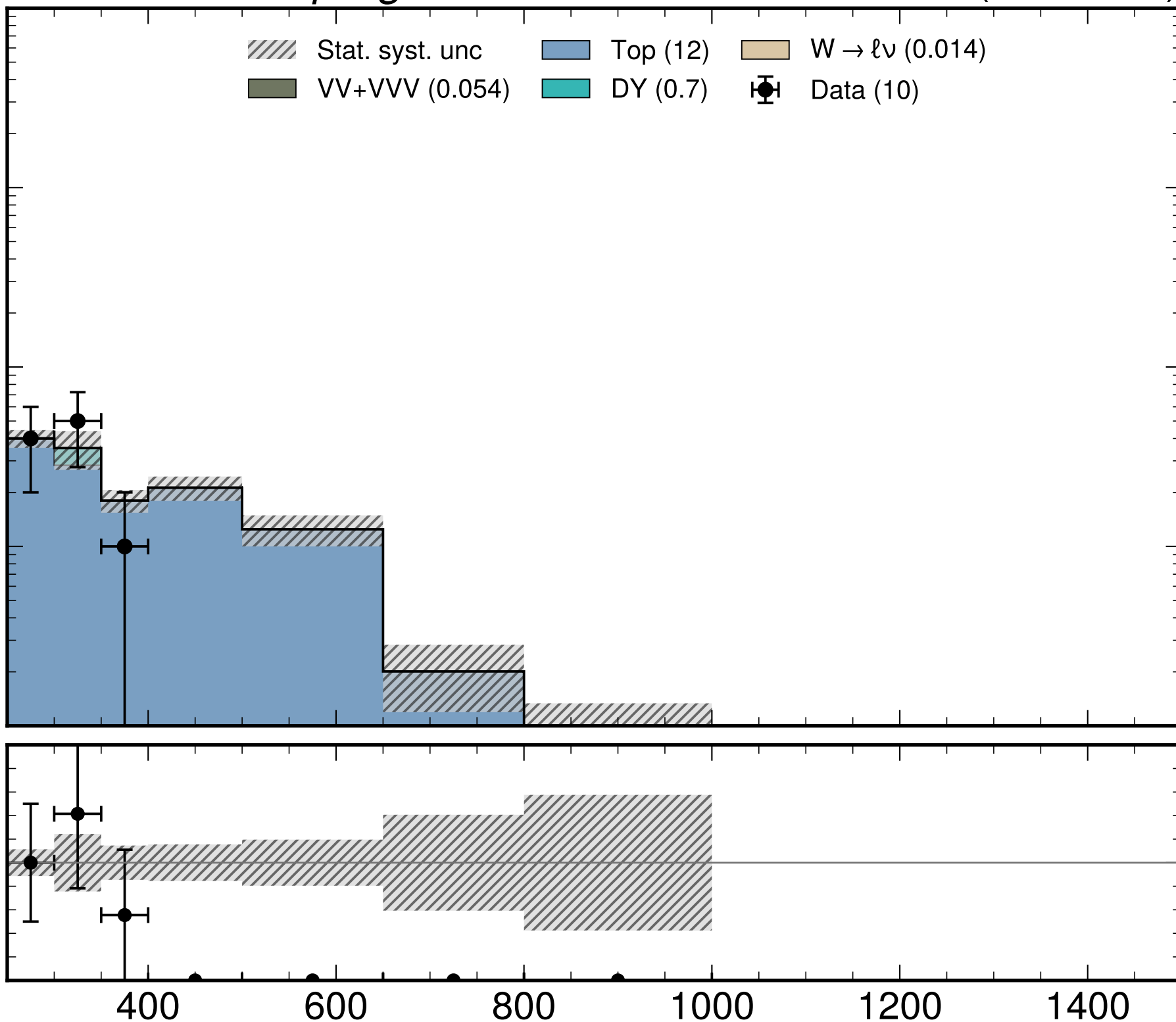
Events



Data/MC

10⁻¹
2
1
0

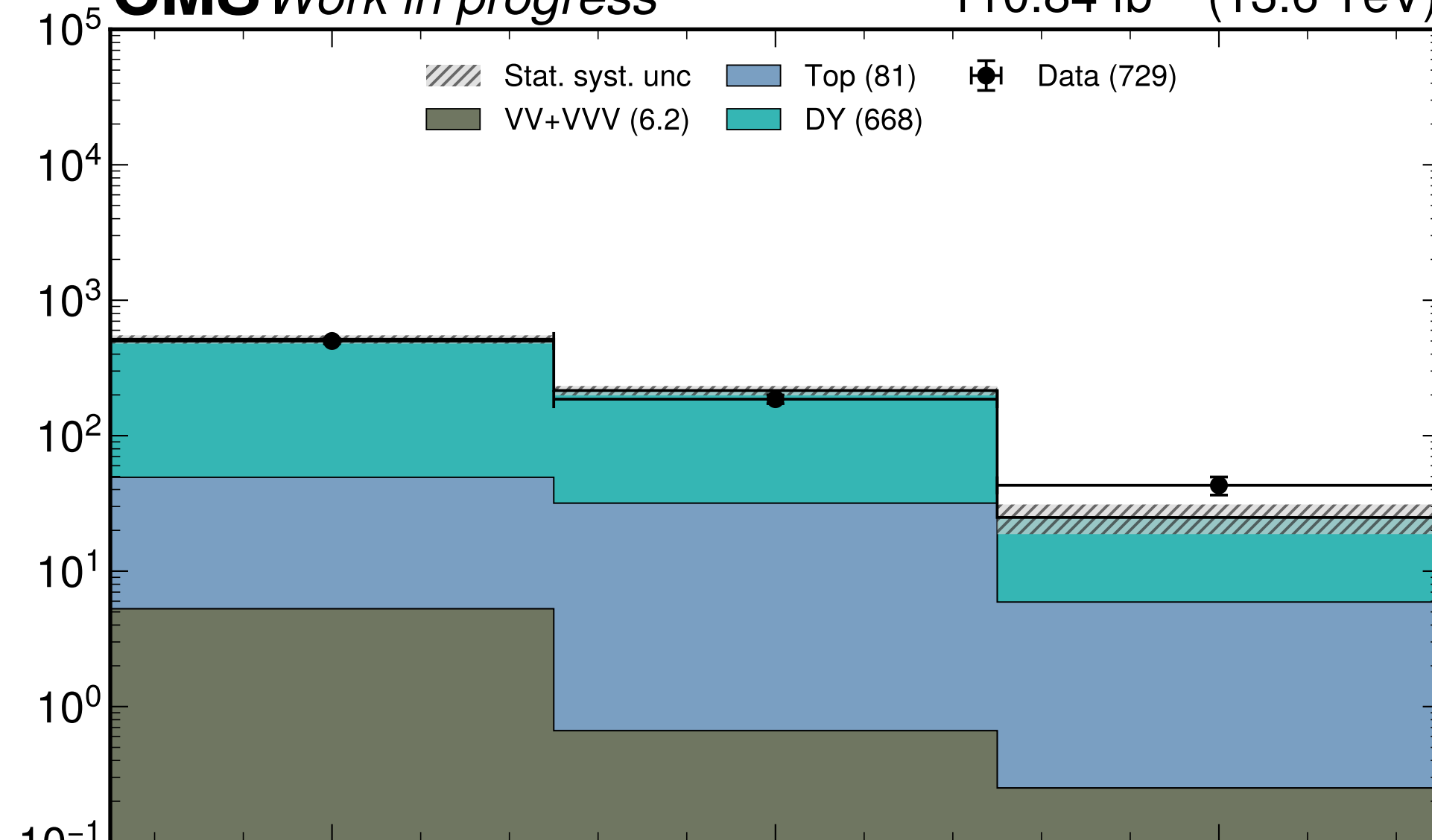
p_T^{miss} (GeV)



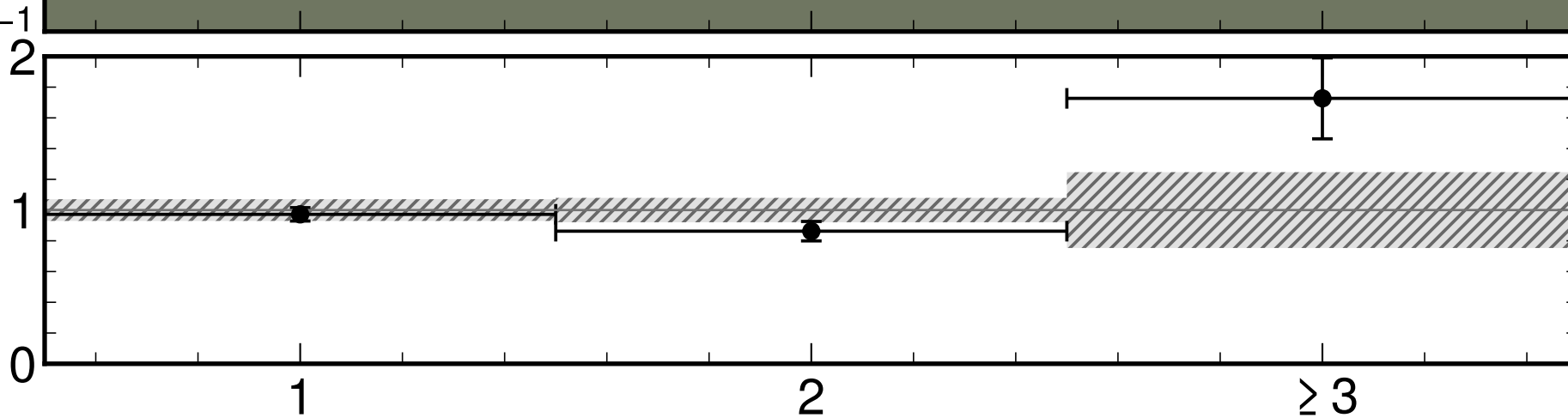
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

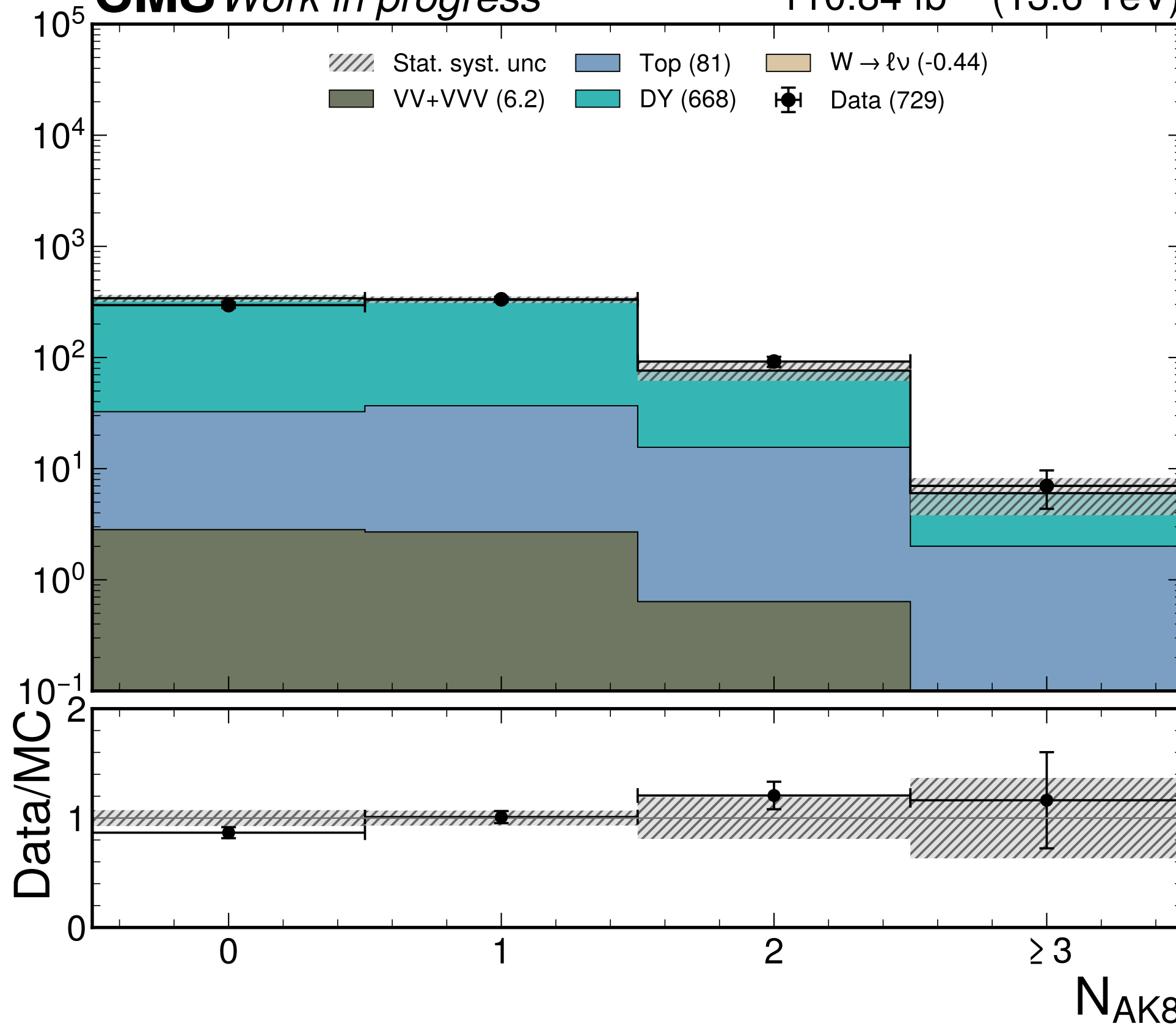
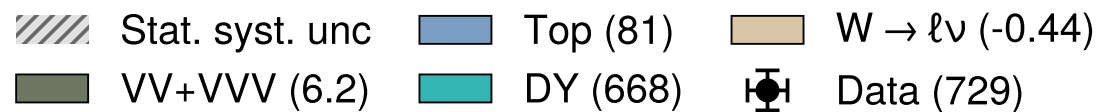


N_b

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

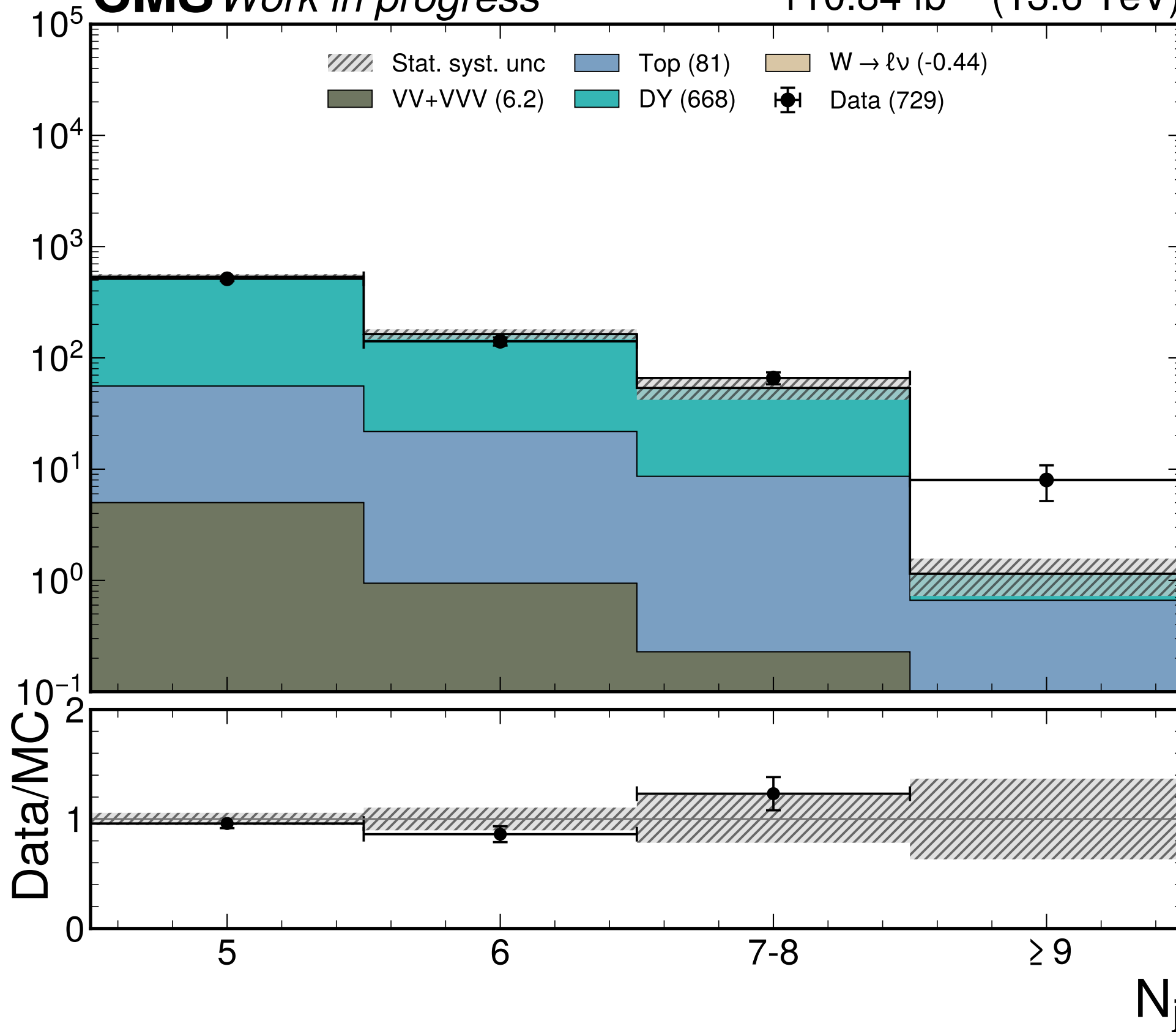
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

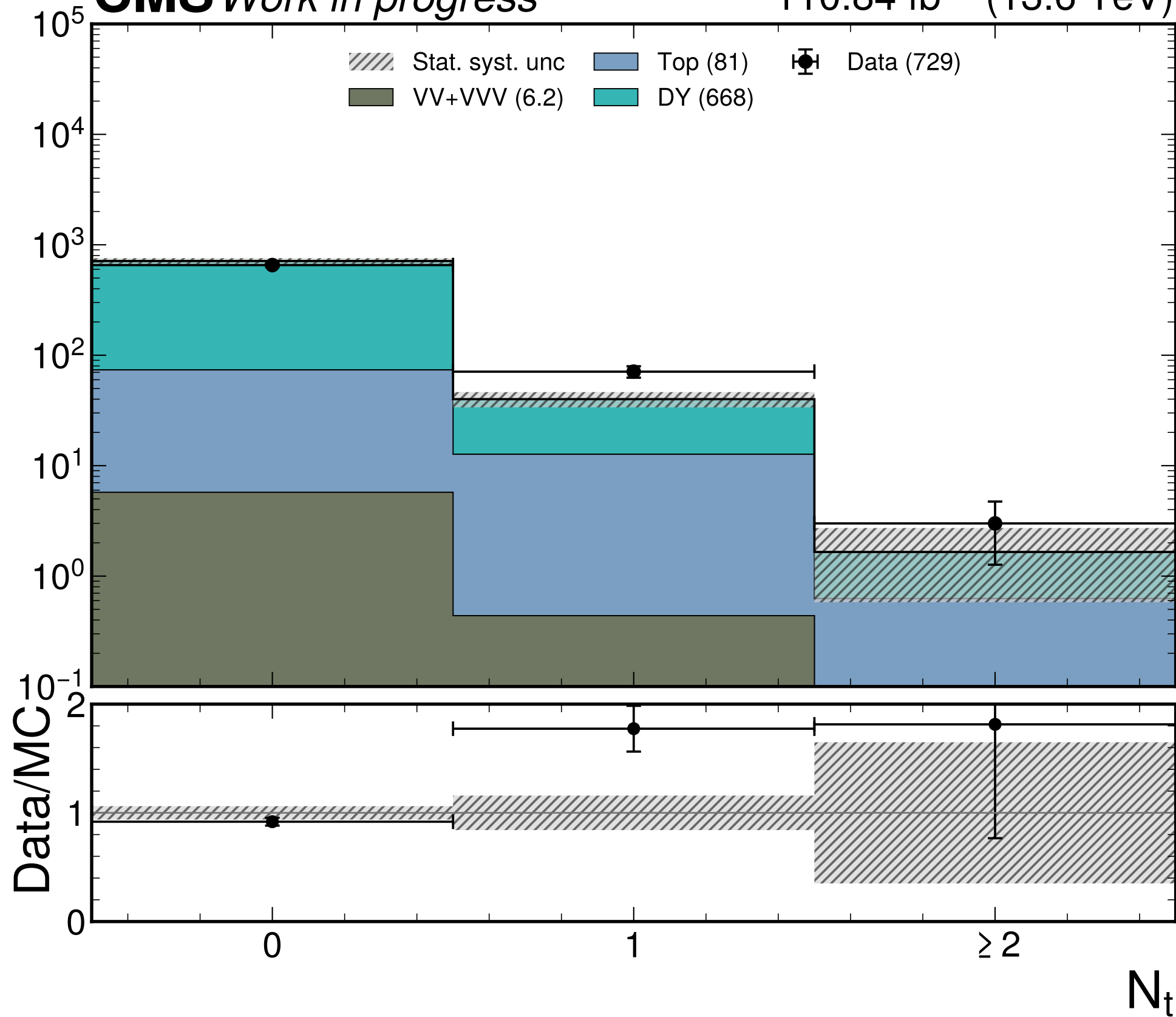
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

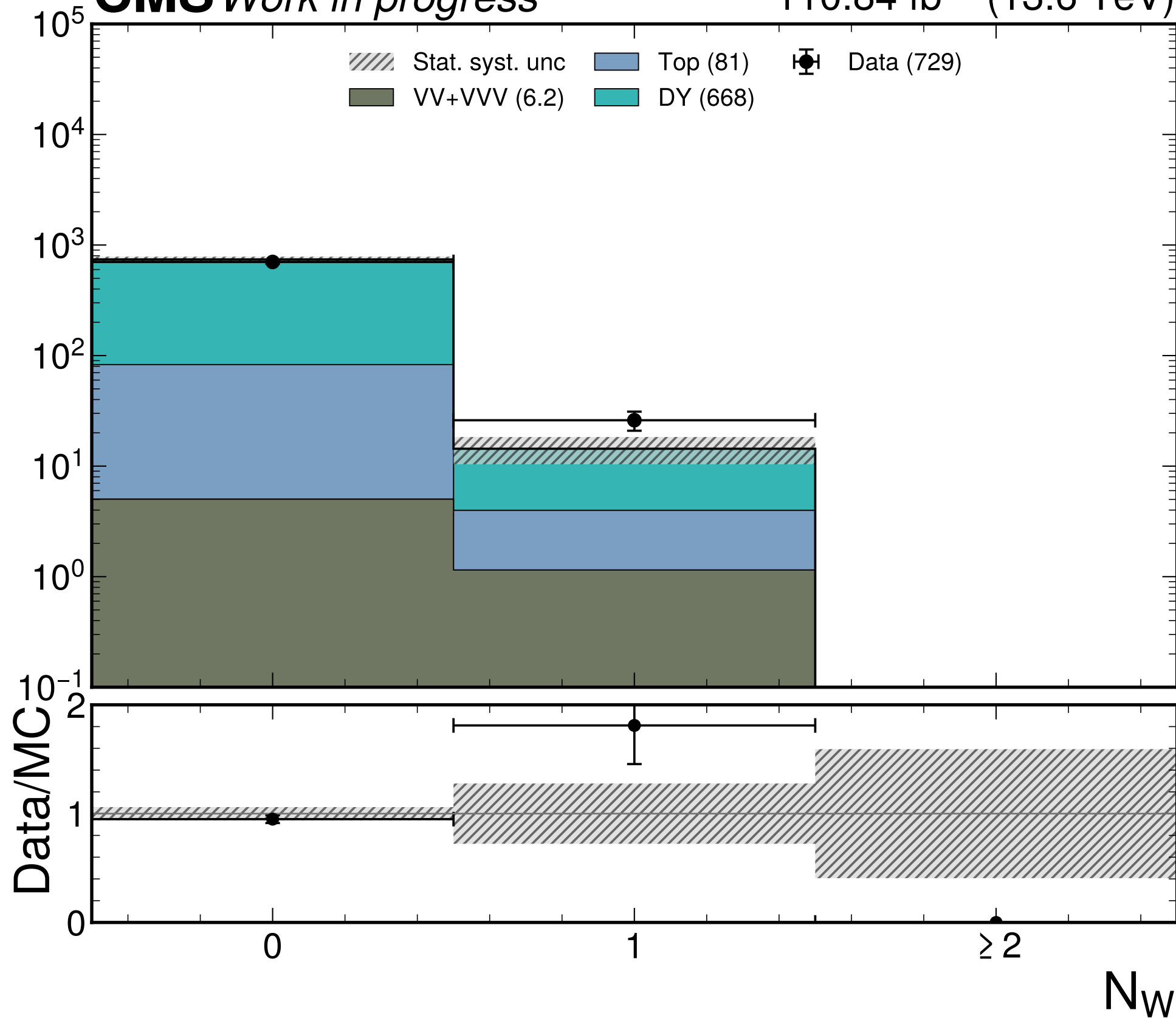
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

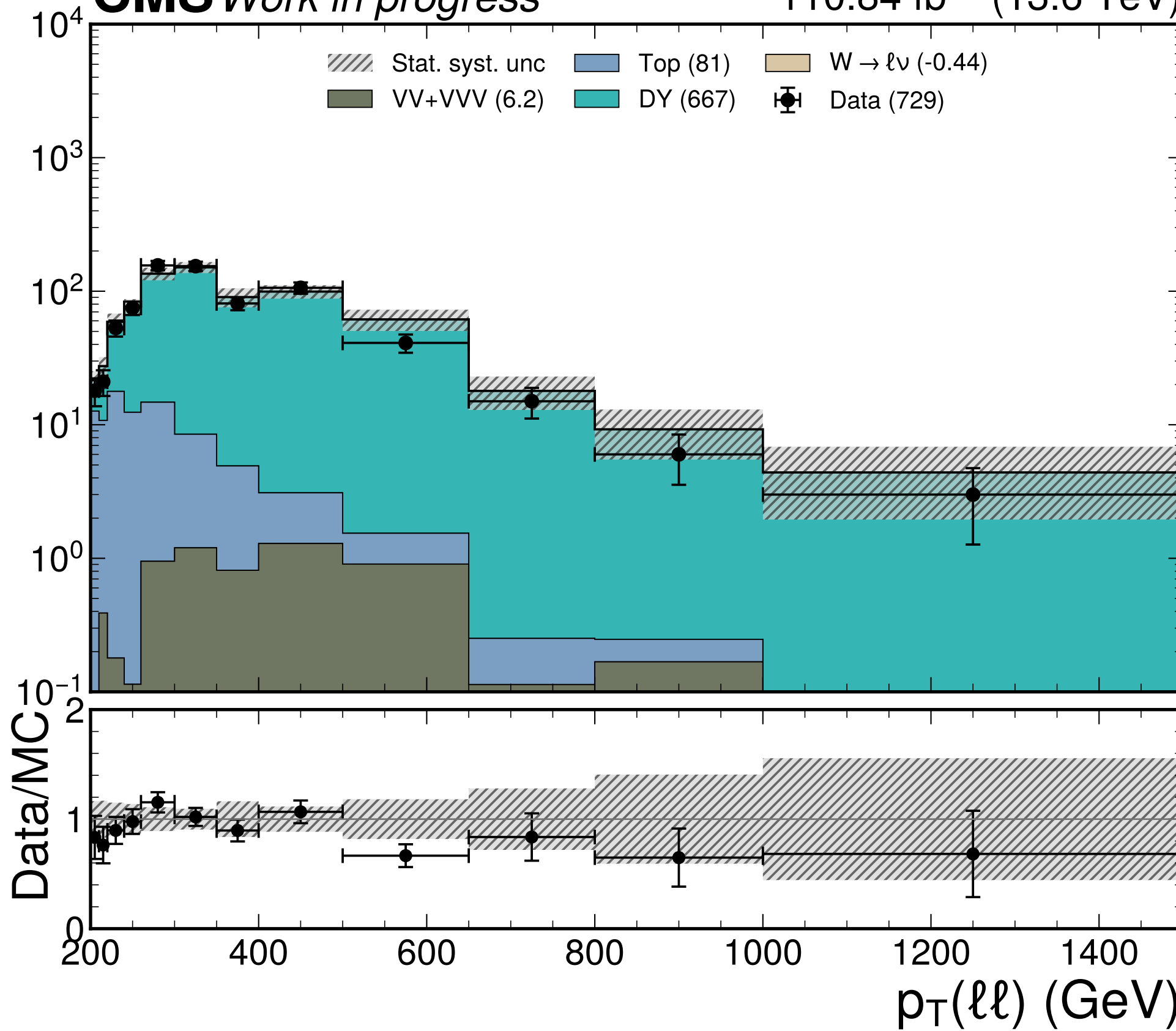
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

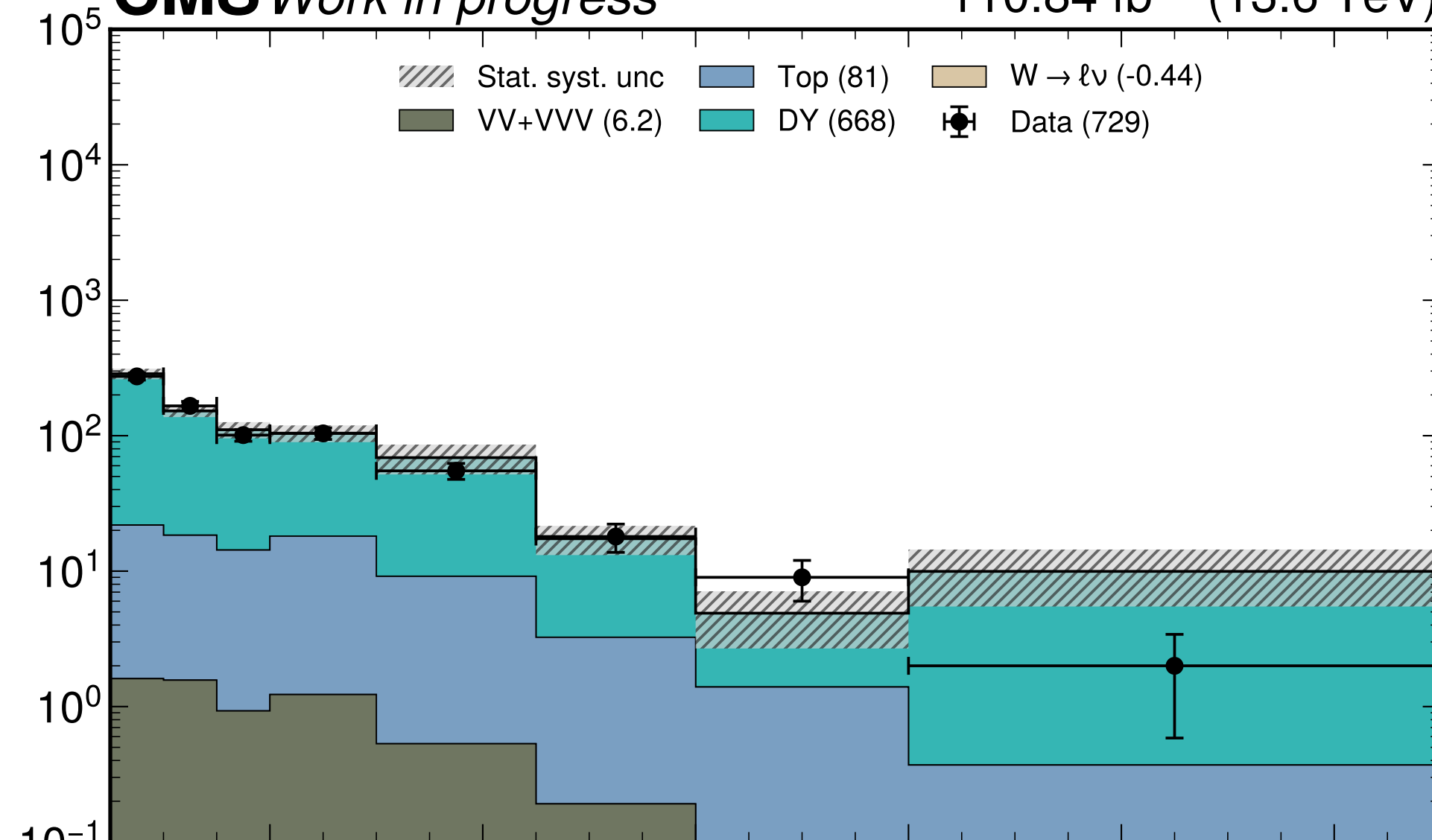
Events



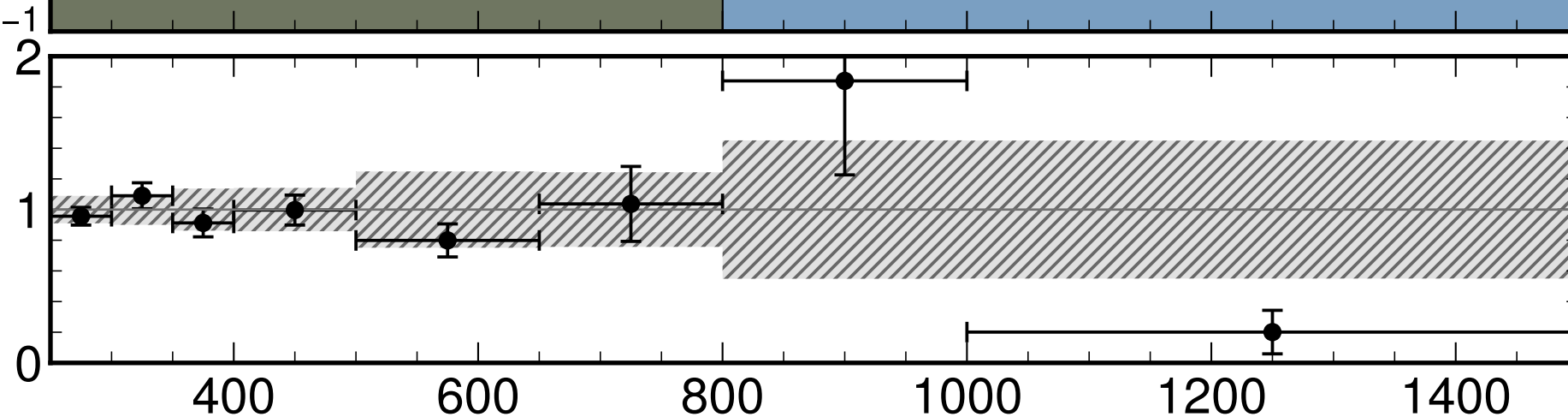
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



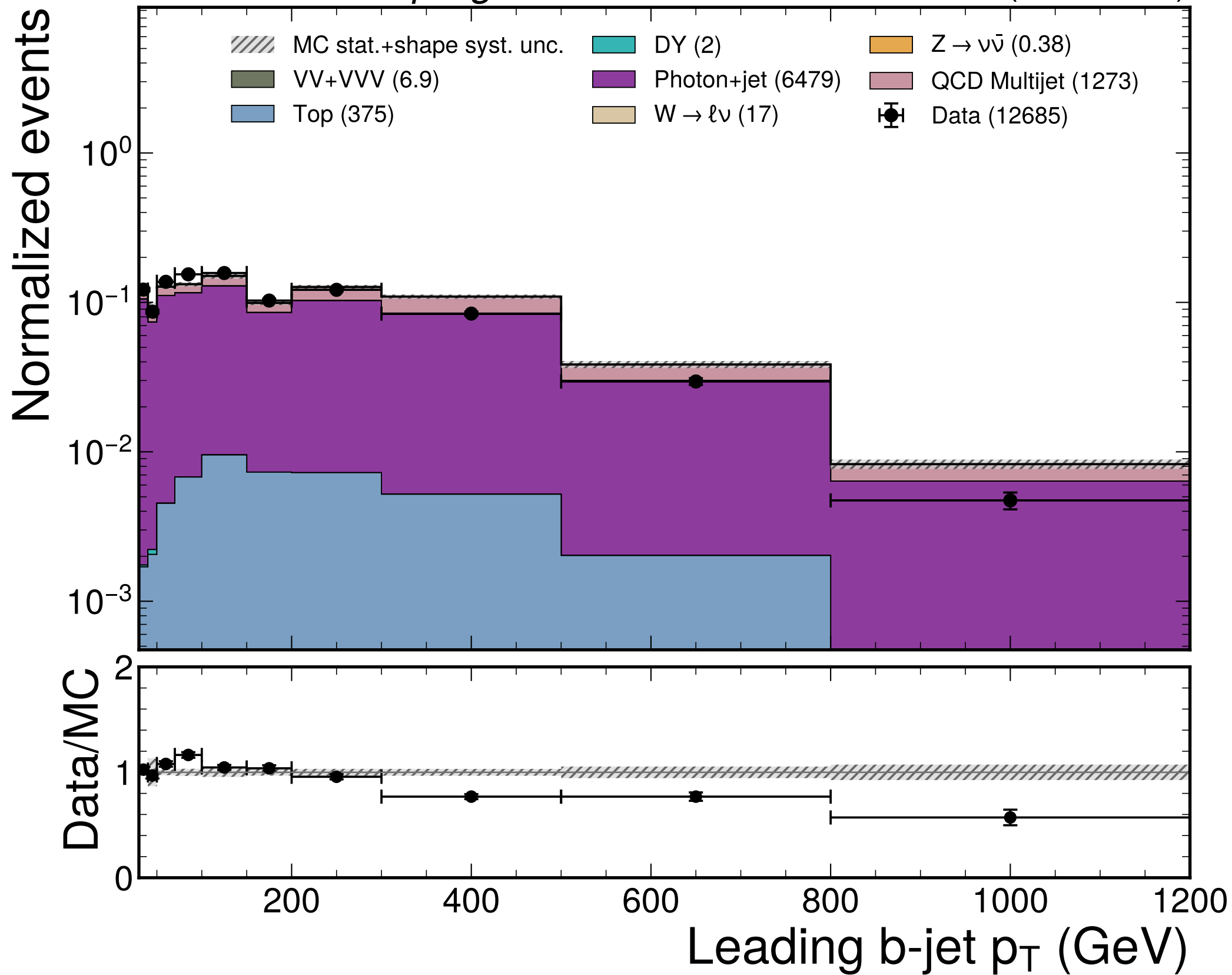
Data/MC



U_T (GeV)

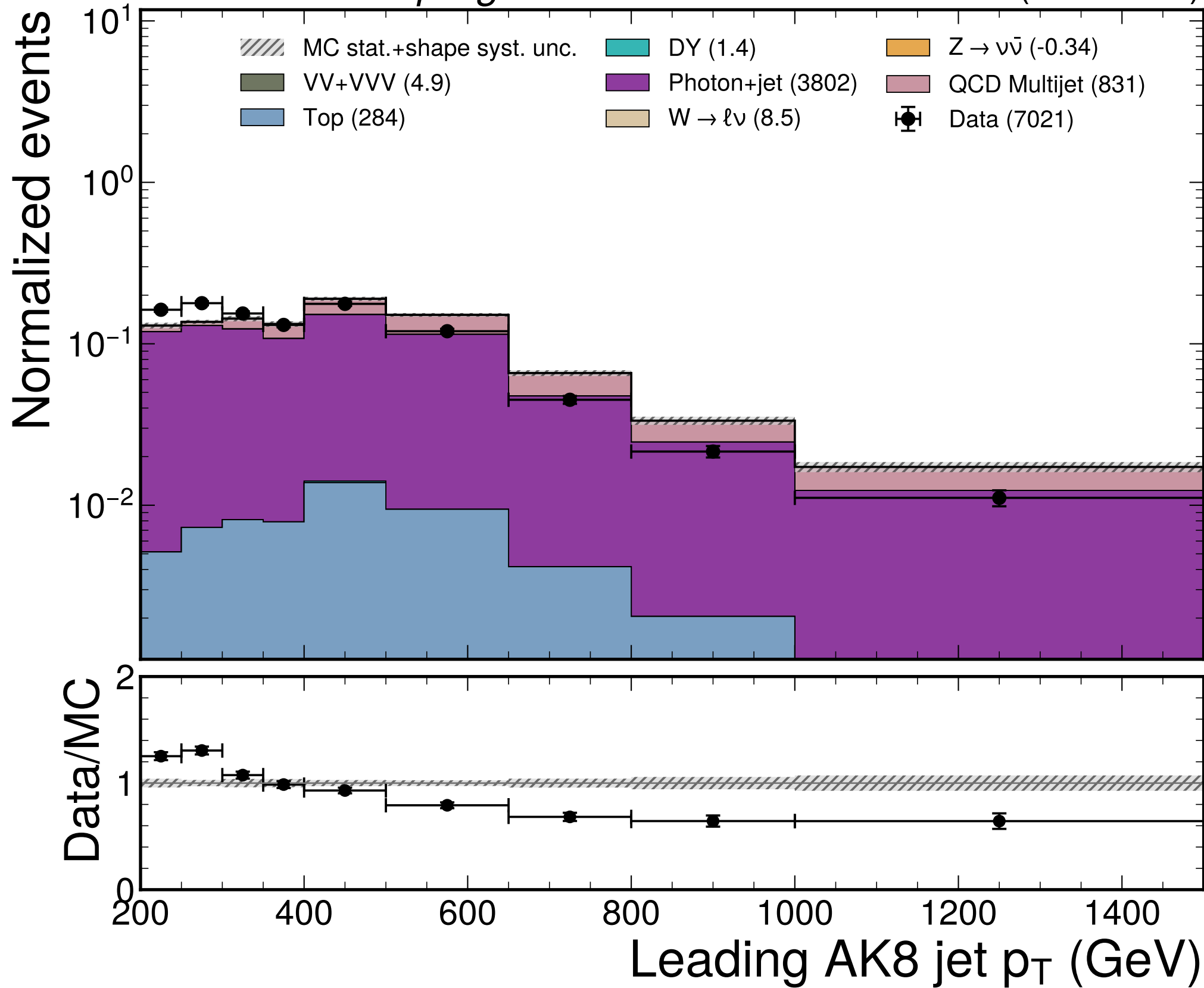
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



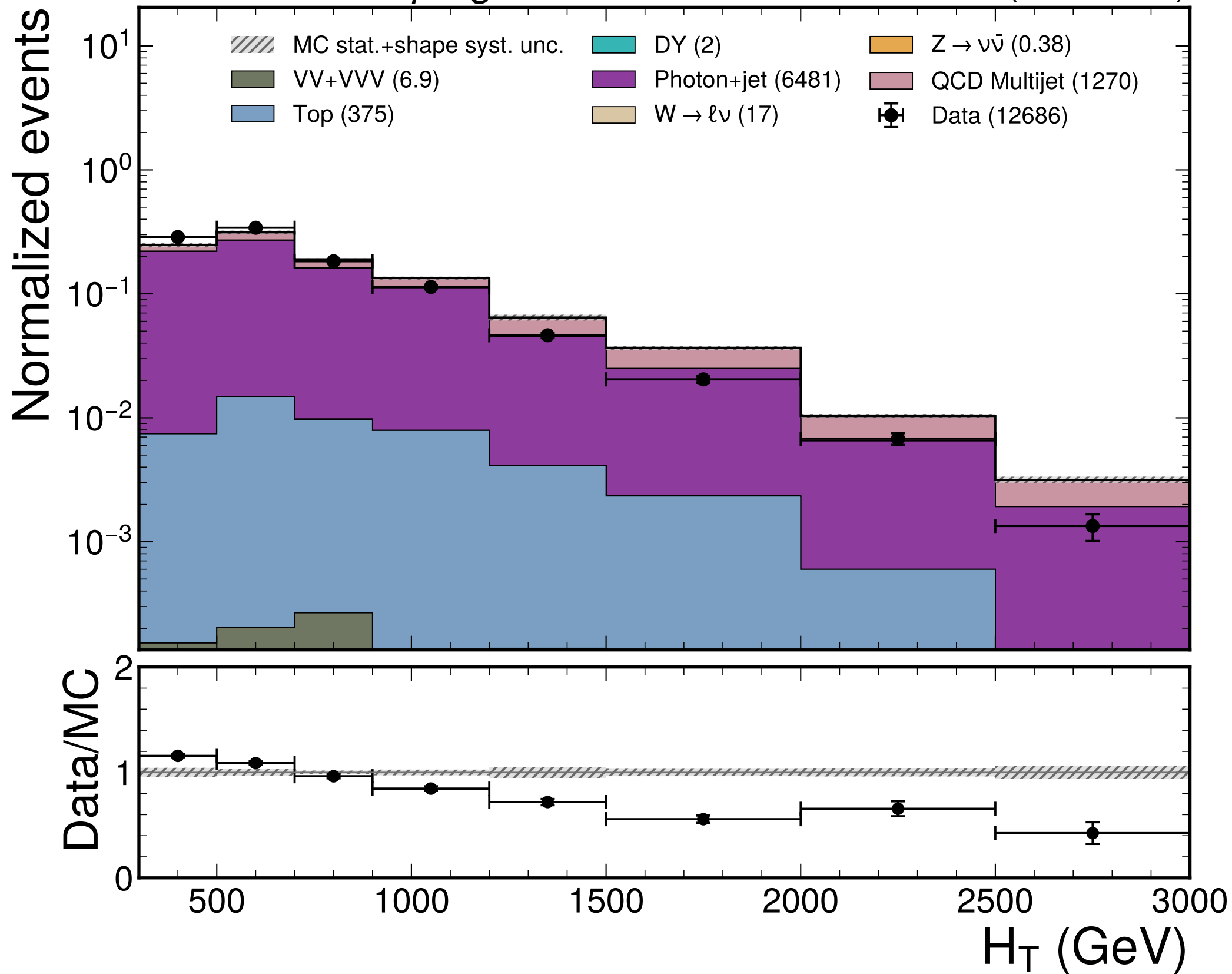
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



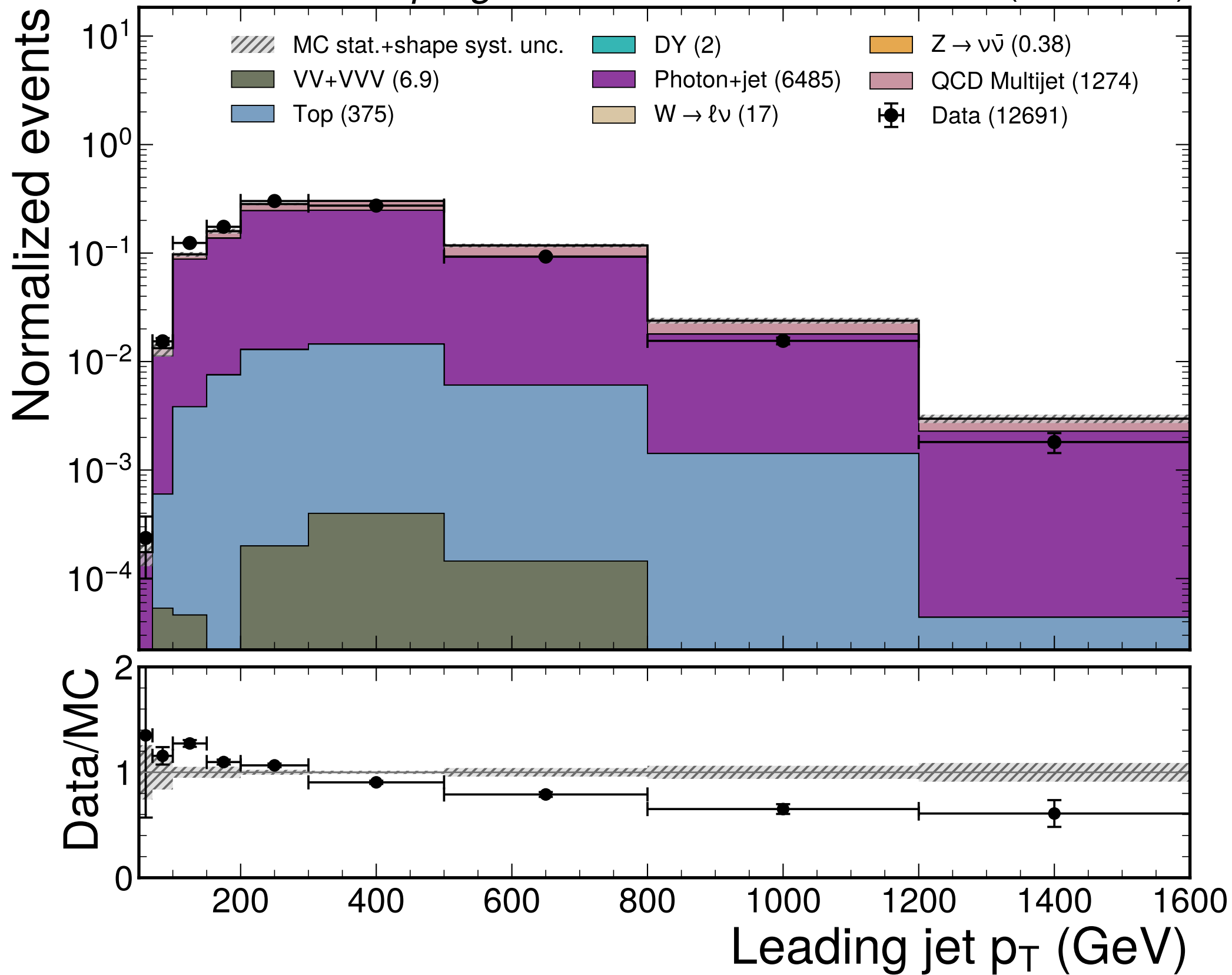
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



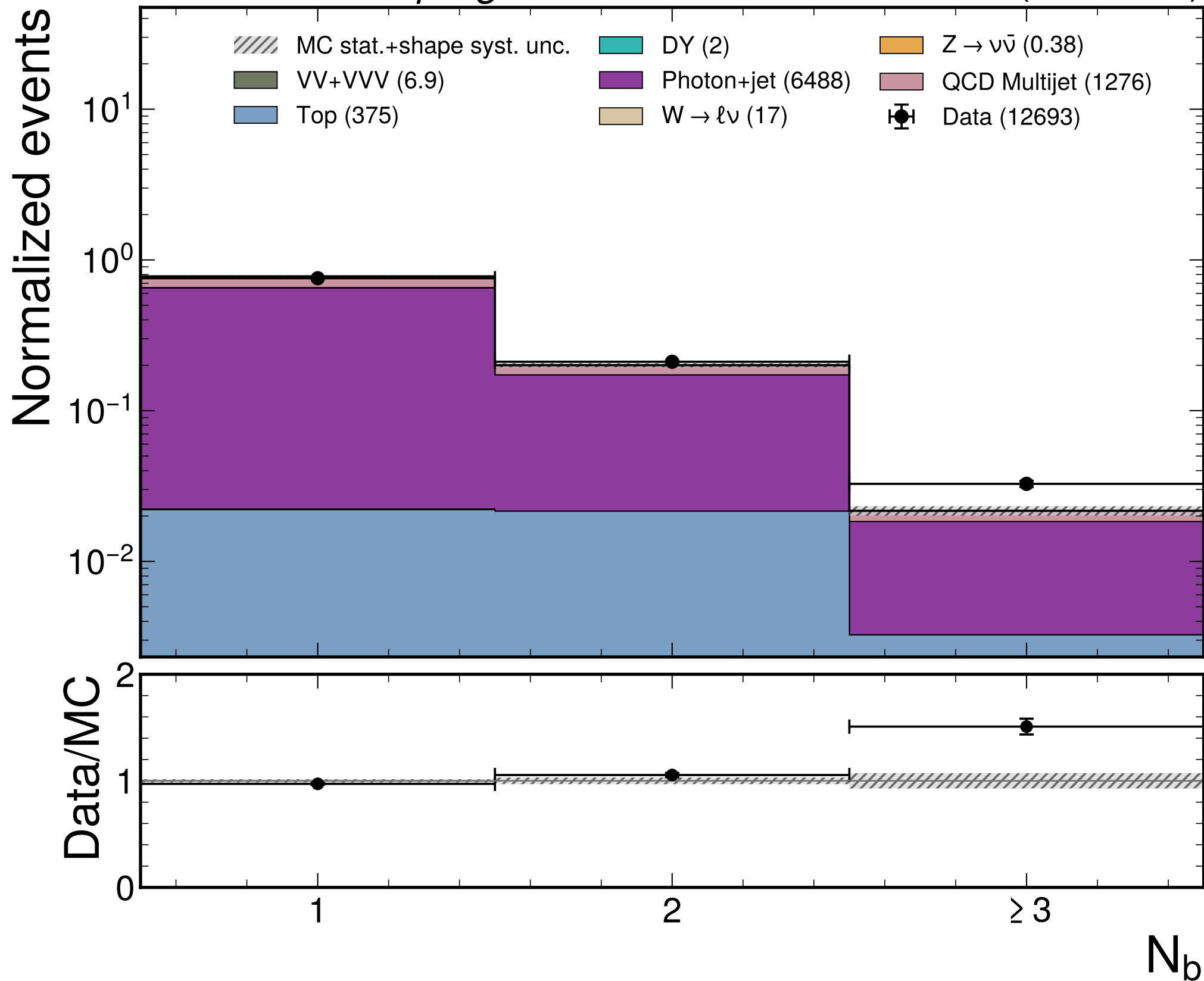
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



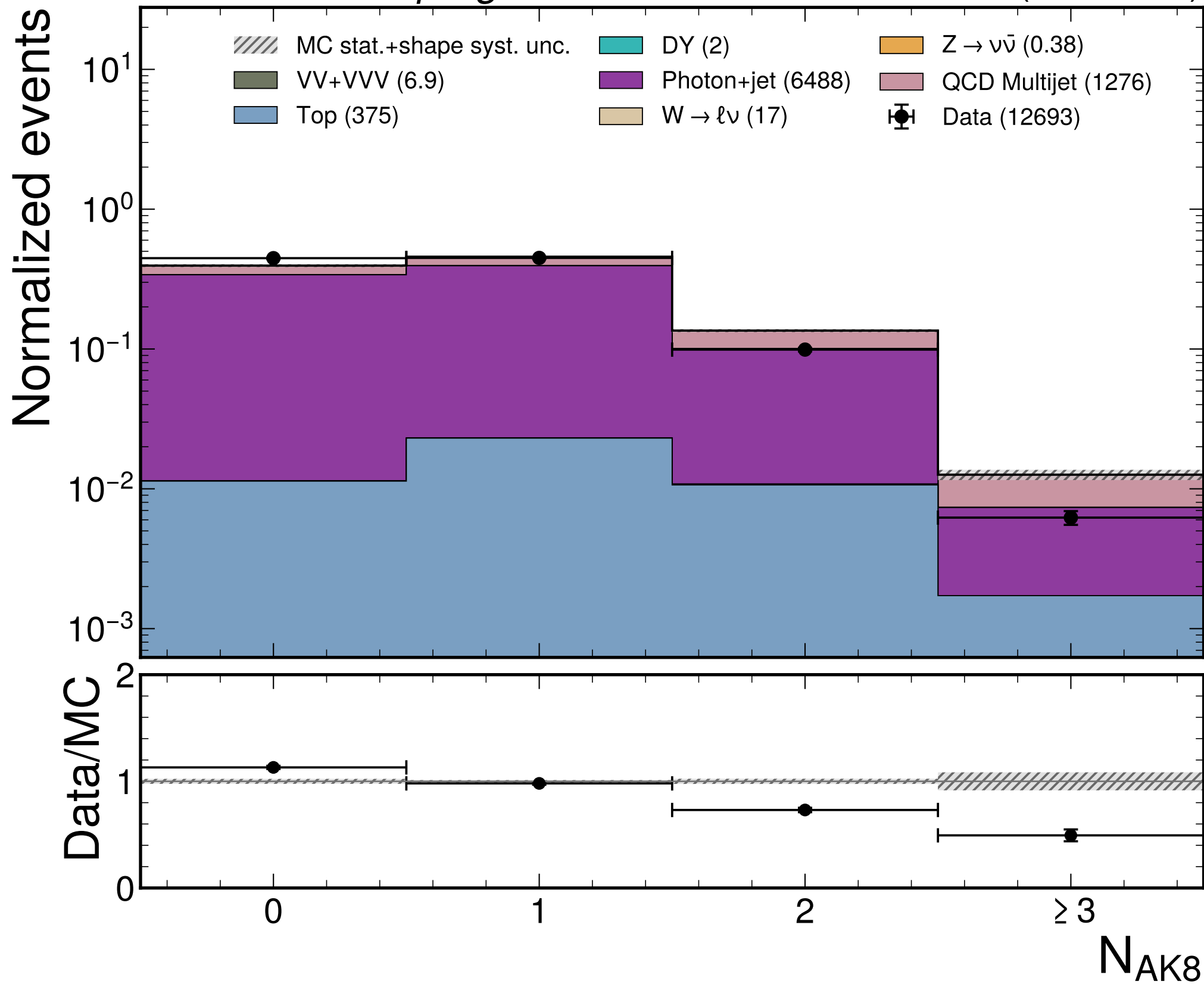
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



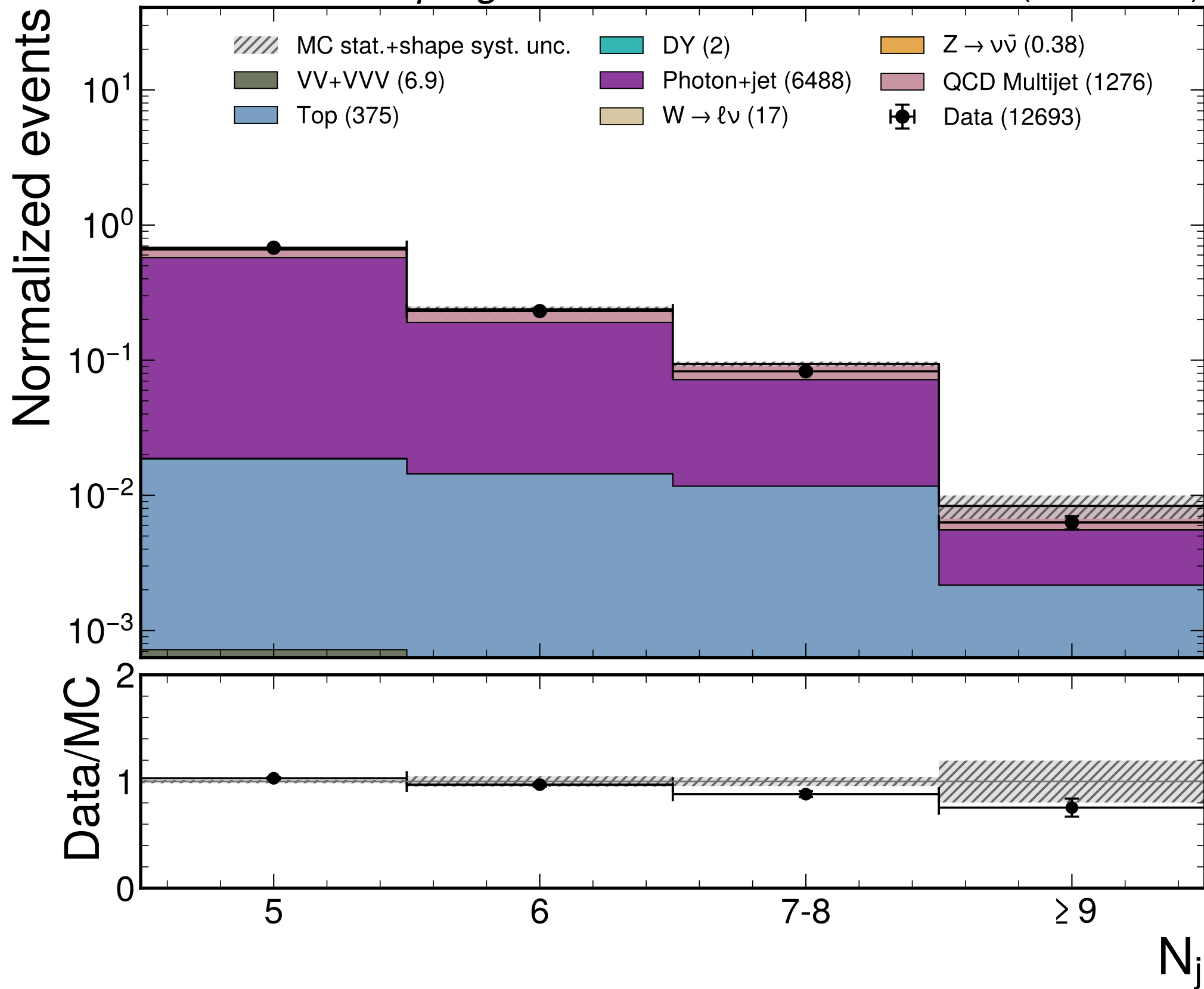
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



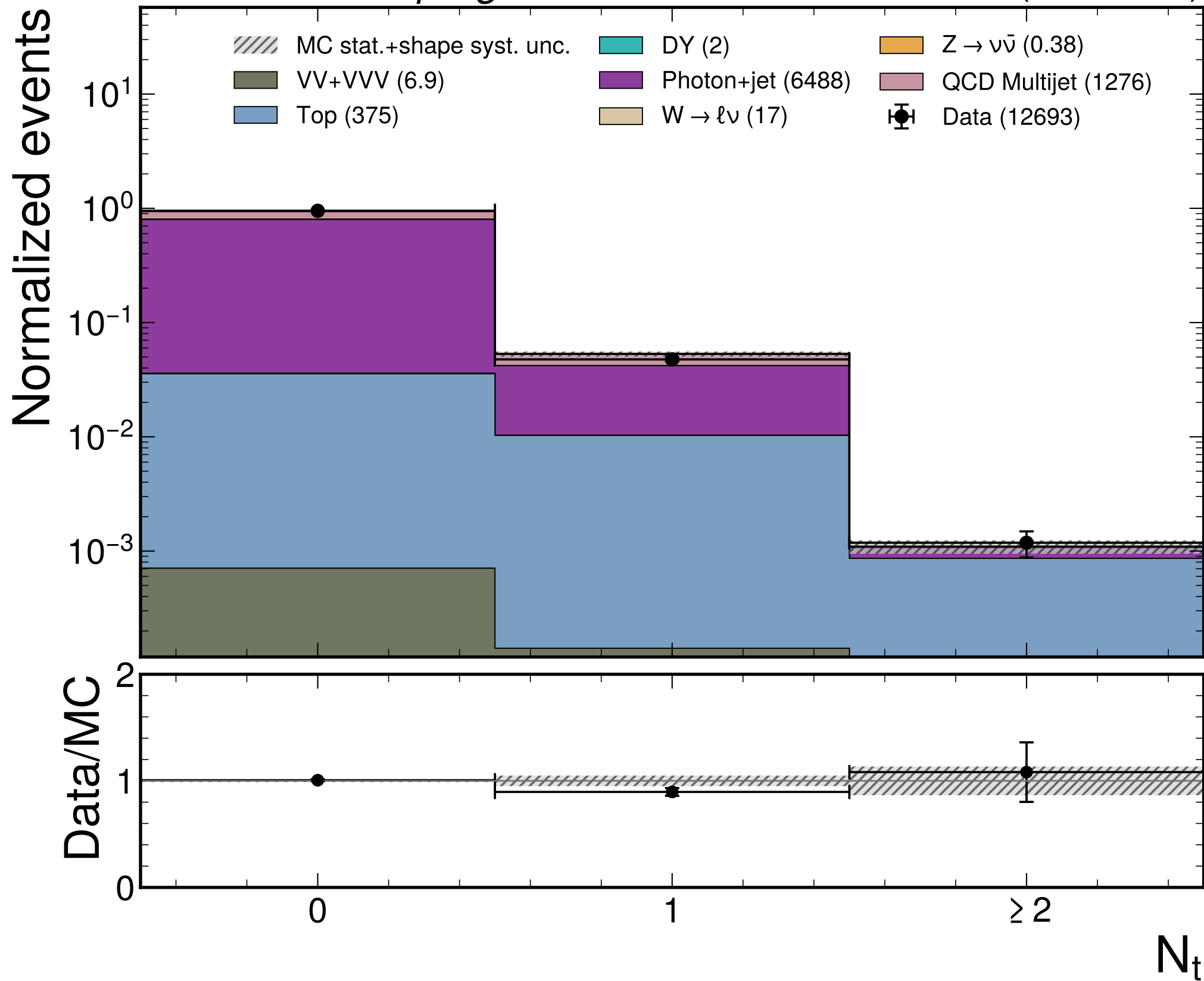
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



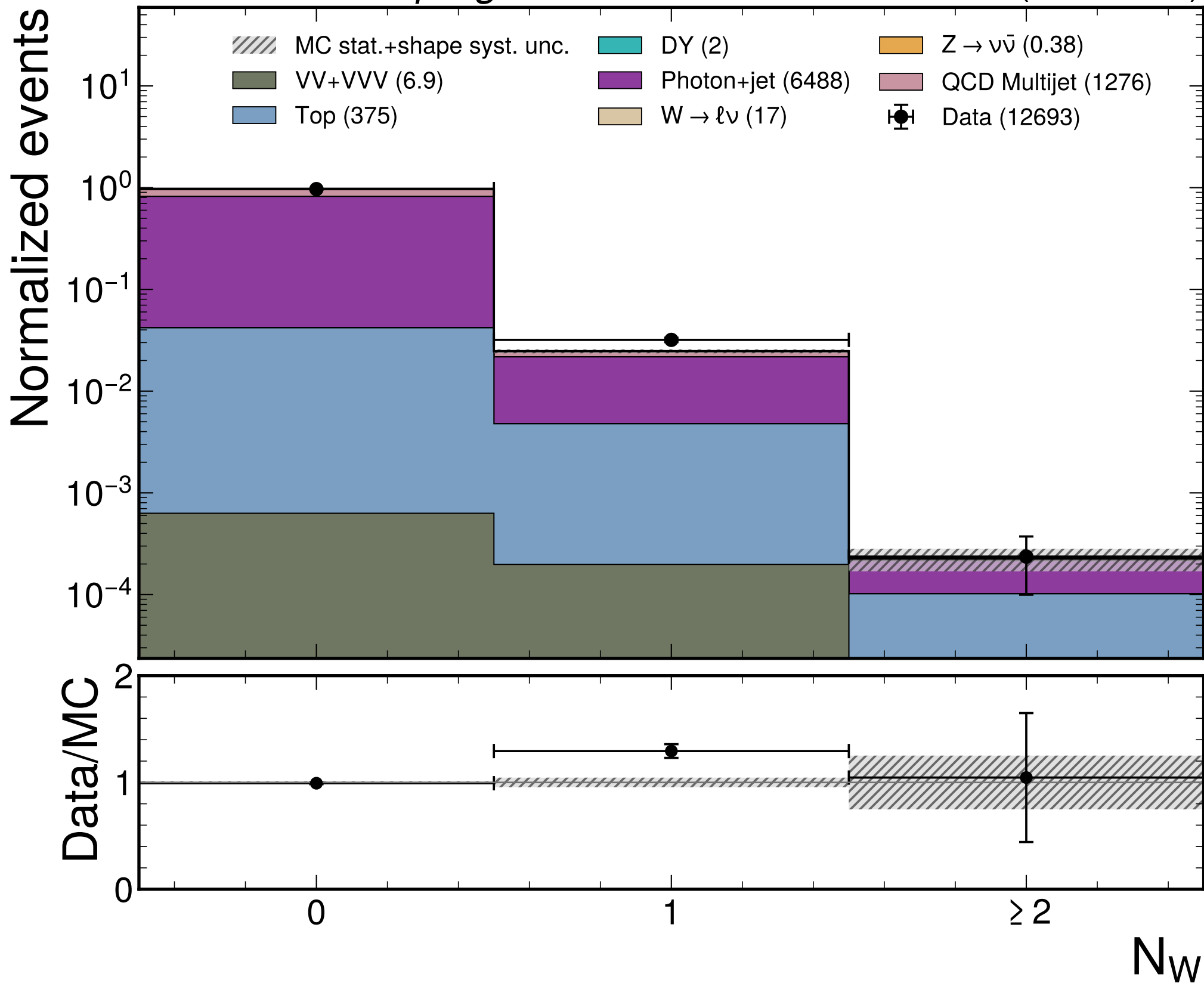
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



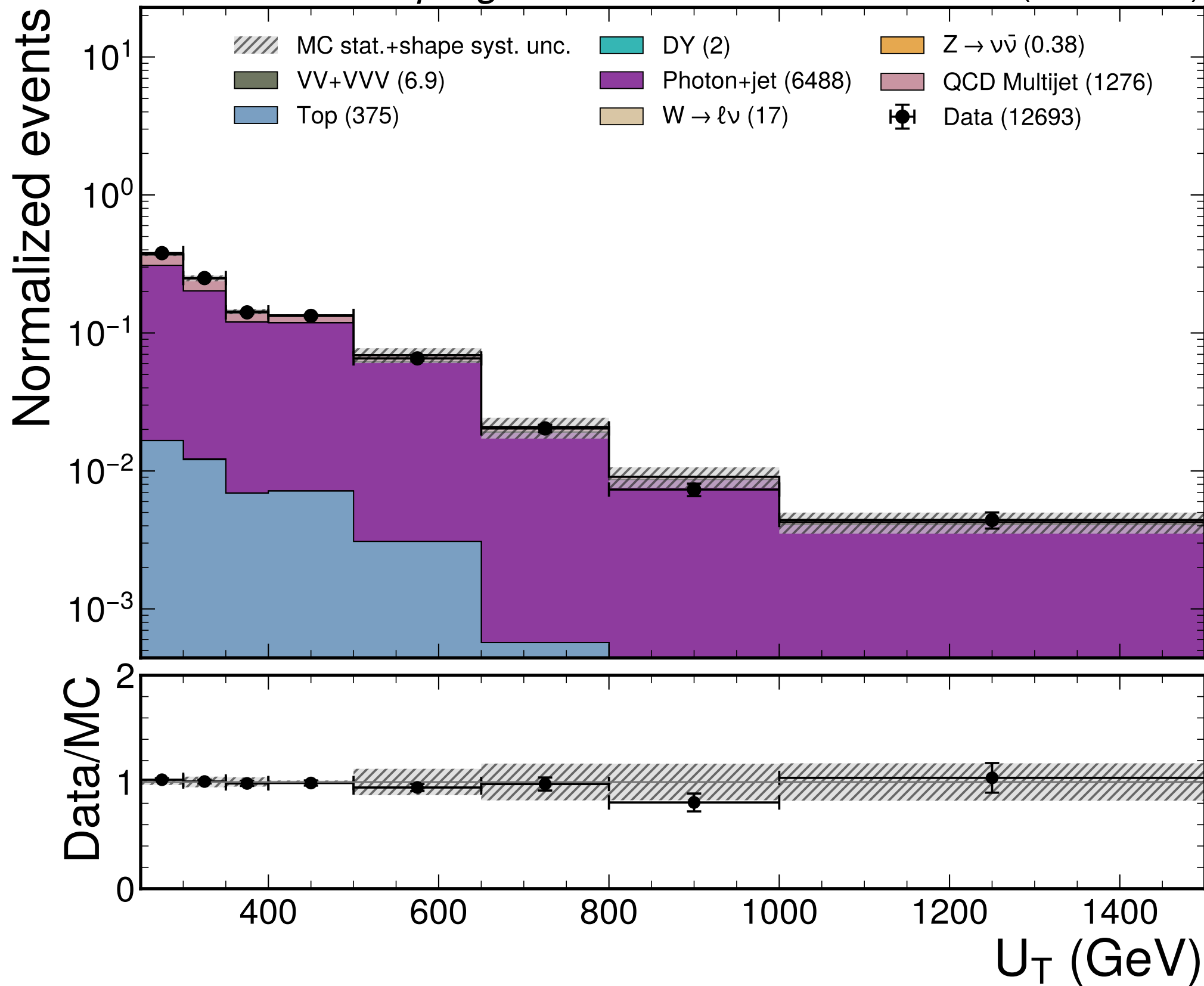
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

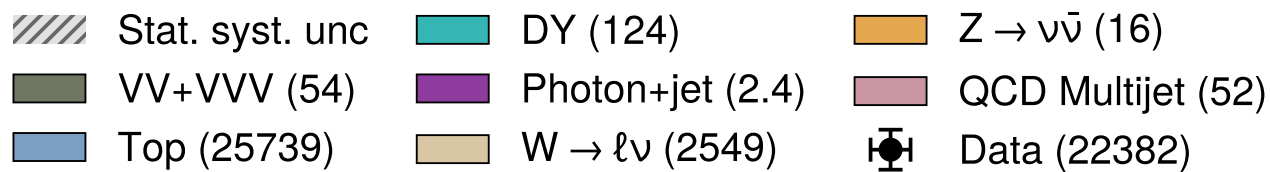
110.84 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻²

2

1

0

200

400

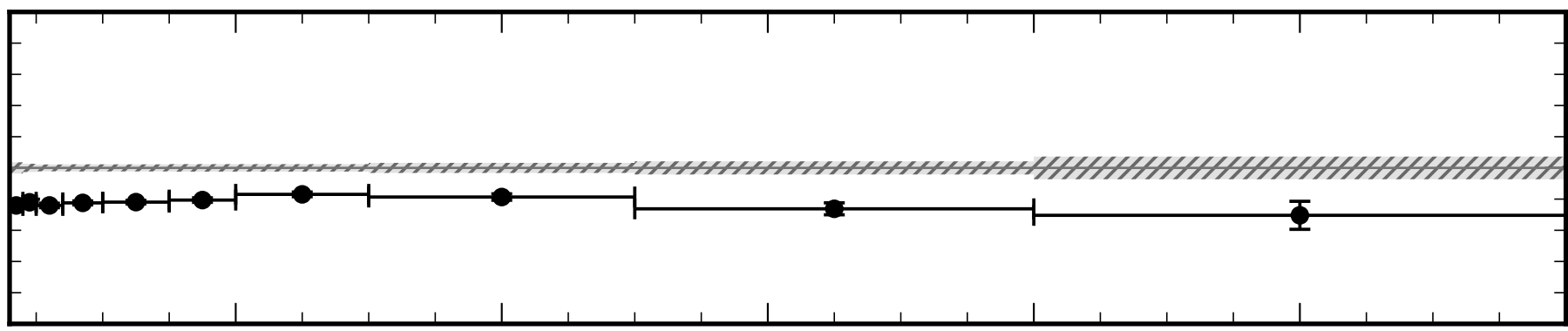
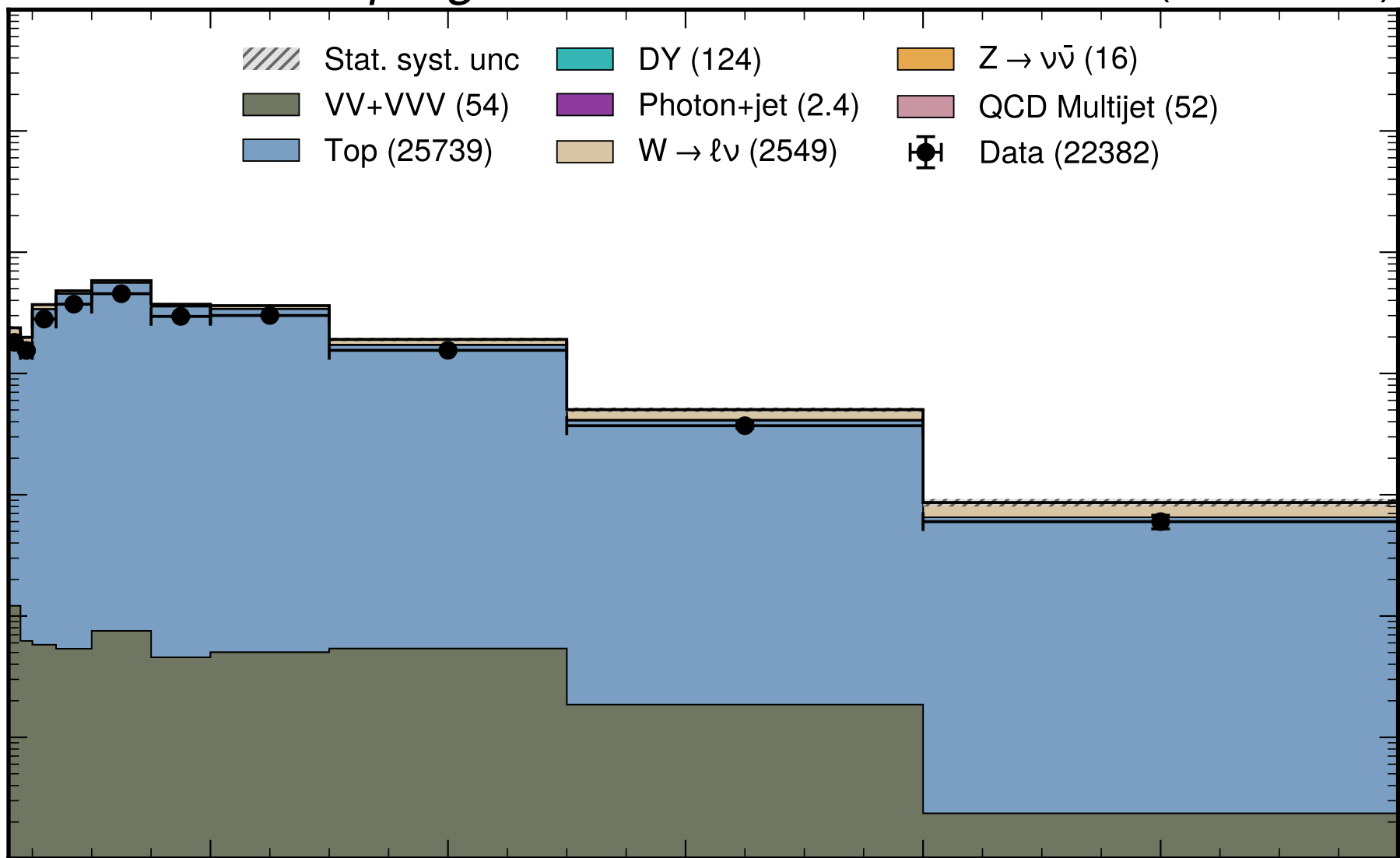
600

800

1000

1200

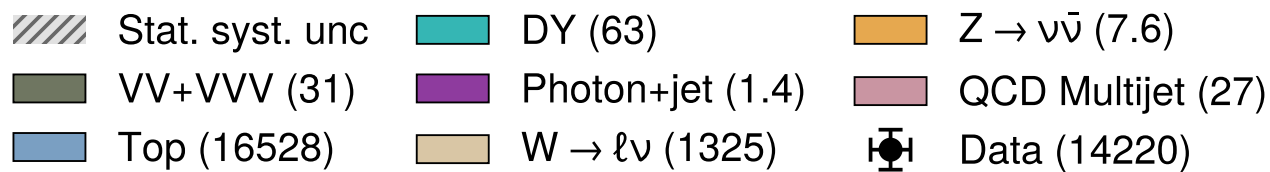
Leading b-jet p_T (GeV)



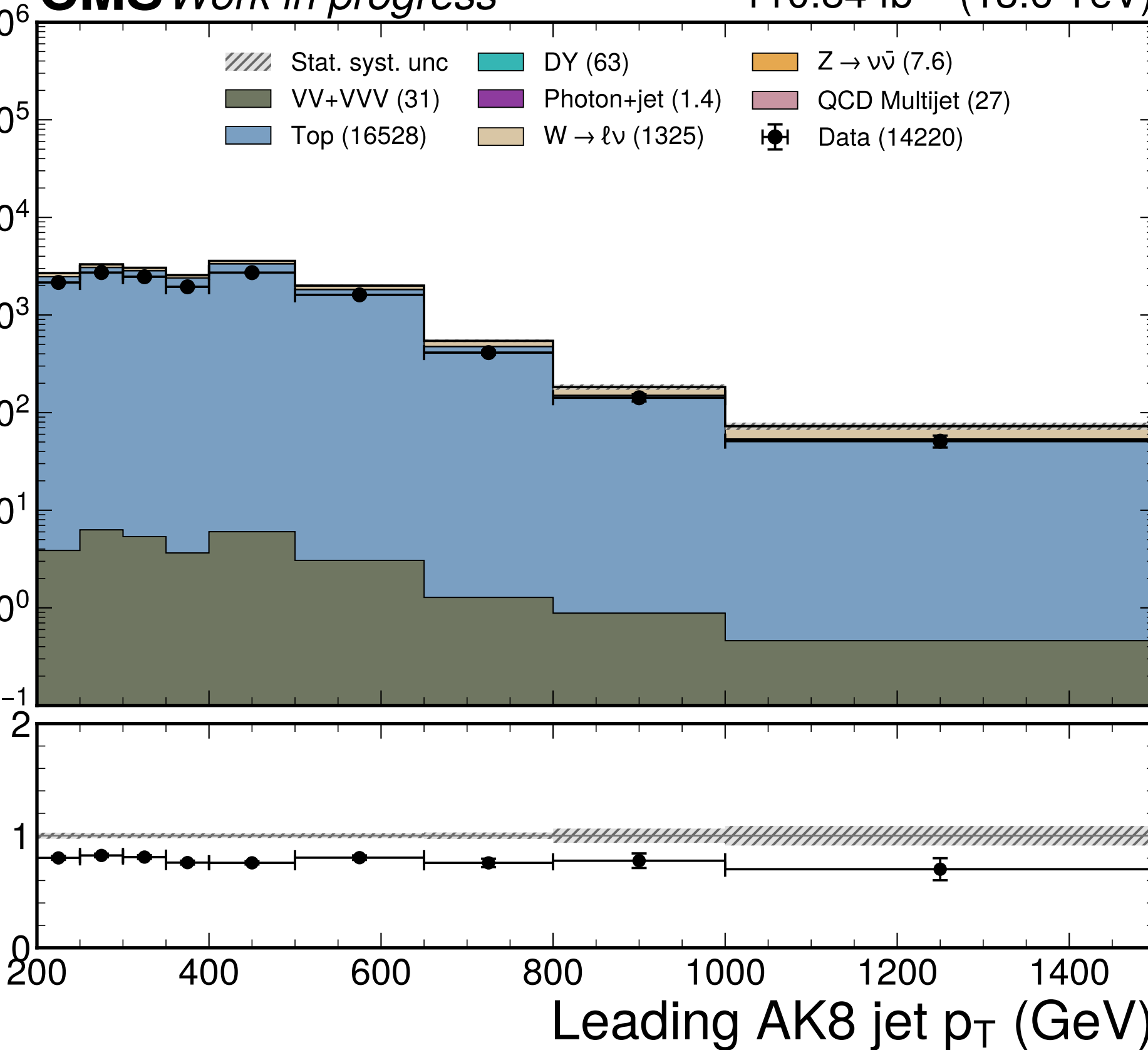
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events

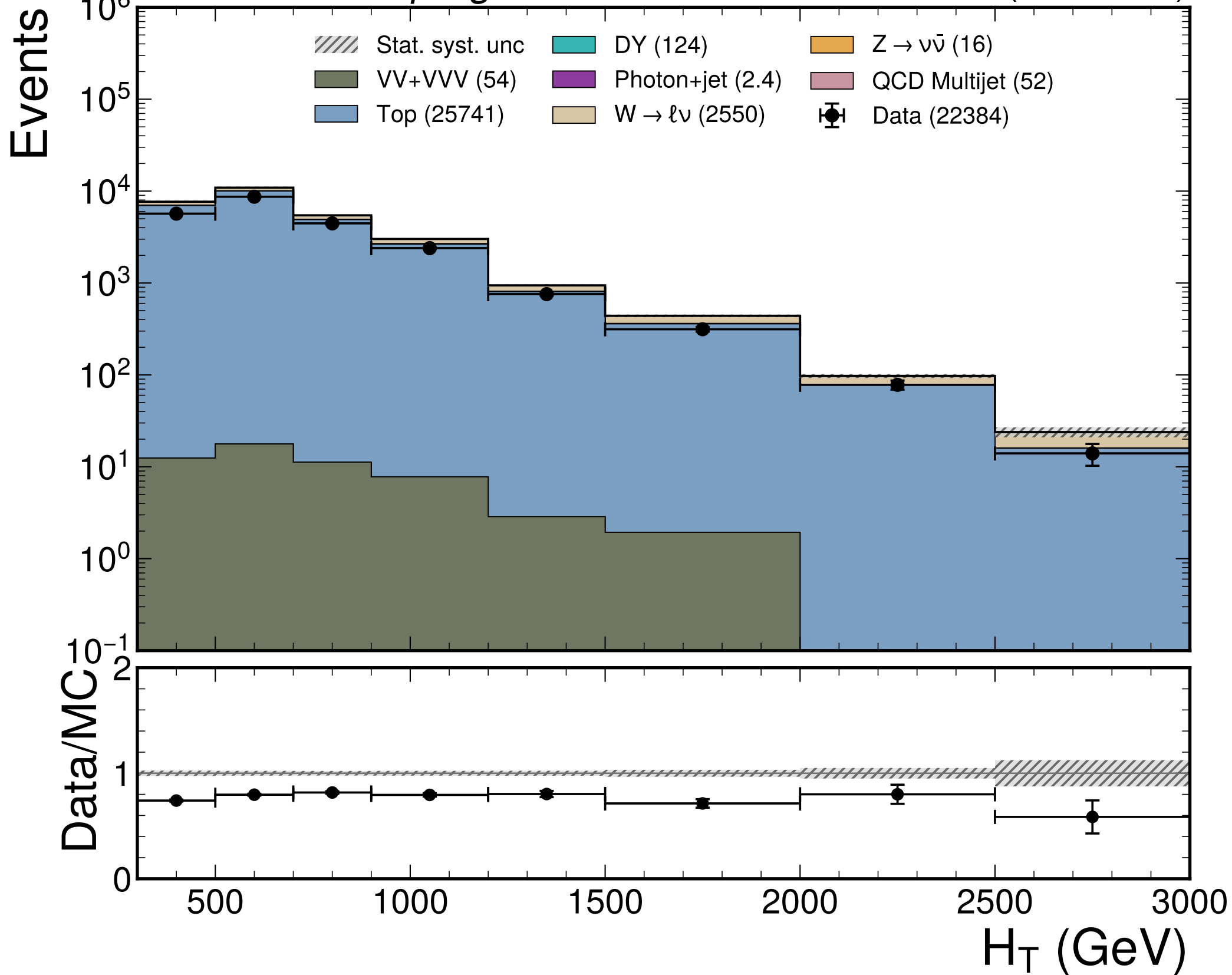


Data/MC



CMS *Work in progress*

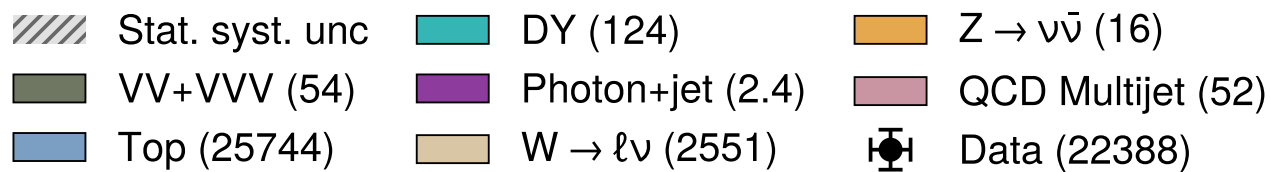
110.84 fb⁻¹ (13.6 TeV)



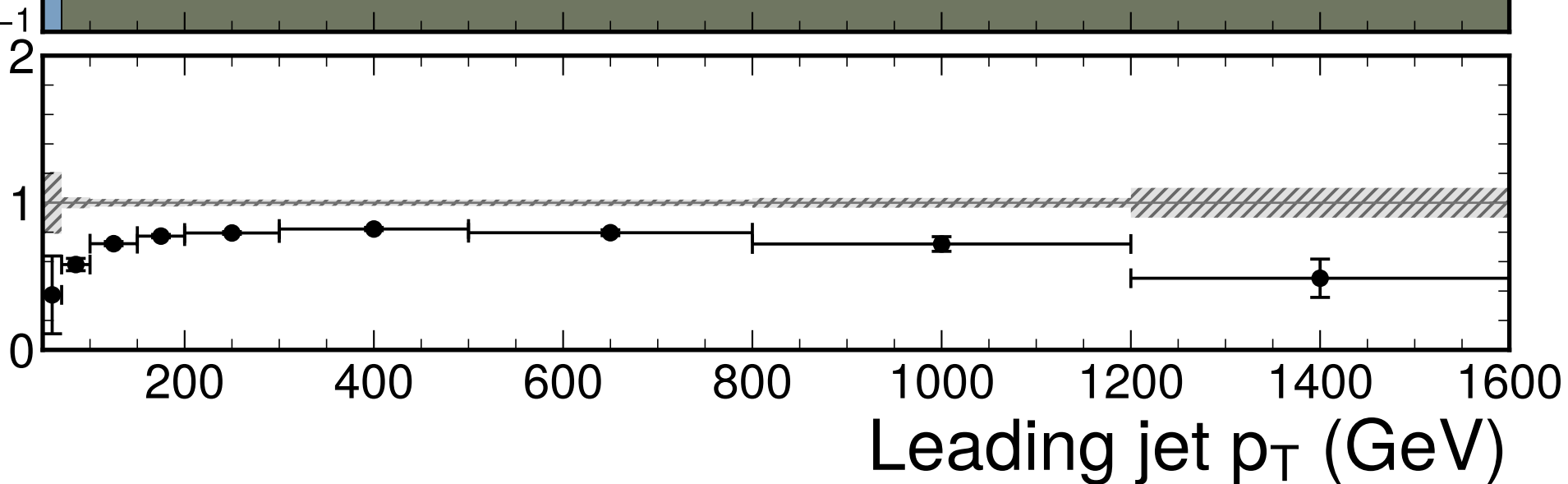
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



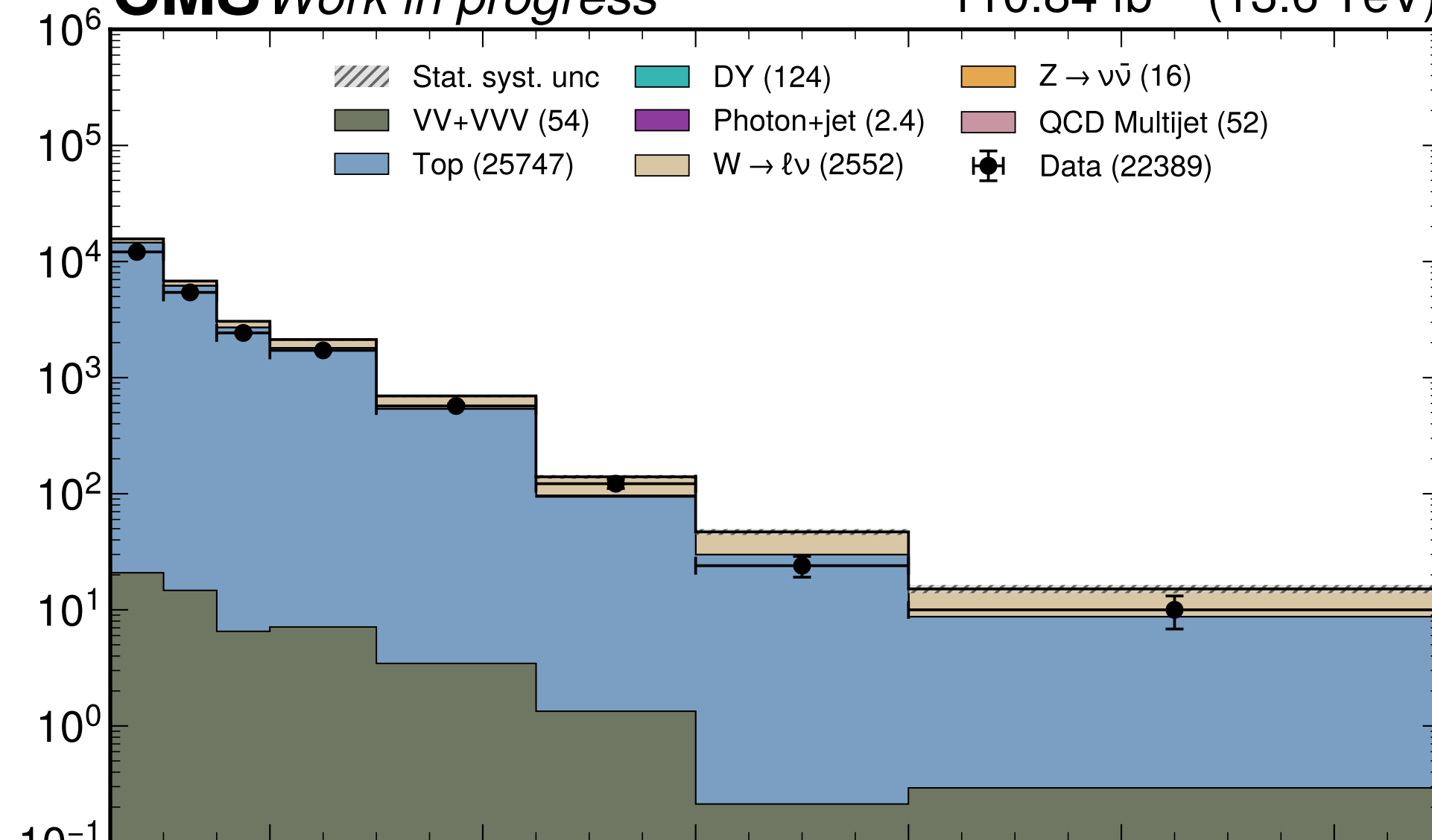
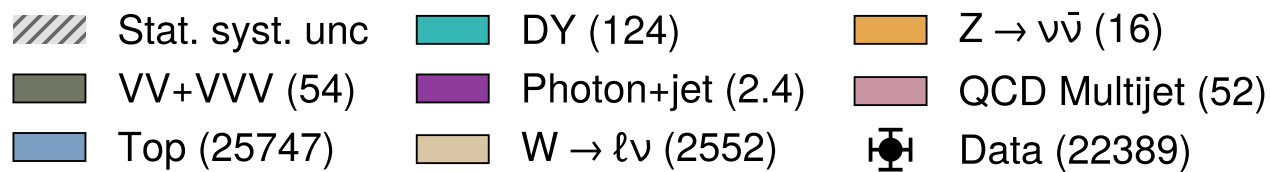
Data/MC



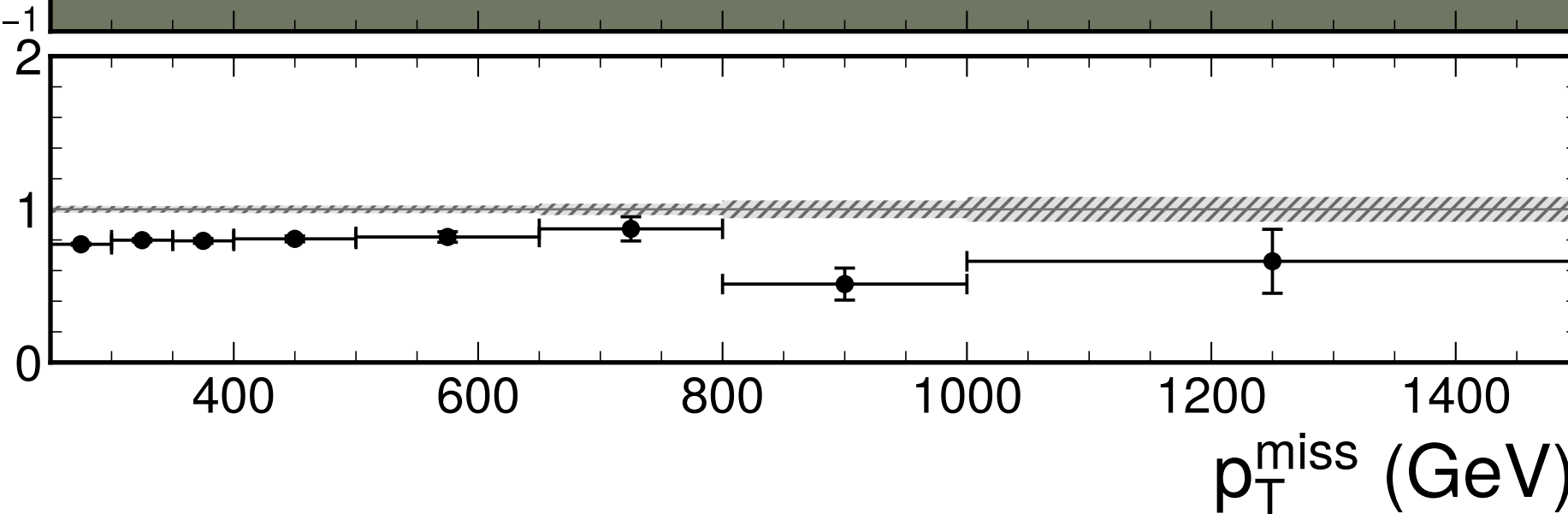
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



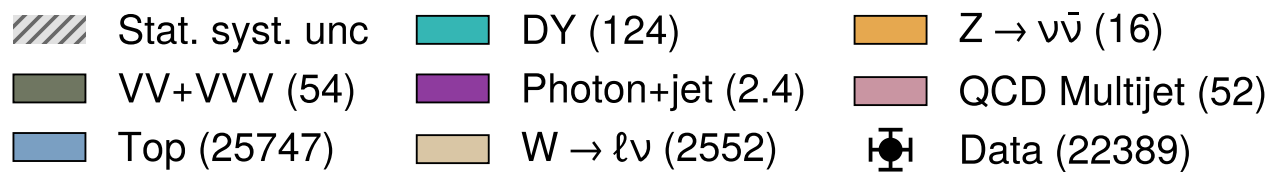
Data/MC



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

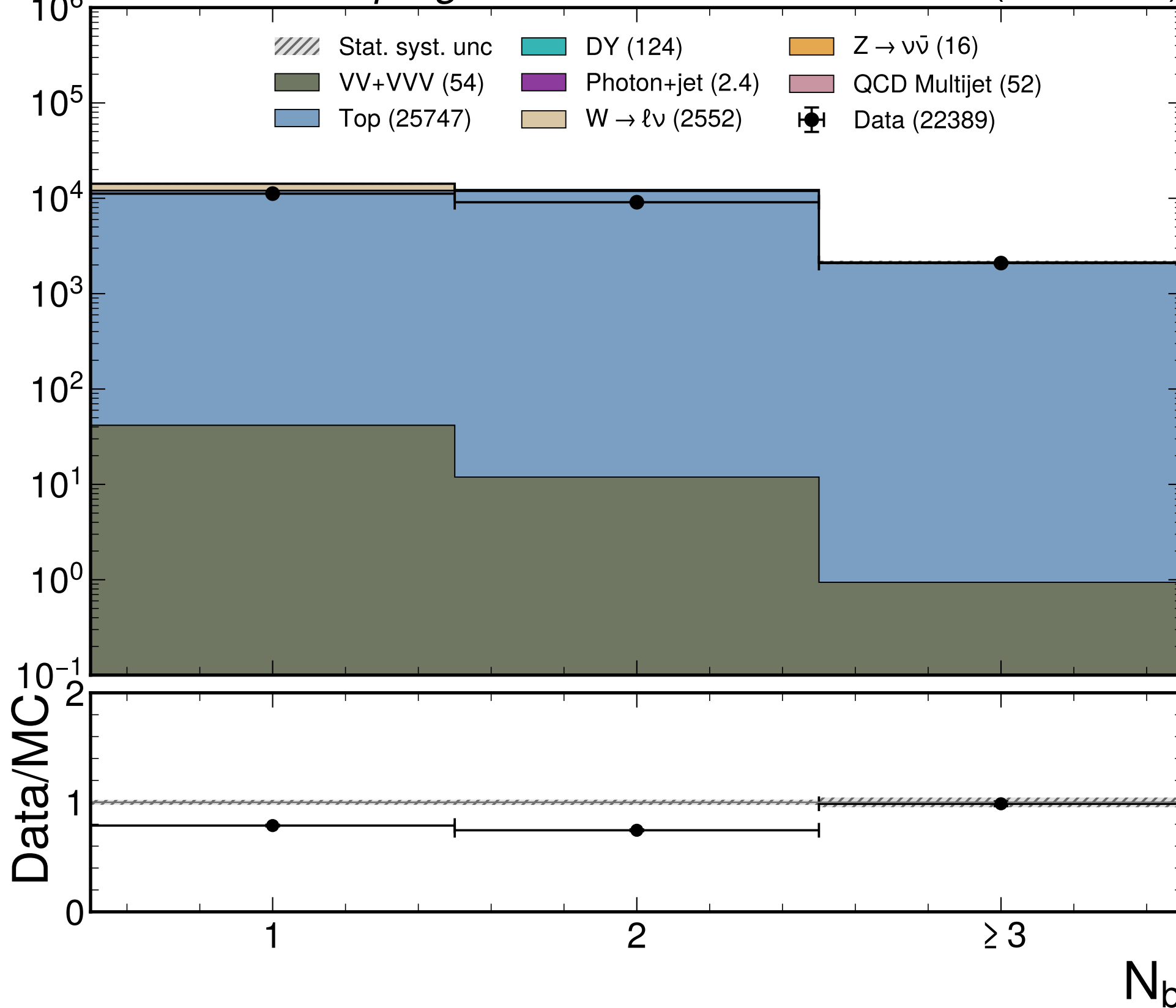
10⁻¹
2
1
0

1

2

≥ 3

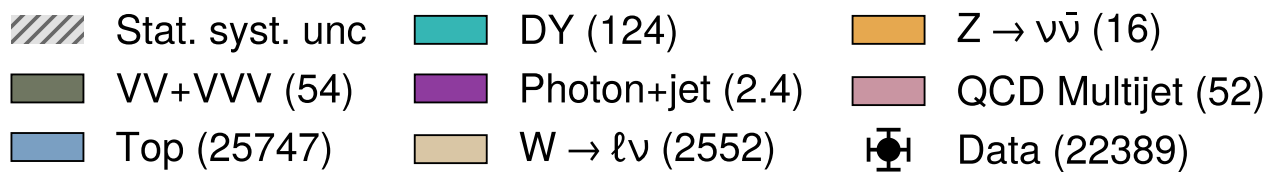
N_b



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

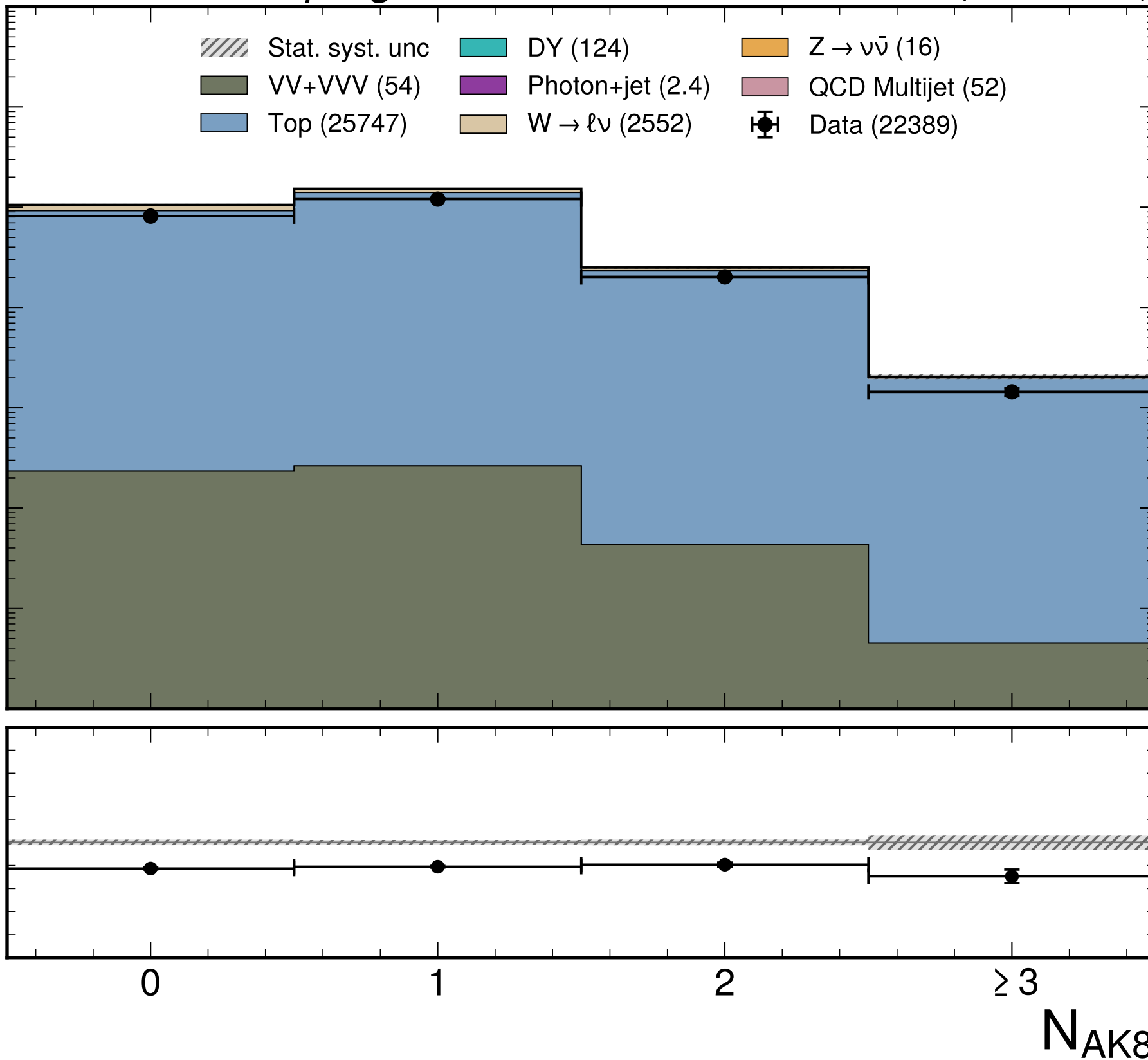
0

1

2

≥ 3

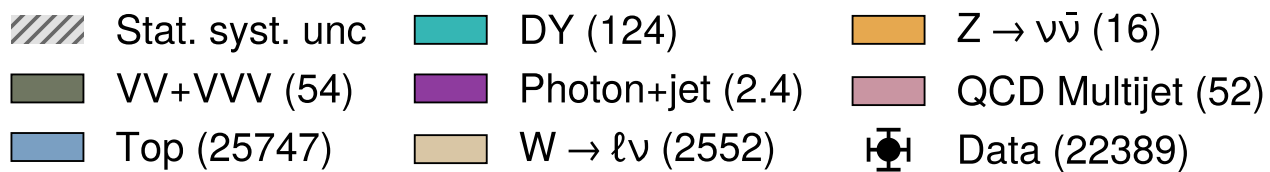
N_{AK8}



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

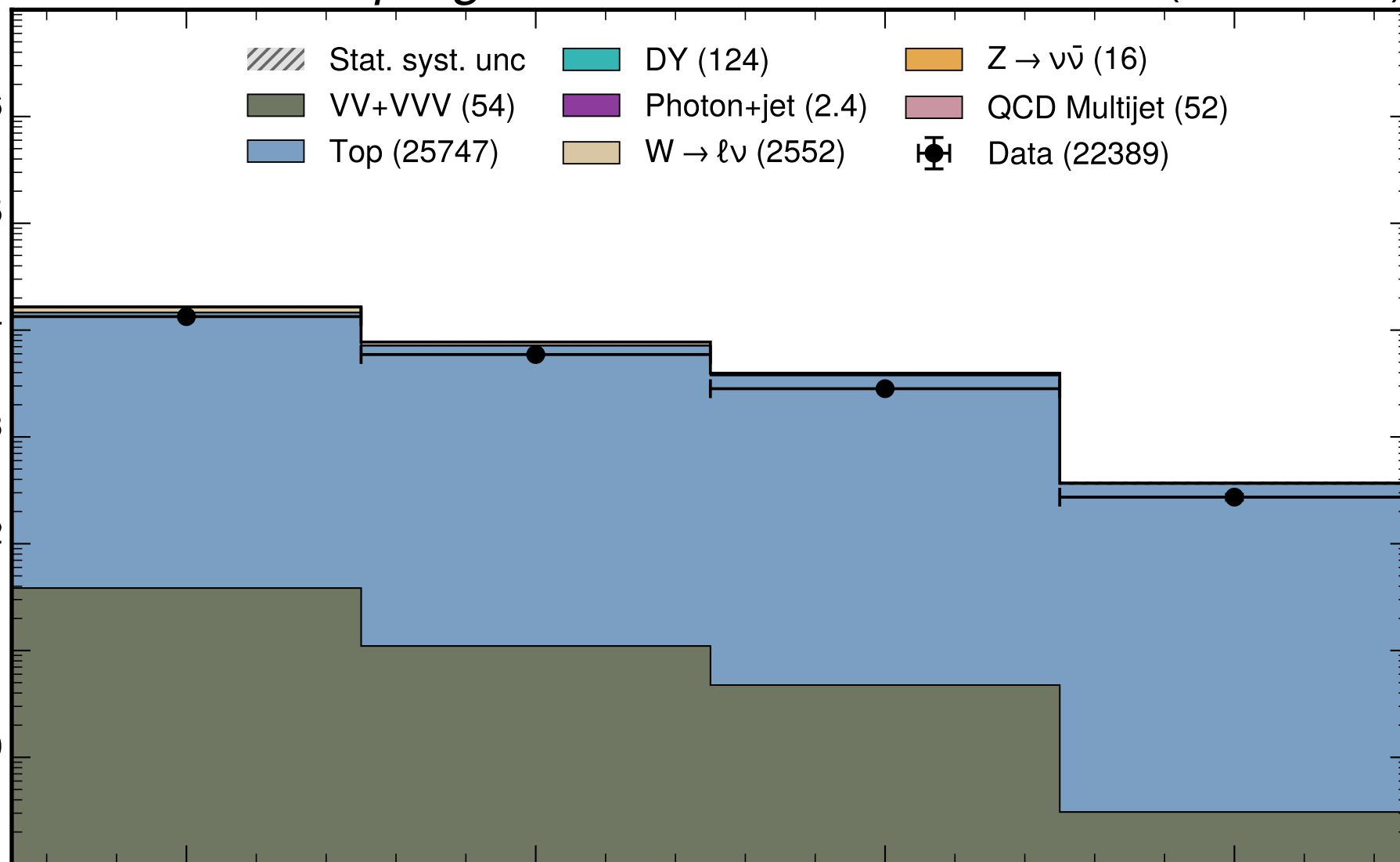
5

6

7-8

≥ 9

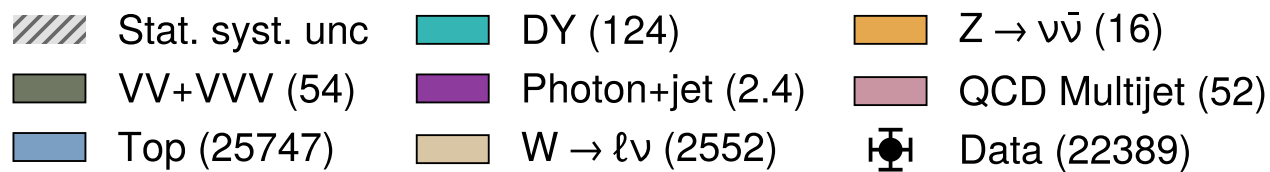
N_j



CMS *Work in progress*

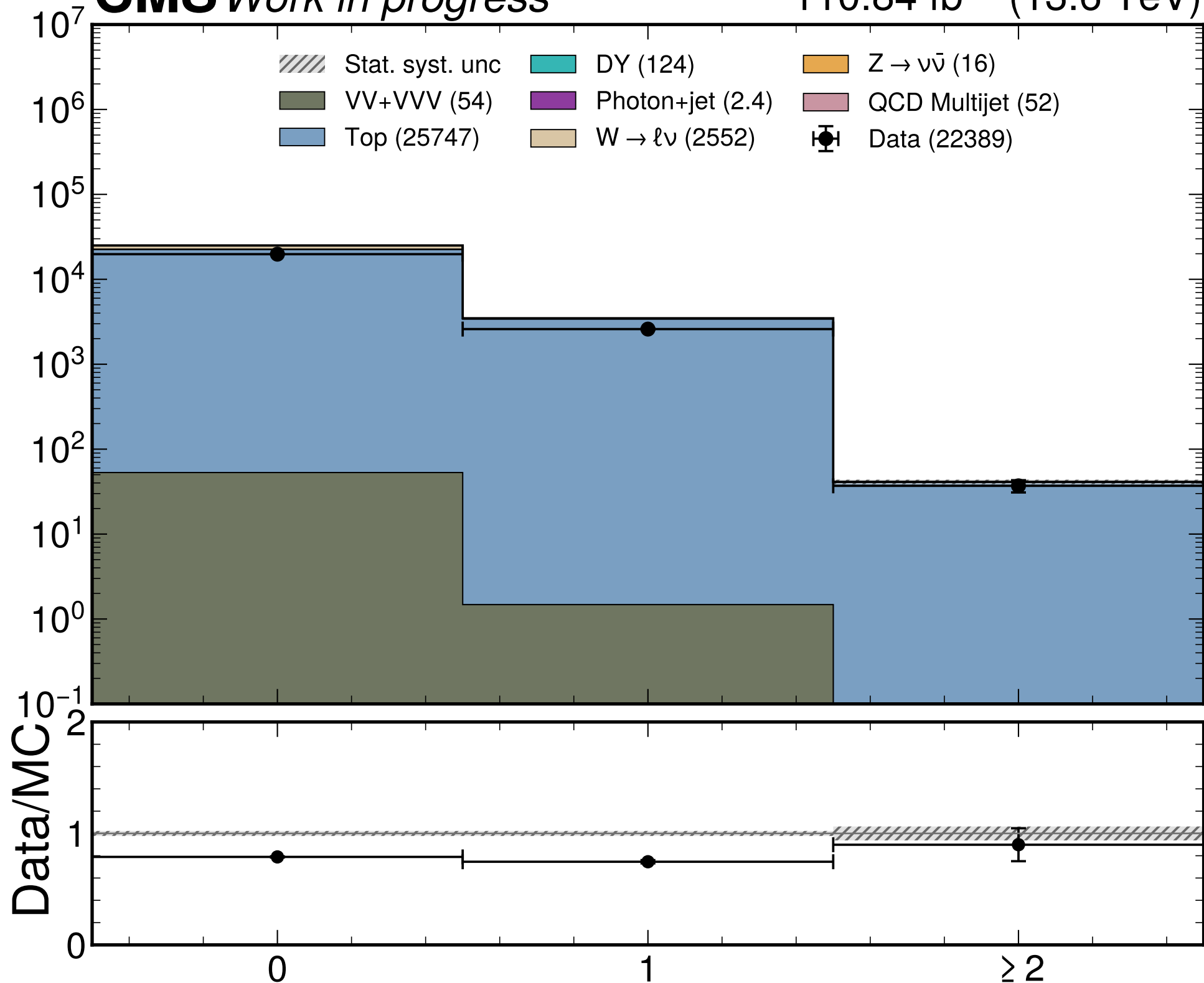
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

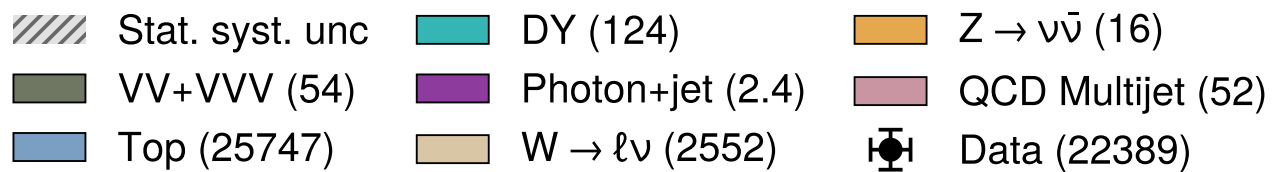
N_t



CMS *Work in progress*

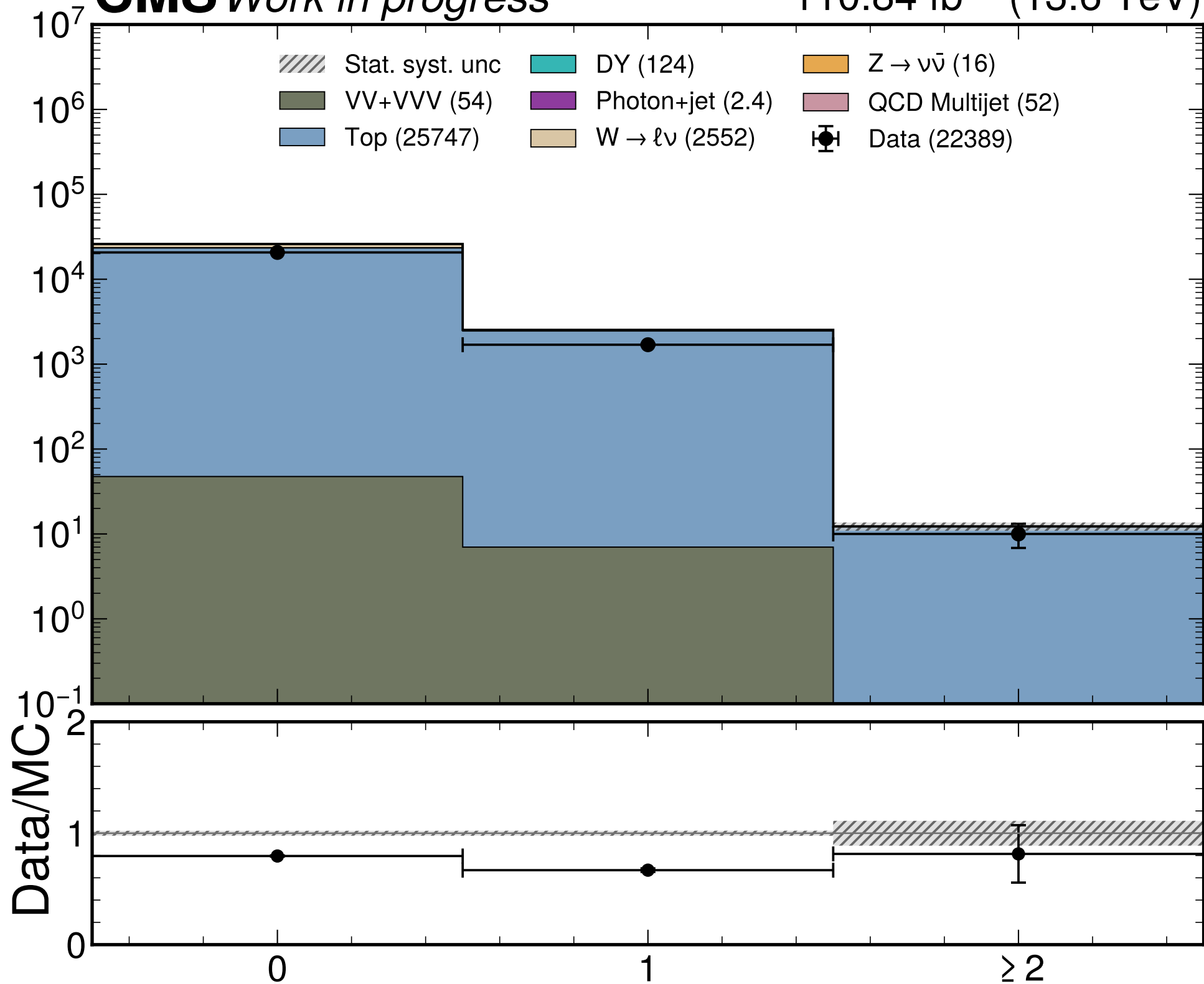
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

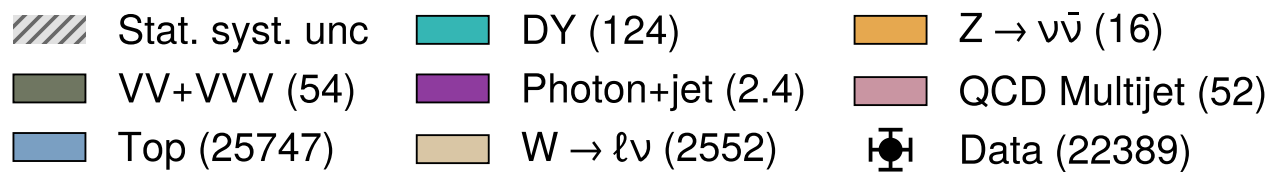
N_W



CMS *Work in progress*

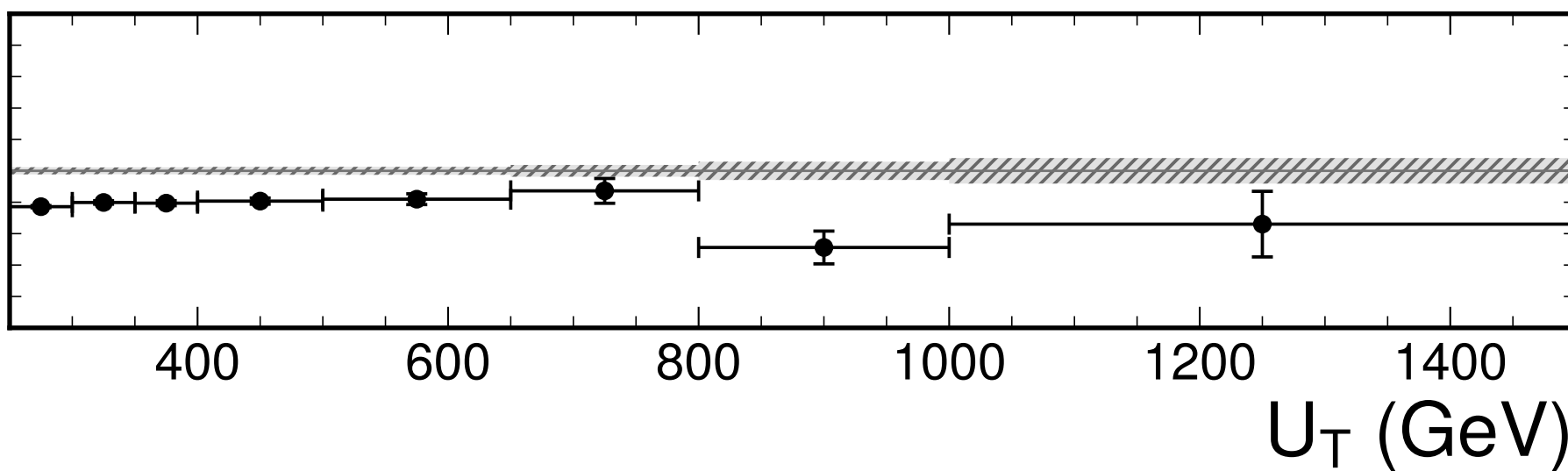
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

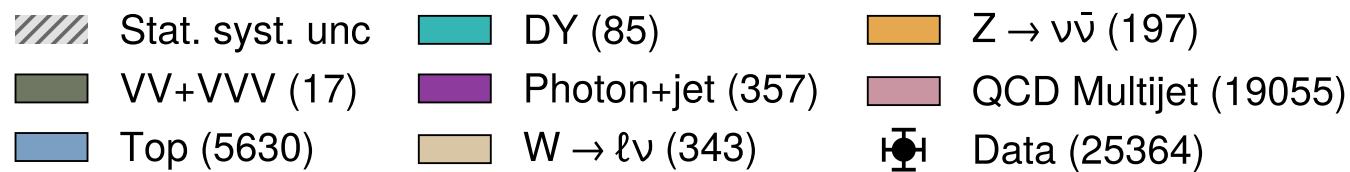
10⁻¹
2
1
0



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

200

400

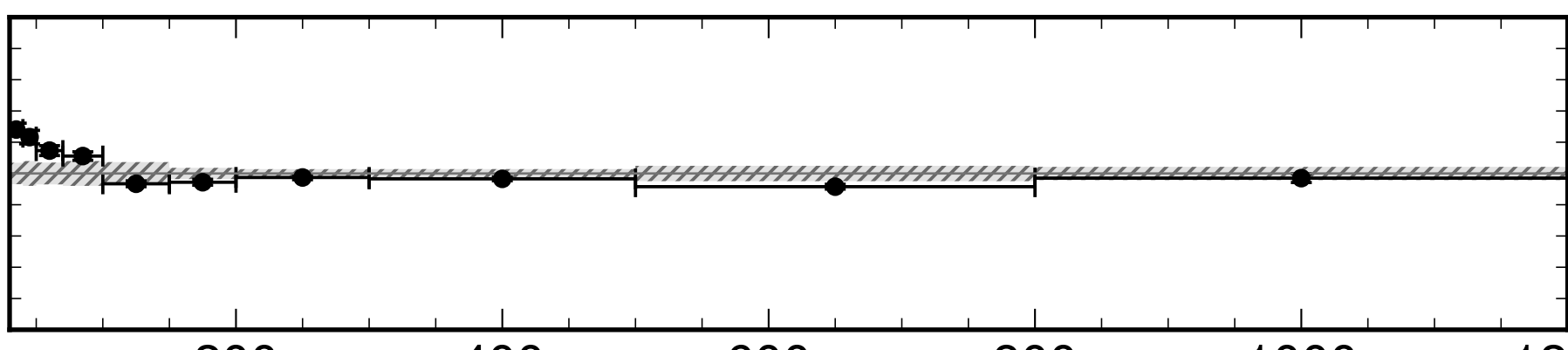
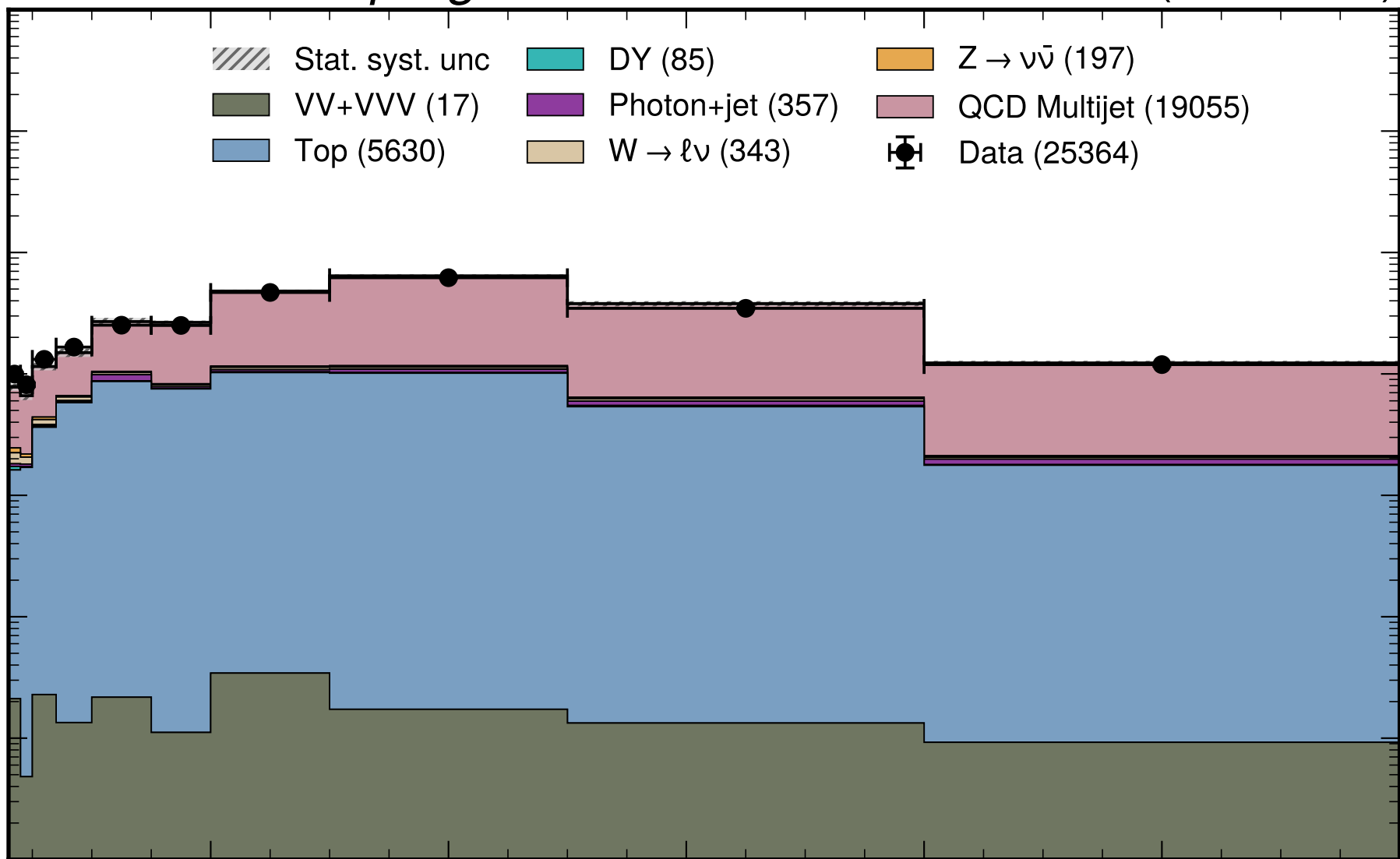
600

800

1000

1200

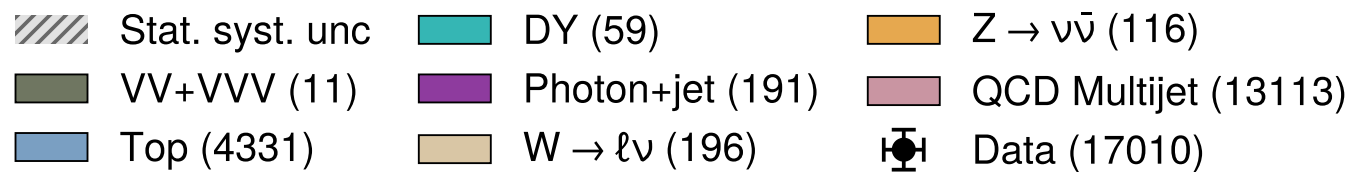
Leading b-jet p_T (GeV)



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻²

200

400

600

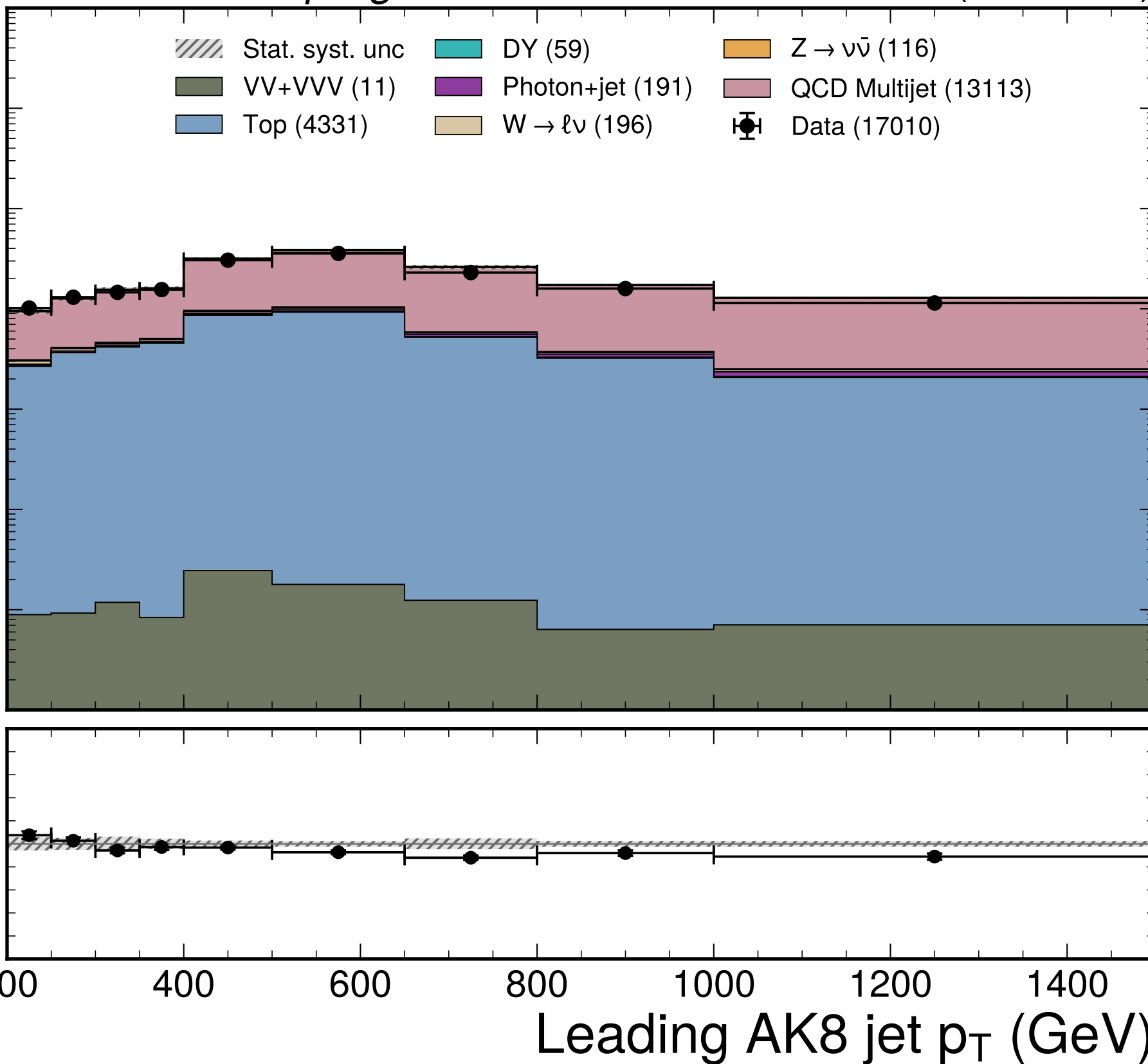
800

1000

1200

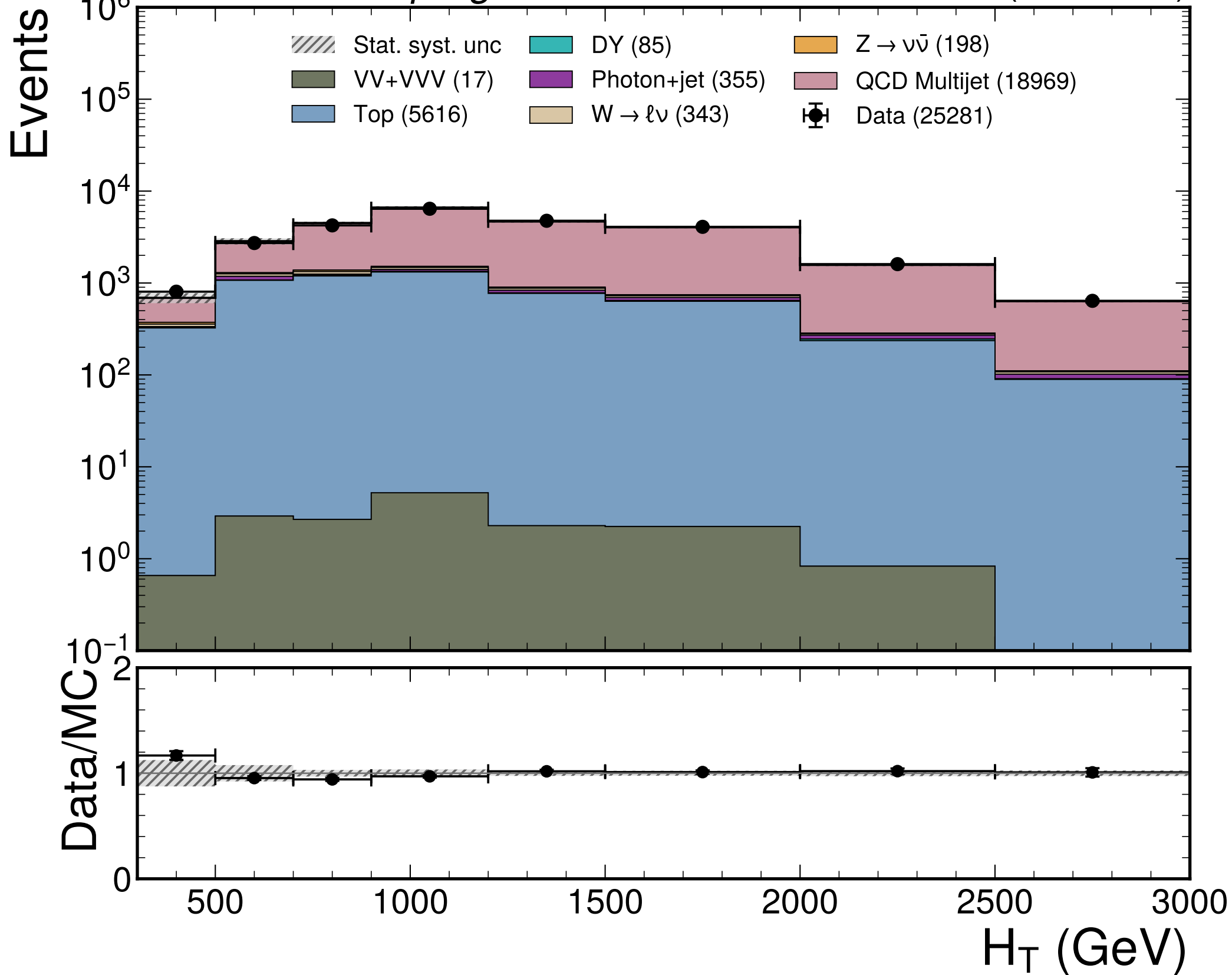
1400

Leading AK8 jet p_T (GeV)



CMS *Work in progress*

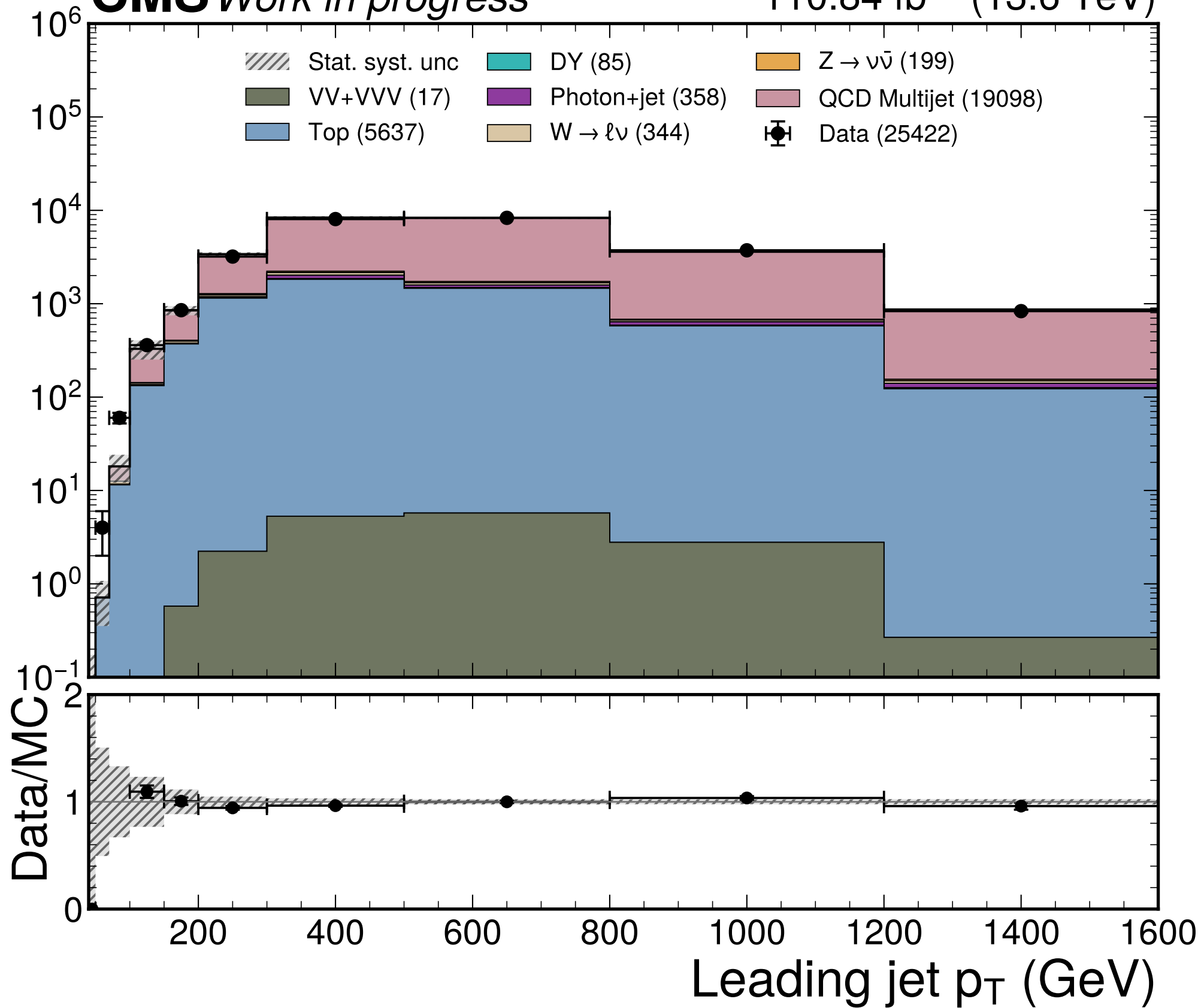
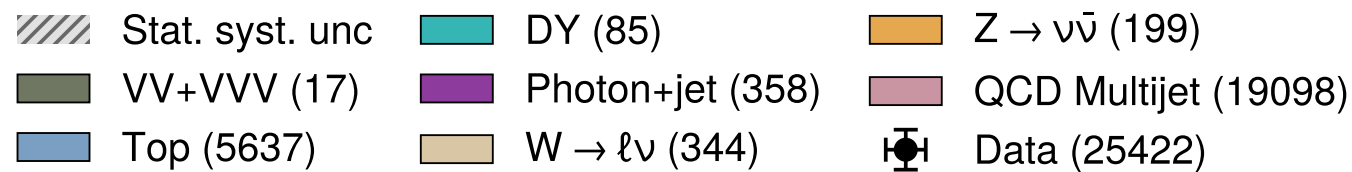
110.84 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

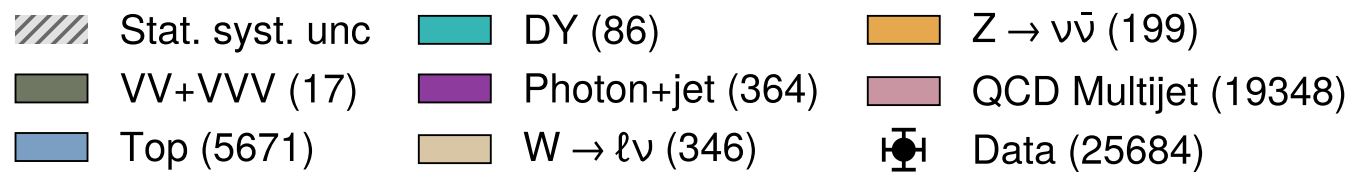
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2

400

600

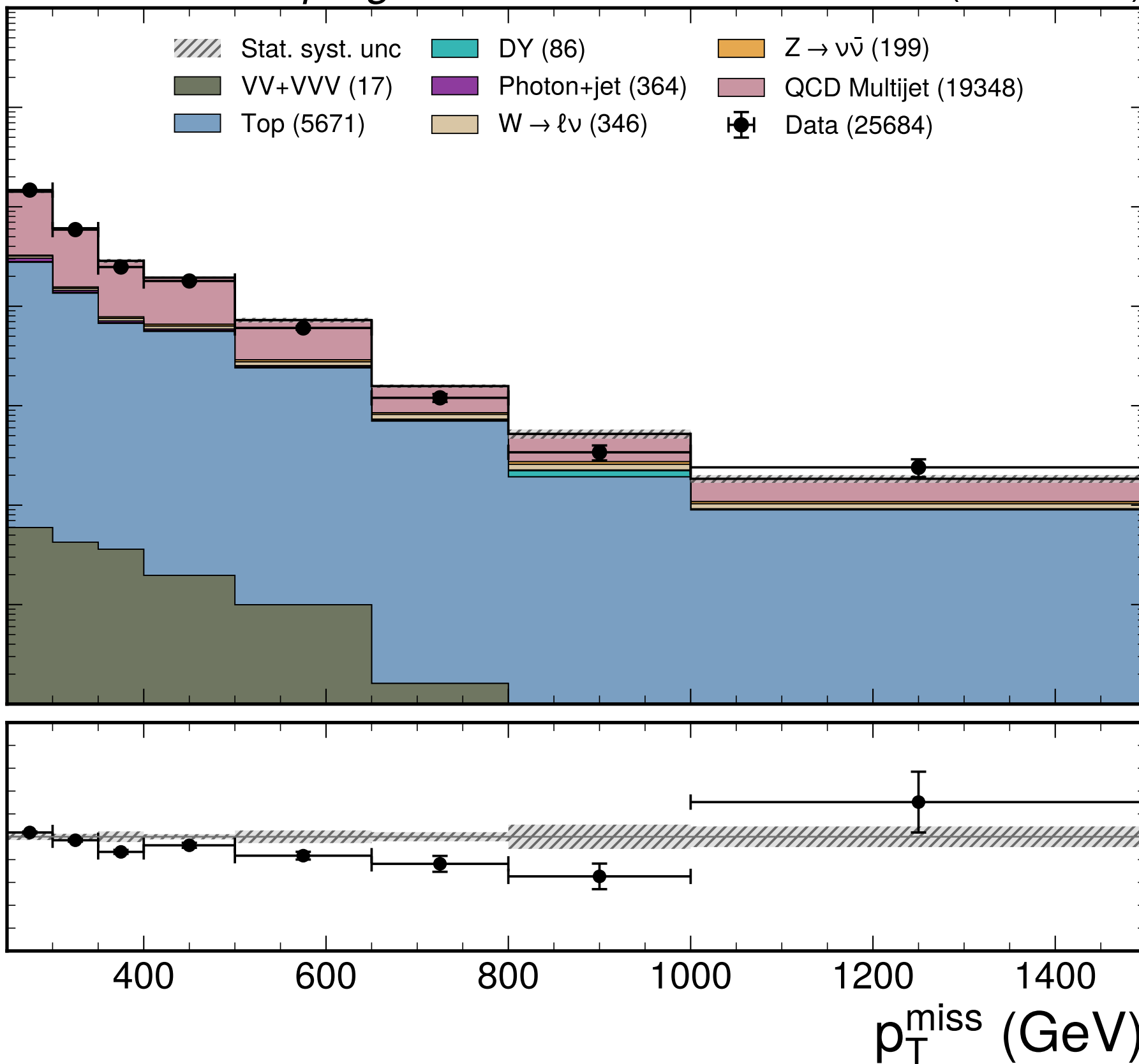
800

1000

1200

1400

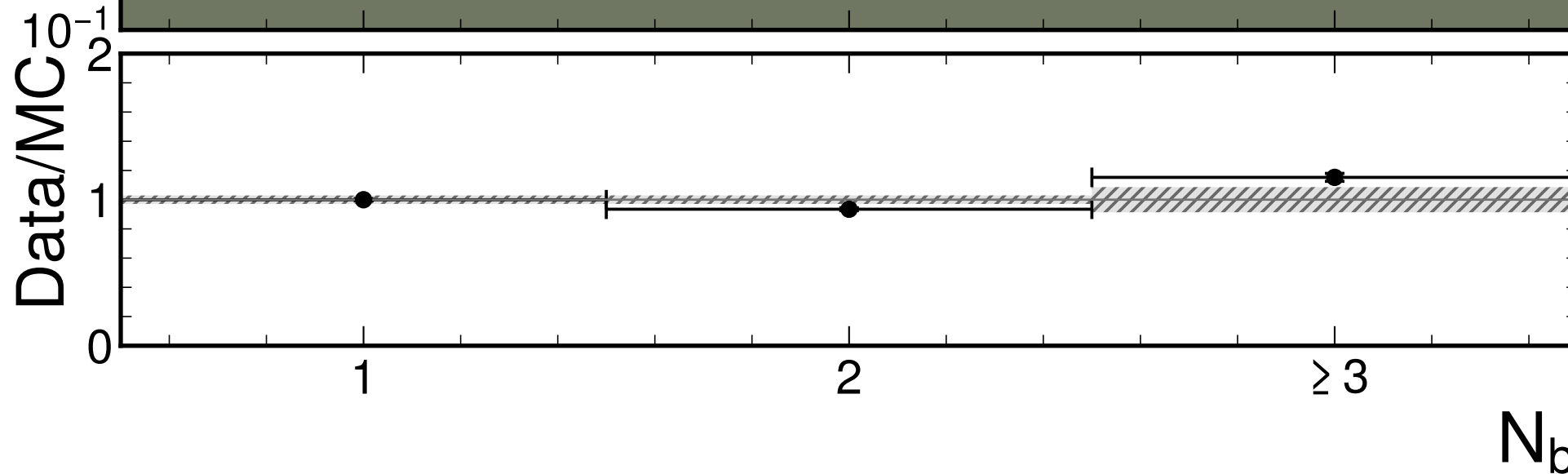
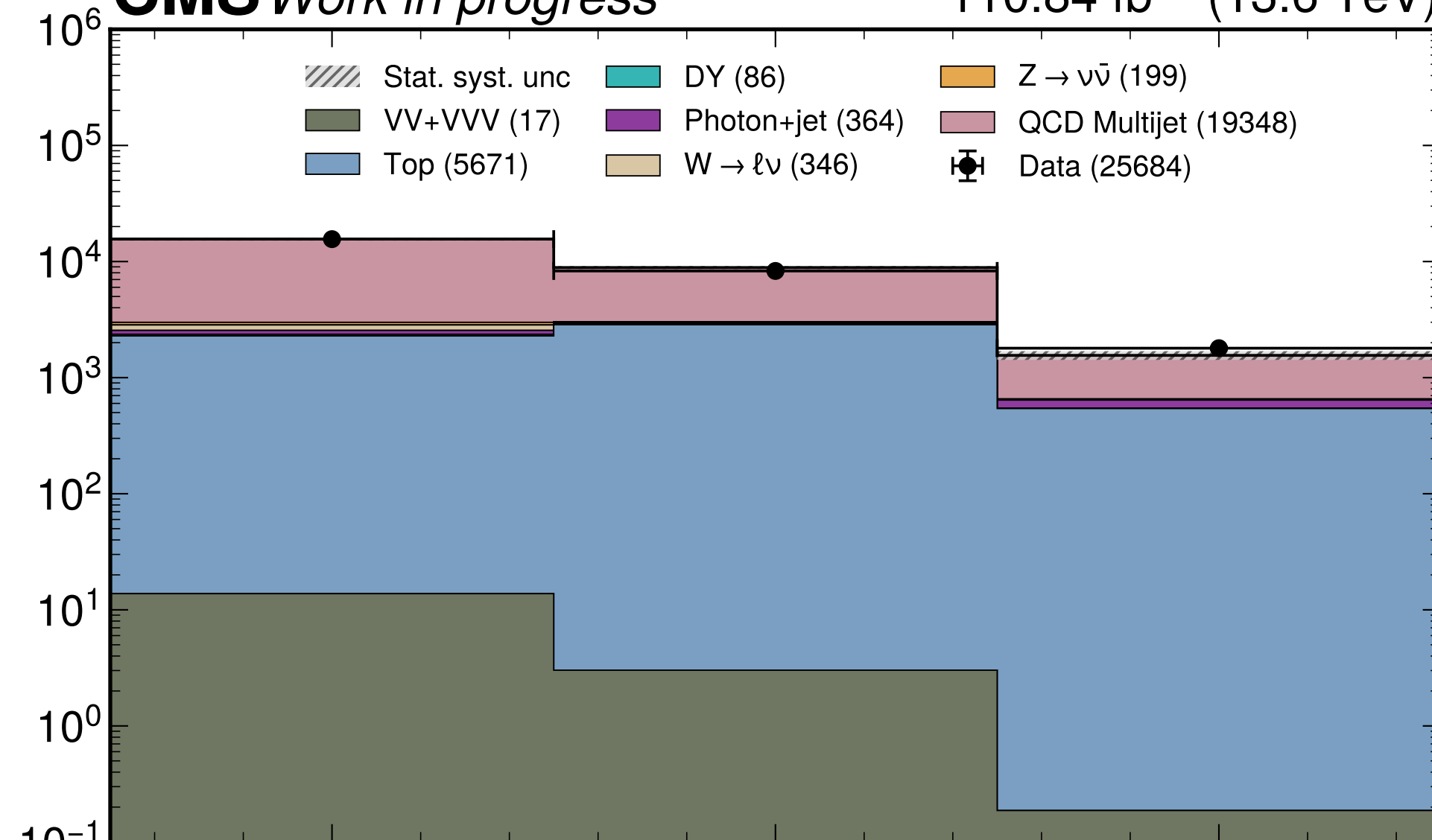
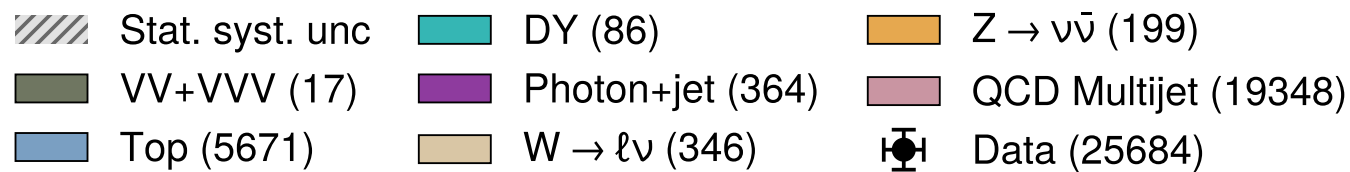
p_T^{miss} (GeV)



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

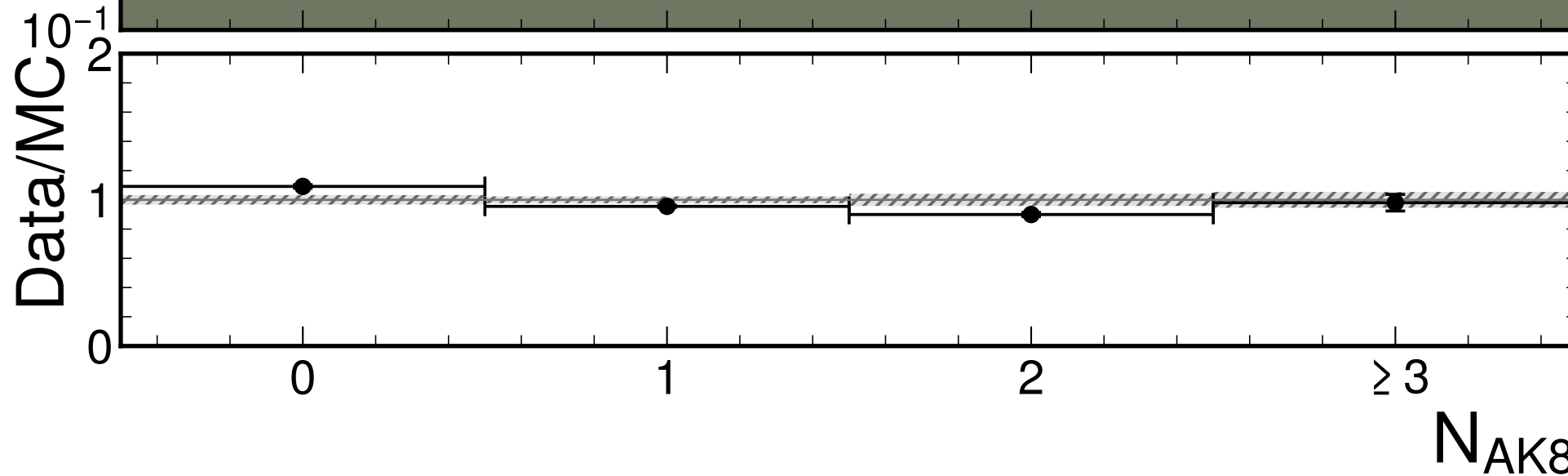
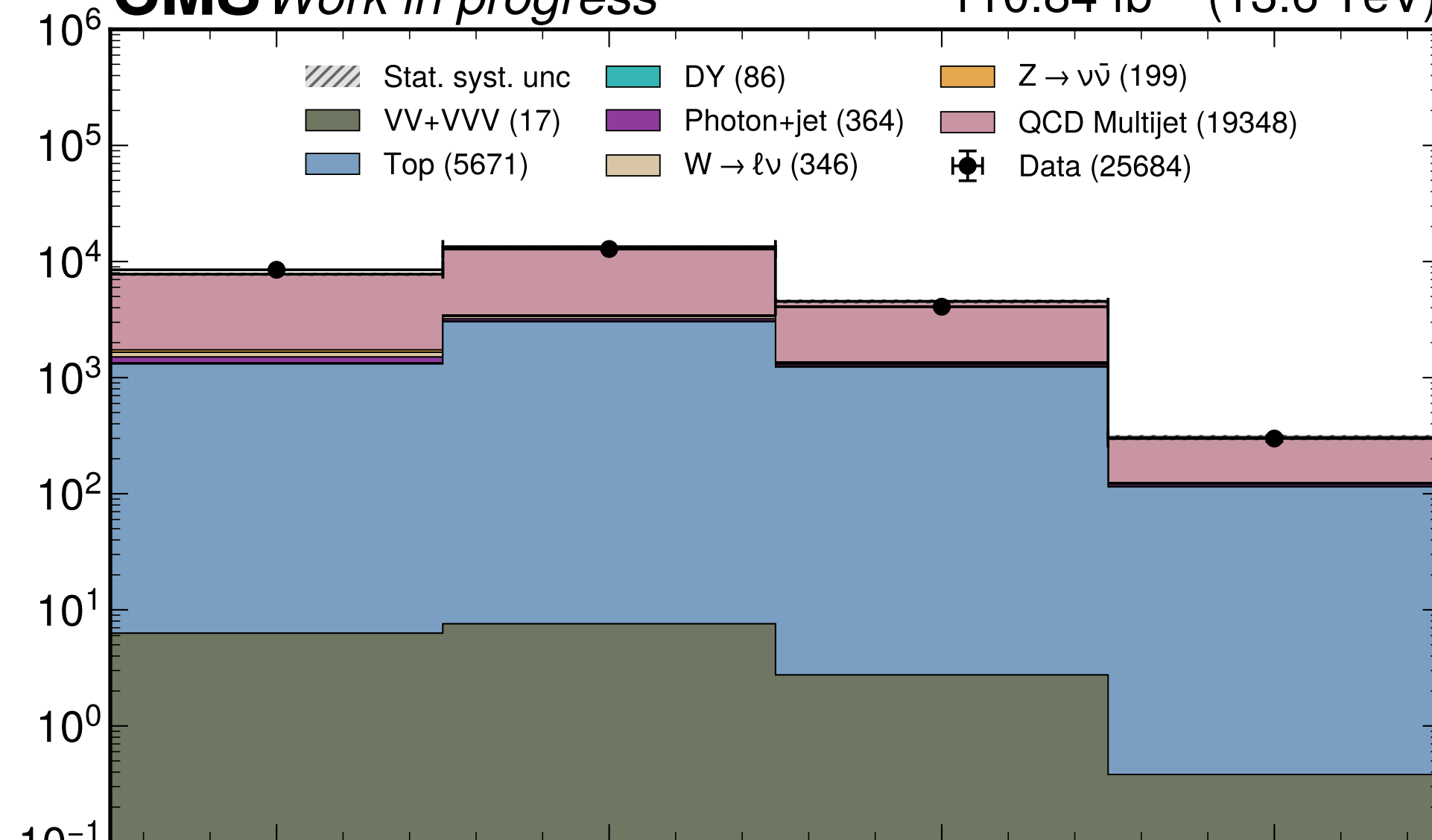
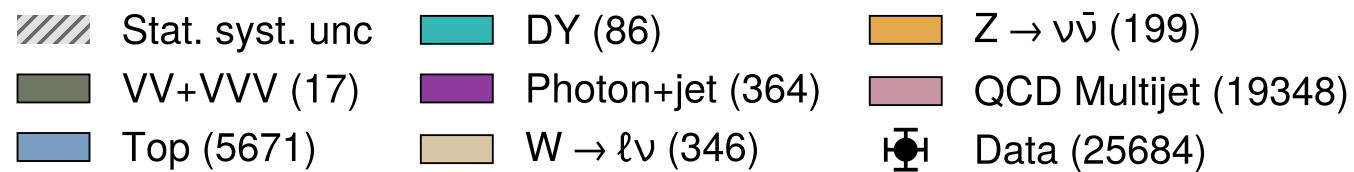
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

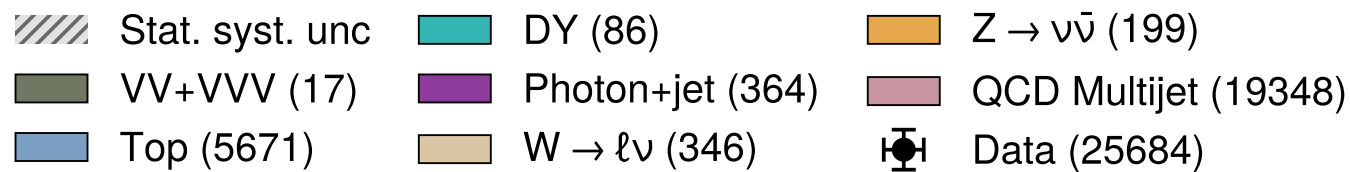
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

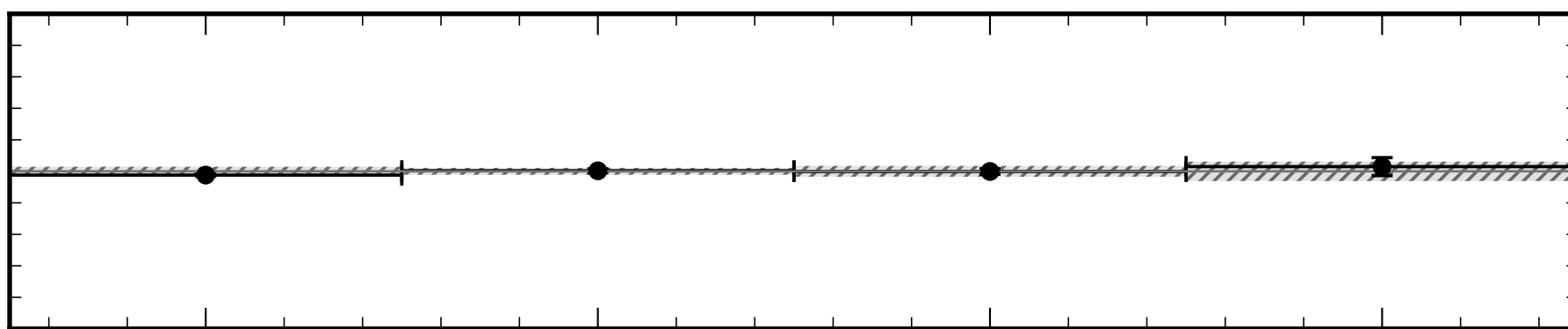
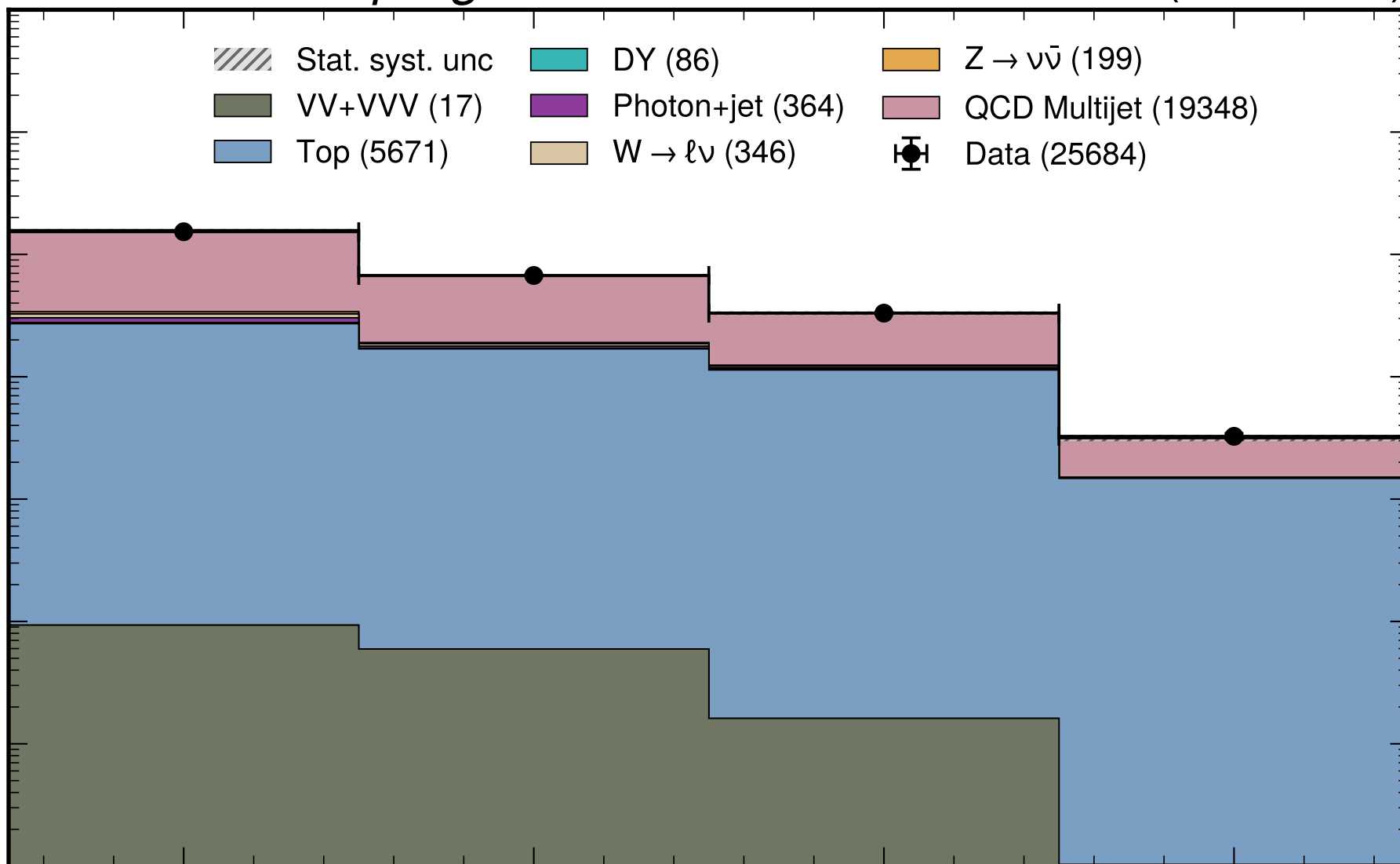
5

6

7-8

≥ 9

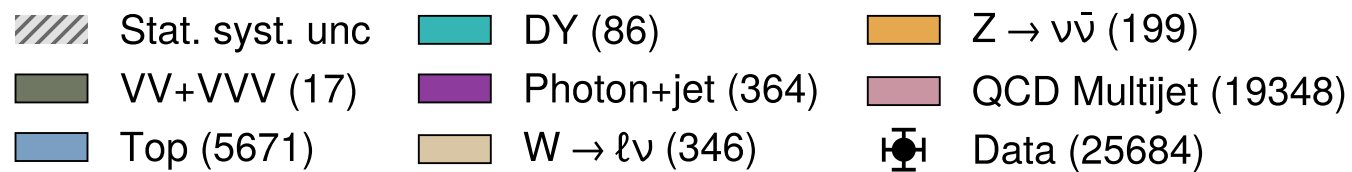
N_j



CMS *Work in progress*

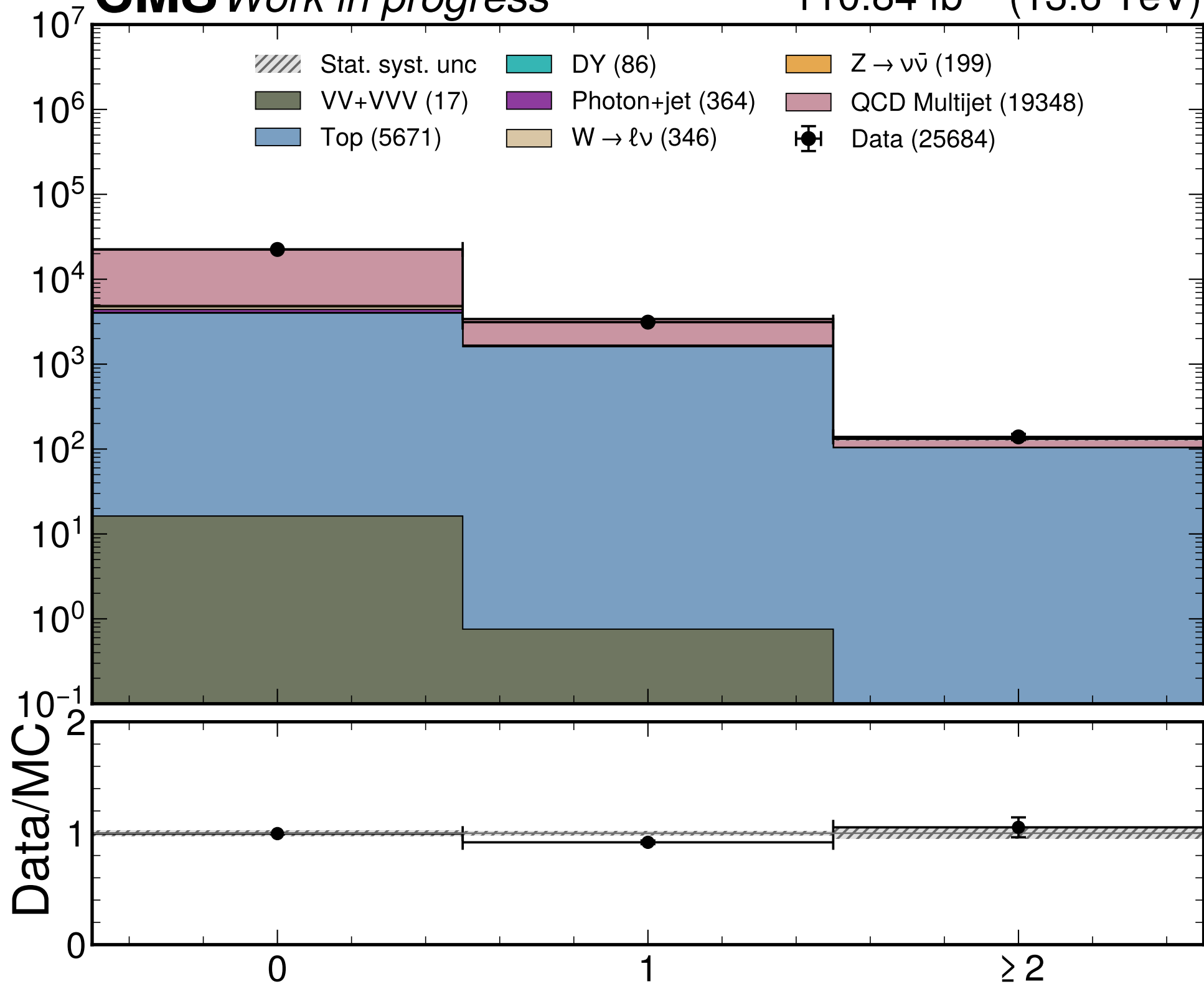
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

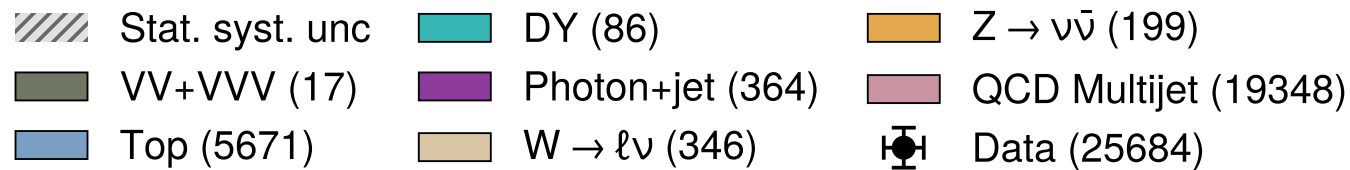
N_t



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

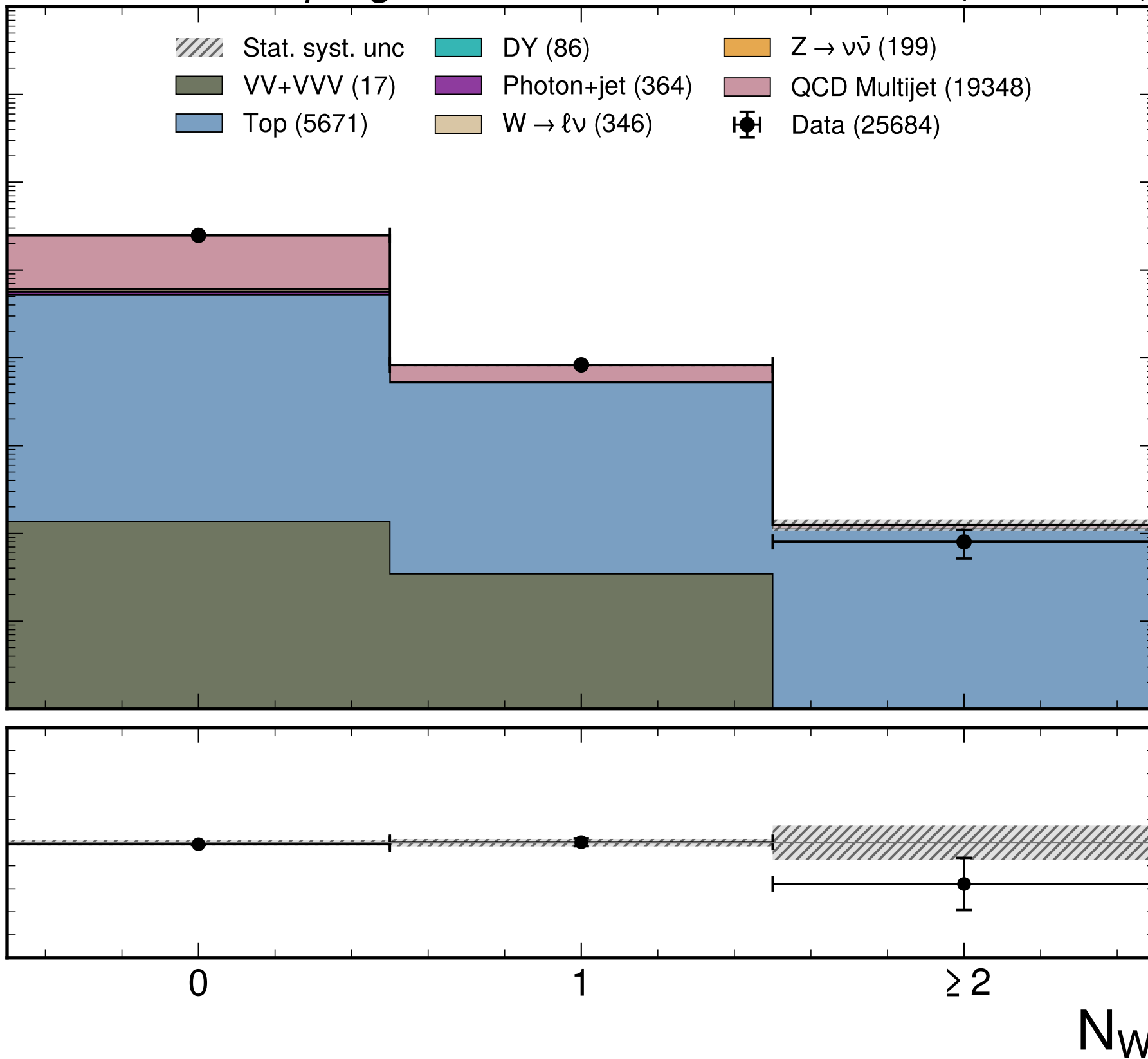
10⁻¹
2
1
0

0

1

≥ 2

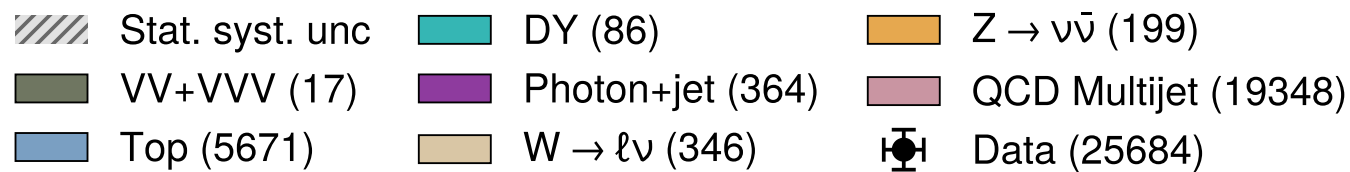
N_W



CMS *Work in progress*

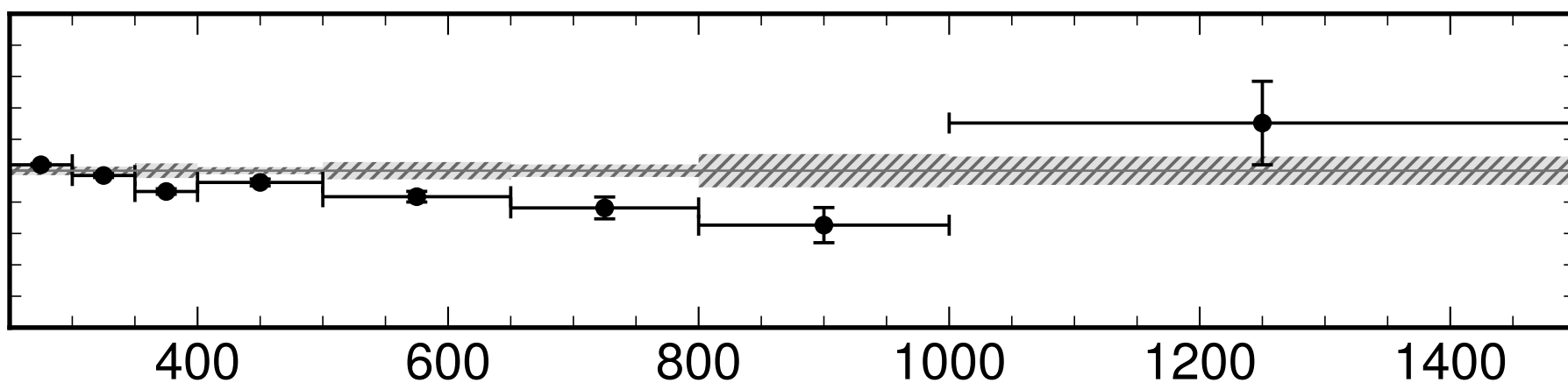
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

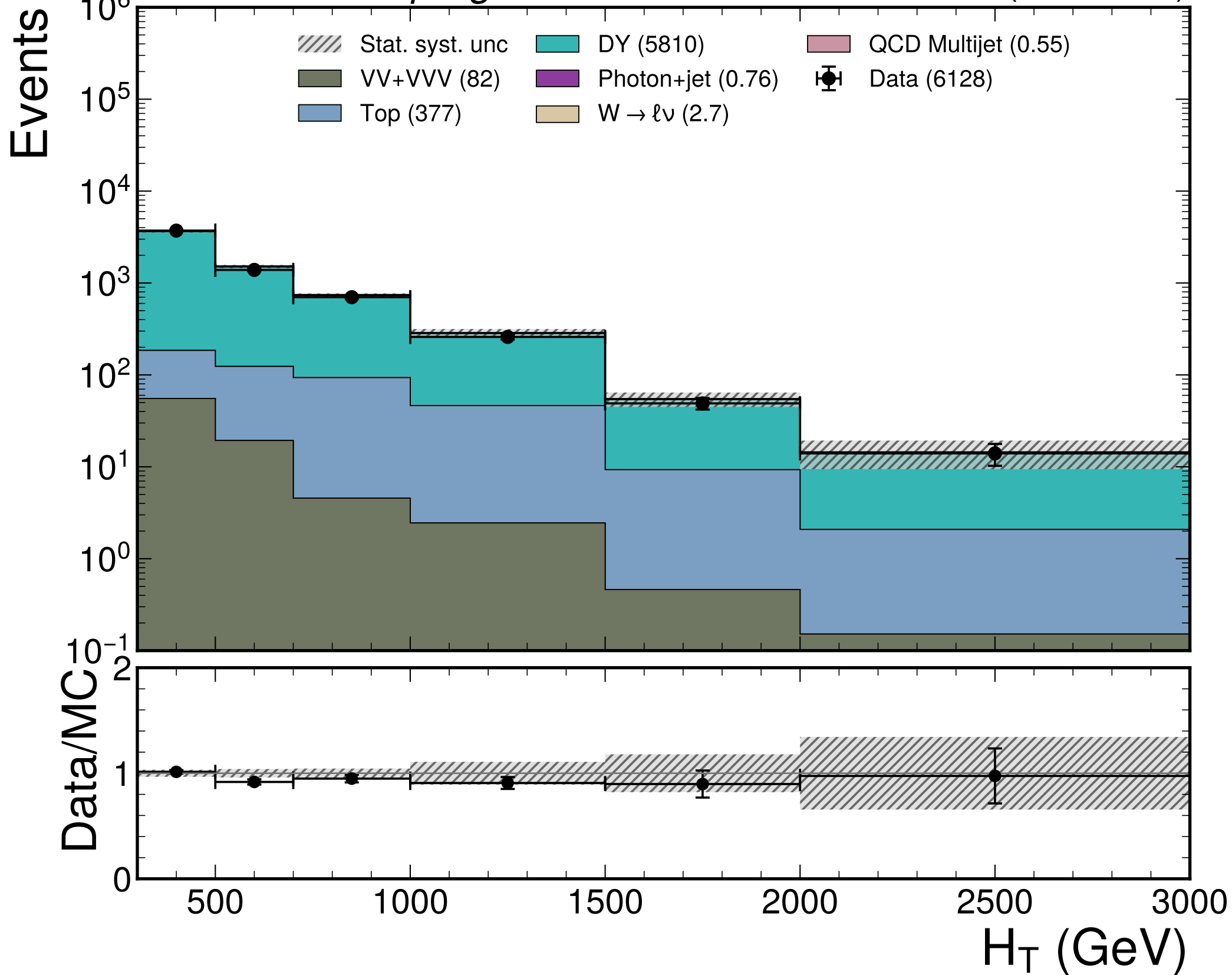
10⁻¹
2
1
0



U_T (GeV)

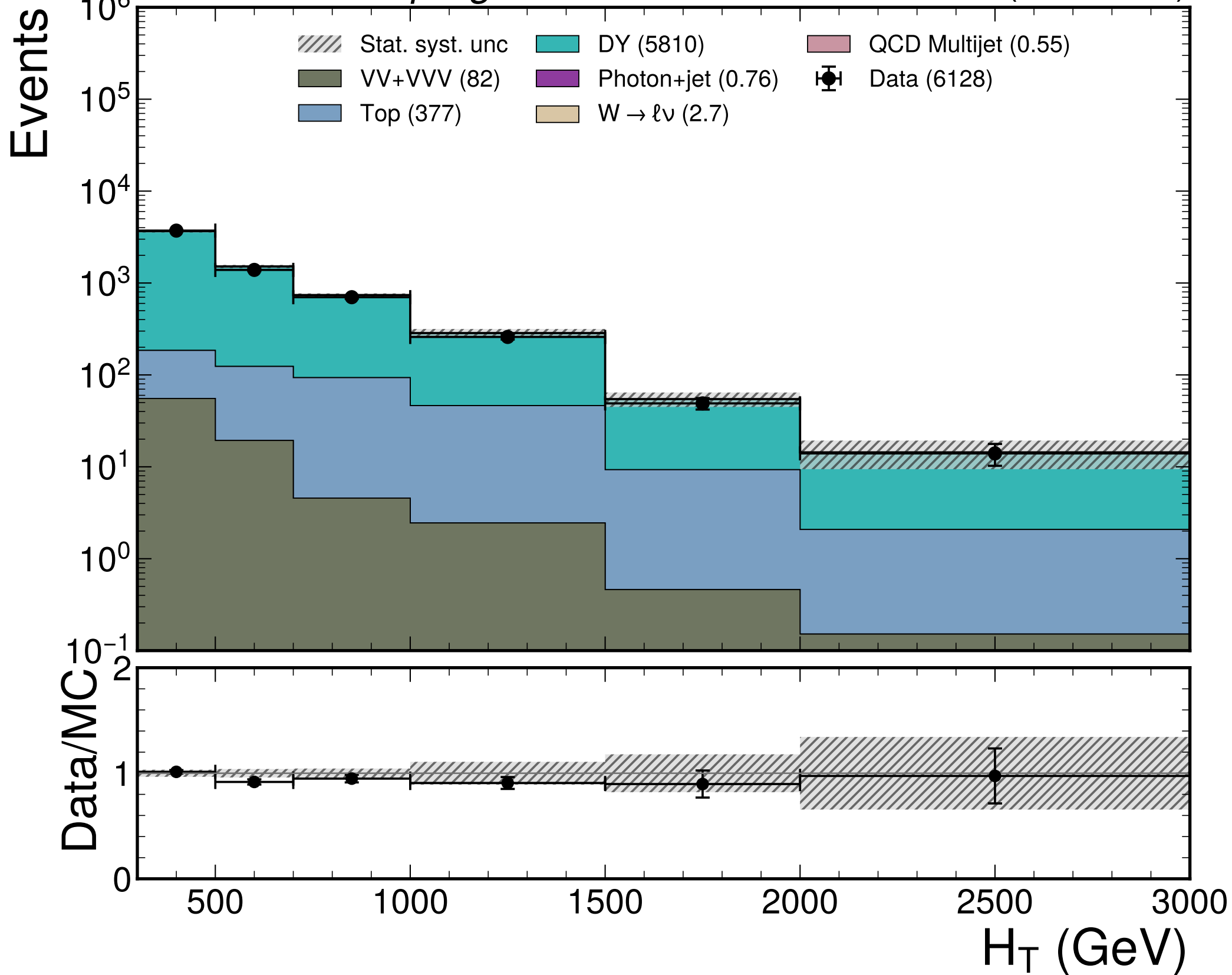
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



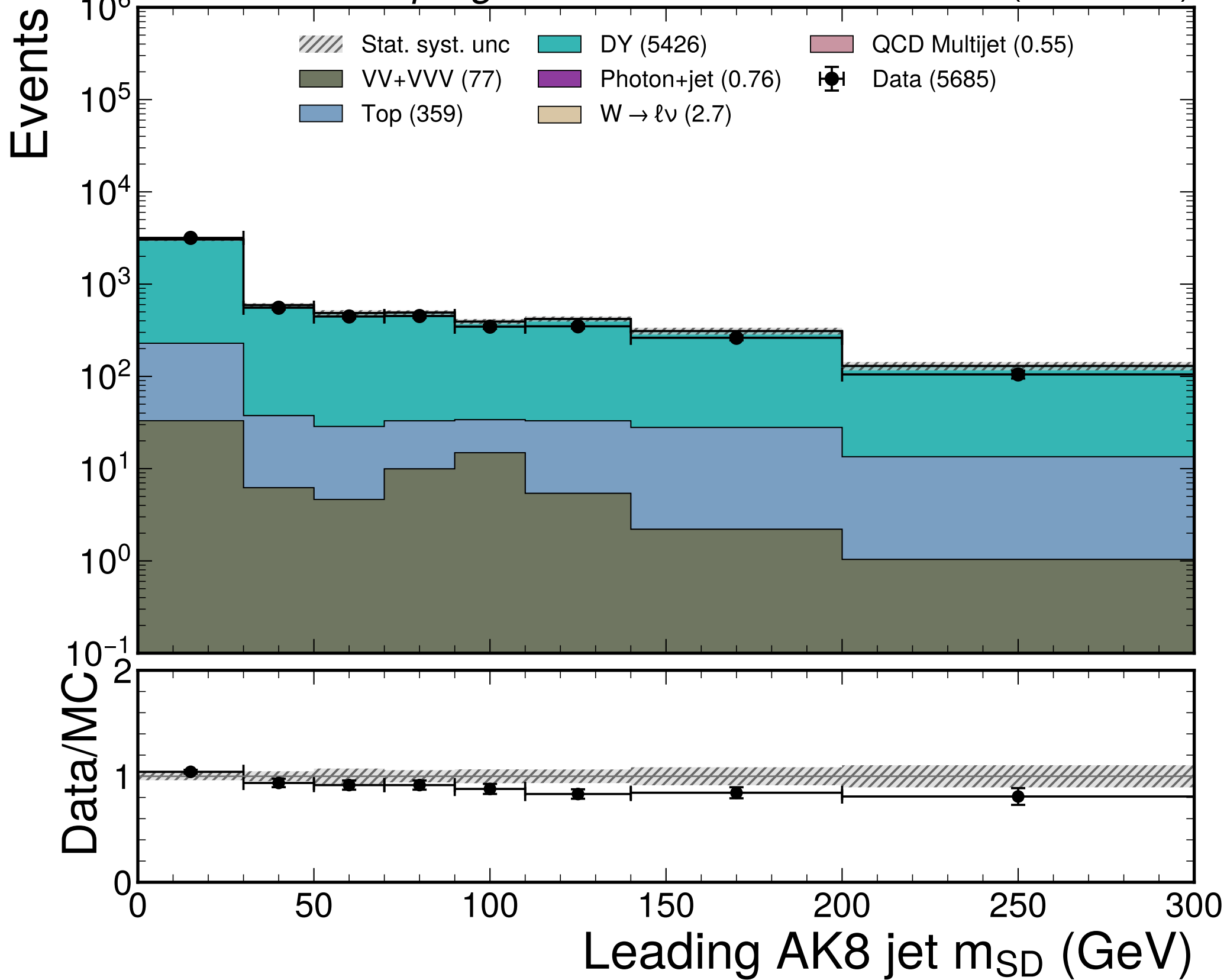
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

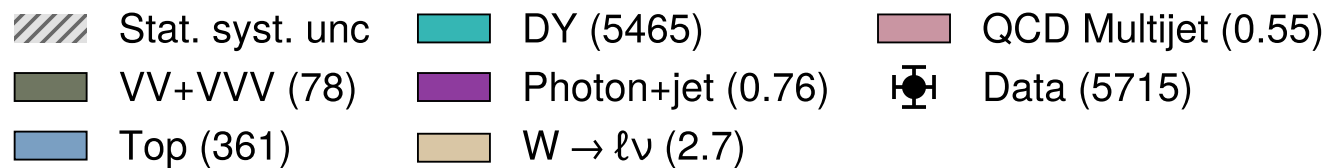
110.84 fb⁻¹ (13.6 TeV)



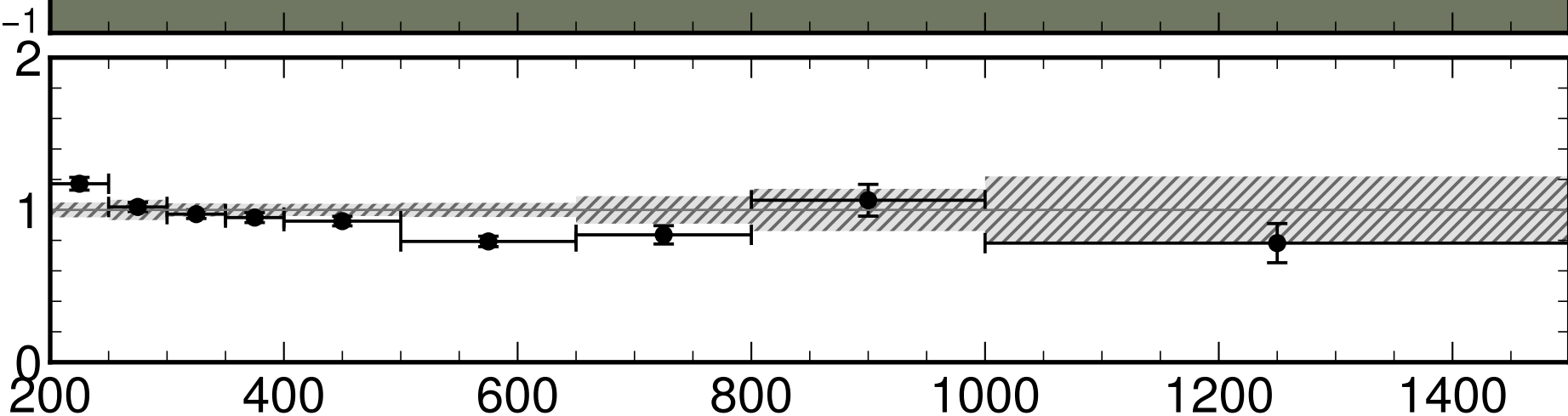
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

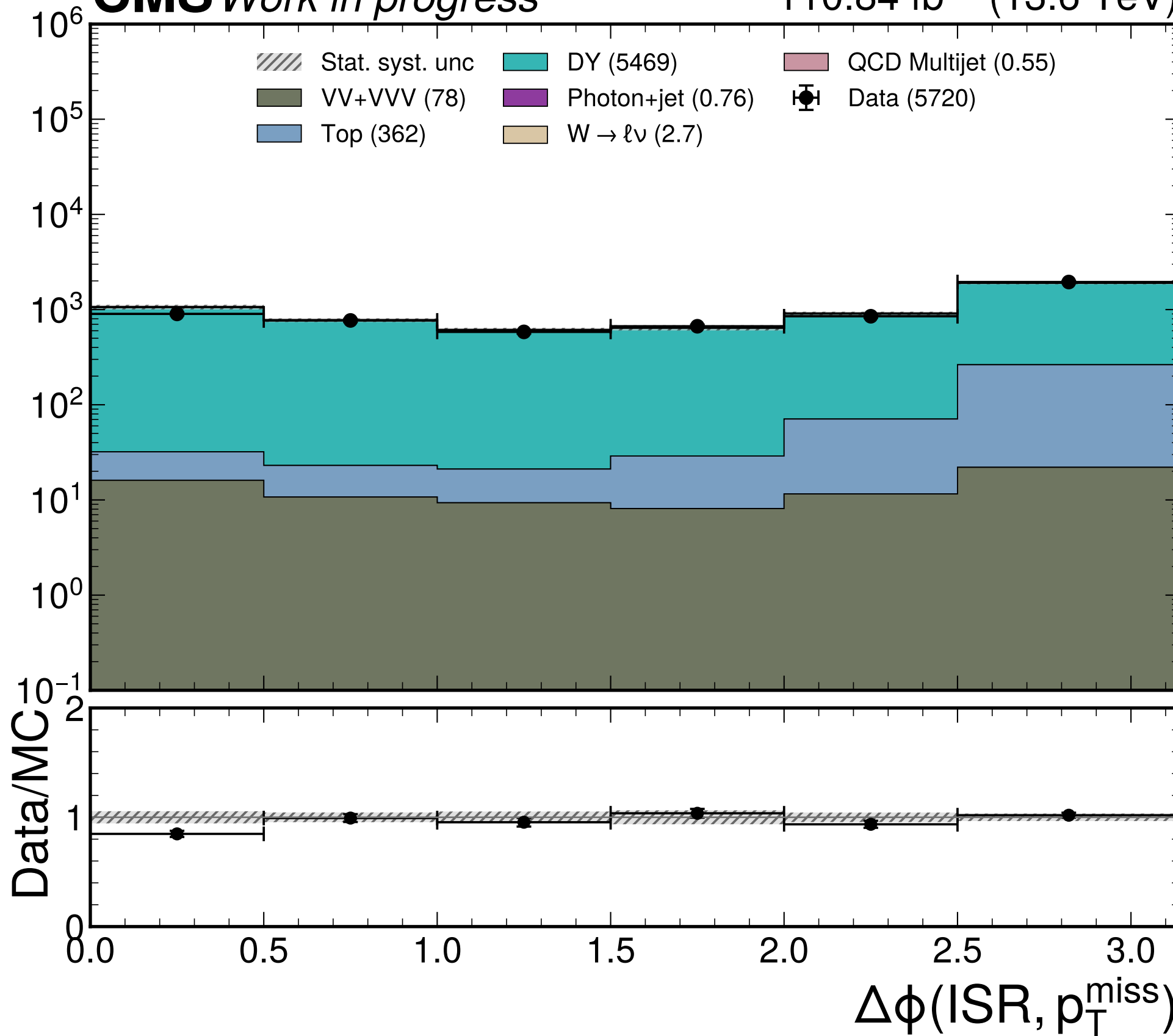
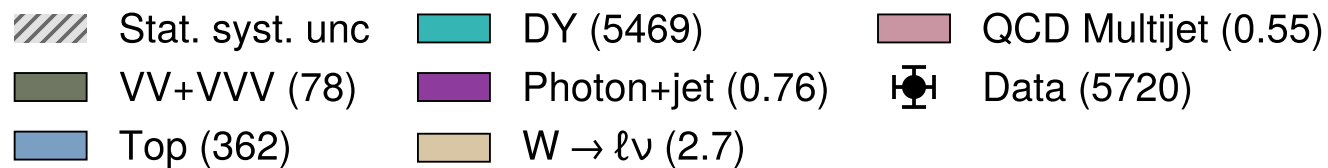


Leading AK8 jet p_T (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

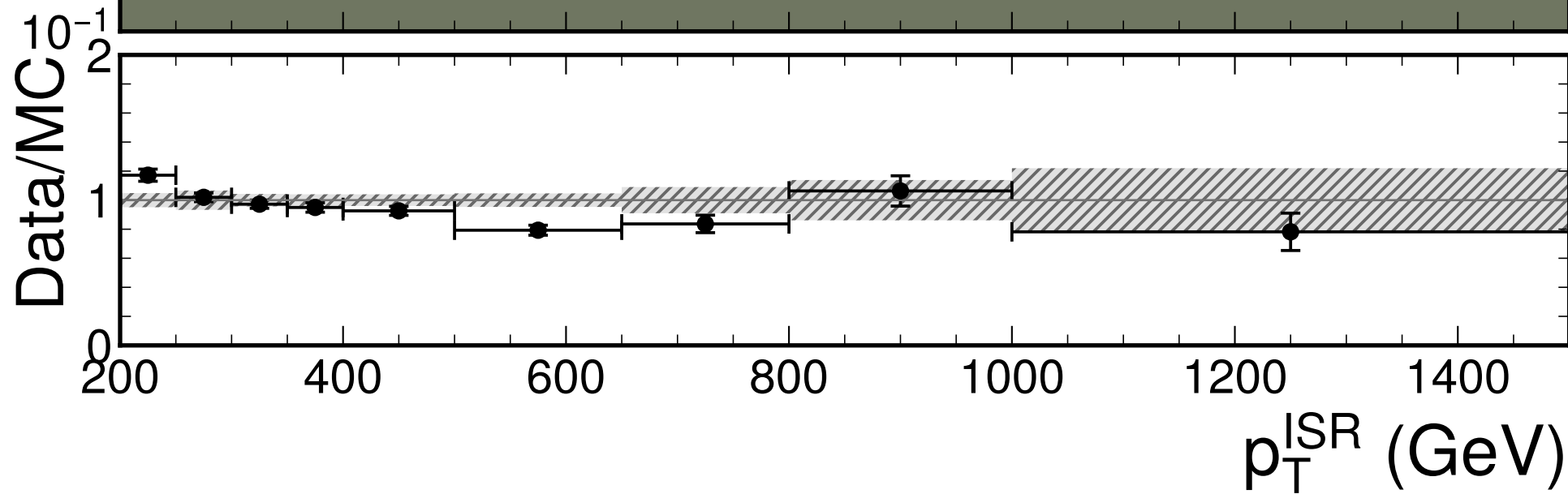
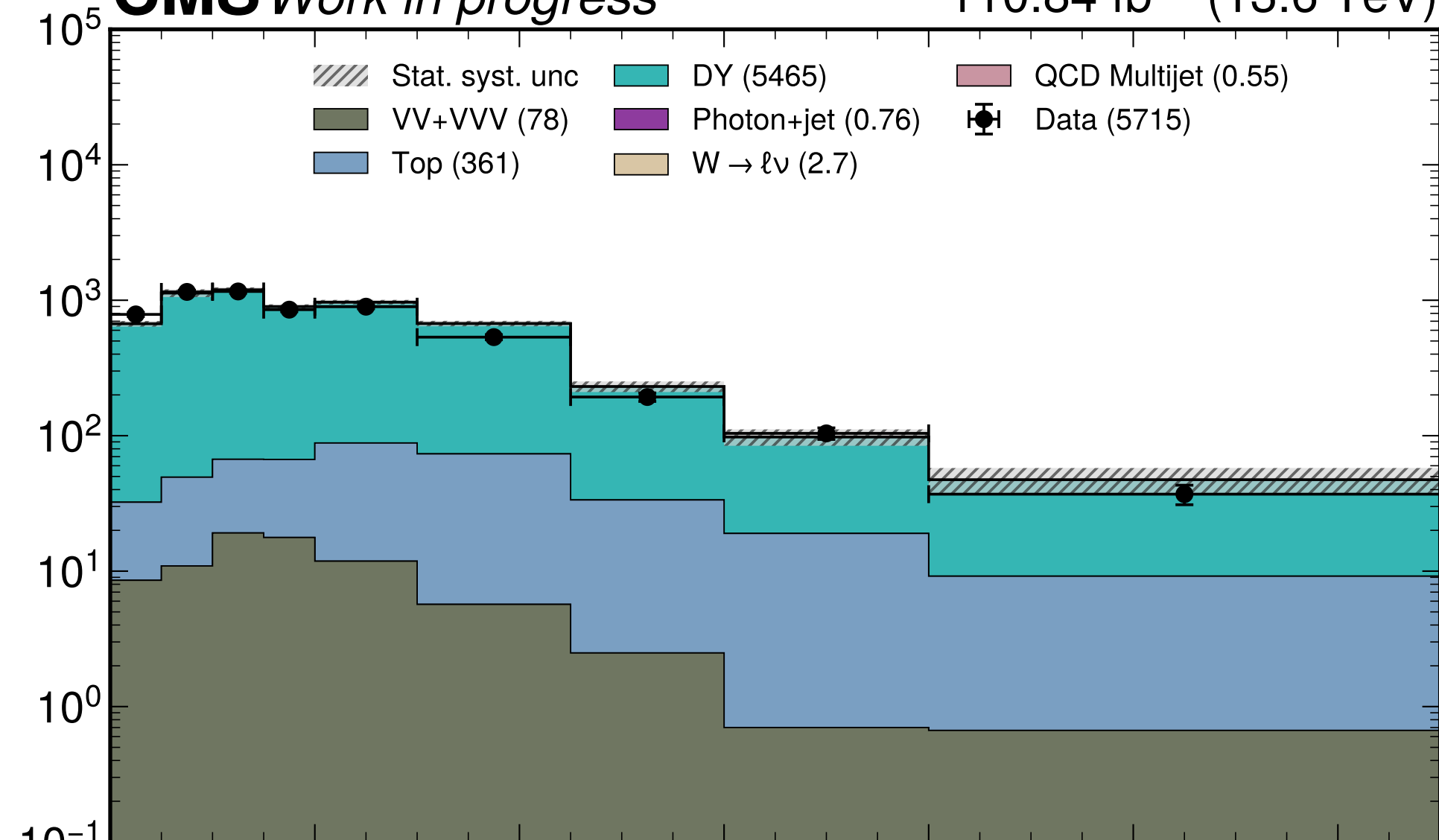
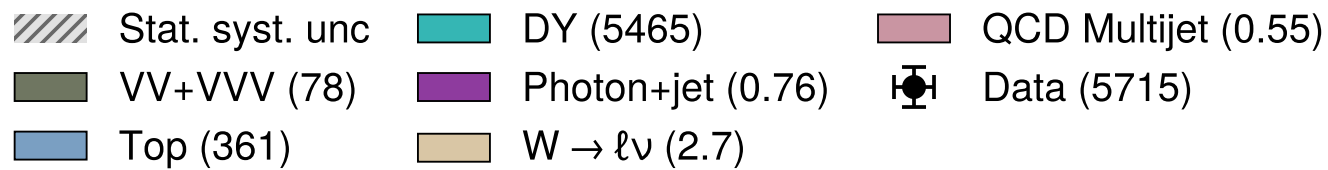
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

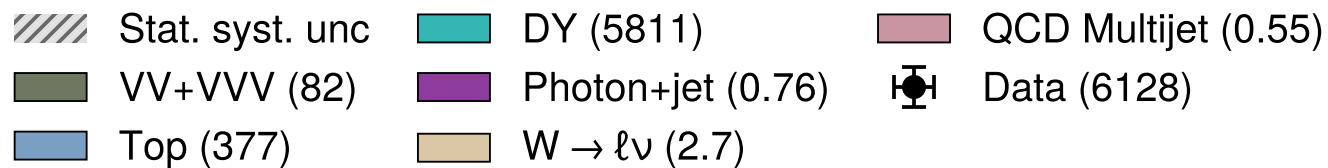
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

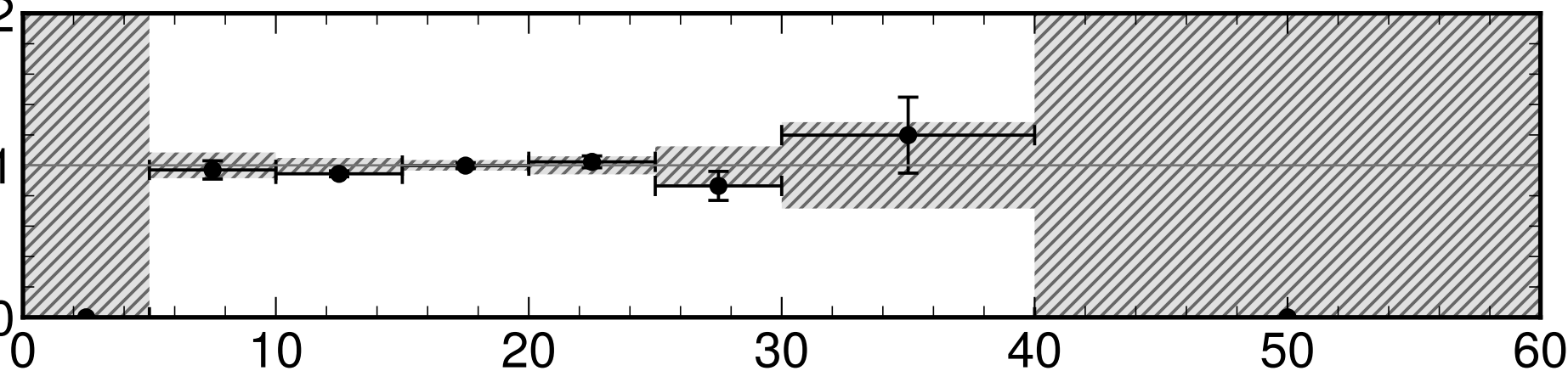
Events



Data/MC

10⁻¹

0

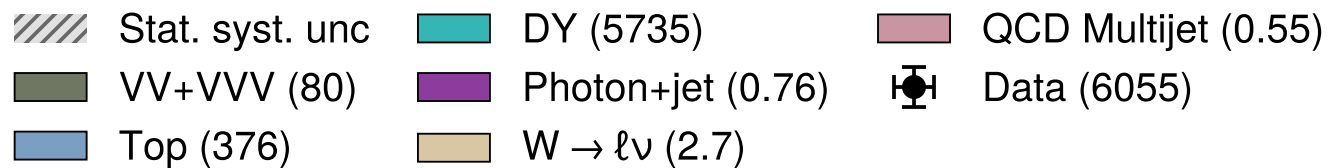


$p_T^{\text{miss}}/\sqrt{H_T}$

CMS *Work in progress*

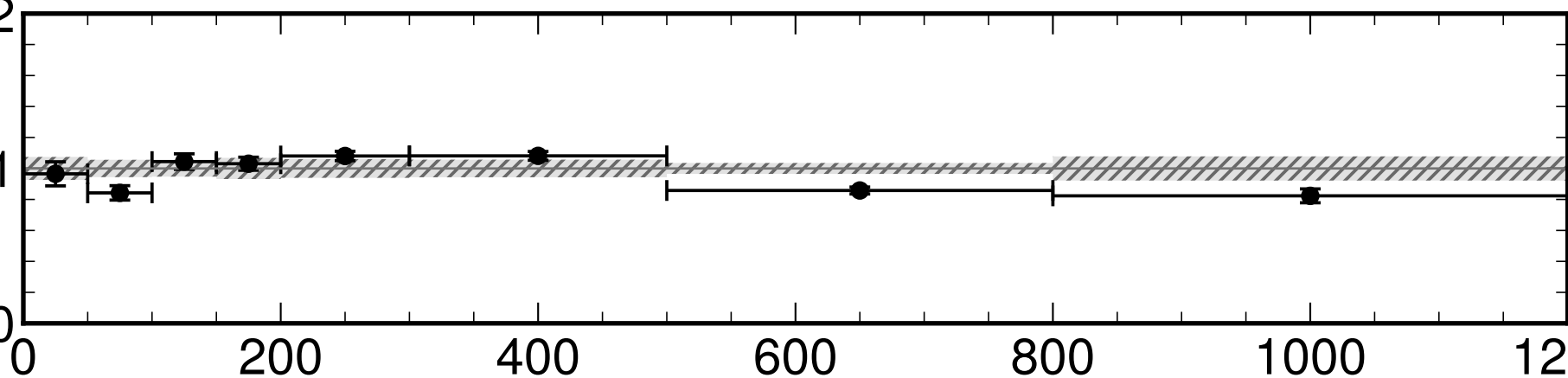
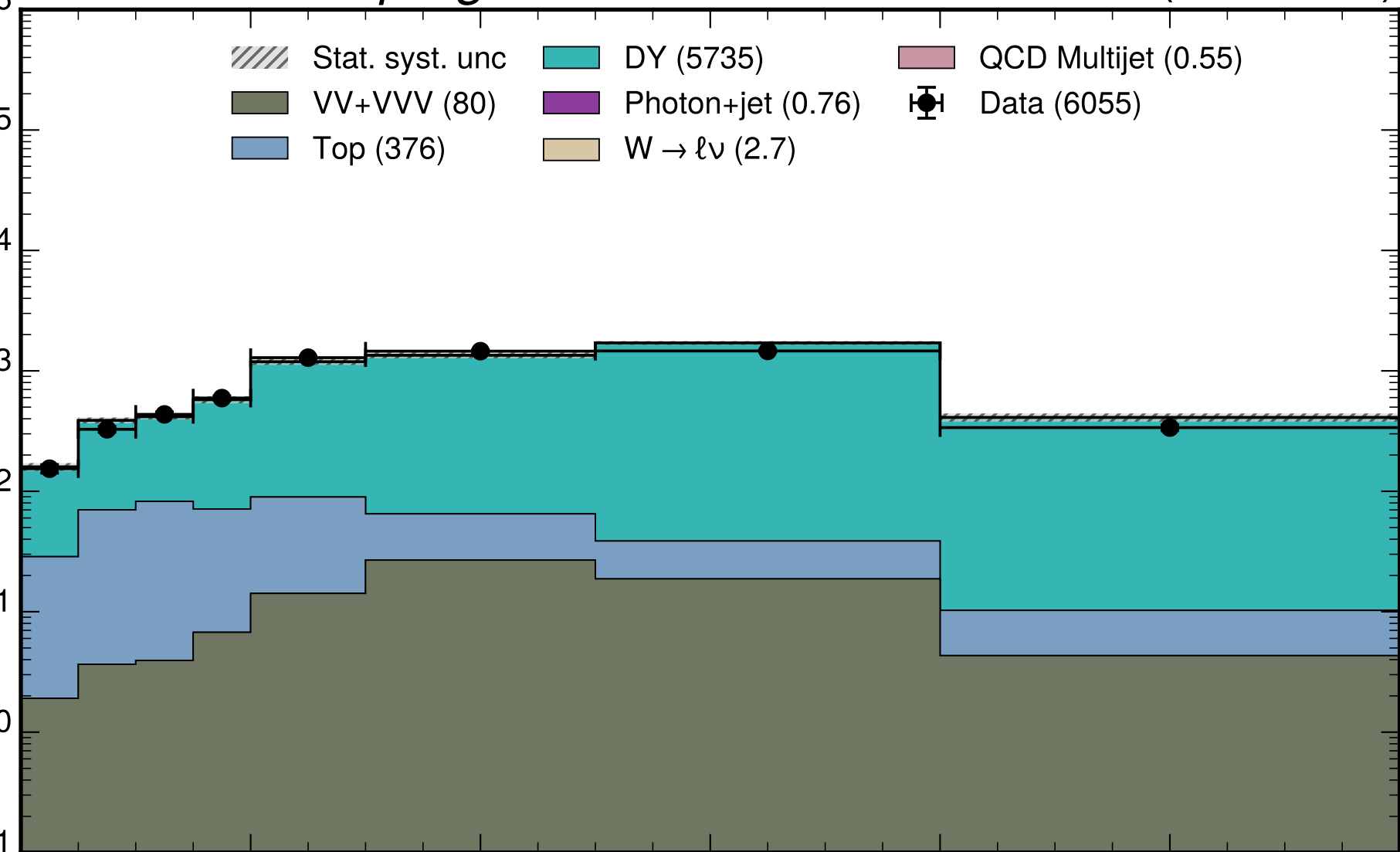
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

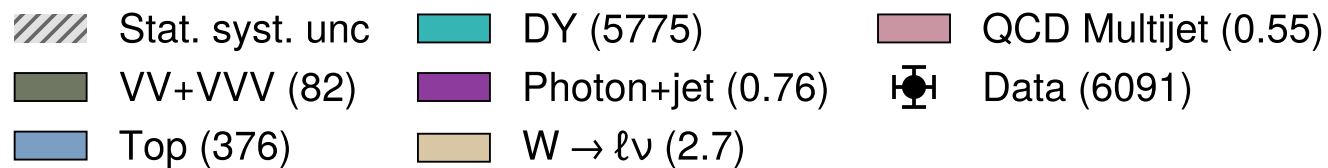


m_T^b (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



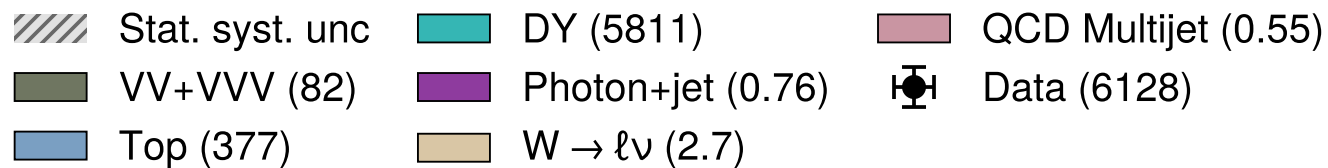
Data/MC

p_T^b (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

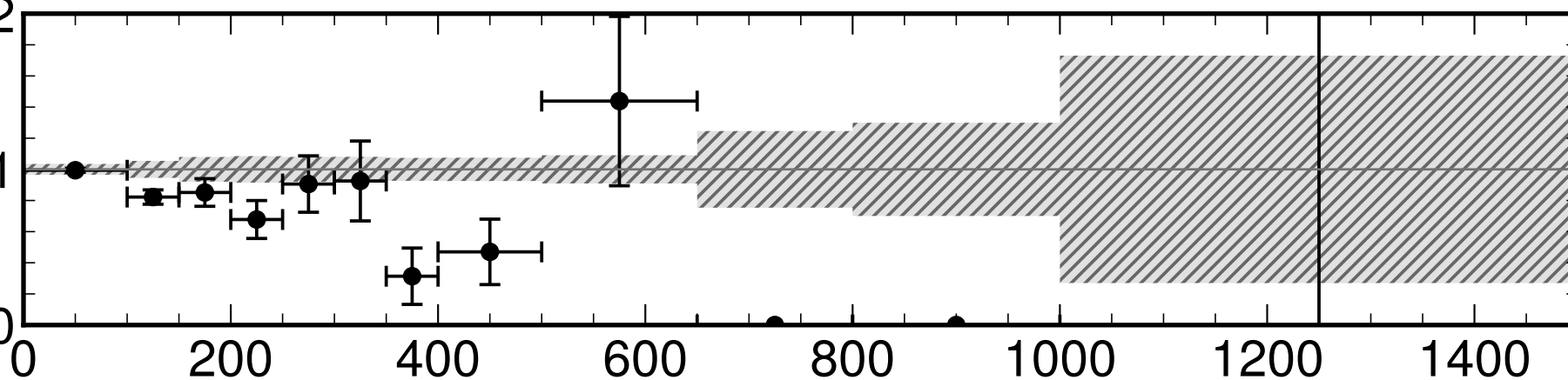
Events



Data/MC

10⁻²

0

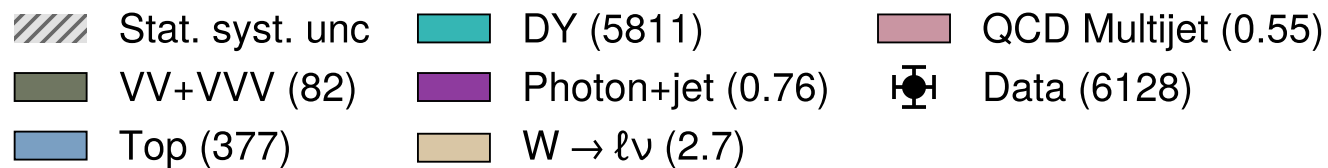


p_T^{miss} (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

70

80

90

100

110

120

130

140

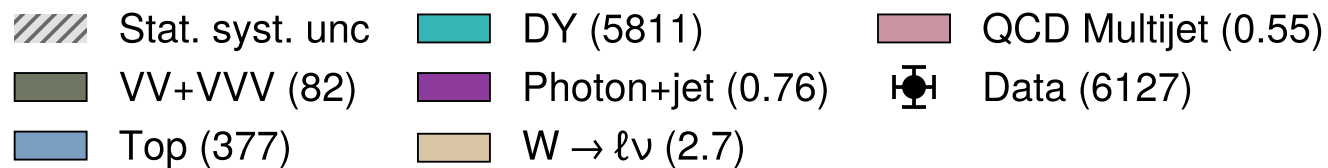
150

$m_{\ell\ell}$ (GeV)

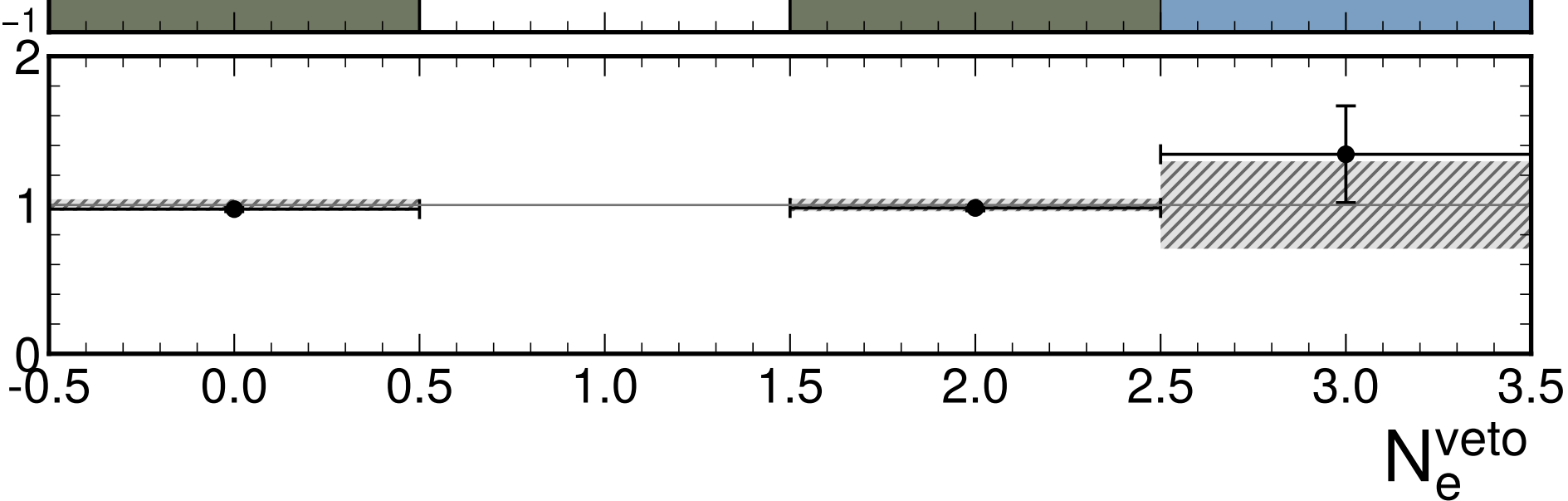
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



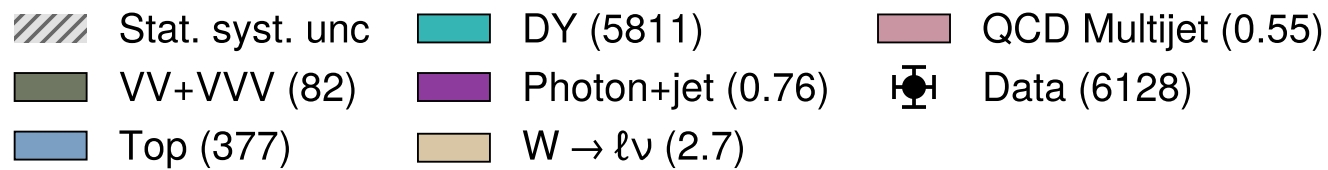
Data/MC



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

0

1

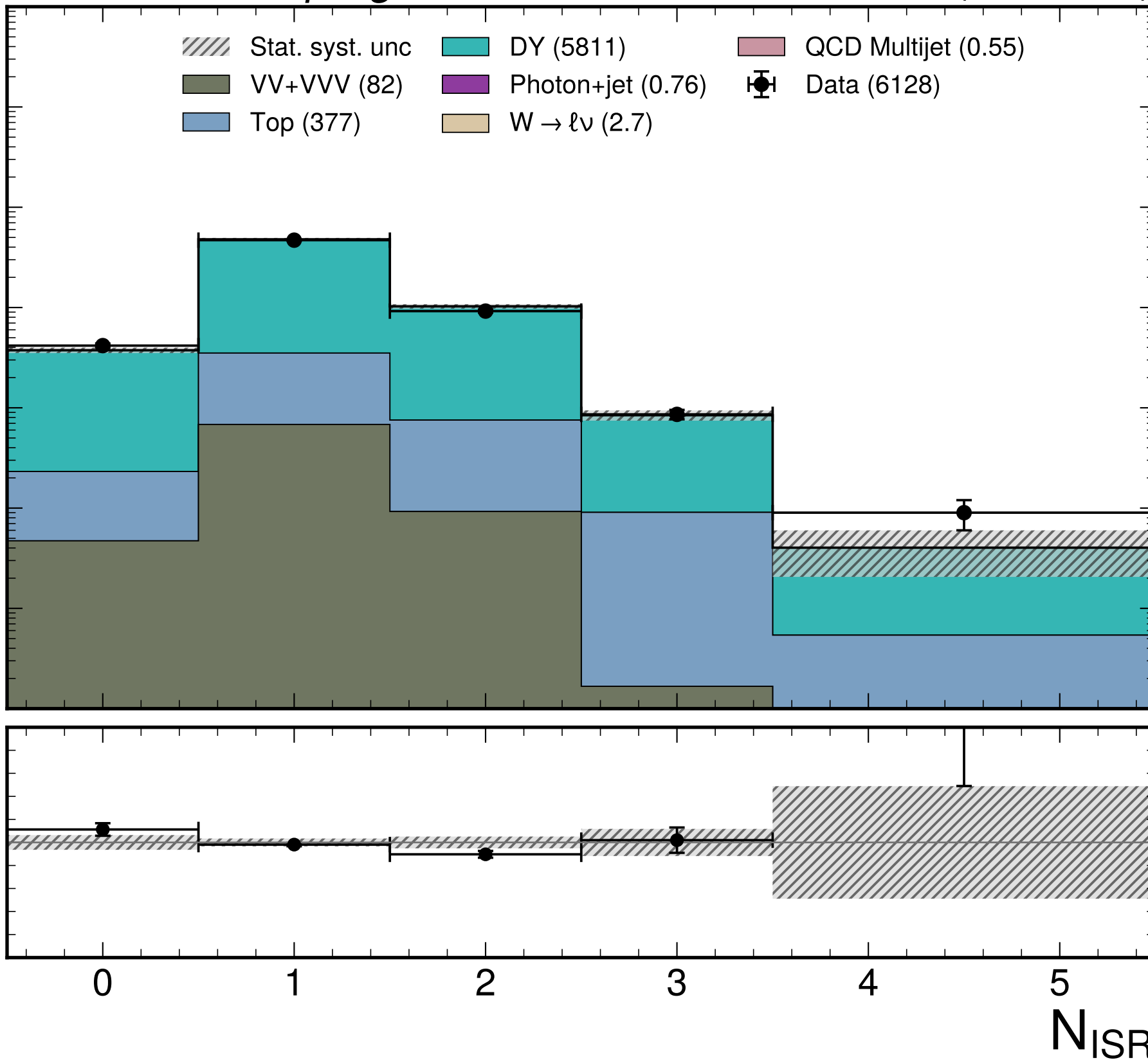
2

3

4

5

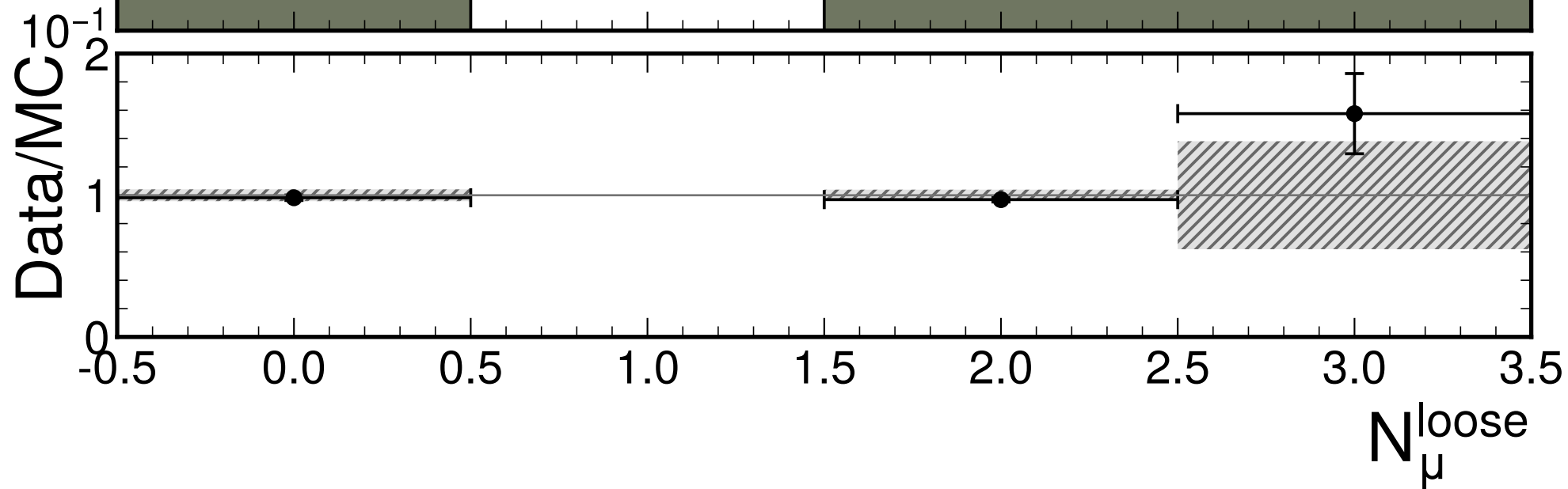
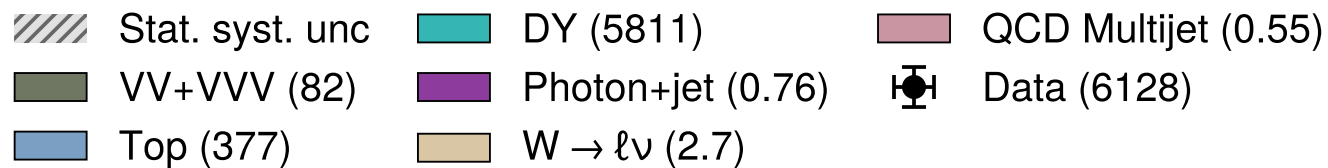
N_{ISR}



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

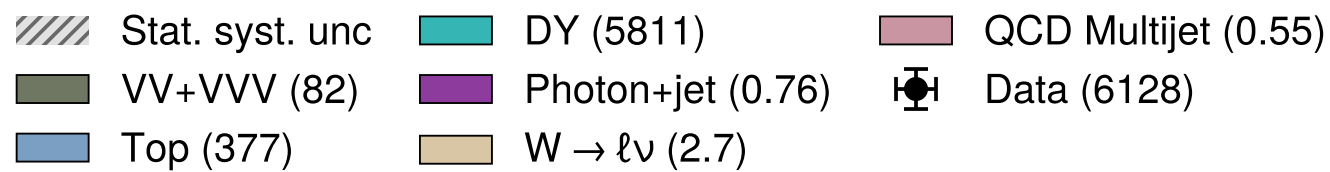
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

2

0

1

2

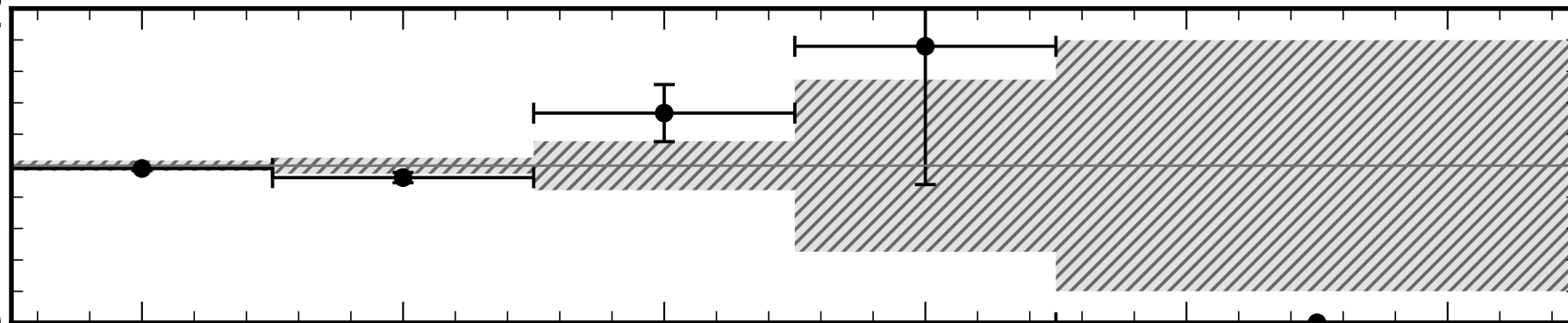
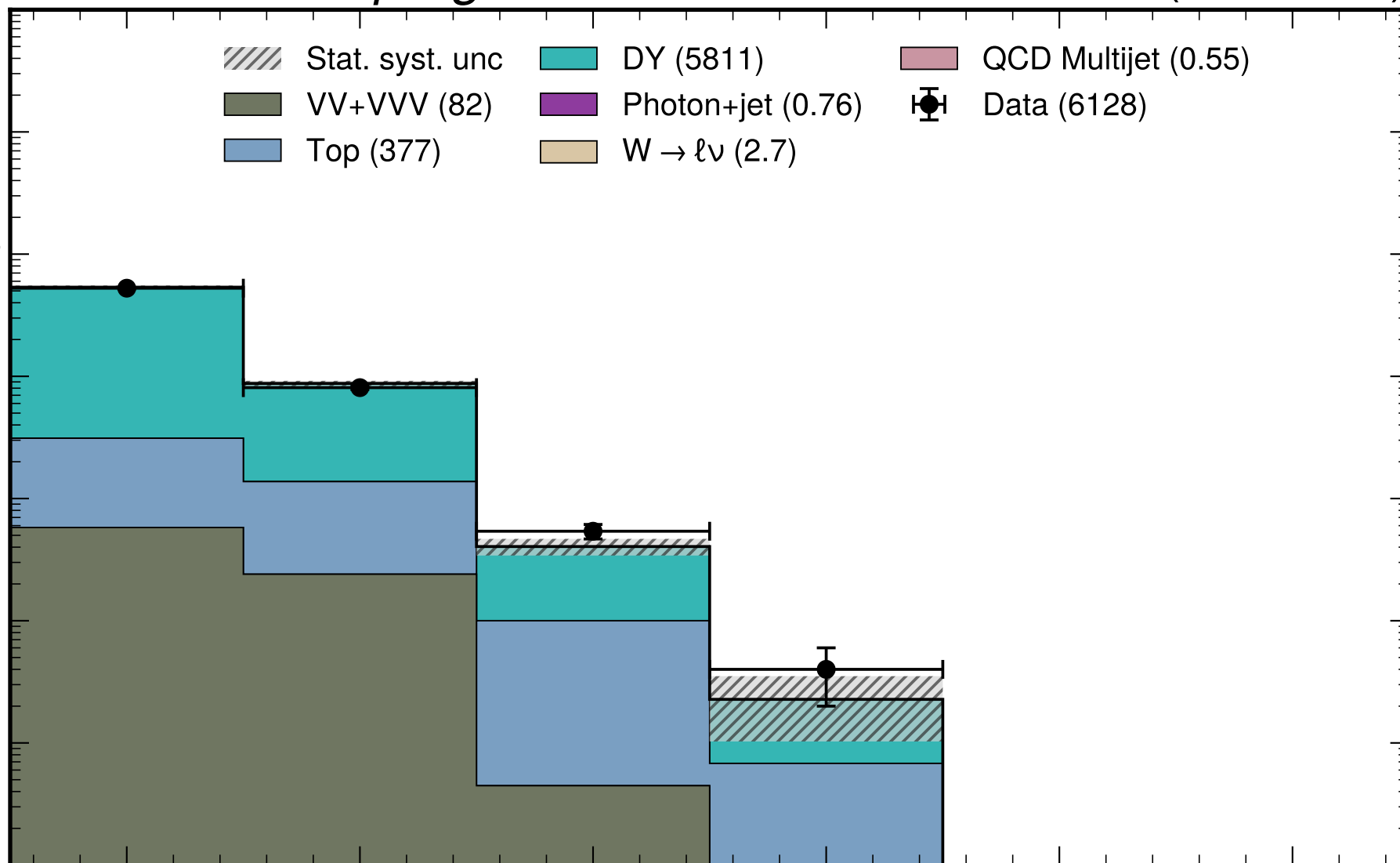
3

4

5

6

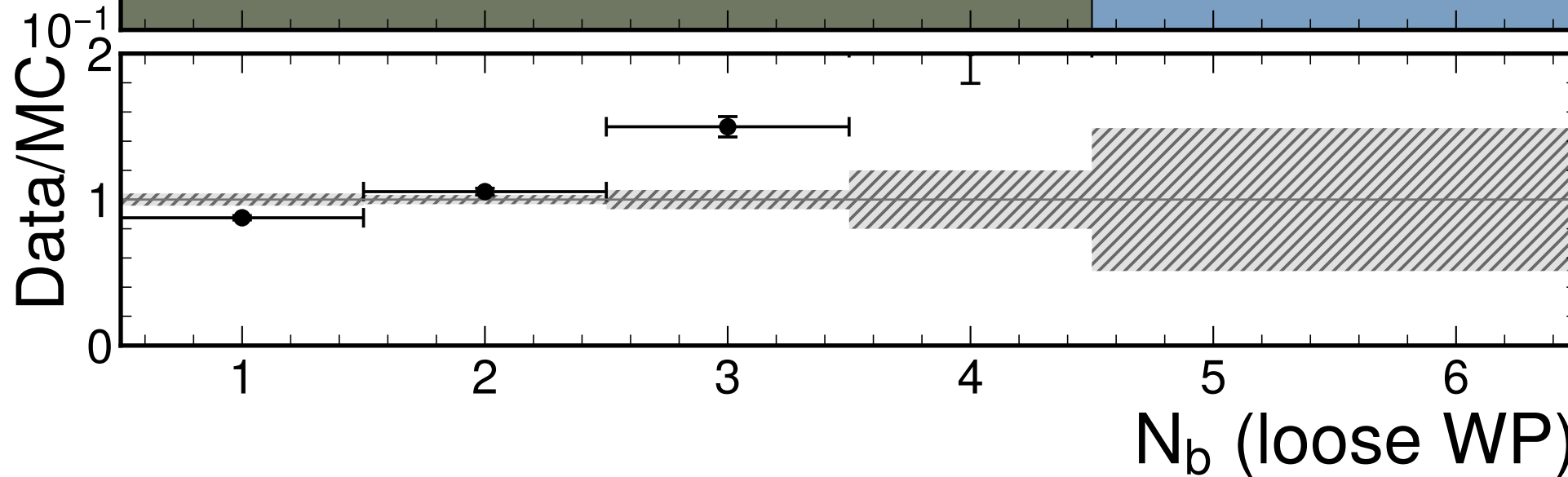
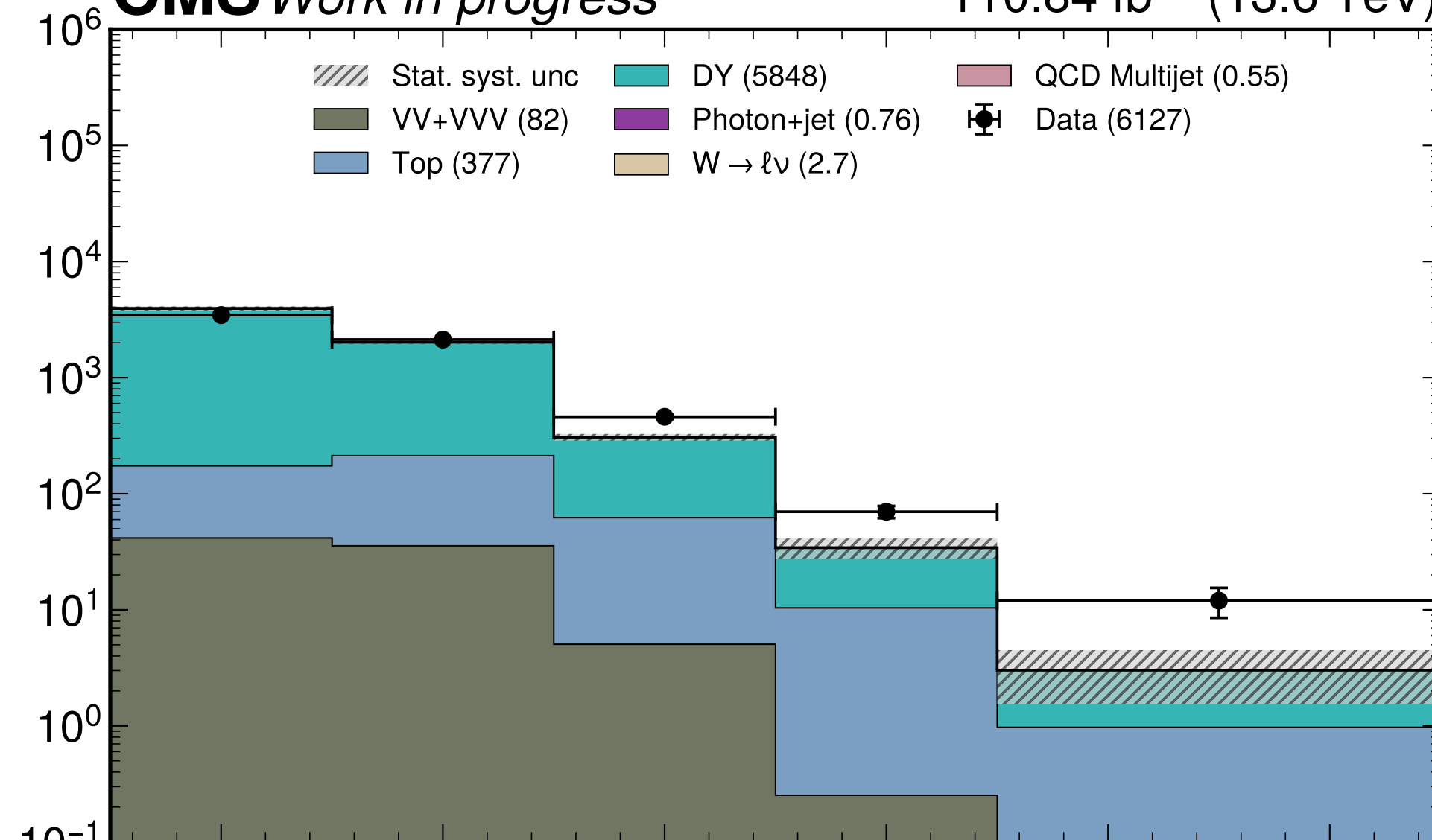
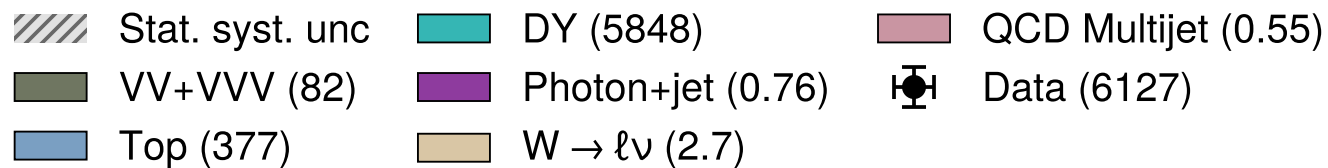
N_b



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

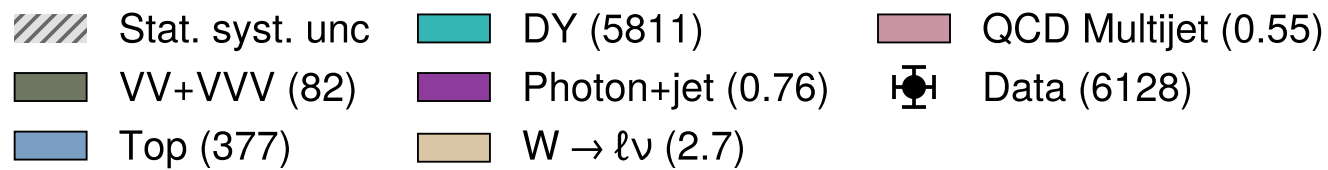
Events



CMS *Work in progress*

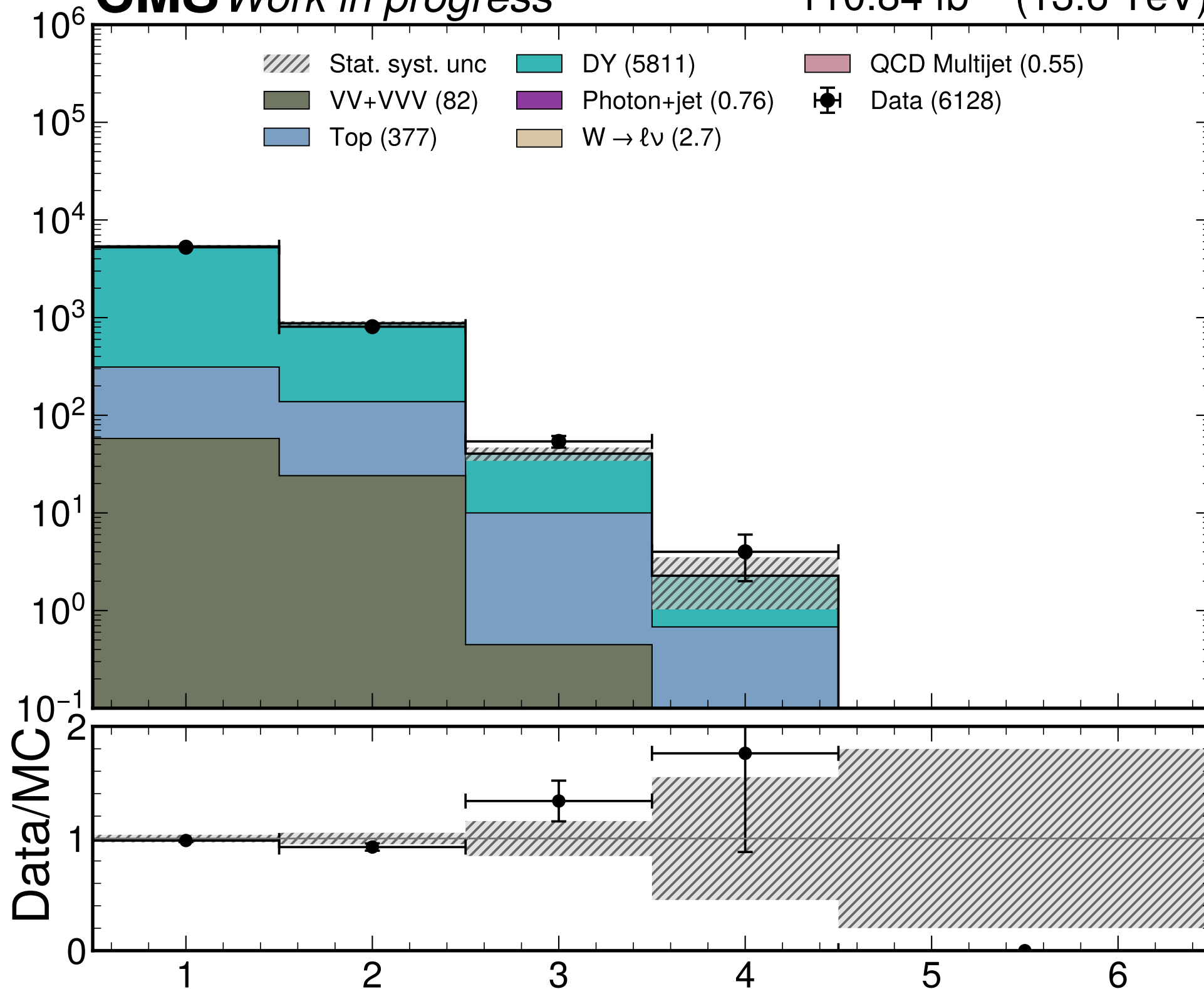
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

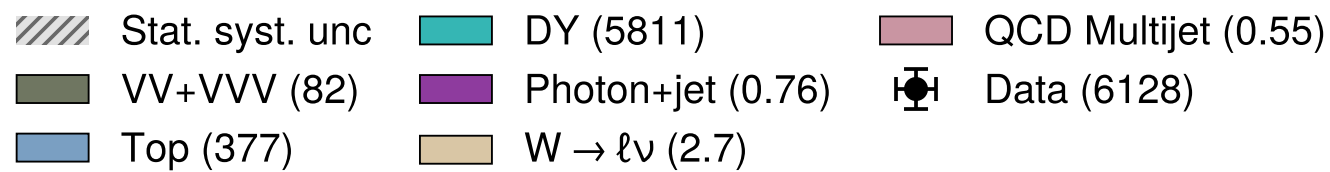
N_b



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

0

2

4

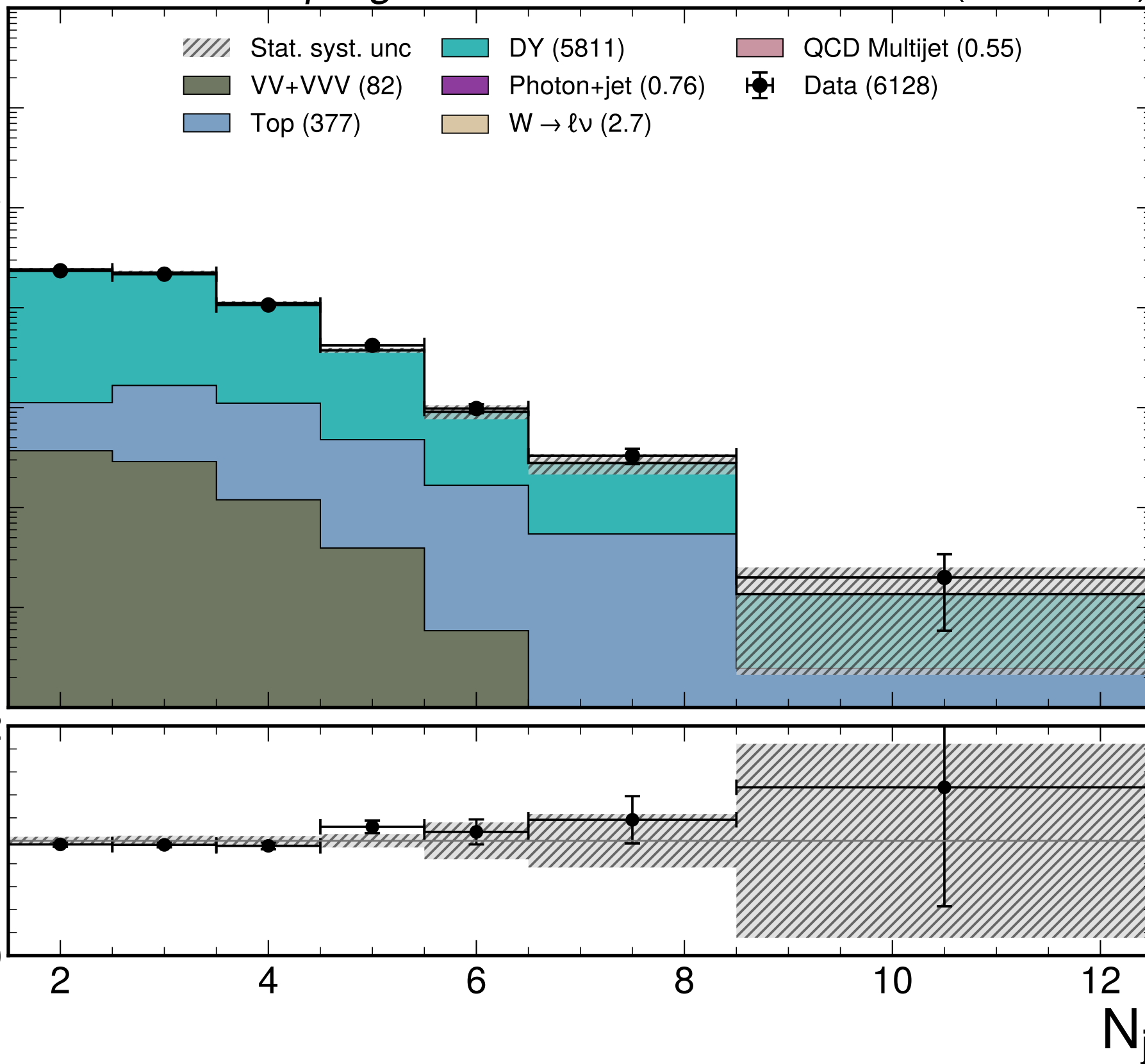
6

8

10

12

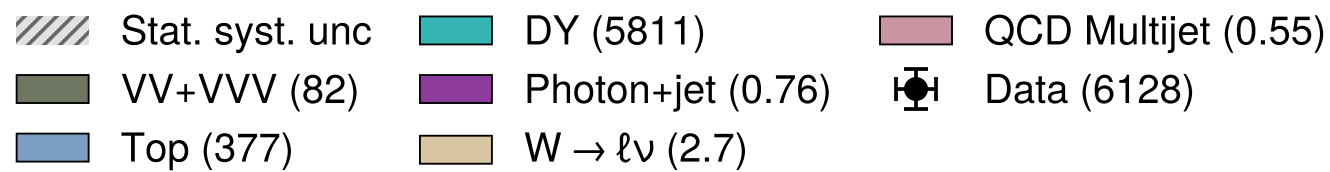
N_j



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹

0

2

4

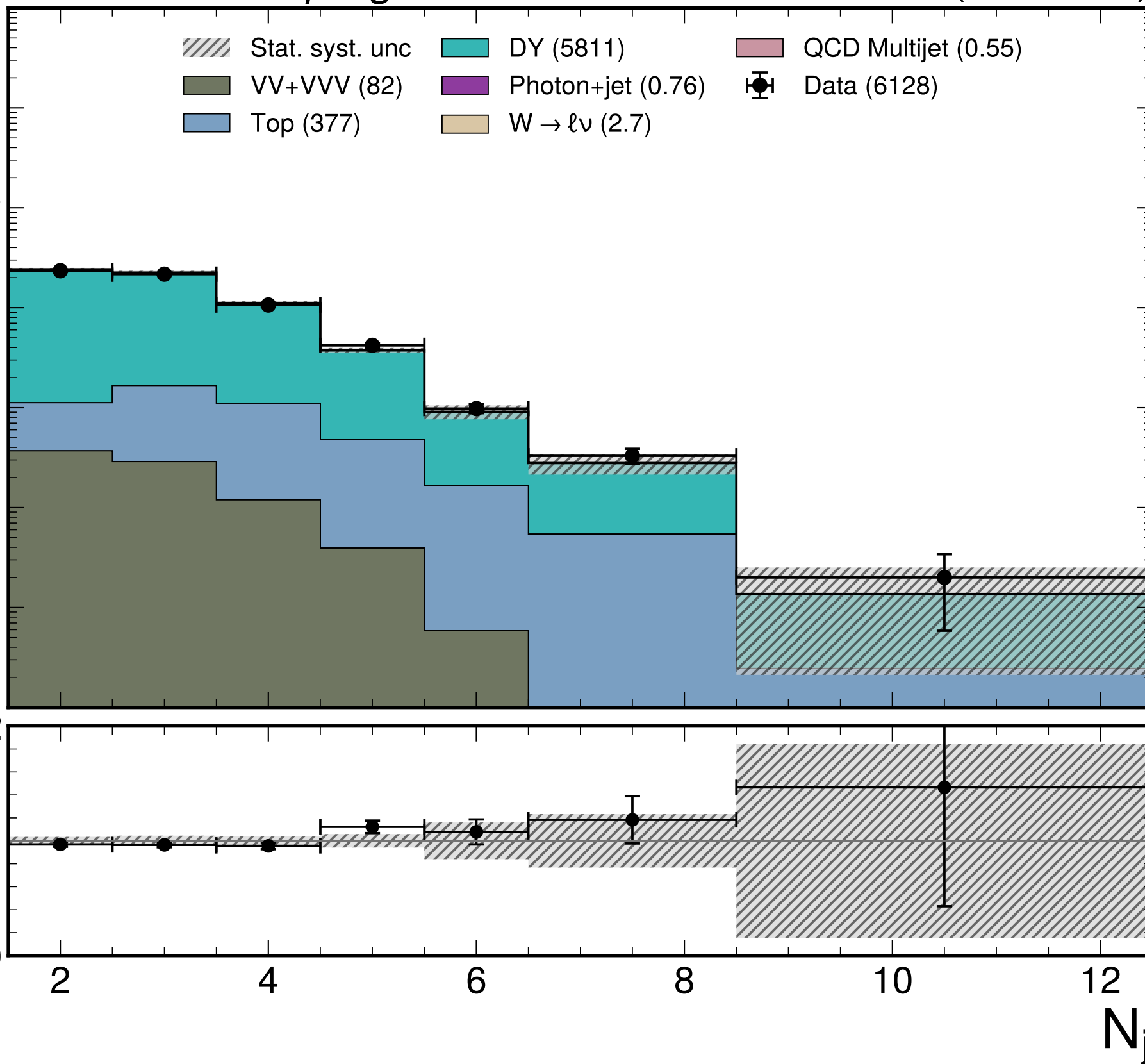
6

8

10

12

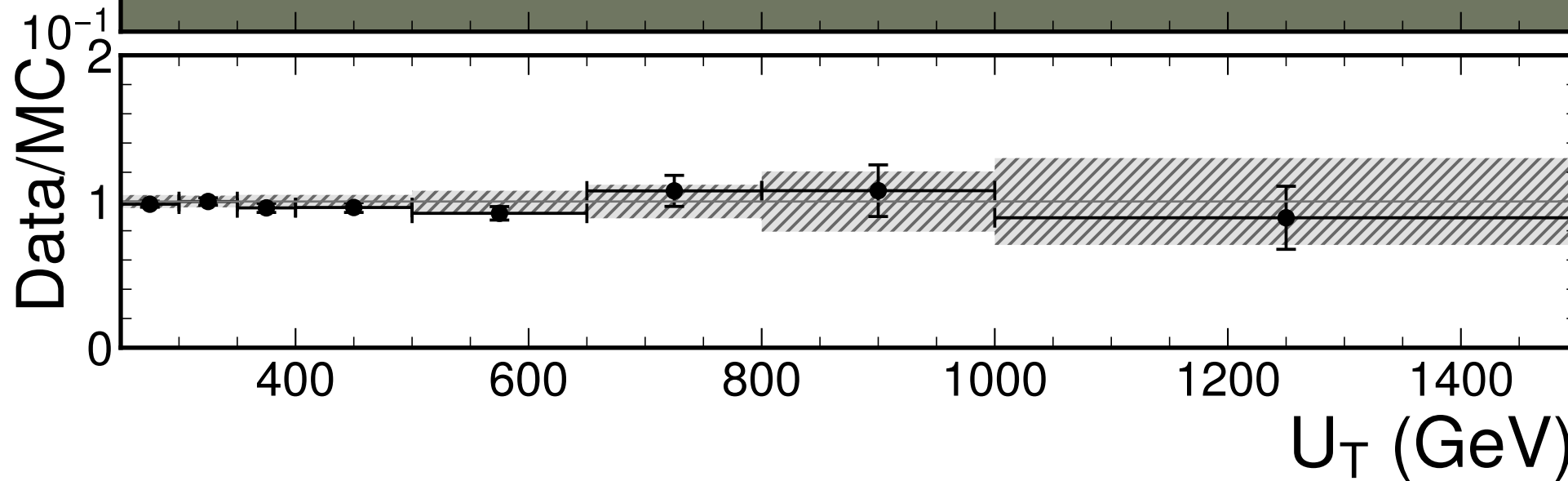
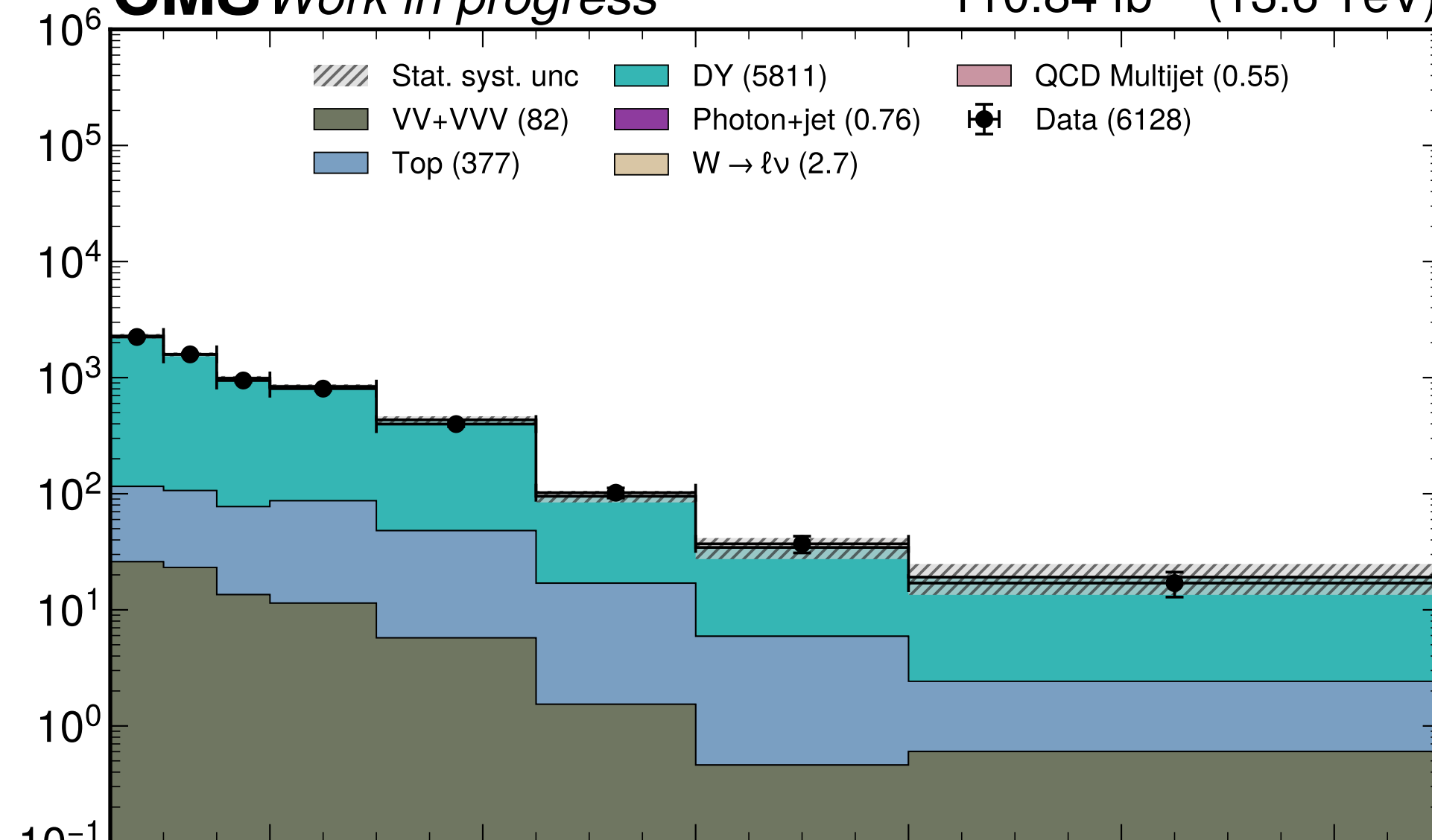
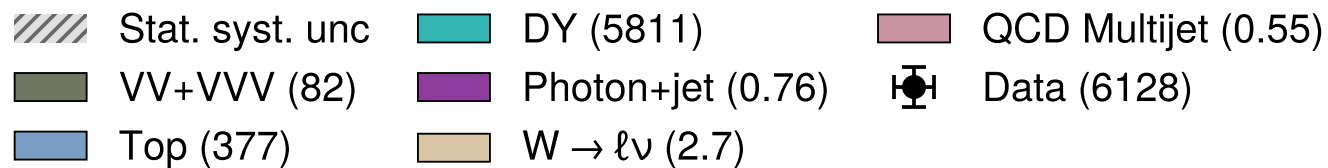
N_j



CMS *Work in progress*

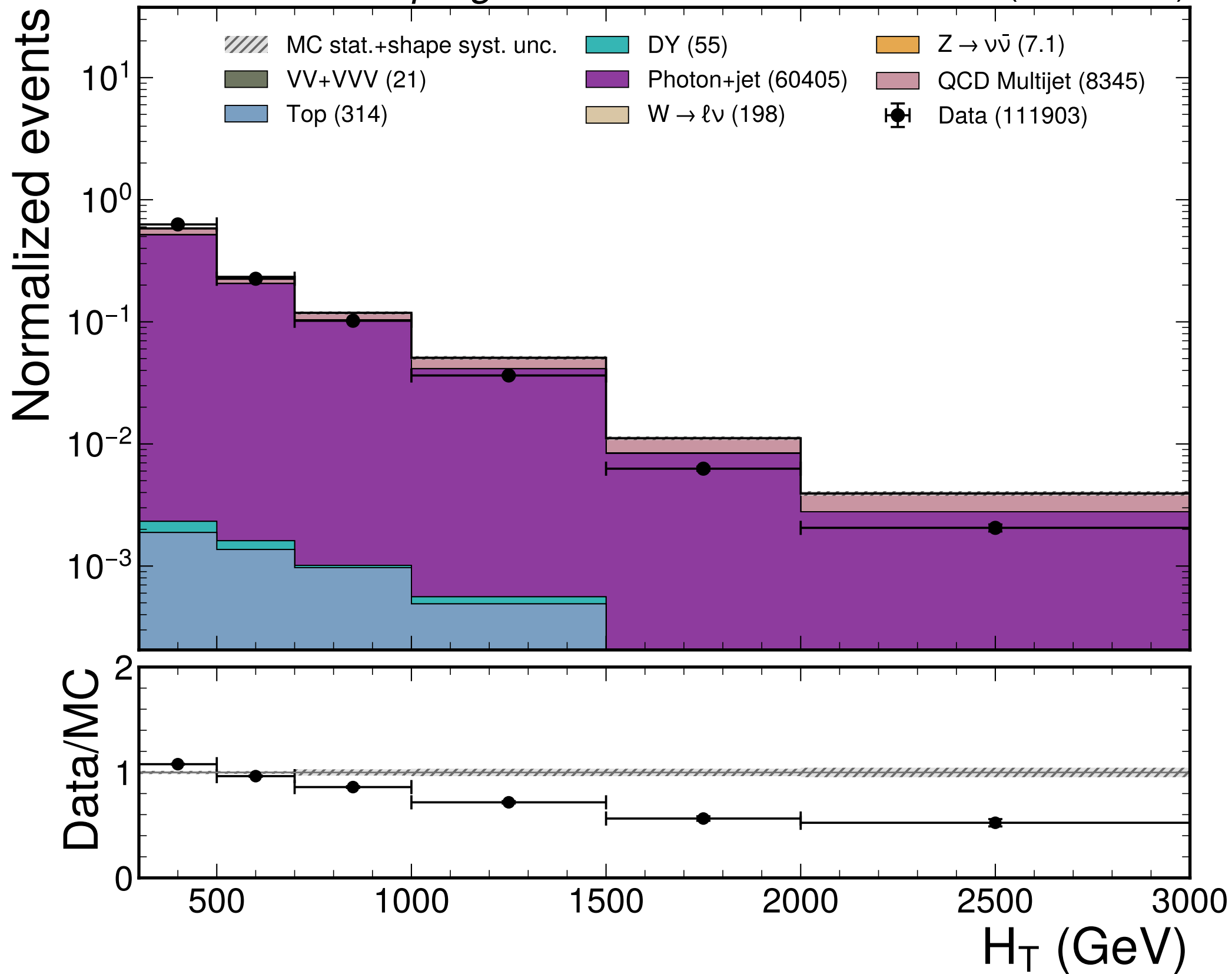
110.84 fb⁻¹ (13.6 TeV)

Events



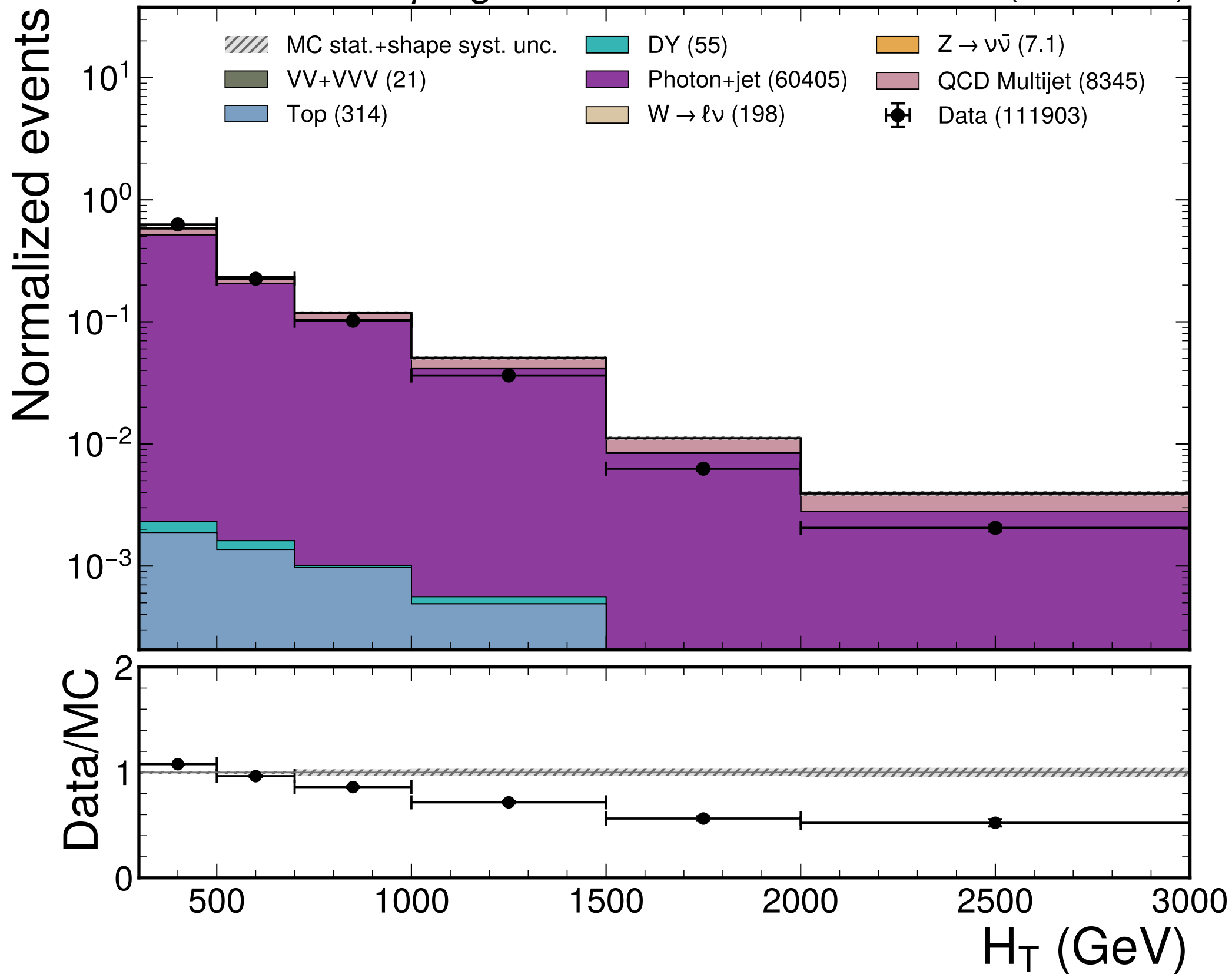
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



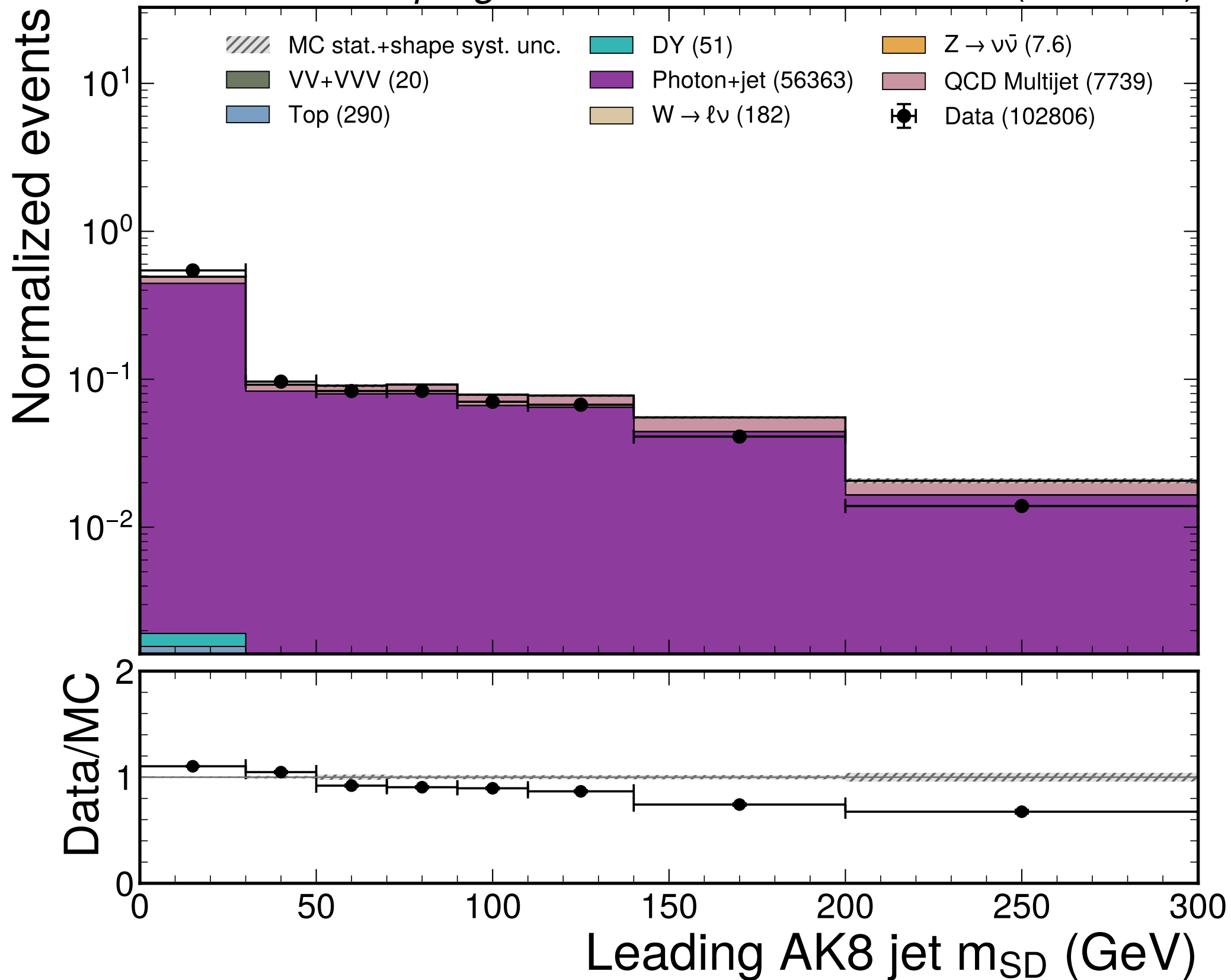
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



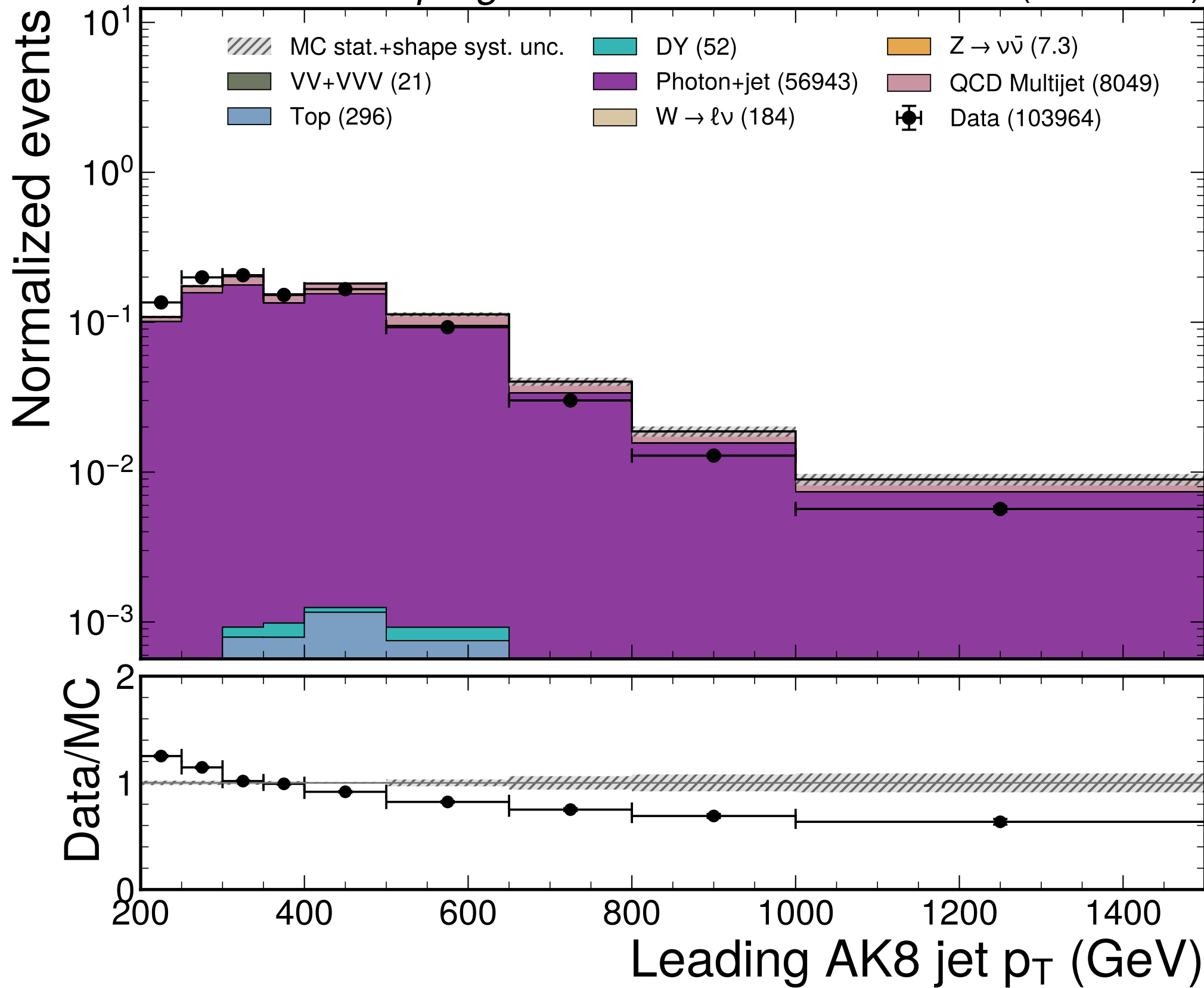
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



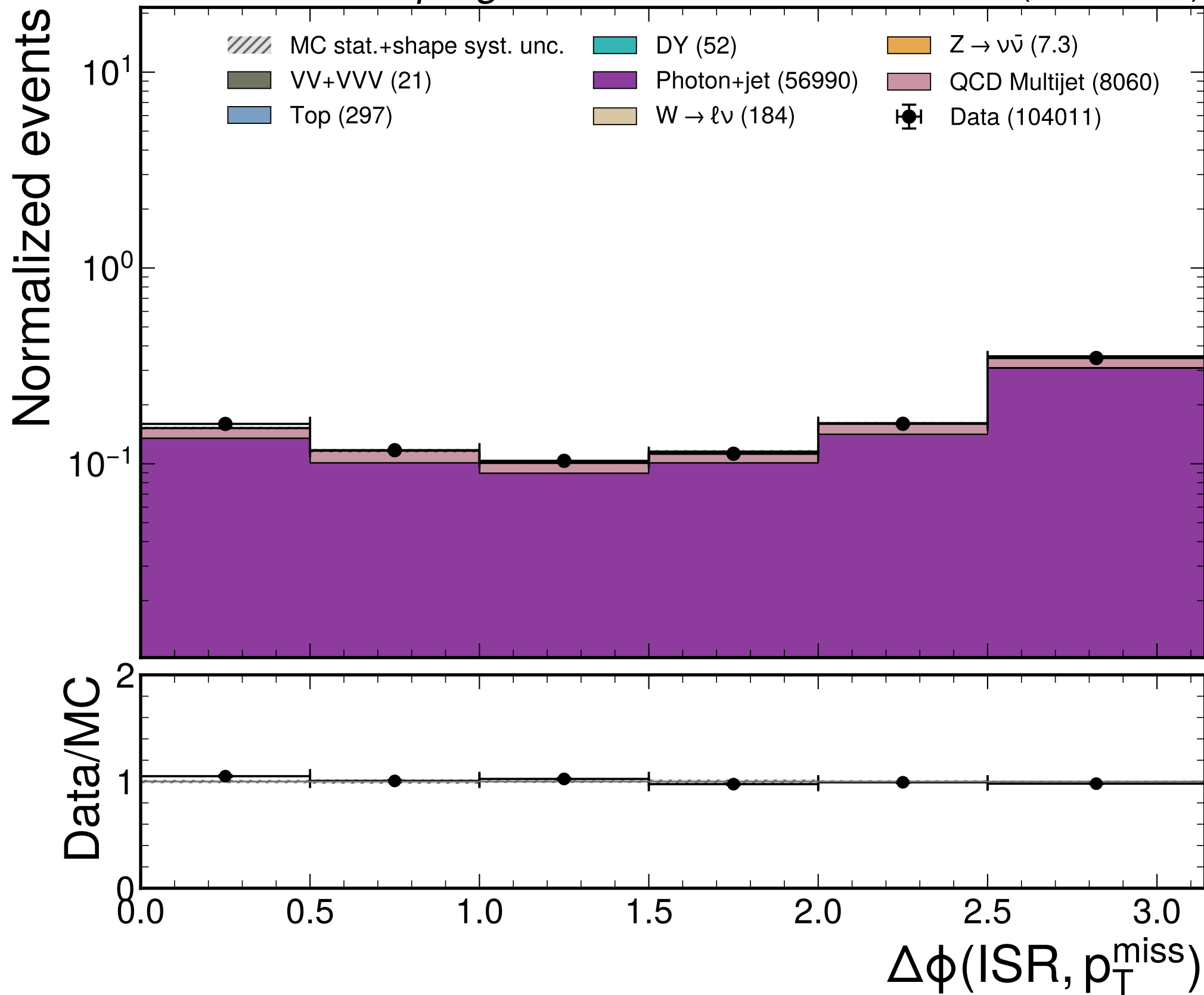
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



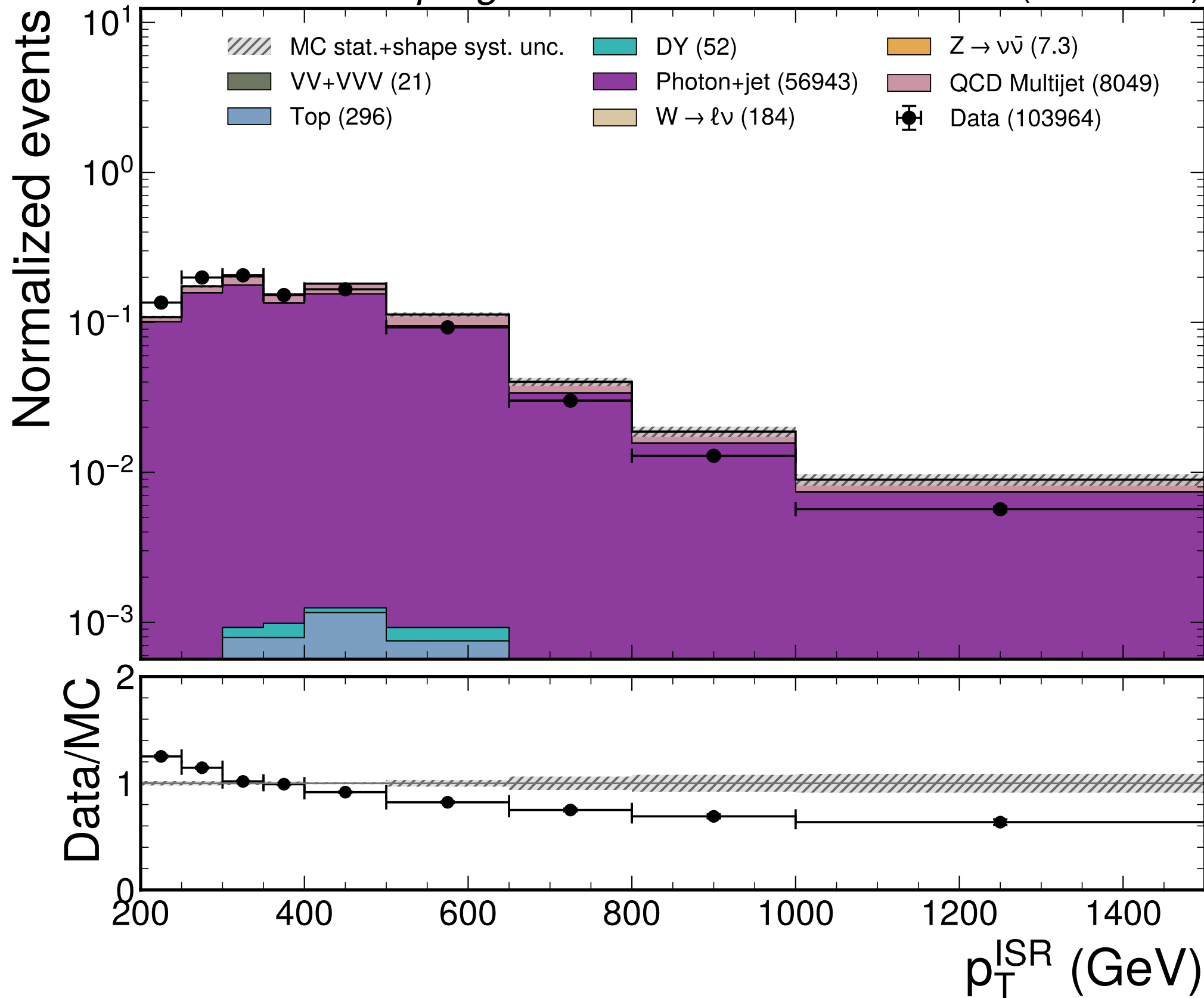
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



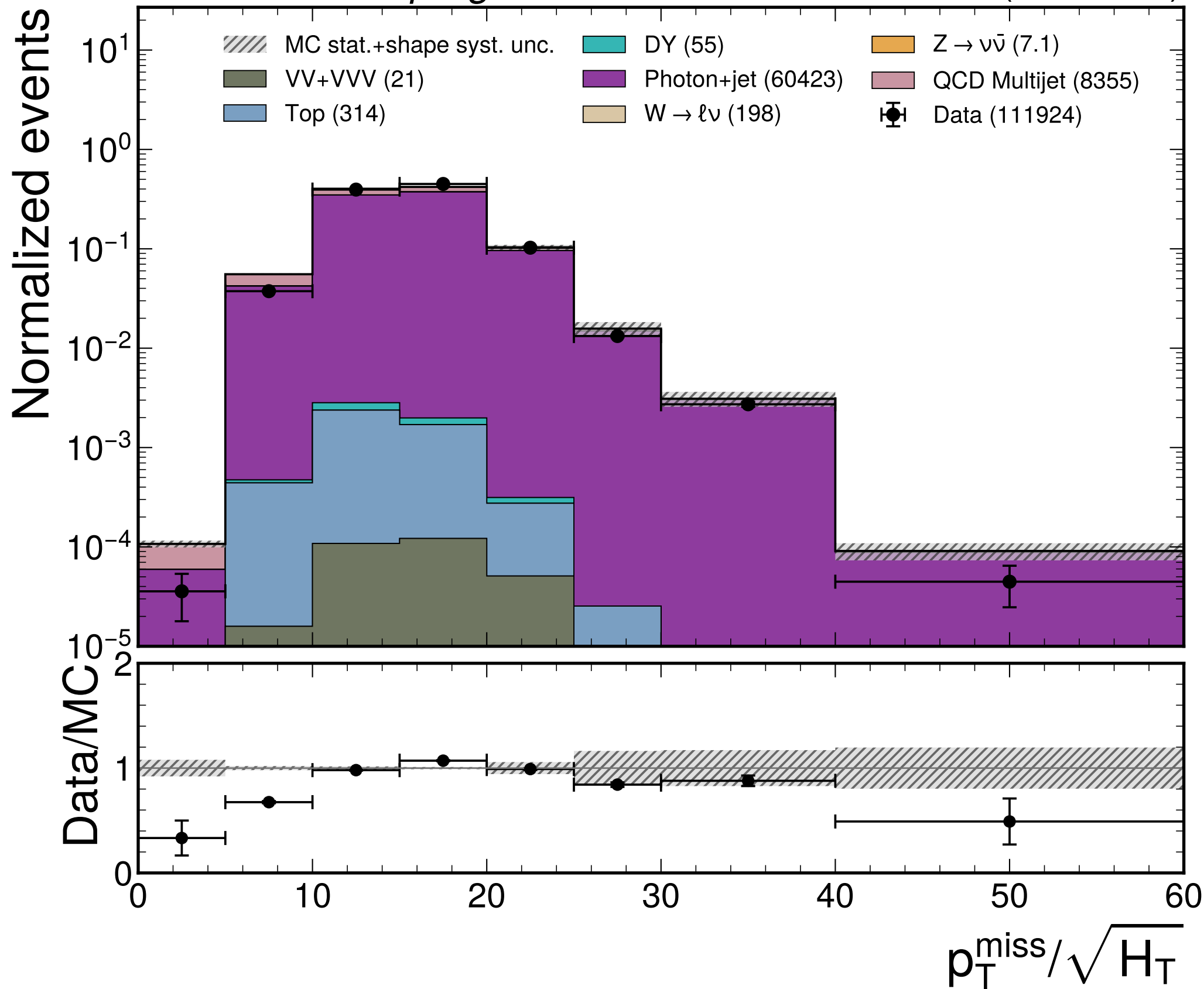
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



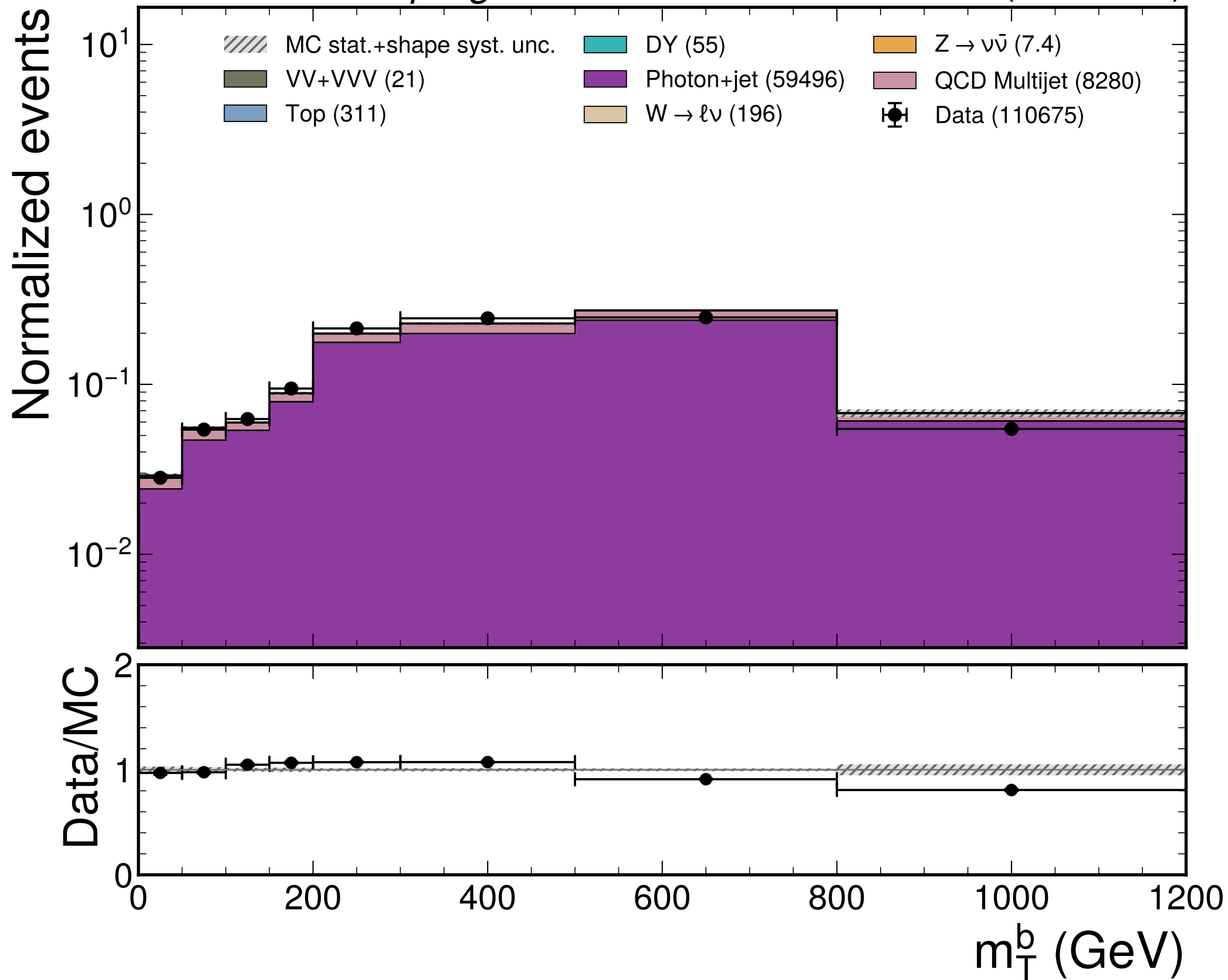
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



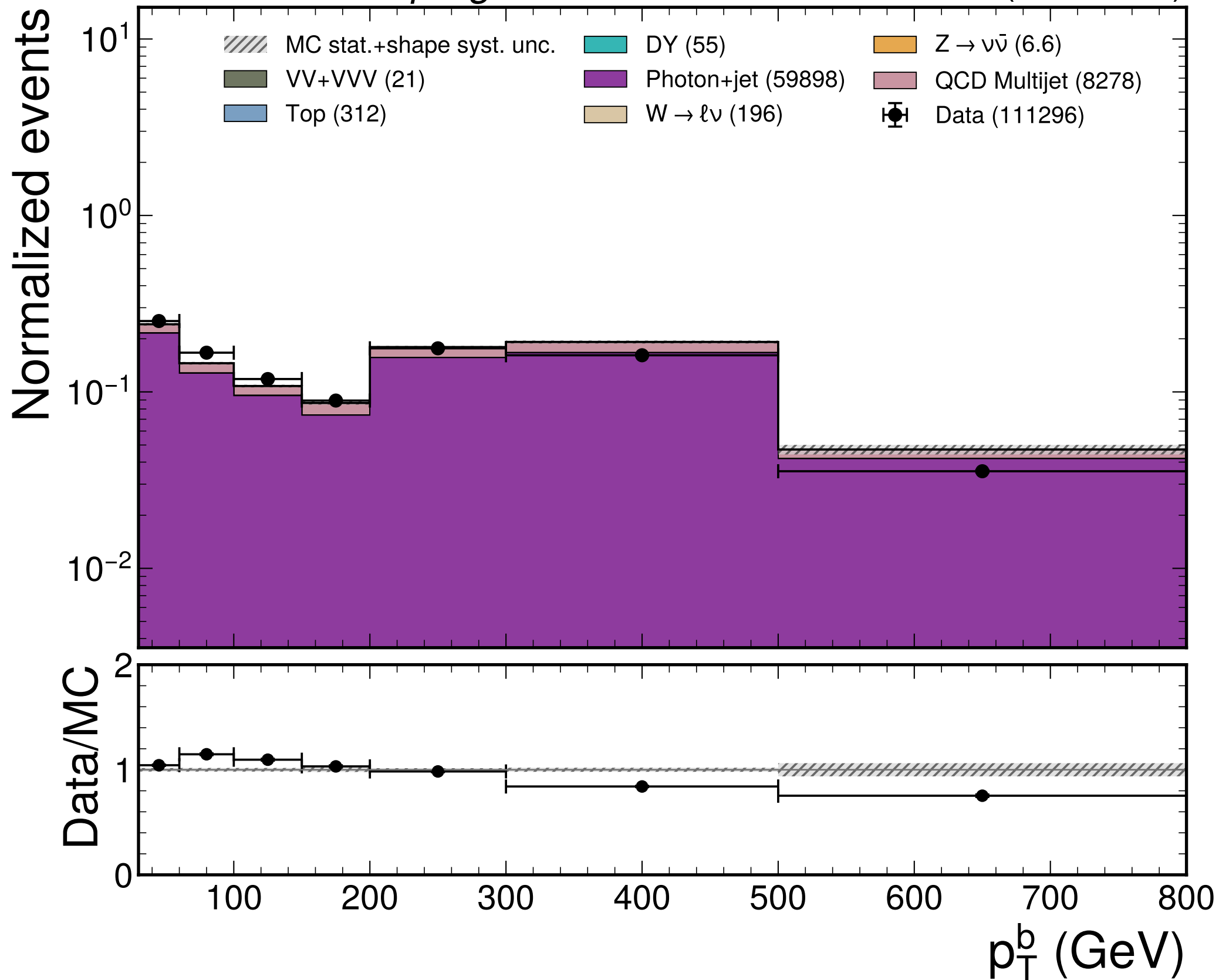
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



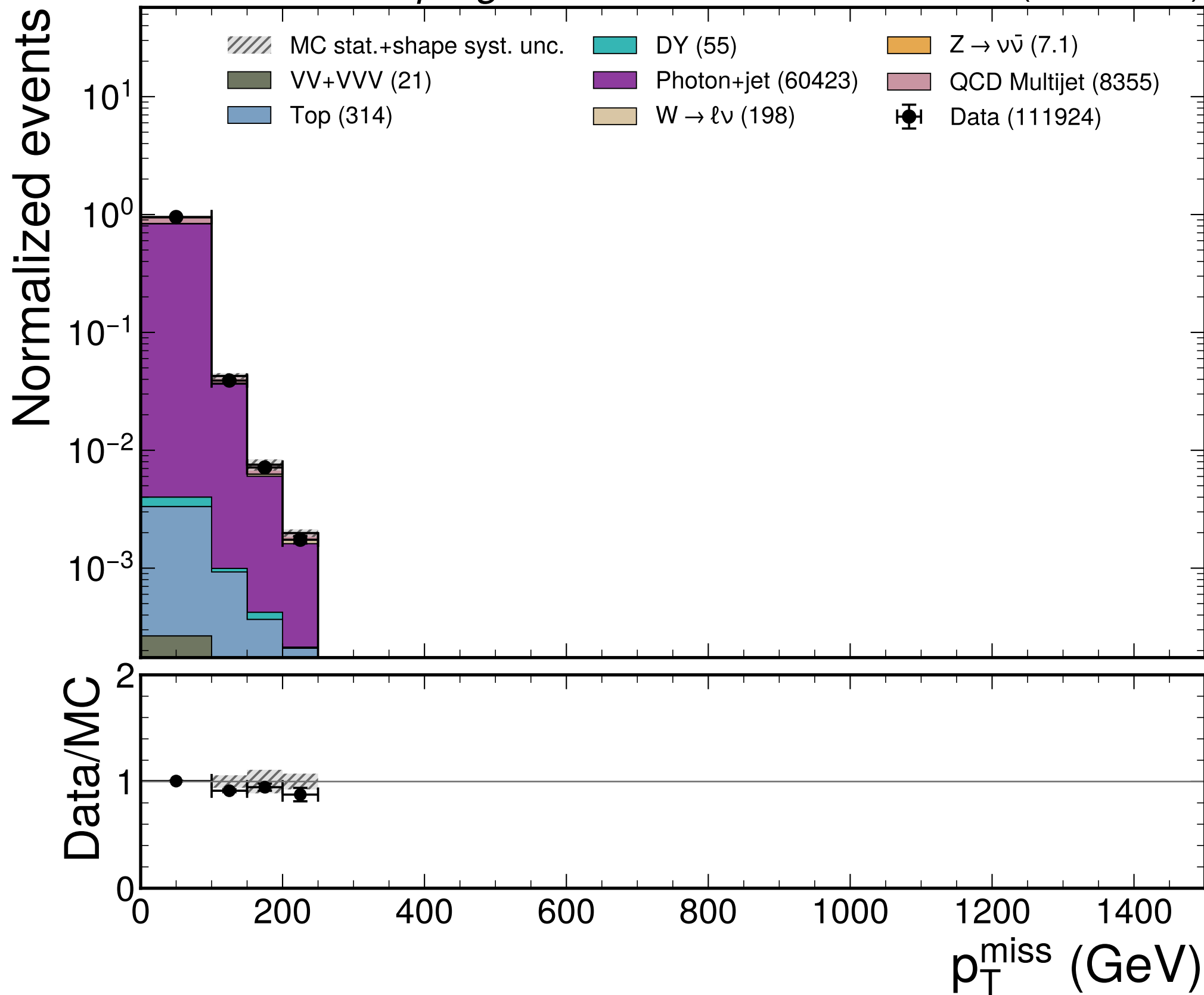
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



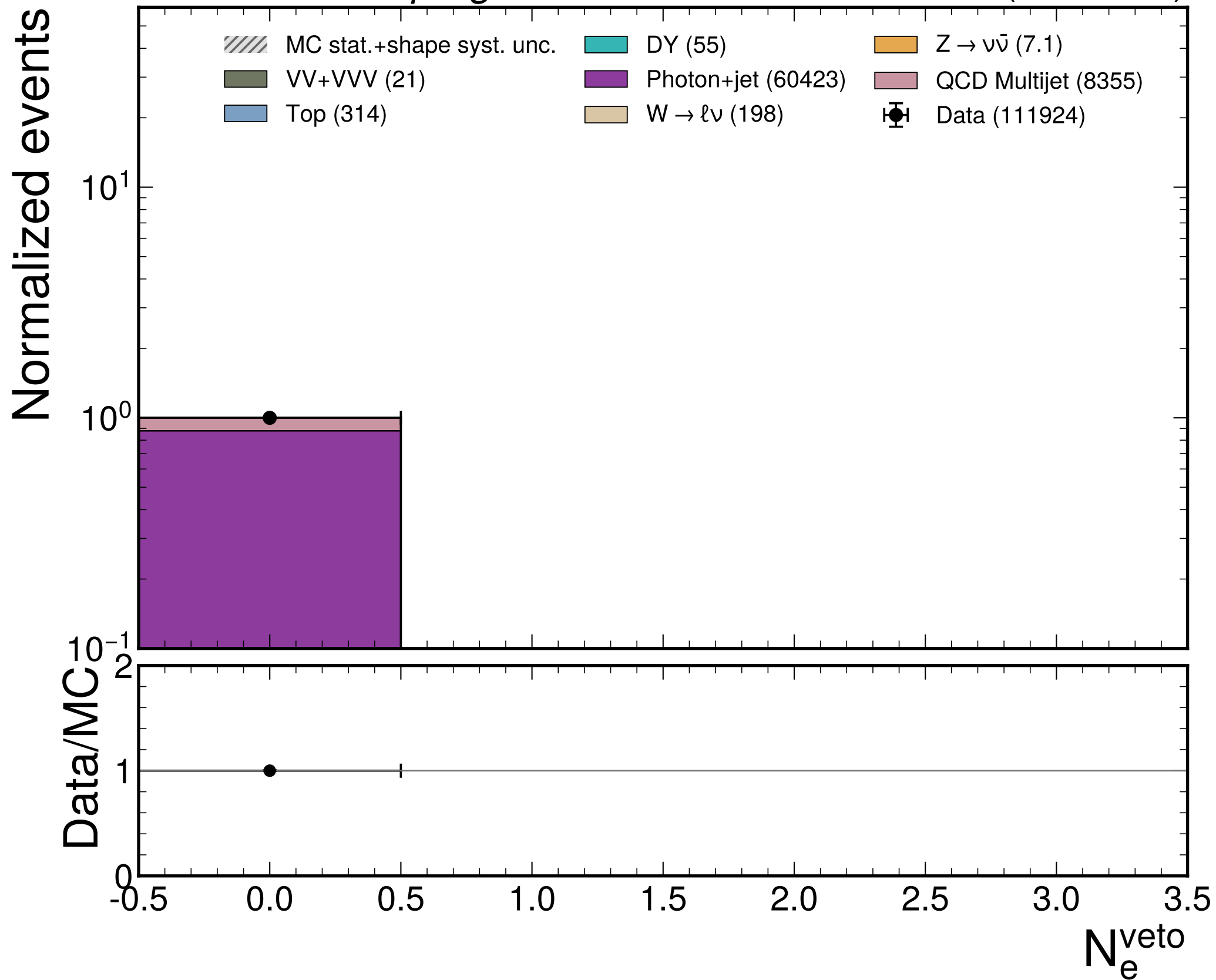
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



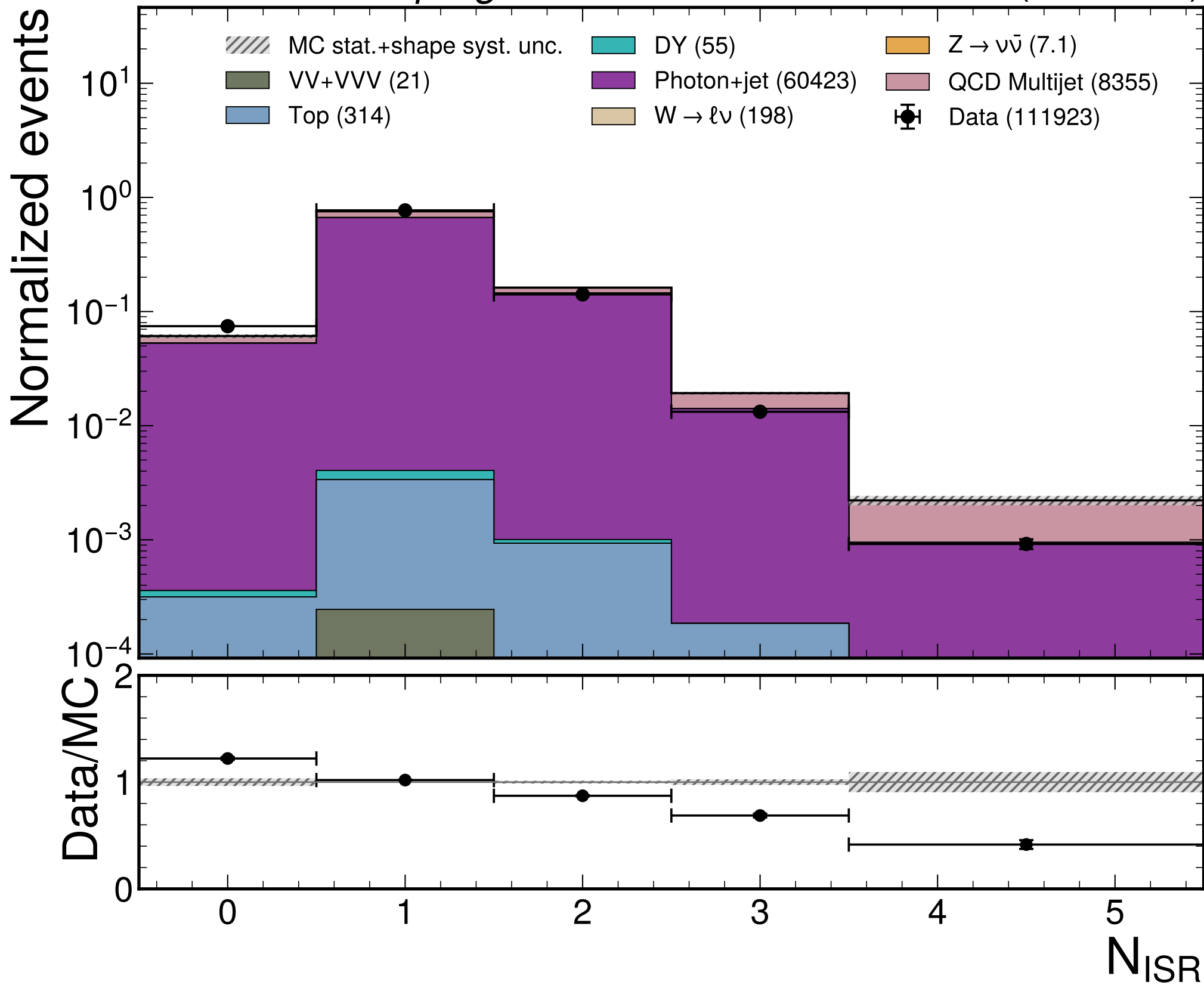
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



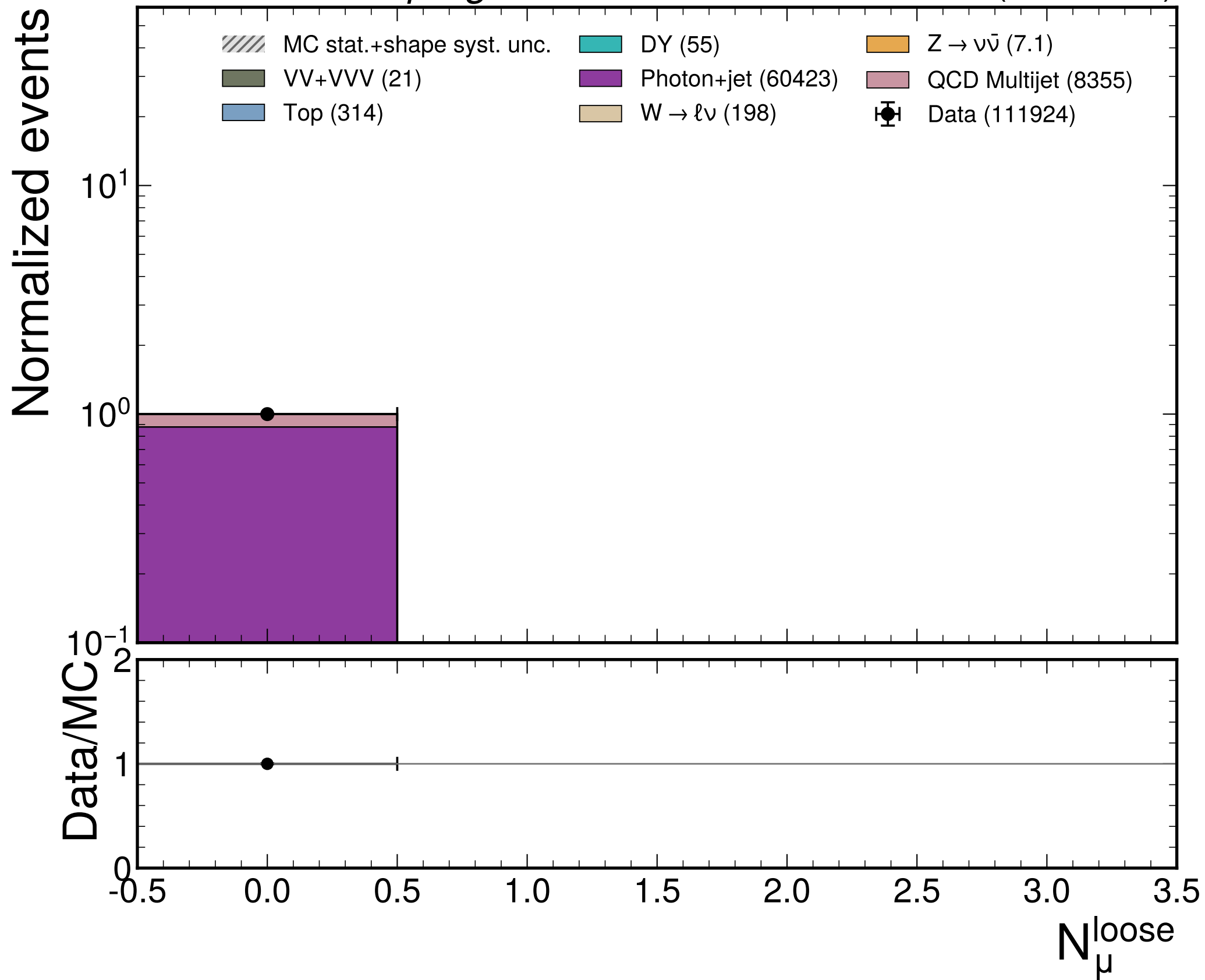
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



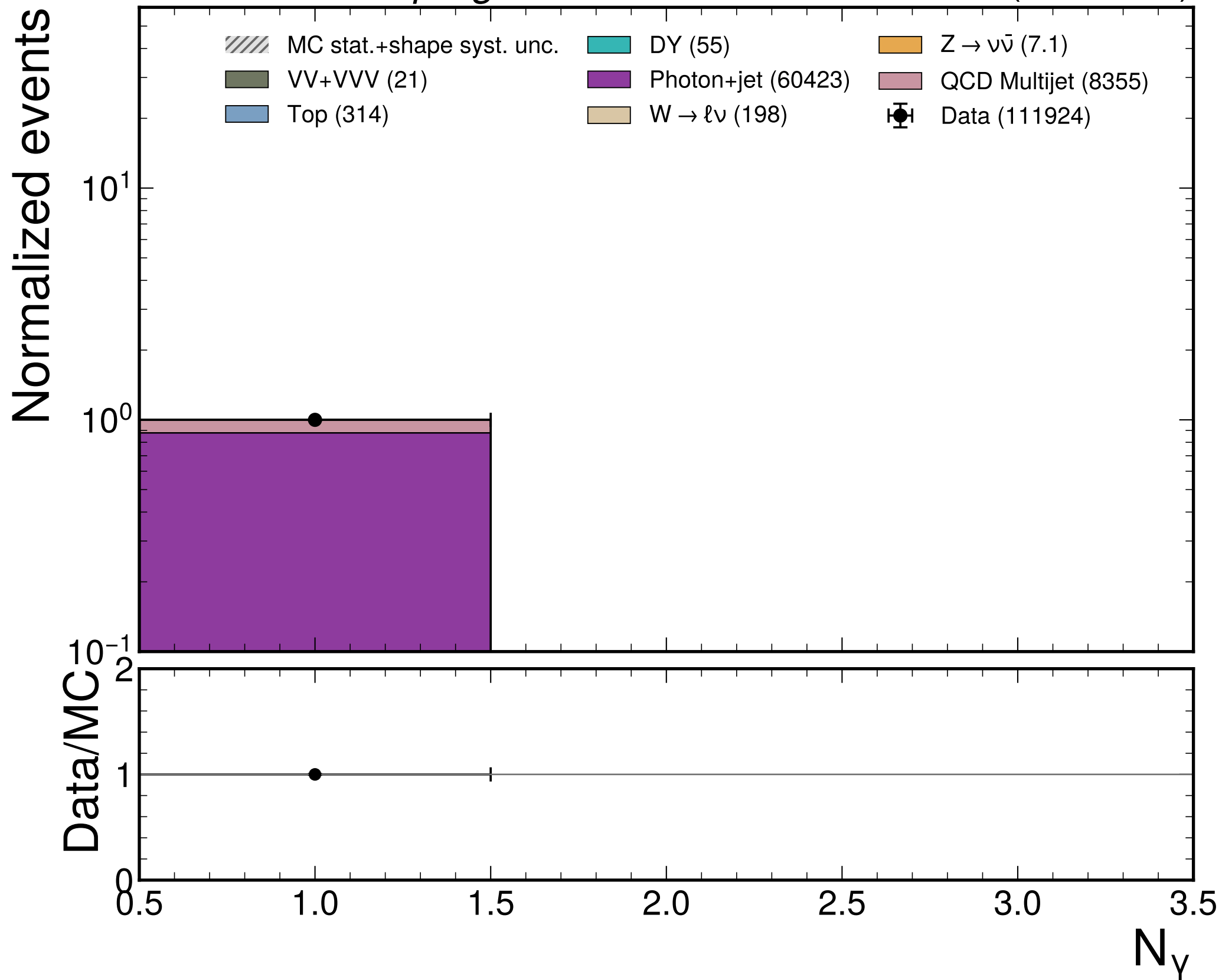
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



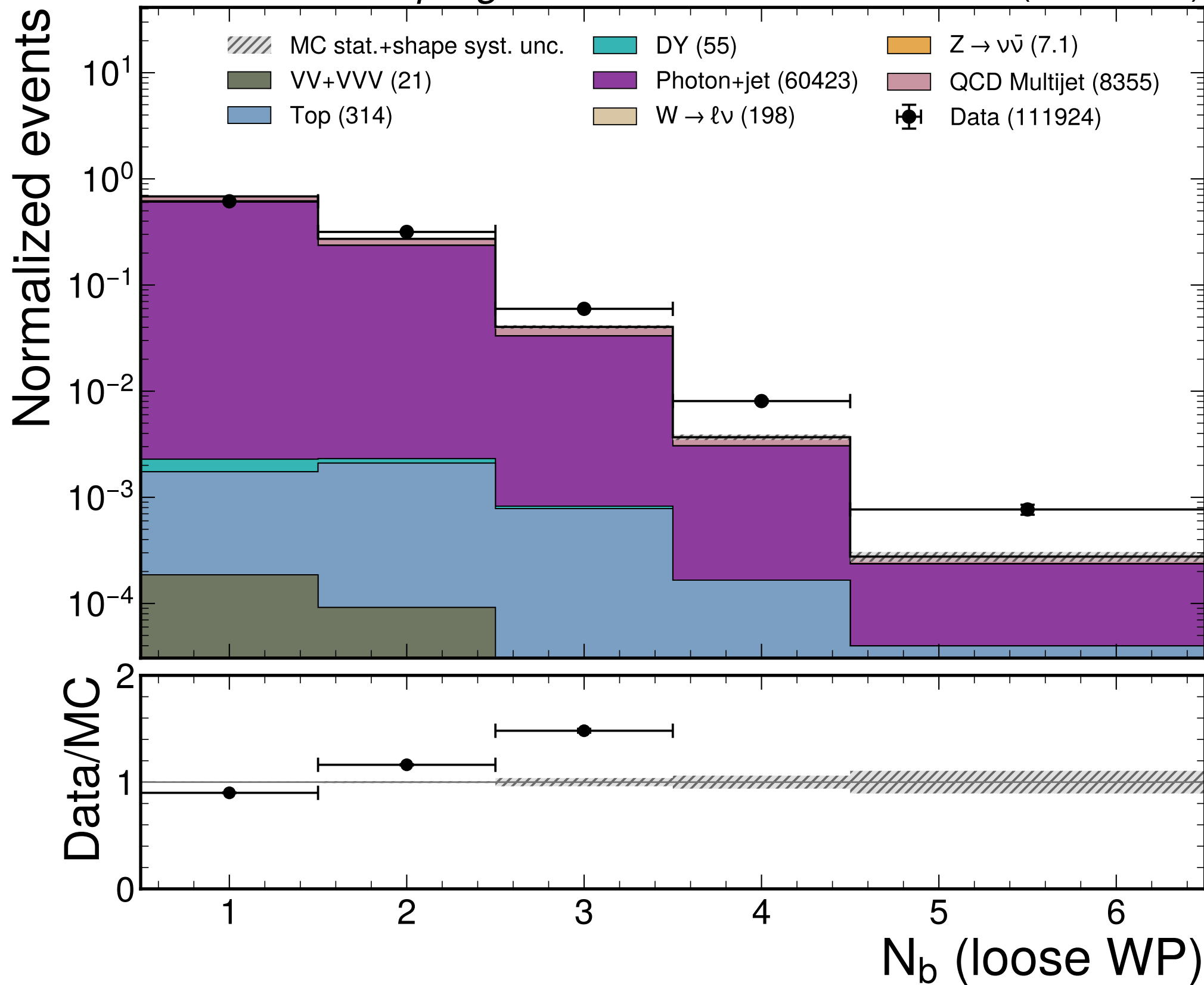
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



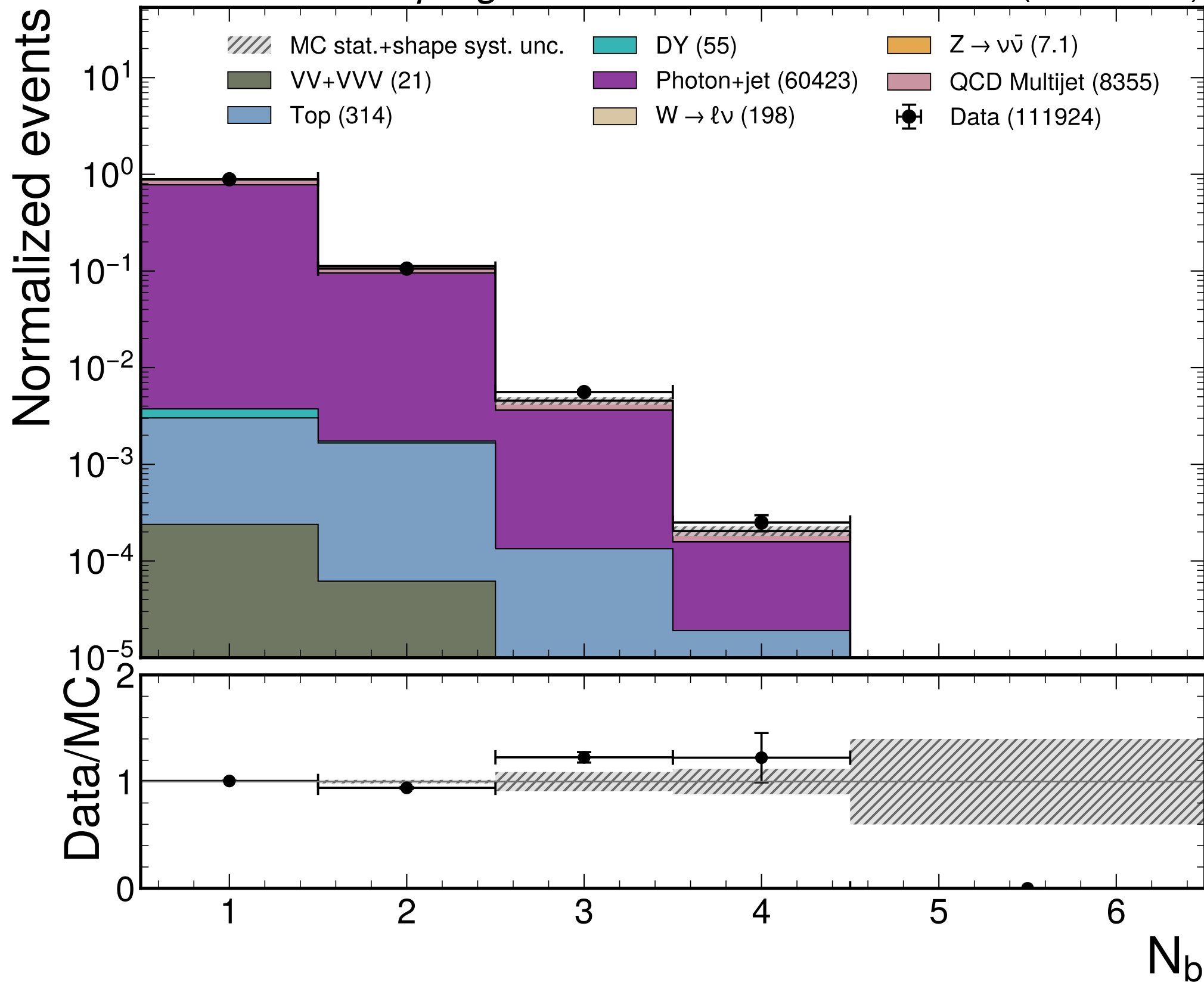
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



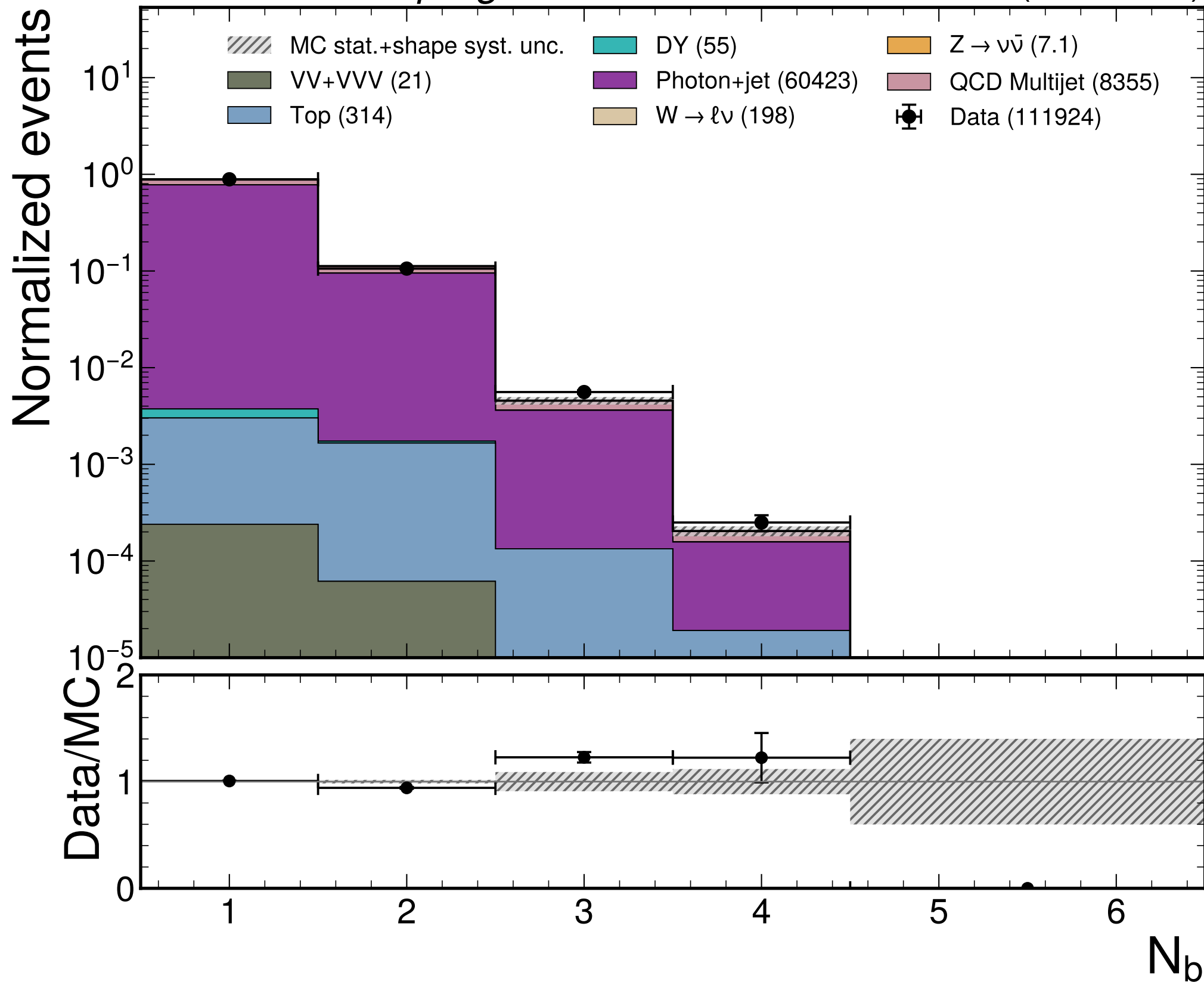
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



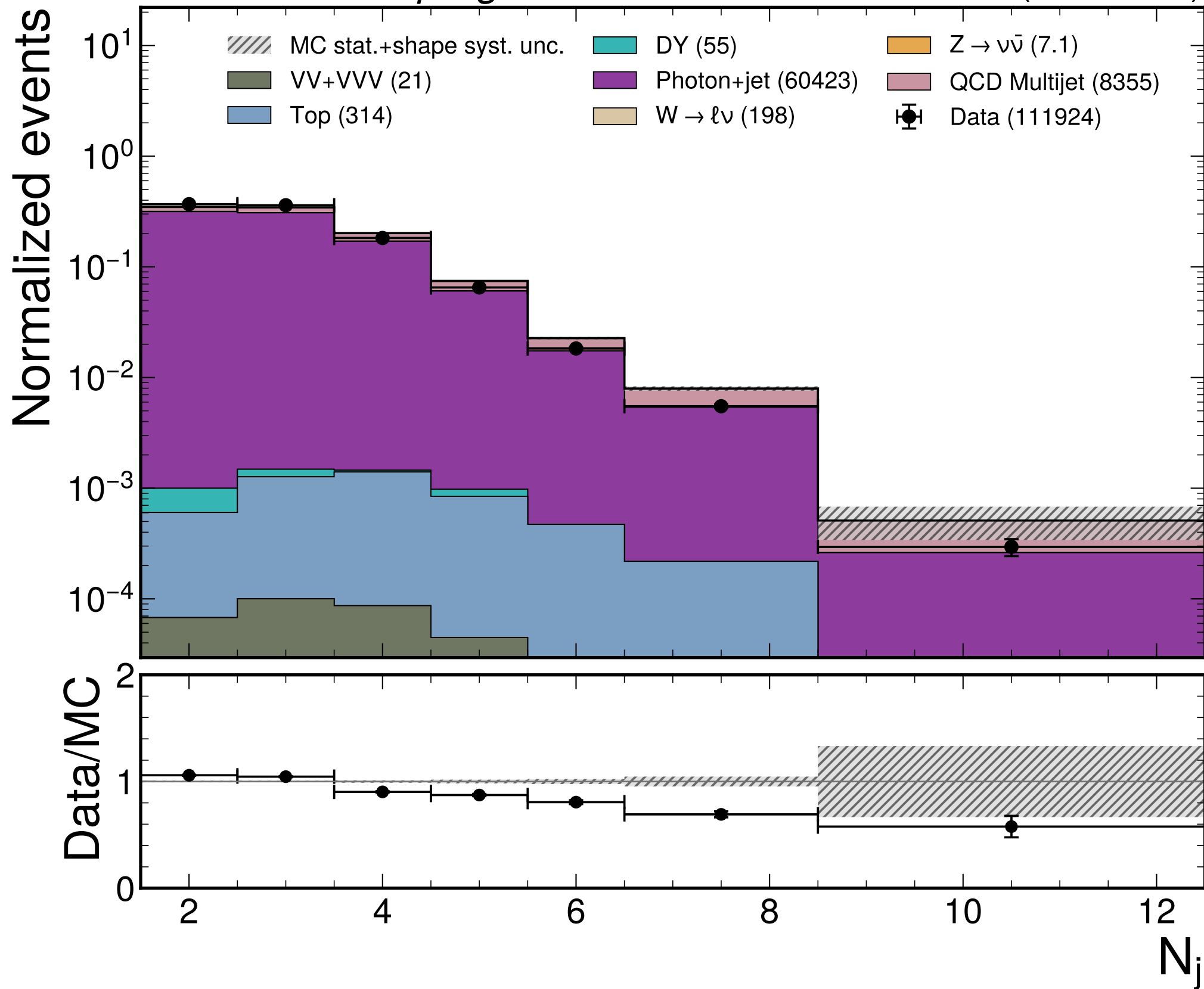
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



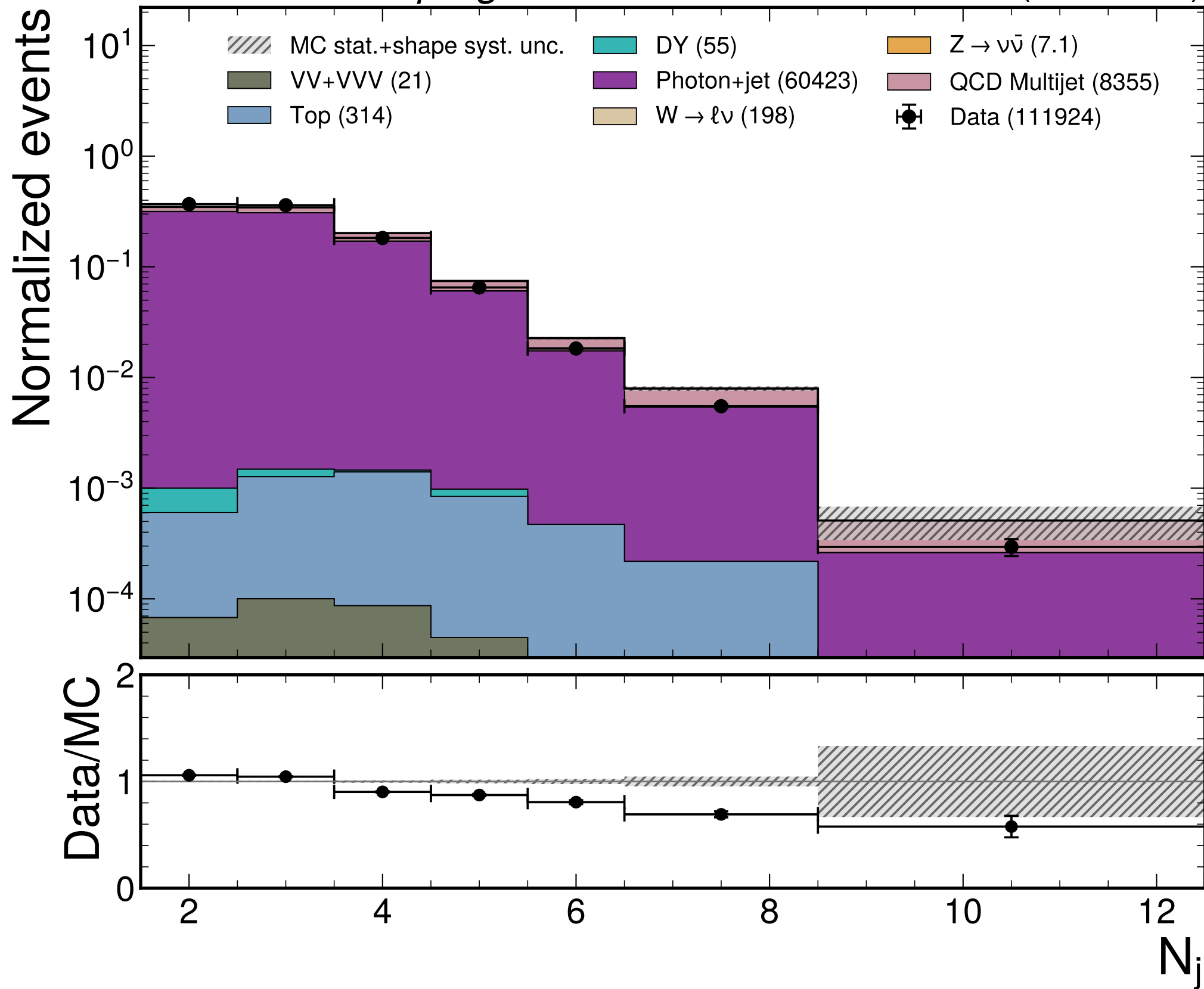
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



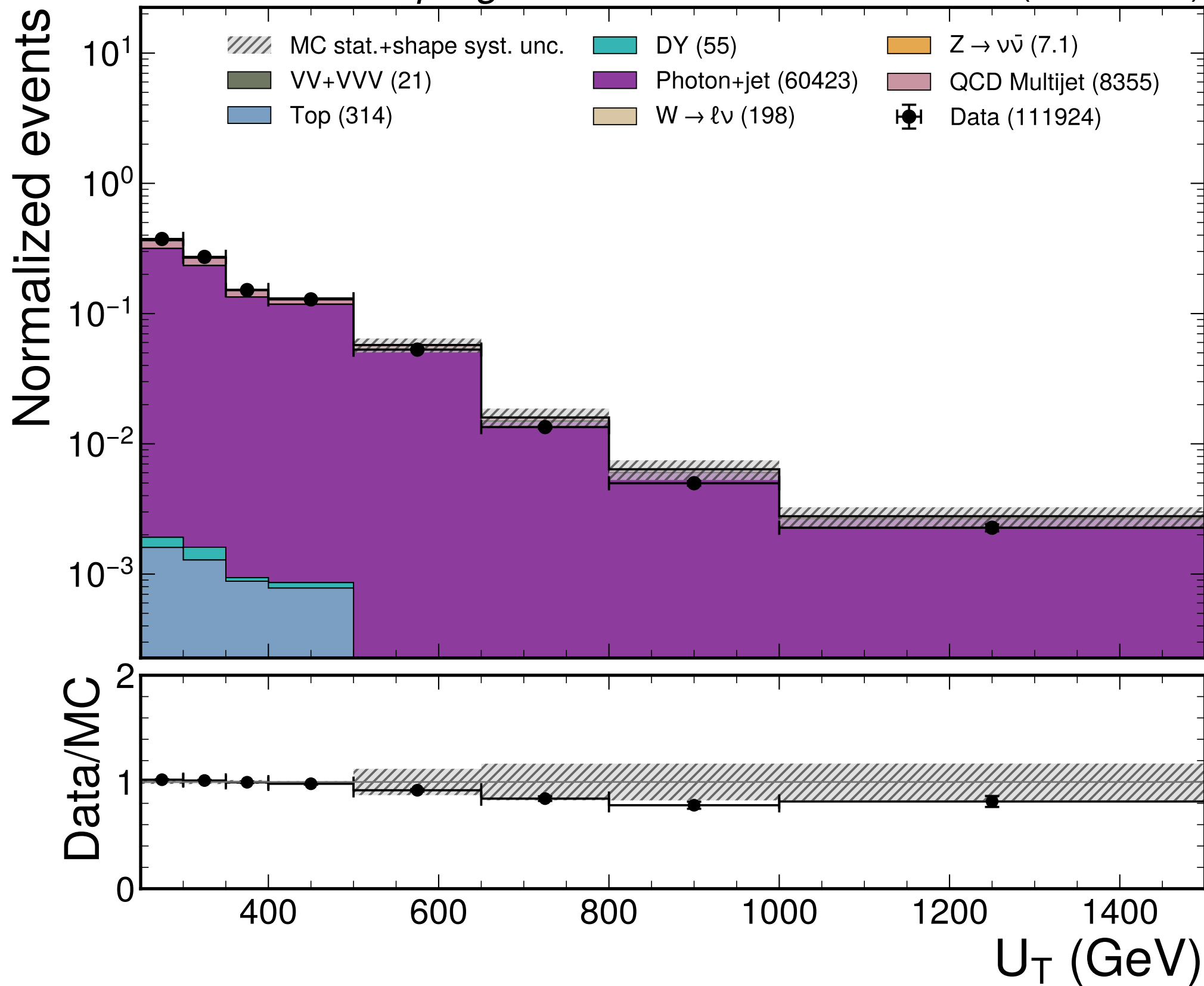
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)



CMS *Work in progress*

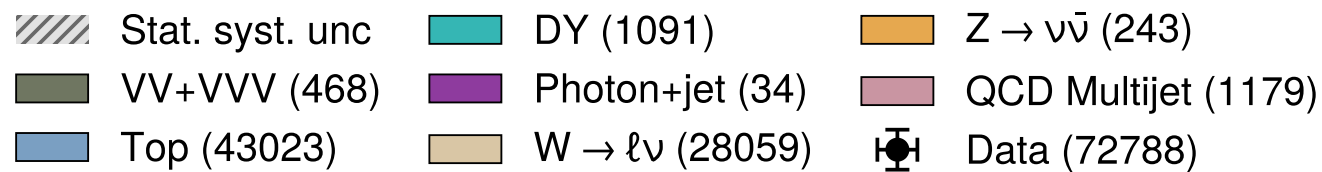
110.84 fb⁻¹ (13.6 TeV)



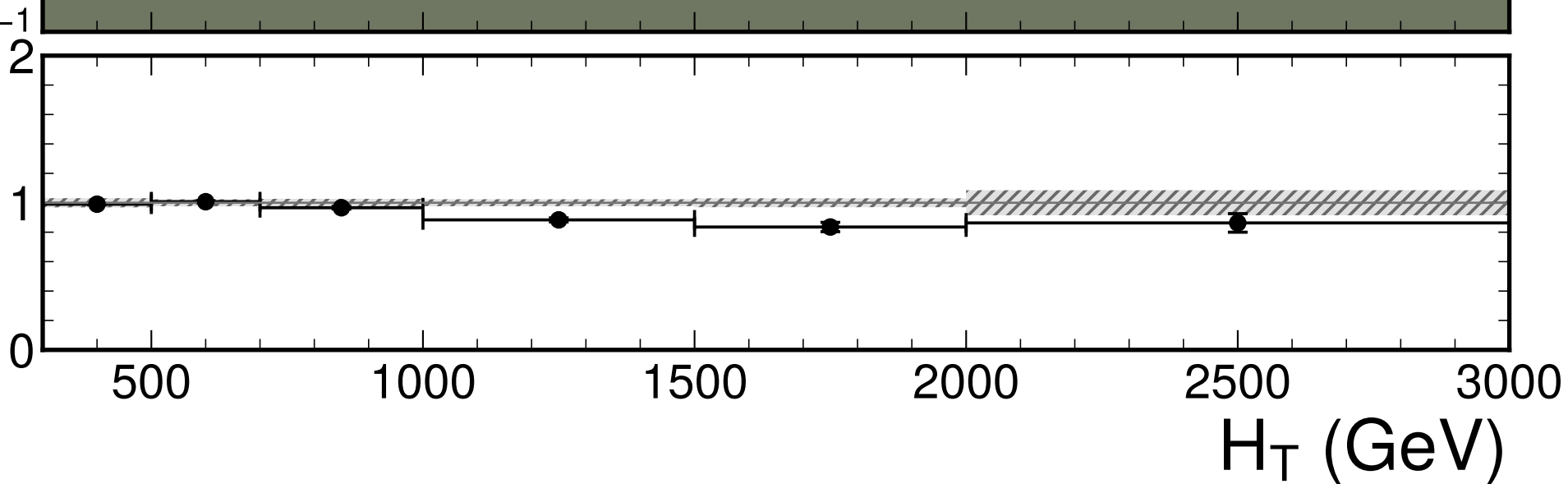
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



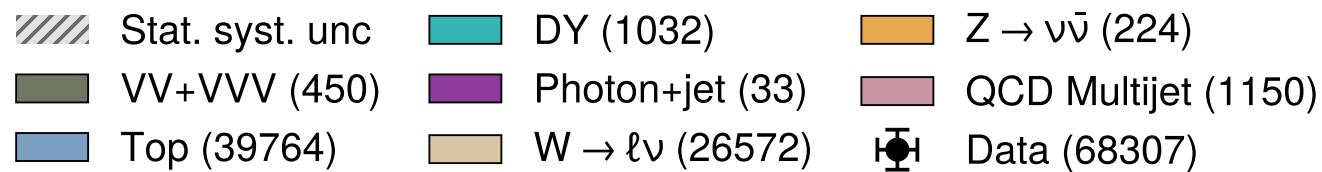
Data/MC



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

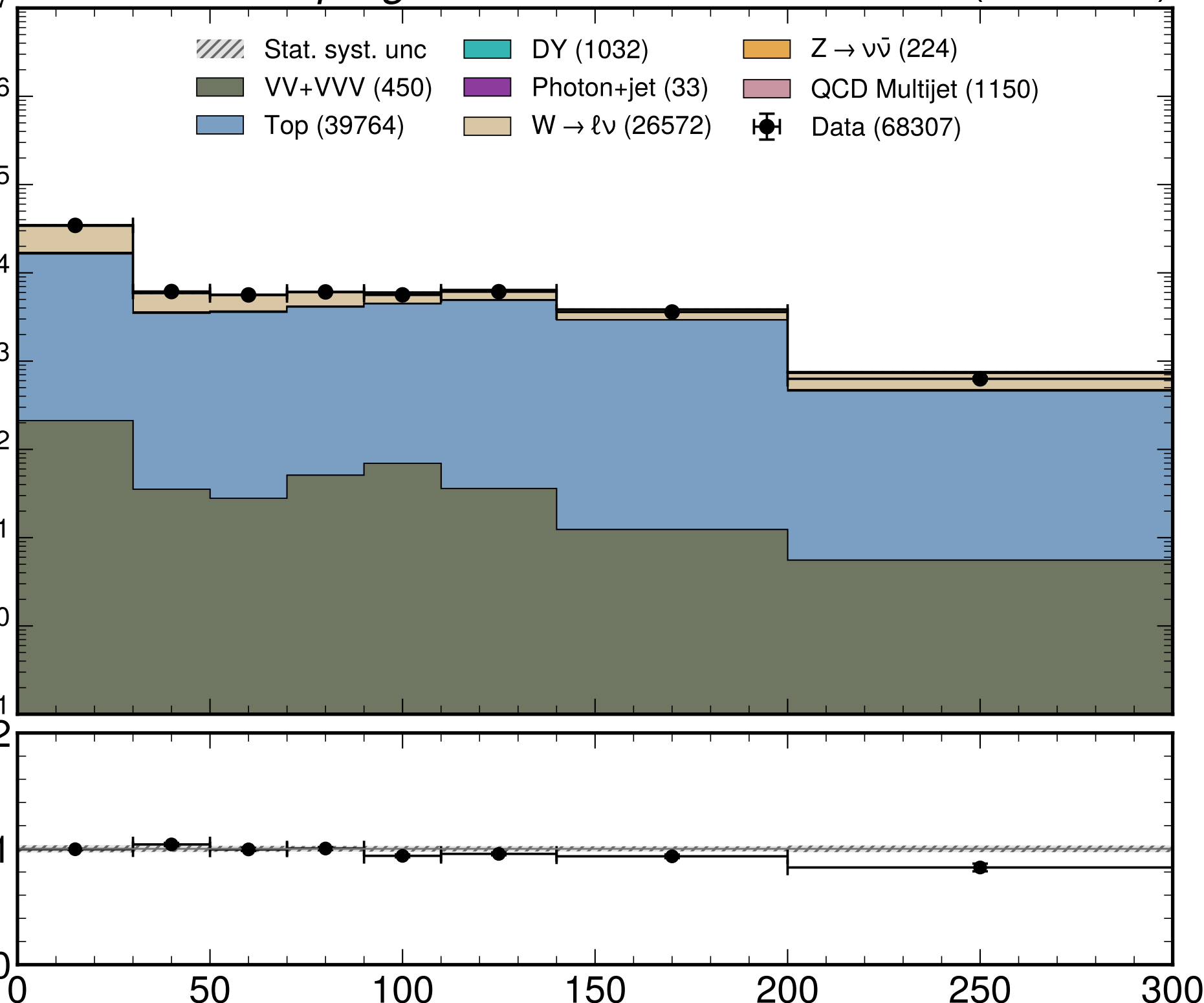
Events



Data/MC

10⁻¹
2
1
0

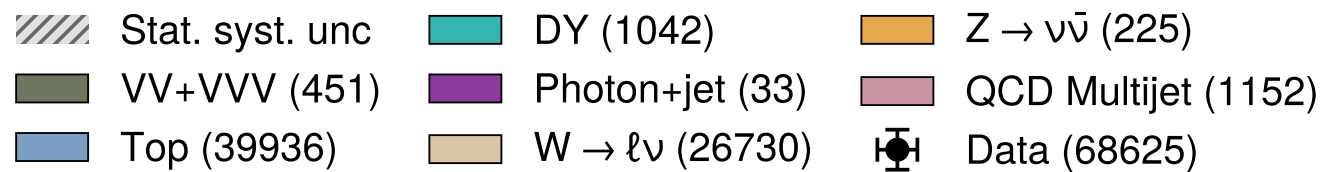
Leading AK8 jet m_{SD} (GeV)



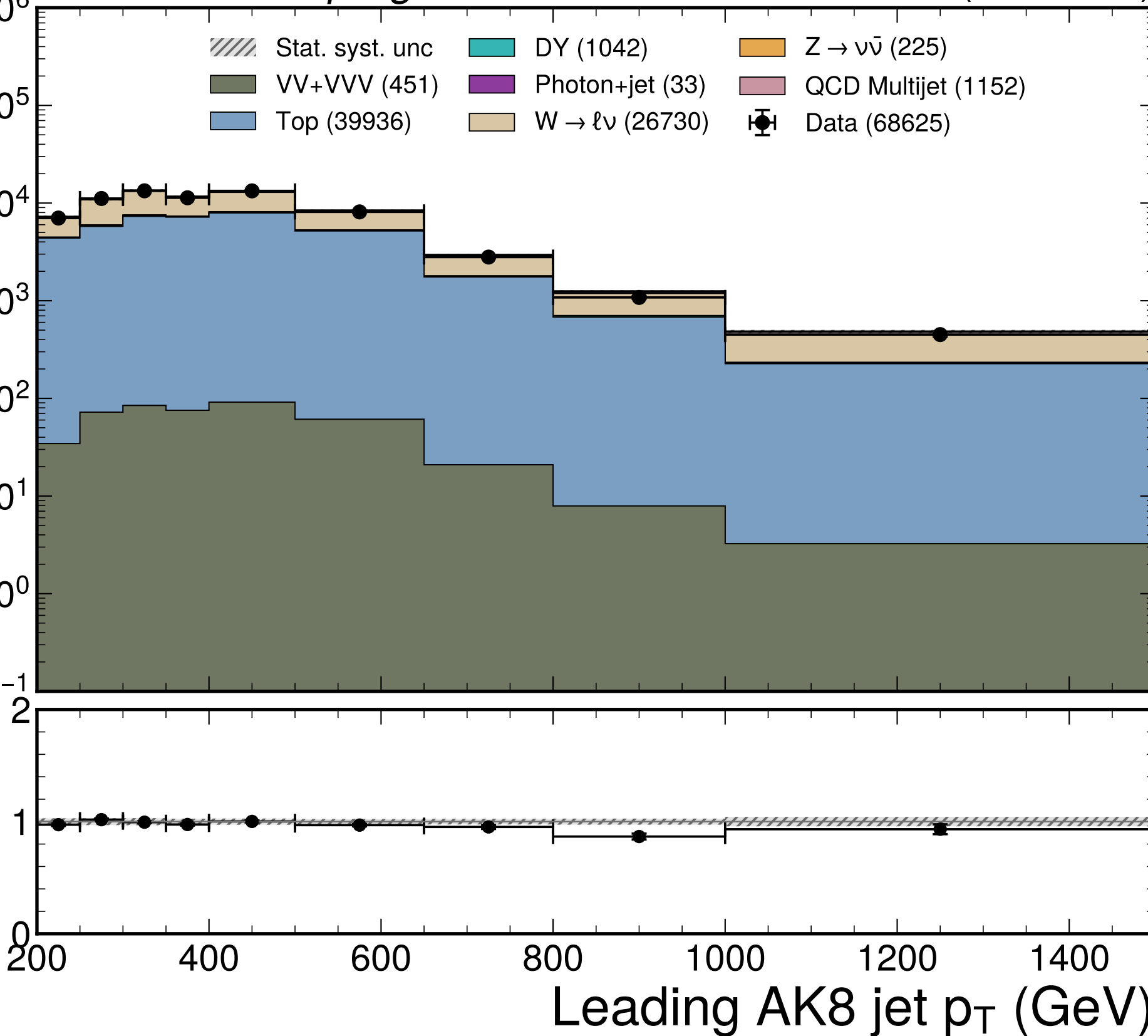
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



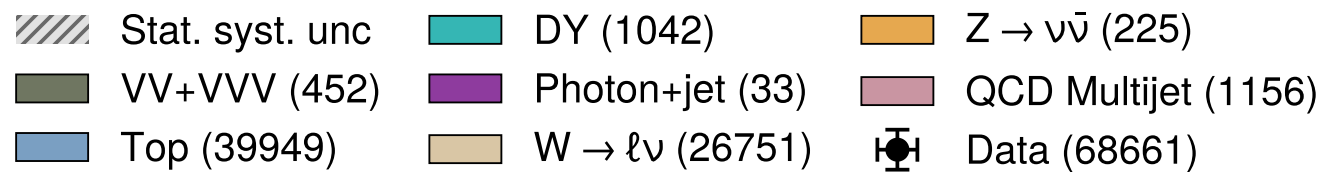
Data/MC



CMS *Work in progress*

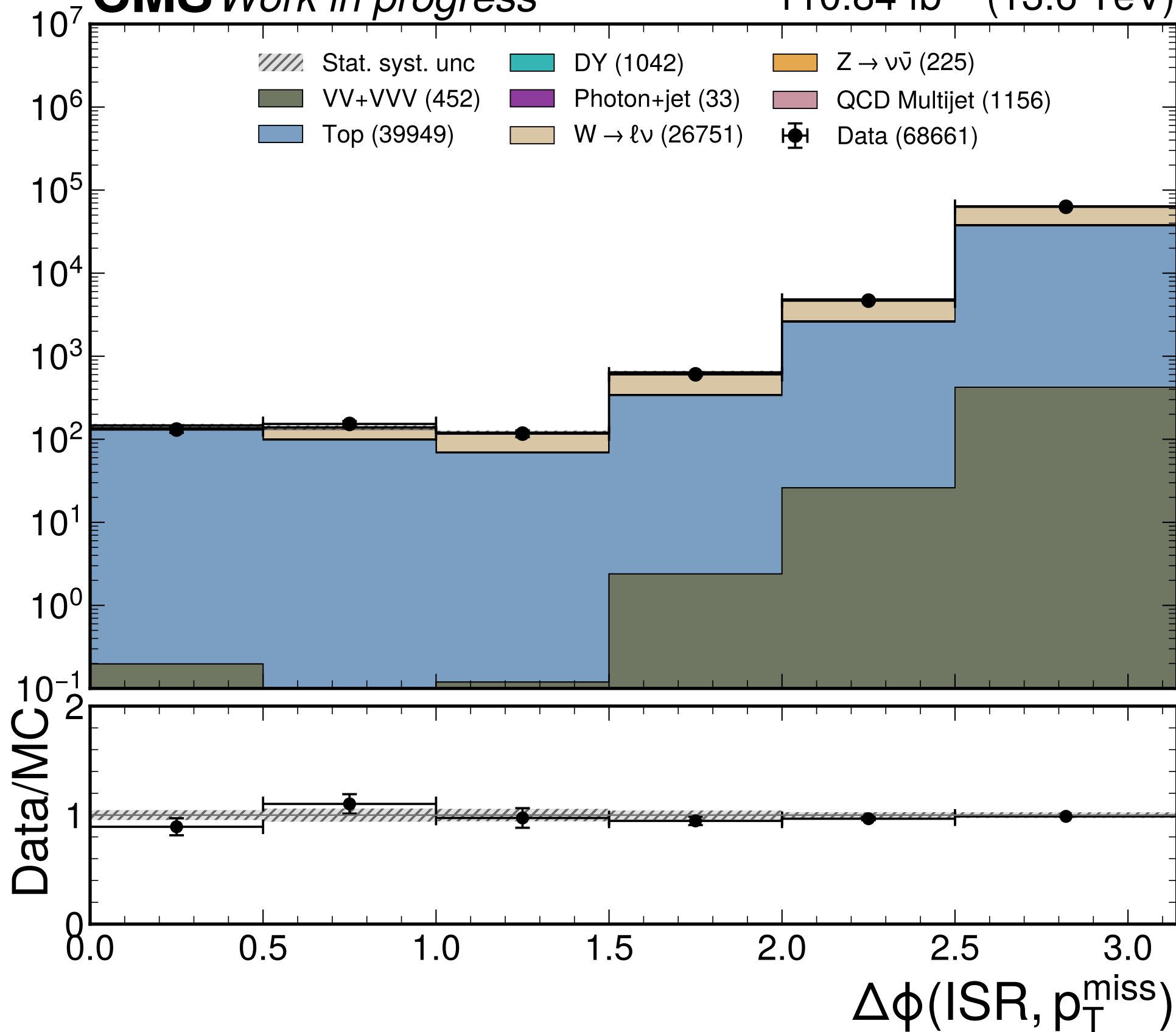
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

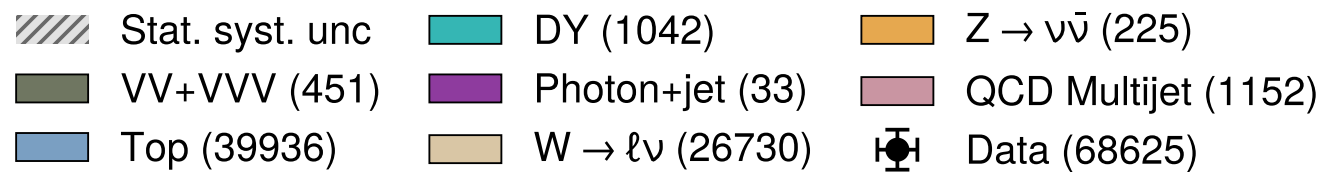
$\Delta\phi(\text{ISR}, p_{\text{T}}^{\text{miss}})$



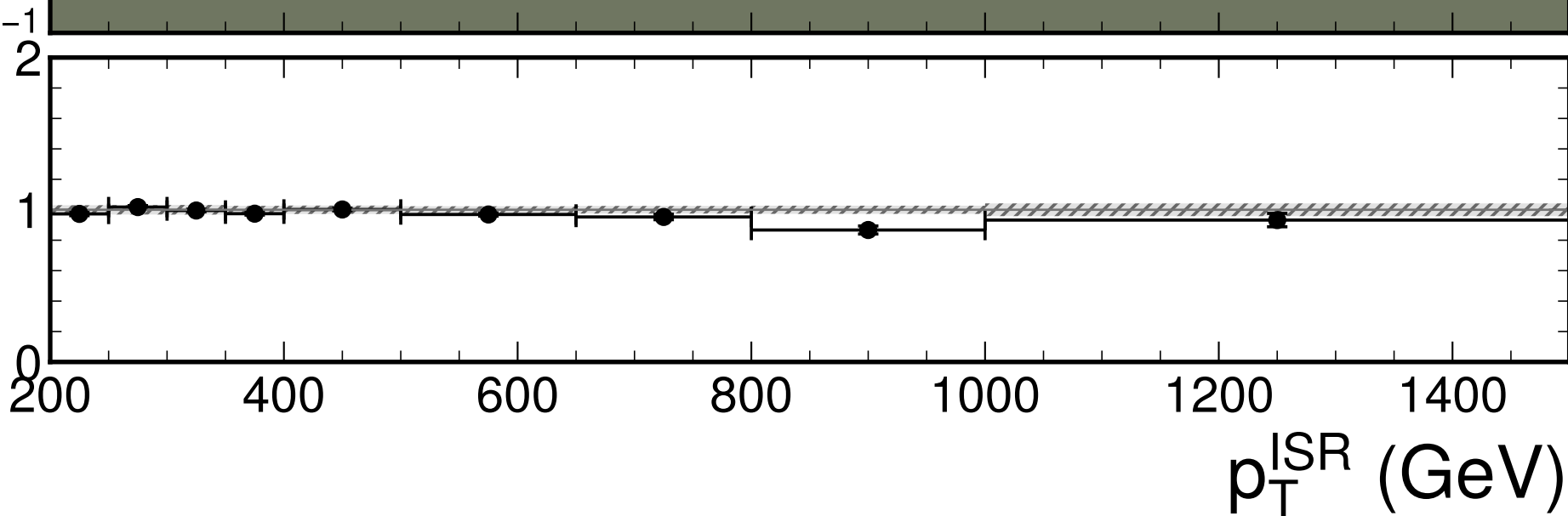
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



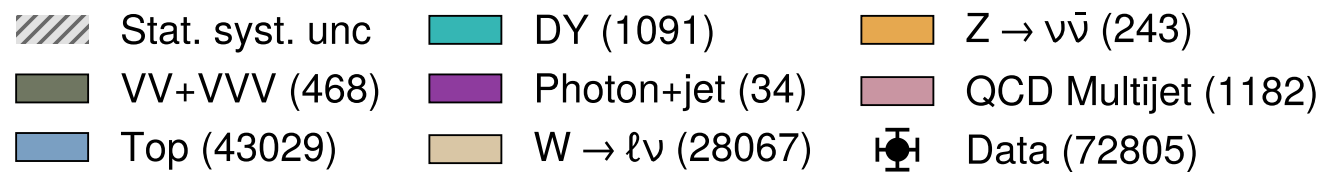
Data/MC



CMS *Work in progress*

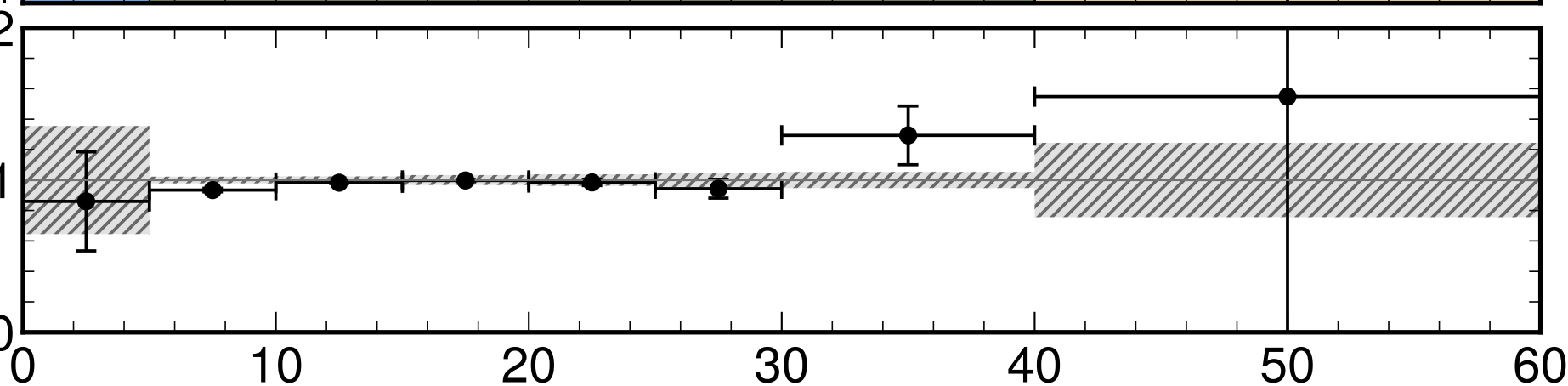
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2

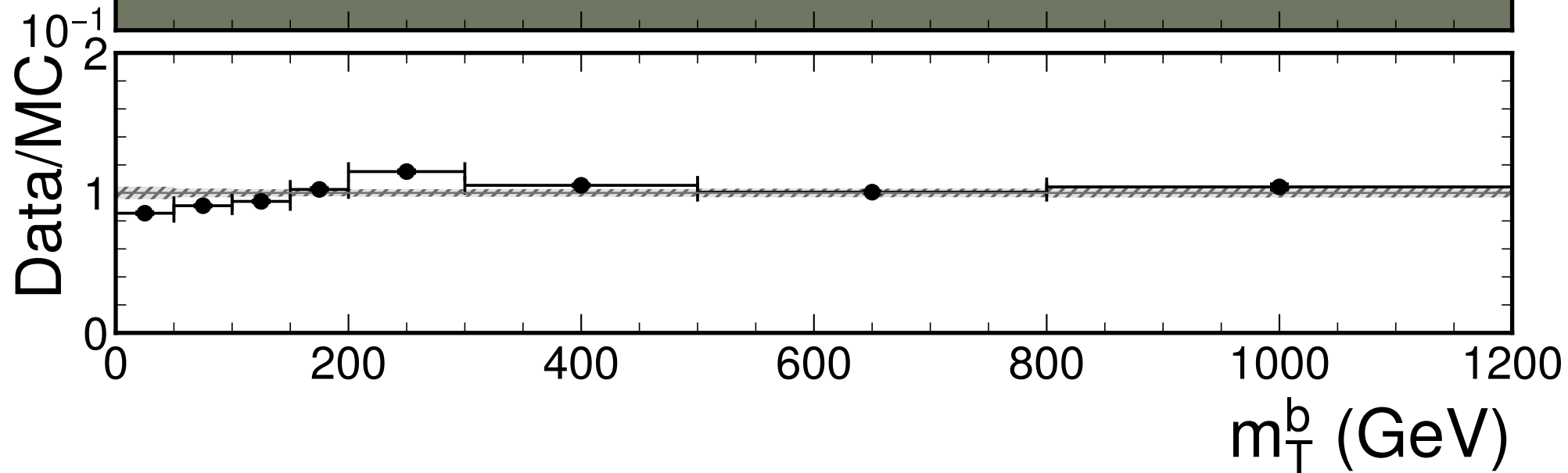
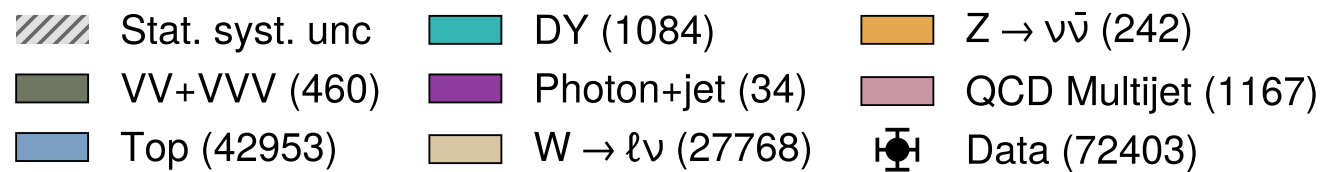


$p_T^{\text{miss}}/\sqrt{H_T}$

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

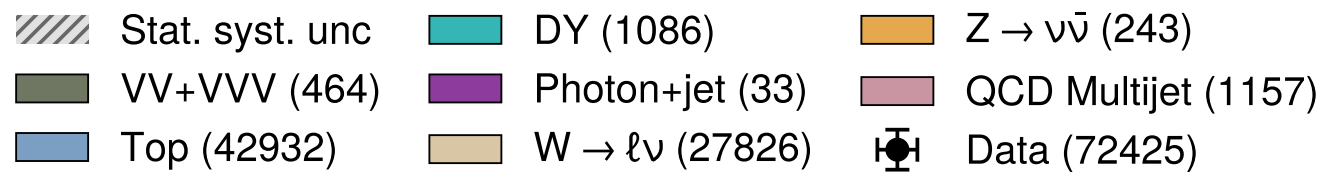
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



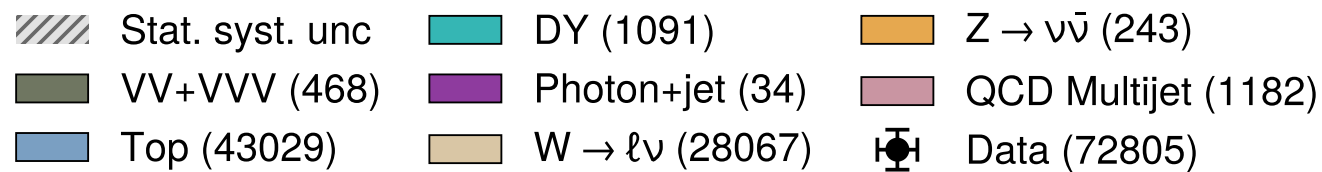
Data/MC

p_T^b (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻²

0

400

600

800

1000

1200

1400

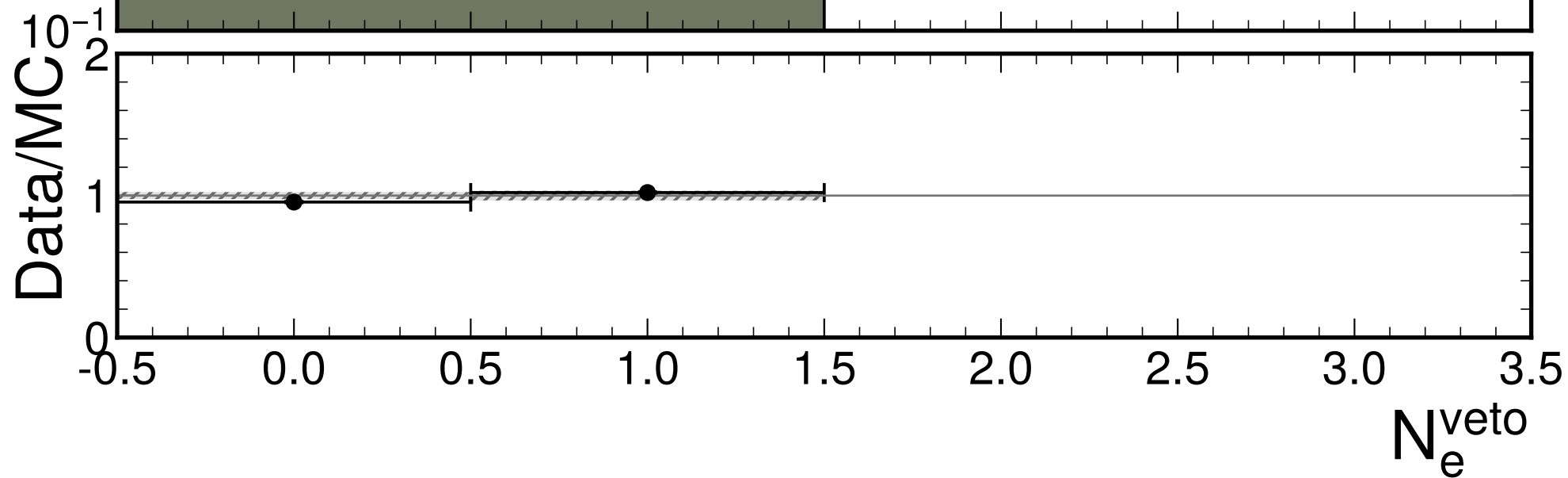
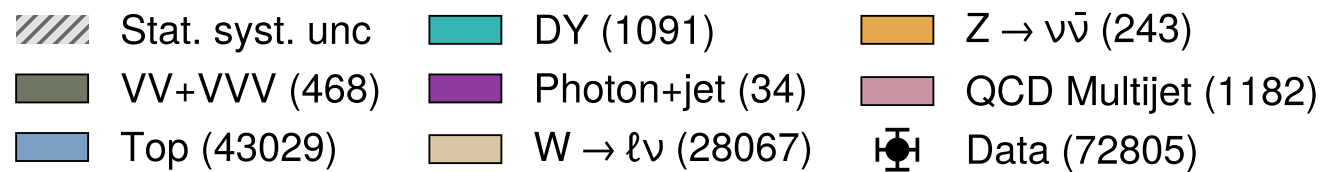
p_T^{miss} (GeV)

Detailed description: The bottom plot shows the ratio Data/MC on a logarithmic y-axis (from 0.1 to 10) against p_T^{miss} (GeV) on a linear x-axis (from 0 to 1500). A horizontal line at y=1 represents the expected ratio. Data points with error bars are shown for various components, mostly clustered around the y=1 line. A hatched band indicates the statistical and systematic uncertainty.

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

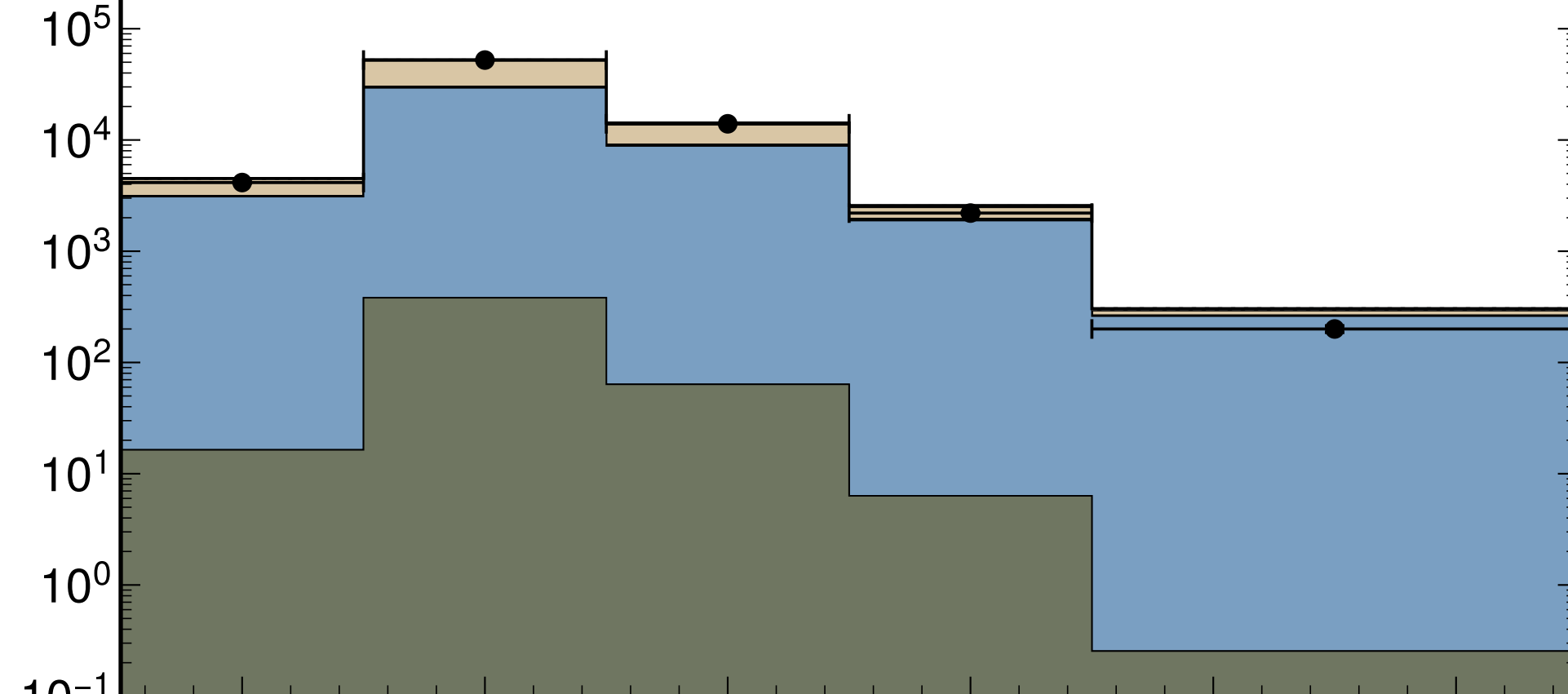
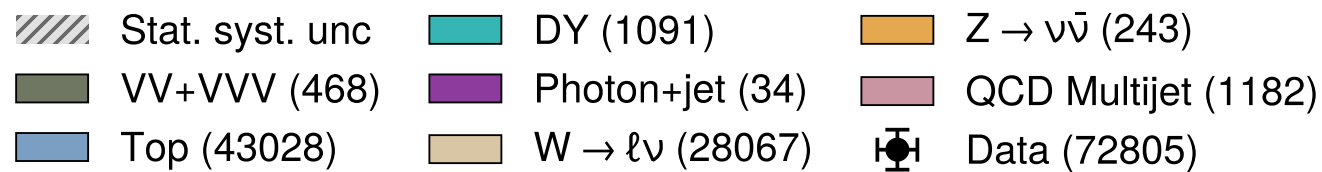
Events



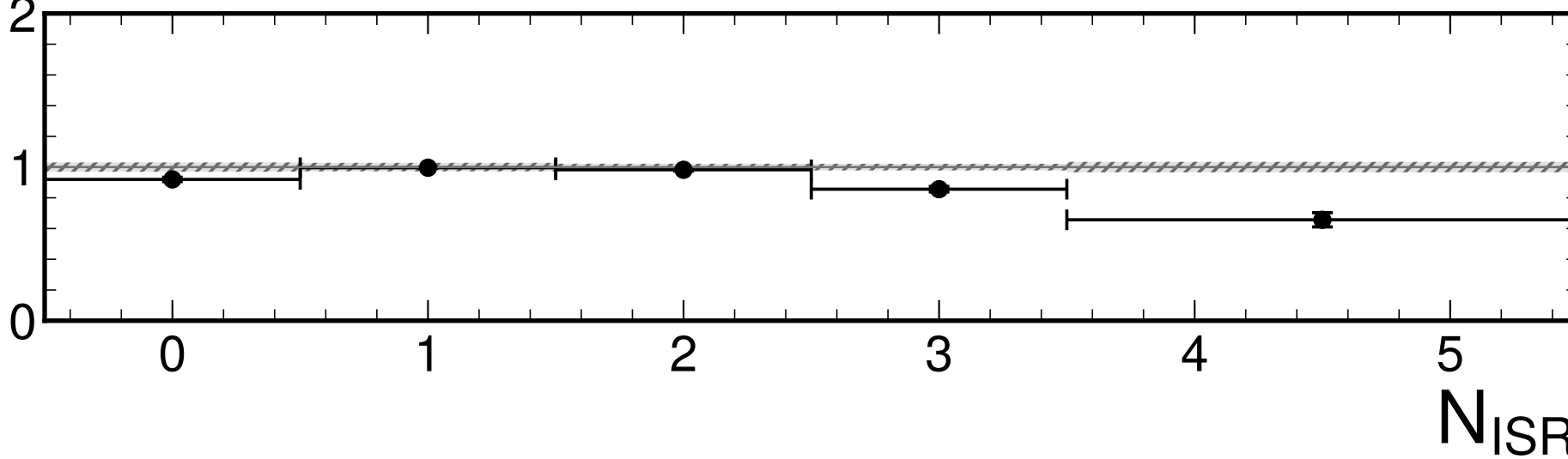
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



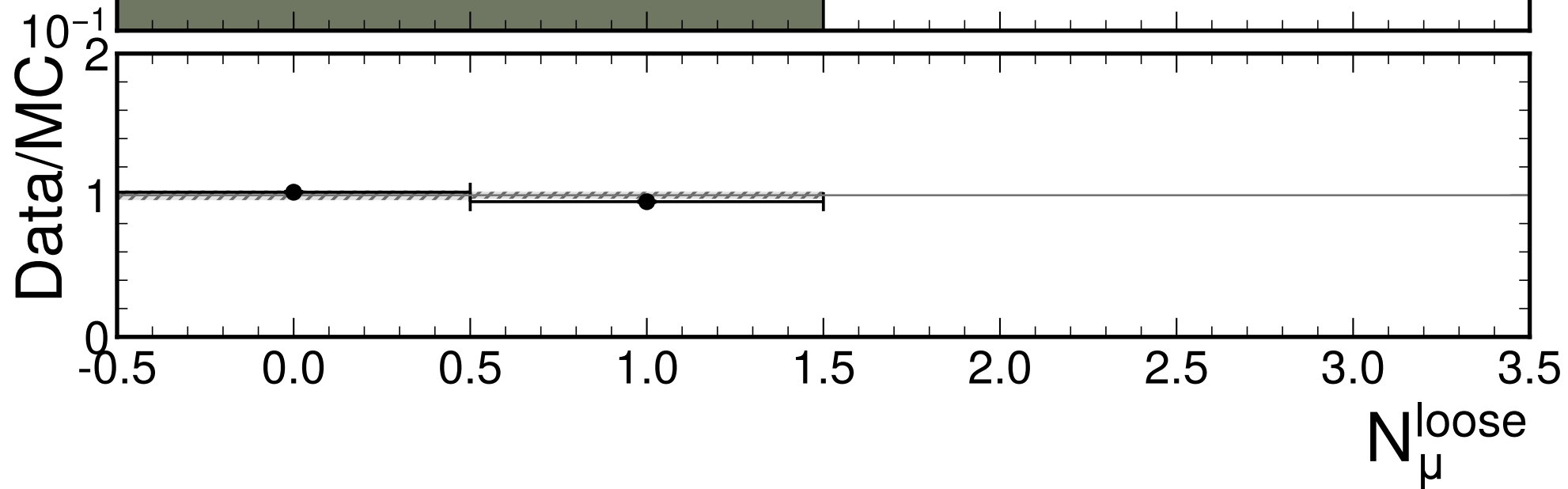
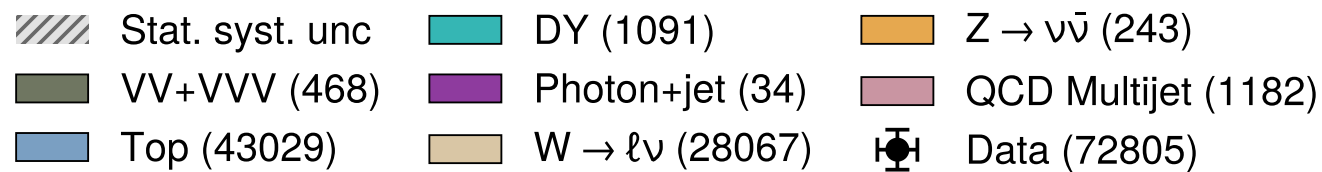
Data/MC



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

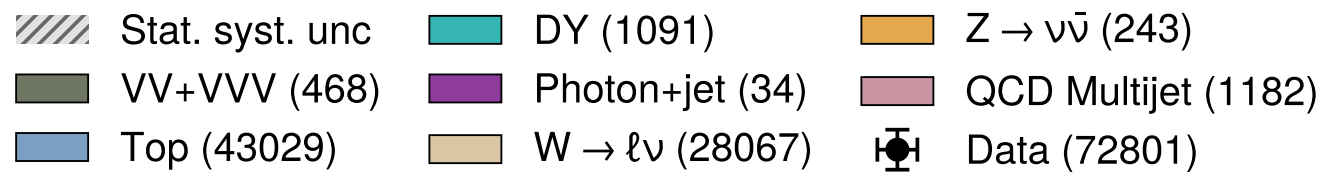
Events



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

1

2

3

4

5

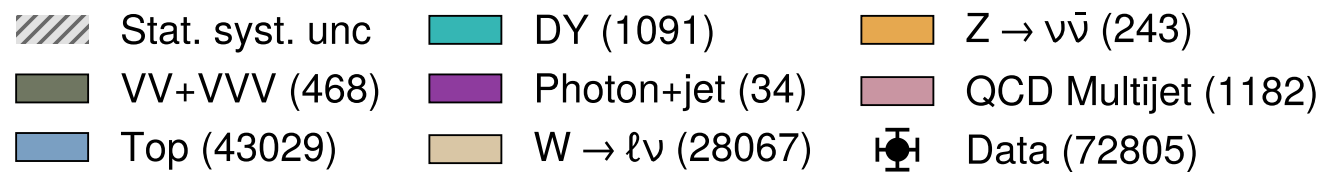
6

N_b (loose WP)

CMS *Work in progress*

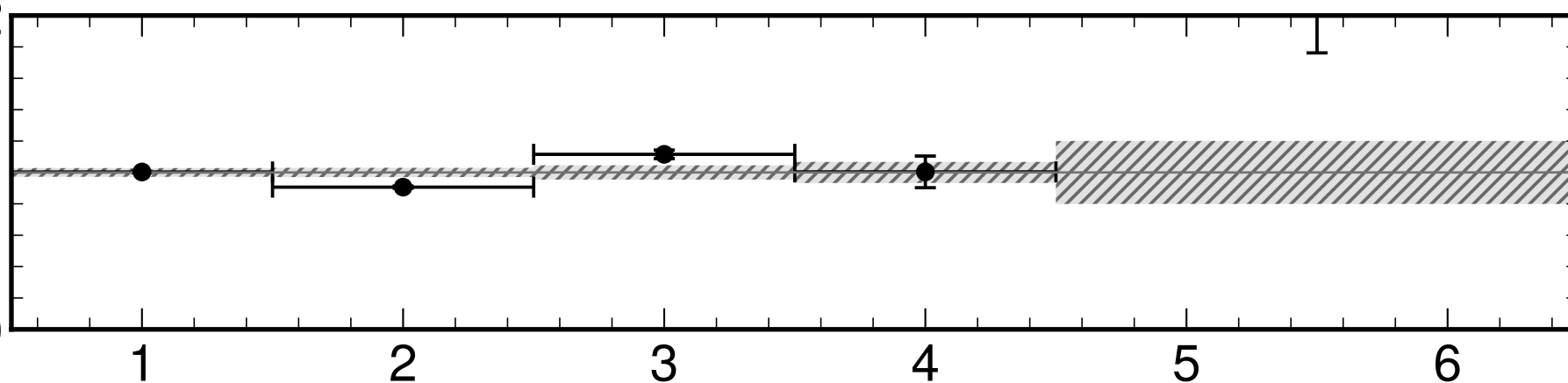
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

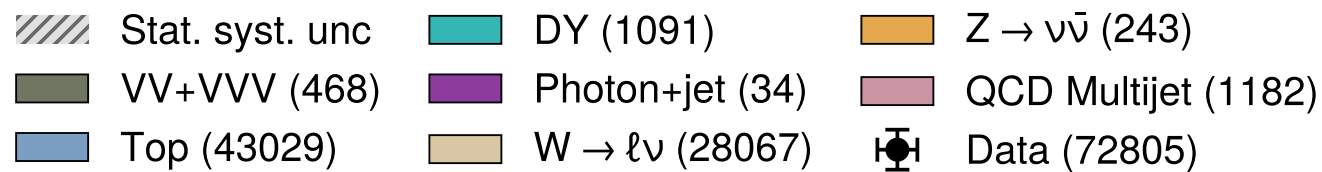


N_b

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

10⁻¹
2
1
0

2

4

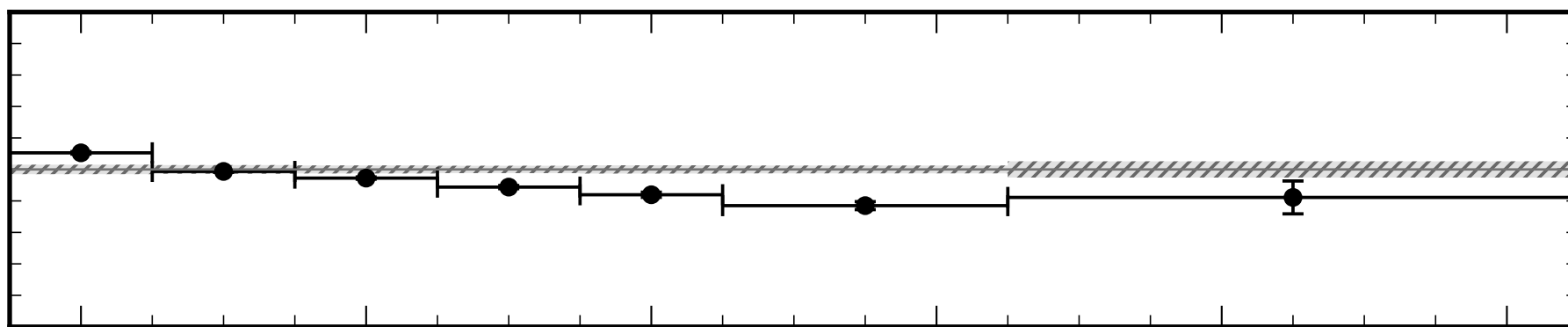
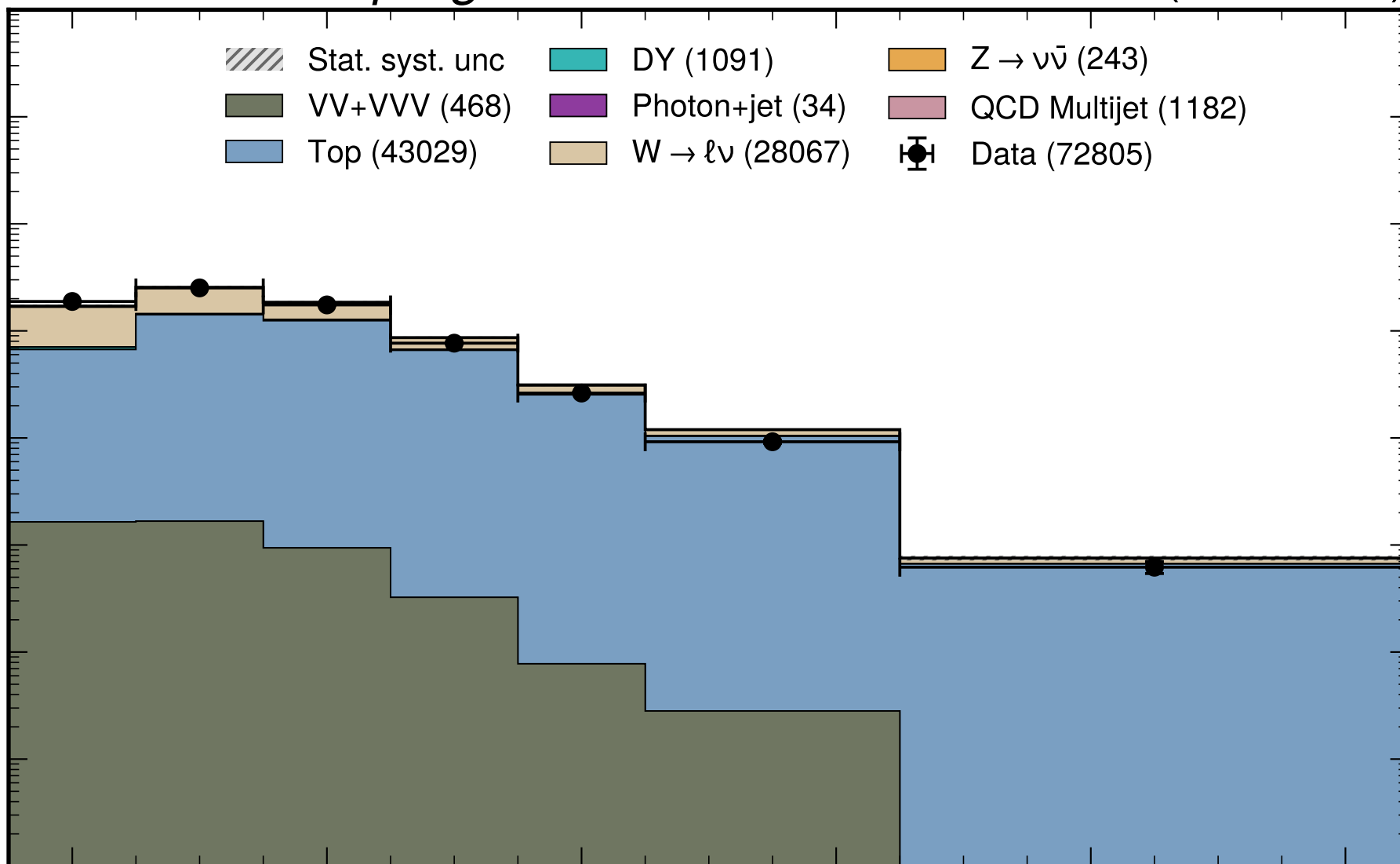
6

8

10

12

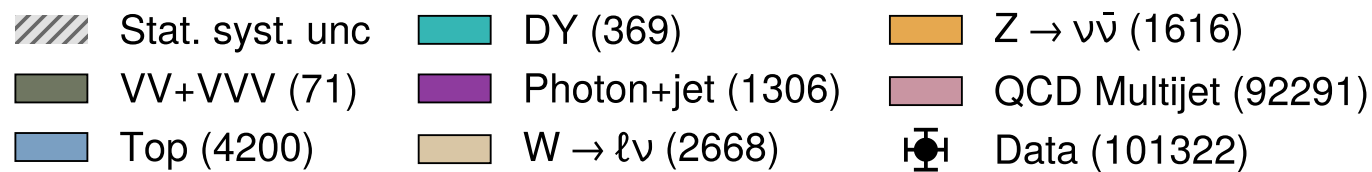
N_j



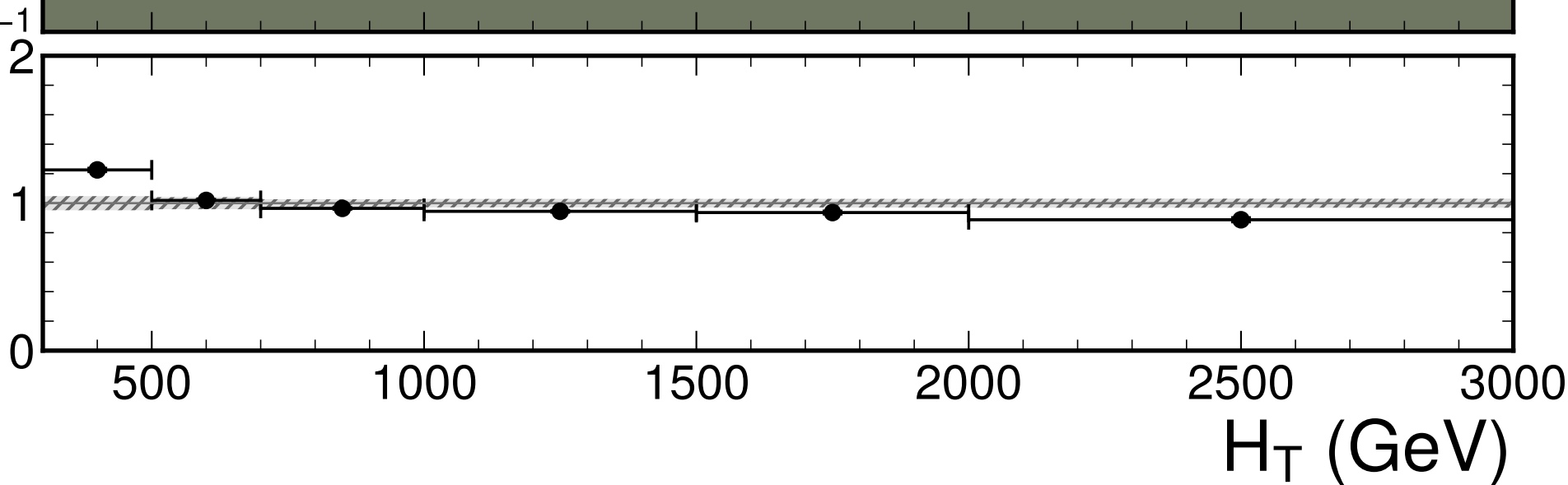
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



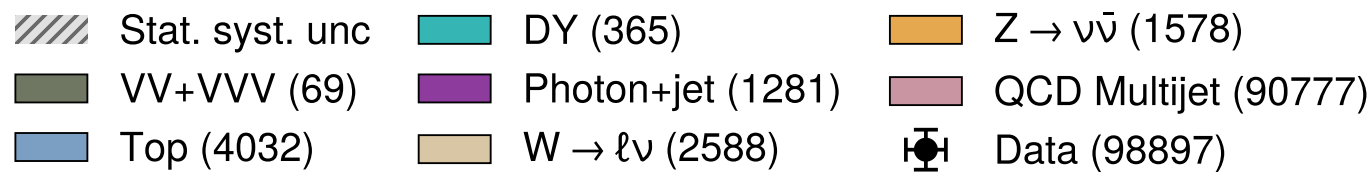
Data/MC



CMS *Work in progress*

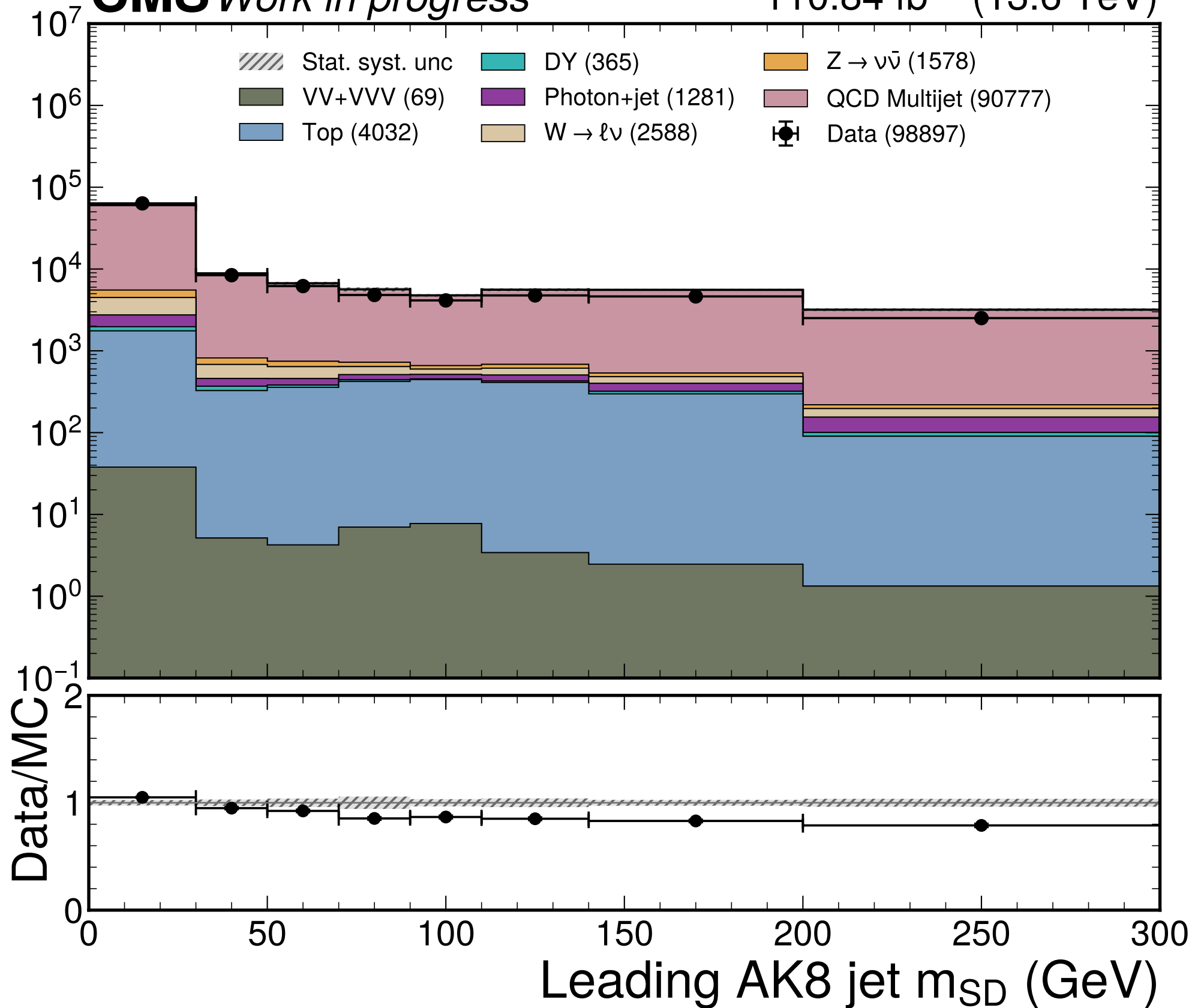
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

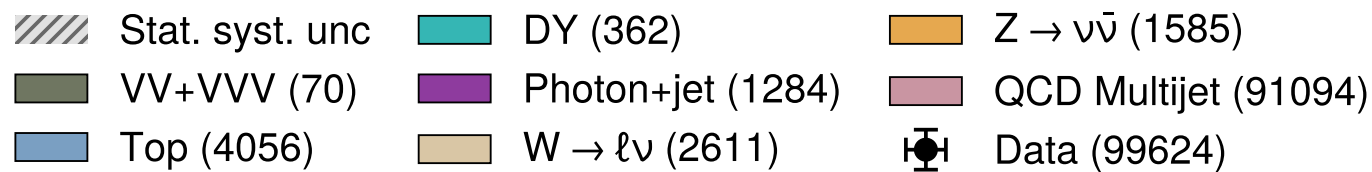
Leading AK8 jet m_{SD} (GeV)



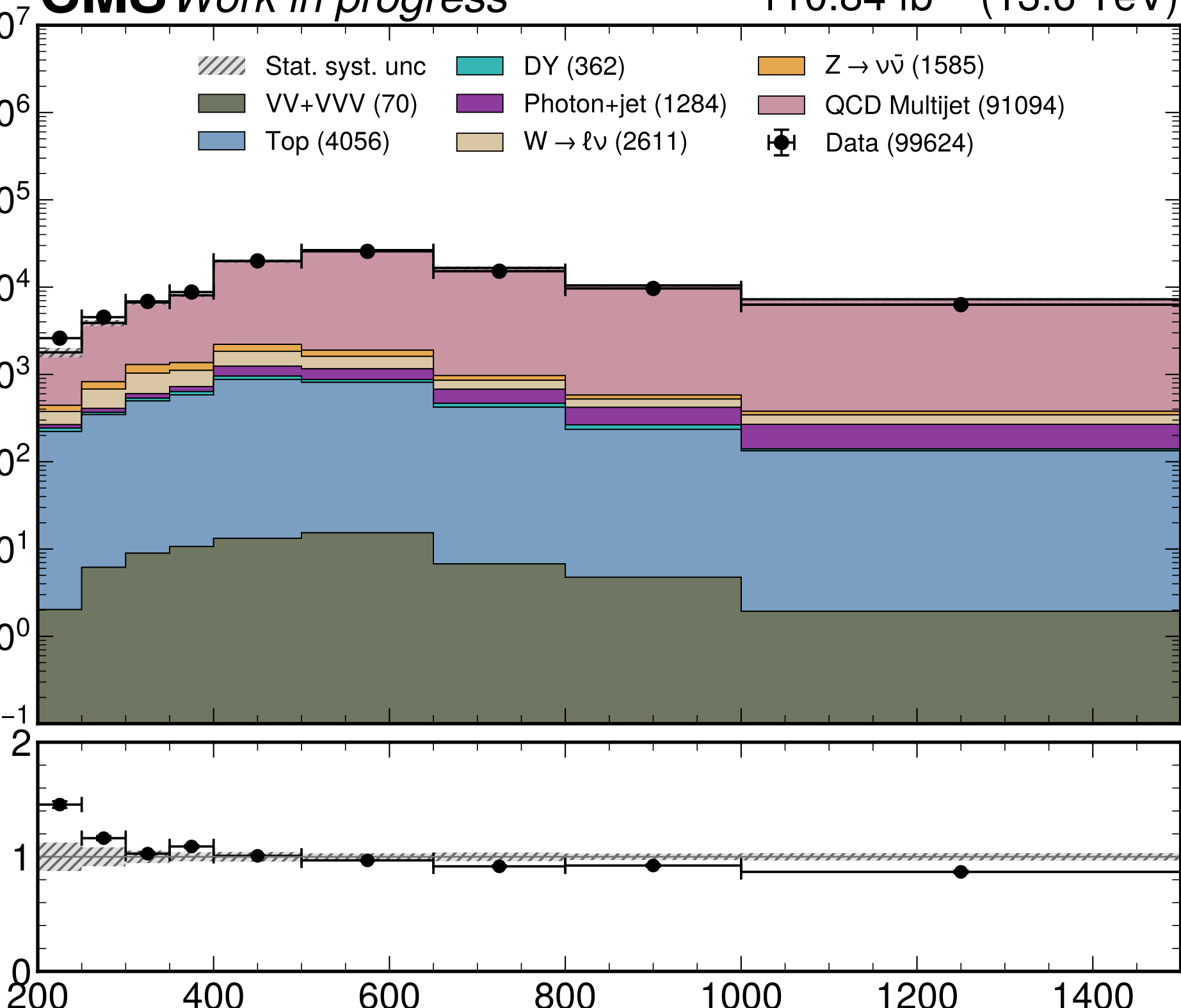
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

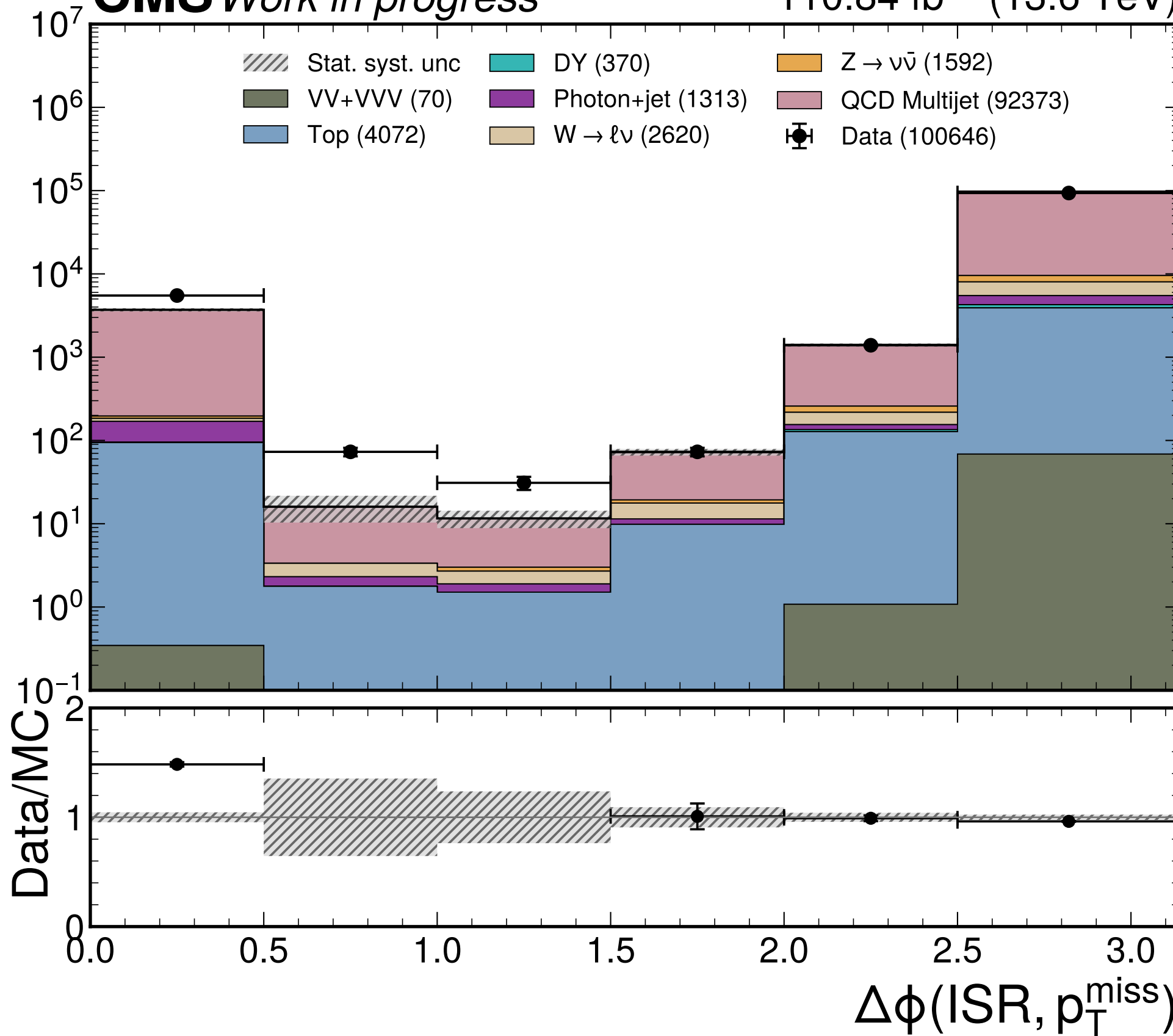
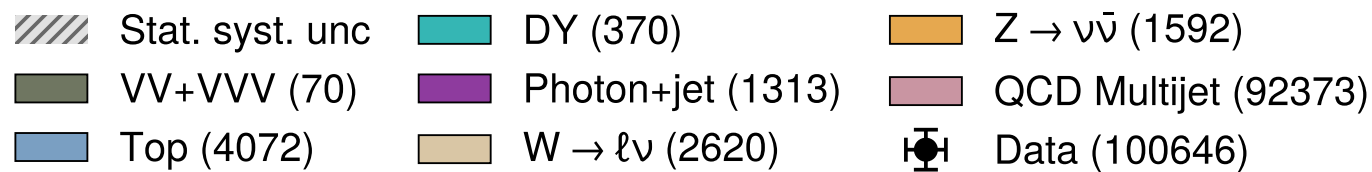


Leading AK8 jet p_T (GeV)

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

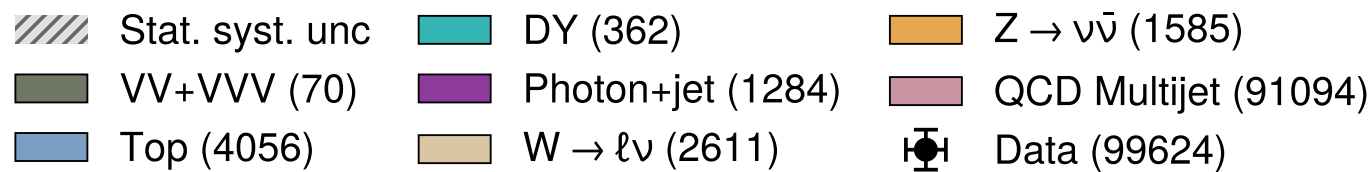
Events



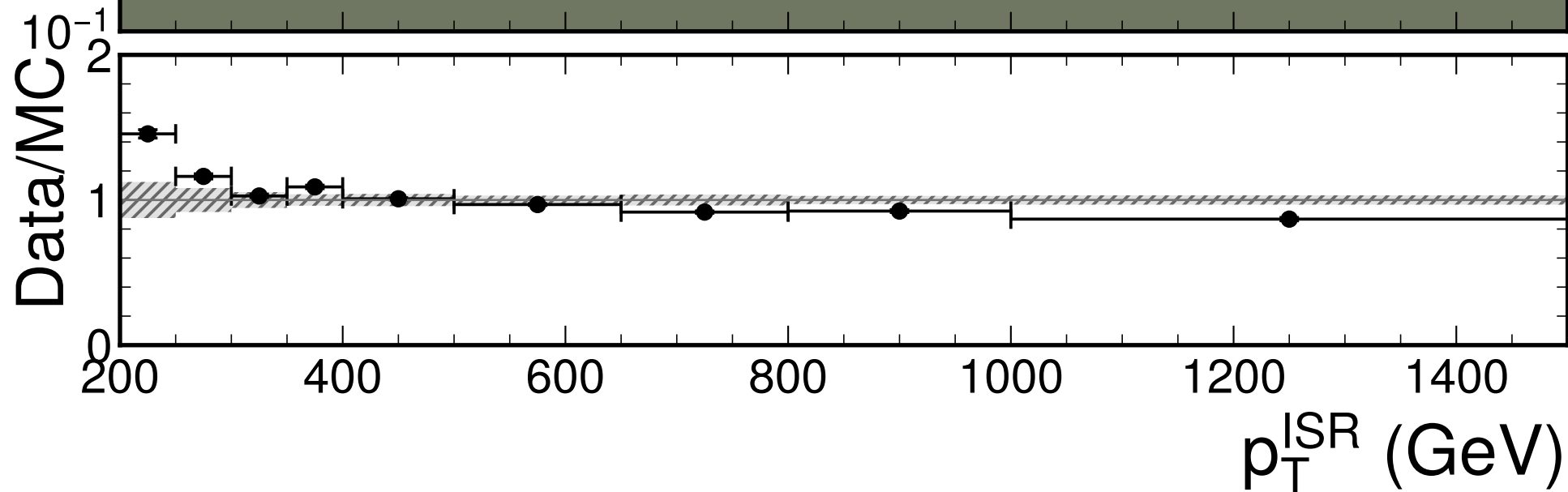
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



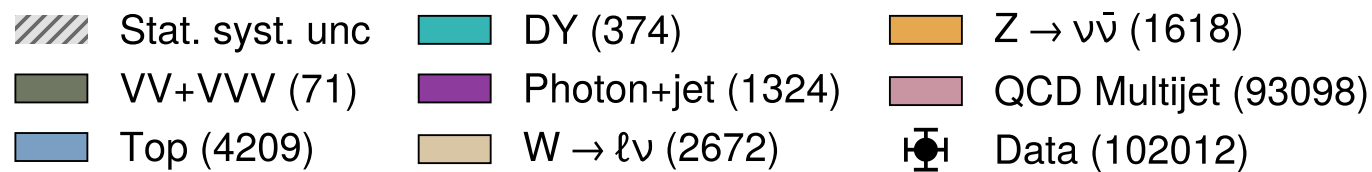
Data/MC



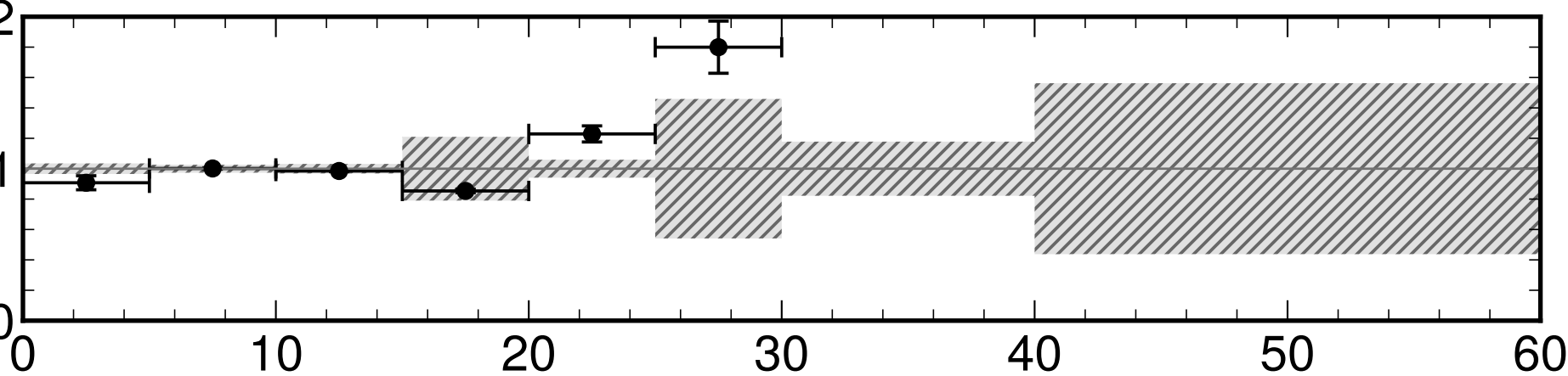
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

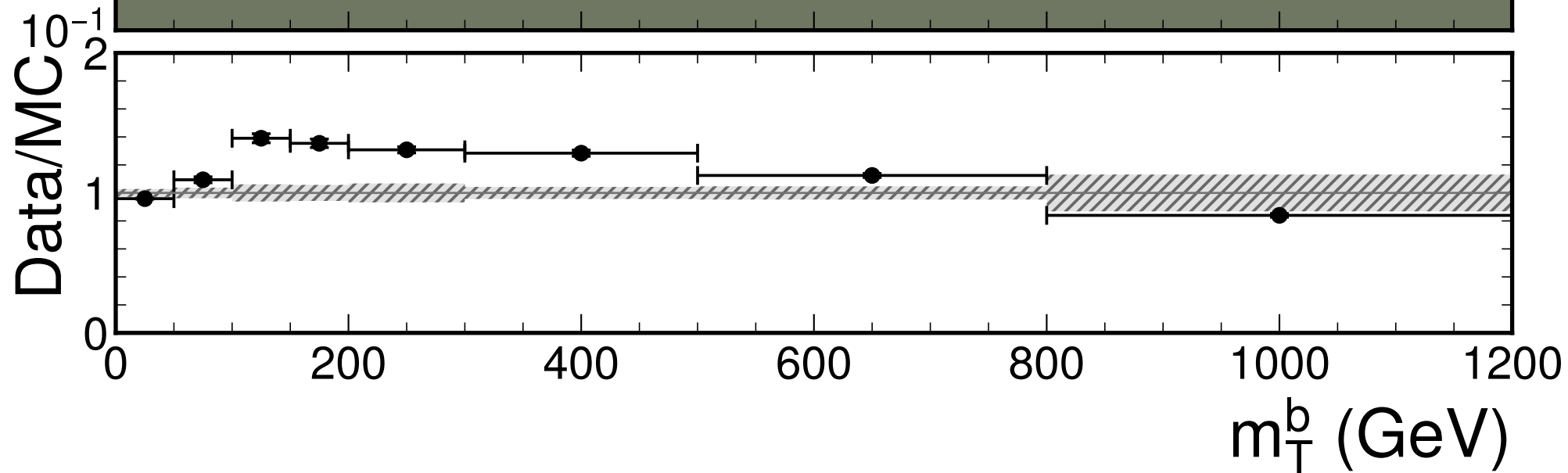
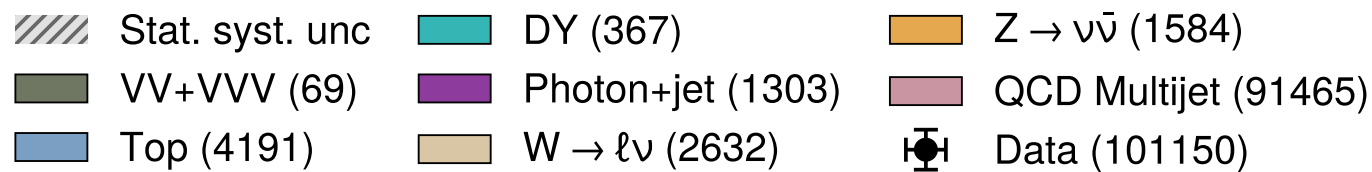


$p_T^{\text{miss}}/\sqrt{H_T}$

CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

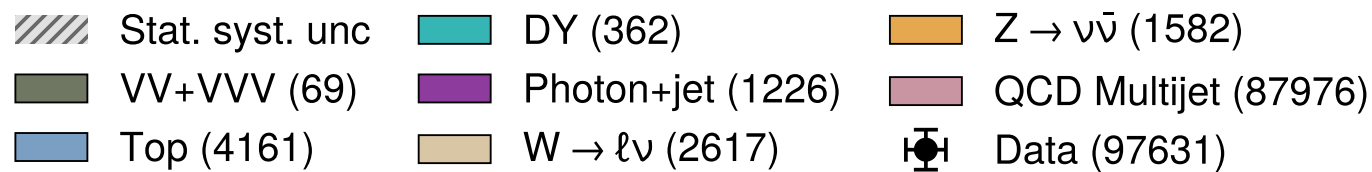
Events



CMS *Work in progress*

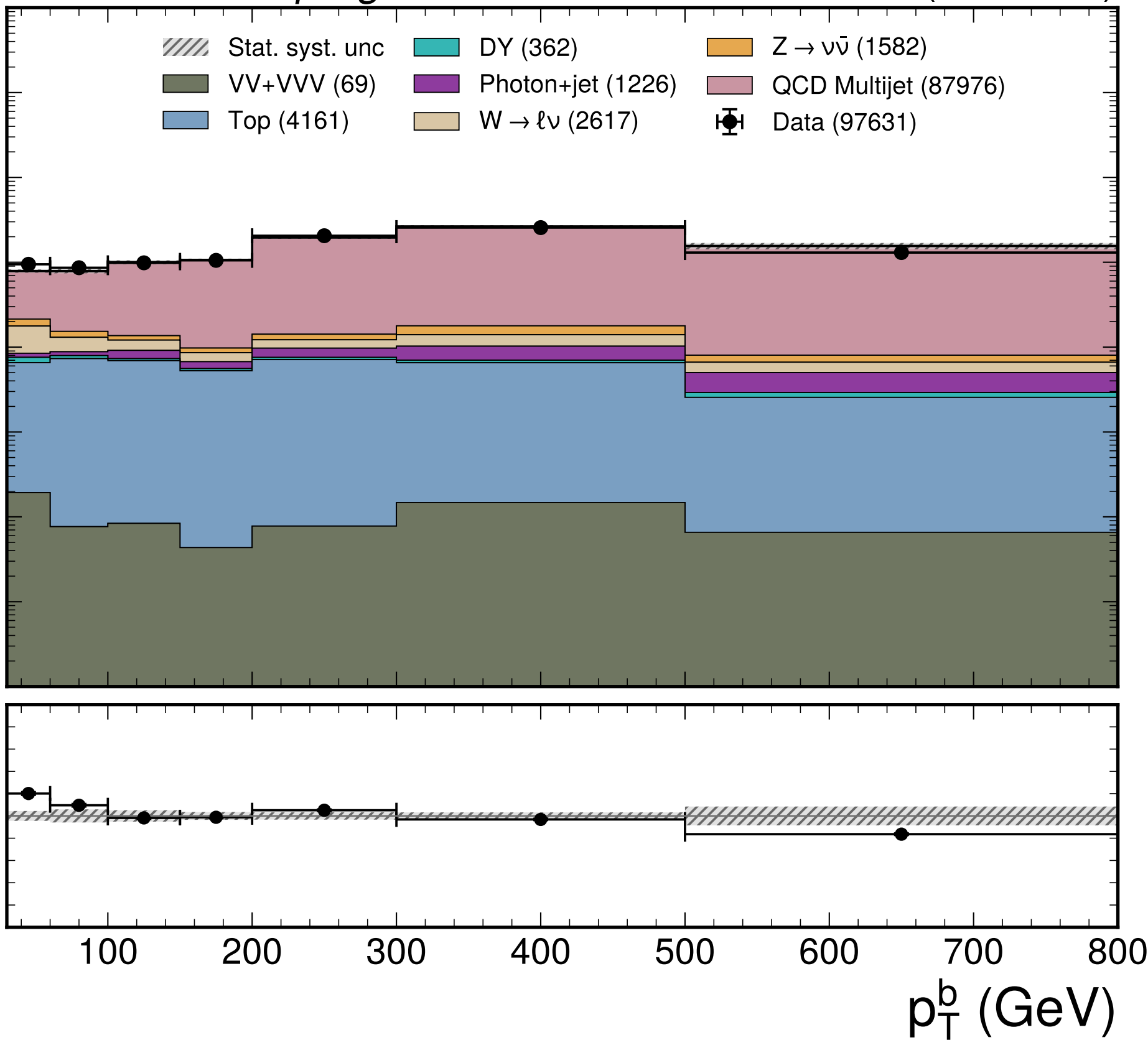
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

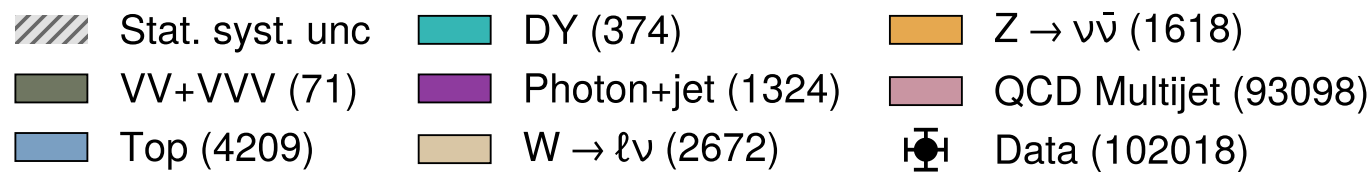
p_T^b (GeV)



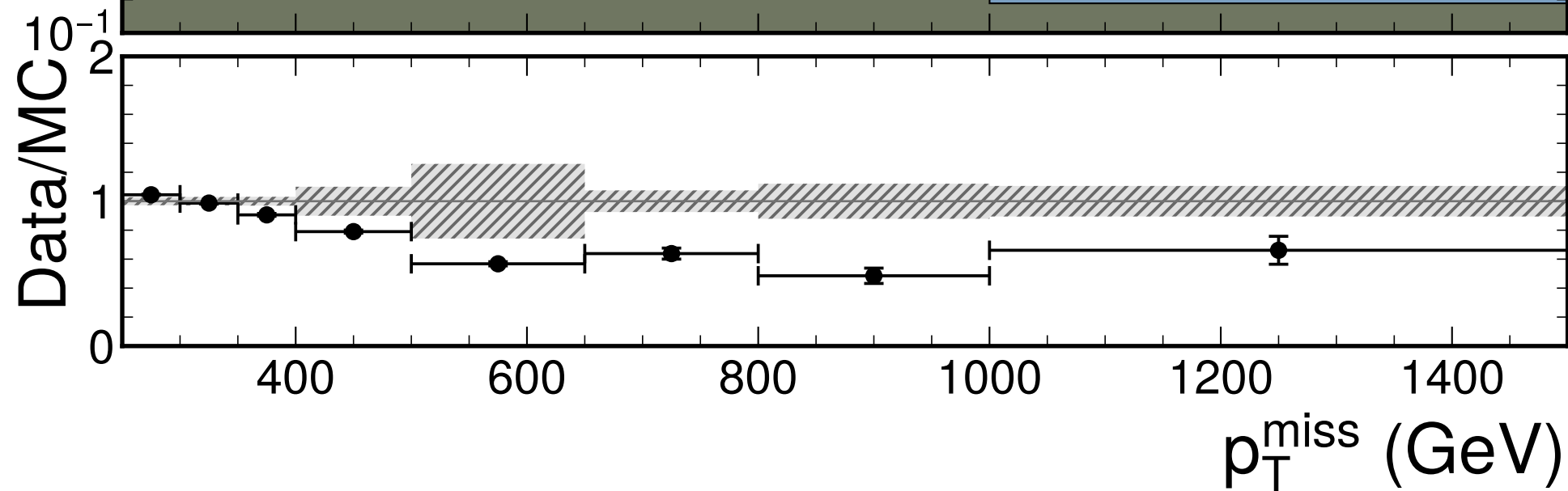
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



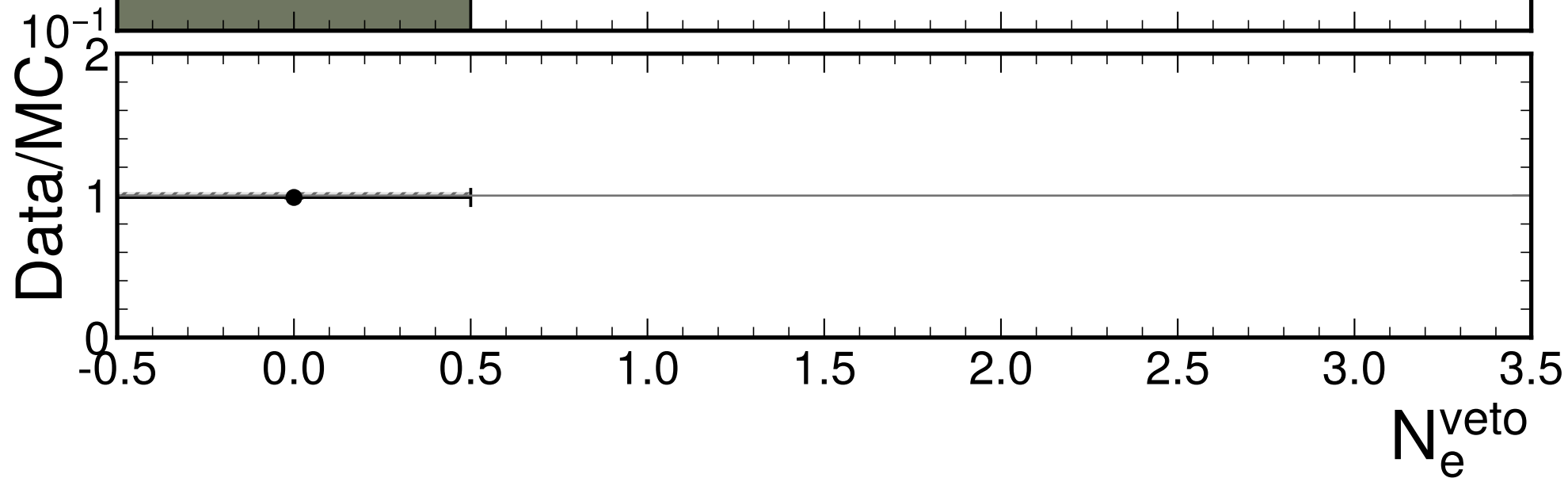
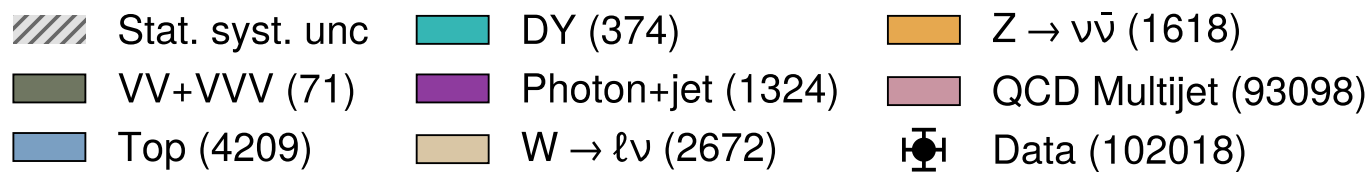
Data/MC



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

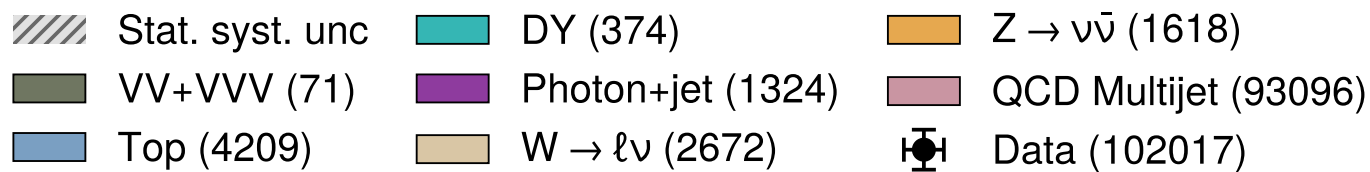
Events



CMS *Work in progress*

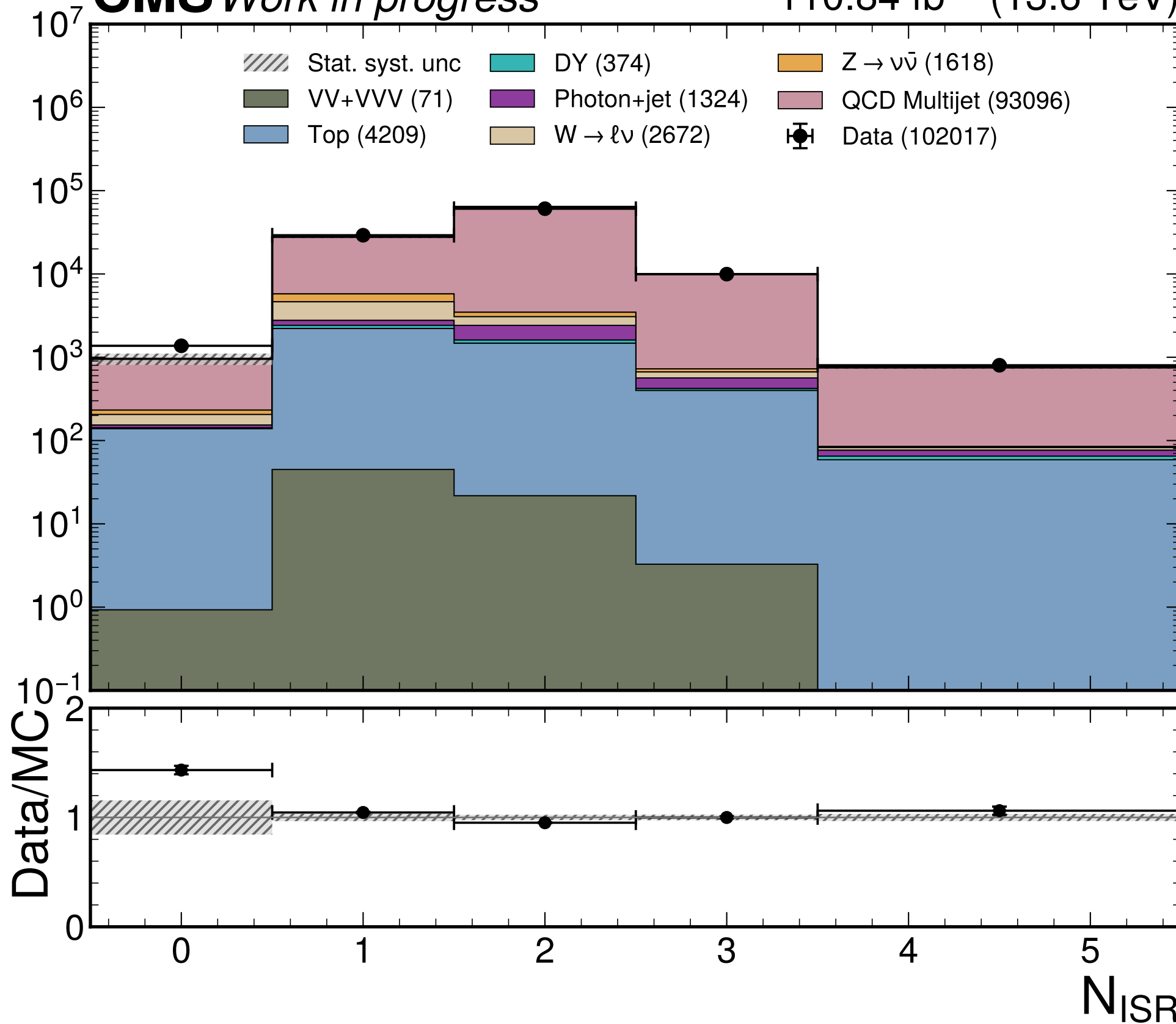
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

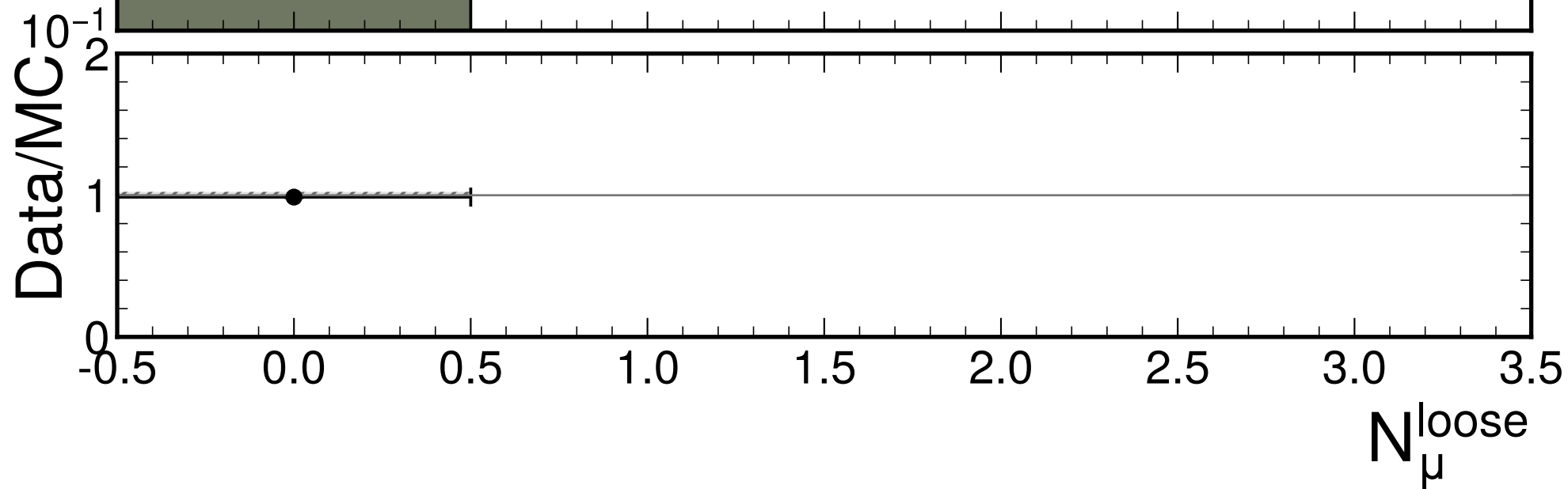
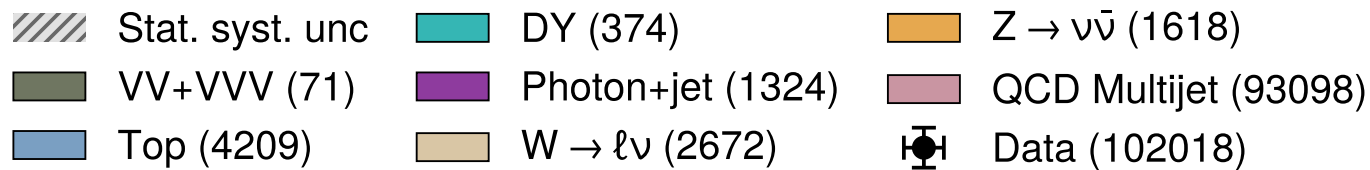
N_{ISR}



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

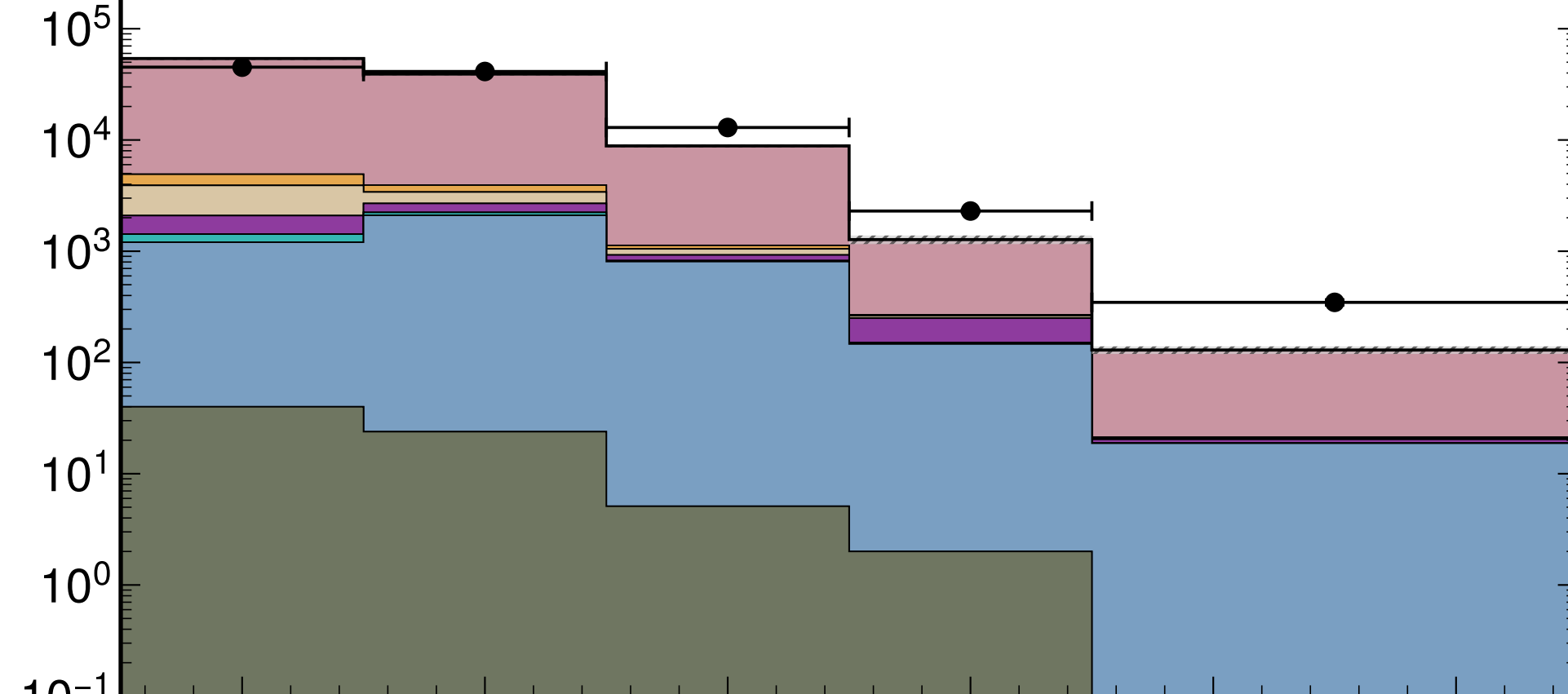
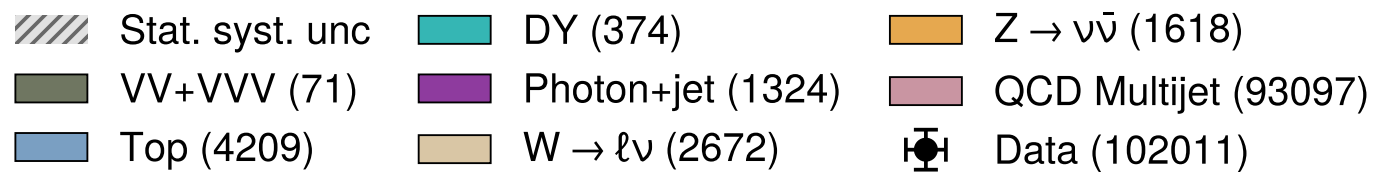
Events



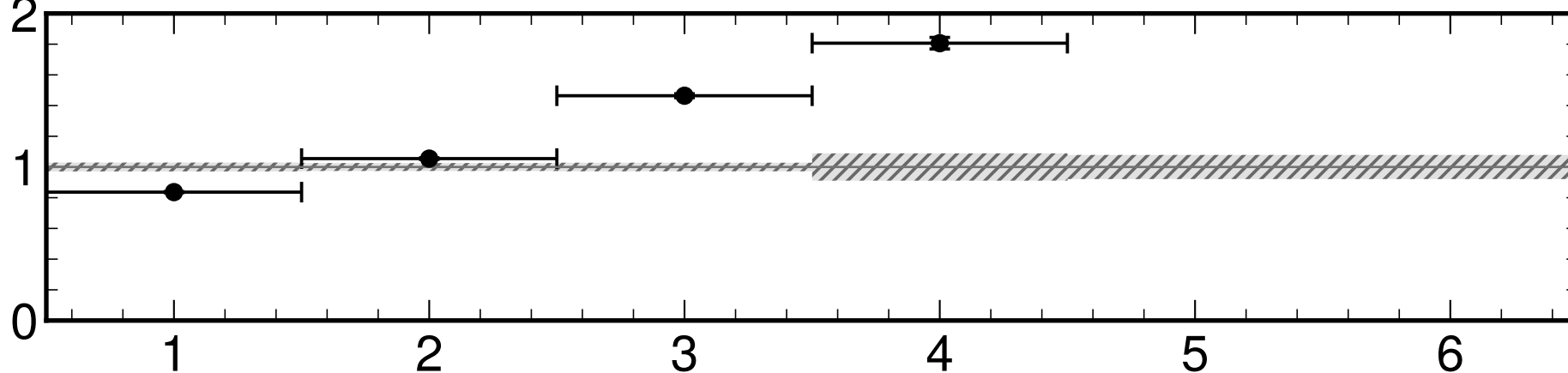
CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

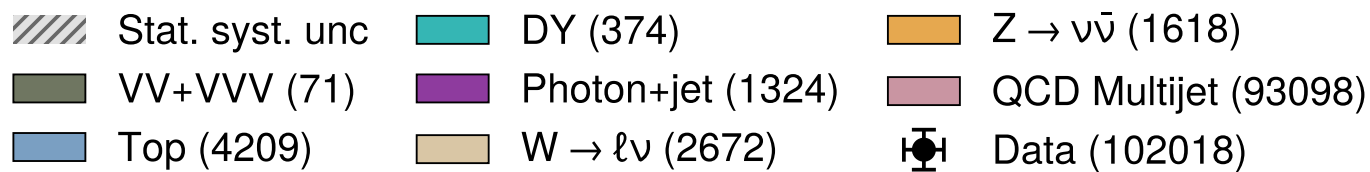


N_b (loose WP)

CMS *Work in progress*

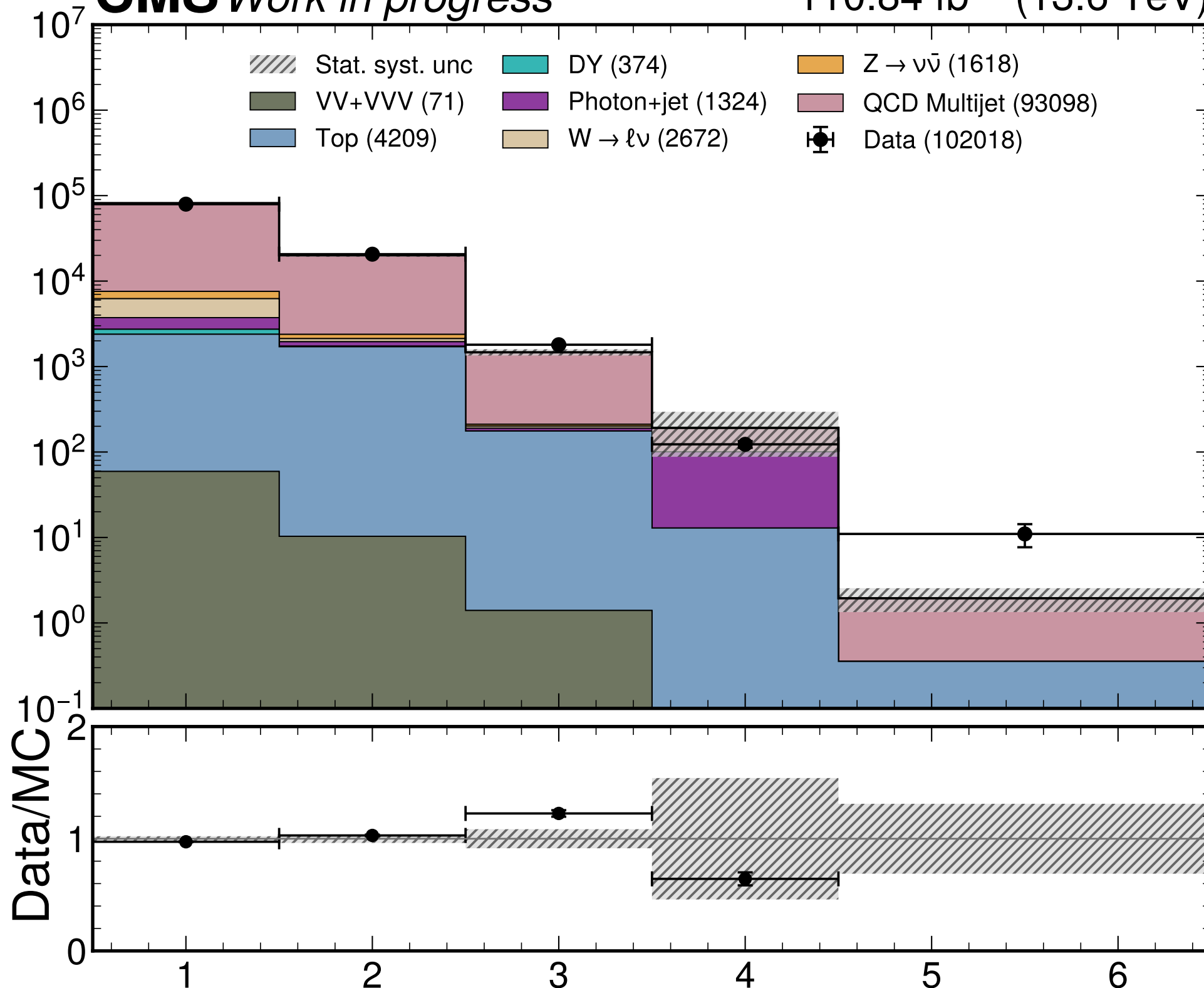
110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

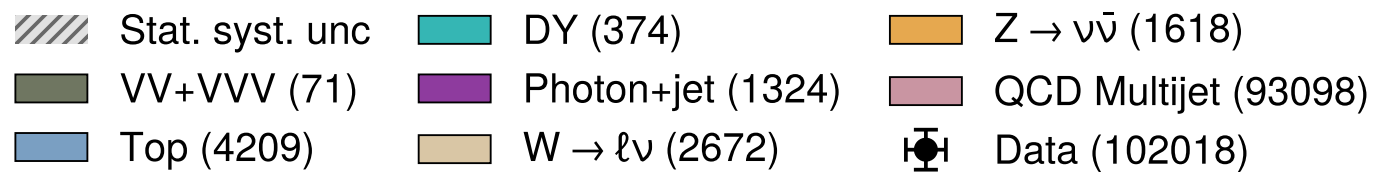
N_b



CMS *Work in progress*

110.84 fb⁻¹ (13.6 TeV)

Events



Data/MC

N_j

